Timesintelli Storm SOC

Design SPEC

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Revision History

Version Number	Date	Summary
0.1	2023/6/1	Initial revision.
0.2	2023/08/10	1. Add bus architecture info 2. Update Clock&Reset structure diagrams. 3. Change usb address 4. Add effuse architecture 5. Update twod and vo module content 6. Update Audio CODEC content 7. Add LDO content 8. Update PMU content 9. Fix typos.

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1. Product Introduction

1.1. Overview

The Storm SOC chip is a high integrity, high performance and ultra-low power consumption SOC chip based on high performance processor CPU targeting for a variety of mid-end camera/vision applications. It integrates the MCU subsystem, Video subsystem, as well as the Audio Subsystem. Besides, a full range of peripherals ensure the various camera/vision application requirements.

1.2. Features

- Operating condition
- ○ Ambient operating temperature: -40℃ ~ 85℃
- ○ Junction temperature: -40℃ ~ 125℃
- ○ ESD: 2kV (HBM)
- ○ Latch-up immunity: 100 mA
- ○ Power consumption
- Peak power <1W @ 85℃
- Deep sleep current (RTC + wake-up IO) < 50uA
- Process & Package:
- ○ SMIC 40nm Low-leakage process, 1P7M
- ○ Package 1： QFN100, body size: 12*12, pin pitch 0.4 mm. SiP with DDR2 or LPDDR2 SDRAM KGD
- ○ Package 2： QFN128, body size: 12.3*12.3 TBD, pin pitch 0.35 mm. SiP with DDR2 SDRAM KGD， without MIPI DSI interface
- ○ Package 3： QFN128M, body size: 12.3*12.3 TBD, pin pitch 0.35 mm. SiP with DDR2 SDRAM KGD， with MIPI DSI interface
- Power supply
- ○ Independent digital supply is available through DVDD11* (1.0V ~ 1.2V) pin for digital core for DVFS
- ○ Independent analog supply is available through DVDD33* (1.7V ~ 3.6V) pin for better noise filtering and shielding. Typical power supply on the PCB will be 1.8V, 2.5V and 3.3V.
- ○ Separate multiple power domains and different logical power modes
- Clock scheme
- ○ Built-in 12MHz oscillators that can work with an external crystal ,
- 12MHz as reference clock for main clock
- 32768Hz for low frequency operation, e.g. RTC
- ○ 3 PLLs for internal high-speed clock generation
- ○ Course-grained and fine-grained clock gating, from cluster level to subsystem level and block level

- System reset sources
- Built-in Power-On Reset (POR)
- External low-level valid Reset pin
- Watchdog Reset
- Software Reset
- Exit from stand-by mode
- Functional features
- CPU subsystem
- High performance RISC processor
- Up to 600MHz
- SIMD and FPU support
- Cache
- 16KB L1 Instruction Cache & 16KB L1 Data Cache
- Support Linux & RTOS.
- 32KB Boot ROM(Mask ROM)
- Interrupt Control: Generic Interrupt Controller
- System Debug: Core-sight subsystem.
- Memory subsystem
- 64KB on-chip System SRAM
- DRAM
- Support SiP DDR2 or LPDDR2 KGD in QFN package
- Operating range of 200Mbps to 1066Mbps
- 16-bit data-bus width, support 512Mbits SDRAM
- Built-in True RNG
- Compliant with NIST SP800-90c and BSI AIS 20/31
- 128-bit random number generation
- Used for ECC encrypt application that require a True Random number as well as interface scramble generation
- Audio/voice subsystem
- Audio codec
- Audio ADC
- Support 1channel ADC, the channel is programmable to work in single-ended or differential configurations.
- 1 channel support line-in for reference audio input
- 1 channel support Analog MIC input with PGA for each channel from 0 ~ 20dB, 4dB/step
- 24-bit Sigma-delta structure
- Support 8kHz ~ 96 kHz sampling rate
- SNR: 97 dB (A-weighted), THDN: -90dB@16KSPS, -3dBFs
- Audio DAC
- 2 channels 16bit/24bit with line outputs, support stereo/mono mode
- 2 channels with Stereo/Mono Mode, +6dB~-54dB, 2dB step
- SNR: 100 dB (A-weighted), THDN: -90dB@-3dBFs
- Support 8 ~ 96kHz sampling rate
- Master clock: 4 ~ 27MHz or 256*fs, with embedded fractional PLL for generation

- System peripherals
- DMA
- 8 independent channels, single bus master interfaces
- Support various transfer types between memory and peripherals
- Support peripheral hardware handshake interfaces, the allocation of the handshake interfaces and peripherals are software configurable
- Support 3 interrupts of GPIO/Timer interrupts as a trigger of DMA data transfer
- Support multi-block data transfer, the max block size is 4095 in terms of source transfer width
- Support burst transaction size up to 64
- Timer/PWM
- Support 12 independent 32-bit timer
- Each timer is configured to count up or count down
- Support 3 working mode: normal timer mode, capture mode and compare mode;
- Configurable counting resolution by pre-scaler
- Independent trigger input for timer to work in capture mode
- Select-able clock source: bus clock, OSC clock or RTC clock
- Watch-dog Timer
- 32-bit timer to prevent system lock-up
- A system reset is generated upon timer expiration
- Programmable divider clock source
- RTC
- Support real-time date (Day-Month-Year-Day of week) and time (Hours-Minutes-Seconds)
- Programmable alarm that can be used to generate internal interrupts for CPU wake-up or event notification
- +/- 5ppm of 32768Hz with calibration
- External peripherals
- USB 2.0 * 1
- Support following speed modes
- High-speed (480 Mbps)
- Full-speed (12 Mbps)
- Low-speed (1.5 Mbps)
- Support OTG 1.3 (SRP and HNP)
- Support 8 end-points (including EP0)
- Support all transfer modes
- control transfer
- bulk transfer
- isochronous transfer
- interrupt transfer
- Support 8 internal DMA channels
- SDIO/SD/eMMC * 2
- 1/4-bit MMC/SD/SDIO modes
- Complies with MMC 4.3, SD2.0 and SDIO 2.0 specifications
- Up to 50MHz data transfer rate
- Support card detect and write protection

- Integrated DMA controller
- SPI * 2
- X1 interface, support up to 2 chip selects
- Support master/slave mode
- Max master clock frequency is 60MHz
- Support configurable clock parity and phase
- Support 4-32bit transfer word width
- UART * 4
- 2 UART support 4-lines including hardware RTS/CTS flow control, while the other 2 support 2-line operation
- Baudrate from 300bps to 3.6864 Mbps
- Programmable serial interface characteristics
- 5/6/7/8 bit characters
- Even, odd, mark, space, or no parity bit generation and detection
- 1/1.5/2 bit stop bit generation
- Support false start bit detection and line break generation/detection
- I2C * 2
- Complies with Philips I2C specification version 2.1
- Support master mode and slave mode
- Support standard mode (100Kb/s) and fast mode (up to 400Kb/s)
- Support 7-bit and 10-bit device addressing modes
- EMAC
- Support 10Mbps/100Mbps
- Support MII/RMII interface
- Provide 50MHz RMII reference clock output to save BOM cost
- GPIO
- Fail-safe IO terminals
- Fully software configurable
- have pull-up and pull-down resistors enabled
- output or input pins
- Open-drain or push-pull
- Support to be used as interrupt inputs
- SAR ADC
- 8 channel, 12-bit resolution
- 2M sample rate
- Support 1.8V or 3.3V reference voltage
- Support voltage/temperature measurement
- Support DMA operation
- Multiple camera interfaces(1MIPI/1DVP)
- Support MIPI RX
- 1 2-lane MIPI D-PHY interface , up to 900Mbit/s per lane
- RAW8,RAW10, RAW12 data type parsing
- YUV420, YUV420(Legacy), YUV420(CSPS), YUV422 of 8-bits and 10-bits
- Maximum input 1296P@60fps@2lane

- Support DVP Interface
- Support BT.601,BT.656,BT.1120 protocol
- Support YUV422 (8bit) data format
- Compatibility with mainstream HD CMOS sensors provided by Sony, OmniVision, SmartSense and Galaxycore etc.
- ISP
- Maximum input 1296P@30fps
- Test Pattern Generator(TPG)
- Black Level Compensation (BLC)
- Defect Pixel Correction (DPC)
- Lens Shading Correction (De-Vignetting)
- Digital Gain
- 2 Stage Adaptive Noise Filter
- Enhanced Color Interpolation (Demosaic)
- Chromatic Aberration Correction (CAC)
- Color Correction Matrix (CCM)
- Color Space Conversion (CSM)
- Gamma Correction
- Auto Focus Measurement (AF)
- Auto White Balancing (AWB)
- Auto Exposure Measurement (AE)
- Histogram Calculation
- Anti-flicker
- Cropping
- Color Processing (Contrast, Saturation, Brightness, Hue) (CPROC)
- Sharpening/Blurring Filter
- Wide Dynamic Range Tone Mapping (WDR)
- Multi-exposure HDR(DOL/Stagger, Stagger output)
- Advanced 2DNR to remove spatial noise
- 3DNR to remove temporal noise
- H.264/JPEG encoding/decoding
- Up to 1080P@60fps image processing ability
- Support multi-stream encoding/decoding
- Efficient customized 2-D accelerator
- Up/Down Scale
- Support DPI interface
- YUV4:2:0/YUV 4:2:2 input
- OSD layer alpha-bending with scale layer
- YUV4:2:0/RGB888/RGBA output
- Fill and Copy
- 90,180,270 rotation
- Draw Rectangle
- LCD display controller
- Display interface

- Support MIPI DSI interface with 2lanes, up to 1.2Gbit/s per lane
- Support parallel 24-bit RGB interface, up to 1080x1920@30fps
- Support 8-bit 8080 mcu interface
- Support serial 8-bit RGB interface

○ Display control

- Support up to 4 planes
- Support YUV420/YUV422/RGB24 data format input
- Layer0 & Layer1 & Layer2

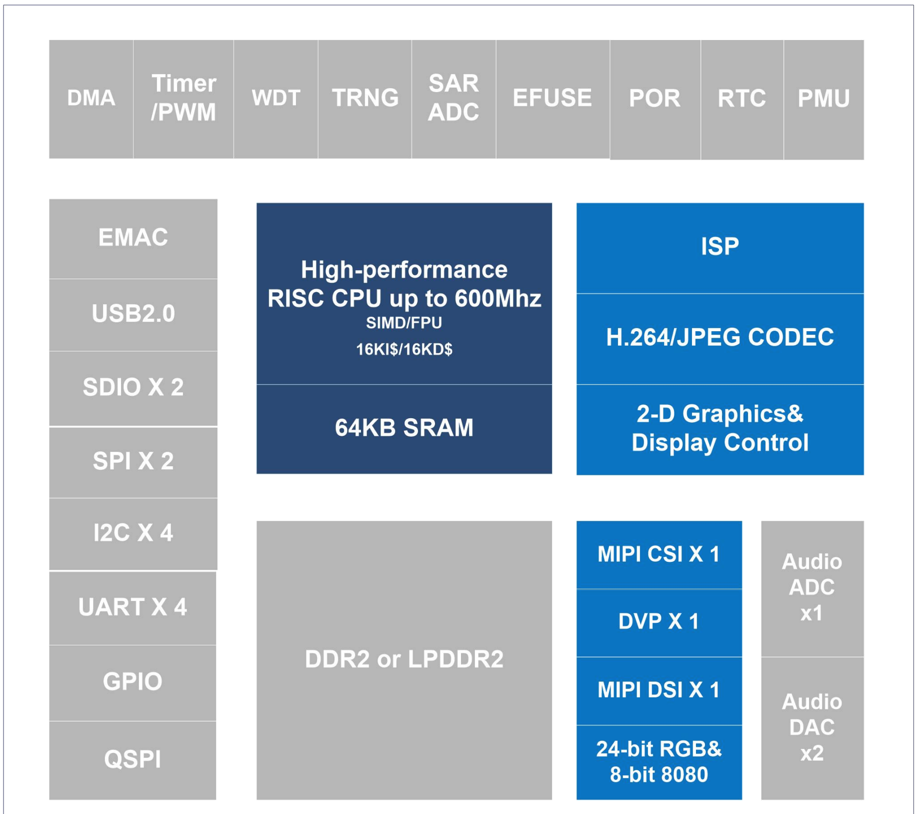
○ Support YUV420/YUV422 data format input

○ Support configurable start pixel and start line

- OSD layer

○ Support ARGB, RGB-24bit, RGB-16bit, ARGB-32bit, ARGB4444, ARGB1555, Monochrome

○ OSD layer support configurable global alpha blending or pixel by pixel alpha blending



Storm SOC Block Diagram

Figure 1- 1 Storm block diagram

1.3. Memory Map

Below is the system memory map.

Table 1- 1 System Memory Map

Block	Start Address	End Address	Size(Byte)
ROM	0x0000_0000	0x0000_7FFF	32K
GIC	0x0040_0000	0x0040_2FFF	12K
DAP	0x0040_3000	0x0041_2FFF	64K
L2SRAM	0x1000_0000	0x1000_FFFF	64K
QSPI	0x3000_0000	0x30FF_FFFF	16M
Reserved	0x4010_0000	0x401F_FFFF	1M
SDIO0	0x4020_0000	0x402F_FFFF	1M
SDIO1	0x4030_0000	0x403F_FFFF	1M
DMAC	0x4040_0000	0x404F_FFFF	1M
EMAC	0x4050_0000	0x405F_FFFF	1M
I2C0	0x9000_0000	0x9000_0FFF	4K
I2C1	0x9000_1000	0x9000_1FFF	4K
SPI0	0x9000_2000	0x9000_2FFF	4K
SPI1	0x9000_3000	0x9000_3FFF	4K
UART0	0x9000_4000	0x9000_4FFF	4K
UART1	0x9000_5000	0x9000_5FFF	4K
UART2	0x9000_6000	0x9000_6FFF	4K
UART3	0x9000_7000	0x9000_7FFF	4K
SARADC	0x9000_8000	0x9000_8FFF	4K
DDRPHY PUBL	0x9000_9000	0x9000_9FFF	4K
QSPI	0x9000_A000	0x9000_AFFF	4K
I2C2	0x9000_D000	0x9000_DFFF	4K
I2C3	0x9000_E000	0x9000_EFFF	4K
DDR Ctrl	0x9001_0000	0x9001_2FFF	12K
RTC/PMU	0x9100_0000	0x9100_0FFF	4K
GPIO	0x9100_1000	0x9100_1FFF	4K
WDT	0x9100_2000	0x9100_2FFF	4K
TRNG	0x9100_3000	0x9100_3FFF	4K
Reserved	0x9100_4000	0x9100_4FFF	4K
SYSCTRL	0x9100_5000	0x9100_5FFF	4K
TIMER	0x9100_6000	0x9100_6FFF	4K
Efuse	0x9100_8000	0x9100_FFFF	32K
GPIO2	0x9101_0000	0x9101_0FFF	4K
GPIO3	0x9101_1000	0x9101_1FFF	4K
AUDIO CODEC	0x9201_0000	0x9201_0FFF	4K
I2S_ADC	0x9201_1000	0x9201_1FFF	4K
I2S_DAC	0x9201_2000	0x9201_2FFF	4K
Reserved	0x9201_3000	0x9201_3FFF	4K
Reserved	0x9201_4000	0x9201_4FFF	4K
2D	0x9300_0000	0x9300_EFFF	60K
VSSC	0x9300_F000	0x9300_FFFF	4K

VO	0x9301_0000	0x9301_EFFF	60K
Reserved	0x9301_F000	0x9301_FFFF	4K
CSIx2	0x9302_0000	0x9302_0FFF	4K
Video_IN0_0	0x9302_1000	0x9302_1FFF	4K
Video_IN1_0	0x9302_3000	0x9302_3FFF	4K
DSIx2	0x9302_5000	0x9302_5FFF	4K
VPU_Encoder	0x9302_4000	0x9302_4FFF	4K
VPU_Decoder	0x9302_F000	0x9302_FFFF	4K
ISP	0x9303_0000	0x9303_FFFF	64K
USB	0xA000_0000	0xA000_FFFF	64K
DDR	0xD000_0000	0xD3FF_FFFF	64M

1.4. Boot Mode

At startup, pin of BOOT_SEL1 and BOOT_SEL0 are sampled and used to select boot order from different media:

Table 1-2 Boot order definition

{BOOT_SEL1, BOOT_SEL0}	Boot order
0x0	UART-> NOR-> NAND-> EMMC
0x1	NOR->NAND-> EMMC->UART
0x2	NAND-> EMMC->UART-> NOR
0x3	EMMC->UART-> NOR-> NAND

The boot loader is located in boot ROM.

1.5. Interrupt Vector Table

Table 1-3 Interrupt vector table

Interrupt Number	Vector Address	Attribute	Description
0	0x00	Maskable	DMA_INT
1	0x04	Maskable	SDIO0_INT
2	0x08	Maskable	SDIO1_INT
3	0x0C	Maskable	USB_INT
4	0x10	Maskable	USB_DMAINT
5	0x14	Maskable	Reserved
6	0x18	Maskable	Reserved
7	0x1C	Maskable	EMAC_INT
8	0x20	Maskable	QSPI_INT
9	0x24	Maskable	I2C0_INT
10	0x28	Maskable	I2C1_INT

11	0x2C	Maskable	I2C2_INT
12	0x30	Maskable	I2C3_INT
13	0x34	Maskable	UART0_INT
14	0x38	Maskable	UART1_INT
15	0x3C	Maskable	UART2_INT
16	0x40	Maskable	UART3_INT
17	0x44	Maskable	SPI0_INT
18	0x48	Maskable	SPI1_INT
19	0x4C	Maskable	SARADC_INT
20	0x50	Maskable	Reserved
21	0x54	Maskable	Reserved
22	0x58	Maskable	Reserved
23	0x5C	Maskable	EFUSE_INT
24	0x60	Maskable	TRNG_INT
25	0x64	Maskable	WDT_INT
26	0x68	Maskable	GPIO_INT0
27	0x6C	Maskable	GPIO_INT1
28	0x70	Maskable	TMR1_INT
29	0x74	Maskable	TMR2_INT
30	0x78	Maskable	TMR3_INT
31	0x7C	Maskable	TMR4_INT
32	0x80	Maskable	TMR5_INT
33	0x84	Maskable	TMR6_INT
34	0x88	Maskable	TMR7_INT
35	0x8C	Maskable	TMR8_INT
36	0x90	Maskable	Reserved
37	0x94	Maskable	ADC_I2S RX
38	0x98	Maskable	DAC_I2S TX
39	0x9C	Maskable	I2S0_TX_INT
40	0xA0	Maskable	I2S1_RX_INT
41	0xA4	Maskable	TMR9_INT
42	0xA8	Maskable	TMR10_INT
43	0xAC	Maskable	TMR11_INT
44	0xB0	Maskable	TMR12_INT
45	0xB4	Maskable	Reserved
46	0xB8	Maskable	Reserved
47	0xBC	Maskable	RTC_INT
48	0xC0	Maskable	VPU_ENC_INT
49	0xC4	Maskable	VPU_DEC_INT
50	0xC8	Maskable	VSSC_INT
51	0xCC	Maskable	2D_FIL_INT
52	0xD0	Maskable	2D_CPY_INT
53	0xD4	Maskable	2D_TRANS_INT

54	0xD8	Maskable	ISP_2D_INT
55	0xDC	Maskable	VIN_CTRL1_INT
56	0xE0	Maskable	VIN_CTRL0_INT|MIPI_CSI_INT
57	0xE4	Maskable	VO_CTRL_INT|MIPI_DSI_INT
58	0xE8	Maskable	TD_INT
59	0xEC	Maskable	GPIO2_INT0
60	0xF0	Maskable	GPIO2_INT1
61	0xF4	Maskable	GPIO3_INT0
62	0xF8	Maskable	GPIO3_INT1
63	0xFC	Maskable	Reserved
64	0xE0	Maskable	Reserved
65	0xE4	Maskable	Reserved
66	0xE8	Maskable	Reserved
67	0xEC	Maskable	Reserved
68	0xF0	Maskable	Reserved
69	0xF4	Maskable	CTI_IRQ
70	0xF5	Maskable	A5_PMUIRQ

1.6. Bus Architecture

1.6.1. Overview

The bus architecture of Storm SOC are as following:

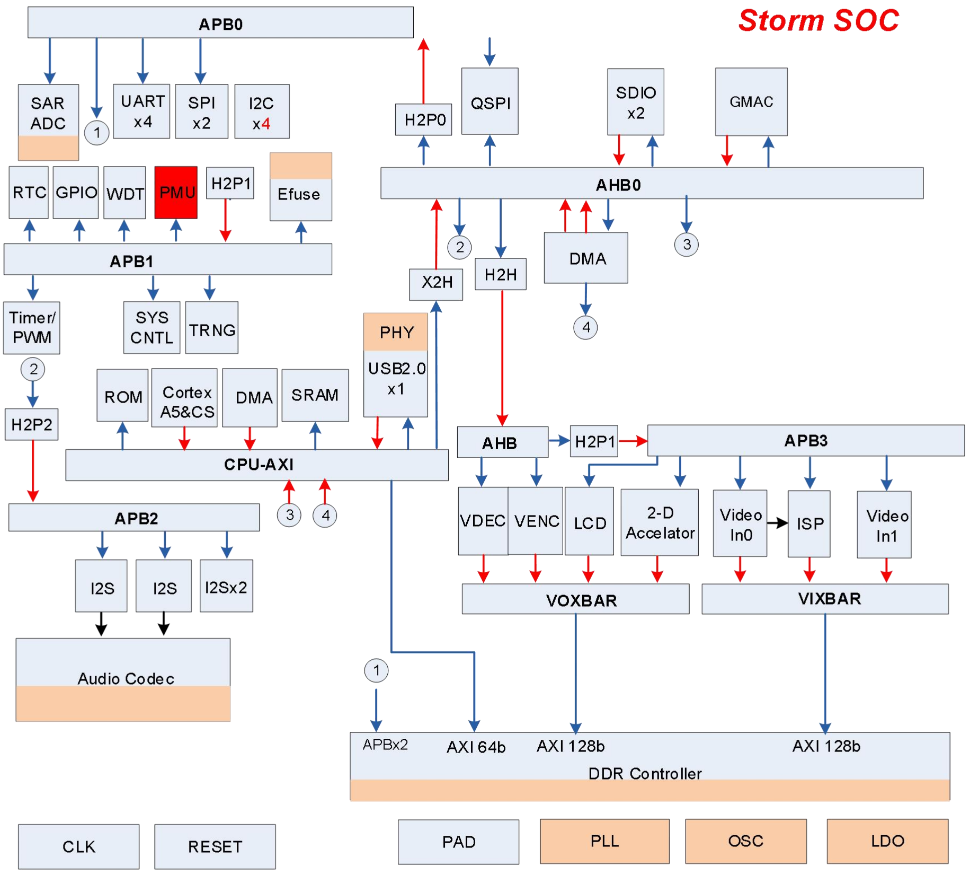


Figure 1-2 Storm Bus Architecture

1.6.2. MCU AXI Bus

1.6.2.1. MCU AXI Access Relationship

The access relationship between Masters and Slaves is as the following table

Table 1-4MCU AXI Access Relationship

	CPU Data/Instr	CPU DEBUG	AXI DMA	USB	H2X
ROM	√	√	√	√	√
DDR	√	√	√	√	√

L2 SRAM	√	√	√	√	√
X2H	√	√	√	√	

1.6.2.2. MCU AXI Memory Map

Table 1- 5MCUAXI Memory Map

Sub	Block	Start Address	End Address	Size (Byte)
AXI	ROM	0x0000_0000	0x0000_7FFF	32K
	GIC	0x0040_0000	0x0040_2FFF	12K
	DAP	0x0040_3000	0x0040_FFFF	64K
	L2SRAM	0x1000_0000	0x1000_FFFF	64K
	X2H	0x3000_0000	0x9400_0000	1632M
	USB	0xA000_0000	0xA000_FFFF	64K
	DMA	0xA100_0000	0xA10F_FFFF	1M
	DDR	0xD000_0000	0XEFFF_FFFF	512M

1.6.3. MCU AHB0 Bus

1.6.3.1. MCU AHB0 Access Relationship

The access relationship between Masters and Slavers is as the following table

Table 1- 6MCU AHB0 Access Relationship

			DMAC (M0)	SDIO0	SDIO1		CPU X2H	DBG0	DBG1
H2P0			√	√	√		√	√	√
H2P1							√	√	√
H2P2			√				√	√	√
H2P3			√				√	√	√
H2H(AHB1)				√	√		√	√	√
QSPI			√	√	√		√	√	√
USB							√	√	√
SDIO0							√	√	√
SDIO1							√	√	√
DMAC							√	√	√

1.6.3.2. MCU AHB0 Memory Map

Table 1- 7MCU AHB0 Memory Map

Sub	Block	Start Address	End Address	Size (Byte)
ROM	ROM	0x0000_0000	0x0000_7FFF	32K
	QSPI	0x3000_0000	0x30FF_FFFF	16M
	SDIO0	0x4020_0000	0x402F_FFFF	1M
	SDIO1	0x4030_0000	0x403F_FFFF	1M
	DMAC	0x4040_0000	0x404F_FFFF	1M
	EMAC	0x4050_0000	0x405F_FFFF	1M
	H2P0	0x9000_0000	0x9001_FFFF	128K
	H2P1	0x9100_0000	0x911F_FFFF	2M
	H2P2	0x9200_0000	0x9201_FFFF	128K
	H2H(Video)	0x9300_0000	0x9303_FFFF	256K
	DDR	0xD000_0000	0xEFFF_FFFF	512M

1.6.4. MCU APB Bus

1.6.4.1. MCU APB0 Memory Map

Table 1- 8MCU APB0 Memory Map

Sub	Block	Start Address	End Address	Size (Byte)
APB0	I2C0	0x9000_0000	0x9000_0FFF	4K
	I2C1	0x9000_1000	0x9000_1FFF	4K
	SPI0	0x9000_2000	0x9000_2FFF	4K
	SPI1	0x9000_3000	0x9000_3FFF	4K
	UART0	0x9000_4000	0x9000_4FFF	4K
	UART1	0x9000_5000	0x9000_5FFF	4K
	UART2	0x9000_6000	0x9000_6FFF	4K
	UART3	0x9000_7000	0x9000_7FFF	4K
	SARADC	0x9000_8000	0x9000_8FFF	4K
	PUBL	0x9000_9000	0x9000_9FFF	4K
	QSPI	0x9000_A000	0x9000_AFFF	4K
	I2C2	0x9000_D000	0x9000_DFFF	4K
	I2C3	0x9000_E000	0x9000_EFFF	4K
	DDRC	0x9001_0000	0x9001_2FFF	12K

1.6.4.2. MCU APB1 Memory Map

Table 1- 9MCU APB1 Memory Map

Sub	Block	Start Address	End Address	Size (Byte)
APB1	RTC/PMU	0x9100_0000	0x9100_0FFF	4K
	GPIO	0x9100_1000	0x9100_1FFF	4K
	WDT	0x9100_2000	0x9100_2FFF	4K
	TRNG	0x9100_3000	0x9100_3FFF	4K
	SYSCTRL	0x9100_5000	0x9100_5FFF	4K
	TIMER	0x9100_6000	0x9100_6FFF	4K
	Efuse CTRL	0x9100_8000	0x9100_FFFF	32K
	GPIO2	0x9101_0000	0x9101_0FFF	4K
	GPIO3	0x9101_1000	0x9101_1FFF	4K

1.6.5. Audio APB Bus

Table 1- 10Audio APB2 Memory Map

Sub	Block	Start Address	End Address	Size (Byte)
	AUDIO-CODEC	0x9201_0000	0x9201_0FFF	4K
	I2S_ADC	0x9201_1000	0x9201_1FFF	4K
	I2S_DAC	0x9201_2000	0x9201_2FFF	4K
	I2S0	0x9201_3000	0x9201_3FFF	4K
	I2S1	0x9201_4000	0x9201_4FFF	4K

1.6.6. Video AHB Bus

Table 1- 11Video AHB Memory Map

Sub	Block	Start Address	End Address	Size (Byte)
AHB	TD	0x9300_0000	0x9300_EFFF	60K
	VSS_CONFIG	0x9300_F000	0x9300_FFFF	4K
	VO	0x9301_0000	0x9301_EFFF	60K
	UVC	0x9301_F000	0x9301_FFFF	4K
	H2P	0x9302_0000	0x9302_EFFF	60K
	VPU_Decoder	0x9302_F000	0x9302_FFFF	4K
	ISP	0x9303_0000	0x9303_FFFF	64K

1.6.7. Video APB Bus

Table 1- 12Video APB Memory Map

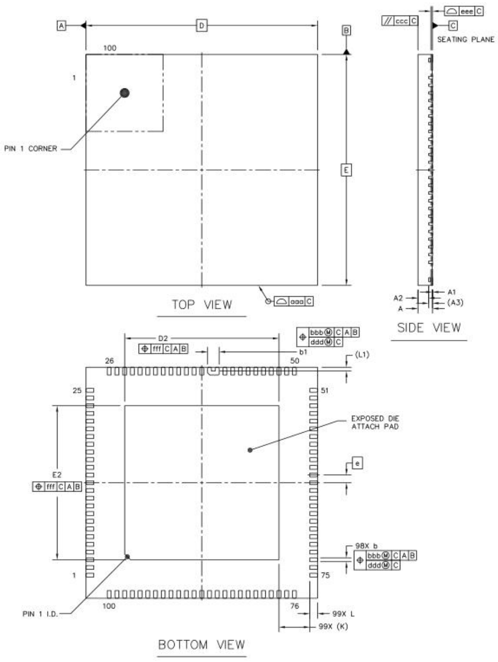
Sub	Block	Start Address	End Address	Size (Byte)
APB	CSIx2	0x9302_0000	0x9302_0FFF	4K
	Video_IN0_0	0x9302_1000	0x9302_11FF	0.5K
	Video_IN0_1	0x9302_1200	0x9302_1FFF	3.5K
	Video_IN1_0	0x9302_2000	0x9302_21FF	0.5K
	Video_IN1_1	0x9302_2200	0x9302_2FFF	3.5K
	DSIx2	0x9302_3000	0x9302_3FFF	4K
	VPU_Encoder	0x9302_4000	0x9302_4FFF	4K

2. Package and Pinout Information

Storm is currently available in package of QFN100，QFN128L，QFN128

2.1. Pacakge 1：QFN100

Package body size: (12x12x0.7mm)
Pin Pitch:　(0.203mm)



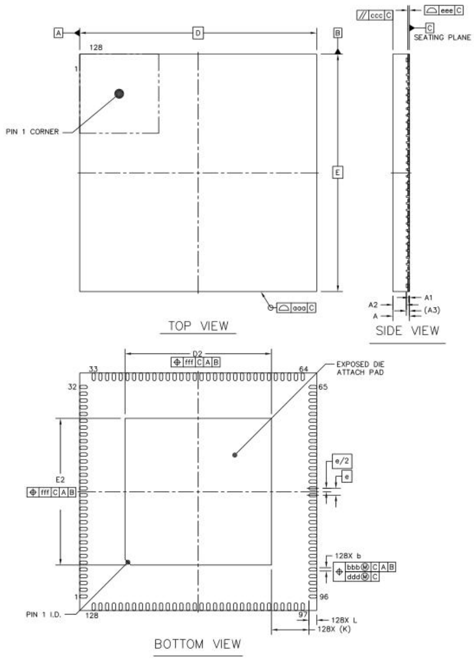
		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.7	0.75	0.8
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.55	---
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.15	0.2	0.25
		b1	0.55	0.6	0.65
BODY SIZE	X	D		12 BSC	
	Y	E		12 BSC	
LEAD PITCH		e		0.4 BSC	
EP SIZE	X	D2	7.9	8	8.1
	Y	E2	7.9	8	8.1
LEAD LENGTH		L	0.3	0.4	0.5
LEAD EDGE TO PACKAGE EDGE		L1		0.2 REF	
LEAD TIP TO EXPOSED PAD EDGE		K		1.6 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
LEAD OFFSET		bbb		0.07	
		ddd		0.05	
EXPOSED PAD OFFSET		fff		0.1	

NOTES
1.REFER TO JEDEC MO-220;
2.COPLANARITY APPLIES TO LEADS, CORNER LEADS AND DIE ATTACH PAD;
3.BAN TO USE THE LEVEL 1 ENVIRONMENT-RELATED SUBSTANCES OF JCET PRESCRIBING;
4.FINISH: Cu/EP • Sn8~20s

Figure 2- 1 Package outline and mechanical data of QFN100 package

2.2. Pacakge 2：QFN128 without MIPI DSI

Package body size: (12.3x12.3x0.85mm)
Pin Pitch:　(0.152)



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.7	---
L/F THICKNESS		A3	0.152 REF		
LEAD WIDTH		b	0.12	0.17	0.22
BODY SIZE	X	D	12.3 BSC		
	Y	E	12.3 BSC		
LEAD PITCH		e	0.35 BSC		
EP SIZE	X	D2	7.5	7.6	7.7
	Y	E2	7.5	7.6	7.7
LEAD LENGTH		L	0.3	0.4	0.5
LEAD TIP TO EXPOSED PAD EDGE		K	1.95 REF		
PACKAGE EDGE TOLERANCE		aaa	0.1		
MOLD FLATNESS		ccc	0.1		
COPLANARITY		eee	0.08		
LEAD OFFSET		bbb	0.07		
		ddd	0.05		
EXPOSED PAD OFFSET		fff	0.1		

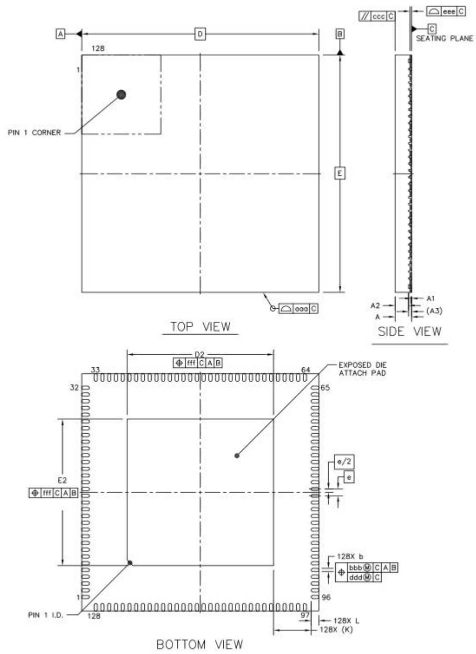
NOTES
1.REFER TO JEDEC MO-220;
2.COPLANARITY APPLIES TO LEADS, CORNER LEADS AND DIE ATTACH PAD;
3.BAN TO USE THE LEVEL 1 ENVIRONMENT-RELATED SUBSTANCES OF JCET PRESCRIBING;
4.FINISH: Cu/EP • Sn8~20s

Figure 2- 2 Package outline and mechanical data of QFN128 package

2.3. Pacakge 3：QFN128 with MIPI DSI

Package body size: (12.3x12.3x0.85mm)

Pin Pitch:　(0.152)



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.7	---
L/F THICKNESS		A3	0.152 REF		
LEAD WIDTH		b	0.12	0.17	0.22
BODY SIZE	X	D	12.3 BSC		
	Y	E	12.3 BSC		
LEAD PITCH		e	0.35 BSC		
EP SIZE	X	D2	7.5	7.6	7.7
	Y	E2	7.5	7.6	7.7
LEAD LENGTH		L	0.3	0.4	0.5
LEAD TIP TO EXPOSED PAD EDGE		K	1.95 REF		
PACKAGE EDGE TOLERANCE		aaa	0.1		
MOLD FLATNESS		ccc	0.1		
COPLANARITY		eee	0.08		
LEAD OFFSET		bbb	0.07		
		ddd	0.05		
EXPOSED PAD OFFSET		fff	0.1		

NOTES
1.REFER TO JEDEC MO-220;
2.COPLANARITY APPLIES TO LEADS, CORNER LEADS AND DIE ATTACH PAD;
3.BAN TO USE THE LEVEL 1 ENVIRONMENT-RELATED SUBSTANCES OF JCET PRESCRIBING;
4.FINISH: Cu/EP • Sn8~20s

Figure 2- 3 Package outline and mechanical data of QFN package

2.4. Pinout information

Table 2- 1 List of QFN package pin

Pin number			Pin name	Pin type	I/O structure	Ball functions	
QFN128	QFN128M	QFN100				Alternate functions	Additional functions
1	1	1	CODEC_AVDD	A		-	-
2	2	2	CODEC_VMID	D		-	-
3	3	3	CODEC_ROUT	A		-	-
4	4	4	XTAL32K_IN	A		-	-
5	5	5	XTAL32K_OUT	A		-	-
6	6	6	PMU_XTALIN	A		-	-
7	7	7	PMU_XTALOUT	A		-	-
8	8	8	PMU_VDDIO	P		-	-
9	9	9	UART0_RXD	D		UART0_RXD,GPIO3[11],TRIG1, I2C3_SDA	-
10	10	10	UART0_TXD	D		UART0_TXD,GPIO3[10], LCD_RESET,DRVVBUS_USB0, I2C3_SCL	-
11	11	11	PWM1	D		PWM1,GPIO3[9],I2C1_SDA	-
12	12	12	PMU_VDD	P		-	-
13	13	13	PWM0	D		PWM0,GPIO3[8],I2C1_SCL	-
14	14	14	LDOEN	D		-	-
15	15	15	RESETN	D		-	-
16	16	16	TEST_MODE	D		-	-
17	17	17	DVDD	P		-	-
18	18	18	JTAG_TDO	D		JTAG_TDO,SPI0_MOSI, PWM5,GPIO3[16]	-
19	19	19	JTAG_TMS	D		JTAG_TMS,SPI0_MISO, PWM6,GPIO3[14]	-
20	20	20	JTAG_TDI	D		JTAG_TDI,SPI0_SCLK, PWM7,GPIO3[15]	-
21	21	21	JTAG_TRSTN	D		JTAG_TRSTN,SPI0_CSN0, UART3_RXD,GPIO3[12]	-
22	22	22	JTAG_TCK	D		JTAG_TCK,SPI0_CSN1, UART3_TXD,GPIO3[13]	-
23	23	-	QSPI_DIO0	D		QSPI_DIO0	-
24	24	-	QSPI_CLK	D		QSPI_CLK	-
25	25	-	QSPI_DIO1	D		QSPI_DIO1	-
26	26	-	QSPI_CSN0	D		QSPI_CSN0	-
27	27	-	QSPI_DIO2	D		QSPI_DIO2	-

28	28	-	QSPI_DIO3	D		QSPI_DIO3	-
29	29	23	VDDIO_A	P		-	-
30	30	24	SDIO0_CLK	D		SDIO0_CLK,GPIO3[3], I2C1_SDA,UART3_RTSN, EMAC0_REFCLK	-
31	31	25	SDIO0_CMD	D		SDIO0_CMD,GPIO3[2], JTAG_TCK,SPI1_CSN0, UART1_TXD,EMAC0_TXD[0]	-
32	32	26	SDIO0_DATA0	D		SDIO0_DATA[0],GPIO3[4], JTAG_TMS,SPI1_CSN1, UART1_RXD,EMAC0_TXD[1]	-
33	33	27	SDIO0_DATA1	D		SDIO0_DATA[1],GPIO3[5], JTAG_TDO,SPI1_MISO, EMAC0_TXEN	-
34	34	28	SDIO0_DATA2	D		SDIO0_DATA[2],GPIO3[0], JTAG_TDI,SPI1_MOSI, EMAC0_MDC	-
35	35	29	SDIO0_DATA3	D		SDIO0_DATA[3],GPIO3[1], JTAG_TRSTN,SPI1_SCLK, EMAC0_MDIO	-
36	36	30	GPIOE	D		GPIO3[24],SPI1_SCLK,PWM4, EMAC0_RXDV	-
37	37	31	GPIOD	D		GPIO2[31],EMAC0_TXEN, SPI1_CSN0,PWM3,I2C2_SCL, EMAC0_RXD[0]	-
38	38	32	GPIOC	D		GPIO2[30],EMAC0_COL, SPI1_CSN1,PWM2,I2C2_SDA, EMAC0_RXD[1]	-
39	39	33	GPIOB	D		GPIO1[31],EMAC0_RXDV, SPI1_MISO,PWM1, UART0_TXD, EMAC0_PHYOSC	-
40	40	34	GPIOA	D		GPIO1[30],EMAC0_MDC, SPI1_MOSI,PWM0, UART0_RXD	-
41	41	35	BOOT_SEL	D		BOOT_SEL0,GPIO3[17]	-
42	42	36	BOOT_SEL_1	D		BOOT_SEL1,GPIO3[21]	-
43	43	37	DDR_VDDQ	P		LCD_VSYNC,EMAC0_PHYOSC, SDIO1_WP,GPIO1[25], DP_WR,DBIB_DCX	-
44	44	38	DDR_DVDD	P		LCD_HSYNC,EMAC0_RXDV, GPIO1[26], DP_RS,DBIB_WRX	-

45	45	39	DDR_VREF0	P		LCD_DEN,SENSOR0_PWD, GPIO1[27],DP_TE,DBIB_CSX	-
46	46	40	DDR_VREF1	P		-	-
47	47	41	DDR_DVDD	P		-	-
48	48	42	DDR_VDDQ	P		-	-
49	49	43	LCD_VSYNC	D		LCD_VSYNC,EMAC0_PHYOSC, SDIO1_WP,GPIO1[25], DP_WR,DBIB_DCX	-
50	50	44	LCD_HSYNC	D		LCD_HSYNC,EMAC0_RXDV, GPIO1[26],DP_RS, DBIB_WRX	-
51	51	45	LCD_DEN	D		LCD_DEN,SENSOR0_PWD, GPIO1[27],DP_TE, DBIB_CSX	-
52	52	46	LCD_PCLK	D		LCD_PCLK,EMAC0_MDIO, SENSOR0_RSTN, SDIO1_CD,GPIO1[24], DP_CS,DBIB_RDX	-
53	-	-	VDDIO	P		-	-
54	53	-	LCD_DOUT0	D		LCD_DOUT[0],EMAC0_MDC, PWM0,GPIO1[0],SPI0_SCLK, DBIB_DATA[0]	-
55	54	-	LCD_DOUT1	D		LCD_DOUT[1],EMAC0_TXEN, PWM1,TRIG0,GPIO1[1], SPI0_MISO,DBIB_DATA[1]	-
56	55	-	LCD_DOUT2	D		LCD_DOUT[2],EMAC0_TXD[1], PWM2,SDIO1_CMD,GPIO1[2], SPI0_MOSI,DBIB_DATA[2]	-
57	56	-	LCD_DOUT3	D		LCD_DOUT[3],EMAC0_TXD[0], PWM3,SDIO1_CLK,GPIO1[3], SPI0_CSN0,DBIB_DATA[3]	-
58	57	-	LCD_DOUT4	D		LCD_DOUT[4], EMAC0_REFCLK,PWM4, SDIO1_DATA[3],GPIO1[4], DBIB_DATA[4]	-
59	58	-	LCD_DOUT5	D		,LCD_DOUT[5], EMAC0_RXD[1],PWM5, SDIO1_DATA[2], GPIO1[5],SPI0_CSN1, DBIB_DATA[5]	-
60	59	-	LCD_DOUT6	D		LCD_DOUT[6],EMAC0_RXD[0], PWM6,SDIO1_DATA[1], GPIO1[6], DBIB_DATA[6]	-

61	60	-	LCD_DOUT7	D		LCD_DOUT[7],EMAC0_RXER, PWM7,SDIO1_DATA[0], GPIO1[7], DBIB_DATA[7]	-
-	61	47	VDDIO_B	P		-	-
62	62	48	LCD_DOUT8	D		LCD_DOUT[8],EMAC0_RXD[2], SDIO1_CD,LCD_DOUT[0], GPIO1[8],DBIB_DATA[0], DBIB_DATA[8]	-
63	63	49	LCD_DOUT9	D		LCD_DOUT[9],EMAC0_RXD[3], SDIO1_WP,LCD_DOUT[1], GPIO1[9],DBIB_DATA[1], DBIB_DATA[9]	-
64	64	50	LCD_DOUT10	D		LCD_DOUT[10], EMAC0_TXD[2], SDIO1_CLK,LCD_DOUT[2], GPIO1[10],DBIB_DATA[2], DBIB_DATA[10]	-
65	65	51	LCD_DOUT11	D		LCD_DOUT[11], EMAC0_TXD[3], SDIO1_CMD,LCD_DOUT[3], GPIO1[11],DBIB_DATA[3], DBIB_DATA[11]	-
66	-	52	LCD_DOUT12	D		LCD_DOUT[12], EMAC0_TXCLK, SDIO1_DATA[0],LCD_DOUT[4], GPIO1[12],DBIB_DATA[4], DBIB_DATA[12]	
67	-	53	LCD_DOUT13	D		LCD_DOUT[13],EMAC0_TXER, SDIO1_DATA[1],LCD_DOUT[5], GPIO1[13],DBIB_DATA[5], DBIB_DATA[13]	
68	-	54	LCD_DOUT14	D		LCD_DOUT[14], EMAC0_RXCLK, SDIO1_DATA[2],LCD_DOUT[6], GPIO1[14],DBIB_DATA[6], DBIB_DATA[14]	
69	-	55	LCD_DOUT15	D		LCD_DOUT[15],EMAC0_CRS, SDIO1_DATA[3],LCD_DOUT[7], GPIO1[15],DBIB_DATA[7], DBIB_DATA[15]	
70	-	56	VDDIO	P			

71	-	-	LCD_DOUT16	D		LCD_DOUT[16], EMAC0_RXD[0], GPIO2[19], DBIB_TE	
72	-	-	LCD_DOUT17	D		LCD_DOUT[17], EMAC0_RXD[1],UART2_RXD, GPIO2[20]	
73	-	-	LCD_DOUT18	D		LCD_DOUT[18], EMAC0_TXD[0],UART2_TXD, GPIO2[21]	
74	-	-	LCD_DOUT19	D		LCD_DOUT[19], EMAC0_TXD[1],I2C3_SDA, UART1_TXD,GPIO2[22]	
75	-	-	LCD_DOUT20	D		LCD_DOUT[20], EMAC0_MDIO,I2C3_SCL, UART1_RXD,GPIO2[23]	
76	66	57	DVDD	P		-	-
77	67	58	MIPI0_DATAN0	D		-	-
78	68	59	MIPI0_DATAP0	D		-	-
79	69	60	MIPI0_AVDD	P		-	-
80	70	61	MIPI0_CLKP	D		-	-
81	71	62	MIPI0_CLKN	D		-	-
82	72	63	MIPI0_DATAN1	D		-	-
83	73	64	MIPI0_DATAP1	D		-	-
-	74	-	MIPI0_AVDD	P		-	-
84	75	64	MIPI0_REXT	D		-	-
-	76	-	MIPI2_DATAN0	D		-	-
-	77	-	MIPI2_DATAP0	D		-	-
85	78	66	MIPI2_AVDD	P		-	-
-	79	-	MIPI2_CLKP	D		-	-
-	80	-	MIPI2_CLKN	D		-	-
-	81	-	MIPI2_DATAN1	D		-	-
-	82	-	MIPI2_DATAP1	D		-	-
-	83	-	MIPI2_REXT	D		-	-
-	84	-	MIPI2_AVDD	P		-	-
86	85	67	PLL_AVDD0	P		-	-
87	86	68	PLL_AVSS0	P		-	-
88	87	-	SENSOR0_CLK	D		SENSOR0_CLK,GPIO3[20]	-
89	88	69	SENSOR0_PWD	D		SENSOR0_PWD,GPIO3[19], UART2_TXD,I2C2_SCL	-
-	-	70	SENSOR0_RSTN	D		SENSOR0_RSTN,GPIO3[18], UART2_RXD,I2C2_SDA	-
-	89	-	ADC_AIN3	A		-	-

90	90	71	ADC_AIN2	A		-	-
91	91	-	SENSOR1_CLK	D		SENSOR1_CLK,GPIO3[23]	-
92	92	72	VDDIO_C	P		-	-
93	93	73	I2C0_SDA	D		I2C0_SDA,GPIO1[29]	-
94	94	74	I2C0_SCL	D		I2C0_SCL,GPIO1[28]	-
95	95	75	PWR_SENSOR1	D		-	-
96	96	76	PWR_SENSOR0	D		-	-
97	97	77	LDO_AVDD	P		-	-
98	98	78	SAR_AIN1	A		-	SAR_AIN1
99	99	79	SAR_AIN0	A		-	SAR_AIN0
100	100	80	USB0_DVDD	P		-	-
101	101	81	USB0_VDD33	P		-	-
102	102	82	USB0_DP0	D		-	USB0_DP0
103	103	83	USB0_DM0	D		-	USB0_DM0
104	104	84	USB0_TXRTURE	D		-	-
105	105	85	USB0_VDD25	P		-	-
106	106	86	DRAM_VDDQ	P		-	-
107	107	87	DRAM_VDDQ	P		-	-
108	108	88	DRAM_VDDQ	P		-	-
109	109	89	USB0_DVDD	P		-	-
110	110	-	DVP0_VSYNC	D		DVP0_VSYNC,GPIO2[29]	-
111	111	90	DVP0_HREF	D		DVP0_HREF,GPIO2[28]	-
112	112	-	DVP0_PCLK	D		DVP0_PCLK,GPIO2[12]	-
113	113	-	DVP0_DIN0	D		DVP0_DIN[0],GPIO2[0], PWM0	-
114	114	-	DVP0_DIN1	D		DVP0_DIN[1],GPIO2[1], PWM1	-
115	115	-	DVP0_DIN2	D		DVP0_DIN[2],GPIO2[2], PWM2	-
116	116	-	DVP0_DIN3	D		DVP0_DIN[3],GPIO2[3], PWM3	-
117	117	-	DVP0_DIN4	D		DVP0_DIN[4],GPIO2[4], PWM4	-
118	118	91	DVP0_DIN5	D		DVP0_DIN[5],QSPI_DIO3, GPIO2[5],I2C1_SCL, PWM5	-
119	119	92	DVP0_DIN6	D		DVP0_DIN[6],QSPI_DIO2, GPIO2[6], I2C1_SDA,PWM6	-
120	120	93	DVP0_DIN7	D		DVP0_DIN[7],QSPI_DIO1, GPIO2[7],PWM7	-
121	121	94	VDDIO_D	P			-

122	122	95	DVP0_DIN8	D		DVP0_DIN[8],QSPI_DIO0, GPIO2[8]	-
123	123	96	DVP0_DIN9	D		DVP0_DIN[9],QSPI_CSN, GPIO2[9]	-
124	124	97	DVP0_DIN10	D		DVP0_DIN[10],QSPI_CLK, GPIO2[10],I2C2_SDA, UART3_RXD	-
125	125	-	DVP0_DIN11	D		DVP0_DIN[11],,GPIO2[11], I2C2_SCL,UART3_TXD, DP_RD	
126	126	98	CODEC_LMICP	D		-	
127	127	99	CODEC_MICBIAS	D		-	
128	128	100	CODEC_LOUT	D		-	

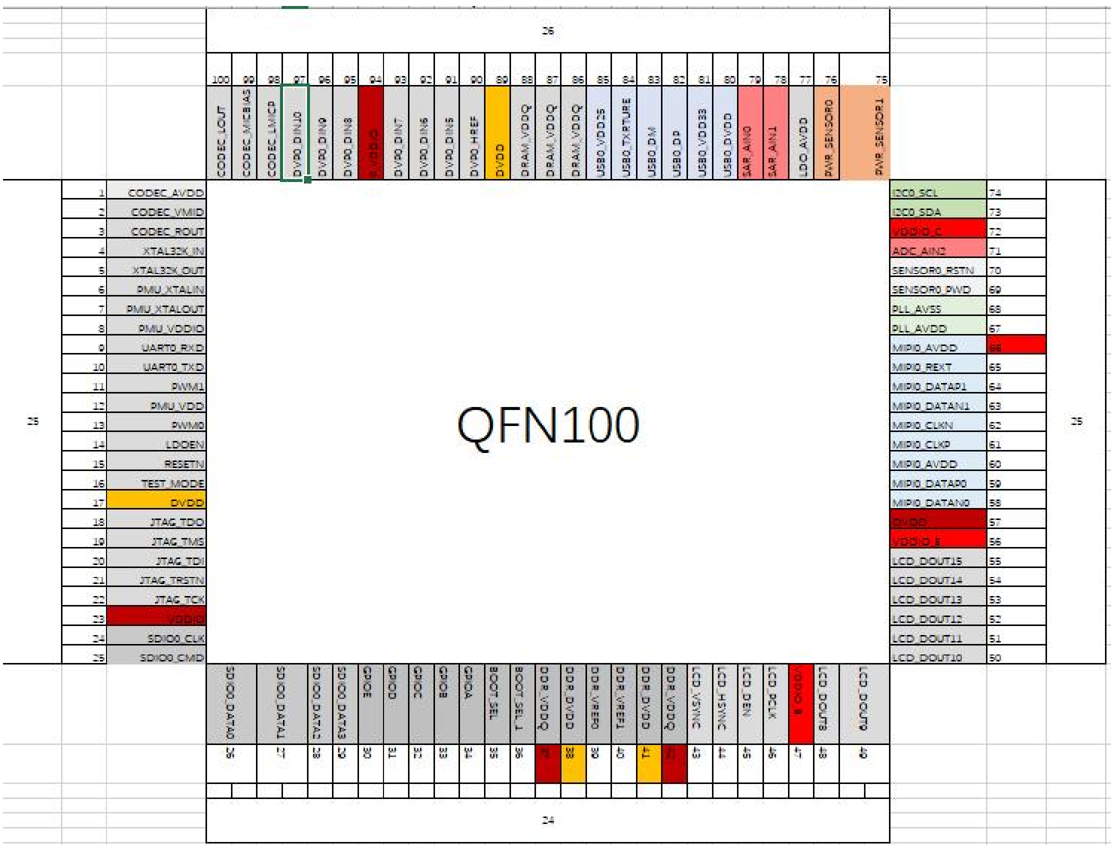


Figure 2- 2 Pin assignment(QFN100)

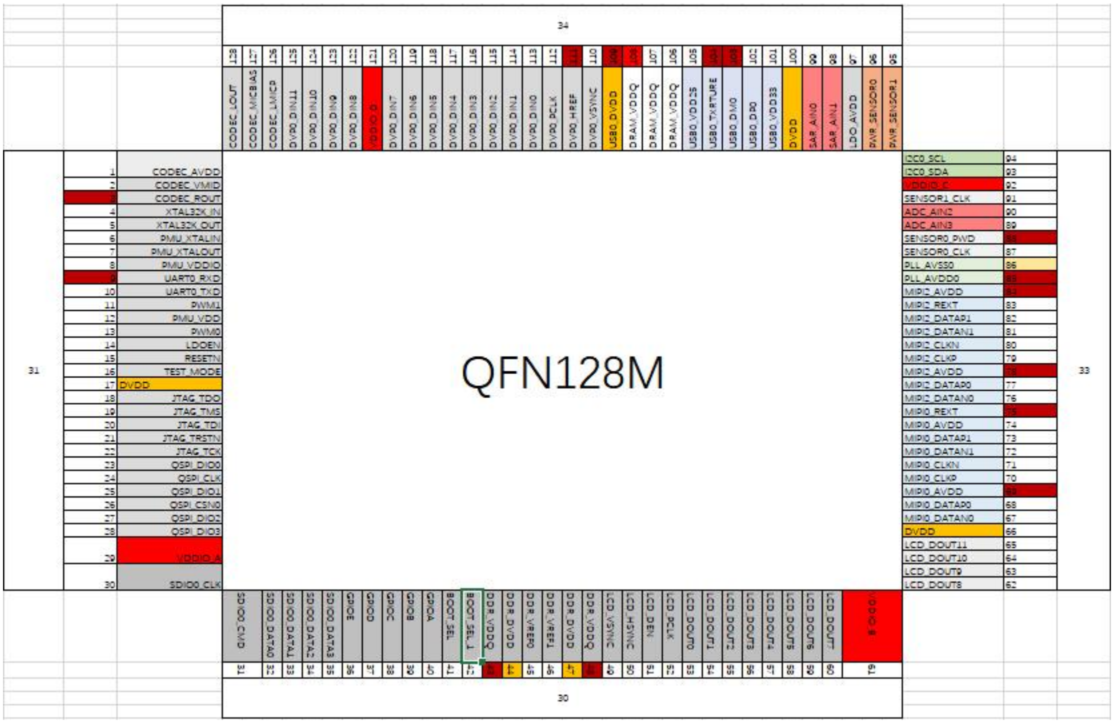


Figure 2 3 Pin assignment(QFN128M)

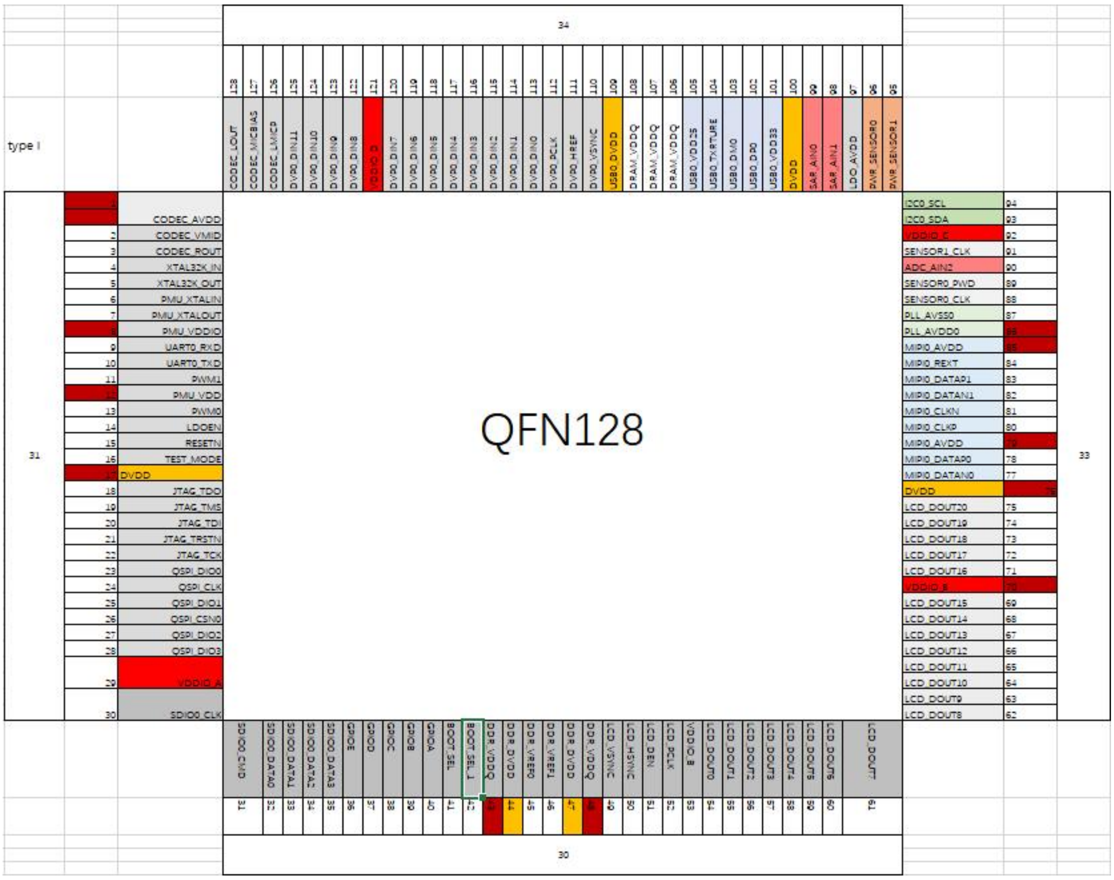


Figure 2 4 Pin assignment(QFN128)

2.5. GPIO definitions

	FUNC_1	FUNC_2	FUNC_3	FUNC_4	FUNC_5	FUNC_6
PAD_name	signal	signal	signal	signal	signal	signal
UART0_RXD	UART0_RXD	GPIO3[11]	TRIG1	-	I2C3_SDA	-
UART0_TXD	UART0_TXD	GPIO3[10]	LCD_RESET	DRVVBUS_USB0	I2C3_SCL	-
PWM1	PWM1	GPIO3[9]	-	I2C1_SDA	-	-
PWM0	PWM0	GPIO3[8]	-	I2C1_SCL	-	-
LDO_EN	-	-	-		-	-
RESETN	-	-	-		-	-
TEST_MODE	-	-	-		-	-
JTAG_TDO	JTAG_TDO	SPI0_MOSI	-	PWM5	-	GPIO3[16]
JTAG_TMS	JTAG_TMS	SPI0_MISO	-	PWM6	-	GPIO3[14]
JTAG_TDI	JTAG_TDI	SPI0_SCLK	-	PWM7	-	GPIO3[15]
JTAG_TRSTN	JTAG_TRSTN	SPI0_CSN0	-	UART3_RXD	-	GPIO3[12]
JTAG_TCK	JTAG_TCK	SPI0_CSN1	-	UART3_TXD	-	GPIO3[13]
QSPI_DIO0	QSPI_DIO0	-	-	-	-	-
QSPI_CLK	QSPI_CLK	-	-	-	-	-
QSPI_DIO1	QSPI_DIO1	-	-	-	-	-
QSPI_CSN	QSPI_CSN	-	-	-	-	-
QSPI_DIO2	QSPI_DIO2	I2C1_SCL	-	-	-	-
QSPI_DIO3	QSPI_DIO3	I2C1_SDA	-	-	-	-
SDIO0_CLK	SDIO0_CLK	GPIO3[3]	I2C1_SDA	UART3_RTSN	-	-
SDIO0_CMD	SDIO0_CMD	GPIO3[2]	JTAG_TCK	-	SPI1_CSN0	UART1_TXD
SDIO0_DATA0	SDIO0_DATA[0]	GPIO3[4]	JTAG_TMS	-	SPI1_CSN1	UART1_RXD
SDIO0_DATA1	SDIO0_DATA[1]	GPIO3[5]	JTAG_TDO	-	SPI1_MISO	-
SDIO0_DATA2	SDIO0_DATA[2]	GPIO3[0]	JTAG_TDI	-	SPI1_MOSI	-
SDIO0_DATA3	SDIO0_DATA[3]	GPIO3[1]	JTAG_TRSTN	-	SPI1_SCLK	-
SDIO0_CD	SDIO0_CD	GPIO3[6]	I2C1_SCL	UART3_CTSN	-	-
SDIO0_WP	SDIO0_WP	GPIO3[7]		-	-	-
GPIOE	GPIO3[24]		SPI1_SCLK	PWM4	-	-
GPIOD	GPIO2[31]	EMAC0_TXEN	SPI1_CSN0	PWM3	I2C2_SCL	-
GPIOC	GPIO2[30]	EMAC0_COL	SPI1_CSN1	PWM2	I2C2_SDA	-
GPIOB	GPIO1[31]	EMAC0_RXDV	SPI1_MISO	PWM1	UART0_TXD	-
GPIOA	GPIO1[30]	EMAC0_MDC	SPI1_MOSI	PWM0	UART0_RXD	-
BOOT_SEL0	BOOT_SEL0	GPIO3[17]	-	-	-	-
BOOT_SEL1	BOOT_SEL1		-	-	GPIO3[21]	-
LCD_VSYNC	LCD_VSYNC	EMAC0_PHYOSC	-	SDIO1_WP	GPIO1[25]	DP_WR
LCD_HSYNC	LCD_HSYNC	EMAC0_RXDV	-	-	GPIO1[26]	DP_RS
LCD_DEN	LCD_DEN	-	SENSOR0_PWD	-	GPIO1[27]	DP_TE
LCD_PCLK	LCD_PCLK	EMAC0_MDIO	SENSOR0_RSTN	SDIO1_CD	GPIO1[24]	DP_CS
LCD_DOUT0	LCD_DOUT[0]	EMAC0_MDC	PWM0	-	GPIO1[0]	SPI0_SCLK

LCD_DOUT1	LCD_DOUT[1]	EMAC0_TXEN	PWM1	TRIG0	GPIO1[1]	SPI0_MISO
LCD_DOUT2	LCD_DOUT[2]	EMAC0_TXD[1]	PWM2	SDIO1_CMD	GPIO1[2]	SPI0_MOSI
LCD_DOUT3	LCD_DOUT[3]	EMAC0_TXD[0]	PWM3	SDIO1_CLK	GPIO1[3]	SPI0_CSN0
LCD_DOUT4	LCD_DOUT[4]	EMAC0_REFCLK	PWM4	SDIO1_DATA[3]	GPIO1[4]	-
LCD_DOUT5	LCD_DOUT[5]	EMAC0_RXD[1]	PWM5	SDIO1_DATA[2]	GPIO1[5]	SPI0_CSN1
LCD_DOUT6	LCD_DOUT[6]	EMAC0_RXD[0]	PWM6	SDIO1_DATA[1]	GPIO1[6]	-
LCD_DOUT7	LCD_DOUT[7]	EMAC0_RXER	PWM7	SDIO1_DATA[0]	GPIO1[7]	-
LCD_DOUT8	LCD_DOUT[8]	EMAC0_RXD[2]	SDIO1_CD	LCD_DOUT[0]	GPIO1[8]	DBIB_DATA[0]
LCD_DOUT9	LCD_DOUT[9]	EMAC0_RXD[3]	SDIO1_WP	LCD_DOUT[1]	GPIO1[9]	DBIB_DATA[1]
LCD_DOUT10	LCD_DOUT[10]	EMAC0_TXD[2]	SDIO1_CLK	LCD_DOUT[2]	GPIO1[10]	DBIB_DATA[2]
LCD_DOUT11	LCD_DOUT[11]	EMAC0_TXD[3]	SDIO1_CMD	LCD_DOUT[3]	GPIO1[11]	DBIB_DATA[3]
LCD_DOUT12	LCD_DOUT[12]	EMAC0_TXCLK	SDIO1_DATA[0]	LCD_DOUT[4]	GPIO1[12]	DBIB_DATA[4]
LCD_DOUT13	LCD_DOUT[13]	EMAC0_TXER	SDIO1_DATA[1]	LCD_DOUT[5]	GPIO1[13]	DBIB_DATA[5]
LCD_DOUT14	LCD_DOUT[14]	EMAC0_RXCLK	SDIO1_DATA[2]	LCD_DOUT[6]	GPIO1[14]	DBIB_DATA[6]
LCD_DOUT15	LCD_DOUT[15]	EMAC0_CRS	SDIO1_DATA[3]	LCD_DOUT[7]	GPIO1[15]	DBIB_DATA[7]
LCD_DOUT16	LCD_DOUT[16]	EMAC0_RXD[0]	-	-	GPIO2[19]	-
LCD_DOUT17	LCD_DOUT[17]	EMAC0_RXD[1]	UART2_RXD	-	GPIO2[20]	-
LCD_DOUT18	LCD_DOUT[18]	EMAC0_TXD[0]	UART2_TXD	-	GPIO2[21]	-
LCD_DOUT19	LCD_DOUT[19]	EMAC0_TXD[1]	I2C3_SDA	UART1_TXD	GPIO2[22]	-
LCD_DOUT20	LCD_DOUT[20]	EMAC0_MDIO	I2C3_SCL	UART1_RXD	GPIO2[23]	-
LCD_DOUT21	LCD_DOUT[21]	-	-	-	GPIO2[24]	-
LCD_DOUT22	LCD_DOUT[22]	-	-	-	GPIO2[25]	-
LCD_DOUT23	LCD_DOUT[23]	-	-	-	GPIO2[26]	-
SENSOR0_CLK	SENSOR0_CLK	GPIO3[20]	-	-	-	-
SENSOR0_PWD	SENSOR0_PWD	GPIO3[19]	UART2_TXD	I2C2_SCL	-	-
SENSOR0_RSTN	SENSOR0_RSTN	GPIO3[18]	UART2_RXD	I2C2_SDA	-	-
SENSOR1_CLK	SENSOR1_CLK	GPIO3[23]	-	-	-	-
SENSOR1_PWD	SENSOR1_PWD	GPIO3[22]	-	-	-	-
I2C0_SDA	I2C0_SDA	GPIO1[29]	-	-	-	-
I2C0_SCL	I2C0_SCL	GPIO1[28]	-	-	-	-
DVP0_VSYNC	DVP0_VSYNC	-	GPIO2[29]	-	-	-
DVP0_HREF	DVP0_HREF	-	GPIO2[28]	-	-	-
DVP0_PCLK	DVP0_PCLK	-	GPIO2[12]	-	-	-
DVP0_DIN0	DVP0_DIN[0]	-	GPIO2[0]	-	PWM0	-
DVP0_DIN1	DVP0_DIN[1]	-	GPIO2[1]	-	PWM1	-
DVP0_DIN2	DVP0_DIN[2]	-	GPIO2[2]	-	PWM2	-
DVP0_DIN3	DVP0_DIN[3]	-	GPIO2[3]	-	PWM3	-
DVP0_DIN4	DVP0_DIN[4]	-	GPIO2[4]	-	PWM4	-
DVP0_DIN5	DVP0_DIN[5]	QSPI_DIO3	GPIO2[5]	I2C1_SCL	PWM5	-
DVP0_DIN6	DVP0_DIN[6]	QSPI_DIO2	GPIO2[6]	I2C1_SDA	PWM6	-
DVP0_DIN7	DVP0_DIN[7]	QSPI_DIO1	GPIO2[7]	-	PWM7	-
DVP0_DIN8	DVP0_DIN[8]	QSPI_DIO0	GPIO2[8]	-	-	-
DVP0_DIN9	DVP0_DIN[9]	QSPI_CSN	GPIO2[9]	-	-	-

DVP0_DIN10	DVP0_DIN[10]	QSPI_CLK	GPIO2[10]	I2C2_SDA	UART3_RXD	-
DVP0_DIN11	DVP0_DIN[11]	-	GPIO2[11]	I2C2_SCL	UART3_TXD	DP_RD

Table 2- 2 Function definition of GPIO pins

(*): Number in parentheses indicates the remap number of the same function pin
(PMU): GPIO pin in Always-On domain
Underlined text indicates the default function after booting
Please refer to XXX for details of pin configuration register

3. Electrical Specification

3.1. Operation Condition

Table 3- 1 Recommended Operation Conditions

	Symbol	Min	Typ	Max	Unit
3.3V I/O Supply Voltage	DVDD33	2.97	3.3	3.63	V
1.8V I/O Supply Voltage	DVDD18	1.62	1.8	1.98	V
Core Supply Voltage	DVDD11	0.99	1.1	1.21	V
DRAM VDD1	DRAM_VDD1	1.7	1.8	1.95	V
DRAM VDD2 VDDQ (LPDDR2)	DRAM_VDD2/ DRAM_VDDQ	1.14	1.2	1.3	V
DRAM VDD2 VDDQ (DDR2)	DRAM_VDD2/ DRAM_VDDQ	1.7	1.8	1.95	V
VDDQ for DDR PHY (LPDDR2)	DDR_VDDQ	1.14	1.2	1.32	V
VDDQ for DDR PHY (DDR2)	DDR_VDDQ	1.7	1.8	1.95	V
VDD for DDR PHY	DDR_VDD	0.99	1.1	1.21	V
MIPI analog power	MP0_AVDD	2.25	2.5	2.75	V
USB PHY analog power supply	USB_VDD25	2.33	2.5	2.75	V
USB PHY analog power supply	USB_VDD330	3.07	3.3	3.63	V
Efuse power supply	EFUSE_AVDD	2.4	2.5	2.6	V
PLL power supply	PLL_AVDD	1	1.1	1.2	V
External High Frequency Crystal Frequency			12		MHz
Junction Temperature		-40	25	125	℃
Ambient Temperature					℃

3.2. Power Consumption

Table 3- 2 Power consumption in different scenarios

Condition	Voltage	BOOT (mA)	FULL_SPEED (mA)	STANDBY (mA)
DVDD33	3.3	T.B.D	T.B.D	T.B.D
DVDD18	1.8	T.B.D	T.B.D	T.B.D
DVDD11	1.1	T.B.D	T.B.D	T.B.D
AVDD25	2.5	T.B.D	T.B.D	T.B.D
DDR_VDDQ/ DRAM_VDDQ(LPDDR2)	1.2	T.B.D	T.B.D	T.B.D
DDR_VDDQ/ DRAM_VDDQ(DDR2)	1.8	T.B.D	T.B.D	T.B.D
Total		T.B.D	T.B.D	T.B.D

3.3. Digital I/O Characteristics

Table 3- 3 DC characteristics for digital IOs

	Symbol	Min	Typ	Max	Unit
Recommend Operation Condition for 3.3V IO application					
Input High Voltage	V_{IH}	2		DVDD33+0.3	V
Input Low Voltage	V_{IL}	-0.3		0.8	V
Output Low voltage	V_{OL}			0.4	V
Output High voltage	V_{OH}	2.4			V
Driver Strength at 2.4V @TT, 25°C, 3.3V/1.1V	I_{oh}	15.5		20.6	mA
Driver Strength at 0.4V @TT, 25°C, 3.3V/1.1V	I_{ol}	9.3		12.3	mA
Recommend Operation Condition for 1.8V IO application					
Input High Voltage	V_{IH}	DVDD18*0.65		DVDD18+0.3	V
Input Low Voltage	V_{IL}	-0.3		DVDD18*0.35	V
Output Low voltage	V_{OL}			0.4	V
Output High voltage	V_{OH}	1.35			V
Driver Strength at 1.3V @TT, 25°C, 1.8V/1.1V	I_{oh}	4.7V		6.3	mA
Driver Strength at 0.4V @TT, 25°C, 1.8V/1.1V	I_{ol}	5.2		7	mA
Input Leakage Current				±10	uA

Tri-State Output leakage				±10	uA

Table 3- 4 AC characteristics for digital IOs

	Symbol	Min	Typ	Max	Unit
3.3V IO application					
Threshold point	V_T	1.34	1.47	1.72	V
Schmitt trigger L to H threshold point	V_{T+}	1.60	1.74	1.96	V
Schmitt trigger H to L threshold point	V_{T-}	1.15	1.28	1.50	V
Pull Up Resistor	Rpu	27	40	64	KΩ
Pull Down Resistor	Rpd	31	46	78	KΩ
1.8V IO application					
Threshold point	V_T	0.73	0.83	0.97	V
Schmitt trigger L to H threshold point	V_{T+}	0.93	1.04	1.13	V
Schmitt trigger H to L threshold point	V_{T-}	0.59	0.68	0.81	V
Pull Up Resistor	Rpu	53	89	163	KΩ
Pull Down Resistor	Rpd	54	96	189	KΩ

3.4. 1.1V Power on reset

Table 3- 5 1.1V POR characteristics

	Symbol	Min	Typ	Max	Unit
1.1V Supply Voltage	DVDD11	0.99	1.1	1.21	V
1.1V Power Up trigger level		0.6			V
Hysteresis			150		mV
Rising time of 1.1V Supply		500ns		10ms	
Reset time after trigger			15		us

3.5. SAR ADC

Table 3- 6 SAC-ADC Characteristics

	Symbol	Min	Typ	Max	Unit
Analog In					
Input Range	AIN[7:0]	0.01*Vref		0.99*Vref	V
Input Capacitance	C_{IN}		1		pF
Positive Reference	Vref	1.8V		3.3V	V
Power Supply					

Analog Supply Voltage	VDDA	2.97	3.3	3.63	V
Resolution			12		Bits
Integral Nonlinearity	INL		+/- 2 LSB@1Msps, +/- 2.2 LSB@2Msps,		LSB
Differential Nonlinearity	DNL		+/- 1 LSB@1Msps, +/- 1.3 LSB@2Msps,		LSB
Sampling Clock	SOC			2	MHz
Main Clock Frequency	CLK			22	MHz
Data Latency			12		CLK Cycle
Signal to Noise Ration plus Distortion (Up to 5th harmonic)	SINAD		Fin=10KHz/1Msps：65dB, Fin=10KHz/2Msps：62dB		dB
Spurious-Free Dynamic Range*	SFDR				dB
Total Harmonic Distortion (First 5 harmonics)	THD		Fin=10KHz/1Msps：-72dB, Fin=10KHz/2Msps：-68dB		dB
Effective Number of Bits	ENOB		Fin=10KHz/1Msps：10.5bit, Fin=10KHz/2Msps：10.2bit		Bits

3.6. Audio CODEC

Table 3- 7 Audio Codec characteristics

	Symbol	Min	Typ	Max	Unit
Line-in					
Dynamic Range @-60dB,A-weight	DR		90		dB
Total Harmonic Distortion +N	THDN		-87		dB
S/N (A-weighted)	SNR		97		dB
Channel separation			92		dB
Channel matching			0.2		dB
Full scale input	VFS		0.9		V_{rms}
Input impedance		4	10		KΩ
Capacitive Load			10		pF
Mic Bias output					

Output Voltage	V_{bias}		0.75*AVDD 或 0.9*AVDD		V
Max Output Current	I_{bias}		3		mA
Capacitive Load				50	pF
Line-outoutput					
Full scale output			1		V_{rms}
Dynamic Rang (-60dB input, A-weighted)			98		dB
Total Harmonic Distortion +N @-3dBFs 1K Sine input	THDN		-90		dB
S/N (A-weighted)	SNR		98		dB
Channel separation			98		dB
Channel matching			0.2		dB
Load resistance			10		KΩ
Load capacitance				50	pF
Power su					
Analog Supply Voltage	AVDD	2.97	3.3	3.63	V
Reference Voltage	VMID	0.5*AVDD*0.97	0.5*AVDD	0.5*AVDD*1.03	V

3.7. USB2.0

Table 3- 8 USB Electrical Specifications

	Symbol	Min	Typ	Max	Unit
Power Supply					
Digital Power Supply	USB_DVDD	-7%	3.3	+10%	V
3.3V Analog Power Supply	USB_VDD33	-7%	1.1	+10%	V
2.5V Analog Power Supply	USB_VDD25	-7%	2.5	+10%	V
USB I/O					
USB 5V Signal	VBUS	0		5.25	V
USB ID	USB_ID	0		3.3	V
USB Data	DP/DM	0		3.3 or 5.25*	V

* In a 5V short condition

4. System Function

4.1. Reset

Storm has following reset sources:

- Power on reset (POR)
- External RESETN pin reset
- Global software reset
- Power down reset from PMU
- Watch dog timer (WDT) reset
- Audio subsystem correlated block software reset
- MCU subsystem correlated block software reset

4.1.1. Reset Architecture Diagram

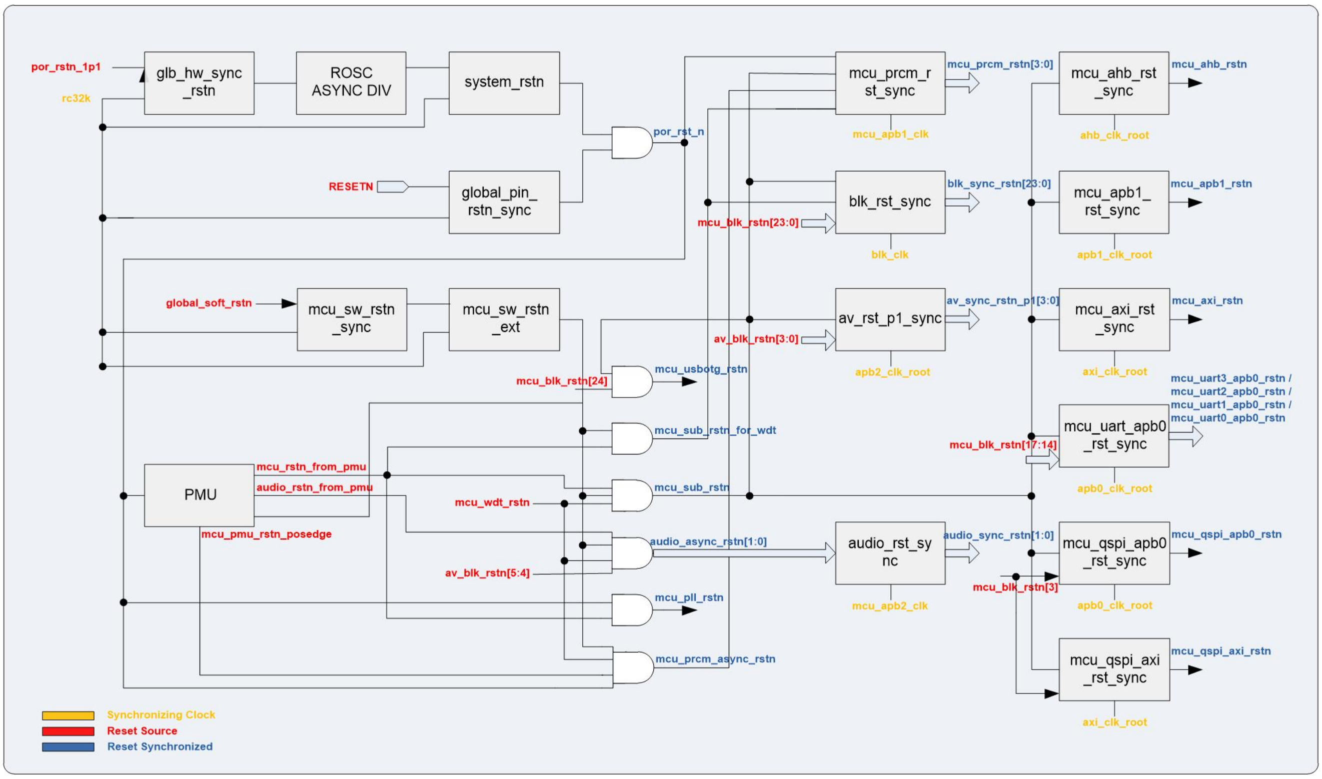


Figure 4- 1 Reset Architecture Diagram

4.1.2. Power on Reset (POR)

Storm has two power on reset separately responsible 1.1V power domain and 3.3V power domain. When both of these 2 power on reset are released, will the pmu start power-on sequence.

4.1.3. External RESETN Pin Reset

The whole system of Storm can be reset by inputting a valid low level into external RESETN pin. The reset state will be holding until RESETN pin pulls out to a high level.

4.1.4. Global Software Reset

Writing 0xAA55A5A5 to Global Software Reset Register will cause the global reset and it will be released by firmware automatically. When global software reset enable, it resets whole chip except clock generate and related control registers. Write other value will take no effect.

User can refer to GLOBAL_SOFT_RST Registers in chapter nine.

4.1.5. Power Down Reset

Storm support power down control for separate multiple power domain. There are three independent subsystems whose power domain can be power on or power down due to the scenario of power management. User can refer to chapter twelve for detail of power management.

The three subsystems mentioned above are MCU subsystem, audio subsystem and computing subsystem, as the result, these three subsystems can be reset once their power domains are power down.

4.1.6. Watch Dog Timer (WDT) Reset

In normal operation, the operating system reloads the watchdog at regular intervals before the timer overflow occurs. A timer overflow should generate an internal reset of the system including MCU subsystem, audio subsystem and computing subsystem. An indication that a watchdog reset occurred should be available in the form of a register bit, which can be read and must be explicitly cleared by firmware.

4.1.7. Block Software Reset

The blocks in MCU subsystem and audio subsystem can be reset by setting high level to the correlated bit of MCU_BLK_RST Register or it of AUDIO_VIDEO_BLK_RST Register. The reset can be released by setting low level to the same bit.

User can refer to MCU_BLK_RST Registers in chapter 4.6.

4.2. Clock

4.2.1. Overview

Storm has following clocks:

- External 12MHz crystal clock
- External 32KHz crystal clock
- 1 MCU PLL clock, maximum frequency up to 1.2GHz, which can also be shared by VPU
- 1 DDR PLL provides frequency for DDR Controller as well as the reference clock to DDRPHY built-in PLL
- 1 Video PLL, maximum frequency up to 594Mhz, provide various clock frequency to video subsystem
- Frequency division clock of MCU PLL clock, including AXI bus clock, AHB bus clock, APB bus clock, SDIO clock and UART decimal frequency division clock
- Built-in Audio Codec PLL clock, maximum frequency 200MHz
- Built-in USB PLL clock, frequency is 480MHz
- Eight frequency division clock of USB PLL clock for USB OTG controller use
- JTAG TCK clock, maximum frequency should below half of the AHB bus clock

4.2.2. Clock Architecture Diagram

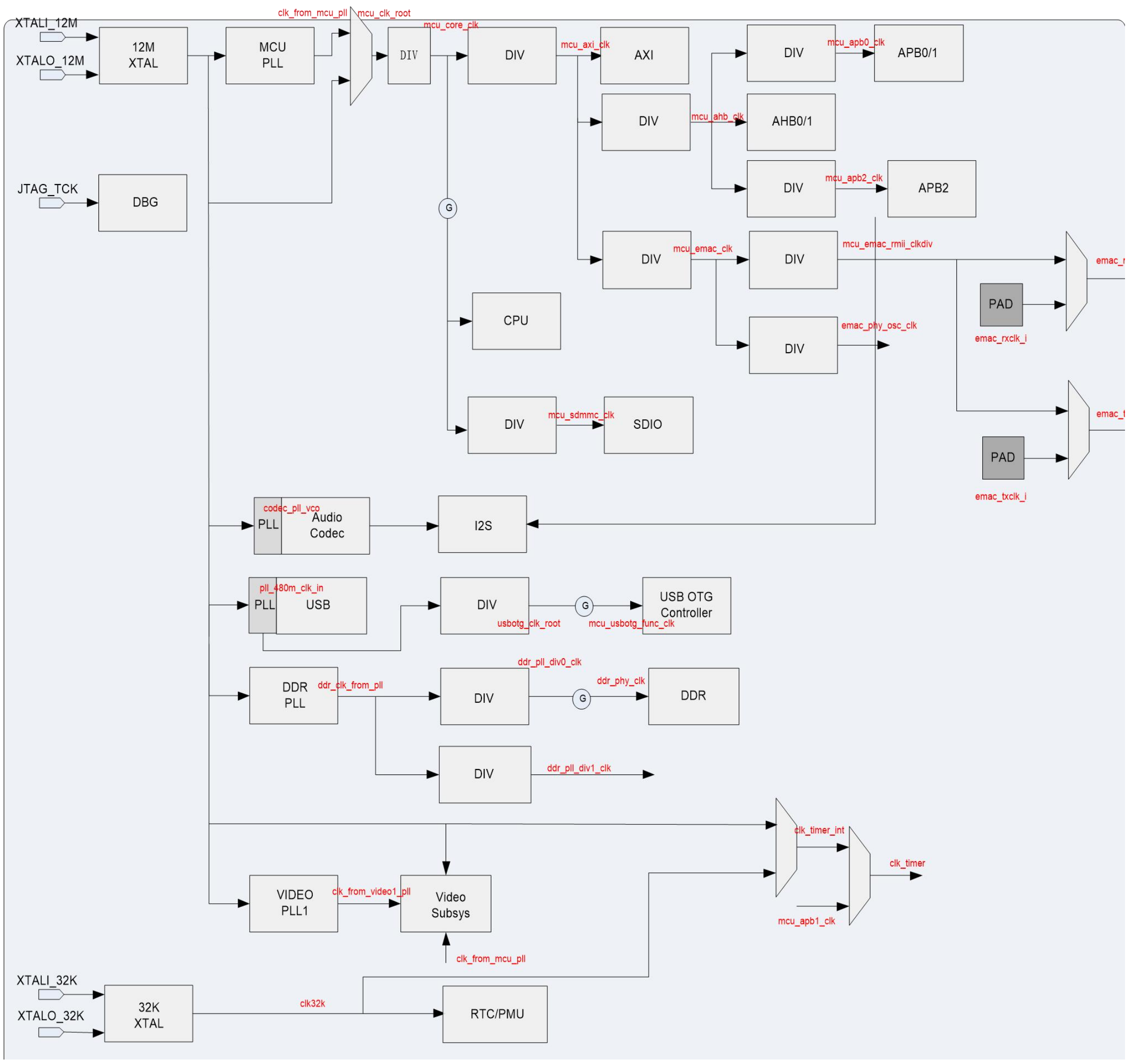


Figure 4- 2 Clock Architecture Diagram

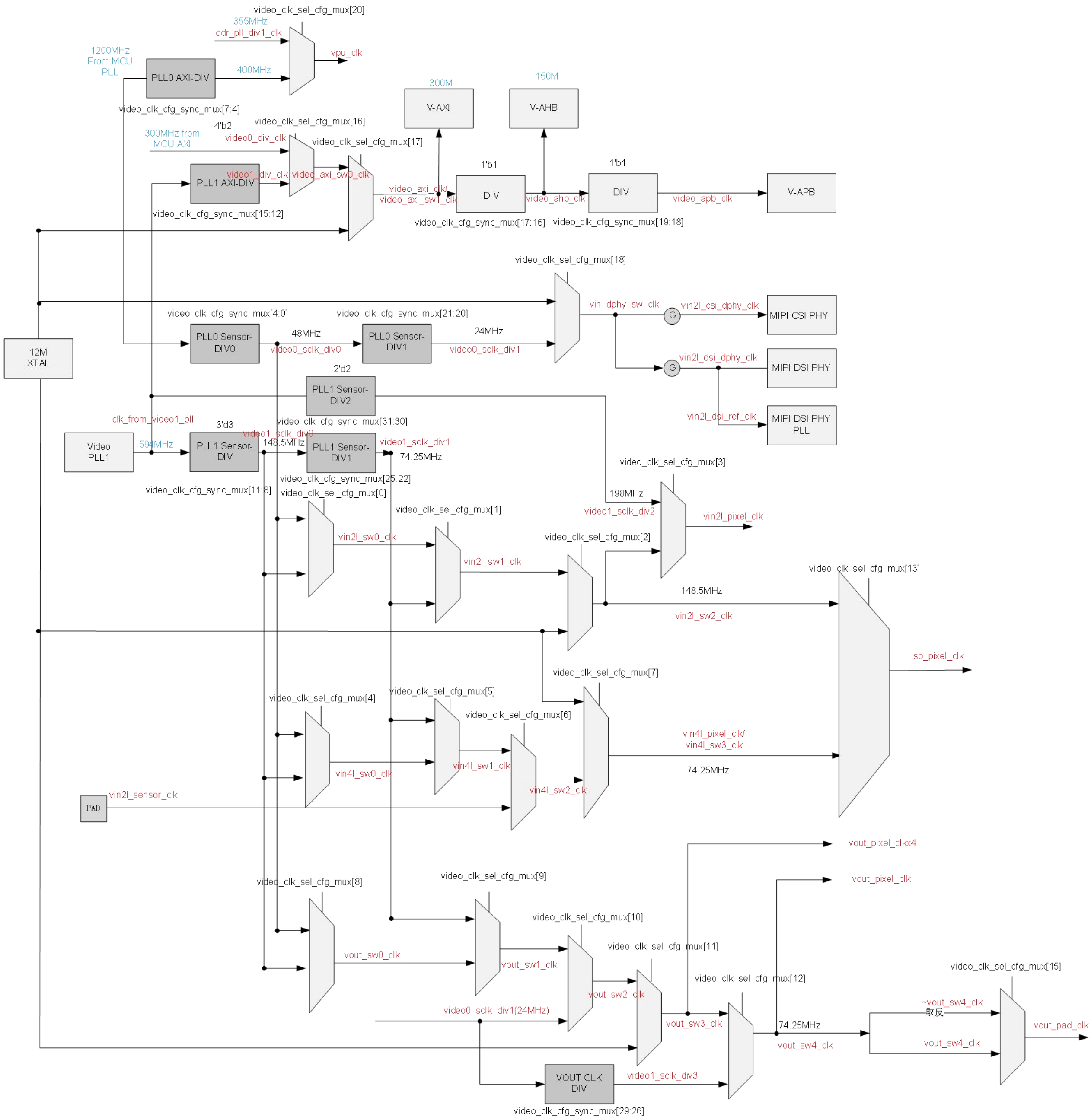


Figure 4- 3Video Clock Architecture Diagram

4.2.3. Source Clock

Storm has two external crystal clocks and three built-in PLL macro clocks. All of them are used as the source clocks in different subsystems. The brief introduction of these clocks shown below:

- External 12MHz crystal clock

The 12MHz crystal clock should work with external crystal and it's always on under 3.3V power domain. It's the default MCU subsystem clock after power on reset, consequently user can switch it to

MCU PLL clock as well as the PLL clock is locked after inquiring the lock status by software.

The 12MHz crystal clock is also the reference clocks for both Built-in Audio Codec PLL macro and Built-in USB PLL macro.

- External 32KHz crystal clock

The 32KHz crystal clock should work with external crystal. Depending on the power switch internal, it provide clock for correlated logic of RTC. It is allowed to not put external crystal outside chip. In this circumstances software should not choose it as 32KHz clock source and choose 32KHz RC clock instead. By default, 32KHz crystal is not used and not connected to the package pins.

- MCU PLL clock

The MCU PLL clock generates high speed frequency doubling clock for MCU subsystem and it works as the source clock for AXI bus clock, AHB bus clock, APB bus clock, SDIO clock and UART decimal frequency division clock.

The reference clock of MCU PLL clock is 12MHz crystal clock, and it provides a frequency range from 62.5MHz to 1500MHz. The maximum frequency is 900MHz currently due to CPU core fetching process limited.

- DDR PLL clock

The DDR PLL clock generates high speed frequency for DDR Controller and provides reference clock to DDR PHY built-in PLL. The frequency that DDR PLL generates equals to the DDR3/2 device datarate dividing by 4.

For example, DDR3 device runs at 1333Mbps datarate, then the clock generated by DDR PLL is 333MHz.

- Video PLL clock

Two Video PLLs generate various clock frequency that satisfies the requirement of all modules suchasvideoinput and display modules in video subsystems. Please refer to Figure 7-2 above.

- Audio Codec PLL clock

The Audio Codec PLL clock works as the source clock for both PLLADCLK and PLLDACLK. The reference clock of Audio Codec PLL clock is 12MHz crystal clock and its general frequency could be configured to 98.304MHz through PLL configuration control register.

So PLLADCLK would be 128*fs (fs=8K/12K/16K/24K/32K/48K/96KSPS) and PLLDACLK would be 256*fs (fs= 8K/12K/16K/24K/32K/48KSPS).

PLLDACLK is sent to I2S through the frequency divider with two frequency division or four frequency division.

- USB PLL clock

The reference clock of USB PLL clock is 12MHz crystal clock and it provides a 480MHz source clock to eight frequency division clock which is propagated to USB OTG controller.

- JTAG TCK clock

The JTAG TCK clock is input from JTAG_TCK PAD by alternative function. It could either be the JATG clock for debugger or the TCK clock for test use.

4.2.4. Generated Clock

- AXI bus clock
  The AXI bus clock domain uses the generated clock divided from MCU PLL clock and its frequency division ratio ranged from 1 to 4. User can configure it referring to MCU Clock Control Registers in chapter nine.

- AHB bus clock
  The AHB0/1 bus clock domain uses the generated clock divided from AXI bus clock and its frequency division ratio ranged from 1 to 4. User can configure it referring to MCU Clock Control Registers in chapter nine.

- APB0/1 bus clock
  The APB0/1 bus clock domain uses the generated clock divided from AHB bus clock and its frequency division ratio ranged from 1 to 4. User can configure it referring to MCU Clock Control Registers in chapter nine.

- APB2 bus clock
  The APB2 bus clock domain uses the generated clock divided from AHB bus clock and its frequency division ratio ranged from 1 to 4. User can configure it referring to MCU Clock Control Registers in chapter nine.

- SDIO clock
  The SDIO clock uses the generated clock divided from MCU PLL clock and its frequency division ratio ranged from 1 to 16. User can configure it referring to MCU Clock Control Registers in chapter nine.

- USB OTG controller clock
  The USB OTG controller clock uses the generated clock divided from USB PLL clock. It is eight frequency division internal USB PHY.

4.2.5. Clock Distribution

The main clocks of the system are listed in following tables:

Table 4- 1 System Main Clocks

Clock Name	Default Freq (MHz)	Source	Comment
mclk_12m	12	12M XTAL	External crystal clock
clk_from_mcu_pll	1200	MCU PLL	MCU PLL clock
mcu_clk_root	1200	mclk_12 or clk_from mcu_pll	Mcu root clock
mcu_core_clk	600	mcu_clk_root	Same frequency as mcu root clock
mcu_axi_clk	300	mcu_core_clk /2	frequency division ratio:1~4
mcu_ahb_clk	150	mcu_axi_clk/2	frequency division ratio:1~4
mcu_apb0_clk	75	mcu_ahb_clk /2	frequency division ratio:1~4
mcu_apb2_clk	75	mcu_ahb_clk /2	frequency division ratio:1~4
mcu_sdmmc_clk	150	mcu_core_clk /4	frequency division ratio:1~16
mcu_emac_clk	100	mcu_axi_clk/3	
mcu_osc_phy_clk	50	mcu_emac_clk/2	
codec_pll_vco	98.304	Audio Codec PLL	Built-in Audio CODEC PLL clock
pll_480m_clk_in	480	USB PLL	Built-in USB PLL clock
mcu_usbotg_func_clk	60	pll_480m_clk_in/8	div8clk of USB PLL clock
ddr_clk_from_pll	1066	DDR PLL Source	DDR PLL clock source
ddr_pll_div1_clk	355	DDR PLL/3	VPU clock source from DDR PLL
ddr_phy_clk	533	DDR PLL/2	DDR phy clock
vpu_clk	355	DDR PLL/3 or MCU PLL/3	VPU clock
video_axi_clk	297	clk_from_video1_pll/2	
video_ahb_clk	148.5	video_axi_clk/2	
video_apb_clk	74.25	video_ahb_clk/2	
clk_from_video1_pll	594	VIDEO1 PLL	VIDEO1 PLL clock
vin2l_csi_dphy_clk	24	video0_clk_div1	
vin2l_pixel_clk	197.94	video1_clk_div0	
isp_pixel_clk	148.5	video1_clk_div0	
vin4l_pixel_clk	74.25	video1_clk_div0	
vo_pad_clk	74.25	video1_clk_div0	
vout_pixel_clkx4	72.23	mcu_pll_clk/n	

Clock Name	Default Freq (MHz)	Source	Comment
osc32k	32768	32K XTAL/RC 32K	External crystal clock or RC32K clock in 3.3V domain
jtag_tck	20M	JTAG clock	Input from JTAG_TCK PAD

4.2.6. Clock Gating Scheme

The whole clock network use both Course-grained and fine-grained clock gating, from cluster level to subsystem level and block level. All the course-grained clock gating could set enable or disable by software which user may refer to MCU Clock Gating Control Registers in chapter nine.

4.2.7. Phase Locked Loop (PLL)

The figure below shows the pin list of PLL macro used in Columbus, which could generate stable high-speed clock from a slower reference clock with single power supply. It consists of a Phase Frequency Detector (PFD), a Low Pass Filter (LPF), a Voltage Controlled Oscillator (VCO) and other associated circuits. All fundamental building blocks as well as fully programmable dividers are integrated in the core.

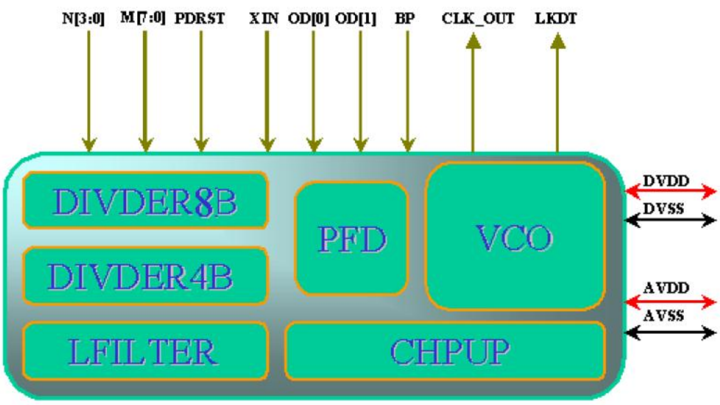


Figure 4- 4 PLL Macro

The XIN is the reference input clock from Crystal Oscillator or other clock sources, and CLK_OUT is the output clock.

8-bit M an 4-bit N are feedback and input divider controls, while 2-bit OD is for output divider control, as below:

CLK_OUT = XIN * (M/N) / (2^OD)

Note that the certain limitation applies for the divider, including:

1. XIN/N should be within [1MHz, 50MHz];
2. CLK_OUT*(2^OD) should be within [500MHz, 1500MHz];
3. M is not less than 2 and N is not less than 1;

BP is a high valid signal to bypass PLL while PDRST is used to power down analogy blocks and reset digital flip-flops which should be 0 during the normal operation. After power-on PLL or change M/N settings, PDRST is required to be active for at least 10ns before PLL could get locked, as shown in the below timing diagram.

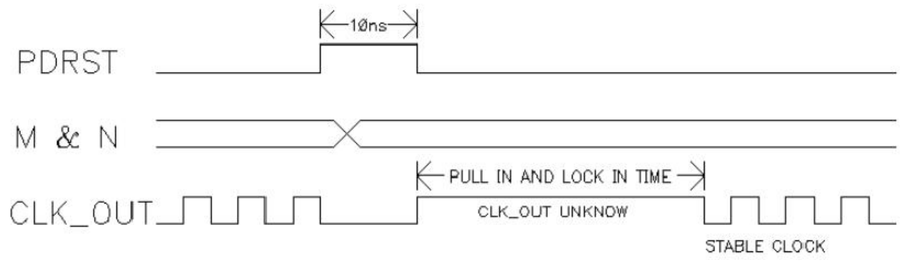


Figure 4- 5 PDRST Timing Diagram

The impact on CLK_OUT by PDRST and BP is shown as below table:

Table 4- 2 PDRST and BP impact on CLK_OUT

PDRST	**BP**	**CLK_OUT**
1	X	0
0	1	XIN
0	0	CLK_OUT

These control inputs are SW configurable through system registers, as specified in section 4.6 in more details.

4.3. Main CPU Sub-system

The main CPU sub-system is as following：

- High performance RISC processor
  - Up to 600MHz
  - SIMD and FPU support
  - Cache
- 16KB L1 Instruction Cache & 16KB L1 Data Cache
  - Support Linux & RTOS.
- 32KB Boot ROM(Mask ROM)
- 64KB System SRAM
- Interrupt Control: Generic Interrupt Controller

- System Debug: Core-sight subsystem.

4.4. Interrupt Scheme

The interrupt scheme is described in Table 4- 3.

4.5. System Control

The system registers are used to control or monitor various functional blocks of the system. All these registers are packed together and accessible through APB bus with base address at 0x9100_5000.

Table 4- 4 System control register summary

Name	Offset	Description
MCU_PLL_CFG	0x0000	MCU PLL configuration
MCU_CLK_CFG	0x0004	MCU subsystem clock configuration
DDR_PLL_CFG	0x0008	DDR PLL configuration
VIDEO1_PLL_CFG	0x0010	Video1 PLL configuration
VIDEO_CLK_DIV_CFG	0x0014	Video subsystem clock configuration
VIDEO_CLK_SEL_CFG	0x0018	Video subsystem clock configuration
GLOBAL_SOFT_RST	0x0020	Global software reset
MCU_BLK_RST	0x0024	Software reset for MCU subsystem
AUDIO_VIDEO_BLK_RST	0x0028	Software reset for audio and video subsystem
DDR_RESET_RELEASE	0x002C	DDR software reset
MCU_BLK_CG	0x0030	Clock gating for blocks in MCU subsystem
AUDIO_VIDEO_BLK_CG	0x0034	Clock gating for blocks in audio and video subsystem
BOOT_STATUS	0x0040	BOOT status
BOOT_MODE	0x0044	BOOT mode indication
PMU_MISC_CTRL	0x0048	PMU misc control signals
ATE_TEST_MODE	0x004C	FT test mode indication
MCU_CHIPID	0x0060	Chip ID
MCU_AHB_HPROT_CTRL	0x0074	HPPROT control for MCU subsystem blocks
DMA_HS_SEL0	0x0078	DMA Hand Shake select
DMA_HS_SEL 1	0x007c	DMA Hand Shake select
DMA_HS_SEL 2	0x0080	DMA Hand Shake select
DMA_HS_SEL 3	0x0084	DMA Hand Shake select

DMA_PERI_EN	0x0088	MCU DMA Peripheral Use Enable
MCU_DMA_HW_REQ_INUSE_STATUS	0x008C	MCU DMA HW Request In-Use Status
MCU_DMA_EXT_INT_SRC_SEL	0x0090	MCU DMA External trigger Source Selection
TRNG_EXT_CTRL	0x009C	TRNG block zerorize control
PIN_FUNC_SEL0	0x00a0	GPIO pin function select
PIN_FUNC_SEL1	0x00a4	GPIO pin function select
PIN_FUNC_SEL2	0x00a8	GPIO pin function select
PIN_FUNC_SEL3	0x00ac	GPIO pin function select
PIN_FUNC_SEL4	0x00b0	GPIO pin function select
PIN_FUNC_SEL5	0x00b4	GPIO pin function select
PIN_FUNC_SEL6	0x00b8	GPIO pin function select
PIN_FUNC_SEL7	0x00bc	GPIO pin function select
PIN_FUNC_SEL8	0x0110	GPIO pin function select
PIN_FUNC_SEL9	0x0114	GPIO pin function select
PIN_FUNC_SEL10	0x0118	GPIO pin function select
PIN_REMAP_CTRL	0x011c	Software remap of peripheral I/O functions
PIN_PULLUP_CTRL	0x00c0	IO pull-up control
PIN_PULLUP_CTRL1	0x00c4	IO pull-up control
PIN_PULLUP_CTRL2	0x00c8	IO pull-up control
PIN_PULLDIS_CTRL	0x00d0	IO pull disable control
PIN_PULLDIS_CTRL1	0x00d4	IO pull disable control
PIN_PULLDIS_CTRL2	0x00d8	IO pull disable control
PIN_DS_CTRL	0x00e0	IO driving strength control
PIN_DS_CTRL1	0x00e4	IO driving strength control
PIN_DS_CTRL2	0x00e8	IO driving strength control
PIN_SL_CTRL	0x00f0	IO slew rate control
PIN_SL_CTRL1	0x00f4	IO slew rate control
PIN_SL_CTRL2	0x00f8	IO slew rate control
PIN_OD_CTRL	0x0100	IO open drain control
PIN_OD_CTRL1	0x0104	IO open drain control
PIN_OD_CTRL2	0x0108	IO open drain control
MCU_AUX_DBG0	0x0120	Auxiliary register for debugging
MCU_AUX_DBG1	0x0124	Auxiliary register for debugging
MCU_AUX_DBG2	0x0128	Auxiliary register for debugging
MCU_AUX_DBG3	0x012C	Auxiliary register for debugging
USBPHY_TUNE	0x0130	USB PHY tuning control
USBPHY_CFG	0x0134	USB PHY configuration
Reserved	0x0138	
Reserved	0x013c	
USBPHY_TEST_CTRL	0x0140	USB PHY test control
ECC_AUX1	0x0150	Auxiliary control for ECC module

USB_INT_EN	0x0160	USB ID line interrupt enable
USB_RAW_INT	0x0164	USB ID line raw interrupt status
USB_INT	0x0168	USB ID line interrupt status
USB_INT_CLR	0x016C	USB ID line interrupt status clear
ID_DEBOUNCE_CNT	0x0170	USB ID input signal de-bounce count value
Reserved	0x0174	
USB_OTG_CTRL	0x0178	USB OTG control
EMAC_CFG	0x0190	Ethernet MAC-PHY interface configuration
EMAC_TIMESTAMP_LSB	0x0194	Emac timestamp output value
EMAC_TIMESTAMP_MSB	0x0198	Emac timestamp output value
Reserved	0x01A0	
Reserved	0x01A4	
I2S_CFG	0x01A8	I2S configuration
MISC_OTP_STATUS	0x01B0	OTP controller status
EXT_INT_CFG	0x01C0	WDT interrupt mask control
Reserved	0x01C4	
QSPI_DLL_DLY_SEL0	0x01E0	QSPI DLL configuration
QSPI_DLL_DLY_SEL1	0x01E4	QSPI DLL configuration
QSPI_DLL_DLY_SEL2	0x01E8	QSPI DLL configuration
QSPI_DLL_DLY_SEL3	0x01EC	QSPI DLL configuration

4.5.1. Clock Control Registers

4.5.1.1. MCU_PLL_CFG

Offset: 0x0000

Bits	Name	R/W	Reset	Description
[3:0]	PLL_N	RW	0x1	PLL input divider N, 1200MHz by default
[11:4]	PLL_M	RW	0x64	PLL feedback divider M, 1200MHz by default
[13:12]	PLL_OD	RW	0x0	PLL output divider OD
[14]	PLL_BYP	RW	0x0	0: clock output comes from the PLL. 1: PLL is bypassed; clock output comes from clock input directly.
[15]	PLL_PWDN	RW	0x0	0: PLL is power on 1: PLL is power down
[29:16]	Reserved			
[30]	PLL_LKDT	RO	0x0	PLL LOCK status from PLL IP. Optional reference value for SW to poll if needed.
[31]	PLL_LOCK	RO	0x0	Clock counted LOCK status

Note:

1. When changing the PLL configuration, the software should bypass the PLL clock first, and wait 500us for the PLL locked, then switch the system clock to the PLL clock.

4.5.1.2. MCU_CLK_CFG

Offset: 0x0004

Bits	Name	R/W	Reset	Description
[1:0]	mcu_axi_div	RW	0x01	AXI bus clock divider mcu_axi_clk = pll_clk/(bit[1:0]+1) Default: 0x01 (1/2 of pll_clk)
[3:2]	mcu_ahb_div	RW	0x01	AHB clock divider mcu_ahb_clk = axi_clk/(bit[3:2]+1) Default: 0x01(1/2 of axi_clk)
[5:4]	mcu_apb_div	RW	0x01	APB0/APB1 clock divider mcu_apb0_clk=mcu_ahb_clk/(bit[5:4]+1) Default: 0x01 (1/2 of mcu_ahb_clk)
[7:6]	audio_apb_div	RW	0x01	APB2 clock divider apb2_clk=mcu_ahb_clk/(bit[7:6]+1) Default: 0x01 (1/2 of mcu_ahb_clk)

Bits	Name	R/W	Reset	Description
[11:8]	sdmmc_clk_div	RW	0x03	Clock divider for SDMMC moule sdmmc_clk=pll_clk/(bit[11:8]+1) Default: 0x03(1/4 of pll_clk)
[13:12]	arm_rom_div	RW	0x01	Clock divider for ROM rom_clk= axi_clk/(bit[13:12]+1) Default:0x01(1/2 of mcu_axi_clk)
[15:14]	timer_clk_sel	RW	0x0	Timer clock source selection: 2'b00: APB clock 2'b01:APB clock 2'b10:12M OSC clock 2'b11:32K OSC clock
[23:16]	uart_clk_div	RW	0x14	Fractional clock divider for UART uart_clk= mcu_axi_clk/(bit[18:16]+bit[23:19]/32)) NOT in use Default: 0x14 (1/4.0625 of mcu_axi_clk)
[24]	uart_clk_div_rst	RW	0x0	uart_clk_divider block reset. 0: reset release 1: divider block reset
[25]	mcu_clk	RW	0x0	mcu_clk source selection control. 0: mcu_clk from rocs_100m; 1: mcu_clk from MCU PLL;
[31:28]	emac_clk_div.	RW	0x07	Clock divider for EMAC module emac_clk= mcu_axi_clk/(bit[31:28]+1). Default: 0x07(1/8 of mcu_axi_clk)

4.5.1.3. **DDR_PLL_CFG**

Offset: 0x0008

Bits	Name	R/W	Reset	Description
[3:0]	PLL_N	RW	0x1	PLL input divider N, default value for 533MHz output
[11:4]	PLL_M	RW	0x59	PLL feedback divider M
[13:12]	PLL_OD	RW	0x0	PLL output divider OD
[14]	PLL_BYP	RW	0x0	0: clock output comes from the PLL. 1: PLL is bypassed; clock output comes from clock input directly.
[15]	PLL_PWDN	RW	0x0	0: PLL is power on 1: PLL is power down
[29:16]	Reserved			

Bits	Name	R/W	Reset	Description
[30]	PLL_LKDT	RO	0x0	PLL LOCK status from PLL IP. Optional reference value for SW to poll if needed.
[31]	PLL_LOCK	RO	0x0	Clock counted LOCK status

Note:

1. When changing the PLL configuration, the software should bypass the PLL clock first, and wait 500us for the PLL locked, then switch the system clock to the PLL clock.

4.5.1.4. **VIDEO1_PLL_CFG**

Offset: 0x0010

Bits	Name	R/W	Reset	Description
[3:0]	PLL_N	RW	0x2	PLL input divider N, default value for 594 MHz output
[11:4]	PLL_M	RW	0x63	PLL feedback divider M
[13:12]	PLL_OD	RW	0x0	PLL output divider OD
[14]	PLL_BYP	RW	0x0	0: clock output comes from the PLL. 1: PLL is bypassed; clock output comes from clock input directly.
[15]	PLL_PWDN	RW	0x0	0: PLL is power on 1: PLL is power down
[29:16]	Reserved			
[30]	PLL_LKDT	RO	0x0	PLL LOCK status from PLL IP. Optional reference value for SW to poll if needed.
[31]	PLL_LOCK	RO	0x0	Clock counted LOCK status

Note:

1. When changing the PLL configuration, the software should bypass the PLL clock first, and wait 500us for the PLL locked, then switch the system clock to the PLL clock.

4.5.1.5. **VIDEO_CLK_DIV_CFG**

Offset: 0x0014

Bits	Name	R/W	Reset	Description
[4:0]	video0_sclk_div0	RW	0x18	video0_sclk_div0=mcu_pll_clk/(bit[4:0]+1) Default: 0x18 (1/25 of mcu_pll_clk which is 1200/25=48MHz)
[7:5]	video0_axi_div	RW	0x02	video0_axi_clk=video0_pll_clk/(bit[7:4]+1) Default: 0x02(1/3 of mcu_pll_clk which is 1200/3=400MHz)

Bits	Name	R/W	Reset	Description
[11:8]	video1_sclk_div0	RW	0x03	video1_sclk_div0=video1_pll_clk/(bit[11:8]+1) Default: 0x03 (1/4 of video1_pll_clk which is 594/4=148.5MHz)
[15:12]	video1_axi_div	RW	0x01	video1_axi_clk=video1_pll_clk/(bit[15:12]+1) Default: 0x01(1/2 of video1_pll_clk which is 594/2=297MHz)
[17:16]	video_ahb_div	RW	0x01	Divider for HCLK in video subsystem video_ahb_clk=video_axi_clk/(bit[17:16]+1) Default: 0x01 (1/2 of video_axi_clk)
[19:18]	video_apb_div	RW	0x01	Divider for PCLK in video subsystem video_apb_clk=video_ahb_clk/(bit[19:18]+1) Default: 0x01 (1/2 of video_ahb_clk)
[21:20]	video0_sclk_div1	RW	0x01	video0_sclk_div1= video0_sclk_div0/(bit[21:20]+1) Default: 0x01 (1/2 of video0_sclk_div0 which is 48/2=24MHz)
[25:22]	video1_sclk_div1	RW	0x01	video1_sclk_div1= video1_sclk_div0/(bit[25:22]+1) Default: 0x01 (1/2 of video1_sclk_div0 which is 148.5/2=74.25MHz)
[29:26]	video0_sclk_div2	RW	0x07	video0_sclk_div2=video0_sclk_div1/(bit[29:26]+1) Default: 0x07 (1/8 of video0_sclk_div1 which is 24/8=3MHz)
[31:30]	video1_sclk_div2	RW	0x02	video1_sclk_div2= video1_pll_clk/(bit[31:30]+1) Default: 0x02 (1/3 of video1_pll_clk which is 594/3=198MHz)

4.5.1.6. VIDEO_CLK_SEL_CFG

Offset: 0x0018

Bits	Name	R/W	Reset	Description
[0]	VIN2L_CLK_SW0	RW	0x0	VIN2L clock switch 0 source selection control. 0: VIN2L clock switch0 source from video1_sclk_div0; 1: VIN2L clock switch0 source from video0_sclk_div0;

Bits	Name	R/W	Reset	Description
[1]	VIN2L_CLK_SW1	RW	0x0	VIN2L clock switch 1 source selection control. 0: VIN2L clock switch1 source from output of VIN2L clock switch 0; 1: VIN2L clock switch1 source from video1_sclk_div1;
[2]	VIN2L_CLK_SW2	RW	0x0	VIN2L pixel clock source selection control. 0: VIN2L clock switch2 source from output of VIN2L clock switch 1; 1: VIN2L clock switch2 source from OSC12M;
[3]	VIN2L_CLK_SW3	RW	0x0	VIN2L pixel clock source selection control. 0: VIN2L clock switch2 source from video1_sclk_div2; 1: VIN2L clock switch2 source from output of VIN2L clock switch 2;
[4]	VIN4L_CLK_SW0	RW	0x0	VIN4L clock switch 0 source selection control. 0: VIN4L clock switch0 source from video1_sclk_div0; 1: VIN4L clock switch0 source from video0_sclk_div0;
[5]	VIN4L_CLK_SW1	RW	0x0	VIN4L clock switch 1 source selection control. 0: VIN4L clock switch1 source from video1_sclk_div1; 1: VIN4L clock switch1 source from output of VIN4L clock switch 0;
[6]	VIN4L_CLK_SW2	RW	0x0	VIN4L clock switch 2 source selection control. 0: VIN4L clock switch2 source from output of VIN4L clock switch 1; 1: VIN4L clock switch2 source from external sensor clock;
[7]	VIN4L_CLK_SW3	RW	0x0	VIN4L pixel clock source selection control. 0: VIN4L clock switch3 source from output of VIN4L clock switch 2; 1: VIN4L clock switch3 source from OSC12M;
[8]	VO_CLK_SW0	RW	0x0	VideoOut clock switch 0 source selection control. 0: VideoOut clock switch0 source from video1_sclk_div0; 1: VideoOut clock switch0 source from video0_sclk_div0;
[9]	VO_CLK_SW1	RW	0x0	VideoOut clock switch 1 source selection control. 0: VideoOut clock switch1 source from video1_sclk_div1; 1: VideoOut clock switch1 source from output of VideoOut clock switch 0;

Bits	Name	R/W	Reset	Description
[10]	VO_CLK_SW2	RW	0x0	VideoOut clock switch 2 source selection control. 0: VideoOut clock switch2 source from output of VideoOut clock switch 1; 1: VideoOut clock switch2 source from video0_sclk_div1;
[11]	VO_CLK_SW3 (VO_PIXEL_CLK X4)	RW	0x0	VideoOut clock switch 3 (VO_PIXEL_CLKX4) source selection control. 0: VideoOut clock switch3 source from output of VideoOut clock switch 2; 1: VideoOut clock switch3 source from OSC12Mhz;
[12]	VO_CLK_SW4 (VO_PIXEL_CLK)	RW	0x0	VideoOut clock switch 4 (VO_PIXEL_CLK) source selection control. 0: VideoOut clock switch4 source from output of VideoOut clock switch 3; 1: VideoOut clock switch4 source from video0_sclk_div2;
[13]	ISP_CLK_SW	RW	0x0	ISP pixel clock source selection control. 0: ISP clock switch source from VIN2L clock switch 2; 1: ISP clock switch source from vin4l_pixel_clk;
[14]	Reserved			
[15]	DISP_CLK_SW	RW	0x0	DISPLAY clock (output to LCD_PCLK pin) source selection control. 0:DISPLAY clock switch1 source from Video Out pixel clock; 1: DISPLAY clock switch1 source from invert of Video Out pixel clock;
[16]	VIDEO_XCLK_SW0	RW	0x0	Video subsystem AXI clock switch 0 source selection control. 0: Video AXI clock switch0 source from mcu axi clock; 1: Video AXI clock switch0 source from video1 PLL AXI divider;
[17]	VIDEO_XCLK_SW1	RW	0x0	Bit: Video AXI clock switch 1 source selection control. 0: Video AXI clock switch1 source from output of Video AXI clock switch 0; 1: Video AXI clock switch1 source from OSC12Mhz;

Bits	Name	R/W	Reset	Description
[18]	VIN_DPHY_CLK_SW	RW	0x0	VIN2L_CSI_DPHY_CLK source selection control. 0: VideoPhy clock switch source from video0_sclk_div1(default to 24MHz); 1: VideoPhy clock switch source from OSC12Mhz;
[19]	Reserved			
[20]	VPU_CLK_SEL	RW	0x1	VPU clock source select 0: Clock from MCU PLL/3, which is 400MHz 1: Clock from DDR PLL/3, which is 355MHz
[31:21]	Reserved			

4.5.1.7. DDR_CLK_CFG

Offset: 0x001C

Bits	Name	R/W	Reset	Description
[1:0]	ddr_pll_div0	RW	0x1	ddr_pll_div0_clk = ddr_clk_from_pll/(ddr_pll_div0+1) It is used for ddr subsystem.By default it is 1066/2=533Mhz
[3:2]	ddr_pll_div0	RW	0x2	ddr_pll_div1_clk = ddr_clk_from_pll/(ddr_pll_div1+1) It is used as a back-up clock source for vpu.By default it is 1066/3=355Mhz
[31:4]	Reserved	RO	0x0	Reserved.

4.5.1.8. MCU_BLK_CG

Offset: 0x0030

Bits	Name	R/W	Reset	Description
[0]	SDMMC0_CG	RW	0x0	1: clock gating off the SDMMC0 module;
[1]	SDMMC1_CG	RW	0x0	1: clock gating off the SDMMC1 module;
[2]	DMA_CG	RW	0x0	1: clock gating off the DMA module;
[3]	QSPI_CG	RW	0x0	1: clock gating off the QSPI module;
[4]	DDR whole CG	RW	0x0	1: clock gating off the DDRC & DDR all Channel.
[5]	DDR apb_clk CG	RW	0x0	1: clock gating off the DDR apb bus
[6]	DDR Channel 0 CG	RW	0x0	1: clock gating off the DDR Channel 0

[7]	DDR Channel 1 CG	RW	0x0	1: clock gating off the DDR Channel 1
[8]	DDR Channel 2 CG	RW	0x0	1: clock gating off the DDR Channel 2
[9]	I2C0_CG	RW	0x0	1: clock gating off the I2C0 module;
[10]	I2C1_CG	RW	0x0	1: clock gating off the I2C1 module;
[11]	SPI0_CG	RW	0x0	1: clock gating off SPI0 & OTP module;
[12]	SPI1_CG	RW	0x0	1: clock gating off SPI1 module;
[13]	SAR ADC_CG	RW	0x0	1: clock gating off SAR ADC controller;
[14]	UART0_CG	RW	0x0	1: clock gating off UART0 module;
[15]	UART1_CG	RW	0x0	1: clock gating off UART1 module;
[16]	UART2_CG	RW	0x0	1: clock gating off UART2 module;
[17]	UART3_CG	RW	0x0	1: clock gating off UART3 module;
[18]	WDT_CG	RW	0x0	1: clock gating off WDT module;
[19]	TIMER_CG	RW	0x0	1: clock gating off Timer module;
[20]	GPIO_CG	RW	0x0	1: clock gating off GPIO module;
[21]	Reserved	RW	0x0	Reserved
[22]	Reserved	RW	0x0	Reserved
[23]	Reserved	RW	0x0	Reserved
[24]	USBOTG_CG	RW	0x0	1: clock gating off USBCTRL module;
[25]	Reserved	RW	0x0	Reserved
[27]	GPIO2_CG	RW	0x0	1: clock gating off GPIO2 module;
[28]	GPIO3_CG	RW	0x0	1: clock gating off GPIO3 module;
[29]	GPIO4_CG	RW	0x0	1: clock gating off GPIO4 module;
[30]	EMAC_CG	RW	0x0	1: clock gating off EMAC module;
[31]	Reserved			

4.5.1.9. AUDIO_VIDEO_BLK_CG

Offset: 0x0034

Bits	Name	R/W	Reset	Description
[0]	I2S_ADC_CG	RW	0x0	1: clock gating off the Audio I2S ADC module;
[1]	I2S_DAC_CG	RW	0x0	1: clock gating off the Audio I2S DAC module;
[2]	Reserved	RW	0x0	Reserved
[3]	Reserved	RW	0x0	Reserved
[4]	CODEC_CG	RW	0x0	1: clock gating off the Audio Codec module;
[7:5]	Reserved			
[8]	I2C2_CG	RW	0x0	1: clock gating off the I2C2;
[9]	I2C3_CG	RW	0x0	1: clock gating off the I2C3;

[15:10]	Reserved			
[16]	VIN0_CG	RW	0x0	1: clock gating off the Video In (CSI Ctrl) module;
[17]	VIN1_CG	RW	0x0	1: clock gating off the Video In (DVP) module;
[18]	ISP_CG	RW	0x0	1: clock gating off the ISP module;
[19]	VPU_CG	RW	0x0	1: clock gating off the Video VPU module;
[20]	TD_CG	RW	0x0	1: clock gating off the Video 2D module;
[21]	VO_CG	RW	0x0	1: clock gating off the Video Out module;
[22]	DSIC_CG	RW	0x0	1: clock gating off the DSI Ctrl module;
[23]	MIPICSI_CG	RW	0x0	1: clock gating off the MIPI CSI DPHY module;
[24]	MIPIDSI_CG	RW	0x0	1: clock gating off the MIPI DSI DPHY module;
[31:25]	Reserved			

4.5.2. Software Reset Registers

4.5.2.1. GLOBAL_SOFT_RST

Offset: 0x0020

Bits	Name	R/W	Reset	Description
[31:0]	SW_GLB_RST	RW	0x0	Software Global Reset. Write 0xAA55A5A5 to this register will cause the global reset. When enable, it resets whole chip except clock generate and related control registers. Write other value has no effect.

4.5.2.2. MCU_BLK_RST

Offset: 0x0024

Bits	Name	R/W	Reset	Description
[0]	SDMMC0_ SRST	RW	0x1	0: Reset SDMMC0 module;
[1]	SDMMC1_ SRST	RW	0x1	0: Reset SDMMC1 module;
[2]	DMA _SRST	RW	0x1	0: Reset DMA module;
[3]	QSPI _SRST	RW	0x1	0: Reset QSPI module;
[4]	Reserved	RW	0x1	Reserved
[5]	Reserved	RW	0x1	Reserved
[6]	Reserved	RW	0x1	Reserved
[7]	Reserved	RW	0x1	Reserved
[8]	EFUSE_SRST	RW	0x1	0: Reset EFUSE module
[9]	I2C0 _SRST	RW	0x1	0: Reset I2C0 module;

[10]	I2C1 _SRST	RW	0x1	0: Reset I2C1 module;
[11]	SPI0 _SRST	RW	0x1	0: Reset SPI0 module;
[12]	SPI1 _SRST	RW	0x1	0: Reset SPI1 module;
[13]	SAR ADC _SRST	RW	0x1	0: Reset SAR ADC controller;
[14]	UART0 _SRST	RW	0x1	0: Reset UART0 module;
[15]	UART1_ SRST	RW	0x1	0: Reset UART1 module;
[16]	UART2_ SRST	RW	0x1	0: Reset UART2 module;
[17]	UART3_ SRST	RW	0x1	0: Reset UART3 module;
[18]	WDT_ SRST	RW	0x1	0: Reset WDT module;
[19]	TIMER_ SRST	RW	0x1	0: Reset Timer module;
[20]	GPIO_ SRST	RW	0x1	0: Reset GPIO module;
[21]	TRNG_ SRST	RW	0x1	0: Reset TRNG module;
[22]	Reserved	RW	0x1	Reserved
[23]	Reserved	RW	0x1	Reserved
[24]	USBOTG_ SRST	RW	0x1	0: Reset USBCTRL module;
[25]	Reserved	RW	0x0	Reserved
[26]	Reserved	RW	0x0	
[27]	GPIO2_ SRST	RW	0x1	0: Reset GPIO2 module;
[28]	GPIO3_ SRST	RW	0x1	0: Reset GPIO3 module;
[29]	GPIO4_ SRST	RW	0x1	0: Reset GPIO4 module;
[30]	EMAC_SRST	RW	0x1	0: Reset EMAC module & MCU2TF H2X module
[31]	CS_SRST	RW	0x1	0: Reset CoreSight module;

4.5.2.3. AUDIO_VIDEO_BLK_RST

Offset: 0x0028

Bits	Name	R/W	Reset	Description
[0]	I2S_ADC_SRST	RW	0x1	0: Reset I2S_ADC module;
[1]	I2S_DAC_SRST	RW	0x1	0: Reset I2S_DAC module;
[2]	I2S0_SRST	RW	0x1	0: Reset I2S0 module;
[3]	I2S1_SRST	RW	0x1	0: Reset I2S1 module;
[4]	CODEC_SRST	RW	0x1	0: Reset Audio Codec module;
[7:5]	Reserved			
[8]	I2C2_SRST	RW	0x1	0: Reset I2C2;
[9]	I2C3_SRST	RW	0x1	0: Reset I2C3;
[15:10]	Reserved			
[16]	VIN0_SRST	RW	0x1	0: Reset Video In (CSI Ctrl) module;
[17]	VIN1_SRST	RW	0x1	0: Reset Video In (DVP) module;
[18]	ISP_SRST	RW	0x1	0: Reset ISP module;

[19]	VPU_SRST	RW	0x1	0: Reset Video VPU module;
[20]	TD_SRST	RW	0x1	0: Reset Video 2D module;
[21]	VO_SRST	RW	0x1	0: Reset Video OUT module;
[22]	DSI_SRST	RW	0x1	0: Reset DSI Ctrl module;
[31:23]	Reserved			

4.5.2.4. DDR_RESET_RELEASE

Offset: 0x002C

Bits	Name	R/W	Reset	Description
[0]	DDRC_APB_SRST	RW	0x0	DDRC APB Configure Reset Release. 0:Reset; 1:Release
[1]	Reserved			
[2]	DDRC_AXI1_SRST	RW	0x0	DDRC AXI1 Configure Reset Release. 0:Reset; 1:Release
[3]	DDRC_PHY_SRST	RW	0x0	DDRC/PHY Reset Release. 0:Reset; 1:Release
[31:4]	Reserved			

4.5.3. GPIO Pinmux Registers

The number in ( ) indicate the peripheral IO remap as configured by PIN_REMAP_CTRL

Offset: 0x00A0

Bits	Name	R/W	Reset	Description
[2:0]	LCD_DEN_FUNC	RW	0x0	LCD_DEN pin function select 0x0: LCD_DEN 0x2: SENSOR_PWD 0x4: GPIO1_27 0x6: DBIB_CSX

[6:4]	LCD_HSYNC_FUNC	RW	0x0	0x0: LCD_HSYNC 0x1: EMAC_RXDV 0x4: GPIO1_26 0x6: DBIB_WRX
[10:8]	LCD_VSYNC_FUNC	RW	0x0	0x0: LCD_VSYNC 0x1: EMAC0_PHYOSC 0x3: SDIO1_WP 0x4: GPIO1_25 0x6: DBIB_DCX
[14:12]	LCD_PCLK_FUNC	RW	0x0	0x0: LCD_PCLK 0x1: EMAC0_MDIO 0x2: SENSOR0_RSTN 0x3: SDIO1_CD (0) 0x4: GPIO1_24 0x6: DBIB_RDX
[18:16]	I2C0_SDA_FUNC	RW	0x0	0x0: I2C0_SDA 0x1: GPIO1_29
[22:20]	I2C0_SCL_FUNC	RW	0x0	0x0:I2C0_SCL 0x1: GPIO1_28

4.5.3.1. **PIN_FUNC_SEL1**

Offset: 0x00A4

Bits	Name	R/W	Reset	Description
[15:0]	Reserved			
[18:16]	LCD_DATA15_FUNC*	RW	0x0	0x0: LCD_DATA15 0x1:EMAC0_CRS 0x2:SDIO1_DATA3 0x3:LCD_DATA7 0x4: GPIO1_15 0x5:DBIB_DATA7 0x6:DBIB_DATA15

[22:20]	LCD_DATA14_FUNC*	RW	0x0	0x0:LCD_DATA14 0x1:EMAC0_RXCLK 0x2:SDIO1_DATA2 0x3:LCD_DATA6 0x4: GPIO1_14 0x5:DBIB_DATA6 0x6:DBIB_DATA14
[26:24]	LCD_DATA13_FUNC*	RW	0x0	0x0:LCD_DATA13 0x1:EMAC0_TXER 0x2:SDIO1_DATA1 0x3:LCD_DATA5 0x4:GPIO1_13 0x5:DBIB_DATA5 0x6:DBIB_DATA13
[30:28]	LCD_DATA12_FUNC*	RW	0x0	0x0: LCD_DATA12 0x1:EMAC0_TXCLK 0x2:SDIO1_DATA0 0x3:LCD_DATA4 0x4: GPIO1_12 0x5: DBIB_DATA4 0x6:DBIB_DATA12

* Pins are only available in BGA package

4.5.3.2. **PIN_FUNC_SEL2**

Offset: 0x00A8

Bits	Name	R/W	Reset	Description
[2:0]	LCD_DATA11_FUNC*	RW	0x0	0x0:LCD_DATA11 0x1:EMAC0_TXD3 0x2:SDIO1_CMD 0x3:LCD_DATA3 0x4:GPIO1_11 0x5:DBIB_DATA3 0x6:DBIB_DATA11

[6:4]	LCD_DATA10_FUNC*	RW	0x0	0x0:LCD_DATA10 0x1:EMAC0_TXD2 0x2:SDIO1_CLK 0x3:LCD_DATA2 0x4: GPIO1_10 0x5:DBIB_DATA2 0x6:DBIB_DATA10
[10:8]	LCD_DATA9_FUNC*	RW	0x0	0x0:LCD_DATA9 0x1:EMAC0_RXD3 0x2:SDIO1_WP 0x3:LCD_DATA1 0x4:GPIO1_9 0x5:DBIB_DATA1 0x6:DBIB_DATA9
[14:12]	LCD_DATA8_FUNC*	RW	0x0	0x0: LCD_DATA8 0x1:EMAC0_RXD2 0x2:SDIO1_CD 0x3:LCD_DATA0 0x4: GPIO1_8 0x5:DBIB_DATA0 0x6:DBIB_DATA8
[18:16]	LCD_DATA7_FUNC	RW	0x0	0x0: LCD_DATA7 0x1: EMAC0_RXER 0x2: PWM7 0x3: SDIO1_DATA0 0x4: GPIO1_7 0x6: DBIB_DATA7
[22:20]	LCD_DATA6_FUNC	RW	0x0	0x0: LCD_DATA6 0x1: EMAC0_RXD0 0x2: PWM6 0x3: SDIO1_DATA1 0x4: GPIO1_6 0x6: DBIB_DATA6
[26:24]	LCD_DATA5_FUNC	RW	0x0	0x0: LCD_DATA5 0x1: EMAC0_RXD1 0x2: PWM5 0x3: SDIO1_DATA2 0x4: GPIO1_5 0x5: SPI0_CSN1 0x6: DBIB_DATA5

[30:28]	LCD_DATA4_FUNC	RW	0x0	0x0: LCD_DATA4 0x1: EMAC0_COL (MII) / EMAC0_REFCLK(RMII) 0x2:PWM4 0x3:SDIO1_DATA3 0x4: GPIO1_4 0x6:DBIB_DATA4

* Pins are only available in BGA package

4.5.3.3. PIN_FUNC_SEL3

Offset: 0x00AC

Bits	Name	R/W	Reset	Description
[2:0]	LCD_DATA3_FUNC	RW	0x0	0x0: LCD_DATA3 0x1: EMAC0_TXD0 0x2: PWM3 0x3: SDIO1_CLK 0x4: GPIO1_3 0x5: SPI0_CSN0 0x6: DBIB_DATA3
[6:4]	LCD_DATA2_FUNC	RW	0x0	0x0: LCD_DATA2 0x1: EMAC0_TXD1 0x2: PWM2 0x3: SDIO1_CMD 0x4: GPIO1_2 0x5: SPI0_MOSI 0x6:DBIB_DATA2
[10:8]	LCD_DATA1_FUNC	RW	0x0	0x0: LCD_DATA1 0x1: EMAC0_TXEN 0x2: PWM1 0x3: TRIG0 0x4: GPIO1_1 0x5: SPI0_MISO 0x6: DBIB_DATA1
[14:12]	LCD_DATA0_FUNC	RW	0x0	0x0: LCD_DATA0 0x1: EMAC0_MDC 0x2: PWM0 0x4: GPIO1_0 0x5: SPI0_SCLK 0x6:DBIB_DATA0

[18:16]	DVP0_PCLK_FUNC	RW	0x0	0x0: DVP0_PCLK 0x2: GPIO2_11 0x3: I2C2_SDA
[22:20]	DVP0_DIN11_FUNC*	RW	0x0	0x0: DVP0_DIN11 0x2: GPIO2_11 0x3: I2C2_SCL
[26:24]	DVP0_DIN10_FUNC*	RW	0x0	0x0: DVP0_DIN10 0x1: QSPI_CLK 0x2: GPIO2_10 0x3:I2C2_SDA 0x4:UART3_RXD
[30:28]	DVP0_DIN9_FUNC*	RW	0x0	0x0: DVP0_DIN9 0x1:QSPI_CSN 0x2: GPIO2_9

* Pins are only available in BGA package

4.5.3.4. **PIN_FUNC_SEL4**

Offset: 0x00B0

Bits	Name	R/W	Reset	Description
[2:0]	DVP0_DIN8_FUNC*	RW	0x0	0x0: DVP0_DIN8 0x1: QSPI_DIO0 0x2: GPIO2_8
[6:4]	DVP0_ DIN7_FUNC	RW	0x0	0x0: DVP0_DIN7 0x1: QSPI_DIO1 0x2: GPIO2_7 0x4: PWM7
[10:8]	DVP0_ DIN6_FUNC	RW	0x0	0x0: DVP0_DIN6 0x1: QSPI_DIO2 0x2: GPIO2_6 0x3: I2C1_SDA 0x4: PWM6
[14:12]	DVP0_ DIN5_FUNC	RW	0x0	0x0: DVP0_DIN5 0x1: QSPI_DIO3 0x2: GPIO2_5 0x3: I2C1_SCL 0x4: PWM5
[18:16]	DVP0_ DIN4_FUNC	RW	0x0	0x0: DVP0_DIN4 0x2: GPIO2_4 0x4: PWM4

[22:20]	DVP0_ DIN3_FUNC	RW	0x0	0x0: DVP0_DIN3 0x2:GPIO2_3 0x4: PWM3
[26:24]	DVP0_ DIN2_FUNC	RW	0x0	0x2: DVP0_DIN2 0x0:GPIO2_2 0x4: PWM2
[30:28]	DVP0_ DIN1_FUNC	RW	0x0	0x2: DVP0_DIN1 0x0:GPIO2_1 0x4: PWM1

* Pins are only available in BGA package

4.5.3.5. PIN_FUNC_SEL5

Offset: 0x00B4

Bits	Name	R/W	Reset	Description
[2:0]	DVP0_DIN0_FUNC	RW	0x0	0x0: DVP0_DIN0 0x1: GPIO2_0 0x4: PWM0
[15:3]	-	-	-	-
[18:16]	GPIOF_FUNC*	RW	0x0	0x0:GPIO2_27
[22:20]	LCD_DATA23_FUNC	RW	0x0	0x0: LCD_DOUT23 0x4: GPIO2_26
[26:24]	LCD_DATA22_FUNC	RW	0x0	0x0: LCD_DOUT22 0x4: GPIO2_25
[30:28]	LCD_DATA21_FUNC	RW	0x0	0x0: LCD_DOUT21 0x3: UART1_TXD 0x4: GPIO2_24

* Pins are only available in BGA package

4.5.3.6. PIN_FUNC_SEL6

Offset: 0x00B8

Bits	Name	R/W	Reset	Description
[2:0]	LCD_DATA20_FUNC	RW	0x0	0x0: LCD_DOUT20 0x1: EMAC0_MDIO 0x2: I2C3_SCL 0x3: UART1_RXD 0x4: GPIO2_23
[6:4]	LCD_DATA19_FUNC	RW	0x0	0x0: LCD_DOUT19 0x1: EMAC0_TXD1 0x2: I2C3_SDA 0x4: GPIO2_22
[10:8]	LCD_DATA18_FUNC	RW	0x0	0x0: LCD_DOUT18 0x1: EMAC0_TXD0 0x2: UART2_TXD 0x4: GPIO2_21
[14:12]	LCD_DATA17_FUNC	RW	0x0	0x0: LCD_DOUT17 0x1: EMAC0_RXD1 0x2: UART2_RXD 0x4: GPIO2_20
[18:16]	LCD_DATA16_FUNC	RW	0x0	0x0: LCD_DOUT16 0x1: EMAC0_RXD0 0x4: GPIO2_19

* Pins are only available in BGA package

4.5.3.7. PIN_FUNC_SEL7

Offset: 0x00BC

Bits	Name	R/W	Reset	Description
[2:0]	SDIO0_CLK_FUNC	RW	0x1	0x0: SDIO0_CLK 0x1: GPIO3_3 0x2: I2C1_SDA 0x3: UART3_RTSN 0x6: EMAC0_REFCLK
[6:4]	SDIO0_CMD_FUNC	RW	0x1	0x0: SDIO0_CMD 0x1: GPIO3_2 0x2: JTAG_TCK 0x4: SPI1_CSN0 0x5: UART1_RXD 0x6: EMAC0_TXD0

[10:8]	SDIO0_DATA3_FUNC	RW	0x1	0x0: SDIO0_DATA3 0x1: GPIO3_1 0x2: JTAG_TRSTN 0x4: SPI1_SCLK 0x6: EMAC0_MIDO
[14:12]	SDIO0_DATA2_FUNC	RW	0x1	0x0: SDIO0_DATA2 0x1: GPIO3_0 0x2: JTAG_TDI 0x4: SPI1_MOSI 0x6: EMAC0_MDC
[18:16]	SDIO0_DATA1_FUNC	RW	0x1	0x1: SDIO0_DATA1 0x0: GPIO3_5 0x2: JTAG_TDO 0x4: SPI1_MISO 0x6: EMAC0_TXEN
[22:20]	SDIO0_DATA0_FUNC	RW	0x1	0x0: SDIO0_DATA0 0x1: GPIO3_4 0x2: JTAG_TMS 0x4: SPI1_CSN1 0x5: UART1_RXD 0x6: EMAC0_TXD1
[26:24]	SDIO0_CD_FUNC	RW	0x1	0x1: SDIO0_CD 0x0: GPIO3_6 0x2: I2C1_SCL 0x3: UART3_CTSN
[30:28]	SDIO0_WP_FUNC	RW	0x1	0x1: SDIO0_WP 0x0: GPIO3_7

4.5.3.8. PIN_FUNC_SEL8

Offset: 0x0110

Bits	Name	R/W	Reset	Description
[2:0]	GPIOA_FUNC	RW	0x0	0x0: GPIO1_30 0x1: EMAC0_MDC 0x2: SPI1_MOSI 0x3: PWM0 0x4: UART0_RXD

[6:4]	GPIOB_FUNC	RW	0x0	0x0: GPIO1_31 0x1: EMAC0_RXDV 0x2: SPI1_MISO 0x3: PWM1 0x4: UART0_TXD 0x6: EMAC0_PHYOSC
[10:8]	GPIOC_FUNC	RW	0x0	0x0: GPIO2_30 0x1: EMAC0_COL 0x2: SPI1_CSN1 0x3: PWM2 0x4: I2C2_SDA 0x6: EMAC0_RXD1
[14:12]	GPIOD_FUNC	RW	0x0	0x0: GPIO2_31 0x1: EMAC0_TXEN 0x2: SPI1_CSN0 0x3: PWM3 0x4: I2C2_SCL 0x6: EMAC0_RXD0
[18:16]	GPIOE_FUNC	RW	0x0	0x0: GPIO3_24 0x2: SPI1_SCLK 0x3: PWM4 0x6: EMAC0_RXDV
[22:20]		RW	0x0	
[26:24]	JTAG_TCK_FUNC	RW	0x0	0x0: JTAG_TCK 0x1: SPI0_CSN1 0x3: UART3_TXD 0x5: GPIO3_13
[30:28]	JTAG_TDI_FUNC	RW	0x0	0x0: JTAG_TDI (0) 0x1: SPI0_SCLK 0x3: PWM7 0x2: GPIO3_15

4.5.3.9. **PIN_FUNC_SEL9**

Offset: 0x0114

Bits	Name	R/W	Reset	Description
[2:0]	JTAG_TMS_FUNC	RW	0x0	0x0: JTAG_TMS 0x1: SPI0_MISO 0x2: GPIO3_14 0x3: UART3_CTSN 0x5: DMIC_CLK

[6:4]	JTAG_TRSTN_FUNC	RW	0x0	0x0: JTAG_TRSTN 0x1: SPI1_CSN0 0x3: UART3_RXD 0x5: GPIO3_12
[10:8]	JTAG_TDO_FUNC	RW	0x0	0x0: JTAG_TDO 0x1: SPI1_MOSI 0x3: PWM5 0x5: GPIO3_16
[14:12]	UART0_TXD_FUNC	RW	0x0	0x0:UART0_TXD 0x1:GPIO3_10 0x2: LCD_RESET 0x3:DRVVBUS_USB0 0x4: I2C3_SCL
[18:16]	UART0_RXD_FUNC	RW	0x0	0x0:GPIO3_11 0x1:UART0_RXD 0x2:TRIG1 0x4: I2C3_SDA
[22:20]	PWM0_FUNC	RW	0x0	0x0: PWM0 0x1: GPIO3_8 0x3: I2C1_SCL
[26:24]	PWM1_FUNC	RW	0x0	0x1: PWM1 0x0: GPIO3_9 0x3: I2C1_SDA
[30:28]	BOOT_SEL0_FUNC	RW	0x0	0x0: BOOT_SEL0 0x1: GPIO3_17

4.5.3.10. **PIN_FUNC_SEL10**

Offset: 0x0118

Bits	Name	R/W	Reset	Description
[2:0]	SENSOR0_RSTN_FUNC	RW	0x0	0x0: SENSOR0_RSTN 0x1: GPIO3_18 0x2: UART0_RXD 0x3: I2C2_SDA
[6:4]	SENSOR0_PWD_FUNC	RW	0x0	0x0: SENSOR0_PWD 0x1: GPIO3_19 0x2: UART2_TXD 0x3: I2C2_SCL
[10:8]	SENSOR0_CLK_FUNC	RW	0x2	0x0: SENSOR0_CLK 0x1: GPIO3_20

[14:12]	-	-	-	-
[18:16]	SENSOR1_PWD_FUNC*	RW	0x0	0x0: SENSOR1_PWD 0x1: GPIO3_22
[22:20]	SENSOR1_CLK_FUNC*	RW	0x0	0x1: SENSOR1_CLK 0x0: GPIO3_23
[26:24]	DVP0_HREF_FUNC	RW	0x0	0x0: DVP0_HREF 0x2: GPIO2_28
[30:28]	DVP0_VSYNC_FUNC	RW	0x0	0x0: DVP0_VSYNC 0x2: GPIO2_29

4.5.3.11. **PIN_FUNC_SEL11**

Offset: 0x010c

Bits	Name	R/W	Reset	Description
[2:0]	QSPI_DIO0_FUN	RW	0x0	0x00: QSPI_DIO0
[6:4]	QSPI_CLK_FUN	RW	0x0	0x00: QSPI_CLK
[10:8]	QSPI_CSN0_FUN	RW	0x0	0x00: QSPI_CSN0
[14:12]	QSPI_DIO1_FUN	RW	0x0	0x00: QSPI_DIO1
[18:16]	QSPI_DIO3_FUN	RW	0x0	0x00: QSPI_DIO3 0x01: I2C1_SDA
[22:20]	QSPI_DIO2_FUN	RW	0x0	0x00: QSPI_DIO2 0x01: I2C1_SCL

* Pins are only available in BGA package

4.5.3.12. **PIN_REMAP_CTRL**

Offset: 0x011C

Bits	Name	R/W	Reset	Description
[1 :0]	DBIB_RMP	RW	0x0	0x0: DBIB_RMP0 0x1: DBIB_RMP1
[3 :2]	EMAC0_RMP	RW	0x0	0x0: EMAC0_RMP0(MII) 0x1: EMAC0_RMP1(RMII) 0x2: EMAC0_RMP2(RMII)
[5 :4]	I2C0_RMP	RW	0x0	0x0: I2C0_RMP
[7 :6]	I2C1_RMP	RW	0x0	0x0: I2C1_RMP0 0x1: I2C1_RMP1 0x2: I2C1_RMP2

[9 :8]	I2C2_RMP	RW	0x0	0x0: I2C2_RMP0 0x1: I2C2_RMP0
[11:10]	I2C3_RMP	RW	0x0	0x0: I2C3_RMP0 0x1: I2C3_RMP0
[13:12]	JTAG_RMP	RW	0x0	0x0: JTAG_RMP0 0x1: JTAG_RMP1
[15:14]	LCD_RMP	RW	0x0	0x0: LCD_RMP0 0x1: LCD_RMP1
[17:16]	PWM0_RMP	RW	0x0	0x0: PWM0_RMP0 0x1: PWM0_RMP1 0x2: PWM0_RMP2
[19:18]	PWM1_RMP	RW	0x0	0x0: PWM1_RMP0 0x1: PWM1_RMP1 0x2: PWM1_RMP2
[21:20]	PWM2_RMP	RW	0x0	0x0: PWM2_RMP0 0x1: PWM2_RMP1 0x2: PWM2_RMP2
[23:22]	PWM3_RMP	RW	0x0	0x0: PWM3_RMP0 0x1: PWM3_RMP1 0x2: PWM3_RMP2
[25:24]	PWM4_RMP	RW	0x0	0x0: PWM4_RMP0 0x1: PWM4_RMP1 0x2: PWM4_RMP2
[27:26]	PWM5_RMP	RW	0x0	0x0: PWM5_RMP0 0x1: PWM5_RMP1 0x2: PWM5_RMP2
[29:28]	PWM6_RMP	RW	0x0	0x0: PWM6_RMP0 0x1: PWM6_RMP1 0x2: PWM6_RMP2
[31:30]	PWM7_RMP	RW	0x0	0x0: PWM7_RMP0 0x1: PWM7_RMP1 0x2: PWM7_RMP2

4.5.3.13. **PIN_REMAP_CTRL1**

Offset: 0x0138

[1 :0]	QSPI_RMP	RW	0x0	0x0: QSPI_RMP1 0x1: QSPI_RMP0
[3 :2]	SDIO0_RMP	RW	0x0	0x0: SDIO0_RMP
[5 :4]	SDIO1_RMP	RW	0x0	0x0: SDIO1_RMP 0x1: SDIO1_RMP1

[7 :6]	SENSOR0_RMP	RW	0x0	0x0: SENSOR0_RMP0 0x1: SENSOR0_RMP1
[9 :8]	SPI0_RMP	RW	0x0	0x0: SPI0_RMP0 0x1: SPI0_RMP1
[11:10]	SPI1_RMP	RW	0x0	0x0: SPI1_RMP0 0x1: SPI1_RMP1
[13:12]	UART0_RMP	RW	0x0	0x0: UART0_RMP0 0x1: UART0_RMP1
[15:14]	UART1_RMP	RW	0x0	0x0: UART1_RMP0 0x1: UART1_RMP1
[17:16]	UART2_RMP	RW	0x0	0x0: UART2_RMP0 0x1: UART2_RMP1
[19:18]	UART3_RMP	RW	0x0	0x0: UART3_RMP0 0x1: UART3_RMP1
[31:20]	reserve	-	-	-

All GPIO pads use 3-state bi-direction I/O pads with

- Controlled driving strength of middle level, slew rate, Schmitt trigger and
- Controllable pull-down, pull-up resistor and
- 3.3V/1.8V input and output.

4.5.4. GPIO Control Registers

All GPIO pads use 3-state bi-direction I/O pads with

- Controlled driving strength of middle level, slew rate, Schmitt trigger and
- Controllable pull-down, pull-up resistor and
- 3.3V/1.8V input and output.

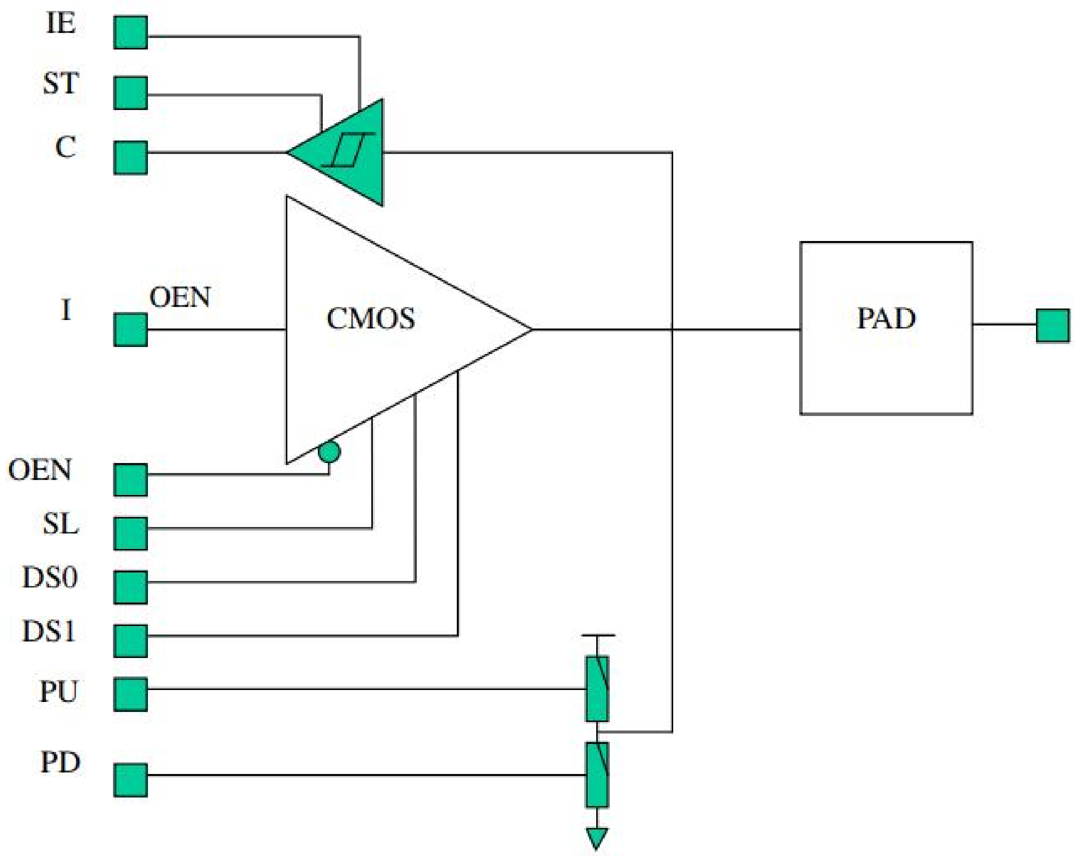


Figure 4- 6 GPIO PAD structure

GPIO Pull up/Pull down features can be configured by the two registers such as PIN_PULLDIS_CTRL [x] and PIN_PULLUP_CTRL [x].

When PIN_PULLDIS_CTRL [x]=0 & PIN_PULLUP_CTRL [x]=1,
GPIO has internal pull-up resistor activated
When PIN_PULLDIS_CTRL [x]=0 & PIN_PULLUP_CTRL [x]=0,
GPIO has internal pull-down resistor activated

When PIN_PULLDIS_CTRL [x]=1,
GPIO has internal pull resistor disabled.

Table 4- 5 Driving strength setting condition

	@SS, 125C DVDDIO=2.97V DVDD=0.97V Ioh @ 2.4V	@SS, 125C DVDDIO=2.97V DVDD=0.97V Iol @ 0.4V	@TT,25C DVDDIO=3.3V DVDD=1.1V Ioh @ 2.4V	@TT,25C DVDDIO=3.3V DVDD=1.1V Iol @ 0.4V
DS1=0, DS0=0	7.0mA	5.7mA	15.5mA	9.3mA
DS1=0. DS0=1	9.3mA	7.5mA	20.6mA	12.3mA

4.5.4.1. PIN_PULLUP_CTRL

Offset: 0x00C0

Bits	Name	R/W	Reset	Description
[0]	SDIO0_CLK	RW	0x0	5. 6. IO pin pull up/pull down control; Default: 0x20000 7. 0: pull down is selected when pull logic is active (corresponding bit in PIN_PULLDIS is L) 8. 1: pull up is selected when pull logic is active (corresponding bit in PIN_PULLDIS is L)
[1]	SDIO0_CMD	RW	0x0	
[2]	SDIO0_DATA3	RW	0x0	
[3]	SDIO0_DATA2	RW	0x0	
[4]	SDIO0_DATA1	RW	0x0	
[5]	SDIO0_DATA0	RW	0x0	
[6]	SDIO0_CD	RW	0x0	
[7]	SDIO0_WP	RW	0x0	
[8]	QSPI_CLK	RW	0x0	
[9]	QSPI_DIO0	RW	0x0	
[10]	QSPI_DIO1	RW	0x0	
[11]	QSPI_DIO2	RW	0x0	
[12]	QSPI_DIO3	RW	0x0	
[13]	QSPI_CSN0	RW	0x0	
[14]	JTAG_TCK	RW	0x0	
[15]	JTAG_TDI	RW	0x0	
[16]	JTAG_TMS	RW	0x0	
[17]	JTAG_TRSTND	RW	0x1	
[18]	JTAG_TDO	RW	0x0	
[19]	UART0_TXD	RW	0x0	
[20]	UART0_RXD	RW	0x0	
[21]	PWM0	RW	0x0	
[22]	PWM1	RW	0x0	
[23]	Reserved	RW	0x0	
[24]	BOOT_SEL	RW	0x0	
[25]	I2C0_SCL	RW	0x0	
[26]	I2C0_SDA	RW	0x0	
[27]	GPIOA	RW	0x0	
[28]	GPIOB	RW	0x0	
[29]	GPIOC	RW	0x0	
[30]	GPIOD	RW	0x0	
[31]	GPIOE	RW	0x0	

8.1.1.1. **VS_PIN_PULLUP_CTRL1**

Offset: 0x00C4

Bits	Name	R/W	Reset	Description
[0]	LCD_DEN	RW	0x0	9. 10. IO pin pull up/pull down control for video output block Default: 0x00000000 11. 0: pull down is selected when pull logic is active (corresponding bit in PIN_PULLDIS is L) 12. 1: pull up is selected when pull logic is active (corresponding bit in PIN_PULLDIS is L) 13.
[1]	LCD_HSYNC	RW	0x0	
[2]	LCD_VSYNC	RW	0x0	
[3]	LCD_PCLK	RW	0x0	
[4]	LCD_DATA23	RW	0x0	
[5]	LCD_DATA22	RW	0x0	
[6]	LCD_DATA21	RW	0x0	
[7]	LCD_DATA20	RW	0x0	
[8]	LCD_DATA19	RW	0x0	
[9]	LCD_DATA18	RW	0x0	
[10]	LCD_DATA17	RW	0x0	
[11]	LCD_DATA16	RW	0x0	
[12]	LCD_DATA15	RW	0x0	
[13]	LCD_DATA14	RW	0x0	
[14]	LCD_DATA13	RW	0x0	
[15]	LCD_DATA12	RW	0x0	
[16]	LCD_DATA11	RW	0x0	
[17]	LCD_DATA10	RW	0x0	
[18]	LCD_DATA9	RW	0x0	
[19]	LCD_DATA8	RW	0x0	
[20]	LCD_DATA7	RW	0x0	
[21]	LCD_DATA6	RW	0x0	
[22]	LCD_DATA5	RW	0x0	
[23]	LCD_DATA4	RW	0x0	
[24]	LCD_DATA3	RW	0x0	
[25]	LCD_DATA2	RW	0x0	
[26]	LCD_DATA1	RW	0x0	
[27]	LCD_DATA0	RW	0x0	
[28]	SENSOR0_RSTN	RW	0x0	
[29]	SENSOR0_PWD	RW	0x0	
[30]	SENSOR0_CLK	RW	0x0	
[31]	SENSOR1_RSTN	RW	0x0	

13.1.1.1. **VS_PIN_PULLUP_CTRL2**

Offset: 0x00C8

Bits	Name	R/W	Reset	Description
[0]	DVP0_PCLK	RW	0x0	14. IO pin pull up/pull down control for video output block Default: 0x00000000 15. 0: pull down is selected when pull logic is active (corresponding bit in PIN_PULLDIS is L) 16. 1: pull up is selected when pull logic is active (corresponding bit in PIN_PULLDIS is L) 17.
[1]	DVP0_DIN11	RW	0x0	
[2]	DVP0_DIN10	RW	0x0	
[3]	DVP0_DIN9	RW	0x0	
[4]	DVP0_DIN8	RW	0x0	
[5]	DVP0_DIN7	RW	0x0	
[6]	DVP0_DIN6	RW	0x0	
[7]	DVP0_DIN5	RW	0x0	
[8]	DVP0_DIN4	RW	0x0	
[9]	DVP0_DIN3	RW	0x0	
[10]	DVP0_DIN2	RW	0x0	
[11]	DVP0_DIN1	RW	0x0	
[12]	DVP0_DIN0	RW	0x0	
[13]	SDIO1_DATA3	RW	0x0	
[14]	SDIO1_DATA2	RW	0x0	
[15]	SDIO1_DATA1	RW	0x0	
[16]	EMAC0_TXEN	RW	0x0	
[17]	EMAC0_COL	RW	0x0	
[18]	EMAC0_RXDV	RW	0x0	
[19]	EMAC0_MDC	RW	0x0	
[20]	EMAC0_MDIO	RW	0x0	
[21]	EMAC0_TXD1	RW	0x0	
[22]	EMAC0_TXD0	RW	0x0	
[23]	EMAC0_RXD1	RW	0x0	
[24]	EMAC0_RXD0	RW	0x0	
[25]	SDIO1_CMD	RW	0x0	
[26]	SDIO1_CLK	RW	0x0	
[27]	SDIO1_DATA0	RW	0x0	
[28]	DVP0_HREF	RW	0x0	
[29]	DVP0_VSYNC	RW	0x0	
[30]	SENSOR1_PWD	RW	0x0	
[31]	SENSOR1_CLK	RW	0x0	

17.1.1.1. **PIN_PULLDIS_CTRL**

Offset: 0x00d0

Bits	Name	R/W	Reset	Description
[0]	SDIO0_CLK	RW	0x0	18. IO pin pull up/pull function disable control; 19. Default: 0x0004_3f00 20. 0: pull up/pull down is active 21. 1: pull up/pull down is disabled
[1]	SDIO0_CMD	RW	0x0	
[2]	SDIO0_DATA3	RW	0x0	
[3]	SDIO0_DATA2	RW	0x0	
[4]	SDIO0_DATA1	RW	0x0	
[5]	SDIO0_DATA0	RW	0x0	
[6]	SDIO0_CD	RW	0x0	
[7]	SDIO0_WP	RW	0x0	
[8]	QSPI_CLK	RW	0x1	
[9]	QSPI_DIO0	RW	0x1	
[10]	QSPI_DIO1	RW	0x1	
[11]	QSPI_DIO2	RW	0x1	
[12]	QSPI_DIO3	RW	0x1	
[13]	QSPI_CSN0	RW	0x1	
[14]	JTAG_TCK	RW	0x0	
[15]	JTAG_TDI	RW	0x0	
[16]	JTAG_TMS	RW	0x0	
[17]	JTAG_TRSTND	RW	0x0	
[18]	JTAG_TDO	RW	0x1	
[19]	UART0_TXD	RW	0x0	
[20]	UART0_RXD	RW	0x0	
[21]	PWM0	RW	0x0	
[22]	PWM1	RW	0x0	
[23]	Reserved	RW	0x0	
[24]	BOOT_SEL	RW	0x0	
[25]	I2C0_SCL	RW	0x0	
[26]	I2C0_SDA	RW	0x0	
[27]	GPIOA	RW	0x0	
[28]	GPIOB	RW	0x0	
[29]	GPIOC	RW	0x0	
[30]	GPIOD	RW	0x0	
[31]	GPIOE	RW	0x0	

21.1.1.1. **VS_PIN_ PULLDIS_CTRL1**

Offset: 0x00d4

Bits	Name	R/W	Reset	Description
[0]	LCD_DEN	RW	0x0	22. IO pin pull up/pull function disable control; 23. Default: 0x00000000 24. 0: pull up/pull down is active 25. 1: pull up/pull down is disabled 26.
[1]	LCD_HSYNC	RW	0x0	
[2]	LCD_VSYNC	RW	0x0	
[3]	LCD_PCLK	RW	0x0	
[4]	LCD_DATA23	RW	0x0	
[5]	LCD_DATA22	RW	0x0	
[6]	LCD_DATA21	RW	0x0	
[7]	LCD_DATA20	RW	0x0	
[8]	LCD_DATA19	RW	0x0	
[9]	LCD_DATA18	RW	0x0	
[10]	LCD_DATA17	RW	0x0	
[11]	LCD_DATA16	RW	0x0	
[12]	LCD_DATA15	RW	0x0	
[13]	LCD_DATA14	RW	0x0	
[14]	LCD_DATA13	RW	0x0	
[15]	LCD_DATA12	RW	0x0	
[16]	LCD_DATA11	RW	0x0	
[17]	LCD_DATA10	RW	0x0	
[18]	LCD_DATA9	RW	0x0	
[19]	LCD_DATA8	RW	0x0	
[20]	LCD_DATA7	RW	0x0	
[21]	LCD_DATA6	RW	0x0	
[22]	LCD_DATA5	RW	0x0	
[23]	LCD_DATA4	RW	0x0	
[24]	LCD_DATA3	RW	0x0	
[25]	LCD_DATA2	RW	0x0	
[26]	LCD_DATA1	RW	0x0	
[27]	LCD_DATA0	RW	0x0	
[28]	SENSOR0_RSTN	RW	0x0	
[29]	SENSOR0_PWD	RW	0x0	
[30]	SENSOR0_CLK	RW	0x0	
[31]	SENSOR1_RSTN	RW	0x0	

26.1.1.1. **VS_PIN_ PULLDIS_CTRL2**

Offset: 0x00d8

Bits	Name	R/W	Reset	Description
[0]	DVP0_PCLK	RW	0x0	27. IO pin pull up/pull function disable control; 28. Default: 0x00000000 29. 0: pull up/pull down is active 30. 1: pull up/pull down is disabled 31.
[1]	DVP0_DIN11	RW	0x0	
[2]	DVP0_DIN10	RW	0x0	
[3]	DVP0_DIN9	RW	0x0	
[4]	DVP0_DIN8	RW	0x0	
[5]	DVP0_DIN7	RW	0x0	
[6]	DVP0_DIN6	RW	0x0	
[7]	DVP0_DIN5	RW	0x0	
[8]	DVP0_DIN4	RW	0x0	
[9]	DVP0_DIN3	RW	0x0	
[10]	DVP0_DIN2	RW	0x0	
[11]	DVP0_DIN1	RW	0x0	
[12]	DVP0_DIN0	RW	0x0	
[13]	SDIO1_DATA3	RW	0x0	
[14]	SDIO1_DATA2	RW	0x0	
[15]	SDIO1_DATA1	RW	0x0	
[16]	EMAC0_TXEN	RW	0x0	
[17]	EMAC0_COL	RW	0x0	
[18]	EMAC0_RXDV	RW	0x0	
[19]	EMAC0_MDC	RW	0x0	
[20]	EMAC0_MDIO	RW	0x0	
[21]	EMAC0_TXD1	RW	0x0	
[22]	EMAC0_TXD0	RW	0x0	
[23]	EMAC0_RXD1	RW	0x0	
[24]	EMAC0_RXD0	RW	0x0	
[25]	SDIO1_CMD	RW	0x0	
[26]	SDIO1_CLK	RW	0x0	
[27]	SDIO1_DATA0	RW	0x0	
[28]	DVP0_HREF	RW	0x0	
[29]	DVP0_VSYNC	RW	0x0	
[30]	SENSOR1_PWD	RW	0x0	
[31]	SENSOR1_CLK	RW	0x0	

31.1.1.1. **PIN_DS_CTRL**

Offset: 0x00E0

Bits	Name	R/W	Reset	Description
[0]	SDIO0_CLK	RW	0x0	32. I/O pin driving strength control DS0; Default0000 33. 0: DS1=1, DS0=0; 34. 1: DS1=1, DS0=1; 35.
[1]	SDIO0_CMD	RW	0x0	
[2]	SDIO0_DATA3	RW	0x0	
[3]	SDIO0_DATA2	RW	0x0	
[4]	SDIO0_DATA1	RW	0x0	
[5]	SDIO0_DATA0	RW	0x0	
[6]	SDIO0_CD	RW	0x0	
[7]	SDIO0_WP	RW	0x0	
[8]	QSPI_CLK	RW	0x0	
[9]	QSPI_DIO0	RW	0x0	
[10]	QSPI_DIO1	RW	0x0	
[11]	QSPI_DIO2	RW	0x0	
[12]	QSPI_DIO3	RW	0x0	
[13]	QSPI_CSN0	RW	0x0	
[14]	JTAG_TCK	RW	0x0	
[15]	JTAG_TDI	RW	0x0	
[16]	JTAG_TMS	RW	0x0	
[17]	JTAG_TRSTND	RW	0x0	
[18]	JTAG_TDO	RW	0x0	
[19]	UART0_TXD	RW	0x0	
[20]	UART0_RXD	RW	0x0	
[21]	PWM0	RW	0x0	
[22]	PWM1	RW	0x0	
[23]	Reserved	RW	0x0	
[24]	BOOT_SEL	RW	0x0	
[25]	I2C0_SCL	RW	0x0	
[26]	I2C0_SDA	RW	0x0	
[27]	GPIOA	RW	0x0	
[28]	GPIOB	RW	0x0	
[29]	GPIOC	RW	0x0	
[30]	GPIOD	RW	0x0	
[31]	GPIOE	RW	0x0	

35.1.1.1. **VS_PIN_ DS _CTRL1**

Offset: 0x00E4

Bits	Name	R/W	Reset	Description
[0]	LCD_DEN	RW	0x1	36. 37. I I/O pin driving strength control DS0; Default0000 38. 0: DS1=1, DS0=0; 39. 1: DS1=1, DS0=1; 40.
[1]	LCD_HSYNC	RW	0x1	
[2]	LCD_VSYNC	RW	0x1	
[3]	LCD_PCLK	RW	0x1	
[4]	LCD_DATA23	RW	0x1	
[5]	LCD_DATA22	RW	0x1	
[6]	LCD_DATA21	RW	0x1	
[7]	LCD_DATA20	RW	0x1	
[8]	LCD_DATA19	RW	0x1	
[9]	LCD_DATA18	RW	0x1	
[10]	LCD_DATA17	RW	0x1	
[11]	LCD_DATA16	RW	0x1	
[12]	LCD_DATA15	RW	0x1	
[13]	LCD_DATA14	RW	0x1	
[14]	LCD_DATA13	RW	0x1	
[15]	LCD_DATA12	RW	0x1	
[16]	LCD_DATA11	RW	0x1	
[17]	LCD_DATA10	RW	0x0	
[18]	LCD_DATA9	RW	0x0	
[19]	LCD_DATA8	RW	0x1	
[20]	LCD_DATA7	RW	0x1	
[21]	LCD_DATA6	RW	0x1	
[22]	LCD_DATA5	RW	0x1	
[23]	LCD_DATA4	RW	0x1	
[24]	LCD_DATA3	RW	0x1	
[25]	LCD_DATA2	RW	0x1	
[26]	LCD_DATA1	RW	0x1	
[27]	LCD_DATA0	RW	0x1	
[28]	SENSOR0_RSTN	RW	0x1	
[29]	SENSOR0_PWD	RW	0x1	
[30]	SENSOR0_CLK	RW	0x1	
[31]	SENSOR1_RSTN	RW	0x1	

40.1.1.1. **VS_PIN_ DS _CTRL2**

Offset: 0x00E8

Bits	Name	R/W	Reset	Description
[0]	DVP0_PCLK	RW	0x0	41. 42. l I/O pin driving strength control DS0; Default0000 43. 0: DS1=1, DS0=0; 44. 1: DS1=1, DS0=1; 45.
[1]	DVP0_DIN11	RW	0x0	
[2]	DVP0_DIN10	RW	0x0	
[3]	DVP0_DIN9	RW	0x0	
[4]	DVP0_DIN8	RW	0x0	
[5]	DVP0_DIN7	RW	0x0	
[6]	DVP0_DIN6	RW	0x0	
[7]	DVP0_DIN5	RW	0x0	
[8]	DVP0_DIN4	RW	0x0	
[9]	DVP0_DIN3	RW	0x0	
[10]	DVP0_DIN2	RW	0x0	
[11]	DVP0_DIN1	RW	0x0	
[12]	DVP0_DIN0	RW	0x0	
[13]	SDIO1_DATA3	RW	0x0	
[14]	SDIO1_DATA2	RW	0x0	
[15]	SDIO1_DATA1	RW	0x0	
[16]	EMAC0_TXEN	RW	0x0	
[17]	EMAC0_COL	RW	0x0	
[18]	EMAC0_RXDV	RW	0x0	
[19]	EMAC0_MDC	RW	0x0	
[20]	EMAC0_MDIO	RW	0x0	
[21]	EMAC0_TXD1	RW	0x0	
[22]	EMAC0_TXD0	RW	0x0	
[23]	EMAC0_RXD1	RW	0x0	
[24]	EMAC0_RXD0	RW	0x0	
[25]	SDIO1_CMD	RW	0x0	
[26]	SDIO1_CLK	RW	0x0	
[27]	SDIO1_DATA0	RW	0x0	
[28]	DVP0_HREF	RW	0x0	
[29]	DVP0_VSYNC	RW	0x0	
[30]	SENSOR1_PWD	RW	0x0	
[31]	SENSOR1_CLK	RW	0x0	

45.1.1.1. **PIN_SL_CTRL**

Offset: 0x00F0

Bits	Name	R/W	Reset	Description
[0]	SDIO0_CLK	RW	0x0	46. Ouptut slew rate control for IO pins; 0x1: Fast 0x0: Slow 47.
[1]	SDIO0_CMD	RW	0x0	
[2]	SDIO0_DATA3	RW	0x0	
[3]	SDIO0_DATA2	RW	0x0	
[4]	SDIO0_DATA1	RW	0x0	
[5]	SDIO0_DATA0	RW	0x0	
[6]	SDIO0_CD	RW	0x0	
[7]	SDIO0_WP	RW	0x0	
[8]	QSPI_CLK	RW	0x0	
[9]	QSPI_DIO0	RW	0x0	
[10]	QSPI_DIO1	RW	0x0	
[11]	QSPI_DIO2	RW	0x0	
[12]	QSPI_DIO3	RW	0x0	
[13]	QSPI_CSN0	RW	0x0	
[14]	JTAG_TCK	RW	0x0	
[15]	JTAG_TDI	RW	0x0	
[16]	JTAG_TMS	RW	0x0	
[17]	JTAG_TRSTND	RW	0x0	
[18]	JTAG_TDO	RW	0x0	
[19]	UART0_TXD	RW	0x0	
[20]	UART0_RXD	RW	0x0	
[21]	PWM0	RW	0x0	
[22]	PWM1	RW	0x0	
[23]	Reserved	RW	0x0	
[24]	BOOT_SEL	RW	0x0	
[25]	I2C0_SCL	RW	0x0	
[26]	I2C0_SDA	RW	0x0	
[27]	GPIOA	RW	0x0	
[28]	GPIOB	RW	0x0	
[29]	GPIOC	RW	0x0	
[30]	GPIOD	RW	0x0	
[31]	GPIOE	RW	0x0	

47.1.1.1. **VS_PIN_SL_CTRL1**

Offset: 0x00F4

Bits	Name	R/W	Reset	Description
[0]	LCD_DEN	RW	0x0	48. Ouptut slew rate control for IO pins; 0x1: Fast 0x0: Slow 49.
[1]	LCD_HSYNC	RW	0x0	
[2]	LCD_VSYNC	RW	0x0	
[3]	LCD_PCLK	RW	0x0	
[4]	LCD_DATA23	RW	0x0	
[5]	LCD_DATA22	RW	0x0	
[6]	LCD_DATA21	RW	0x0	
[7]	LCD_DATA20	RW	0x0	
[8]	LCD_DATA19	RW	0x0	
[9]	LCD_DATA18	RW	0x0	
[10]	LCD_DATA17	RW	0x0	
[11]	LCD_DATA16	RW	0x0	
[12]	LCD_DATA15	RW	0x0	
[13]	LCD_DATA14	RW	0x0	
[14]	LCD_DATA13	RW	0x0	
[15]	LCD_DATA12	RW	0x0	
[16]	LCD_DATA11	RW	0x0	
[17]	LCD_DATA10	RW	0x0	
[18]	LCD_DATA9	RW	0x0	
[19]	LCD_DATA8	RW	0x0	
[20]	LCD_DATA7	RW	0x0	
[21]	LCD_DATA6	RW	0x0	
[22]	LCD_DATA5	RW	0x0	
[23]	LCD_DATA4	RW	0x0	
[24]	LCD_DATA3	RW	0x0	
[25]	LCD_DATA2	RW	0x0	
[26]	LCD_DATA1	RW	0x0	
[27]	LCD_DATA0	RW	0x0	
[28]	SENSOR0_RSTN	RW	0x0	
[29]	SENSOR0_PWD	RW	0x0	
[30]	SENSOR0_CLK	RW	0x0	
[31]	SENSOR1_RSTN	RW	0x0	

49.1.1.1. **VS_PIN_SL_CTRL2**

Offset: 0x00F8

Bits	Name	R/W	Reset	Description
[0]	DVP0_PCLK	RW	0x1	50. Ouptut slew rate control for IO pins; 0x1: Fast 0x0: Slow 51.
[1]	DVP0_DIN11	RW	0x1	
[2]	DVP0_DIN10	RW	0x1	
[3]	DVP0_DIN9	RW	0x1	
[4]	DVP0_DIN8	RW	0x1	
[5]	DVP0_DIN7	RW	0x1	
[6]	DVP0_DIN6	RW	0x1	
[7]	DVP0_DIN5	RW	0x1	
[8]	DVP0_DIN4	RW	0x1	
[9]	DVP0_DIN3	RW	0x1	
[10]	DVP0_DIN2	RW	0x1	
[11]	DVP0_DIN1	RW	0x1	
[12]	DVP0_DIN0	RW	0x1	
[13]	SDIO1_DATA3	RW	0x1	
[14]	SDIO1_DATA2	RW	0x1	
[15]	SDIO1_DATA1	RW	0x1	
[16]	EMAC0_TXEN	RW	0x1	
[17]	EMAC0_COL	RW	0x0	
[18]	EMAC0_RXDV	RW	0x0	
[19]	EMAC0_MDC	RW	0x1	
[20]	EMAC0_MDIO	RW	0x1	
[21]	EMAC0_TXD1	RW	0x1	
[22]	EMAC0_TXD0	RW	0x1	
[23]	EMAC0_RXD1	RW	0x1	
[24]	EMAC0_RXD0	RW	0x1	
[25]	SDIO1_CMD	RW	0x1	
[26]	SDIO1_CLK	RW	0x1	
[27]	SDIO1_DATA0	RW	0x1	
[28]	DVP0_HREF	RW	0x1	
[29]	DVP0_VSYNC	RW	0x1	
[30]	SENSOR1_PWD	RW	0x1	
[31]	SENSOR1_CLK	RW	0x1	

51.1.1.1. **PIN_OD_CTRL**

Offset: 0x0100

Bits	Name	R/W	Reset	Description
[0]	SDIO0_CLK	RW	0x0	52. 53. Open drain select signal when corresponding I/O pin is used as GPIO pin. 54. 0: push pull 55. 1: open drain
[1]	SDIO0_CMD	RW	0x0	
[2]	SDIO0_DATA3	RW	0x0	
[3]	SDIO0_DATA2	RW	0x0	
[4]	SDIO0_DATA1	RW	0x0	
[5]	SDIO0_DATA0	RW	0x0	
[6]	SDIO0_CD	RW	0x0	
[7]	SDIO0_WP	RW	0x0	
[8]	QSPI_CLK	RW	0x0	
[9]	QSPI_DIO0	RW	0x0	
[10]	QSPI_DIO1	RW	0x0	
[11]	QSPI_DIO2	RW	0x0	
[12]	QSPI_DIO3	RW	0x0	
[13]	QSPI_CSN0	RW	0x0	
[14]	JTAG_TCK	RW	0x0	
[15]	JTAG_TDI	RW	0x0	
[16]	JTAG_TMS	RW	0x0	
[17]	JTAG_TRSTND	RW	0x0	
[18]	JTAG_TDO	RW	0x0	
[19]	UART0_TXD	RW	0x0	
[20]	UART0_RXD	RW	0x0	
[21]	PWM0	RW	0x0	
[22]	PWM1	RW	0x0	
[23]	Reserved	RW	0x0	
[24]	BOOT_SEL	RW	0x0	
[25]	I2C0_SCL	RW	0x0	
[26]	I2C0_SDA	RW	0x0	
[27]	GPIOA	RW	0x0	
[28]	GPIOB	RW	0x0	
[29]	GPIOC	RW	0x0	
[30]	GPIOD	RW	0x0	
[31]	GPIOE	RW	0x0	

55.1.1.1. **VS_PIN_OD_CTRL1**

Offset: 0x0104

Bits	Name	R/W	Reset	Description
[0]	LCD_DEN	RW	0x0	56. 57. Open drain select signal when corresponding I/O pin is used as GPIO pin. 58. 0: push pull 59. 1: open drain 60.
[1]	LCD_HSYNC	RW	0x0	
[2]	LCD_VSYNC	RW	0x0	
[3]	LCD_PCLK	RW	0x0	
[4]	LCD_DATA23	RW	0x0	
[5]	LCD_DATA22	RW	0x0	
[6]	LCD_DATA21	RW	0x0	
[7]	LCD_DATA20	RW	0x0	
[8]	LCD_DATA19	RW	0x0	
[9]	LCD_DATA18	RW	0x0	
[10]	LCD_DATA17	RW	0x0	
[11]	LCD_DATA16	RW	0x0	
[12]	LCD_DATA15	RW	0x0	
[13]	LCD_DATA14	RW	0x0	
[14]	LCD_DATA13	RW	0x0	
[15]	LCD_DATA12	RW	0x0	
[16]	LCD_DATA11	RW	0x0	
[17]	LCD_DATA10	RW	0x0	
[18]	LCD_DATA9	RW	0x0	
[19]	LCD_DATA8	RW	0x0	
[20]	LCD_DATA7	RW	0x0	
[21]	LCD_DATA6	RW	0x0	
[22]	LCD_DATA5	RW	0x0	
[23]	LCD_DATA4	RW	0x0	
[24]	LCD_DATA3	RW	0x0	
[25]	LCD_DATA2	RW	0x0	
[26]	LCD_DATA1	RW	0x0	
[27]	LCD_DATA0	RW	0x0	
[28]	SENSOR0_RSTN	RW	0x0	
[29]	SENSOR0_PWD	RW	0x0	
[30]	SENSOR0_CLK	RW	0x0	
[31]	SENSOR1_RSTN	RW	0x0	

60.1.1.1. **VS_PIN_OD_CTRL2**

Offset: 0x0108

Bits	Name	R/W	Reset	Description
[0]	DVP0_PCLK	RW	0x0	61. 62. Open drain select signal when corresponding I/O pin is used as GPIO pin. 63. 0: push pull 64. 1: open drain 65.
[1]	DVP0_DIN11	RW	0x0	
[2]	DVP0_DIN10	RW	0x0	
[3]	DVP0_DIN9	RW	0x0	
[4]	DVP0_DIN8	RW	0x0	
[5]	DVP0_DIN7	RW	0x0	
[6]	DVP0_DIN6	RW	0x0	
[7]	DVP0_DIN5	RW	0x0	
[8]	DVP0_DIN4	RW	0x0	
[9]	DVP0_DIN3	RW	0x0	
[10]	DVP0_DIN2	RW	0x0	
[11]	DVP0_DIN1	RW	0x0	
[12]	DVP0_DIN0	RW	0x0	
[13]	SDIO1_DATA3	RW	0x0	
[14]	SDIO1_DATA2	RW	0x0	
[15]	SDIO1_DATA1	RW	0x0	
[16]	EMAC0_TXEN	RW	0x0	
[17]	EMAC0_COL	RW	0x0	
[18]	EMAC0_RXDV	RW	0x0	
[19]	EMAC0_MDC	RW	0x0	
[20]	EMAC0_MDIO	RW	0x0	
[21]	EMAC0_TXD1	RW	0x0	
[22]	EMAC0_TXD0	RW	0x0	
[23]	EMAC0_RXD1	RW	0x0	
[24]	EMAC0_RXD0	RW	0x0	
[25]	SDIO1_CMD	RW	0x0	
[26]	SDIO1_CLK	RW	0x0	
[27]	SDIO1_DATA0	RW	0x0	
[28]	DVP0_HREF	RW	0x0	
[29]	DVP0_VSYNC	RW	0x0	
[30]	SENSOR1_PWD	RW	0x0	
[31]	SENSOR1_CLK	RW	0x0	

65.1.1. Miscellaneous Control Registers

65.1.1.1. BOOT_STATUS

Offset: 0x0040

Bits	Name	R/W	Reset	Description
[1:0]	BOOT_STATUS	RW	0x0	00: SOC in the status after external hardware reset; 01: SOC in the status after watch dog reset; 10: SOC in the status after MCU_SOFT_RST or GLOBAL_SOFT_RST;
[31:2]	Reserved	RW	0x0	Reserved

65.1.1.2. BOOT_MODE

Offset: 0x0044

Bits	Name	R/W	Reset	Description
[1:0]	BOOT_SEL	RO	0x0	BOOT_SEL; Boot media order select register, corresponding to input of BOOT_SEL after external reset. 0x0: UART-> NOR-> NAND-> EMMC 0x1: NOR->NAND-> EMMC->UART 0x2: NAND-> EMMC->UART-> NOR 0x3: EMMC->UART-> NOR-> NAND It indicates the input value on IO pin of {BOOT_SEL, SENSOR0_CLK} during boot up
[31:2]	Reserved	RW	0x0	Reserved

65.1.1.3. PMU_MISC_CTRL

Offset: 0x0048

Bits	Name	R/W	Reset	Description
[0]	OSC32K_STABLE	RW	0x0	Need software to write 1 to this bit 1s after 1st time system power-up to enable RTC logic. This will also switch pmu logic from 12M OSC clock to 32K OSC clock to save power.
[31:1]	Reserved	RW	0x800_0000	

65.1.1.4. ATE_TEST_MODE

Offset: 0x004C

Bits	Name	R/W	Reset	Description
[4:0]	ATE_MODE	RO	0x0	ATE test mode select which is determined by input value on pins of Bit0: LCD_DATA0 Bit1: LCD_DATA1 Bit2: LCD_DATA2 Bit3: LCD_DATA3 Bit4:TEST_MODE
[31:5]	Reserved			

65.1.1.5. MCU_CHIPID

Offset: 0x0060

Bits	Name	R/W	Reset	Description
[31:0]	CHIPID	RO	0x20240001	Return the ID of this chip

65.1.1.6. **MCU_AHB_HPROT_CTRL**

Offset: 0x0074

Bits	Name	R/W	Reset	Description
[2:0]	HPROT_USB	RW	0x1	HPROT[3:1] for USB AHB interface It is a HPROT control register to drive AHB HPROT[3:1] bus. HPROT[0] = 0: Opcodefetch HPROT[0] = 1: Data access HPROT[1] = 0: User access HPROT[1] = 1: Privileged access HPROT[2] = 0: Not bufferable HPROT[2] = 1: Bufferable HPROT[3] = 0: Not cacheable HPROT[3] = 1: Cacheable The AMBA Specification recommends that the default value of HPROT indicates a non-cached, non-buffered, privileged data access. The is used to indicate such an access. HPROT[0] is tied high because all transfers are data accesses, as there are no opcode fetches.
[7:3]	Reserved			
[10:8]	HPROT_SD0	RW	0x1	HPROT[3:1] for SD0 AHB interface
[14:12]	HPROT_SD1	RW	0x1	HPROT[3:1] for SD1AHB interface
[18:16]	HPROT_EMAC		0x1	HPROT[3:1] for EMAC AHB interface

65.1.1.7. **DMA_HS_SEL0**

Offset: 0x0078

Bits	Name	R/W	Reset	Description
[3:0]	QSPI_TX_HS	RW	0x0	DMA Handshake selection for QSPI TX, Handshake request number 0x0~0xF
[7:4]	QSPI_RX_HS	RW	0x1	DMA Handshake selection for QSPI RX, Handshake request number 0x0~0xF

[11:8]	UART1_TX_HS	RW	0x2	DMA Handshake selection for UART1 TX, Handshake request number 0x0~0xF
[15:12]	UART1_RX_HS	RW	0x3	DMA Handshake selection for UART1 RX, Handshake request number 0x0~0xF
[19:16]	Reserved	RW	0x4	Reserved
[23:20]	Reserved	RW	0x5	Reserved
[27:24]	I2C0_TX_HS	RW	0x6	DMA Handshake selection for I2C0 TX, Handshake request number 0x0~0xF
[31:28]	I2C0_RX_HS	RW	0x7	DMA Handshake selection for I2C0 RX, Handshake request number 0x0~0xF

65.1.1.8. **DMA_HS_SEL1**

Offset: 0x007C

Bits	Name	R/W	Reset	Description
[3:0]	UART2_TX_HS	RW	0x8	DMA Handshake selection for UART2_TX, Handshake request number 0x0~0xF
[7:4]	UART2 _RX_HS	RW	0x9	DMA Handshake selection for UART2 _RX, Handshake request number 0x0~0xF
[11:8]	UART3 _TX_HS	RW	0xA	DMA Handshake selection for UART3 _TX, Handshake request number 0x0~0xF
[15:12]	UART3_ RX_HS	RW	0xB	DMA Handshake selection for UART3_ RX, Handshake request number 0x0~0xF
[19:16]	SPI0_TX_HS	RW	0xC	DMA Handshake selection for SPI0_TX, Handshake request number 0x0~0xF
[23:20]	SPI0_RX_HS	RW	0xD	DMA Handshake selection for SPI0_RX, Handshake request number 0x0~0xF
[27:24]	I2C1_TX_HS	RW	0xE	DMA Handshake selection for I2C1_TX, Handshake request number 0x0~0xF
[31:28]	I2C1_RX_HS	RW	0xF	DMA Handshake selection for I2C1_RX, Handshake request number 0x0~0xF

65.1.1.9. **DMA_HS_SEL2**

Offset: 0x0080

Bits	Name	R/W	Reset	Description
[3:0]	UART0 _TX_HS	RW	0x0	DMA Handshake selection for UART0 TX, Handshake request number 0x0~0xF
[7:4]	UART0_ RX_HS	RW	0x1	DMA Handshake selection for UART0 RX, Handshake request number 0x0~0xF
[11:8]	SPI1_TX_HS	RW	0x2	DMA Handshake selection for SPI1 TX, Handshake request number 0x0~0xF
[15:12]	SPI1_RX_HS	RW	0x3	DMA Handshake selection for SPI1 RX, Handshake request number 0x0~0xF
[19:16]	SHA_IN_HS	RW	0x4	DMA Handshake selection for SHA IN, Handshake request number 0x0~0xF
[23:20]	SARADC_RX_HS	RW	0x5	DMA Handshake selection for SAR ADC RX, Handshake request number 0x0~0xF
[27:24]	ADC_I2S RX_HS	RW	0x6	DMA Handshake selection for ADC I2S RX, Handshake request number 0x0~0xF
[31:28]	DAC_I2S_TX_HS	RW	0x7	DMA Handshake selection for DAC I2S TX, Handshake request number 0x0~0xF

65.1.1.10. **DMA_HS_SEL3**

Offset: 0x0084

Bits	Name	R/W	Reset	Description
[3:0]	I2S0_TX_HS	RW	0x8	DMA Handshake selection for I2S0 TX, Handshake request number 0x0~0xF

[7:4]	I2S1_RX_HS	RW	0x9	DMA Handshake selection for I2S1 RX, Handshake request number 0x0~0xF
[11:8]	I2C2_TX_HS	RW	0xA	DMA Handshake selection for I2C2 TX, Handshake request number 0x0~0xF
[15:12]	I2C2_RX_HS	RW	0xB	DMA Handshake selection for I2C2 RX, Handshake request number 0x0~0xF
[19:16]	EXT_INT0_HS	RW	0xC	DMA Handshake selection for EXT INT0, Handshake request number 0x0~0xF
[23:20]	EXT_INT1_HS	RW	0xD	DMA Handshake selection for EXT INT1, Handshake request number 0x0~0xF
[27:24]	I2C3_TX _HS	RW	0xE	DMA Handshake selection for I2C3 TX, Handshake request number 0x0~0xF
[31:28]	I2C3_RX_HS	RW	0xF	DMA Handshake selection for I2C3 RX, Handshake request number 0x0~0xF

65.1.1.11. **DMA_PERI_EN**

Offset: 0x0088

Bits	Name	R/W	Reset	Description
[0]	QSPI_TX	RW	0x0	66.QSPI TX HW handshake enable 1’b1: QSPI TX DMA HW handshake interface enabled 1’b0: QSPI TX DMA HW handshake interface disabled

[1]	QSPI_RX	RW	0x0	QSPI RX HW handshake enable
[2]	UART1_TX	RW	0x0	UART1 TX HW handshake enable
[3]	UART1_RX	RW	0x0	UART1 RX HW handshake enable
[4]	Reserved	RW	0x0	Reserved
[5]	Reserved	RW	0x0	Reserved
[6]	I2C0_TX	RW	0x0	I2C0 TX HW handshake enable
[7]	I2C0_RX	RW	0x0	I2C0 RX HW handshake enable
[8]	UART2_TX	RW	0x0	UART2 TX HW handshake enable
[9]	UART2_RX	RW	0x0	UART2 RX HW handshake enable
[10]	UART3_TX	RW	0x0	UART3 TX HW handshake enable
[11]	UART3_RX	RW	0x0	UART3 RX HW handshake enable
[12]	SPI0_TX	RW	0x0	SPI0 TX HW handshake enable
[13]	SPI0_RX	RW	0x0	SPI0 RX HW handshake enable
[14]	I2C1_TX	RW	0x0	I2C1 TX HW handshake enable
[15]	I2C1_RX	RW	0x0	I2C1 RX HW handshake enable
[16]	UART0_TX	RW	0x0	UART0 TX HW handshake enable
[17]	UART0_RX	RW	0x0	UART0 RX HW handshake enable
[18]	SPI1_TX	RW	0x0	SPI1 TX HW handshake enable
[19]	SPI1_RX	RW	0x0	SPI1 RX HW handshake enable
[20]	SHA_IN	RW	0x0	SHA IN HW handshake enable
[21]	SARADC_RX	RW	0x0	SARADC RX HW handshake enable
[22]	ADC_I2S	RW	0x0	ADC I2S RX HW handshake enable
[23]	DAC_I2S	RW	0x0	DAC I2S TX HW handshake
[24]	Reserved	RW	0x0	Reserved
[25]	Reserved	RW	0x0	Reserved
[26]	EXT_INT0	RW	0x0	GPIO/TIMER INT0 handshake
[27]	EXT_INT1	RW	0x0	GPIO/TIMER INT1 handshake enable
[28]	I2C2_TX	RW	0x0	I2C2 TX HW handshake enable
[29]	I2C2_RX	RW	0x0	I2C2 RX HW handshake enable
[30]	I2C3_TX	RW	0x0	I2C3 TX HW handshake enable
[31]	I2C3_RX	RW	0x0	I2C3 RX HW handshake enable

66.1.1.1. MCU_DMA_HW_REQ_INUSE_STATUS

Offset: 0x008C

Bits	Name	R/W	Reset	Description
[15:0]	DMA_HW_REQ_INUSE	RO	0x0	Indicate the In-Use status for 16 HW request handshake channels 'b1: corresponding HW handshake line is in use 'b0: corresponding HW handshake line is available
[31:16]	Reserved			

66.1.1.2. MCU_DMA_EXT_INT_SRC_SEL

Offset: 0x0090

Bits	Name	R/W	Reset	Description
[3:0]	EXT_INT0_SRC	RW	0x0	This register determines the ext interrupt sources that are used as the EXT INT0 handshake signal 4’b0000: GPIO INT0 4’b0001: GPIO INT1 4’b0010: GPIO2 INT0 4’b0011: GPIO 2 INT1 4’b0100: GPIO3 INT0 4’b0101: GPIO3 INT1 4’b0110: GPIO4 INT0 4’b0111: GPIO4 INT1 4’b1000: Timer 0 INT 4’b1001: Timer 1 INT 4’b1010: Timer 2 INT 4’b1011: Timer 3 INT 4’b1100: Timer 4 INT 4’b1101: Timer 5 INT 4’b1110: Timer 6 INT 4’b1111: Timer 7 INT
[7:4]	EXT_INT1_SRC	RW	0x1	This register determines the ext interrupt sources that are used as the EXT INT1 handshake signal 4’b0000: GPIO Bank A INT0 4’b0001: GPIO Bank A INT1 4’b0010: GPIO2 Bank A INT0 4’b0011: GPIO 2Bank A INT1 4’b0100: GPIO3 Bank A INT0 4’b0101: GPIO3 Bank A INT1 4’b0110: GPIO4 Bank A INT0 4’b0111: GPIO4 Bank A INT1 4’b1000: Timer 0 INT 4’b1001: Timer 1 INT 4’b1010: Timer 2 INT 4’b1011: Timer 3 INT 4’b1100: Timer 4 INT 4’b1101: Timer 5 INT 4’b1110: Timer 6 INT 4’b1111: Timer 7 INT
[31:8]	Reserved			

66.1.1.3. **TRNG_EXT_CTRL**

Offset: 0x009C

Bits	Name	R/W	Reset	Description
[0]	TRNG_EXT_CTRL	RW	0x0	Write 1 to assert core clean input to zerorize TRNG module. Write 0 to release.
[31：0]	Reserved			

66.1.1.4. **MCU_AUX_DBG0**

Offset: 0x0120

Bits	Name	R/W	Reset	Description
[31:0]	MCU_AUX_DBG0	RW	0x0	The auxiliary register for SW debugging purpose

66.1.1.5. **MCU_AUX_DBG1**

Offset: 0x0124

Bits	Name	R/W	Reset	Description
[31:0]	MCU_AUX_DBG1	RW	0x0	The auxiliary register for SW debugging purpose

66.1.1.6. **MCU_AUX_DBG2**

Offset: 0x0128

Bits	Name	R/W	Reset	Description
[31:0]	MCU_AUX_DBG2	RW	0x0	The auxiliary register for SW debugging purpose

66.1.1.7. **MCU_AUX_DBG3**

Offset: 0x0128

Bits	Name	R/W	Reset	Description
[31:0]	MCU_AUX_DBG3	RW	0x0	The auxiliary register for SW debugging purpose

66.1.1.8. **USBPHY_TUNE**

This parameter override signals are designed as strapping options. The default values for these signals are defined based on simulation results, and the USB 2.0 PHY is designed so that you do not have to change these pins from their default setting to center the design.

Offset: 0x0130

Bits	Name	R/W	Reset	Description
[2:0]	COMPDISTUNE	RW	0x4	
[5:3]	OTGTUNE	RW	0x4	
[8:6]	SQRXTUNE	RW	0x3	
[12:9]	TXFSLSTUNE	RW	0x3	
[14:13]	TXPREEMPAMPTUNE	RW	0x0	
[15]	TXPREEMPPULSETUNE	RW	0x0	
[17:16]	TXRISETUNE	RW	0x2	
[21:18]	TXVREFTUNE	RW	0x8	
[23:22]	TXHSXVTUNE	RW	0x3	
[25:24]	TXRESTUNE	RW	0x1	

66.1.1.9. **USBPHY_CFG**

Offset: 0x0134

Bits	Name	R/W	Reset	Description
[2:0]	FSEL	RW	0x2	Reference clock frequency Default 010 (12MHz)
[4:3]	REFCLKSEL	RW	0x2	Reference Clock Select for PLL Block Default 10, PLL uses CLKCORE as reference

[5]	OTGDISABLE	RW	0x0	OTG block disable 0x0: OTG enabled 0x1: OTG disabled
[6]	VBUSVLDEXTSEL	RW	0x0	External VBUS Valid Select. 0x1: Use VBUSVLDEXT as external VBUS valid input 0x0: Use VBUSVLD detected from VBUS input
[7]	VBUSVLDEXT	RW	0x0	Register to override VBUS valid detector result

66.1.1.10. **USBPHY_TEST_CTRL**

Offset: 0x0140

Bits	Name	R/W	Reset	Description
[7:0]	TEST_DIN	RW	0x0	USB PHY test data in
[11:8]	TEST_ADDR	RW	0x0	USB PHY test data address
[12]	TEST_CLK	RW	0x0	USB PHY test clock
[13]	TEST_DATAOUT_SEL	RW	0x0	USB PHY test data out select
[14]	TEST_SIDDQ	RW	0x0	SIDDQ test
[15]	LPBK_EN	RW	0x0	Loopback test
[16]	ATE_RESET	RW	0x0	ATE reset
[18:17]	VATEST_ENB	RW	0x0	Analog test
[19]	TEST_BURNIN	RW	0x0	Burn-in test
[23:20]	TEST_DOUT	RO	0x0	USB PHY test data out

66.1.1.11. **USB_INT_EN**

Offset: 0x0160

Bits	Name	R/W	Reset	Description
[0]	USB_POS_INT_EN	RW	0x0	USB ID pos-edge interrupt enable 0x0, USB ID pos-edge interrupt disable. 0x0, USB ID pos-edge interrupt enable.
[1]	USB_NEG_INT_EN	RW	0x0	USB ID neg-edge interrupt enable 0x0, USB ID neg-edge interrupt disable. 0x0, USB ID neg-edge interrupt enable.
[31:2]	Reserved			

66.1.1.12. **USB_RAW_INT**

Offset: 0x0164

Bits	Name	R/W	Reset	Description
[0]	USB_POS_RAW_INT	RO	0x0	USB ID pos-edge interrupt raw status 0x0, USB ID pos-edge RAW interrupt inactive. 0x1, USB ID pos-edge RAW interrupt active.
[1]	USB_NEG_RAW_INT			USB ID neg-edge interrupt raw status 0x0, USB ID neg-edge RAW interrupt inactive. 0x1, USB ID neg-edge RAW interrupt active.
[31:2]	Reserved			

66.1.1.13. USB_INT

Offset: 0x0168

Bits	Name	R/W	Reset	Description
[0]	USB_POS_INT	RO	0x0	USB ID pos-edge interrupt status 0x0, USB ID pos-edge interrupt inactive. 0x1, USB ID pos-edge interrupt active.
[1]	USB_NEG_INT	RO	0x0	USB ID neg-edge interrupt status 0x0, USB ID neg-edge interrupt inactive. 0x1, USB ID neg-edge interrupt active.
[31:2]	Reserved			

66.1.1.14. USB_INT_CLR

Offset: 0x016C

Bits	Name	R/W	Reset	Description
[0]	USB_POS_INT_CLR	WO	0x0	USB pos-edge ID interrupt clear Write 0x0, No effect. Write 0x1, Clear USB ID pos-edge interrupt. Auto clear register
[1]	USB_NEG_INT_CLR	WO	0x0	USB neg-edge ID interrupt clear Write 0x0, No effect. Write 0x1, Clear USB ID neg-edge interrupt.
[31:2]	Reserved			

66.1.1.15. ID_DEBOUNCE_CNT

Offset: 0x0170

Bits	Name	R/W	Reset	Description

[31:0]	ID_DEBOUNCE_CNT	RW	0x20000	USB ID input de-bounce count value in AHB clock cycles Pulse on USB ID short than this value will be filtered

66.1.1.16. **USB_OTG_CTRL**

Offset: 0x0178

Bits	Name	R/W	Reset	Description
[7]	iddig_reg_en	RW	0x1	IDDIG reg ctrl enable
[6]	vbusvalid_reg_en	RW	0x1	VBUSVALID reg ctrl enable
[5]	avalid_reg_en	RW	0x1	AVALID reg ctrl enable
[4]	sessend_reg_en	RW	0x1	SESSEND reg ctrl enable
[3]	iddig_reg	RW	0x0	IDDIG reg ctrl
[2]	vbusvalid_reg	RW	0x0	VBUSVALID reg ctrl
[1]	avalid_reg	RW	0x0	AVALID reg ctrl
[0]	sessend_reg	RW	0x0	SESSEND reg ctrl

66.1.1.17. **EMAC_CFG**

Offset: 0x0190

Bits	Name	R/W	Reset	Description
[2:0]	EMAC_PHY_INTF_SEL	RW	0x0	EMAC0 PHY interface selected at reset. 3’b000: MII PHY interface selected. 3’b100: RMII PHY interface selected.
[30:3]	Reserved			
[31]	EMAC_PPS	RO	0x0	Pulse per second

66.1.1.18. **EMAC_TIMESTAMP_LSB**

Offset: 0x0194

Bits	Name	R/W	Reset	Description
[31:0]	EMAC_TIMESTAMP_L	RO	0x0	EMAC timestamp output value[31:0]

66.1.1.19. EMAC_TIMESTAMP_MSB

Offset: 0x0198

Bits	Name	R/W	Reset	Description
[31:0]	EMAC_TIMESTAMP_H	RO	0x0	EMAC timestamp output value[63:32]

66.1.1.20. I2S_CFG

Offset: 0x01A8

Bits	Name	R/W	Reset	Description
[0]	I2S0_SCLK_DIV	RW	0x0	To control clock divider for I2S0's SCLK; Default: 0x0 0x0: I2S0_SCLK is generated by dividing Codec's PLLADCLK by 2 0x1: I2S0_SCLK is generated by dividing Codec's PLLADCLK by 4
[1]	I2S_MST_SEL	RW	0x0	I2S Master/Slave select 0x0: I2S works as Slave, I2S1 block is selected 0x1: I2S works as Master, I2S0 block is selected

66.1.1.21. EXT_INT_CFG

Offset: 0x01C0

Bits	Name	R/W	Reset	Description
[3:0]	Reserved			
[4]	WDT_NMI_SEL,	RW	0x1	Te attribution for WDT interrupt 1'b1: WDT INT masked 1'b0: WDT INT not masked
[31:5]	Reserved			

66.1.1.22. QSPI_DLL_DLY_SEL0

Offset: 0x01E0

Bits	Name	R/W	Reset	Description
[31:0]	QSPI_DLL_DLY_31_0	RW	0x1	Set the value of one-hot coded QSPI clock dll delay as QSPI_DLL_DLY[31:0]

66.1.1.23. QSPI_DLL_DLY_SEL1

Offset: 0x01E4

Bits	Name	R/W	Reset	Description
[31:0]	QSPI_DLL_DLY_63_32	RW	0x0	Set the value of one-hot coded QSPI clock dll delay as QSPI_DLL_DLY[63:32]

66.1.1.24. QSPI_DLL_DLY_SEL2

Offset: 0x01E8

Bits	Name	R/W	Reset	Description
[31:0]	QSPI_DLL_DLY_95_64	RW	0x0	Set the value of one-hot coded QSPI clock dll delay as QSPI_DLL_DLY[95:64]

66.1.1.25. QSPI_DLL_DLY_SEL3

Offset: 0x01EC

Bits	Name	R/W	Reset	Description
[23:0]	QSPI_DLL_DLY_119_96	RW	0x0	Set the value of one-hot coded QSPI clock dll delay as QSPI_DLL_DLY[119:96]

66.1.1.26. LDOA_CTRL

Offset: 0x01F0

Bits	Name	R/W	Reset	Description
[31:1]	Reserved	RO	0x0	
[0]	PDLDOA	RW	0x0	LDOA Power Down Control 0： Power On(Default) 1： Power Off

66.1.1.27. **LDOA_CFG**

Offset: 0x01F4

Bits	Name	R/W	Reset	Description
[31:3]	Reserved	RO	0x0	
[2:0]	ACFG	RW	0x0	LDOA Output Voltage Control 000：1.2V 001：1.3V 010：1.4V 011：1.5V 100：1.6V 101：1.7V 110：1.8V 111：1.9V

66.1.1.28. **LDOB_CTRL**

Offset: 0x01F8

Bits	Name	R/W	Reset	Description
[31:1]	Reserved	RO	0x0	
[0]	PDLDOB	RW	0x0	LDOB Power Down Control 0：Power On(Default) 1：Power Off

66.1.1.29. **LDOB_CFG**

Offset: 0x01FC

Bits	Name	R/W	Reset	Description
[31:4]	Reserved	RO	0x0	

[3:0]	BCFG	RW	0x0	LDOB Output Voltage Control 0000：2.0V 0001：2.1V 0010：2.2V 0011：2.3V 0100：2.4V 0101：2.5V 0110：2.6V 0111：2.7V 1000：2.8V 1001：2.9V 1010：3.0V 1011：3.1V 1100：3.2V 1101：3.3V Others：TBD

66.1.1.30. **LDO_BGTRIM**

Offset: 0x0200

Bits	Name	R/W	Reset	Description
[31:3]	Reserved	RO	0x0	
[2:0]	BGTRIM	RW	0x0	LDO Band-Gap Trim Control: 011:9% 010:6% 001:3% 000:0% 100:-3% 101:-6% 110:-9% 111:-12%

66.2. Direct Memory Access Controller(DMAC)

66.2.1. Overview

In MCU Subsystem, it integrates two DMA Controller, one is a DMAC, is a general DMA controller on AHB bus. The other is an axi_dmac, is a general DMA controller on AXI bus.

66.2.2. Features

AHB_DMAC Main features include:

- 8 independent channels and three master interfaces
- Support peripheral hardware handshake interfaces, each of which can be routed to any DMA channels
- Support 3 interrupts out of all GPIO/Timer interrupts as a trigger of DMA data transfers
- FO per channel for source and destination, with FO depth of 256 in bytes for each channel,
- Maximum value of burst transaction size is 16 for each channel
- Maximum block size in source transfer widths is 4095 for each channel
- Support data transfer types including memory to memory, memory to peripheral, peripheral to memory, and peripheral to peripheral.
- Two AHB master interfaces for data transfer and AHB slave interface for configuration
- Support multi-block transfer operation

AXI_DMAC Main features include:

- 1 independent channels and two master interfaces
- Not support peripheral hardware handshake interfaces
- FO per channel for source and destination, with FO depth of 16 in bytes for each channel,
- Maximum value of burst transaction size is 128 for each channel
- Maximum block size in source transfer widths is 31 for each channel
- Support data transfer types including memory to memory, memory to peripheral, peripheral to memory, and peripheral to peripheral.
- Two AHB master interfaces for data transfer and AHB slave interface for configuration
- Support multi-block transfer operation

66.2.3. Functional Description

66.2.3.1. Basic Interface Definition

The following definitions are used through the following of subsection. For the detailed definition of register CTLx fields, refer to Columbus Memory Map and Register .

- Source single transaction size in bytes

src_single_size_bytes = CTLx.SRC_TR_WIDTH/8 (1)

- Source burst transaction size in bytes

*src_burst_size_bytes = CTLx.SRC_MSIZE * src_single_size_bytes (2)*

- Destination single transaction size in bytes

dst_single_size_bytes = CTLx.DST_TR_WIDTH/8 (3)

- Destination burst transaction size in bytes

*dst_burst_size_bytes = CTLx.DEST_MSIZE * dst_single_size_bytes (4)*

- Block size in bytes:

The processor programs the DMAC with the number of data items (block size) of source transfer width (CTLx.SRC_TR_WIDTH) to be transferred by the DMA in a block transfer; this is programmed into the CTLx.BLOCK_TS field. Therefore, the total number of bytes to be transferred in a block is:

*blk_size_bytes_dma = CTLx.BLOCK_TS * src_single_size_bytes (5)*

66.2.3.2. DMA Transfer Hierarchy

Below is the transfer hierarchy for non-memory peripherals.

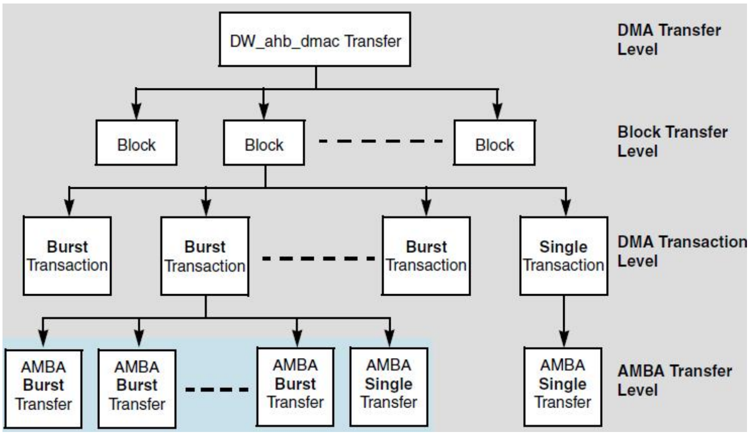


Figure 4.6- 1 DMA Transfer Hierarchy for Non-Memory Peripherals

Below is the transfer hierarchy for memory peripherals.

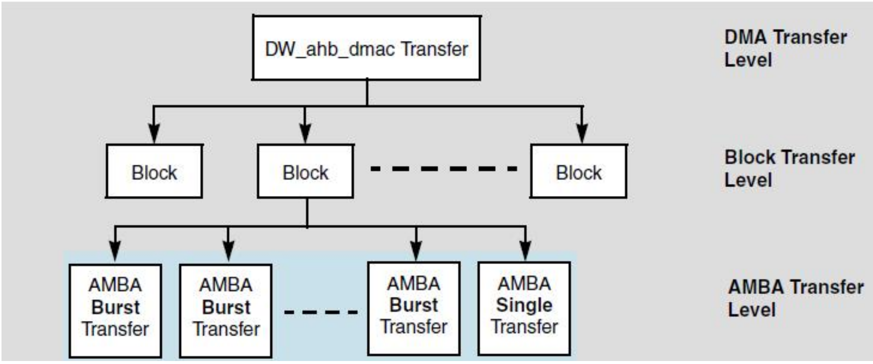


Figure 4.6- 2 DMA Transfer Hierarchy for Memory Peripherals

The detailed definitions of the hierarchy are described as below:

- Block –Block of DMA data, the amount of which is the block length. For transfers between the DMA and memory, a block is broken directly into a sequence of bursts and single transfers. For transfers between the DMA and a non-memory peripheral, a block is broken into a sequence of DMA transactions (single and burst). These are in turn broken into a sequence of AHB transfers.

- Transaction – Basic unit of a DMA transfer, as determined by either the hardware or software handshaking interface. A transaction is relevant for transfers between the DMA and a source or destination peripheral the peripheral is a non-memory device. There are two types of transactions:

  - Single transaction – Length of a single transaction is always 1 and is converted to a single AHB transfer.
  - Burst transaction – Length of a burst transaction is programmed into the DMA. The burst transaction is converted into a sequence of bursts and AHB single transfers. The burst transaction length is under program control and normally bears some relationship to the FO sizes in the DMA and in the source and destination peripherals.

- DMA transfer – Software controls the number of blocks in a DMA transfer. Once the DMA transfer has completed, the hardware within the DMA disables the channel and can generate an interrupt to signal the DMA transfer completion. The channel can then be reprogrammed for a new DMA transfer.
  - Single-block DMA transfer – Consists of a single block.
  - Multi-block DMA transfer – DMA transfer may consist of multiple DMA blocks. Multi-block DMA transfers are supported through block chaining (linked list pointers), auto-reloading channel registers, and contiguous blocks. The source and destination can independently select which method to use.

- Linked lists (block chaining) – Linked list pointer (LLP) points to the location in system memory where the next linked list item (LLI) exists. The LLI is a set of registers that describes the next block (block descriptor) and an LLP register. The DMA fetches the LLI at the beginning of every block when block chaining is enabled. LLI accesses are always 32-bit accesses aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit

- Auto-reloading – DMA automatically reloads the channel registers at the end of each block to the value when the channel was first enabled.
- Contiguous blocks – Address between successive blocks is selected to be a continuation from the end of the previous block.

66.2.3.3. Memory Peripherals

There is no handshaking interface with the DMA for memory peripherals. Once the channel is enabled, the transfer proceeds immediately without waiting for a transaction request.

The CTLx.SRC_MSIZE and CTLx.DEST_MSIZE are properties valid for peripherals with a handshaking interface; they cannot be used for defining the burst length for memory peripherals.

When the peripherals are memory, the DMA uses DMA transfers to move blocks; thus the CTLx.SRC_MSIZE and CTLx.DEST_MSIZE values are not used for memory peripherals. The SRC_MSIZE/DEST_MSIZE limitations are used to accommodate devices that have limited resources, such as a FO. Memory does not normally have limitations similar to the FOs. Therefore:

- Length of burst transfers to memory is always equal to the number of data items available in a channel FO or data items required to complete the block transfer, whichever is smaller.

- Length of burst transfers from memory is always equal to the space available in a channel FO or number of data items required to complete the block transfer, whichever is smaller.

66.2.3.4. Handshake interfaces

Handshaking interfaces are used at the transaction level to control the flow of single or burst transactions. The peripheral uses the handshaking interface to indicate to the DMAC that it is ready to transfer or accept data over the AHB bus.

A non-memory peripheral can request a DMA transfer through the DMAC using one of two types of handshaking interfaces:

- Hardware
- Software

Software selects between the hardware or software handshaking interface on a per-channel basis. Software handshaking is accomplished through memory-mapped registers, while hardware handshaking is accomplished using a dedicated handshaking interface.

Software Handshaking

When the slave peripheral requires the DMA to perform a DMA transaction, it communicates this request by sending an interrupt to the interrupt controller. The interrupt service routine then uses the software registers detailed in “Software Handshaking Registers” to initiate and control a DMA transaction. This group of software registers is used to implement the software handshaking interface.

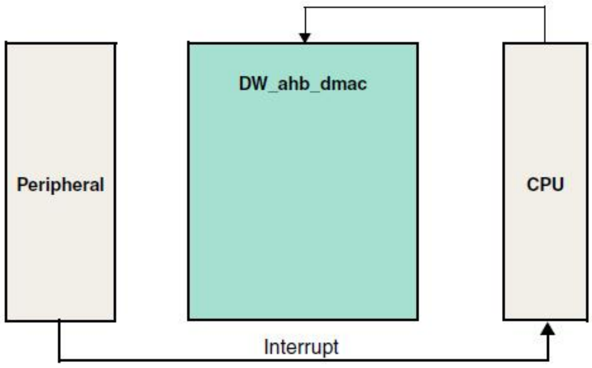


Figure 4.6- 3 DMA Software Handshaking Interface

The HS_SEL_SRC/HS_SEL_DST bit in the CFGx channel configuration register must be set to enable software handshaking.

Hardware Handshaking

DMA tries to efficiently transfer the data using as little of the bus bandwidth as possible. Generally, the DMA tries to transfer the data using burst transactions and, where possible, fill or empty the channel FO in single bursts – provided that the software has not limited the burst length.

There are cases where a DMA block transfer cannot complete using burst transactions. Typically this occurs when the block size is not a multiple of the burst transaction length. In these cases, the block transfer uses burst transactions up to the point where the amount of data left to complete the block is less than the amount of data in a burst transaction. At this point, the DMA samples the “single” status flag and completes the block transfer using single transactions.

The peripheral asserts a single status flag to indicate to the DMA that there is enough data or space to complete a single transaction from or to the source/destination peripheral.

The Single Transaction Region is the time interval where the DMA uses single transactions to complete the block transfer; burst transactions are exclusively used outside this region.

- The source peripheral enters the Single Transaction Region when the number of bytes left to complete in the source block transfer is less than src_burst_size_bytes. : blk_size_bytes/src_burst_size_bytes = integer, then the source never enters this region, and the source block uses burst transactions.

- The destination peripheral enters the Single Transaction Region when the number of bytes left to complete in the destination block transfer is less than dst_burst_size_bytes. : blk_size_bytes/dst_burst_size_bytes = integer, then the destination never enters this region, and the destination block uses burst transactions.

When a source or destination peripheral is in the Single Transaction Region, a burst transaction can still be requested. However, src_burst_size_bytes/ dst_burst_size_bytes is greater than the number of bytes left to complete in the source/destination block transfer at the time that the burst transaction is triggered. In this case, the burst transaction is started and “early-terminated” at block completion without transferring the programmed amount of data – that is, src_burst_size_bytes or dst_burst_size_bytes – but the amount required to complete the block transfer.

The DMA never terminates defined-length bursts early.

The DMA detects a burst when in the Single Transaction Region, the DMA issues an undefined length burst (INCR) of length large enough to complete the block transfer. It does not transfer all programmed src_burst_size_bytes or dst_burst_size_bytes

Figure below illustrates the hardware handshaking interface between a peripheral – whether a destination or source – and the DMA.

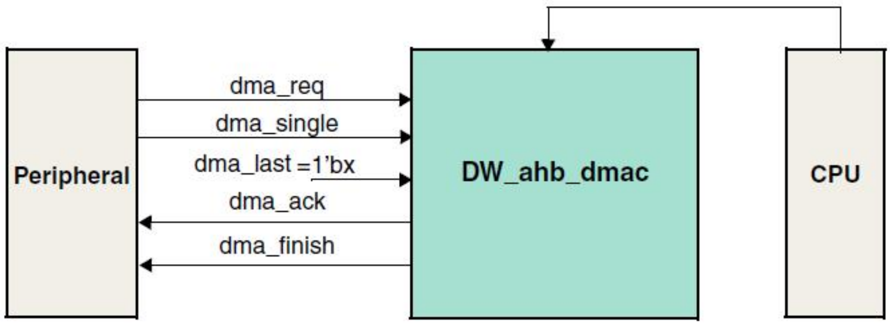


Figure 4.6- 4 DMA Hardware Handshake Interface

The handshaking loop is as follows:
dma_req asserted by peripheral -> dma_ack asserted by DMA -> dma_req de-asserted by peripheral -> dma_ack de-asserted by DMA

The detailed s of DMA hardware handshaking interface signals that are in use are listed in the below table.

Table 4.6- 1 DMA Hardware Handshaking Interface Signals

Signal	Direction	
dma_ack	Output	DMA acknowledge signal to peripheral. The dma_ack signal is asserted after the data phase of the last AHB transfer in the current transaction – single or burst – to the peripheral that has completed. For a single transaction, dma_ack remains asserted until the peripheral de-asserts

		dma_single; dma_ack is de-asserted one hclk cycle later. For a burst transaction, dma_ack remains asserted until the peripheral de-asserts dma_req; dma_ack is de-asserted one hclk cycle later.
dma_finish	input	DMA asserts dma_finish to signal block completion. This has the same timing as dma_ack and forms a handshaking loop with dma_req the last transaction in the block was a burst transaction or with dma_single the last transaction in the block was a single transaction.
dma_req	Input	Burst transaction request from peripheral. The DMA always interprets the dma_req signal as a burst transaction request, regardless of the level of dma_single. This is a level-sensitive signal; once asserted by the peripheral, dma_req must remain asserted until the DMA asserts dma_ack. Upon receiving the dma_ack signal from the DMA to indicate the burst transaction is complete, the peripheral should de-assert the burst request signal, dma_req. Once dma_req is de-asserted by the peripheral, the DMA de-asserts dma_ack. an active level on dma_req is detected in the Single Transaction Region, then the block is completed using an Early-Terminated Burst Transaction.
dma_single	Input	Single transfer status. The dma_single signal is a status signal that is asserted by a destination peripheral when it can accept at least one destination data item; otherwise it is cleared. For a source peripheral, the dma_single signal is again a status signal and is asserted by a source peripheral when it can transmit at least one source data item; otherwise it is cleared. Once asserted, dma_single must remain asserted until dma_ack is asserted, at which time the peripheral should de-assert dma_single. This signal is sampled by the DMA in the Single Transaction Region of the block transfer. Outside of this region, dma_single is ignored and all transactions are burst transactions.

Interrupt Handshake interfaces

DMA supports to use interrupts as handshake signals, which could be satisfy the requirement that external event could trigger the DMA transfer between memory and peripherals or between memory and memory.

As specied below out of in total 16 interrupts including 8 interrupts from timers and 2 interrupts from GPIO, can be configured as the above mentioned trigger events.

Below takes the example of using a GPIO transition to trigger the DMA data transfer between Memory and UART, to illustrate the entire procedures of both hardware and software.

Basically, the GPIO interrupt is used to generate the handshake signal “DMA req” associated with memory, which would trigger the DMA transfer.

1. SW configures the System register as specified in section 4.6 to select the appropriate GPIO interrupt as the handshake signal to DMA;
2. SW configures GPIO controller, to enable the GPIO interrupt could be generated by upon the GPIO transition event;
3. SW configures the GIC to disable the GPIO interrupt from sending to CPU while it can still be routed to DMA handshake interfaces;
4. SW configures DMA, to set the transfer type as “peri-to-peri”, and set other necessary configurations as other normal transfer, including transfer size, source/destination address, handshake signal selection, etc., and enable the used channel
5. Once the trigger event occurs, the DMA transfer could start and automatically finish the entire DMA transfer according to the configuration of DMA ctrl., without any further intervention of CPU.
6. In the interrupt handler that is entered by DMA transfer completeness, an additional operation is required to clear the GPIO interrupt source although it never routes to CPU in the entire procedures.

66.2.4. Registers

66.2.4.1. Register Memory Map

Table below shows the memory map for the DMAC:

Table 4.6- 2 Address Map for DMAC

Addr offset	Name		
0x0	SAR0	Channel 0 Source Address Register	
0x8	DAR0	Channel 0 Destination Address Register	
0x10	LLP0	Channel 0 Linked List Pointer Register	
0x18	CTL0	Channel 0 Control Register	
0x20	Reserved		
0x28	Reserved		
0x30	Reserved		
0x38	Reserved		
0x40	CFG0	Channel 0 Configuration Register	
0x48	SGR0	Channel 0 Source Gather Register	
0x50	DSR0	Channel 0 Destination Scatter Register	
0x58	SAR1	Channel 1 Source Address Register	
0x60	DAR1	Channel 1 Destination Address Register	
0x68	LLP1	Channel 1 Linked List Pointer Register	
0x70	CTL1	Channel 1 Control Register	
0x78	Reserved		
0x80	Reserved		
0x88	Reserved		

0x90	Reserved		
0x98	CFG1	Channel 1 Configuration Register	
0xA0	SGR1	Channel 1 Source Gather Register	
0xA8	DSR1	Channel 1 Destination Scatter Register	
0xB0	SAR2	Channel 2 Source Address Register	
0xB8	DAR2	Channel 2 Destination Address Register	
0xC0	LLP2	Channel 2 Linked List Pointer Register	
0xC8	CTL2	Channel 2 Control Register	
0xD0	Reserved		
0xD8	Reserved		
0xE0	Reserved		
0xE8	Reserved		
0xF0	CFG2	Channel 2 Configuration Register	
0xF8	SGR2	Channel 2 Source Gather Register	
0x100	DSR2	Channel 2 Destination Scatter Register	
0x108	SAR3	Channel 3 Source Address Register	
0x110	DAR3	Channel 3 Destination Address Register	
0x118	LLP3	Channel 3 Linked List Pointer Register	
0x120	CTL3	Channel 3 Control Register	
0x128	Reserved		
0x130	Reserved		
0x138	Reserved		
0x140	Reserved		
0x148	CFG3	Channel 3 Configuration Register	
0x150	SGR3	Channel 3 Source Gather Register	
0x158	DSR3	Channel 3 Destination Scatter Register	
0x160	SAR4	Channel 4 Source Address Register	
0x168	DAR4	Channel 4 Destination Address Register	
0x170	LLP4	Channel 4 Linked List Pointer Register	
0x178	CTL4	Channel 4 Control Register	
0x180	Reserved		
0x188	Reserved		
0x190	Reserved		
0x198	Reserved		
0x1A0	CFG4	Channel 4 Configuration Register	
0x1A8	SGR4	Channel 4 Source Gather Register	
0x1B0	DSR4	Channel 4 Destination Scatter Register	
0x1B8	SAR5	Channel 5 Source Address Register	
0x1C0	DAR5	Channel 5 Destination Address Register	
0x1C8	LLP5	Channel 5 Linked List Pointer Register	
0x1D0	CTL5	Channel 5 Control Register	
0x1D8	Reserved		
0x1E0	Reserved		

0x1E8	Reserved		
0x1F0	Reserved		
0x1F8	CFG5	Channel 5 Configuration Register	
0x200	SGR5	Channel 5 Source Gather Register	
0x208	DSR5	Channel 5 Destination Scatter Register	
0x210	SAR6	Channel 6 Source Address Register	
0x218	DAR6	Channel 6 Destination Address Register	
0x220	LLP6	Channel 6 Linked List Pointer Register	
0x228	CTL6	Channel 6 Control Register	
0x230	Reserved		
0x238	Reserved		
0x240	Reserved		
0x248	Reserved		
0x250	CFG6	Channel 6 Configuration Register	
0x258	SGR6	Channel 6 Source Gather Register	
0x260	DSR6	Channel 6 Destination Scatter Register	
0x268	SAR7	Channel 7 Source Address Register	
0x270	DAR7	Channel 7 Destination Address Register	
0x278	LLP7	Channel 7 Linked List Pointer Register	
0x280	CTL7	Channel 7 Control Register	
0x288	Reserved		
0x290	Reserved		
0x298	Reserved		
0x2A0	Reserved		
0x2A8	CFG7	Channel 7 Destination Address Register	
0x2B0	SGR7	Channel 7 Linked List Pointer Register	
0x2B8	DSR7	Channel 7 Control Register	
0x2C0	RawTfr	Raw Status for Tfr Interrupt	
0x2C8	RawBlock	Raw Status for Block Interrupt	
0x2D0	RawSrcTran	Raw Status for SrcTran Interrupt	
0x2D8	RawDstTran	Raw Status for DstTran Interrupt	
0x2E0	RawErr	Raw Status for Err Interrupt	
0x2E8	StatusTfr	Status for Tfr Interrupt	
0x2F0	StatusBlock	Status for Block Interrupt	
0x2F8	StatusSrcTran	Status for SrcTran Interrupt	
0x300	StatusDstTran	Status for DstTran Interrupt	
0x308	StatusErr	Status for Err Interrupt	
0x310	MaskTfr	Mask for Tfr Interrupt	
0x318	MaskBlock	Mask for Block Interrupt	
0x320	MaskSrcTran	Mask for SrcTran Interrupt	
0x328	MaskDstTran	Mask for DstTran Interrupt	
0x330	MaskErr	Mask for Err Interrupt	
0x338	ClearTfr	Clear for Tfr Interrupt	

0x340	ClearBlock	Clear for Block Interrupt	
0x348	ClearSrcTran	Clear for SrcTran Interrupt	
0x350	ClearDstTran	Clear for DstTran Interrupt	
0x358	ClearErr	Clear for Err Interrupt	
0x360	StatusInt	Status for each interrupt type	
0x368	ReqSrcReg	Source Software Transaction Request Register	
0x370	ReqDstReg	Destination Software Transaction Request Register	
0x378	SglReqSrcReg	Single Source Transaction Request Register	
0x380	SglReqDstReg	Single Destination Transaction Request Register	
0x388	LstSrcReg	Last Source Transaction Request Register	
0x390	LstDstReg	Last Destination Transaction Request Register	
0x398	DmaCfgReg	DMA Configuration Register	
0x3A0	ChEnReg	DMA Channel Enable Register	
0x3A8	DmaIdReg	DMA ID Register	
0x3B0	DmaTestReg	DMA Test Register	
0x3B8	Reserved		
0x3C0	Reserved		
0x3C8	DMA_COMP_PARAMS_6	Component Parameters Register 6	
0x3D0	DMA_COMP_PARAMS_5	Component Parameters Register 5	
0x3D8	DMA_COMP_PARAMS_4	Component Parameters Register 4	
0x3E0	DMA_COMP_PARAMS_3	Component Parameters Register 3	
0x3E8	DMA_COMP_PARAMS_2	Component Parameters Register 2	
0x3F0	DMA_COMP_PARAMS_1	Component Parameters Register 1	
0x3F8	DMA Component ID Register	Component version register.	
0x400 ~ 0xFFFFF	Reserved		

Channel x Source Address Register (x=0,1,2…7)

SARx, x = 0, 1, 2... 7 (Addr: 0x0 + 0x58 * x)

Bit Field	Name	Width	R/W	Default value	
63:32	Reserved				
31:0	SAR	32	RW	0x0	Current Source Address of DMA transfer. Updated after each source transfer. The SINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every source transfer throughout the block transfer.

Channel x Destination Address Register (x=0,1,2…7)

DARx, x = 0, 1, 2... 7 (Addr: 0x8 + 0x58 * x)

Bit Field	Name	Width	R/W	Default value	
63:32	Reserved				
31:0	DAR	32	RW	0x0	Current Destination address of DMA transfer. Updated after each destination transfer. The DINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every destination transfer throughout the block transfer.

66.2.4.2. **Channel x Linked List Pointer Register (x=0,1,2…7)**

LLPx, x = 0, 1, 2... 7 (Addr: 0x10 + 0x58 * x)

Bit Field	Name	Width	R/W	Default value	
63:32	Reserved				
31:2	LOC	30	RW	0x0	Starting Address In Memory of next LLI block chaining is enabled. Note that the two LSBs of the starting address are not stored because the address is assumed to be aligned to a 32-bit boundary. LLI accesses are always 32-bit accesses (Hsize = 2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit.
1:0	Reserved				

66.2.4.3. **Channel x Control Register (x=0,1,2…7)**

CTLx, x = 0, 1, 2... 7 (Addr: 0x18 + 0x58 * x)

Bit Field	Name	Width	R/W	Default value	
63:45	Reserved				

44	DONE	1	RW	0x0	Done bit. status write-back is enabled, the upper word of the control register, CTLn[63:32], is written to the control register location of the Linked List Item (LLI) in system memory at the end of the block transfer with the done bit set. Software can poll the LLI CTLx.DONE bit to see when a block transfer is complete. The LLI CTLx.DONE bit should be cleared when the linked lists are set up in memory prior to enabling the channel. LLI accesses are always 32-bit accesses (Hsize = 2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit.
43:32	BLOCK_TS	12	RW	0x2	Block Transfer Size. When the AHB_DMAC is the flow controller, the user writes this field before the channel is enabled in order to indicate the block size. The number programmed into BLOCK_TS indicates the total number of single transactions to perform for every block transfer; a single transaction is mapped to a single AMBA beat. Width: The width of the single transaction is determined by CTLn.SRC_TR_WIDTH. For further information on setting this field, Once the transfer starts, the read-back value is the total number of data items already read from the source peripheral, regardless of what is the flow controller. When the source or destination peripheral is assigned as the flow controller, then the maximum block size

					that can be read back saturates at 4095, but the actual block size can be greater.
31:29	Reserved				
28	LLP_SRC_EN	1	RW	0x0	Block chaining is enabled on the source side the LLP_SRC_EN field is high and LLPx.LOC is non-zero;
27	LLP_DST_EN	1	RW	0x0	Block chaining is enabled on the destination side the LLP_DST_EN field is high and LLPx.LOC is non-zero.
26:25	SMS	2	RW	0x0	Source Master Select. Identies the Master Interface layer from which the source device (peripheral or memory) is accessed. 00 = AHB master 1 00 is the valid value of this field
24:23	DMS	2	RW	0x0	Destination Master Select. Identies the Master Interface layer where the destination device (peripheral or memory) resides. 00 = AHB master 1 00 is the valid value of this field

22:20	TT_FC	3	RW	0x3	Transfer Type and Flow Control. The following transfer types are supported. • Memory to Memory • Memory to Peripheral • Peripheral to Memory • Peripheral to Peripheral DMAC is allowed to be the flow controller 000: Memory to Memory 001: Memory to Peripheral 010: Peripheral to Memory 011: Peripheral to Peripheral
19	Reserved				
18	DST_SCATTER_EN	1	RW	0x0	Destination scatter enable bit: 0 = Scatter disabled 1 = Scatter enabled Scatter on the destination side is applicable when the CTLn.DINC bit indicates an incrementing or decrementing address control.
17	SRC_GATHER_EN	1	RW	0x0	Source gather enable bit: 0 = Gather disabled 1 = Gather enabled Gather on the source side is applicable when the CTLn.SINC bit indicates an incrementing or decrementing address control.
16:14	SRC_MSIZE	3	RW	0x1	Source Burst Transaction Length. Number of data items, each of width CTLn.SRC_TR_WIDTH, to be read from the source every time a source burst transaction request is made from either the corresponding hardware or software handshaking interface.
13:11	DEST_MSIZE	3	RW	0x1	Destination Burst Transaction Length. Number of data items, each of width CTLn.DST_TR_WIDTH, to be written to the destination every time a destination burst transaction request is made from either the corresponding hardware or software handshaking interface.

10:9	SINC	2	RW	0x0	Source Address Increment. Indicates whether to increment or decrement the source address on every source transfer. the device is fetching data from a source peripheral FO with a fixed address, then set this field to “No change.” 00 = Increment 01 = Decrement 1x = No change NOTE: Incrementing or decrementing is done for alignment to the next CTLn.SRC_TR_WIDTH boundary.
8:7	DINC	2	RW	0x0	Destination Address Increment. Indicates whether to increment or decrement the destination address on every destination transfer. your device is writing data to a destination peripheral FO with a fixed address, then set this field to “No change.” 00 = Increment 01 = Decrement 1x = No change NOTE: Incrementing or decrementing is done for alignment to the next CTLn.DST_TR_WIDTH boundary.
6:4	SRC_TR_WIDTH	3	RW	0x0	Source Transfer Width. Mapped to AHB bus “hsize.” For a non-memory peripheral, typically the peripheral (source) FO width. The valid value are as below: 000: 8-bit 001: 16-bit 010: 32-bit
3:1	DST_TR_WIDTH	3	RW	0x0	Destination Transfer Width. Mapped to AHB bus “hsize.” For a non-memory peripheral, typically rgw peripheral (destination) FO width. The valid value are as below: 000: 8-bit 001: 16-bit 010: 32-bit
0	INT_EN	1	RW	0x1	Interrupt Enable Bit. set, then all interrupt-generating sources are enabled. Functions as a global mask bit for all

					interrupts for the channel; raw* interrupt registers still assert CTLn.INT_EN = 0.

66.2.4.4. **Channel x Configuration Register (x=0,1,2…7)**

CFGx, x = 0, 1, 2... 7 (Addr: 0x40 + 0x58 * x

Bit Field	**Name**	**Width**	**R/W**	**Default value**	
63:47	Reserved				
46:43	DEST_PER	4	RW	0x0	Assigns a hardware handshaking interface (0 - 15) to the destination of channel n the CFGn.HS_SEL_DST field is 0; otherwise, this field is ignored. The channel can then communicate with the destination peripheral connected to that interface through the assigned hardware handshaking interface. NOTE: For correct DMA operation, one peripheral (source or destination) should be assigned to the same handshaking interface.
42:39	SRC_PER	4	RW	0x0	Assigns a hardware handshaking interface (0 - 15) to the source of channel n the CFGn.HS_SEL_SRC field is 0; otherwise, this field is ignored. The channel can then communicate with the source peripheral connected to that interface through the assigned hardware handshaking interface. NOTE: For correct AHB_DMAC operation, one peripheral (source or destination) should be assigned to the same handshaking interface.
38:37	Reserved				

36:34	PROTCTL	3	RW	0x1	Protection Control used to drive the AHB HPROT[3:1] bus. The AMBA Speciation recommends that the default value of HPROT indicates a non-cached, non-buffered, privileged data access. The is used to indicate such an access. HPROT[0] is tied high because all transfers are data accesses, as there are no opcode fetches. There is a one-to-one mapping of these register to the HPROT[3:1] master interface signals.
33	FO_MODE	1	RW	0x0	FO Mode Select. Determines how much space or data needs to be available in the FO before a burst transaction request is serviced. 0 = Space/data available for single AHB transfer of the specied transfer width. 1 = Data available is greater than or equal to half the FO depth for destination transfers and space available is greater than half the fo depth for source transfers. The exceptions are at the end of a burst transaction request or at the end of a block transfer
32	FCMODE	1	RW	0x0	Flow Control Mode. Determines when source transaction requests are serviced when the Destination Peripheral is the flow controller. 0 = Source transaction requests are serviced when they occur. Data pre-fetching is enabled. 1 = Source transaction requests are not serviced until a destination transaction request occurs. In this mode, the amount of data transferred from the source is limited so that it is guaranteed to be transferred to the destination prior to block termination by the destination. Data pre-fetching is disabled.
31	RELOAD_DST	1	RW	0x0	Automatic Destination Reload. The DARn register can be automatically reloaded from its initial value at the end of every block for multi-block transfers.

					A new block transfer is then initiated.
30	RELOAD_SRC	1	RW	0x0	Automatic Source Reload. The SARn register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated.
29:20	MAX_ABRST	10	RW	0x0	Maximum AMBA Burst Length. Maximum AMBA burst length that is used for DMA transfers on this channel. A value of 0 indicates that software is not limiting the maximum AMBA burst length for DMA transfers on this channel.
19	SRC_HS_POL	1	RW	0x0	Source Handshaking Interface Polarity. 0 = Active high 1 = Active low
18	DST_HS_POL	1	RW	0x0	Destination Handshaking Interface Polarity. 0 = Active high 1 = Active low
17:12	Reserved				
11	HS_SEL_SRC	1	RW	0x1	Source Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces – hardware or software – is active for source requests on this channel. 0 = Hardware handshaking interface. Software-initiated transaction requests are ignored. 1 = Software handshaking interface. Hardware-initiated transaction requests are ignored. the source peripheral is memory, then this bit is ignored.
10	HS_SEL_DST	1	RW	0x1	Destination Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces – hardware or software – is active for destination requests on this channel. 0 = Hardware handshaking interface.

					Software-initiated transaction requests are ignored. 1 = Software handshaking interface. Hardware- initiated transaction requests are ignored. the destination peripheral is memory, then this bit is ignored.
9	FO_EMPTY	1	R	0x1	Indicates there is data left in the channel FO. Can be used in conjunction with CFGx.CH_SUSP to cleanly disable a channel. 1 = Channel FO empty 0 = Channel FO not empty
8	CH_SUSP	1	RW	0x0	Channel Suspend. Suspends all DMA data transfers from the source until this bit is cleared. There is no guarantee that the current transaction will complete. Can also be used in conjunction with CFGn.FO_EMPTY to cleanly disable a channel without losing any data. 0 = Not suspended. 1 = Suspend DMA transfer from the source.
7:5	CH_PRIOR	3	RW	Channel Number For example: Chan0=0 Chan1=1	Channel priority. A priority of 7 is the highest priority, and 0 is the lowest. This field must be programmed within the range of 0 ~ 7 A programmed value outside this range will cause erroneous behavior.
4:0	Reserved				

66.2.4.5. Channel x Source Gather Register (x=0,1,2…7)

SGRx, x = 0, 1, 2... 7 (Addr: 0x48 + 0x58 * x)

Bit Field	Name	Width	R/W	Default value	
63:32	Reserved				
31:20	SGC	12	RW	0x0	Source gather count. Source contiguous transfer count between successive gather boundaries.
19:0	SGI	20	RW	0x0	Source gather interval.

66.2.4.6. Channel x Destination Scatter Register (x=0,1,2…7)

DSRx, x = 0, 1, 2... 7 (Addr: 0x48 + 0x58 * x)

Bit Field	Name	Width	R/W	Default value	
63:32	Reserved				
31:20	DSC	12	RW	0x0	Destination scatter count. Destination contiguous transfer count between successive scatter boundaries.
19:0	DSI	20	RW	0x0	Destination scatter interval.

66.2.4.7. Raw Status for Tfr/Block/SrcTran/DstTran/Err Interrupt

RawBlock, RawDstTran, RawErr, RawSrcTran, RawTfr

Bit Field	Name	Width	R/W	Default value	
63:8	Reserved				
7:0	RAW	8	RW	0x0	Raw interrupt status.

66.2.4.8. Status for Tfr/Block/SrcTran/DstTran/Err Interrupt

StatusBlock, StatusDstTran, StatusErr, StatusSrcTran, StatusTfr

Bit Field	Name	Width	R/W	Default value	
63:8	Reserved				
7:0	STATUS	8	R	0x0	Interrupt status.

66.2.4.9. Mask for Tfr/Block/SrcTran/DstTran/Err Interrupt

MaskBlock, MaskDstTran, MaskErr, MaskSrcTran, MaskTfr

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	INT_MASK_WE	8	W	0x0	Interrupt Mask Write Enable 0 = write disabled 1 = write enabled
7:0	INT_MASK	8	RW	0x0	Interrupt Mask 0 = masked 1 = unmasked

66.2.4.10. Clear for Tfr/Block/SrcTran/DstTran/Err Interrupt

ClearBlock, ClearDstTran, ClearErr, ClearSrcTran, ClearTfr

Bit Field	Name	Width	R/W	Default value	
63:8	Reserved				
7:0	CLEAR	8	W	N/A	Interrupt clear. 0 = no effect 1 = clear interrupt

66.2.4.11. Status for Each Interrupt Type

StatusInt (Addr: 0x360)

Bit Field	Name	Width	R/W	Default value	
63:5	Reserved				
4	ERR	1	R	0x0	OR of the contents of StatusErr register.
3	DSTT	1	R	0x0	OR of the contents of StatusDst register.
2	SRCT	1	R	0x0	OR of the contents of StatusSrcTran register.
1	BLOCK	1	R	0x0	OR of the contents of StatusBlock register.
0	TFR	1	R	0x0	OR of the contents of StatusTfr register.

66.2.4.12. Source Software Transaction Request Register

ReqSrcReg (Addr: 0x368)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	SRC_REQ_WE	8	W	0x0	Source request write enable 0 = write disabled 1 = write enabled
7:0	SRC_REQ	8	RW	0x0	Source request

66.2.4.13. Destination Software Transaction Request Register

ReqDstReg (Addr: 0x370)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	DST_REQ_WE	8	W	0x0	Destination request write enable 0 = write disabled 1 = write enabled
7:0	DST_REQ	8	RW	0x0	Destination request

66.2.4.14. Single Source Transaction Request Register

SglReqSrcReg (Addr: 0x378)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	SRC_SGLREQ_WE	8	W	0x0	Single write enable 0 = write disabled 1 = write enabled
7:0	SRC_SGLREQ	8	RW	0x0	Source single request

66.2.4.15. Single Destination Transaction Request Register

SglReqDstReg (Addr: 0x380)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	DST_SGLREQ_WE	8	W	0x0	Destination write enable 0 = write disabled 1 = write enabled
7:0	DST_SGLREQ	8	RW	0x0	Destination single or burst request

66.2.4.16. Last Source Transaction Request Register

LstSrcReg (Addr: 0x388)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				

15:8	LSTSRC_WE	8	W	0x0	Source last transaction request write enable 0 = write disabled 1 = write enabled
7:0	LSTSRC	8	RW	0x0	Source last transaction request 0 = Not last transaction in current block 1 = Last transaction in current block

66.2.4.17. Last Destination Transaction Request Register

LstDstReg (Addr: 0x390)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	LSTDST_WE	8	W	0x0	Destination last transaction request write enable 0 = write disabled 1 = write enabled
7:0	LSTDST	8	RW	0x0	Destination last transaction request 0 = Not last transaction in current block 1 = Last transaction in current block

66.2.4.18. DMA Configuration Register

DmaCfgReg (Addr: 0x398)

Bit Field	Name	Width	R/W	Default value	
63:1	Reserved				
0	DMA_EN	1	RW	0x0	AHB_DMAC Enable bit. 0 = AHB_DMAC Disabled 1 = AHB_DMAC Enabled.

66.2.4.19. DMA Channel Enable Register

ChEnReg (Addr: 0x3A0)

Bit Field	Name	Width	R/W	Default value	
63:16	Reserved				
15:8	CH_EN_WE	8	W	0x0	Channel enable write enable.

7:0	CH_EN	8	RW	0x0	Enables/Disables the channel. Setting this bit enables a channel; clearing this bit disables the channel. 0 = Disable the Channel 1 = Enable the Channel The ChEnReg.CH_EN bit is automatically cleared by hardware to disable the channel after the last AMBA transfer of the DMA transfer to the destination has completed. Software can therefore poll this bit to determine when this channel is free for a new DMA transfer.

66.2.4.20. DMA ID Register

DmaIdReg (Addr: 0x3A8)

Bit Field	Name	Width	R/W	Default value	
63:32	Reserved				
31:0	DMA ID	32	R	0x0	Hardcoded AHB_DMAC Peripheral ID

66.2.4.21. DMA Test Register

DmaTestReg (Addr: 0x3B0)

Bit Field	Name	Width	R/W	Default value	
63:1	Reserved				
0	TEST_SLV_	1	RW	0x0	Puts the AHB slave interface into test mode. In this mode, the readback value of the writable registers always matches the value written. This bit does not allow writing to read-registers. 0 = Normal mode 1 = Test mode

66.2.4.22. Component Parameters Register 6

DMA_COMP_PARAMS_6 (Addr: 0x3C8)

Bit Field	Name	Width	R/W	Default value	
63	Reserved				
62:60	CH7_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH7_FO_DEPTH coreConsultant parameter. 0x3 = 64
59:57	CH7_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH7_SMS coreConsultant parameter. 0x0 = MASTER_1
56:54	CH7_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH7_LMS coreConsultant parameter. 0x0 = MASTER_1
53:51	CH7_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH7_DMS coreConsultant parameter. 0x0 = MASTER_1
50:48	CH7_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH7_MULT_SIZE coreConsultant parameter. 0x2 = 16
47:46	CH7_FC	2	R	0x0	The value of this register is derived from the DMAH_CH7_FC coreConsultant parameter. 0x0 = DMA
45	CH7_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH7_HC_LLP coreConsultant parameter. 0x0 = FALSE
44	CH7_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH7_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
43	CH7_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH7_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE

42	CH7_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH7_LOCK_EN coreConsultant parameter. 0x0 = FALSE
41	CH7_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH7_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE
40	CH7_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH7_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
39	CH7_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH7_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
38	CH7_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH7_STAT_DST coreConsultant parameter. 0x0 = FALSE
37:35	CH7_STW	3	R	0x0	The value of this register is derived from the DMAH_CH7_STW coreConsultant parameter. 0x0 = NO_HARDCODE
34:32	CH7_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH7_DTW coreConsultant parameter. 0x0 = NO_HARDCODE
31:0	Reserved				

66.2.4.23. **Component Parameters Register 5**

DMA_COMP_PARAMS_5 (Addr: 0x3D0)

Bit Field	Name	Width	R/W	Default value	
63	Reserved				
62:60	CH5_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH5_FO_DEPTH coreConsultant parameter. 0x3 = 64

59:57	CH5_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH5_SMS coreConsultant parameter. 0x0 = MASTER_1
56:54	CH5_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH5_LMS coreConsultant parameter. 0x0 = MASTER_1
53:51	CH5_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH5_DMS coreConsultant parameter. 0x0 = MASTER_1
50:48	CH5_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH5_MULT_SIZE coreConsultant parameter. 0x2 = 16
47:46	CH5_FC	2	R	0x0	The value of this register is derived from the DMAH_CH5_FC coreConsultant parameter. 0x0 = DMA
45	CH5_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH5_HC_LLP coreConsultant parameter. 0x0 = FALSE
44	CH5_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH5_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
43	CH5_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH5_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE
42	CH5_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH5_LOCK_EN coreConsultant parameter. 0x0 = FALSE
41	CH5_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH5_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE

40	CH5_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH5_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
39	CH5_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH5_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
38	CH5_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH5_STAT_DST coreConsultant parameter. 0x0 = FALSE
37:35	CH5_STW	3	R	0x0	The value of this register is derived from the DMAH_CH5_STW coreConsultant parameter. 0x0 = NO_HARDCODE
34:32	CH5_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH5_DTW coreConsultant parameter. 0x0 = NO_HARDCODE
31	Reserved				
30:28	CH6_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH6_FO_DEPTH coreConsultant parameter. 0x3 = 64
27:25	CH6_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH6_SMS coreConsultant parameter. 0x0 = MASTER_1
24:22	CH6_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH6_LMS coreConsultant parameter. 0x0 = MASTER_1
21:19	CH6_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH6_DMS coreConsultant parameter. 0x0 = MASTER_1
18:16	CH6_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH6_MULT_SIZE coreConsultant parameter. 0x2 = 16

15:14	CH6_FC	2	R	0x0	The value of this register is derived from the DMAH_CH6_FC coreConsultant parameter. 0x0 = DMA
13	CH6_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH6_HC_LLP coreConsultant parameter. 0x0 = FALSE
12	CH6_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH6_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
11	CH6_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH6_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE
10	CH6_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH6_LOCK_EN coreConsultant parameter. 0x0 = FALSE
9	CH6_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH6_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE
8	CH6_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH6_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
7	CH6_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH6_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
6	CH6_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH6_STAT_DST coreConsultant parameter. 0x0 = FALSE
5:3	CH6_STW	3	R	0x0	The value of this register is derived from the DMAH_CH6_STW coreConsultant parameter. 0x0 = NO_HARDCODE

2:0	CH6_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH6_DTW coreConsultant parameter. 0x0 = NO_HARDCODE

66.2.4.24. **Component Parameters Register 4**

DMA_COMP_PARAMS_4 (Addr: 0x3D8)

Bit Field	**Name**	**Width**	**R/W**	**Default value**	
63	Reserved				
62:60	CH3_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH3_FO_DEPTH coreConsultant parameter. 0x3 = 64
59:57	CH3_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH3_SMS coreConsultant parameter. 0x0 = MASTER_1
56:54	CH3_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH3_LMS coreConsultant parameter. 0x0 = MASTER_1
53:51	CH3_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH3_DMS coreConsultant parameter. 0x0 = MASTER_1
50:48	CH3_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH3_MULT_SIZE coreConsultant parameter. 0x2 = 16
47:46	CH3_FC	2	R	0x0	The value of this register is derived from the DMAH_CH3_FC coreConsultant parameter. 0x0 = DMA
45	CH3_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH3_HC_LLP coreConsultant parameter. 0x0 = FALSE
44	CH3_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH3_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE

43	CH3_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH3_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE
42	CH3_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH3_LOCK_EN coreConsultant parameter. 0x0 = FALSE
41	CH3_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH3_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE
40	CH3_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH3_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
39	CH3_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH3_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
38	CH3_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH3_STAT_DST coreConsultant parameter. 0x0 = FALSE
37:35	CH3_STW	3	R	0x0	The value of this register is derived from the DMAH_CH3_STW coreConsultant parameter. 0x0 = NO_HARDCODE
34:32	CH3_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH3_DTW coreConsultant parameter. 0x0 = NO_HARDCODE
31	Reserved				
30:28	CH4_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH4_FO_DEPTH coreConsultant parameter. 0x3 = 64
27:25	CH4_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH4_SMS coreConsultant parameter.

					0x0 = MASTER_1
24:22	CH4_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH4_LMS coreConsultant parameter. 0x0 = MASTER_1
21:19	CH4_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH4_DMS coreConsultant parameter. 0x0 = MASTER_1
18:16	CH4_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH4_MULT_SIZE coreConsultant parameter. 0x2 = 16
15:14	CH4_FC	2	R	0x0	The value of this register is derived from the DMAH_CH4_FC coreConsultant parameter. 0x0 = DMA
13	CH4_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH4_HC_LLP coreConsultant parameter. 0x0 = FALSE
12	CH4_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH4_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
11	CH4_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH4_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE
10	CH4_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH4_LOCK_EN coreConsultant parameter. 0x0 = FALSE
9	CH4_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH4_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE
8	CH4_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH4_DST_SCA_EN

					coreConsultant parameter. 0x1 = TRUE
7	CH4_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH4_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
6	CH4_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH4_STAT_DST coreConsultant parameter. 0x0 = FALSE
5:3	CH4_STW	3	R	0x0	The value of this register is derived from the DMAH_CH4_STW coreConsultant parameter. 0x0 = NO_HARDCODE
2:0	CH4_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH4_DTW coreConsultant parameter. 0x0 = NO_HARDCODE

66.2.4.25. **Component Parameters Register 3**

DMA_COMP_PARAMS_3 (Addr: 0x3E0)

Bit Field	Name	Width	R/W	Default value	
63	Reserved				
62:60	CH1_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH1_FO_DEPTH coreConsultant parameter. 0x3 = 64
59:57	CH1_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH1_SMS coreConsultant parameter. 0x0 = MASTER_1
56:54	CH1_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH1_LMS coreConsultant parameter. 0x0 = MASTER_1
53:51	CH1_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH1_DMS coreConsultant parameter. 0x0 = MASTER_1

50:48	CH1_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH1_MULT_SIZE coreConsultant parameter. 0x2 = 16
47:46	CH1_FC	2	R	0x0	The value of this register is derived from the DMAH_CH1_FC coreConsultant parameter. 0x0 = DMA
45	CH1_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH1_HC_LLP coreConsultant parameter. 0x0 = FALSE
44	CH1_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH1_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
43	CH1_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH1_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE
42	CH1_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH1_LOCK_EN coreConsultant parameter. 0x0 = FALSE
41	CH1_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH1_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE
40	CH1_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH1_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
39	CH1_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH1_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
38	CH1_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH1_STAT_DST coreConsultant parameter. 0x0 = FALSE

37:35	CH1_STW	3	R	0x0	The value of this register is derived from the DMAH_CH1_STW coreConsultant parameter. 0x0 = NO_HARDCODE
34:32	CH1_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH1_DTW coreConsultant parameter. 0x0 = NO_HARDCODE
31	Reserved				
f	CH2_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH2_FO_DEPTH coreConsultant parameter. 0x3 = 64
27:25	CH2_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH2_SMS coreConsultant parameter. 0x0 = MASTER_1
24:22	CH2_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH2_LMS coreConsultant parameter. 0x0 = MASTER_1
21:19	CH2_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH2_DMS coreConsultant parameter. 0x0 = MASTER_1
18:16	CH2_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH2_MULT_SIZE coreConsultant parameter. 0x2 = 16
15:14	CH2_FC	2	R	0x0	The value of this register is derived from the DMAH_CH2_FC coreConsultant parameter. 0x0 = DMA
13	CH2_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH2_HC_LLP coreConsultant parameter. 0x0 = FALSE
12	CH2_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH2_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
11	CH2_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH2_MULTI_BLK_EN

					coreConsultant parameter. 0x1 = TRUE
10	CH2_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH2_LOCK_EN coreConsultant parameter. 0x0 = FALSE
9	CH2_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH2_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE
8	CH2_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH2_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
7	CH2_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH2_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
6	CH2_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH2_STAT_DST coreConsultant parameter. 0x0 = FALSE
5:3	CH2_STW	3	R	0x0	The value of this register is derived from the DMAH_CH2_STW coreConsultant parameter. 0x0 = NO_HARDCODE
2:0	CH2_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH2_DTW coreConsultant parameter. 0x0 = NO_HARDCODE

66.2.4.26. Component Parameters Register 2

DMA_COMP_PARAMS_2 (Addr: 0x3E8)

Bit Field	Name	Width	R/W	Default value	
63:60	CH7_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH7_MULTI_BLK_TYPE

					coreConsultant parameter. 0x0 = NO_HARDCODE
59:56	CH6_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH6_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
55:52	CH5_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH5_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
51:48	CH4_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH4_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
47:44	CH3_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH3_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
43:40	CH2_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH2_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
39:36	CH1_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH1_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
35:32	CH0_MULTI_BLK_TYP	4	R	0x0	The values of these bit fields are derived from the DMAH_CH0_MULTI_BLK_TYPE coreConsultant parameter. 0x0 = NO_HARDCODE
31	Reserved				
30:28	CH0_FO_DEPTH	3	R	0x3	The value of this register is derived from the DMAH_CH0_FO_DEPTH coreConsultant parameter. 0x3 = 64

27:25	CH0_SMS	3	R	0x0	The value of this register is derived from the DMAH_CH0_SMS coreConsultant parameter. 0x0 = MASTER_1
24:22	CH0_LMS	3	R	0x0	The value of this register is derived from the DMAH_CH0_LMS coreConsultant parameter. 0x0 = MASTER_1
21:19	CH0_DMS	3	R	0x0	The value of this register is derived from the DMAH_CH0_DMS coreConsultant parameter. 0x0 = MASTER_1
18:16	CH0_MAX_MULT_SIZE	3	R	0x2	The value of this register is derived from the DMAH_CH0_MULT_SIZE coreConsultant parameter. 0x2 = 16
15:14	CH0_FC	2	R	0x0	The value of this register is derived from the DMAH_CH0_FC coreConsultant parameter. 0x0 = DMA
13	CH0_HC_LLP	1	R	0x0	The value of this register is derived from the DMAH_CH0_HC_LLP coreConsultant parameter. 0x0 = FALSE
12	CH0_CTL_WB_EN	1	R	0x1	The value of this register is derived from the DMAH_CH0_CTL_WB_EN coreConsultant parameter. 0x1 = TRUE
11	CH0_MULTI_BLK_EN	1	R	0x1	The value of this register is derived from the DMAH_CH0_MULTI_BLK_EN coreConsultant parameter. 0x1 = TRUE
10	CH0_LOCK_EN	1	R	0x0	The value of this register is derived from the DMAH_CH0_LOCK_EN coreConsultant parameter. 0x0 = FALSE
9	CH0_SRC_GAT_EN	1	R	0x1	The value of this register is derived from the DMAH_CH0_SRC_GAT_EN coreConsultant parameter. 0x1 = TRUE

8	CH0_DST_SCA_EN	1	R	0x1	The value of this register is derived from the DMAH_CH0_DST_SCA_EN coreConsultant parameter. 0x1 = TRUE
7	CH0_STAT_SRC	1	R	0x0	The value of this register is derived from the DMAH_CH0_DST_SCA_EN coreConsultant parameter. 0x0 = FALSE
6	CH0_STAT_DST	1	R	0x0	The value of this register is derived from the DMAH_CH0_STAT_DST coreConsultant parameter. 0x0 = FALSE
5:3	CH0_STW	3	R	0x0	The value of this register is derived from the DMAH_CH0_STW coreConsultant parameter. 0x0 = NO_HARDCODE
2:0	CH0_DTW	3	R	0x0	The value of this register is derived from the DMAH_CH0_DTW coreConsultant parameter. 0x0 = NO_HARDCODE

66.2.4.27. **Component Parameters Register 1**

DMA_COMP_PARAMS_1 (Addr: 0x3F0)

Bit Field	**Name**	**Width**	**R/W**	**Default value**	
63:62	Reserved				
61	STATIC_ENDIAN_SELECT	1	R	0x1	The value of this register is derived from the DMAH_STATIC_ENDIAN_SELECT coreConsultant parameter. 1 = TRUE
60	ADD_ENCODED_PARAMS	1	R	0x1	The value of this register is derived from the DMAH_ADD_ENCODED_PARAMS coreConsultant parameter. 1 = TRUE
59:55	NUM_HS_INT	5	R	0x10	The value of this register is derived from the DMAH_NUM_HS_INT coreConsultant parameter. 0x10 = 16

54:53	M1_HDATA_WIDTH	2	R	0x0	The value of this register is derived from the DMAH_M1_HDATA_WIDTH coreConsultant parameter. 0x0 = 32
52:51	M2_HDATA_WIDTH	2	R	0x0	Master port 2 is not existed. Not applied.
50:49	M3_HDATA_WIDTH	2	R	0x0	Master port 3 is not existed. Not applied.
48:47	M4_HDATA_WIDTH	2	R	0x0	Master port 4 is not existed. Not applied.
46:45	S_HDATA_WIDTH	2	R	0x0	The value of this register is derived from the DMAH_S_HDATA_WIDTH coreConsultant parameter. 0x0 = 32
44:43	NUM_MASTER_INT	2	R	0x0	The value of this register is derived from the DMAH_NUM_MASTER_INT coreConsultant parameter 0x0 = 1
42:40	NUM_CHANNELS	3	R	0x7	The value of this register is derived from the DMAH_NUM_CHANNELS core Consultant parameter. 0x7 = 8
39:36	Reserved				
35	MABRST	1	R	0x0	The value of this register is derived from the DMAH_MABRST core Consultant parameter. 0 = FALSE
34:33	INTR_IO	2	R	0x2	The value of this register is derived from the DMAH_INTR_IO core Consultant parameter. 0x2 = COMBINED
32	BIG_ENDIAN	1	R	0x0	The value of this register is derived from the DMAH_BIG_ENDIAN coreConsultant parameter. 0 = FALSE
31:28	CH7_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH7_MAX_BLK_SIZEE core Consultant parameter. 0xa = 4095

27:24	CH6_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH6_MAX_BLK_SIZEE core Consultant parameter. 0xa = 4095
23:20	CH5_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH5_MAX_BLK_SIZEE coreConsultant parameter. 0xa = 4095
19:16	CH4_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH4_MAX_BLK_SIZEE coreConsultant parameter. 0xa = 4095
15:12	CH3_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH3_MAX_BLK_SIZEE coreConsultant parameter. 0xa = 4095
11:8	CH2_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH2_MAX_BLK_SIZEE coreConsultant parameter. 0xa = 4095
7:4	CH1_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH1_MAX_BLK_SIZEE coreConsultant parameter. 0xa = 4095
3:0	CH0_MAX_BLK_SIZE	4	R	0xa	The values of these bit fields are derived from the DMAH_CH0_MAX_BLK_SIZEE coreConsultant parameter. 0xa = 4095

66.2.4.28. Component Version Register

DMA Component ID Register (Addr: 0x3F8)

Bit Field	Name	Width	R/W	Default value	
63:32	DMA_COMP_VERSION	32	R	0x0	Version of the component

31:0	DMA_COMP_TYPE	32	R	0x44571110	Designware Component Type number = 0x44_57_11_10. This assigned unique hex value is constant and is derived from the two ASCII letters "DW" followed by a 32-bit unsigned number.

66.2.5. **Programming Guide**

66.2.5.1. **Application Notes**

In this system, the following peripherals data transfer may depend on DMA. Each channel can select destination peripherals or source peripherals by configuration. When the specified transfer has been finished, the DMAC can be specified to generate the interrupt to CPU.The DMAC is always running on the same frequency of the system bus.DMA controller support hardware handshaking interface for peripheral, no handshaking interface for memory.

66.2.5.2. **Programming Examples**

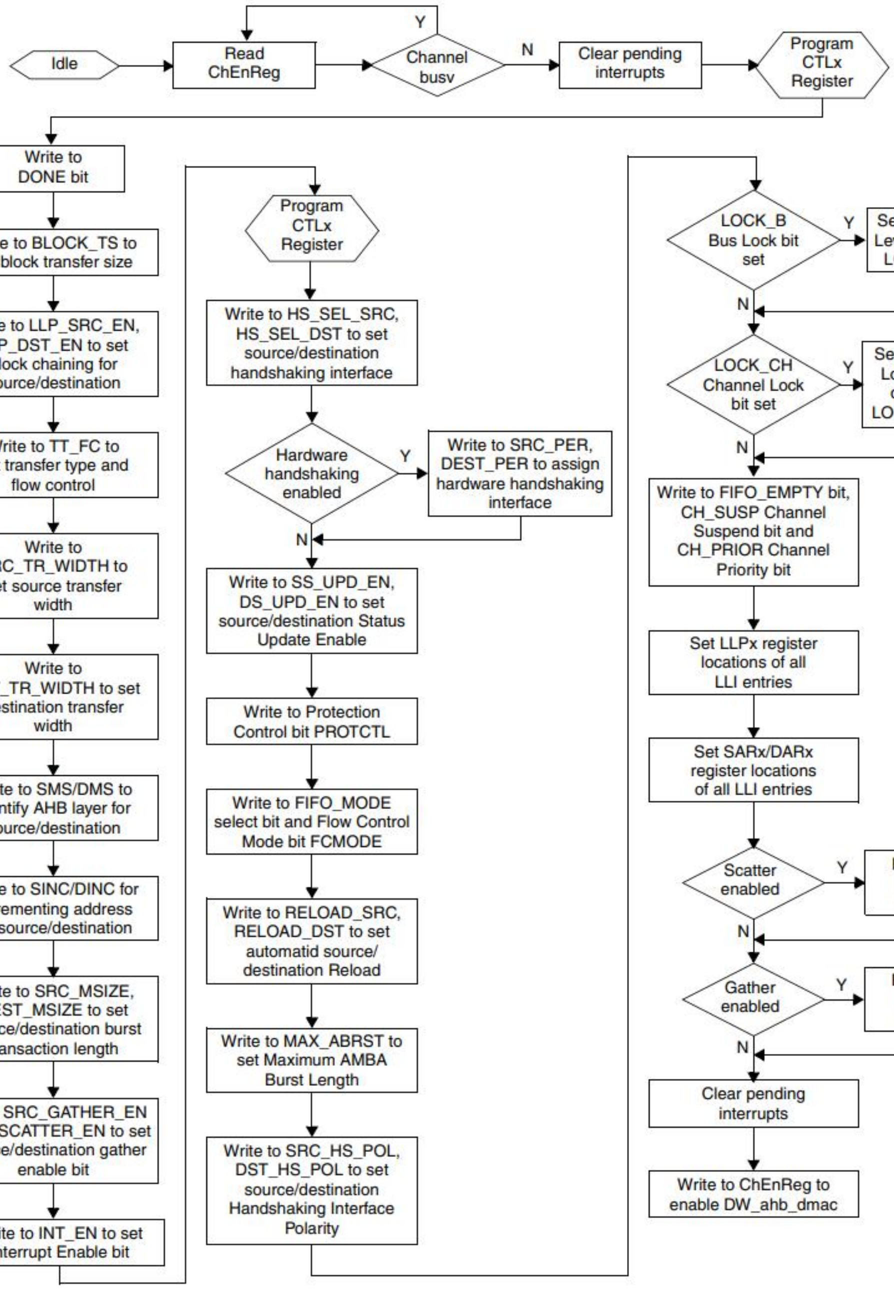


Figure 4- 5 Flowchart for DMA Programming Example

66.2.5.2.1. Programming Example for Linked List Multi-Block Transfer

This section explains the step-by-step programming of the AHB_DMAC. The example demonstrates row 10 of Table 8-1 on page 249 for Multi-Block Transfer with Linked List for Source and Linked List for Destination. This example uses the AHB_DMAC to move four blocks of contiguous data from source to destination memory using the Linked List feature.

1. Set up the chain of Linked List items—otherwise known as block descriptors—in memory.

Write the control information in the LLI.CTLx register location of the block descriptor for each LLI in memory for Channel 1. In the LLI.CTLx register, the following is programmed:

a. Set up the transfer type for a memory-to-memory transfer: - ctlx[22:20] = 3'b000;

b. Set up the transfer characteristics:

i. Transfer width for the source in the SRC_TR_WIDTH field - ctlx[6:4] = 3'b001;

ii. Transfer width for the destination in the DST_TR_WIDTH field - ctlx [3:1] = 3'b001;

iii. Source master layer in the SMS field where the source resides - ctlx[26:25] = 2'b00;

iv. Destination master layer in the DMS field where the destination resides - ctlx[24:23] = 2'b00;

v. Incrementing address for the source in the SINC field - ctlx[10:9] = 2'b00;

vi. Incrementing address for the destination in the DINC field - ctlx[8:7] = 2'b00;

2. Write the channel configuration information into the CFGx register for Channel 1:

a. HS_SEL_SRC/HS_SEL_DST bits select which of the handshaking interfaces—hardware or software—is active for source requests on this channel. - cfgx[11] = 1'b0; - cfgx[10] = 1'b0; These settings are ignored because both the source and destination are memory types.

b. If the hardware handshaking interface is activated for the source or destination peripheral, assign the handshaking interface to the source and destination peripheral by programming the SRC_PER and DEST_PER bits: - cfgx[46:43] = 1'b0; - cfgx[42:39] = 1'b0; These settings are ignored because both the source and destination are memory types.

3. The following For loop, shown as a programming example, sets the following: ❑ LLI.LLPx register locations of all LLI entries in memory (except the last) to non-zero and point to the base address of the next Linked List Item ❑ LLI.SARx/LLI.DARx register locations of all LLI entries in memory point to the start source/destination block address preceding that LLI fetch,

The For statement below configures the LLPx entries:

The For statement below configures the LLPx entries:

```
for(i=0 ; i < 4 ; i=i+1) begin
   if (i == 3)
   llpx = 0;   // end of LLI
else
   llpx = llp_addr + 20; // start of next LLI

 //-: Program SAR
 `AHB_MASTER.write(0, llp_addr, sarx, AhbWord32Attrb, handle[0]);
 //-: Program DAR
 `AHB_MASTER.write(0, (llp_addr + 4), darx, AhbWord32Attrb, handle[0]);

 //-: Program LLP
 `AHB_MASTER.write(0, (llp_addr + 8), llpx, AhbWord32Attrb, handle[0]);

 //-: Program CTL
 `AHB_MASTER.write(0, (llp_addr + 12),  ctlx[31:0], AhbWord32Attrb, handle[0]);
 `AHB_MASTER.write(0, (llp_addr + 16), ctlx[63:32], AhbWord32Attrb, handle[0]);

 // update pointers
 llp_addr = llp_addr + 20; // start of next LLI

 // 4  16-bit words each with scatter/gather interval in each block
 // ( will work only with scatter_gather count of 2)
 sarx = sarx + 24;
 darx = darx + 24;

end
```

4. If Gather is enabled—DMAH_CHx_SRC_GAT_EN = True and CTLx.SRC_GATHER_EN is enabled— program the SGRx register for Channel 1.

5. If Scatter is enabled—DMAH_CHx_DST_SCA_EN = True and CTLx.DST_SCATTER_EN is enabled—program the DSRx register for Channel 1.

6. Clear any pending interrupts on the channel from the previous DMA transfer by writing to the Interrupt Clear registers.

7. Finally, enable the channel by writing a 1 to the ChEnReg.CH_EN bit; the transfer is performed.

66.3. Timer

66.3.1. Overview

Timer module has following features:

- 12 identical but independent programmable 32-bit timers. Each timer can be configured to count up or count down
- Dedicated INT signal output from each timer
- Independent trigger input for timers work in capture mode, it can be multiplexed with GPIO ports
- PWM output from each timer, can be multiplexed with GPIO ports
- PWM output can be pulsed or toggled on overflow or/and match event
- Use APB Clock as functional clock source for all timers, support configurable resolution of timers with pre-scalar circuit by AMBA APB interface.
- Full synchronous R/W access to all registers through APB bus

66.3.2. Functional

66.3.2.1. Operation mode

The general-purpose timer supports 3 functional modes:

- Timer mode

The timer is an upward or downward counter that can be started and stopped at any time. The Timer Counter Register can be loaded when stopped or on the fly(while counting) by directly writing the timer counter register with new timer value or loaded with the value held in the Timer Load Value register by a trigger register write access.

The timer counter register value can be read at any time by a read access.

In the one shot mode, the counter is stopped after counting overflow (counter value remains at zero).

In the auto-reload mode is enabled, the Timer Load Register held value is reloaded after a counting

overflow.
An interrupt can be issued on overflow the overflow interrupt enabled.PWM output is programmed to generate one positive pulse (prescaler duration) or to invert the current value (toggle mode) when an overflow occurs.
In free running mode, counter automatically wraps to 0x0 (count up mode) or 0xFFFFFFFF (count-down mode) after counting overflow.

- Capture mode
  The timer value can be captured and saved when a transition is detected on the module input pin. The module sets the corresponding interrupt flag when an active transition (rising, falling or both) is detected and at the same time the counter value is stored in one of the timer capture registers, TMR_CAP1_VALUE or TMR_CAP2_VALUE.

  Once captured, the following active transitions are ignored till the interrupt flag TMR_CAP_INT_STA is cleared.

- Compare mode
  When Compare mode is enabled, the timer value is permanently compared to the value held in timer match register. The dedicated PWM output pin can be programmed to generate one positive pulse (TIMER clock duration) or to invert the current value (toggle mode) when match event occur. Interrupt IRQ can be issued on match enabled.

66.3.2.2. Clock Prescaler

Prescaler is clocked with the timer clock input and acts as a clock divider for the timer counter. The Timer resolution and interrupt period are dependent on the selected input clock and clock prescale divisor value.

66.3.2.3. PWM output

The timer can be configured to provide a programmable pulse-width modulation output which toggles on specied event. Either overflow or match condition or both can be used to toggle the PWM output pin. In pulse mode, PWM outputs positive pulse for one timer clock cycle for each trigger event. In toggle mode, PWM port can generate Pulse-Width Modulated signals of varying frequency and duty cycle.

66.3.2.4. APB interface

All registers are 32-bit wide, via AMBA-APB interface with 32-bit access (Read/Write). The

Timer function part is also clocked by APB Clock, which enables full synchronous register access by APB bus.

66.3.2.5. Interrupt outputs

One interrupt request per timer module outputs to CPU. It can be configured to assert on occurrence of timer overflow, compare value match or external triggered capture event.

66.3.3. Registers

66.3.3.1. Register Summary

Address space allocated for GTIMER:

Table 4- 3 Timer Address Range

	Address Range
Timer Address Range	0x9100_6000~0x9100_6FFF

Table 4- 4 Timer Register Summary

Name	Address	Description
TMR1_LOAD	0x000	Timer 1 load trigger
TMR1_LOAD_VALUE	0x004	Timer 1 load value
TMR1_COUNT	0x008	Timer 1 current value
TMR1_ CAP1_VALUE	0x00C	Timer 1 captured value
TMR1_ CAP2_VALUE	0x010	Timer 1 captured value
TMR1_COMP_VALUE	0x014	Timer 1 match value
TMR1_CONTROL	0x018	Timer 1 control register
TMR1_IRQ_STA	0x01C	Timer 1 interrupt request status
TMR1_IRQ_RAW	0x020	Timer 1 interrupt request raw status
TMR1_IRQ_EN	0x024	Timer 1 interrupt request enable
TMR1_SYNC_COUNT_OFST	0x028	Timer1 counter offset for synchronization
TMR2_LOAD	0x040	Timer 2 load trigger
TMR2_LOAD_VALUE	0x044	Timer 2 load value
TMR2_COUNT	0x048	Timer 2 current value
TMR2_ CAP1_VALUE	0x04C	Timer 2 captured value
TMR2_ CAP2_VALUE	0x050	Timer 2 captured value
TMR2_COMP_VALUE	0x054	Timer 2 match value
TMR2_CONTROL	0x058	Timer 2 control register
TMR2_IRQ_STA	0x05C	Timer 2 interrupt request status
TMR2_IRQ_RAW	0x060	Timer 2 interrupt request raw status

Name	Address	Description
TMR2_IRQ_EN	0x064	Timer 2 interrupt request enable
TMR2_SYNC_COUNT_OFST	0x068	Timer 2 counter offset for synchronization
TMR3_LOAD	0x080	Timer 3 load trigger
TMR3_LOAD_VALUE	0x084	Timer 3 load value
TMR3_COUNT	0x088	Timer 3 current value
TMR3_ CAP1_VALUE	0x08C	Timer 3 captured value
TMR3_ CAP2_VALUE	0x090	Timer 3 captured value
TMR3_COMP_VALUE	0x094	Timer 3 match value
TMR3_CONTROL	0x098	Timer 3 control register
TMR3_IRQ_STA	0x09C	Timer 3 interrupt request status
TMR3_IRQ_RAW	0x0A0	Timer 3 interrupt request raw status
TMR3_IRQ_EN	0x0A4	Timer 3 interrupt request enable
TMR3_SYNC_COUNT_OFST	0xA8	Timer 3 counter offset for synchronization
TMR4_LOAD	0x0C0	Timer 4 load trigger
TMR4_LOAD_VALUE	0x0C4	Timer 4 load value
TMR4_COUNT	0x0C8	Timer 4 current value
TMR4_ CAP1_VALUE	0x0CC	Timer 4 captured value
TMR4_ CAP2_VALUE	0x0D0	Timer 4 captured value
TMR4_COMP_VALUE	0x0D4	Timer 4 match value
TMR4_CONTROL	0x0D8	Timer 4 control register
TMR4_IRQ_STA	0x0DC	Timer 4 interrupt request status
TMR4_IRQ_RAW	0x0E0	Timer 4 interrupt request raw status
TMR4_IRQ_EN	0x0E4	Timer 4 interrupt request enable
TMR4_SYNC_COUNT_OFST	0xE8	Timer 4 counter offset for synchronization
TMR5_LOAD	0x100	Timer 5 load trigger
TMR5_LOAD_VALUE	0x104	Timer 5 load value
TMR5_COUNT	0x108	Timer 5 current value
TMR5_ CAP1_VALUE	0x10C	Timer 5 captured value
TMR5_ CAP2_VALUE	0x110	Timer 5 captured value
TMR5_COMP_VALUE	0x114	Timer 5 match value
TMR5_CONTROL	0x118	Timer 5 control register
TMR5_IRQ_STA	0x11C	Timer 5 interrupt request status
TMR5_IRQ_RAW	0x120	Timer 5 interrupt request raw status
TMR5_IRQ_EN	0x124	Timer 5 interrupt request enable
TMR5_SYNC_COUNT_OFST	0x128	Timer 5 counter offset for synchronization
TMR6_LOAD	0x140	Timer 6 load trigger
TMR6_LOAD_VALUE	0x144	Timer 6 load value
TMR6_COUNT	0x148	Timer 6 current value
TMR6_ CAP1_VALUE	0x14C	Timer 6 captured value
TMR6_ CAP2_VALUE	0x150	Timer 6 captured value
TMR6_COMP_VALUE	0x154	Timer 6 match value
TMR6_CONTROL	0x158	Timer 6 control register

Name	Address	Description
TMR6_IRQ_STA	0x15C	Timer 6 interrupt request status
TMR6_IRQ_RAW	0x160	Timer 6 interrupt request raw status
TMR6_IRQ_EN	0x164	Timer 6 interrupt request enable
TMR6_SYNC_COUNT_OFST	0x168	Timer 6 counter offset for synchronization
TMR7_LOAD	0x180	Timer 7 load trigger
TMR7_LOAD_VALUE	0x184	Timer 7 load value
TMR7_COUNT	0x188	Timer 7 current value
TMR7_ CAP1_VALUE	0x18C	Timer 7 captured value
TMR7_ CAP2_VALUE	0x190	Timer 7 captured value
TMR7_COMP_VALUE	0x194	Timer 7 match value
TMR7_CONTROL	0x198	Timer 7 control register
TMR7_IRQ_STA	0x19C	Timer 7 interrupt request status
TMR7_IRQ_RAW	0x1A0	Timer 7 interrupt request raw status
TMR7_IRQ_EN	0x1A4	Timer 7 interrupt request enable
TMR7_SYNC_COUNT_OFST	0x1a8	Timer 7 counter offset for synchronization
TMR8_LOAD	0x1C0	Timer 8 load trigger
TMR8_LOAD_VALUE	0x1C4	Timer 8 load value
TMR8_COUNT	0x1C8	Timer 8 current value
TMR8_ CAP1_VALUE	0x1CC	Timer 8 captured value
TMR8_ CAP2_VALUE	0x1D0	Timer 8 captured value
TMR8_COMP_VALUE	0x1D4	Timer 8 match value
TMR8_CONTROL	0x1D8	Timer 8 control register
TMR8_IRQ_STA	0x1DC	Timer 8 interrupt request status
TMR8_IRQ_RAW	0x1E0	Timer 8 interrupt request raw status
TMR8_IRQ_EN	0x1E4	Timer 8 interrupt request enable
TMR8_SYNC_COUNT_OFST	0x1E8	Timer 8 counter offset for synchronization
TMR9_LOAD	0x200	Timer 9 load trigger
TMR9_LOAD_VALUE	0x204	Timer 9 load value
TMR9_COUNT	0x208	Timer 9 current value
TMR9_ CAP1_VALUE	0x20C	Timer 9 captured value
TMR9_ CAP2_VALUE	0x210	Timer 9 captured value
TMR9_COMP_VALUE	0x214	Timer 9 match value
TMR9_CONTROL	0x218	Timer 9 control register
TMR9_IRQ_STA	0x21C	Timer 9 interrupt request status
TMR9_IRQ_RAW	0x220	Timer 9 interrupt request raw status
TMR9_IRQ_EN	0x224	Timer 9 interrupt request enable
TMR9_SYNC_COUNT_OFST	0x240	Timer 9 counter offset for synchronization
TMR10_LOAD	0x244	Timer 10 load trigger
TMR10_LOAD_VALUE	0x248	Timer 10 load value
TMR10_COUNT	0x24C	Timer 10 current value
TMR10_ CAP1_VALUE	0x250	Timer 10 captured value
TMR10_ CAP2_VALUE	0x254	Timer 10 captured value

Name	Address	Description
TMR10_COMP_VALUE	0x258	Timer 10 match value
TMR10_CONTROL	0x25C	Timer 10 control register
TMR10_IRQ_STA	0x260	Timer 10 interrupt request status
TMR10_IRQ_RAW	0x264	Timer 10 interrupt request raw status
TMR10_IRQ_EN	0x268	Timer 10 interrupt request enable
TMR10_SYNC_COUNT_OFST	0x240	Timer 10 counter offset for synchronization
TMR11_LOAD	0x280	Timer 11 load trigger
TMR11_LOAD_VALUE	0x284	Timer 11 load value
TMR11_COUNT	0x288	Timer 11 current value
TMR11_ CAP1_VALUE	0x28C	Timer 11 captured value
TMR11_ CAP2_VALUE	0x290	Timer 11 captured value
TMR11_COMP_VALUE	0x294	Timer 11 match value
TMR11_CONTROL	0x298	Timer 11 control register
TMR11_IRQ_STA	0x29C	Timer 11 interrupt request status
TMR11_IRQ_RAW	0x2A0	Timer 11 interrupt request raw status
TMR11_IRQ_EN	0x2A4	Timer 11 interrupt request enable
TMR11_SYNC_COUNT_OFST	0x2a8	Timer 11 counter offset for synchronization
TMR12_LOAD	0x2C0	Timer 12 load trigger
TMR12_LOAD_VALUE	0x2C4	Timer 12 load value
TMR12_COUNT	0x2C8	Timer 12 current value
TMR12_ CAP1_VALUE	0x2CC	Timer 12 captured value
TMR12_ CAP2_VALUE	0x2D0	Timer 12 captured value
TMR12_COMP_VALUE	0x2D4	Timer 12 match value
TMR12_CONTROL	0x2D8	Timer 12 control register
TMR12_IRQ_STA	0x2DC	Timer 12 interrupt request status
TMR12_IRQ_RAW	0x2E0	Timer 12 interrupt request raw status
TMR12_IRQ_EN	0x2E4	Timer 12 interrupt request enable
TMR12_SYNC_COUNT_OFST	0x2E8	Timer 12 counter offset for synchronization
TMR_SYNC_EN	0x280	Timer counting synchronization enable

66.3.3.2. **Register Description**

- Timer N Load Trigger

  TMR1_LOAD,
  TMR2_LOAD,
  TMR3_LOAD,
  TMR4_LOAD,
  TMR5_LOAD,
  TMR6_LOAD,

TMR7_LOAD,
TMR8_LOAD,
TMR9_LOAD,
TMR10_LOAD,
TMR11_LOAD,
TMR12_LOAD

Table 4- 5 Timer N Load Trigger

	Name	R/W	Reset	
31:1	Reserved			
0	TMR_LOAD	W	0x00	Write to trigger load TMRn_LOAD_VALUE into timer n and clear clock pre-scale counter regardless of Auto-reload setting. 0x1: Load 0x0: No action

- Timer N Load Value

TMR1_LOAD_VALUE,
TMR2_LOAD_VALUE,
TMR3_LOAD_VALUE,
TMR4_LOAD_VALUE,
TMR5_LOAD_VALUE,
TMR6_LOAD_VALUE,
TMR7_LOAD_VALUE,
TMR8_LOAD_VALUE,
TMR9_LOAD_VALUE,
TMR10_LOAD_VALUE,
TMR11_LOAD_VALUE,
TMR12_LOAD_VALUE

Table 4- 6 Timer N Load Value

	Name	R/W	Reset	
31:0	TMR_LOAD_VALUE	R/W	0x00	Value to be loaded into Timer N. This value is loaded on overflow in auto-load mode

- Timer N Current Value

TMR1_COUNT,
TMR2_COUNT,
TMR3_COUNT,
TMR4_COUNT,
TMR5_COUNT,
TMR6_COUNT,

TMR7_COUNT,
TMR8_COUNT,
TMR9_COUNT,
TMR10_COUNT,
TMR11_COUNT,
TMR12_COUNT

The count value can be updated by:
Reset to 0 by software reset TMR_RST
APB directly write to TMRn_COUNT
Reload triggered by TMRn_LOAD
Reload upon overflow
Normal increment or decrement

Table 4- 7 Timer N Current Value

	Name	R/W	Reset	
31:0	TMR_COUNT	R/W	0x00	Current value of Timer N counter. Read access returns current 32-bit value Write access overrides the counter value immediately

- Timer N Captured Value

TMR1_CAP1_VALUE,
TMR2_CAP1_VALUE,
TMR3_CAP1_VALUE,
TMR4_CAP1_VALUE,
TMR5_CAP1_VALUE,
TMR6_CAP1_VALUE,
TMR7_CAP1_VALUE,
TMR8_CAP1_VALUE,
TMR9_CAP1_VALUE,
TMR10_CAP1_VALUE,
TMR11_CAP1_VALUE,
TMR12_CAP1_VALUE

Table 4- 8 Timer N Captured Value

	Name	R/W	Reset	
31:0	TMRn_CAP1_VALUE	R	0x00	Timer N counter value captured on external event trigger for single capture mode

- Timer N 2nd Captured Value

TMR1_CAP2_VALUE,
TMR2_CAP2_VALUE,
TMR3_CAP2_VALUE,

TMR4_CAP2_VALUE,
TMR5_CAP2_VALUE,
TMR6_CAP2_VALUE,
TMR7_CAP2_VALUE,
TMR8_CAP2_VALUE,
TMR9_CAP2_VALUE,
TMR10_CAP2_VALUE,
TMR11_CAP2_VALUE,
TMR12_CAP2_VALUE

Table 4- 9 Timer N 2nd Captured Value

	Name	R/W	Reset	
31:0	TMR_CAP2_VALUE	R	0x00	Timer N counter value captured on external event trigger for second capture mode

- Timer N Match Value

TMR1_COMP_VALUE,
TMR2_COMP_VALUE,
TMR3_COMP_VALUE,
TMR4_COMP_VALUE,
TMR5_COMP_VALUE,
TMR6_COMP_VALUE,
TMR7_COMP_VALUE,
TMR8_COMP_VALUE,
TMR9_COMP_VALUE,
TMR10_COMP_VALUE,
TMR11_COMP_VALUE,
TMR12_COMP_VALUE

Table 4- 10 Timer N Match Value

	Name	R/W	Reset	
31:0	TMR_COMP_VALUE	R/W	0x00	Value to be compared with timer N counter

- Timer N IRQ Status

TMR1_IRQ_STA,
TMR2_IRQ_STA,
TMR3_IRQ_STA,
TMR4_IRQ_STA,
TMR5_IRQ_STA,
TMR6_IRQ_STA,
TMR7_IRQ_STA,
TMR8_IRQ_STA,
TMR9_IRQ_STA,

TMR10_IRQ_STA,
TMR11_IRQ_STA,
TMR12_IRQ_STA

Table 4- 11Timer N IRQ Status

	Name	R/W	Reset	
31:3	Reserved			
2	TMR_CAP_INT_STA	R/W	0x0	Timer n's IRQ status for capture When read 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action 0x1: Clear pending event, any
1	TMR_OVF_INT_STA	R/W	0x0	Timer n's IRQ status for overflow When read 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action 0x1: Clear pending event, any
0	TMR_MAT_INT_ST A	R/W	0x0	Timer n's IRQ status for match When read 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action 0x1: Clear pending event, any

- Timer N IRQ Status Raw

TMR1_IRQ_RAW,
TMR2_IRQ_RAW,
TMR3_IRQ_RAW,
TMR4_IRQ_RAW,
TMR5_IRQ_RAW,
TMR6_IRQ_RAW,
TMR7_IRQ_RAW,
TMR8_IRQ_RAW,
TMR9_IRQ_RAW,
TMR10_IRQ_RAW,
TMR11_IRQ_RAW,
TMR12_IRQ_RAW

Table 4- 12Timer N IRQ Status Raw

	Name	R/W	Reset	
31:3	Reserved	R/W		
2	TMR_CAP_INT_RAW	R/W	0x0	Timer n's IRQ status for capture When read 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action 0x1: Clear pending event, any
1	TMR_OVF_INT_RAW	R/W	0x0	Timer n's IRQ status for overflow When read 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action 0x1: Clear pending event, any
0	TMR_MAT_INT_RAW	R/W	0x0	Timer n's IRQ status for match When read 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action 0x1: Clear pending event, any

- Timer N IRQ Enable

TMR0_IRQ_EN,
TMR1_IRQ_EN,
TMR2_IRQ_EN,
TMR3_IRQ_EN,
TMR4_IRQ_EN,
TMR5_IRQ_EN,
TMR6_IRQ_EN,
TMR7_IRQ_EN,
TMR8_IRQ_EN,
TMR9_IRQ_EN,
TMR10_IRQ_EN,
TMR11_IRQ_EN

Table 4- 13Timer N IRQ Enable

	Name	R/W	Reset	
31:3	Reserved			

	Name	R/W	Reset	
2	TMR_CAP_INT_EN	R/W	0x0	Timer n's IRQ enable for capture When read 0x0: IRQ disabled 0x1: IRQ enabled When write 0x0: IRQ disabled 0x1: IRQ enabled
1	TMR_OVF_INT_EN	R/W	0x0	Timer n's IRQ enable for overflow When read 0x0: IRQ disabled 0x1: IRQ enabled When write 0x0: IRQ disabled 0x1: IRQ enabled
0	TMR_MAT_INT_EN	R/W	0x0	Timer n's IRQ enable for match When read 0x0: IRQ disabled 0x1: IRQ enabled When write 0x0: IRQ disabled 0x1: IRQ enabled

- Timer N Control

TMR1_CONTROL,
TMR0_CONTROL,
TMR1_CONTROL,
TMR2_CONTROL,
TMR3_CONTROL,
TMR4_CONTROL,
TMR5_CONTROL,
TMR6_CONTROL,
TMR7_CONTROL,
TMR8_CONTROL,
TMR9_CONTROL,
TMR10_CONTROL,
TMR11_CONTROL

Table 4- 14Timer N Control

	Name	R/W	Reset	
31:28	Reserved	R/W		
27:17	TMR_CLK_DIV	R/W	0x0	Divisor for clock pre-scaler Frequency divisor is (TMR_CLK_DIV+1) Pre-scaled clock is used to clock time counter

	Name	R/W	Reset	
16	PWM_POL	R/W	0x0	Polarity selection of pulse on PWM port 0x0: Positive 0x1: Negative Valid when TMR_PULSE_TGL = 0x0
15	TMR_SRST	R/W	0x0	Software reset to reset all functional circuit. It also resets all system registers. Read always returns 0 Write 0x1: Synchronous reset active Write 0x0 : Synchronous reset inactive
14	TMR_UP_DWN	R/W	0x0	Timer counting mode 0x0: Count Up intrerrupt occurs upon counting down to 0xFFFFFFFF 0x1: Count Down intrerrupt occurs upon counting down to 0
13	TMR_COMP_EN	R/W	0x0	Compare mode enable 0x0: Disable 0x1: Enable
12	TMR_AUTO_LD	R/W	0x0	Timer counter mode: 0x0: One-shot, timer stops on overflow. Timer can be restarted by reloading. 0x1: Auto-reload It applies to user define mode (TMR_MODE = 0x1)
11	TMR_MODE	R/W	0x0	0x0: Free-running mode 0x1: User define mode
10:9	TMR_PWM_TRG	R/W	0x00	Trigger output mode on PWM output pin 0x0: No trigger 0x1: Trigger on overflow 0x2: Trigger on overflow and Match 0x3: Trigger on Match
8	TMR_PULSE_TGL	R/W	0x0	Output mode on PWM output pin 0x0: Pulse 0x1: Toggle
7	TMR_CAP_MODE	R/W	0x0	0x0: Single capture on the first valid event 0x1: Capture on the second valid event
6:5	TMR_CAP_EVENT	R/W	0x0	Capture event setting 0x0: Capture mode disabled 0x1: Capture on high-to-low transition 0x2: Capture on low-to-high transition 0x3: Capture on both transition edge

	Name	R/W	Reset	
0	TMR_EN	R/W	0x0	Timer Enable 0x1: Timer Enable 0x0: Timer disable, clock counter frozen, all Interrupt flag cleared. Capature state and prescaler counter reset Clock counter can be overwritten by APB bus write or load trigger.

- Timer IRQ Status
  TMR_IRQ_STA[7:0]

Table 4- 15Timer IRQ Status

	Name	R/W	Reset	
31:8	Reserved			
7:0	TMR_IRQ_STA	R	0x0	All timer's IRQ status for quick check When • read Bit[11] for Timer12 Bit[10] for Timer11 Bit[9] for Timer10 Bit[8] for Timer9 Bit[7] for Timer8 Bit[6] for Timer7 Bit[5] for Timer6 Bit[4] for Timer5 Bit[3] for Timer4 Bit[2] for Timer3 Bit[1] for Timer2 Bit[0] for Timer1 0x0: No event pending 0x1: IRQ event pending When write 0x0: No action

66.3.4. Timer usage flow

- Timer mode
  1) Initialize the timer. Set appropriate values for timer mode TMR_UP_DWN, TMR_AUTOLOAD, TMR_MODE
  Configure interrupt and set clock prescaler value by TMR_CLK_DIV

2) Set Load counter value TMRn_LOAD_VALUE
3) Enable timer by setting TMR_EN

- Capture mode

1）Configure capture mode by setting TMR_CAP_MODE, TMR_CAP_EVENT
Note that TMR_CAP_EVENT should be set (change to non-zero value) at least 1 clock cycle after timer is enabled.
2) Enable TMR_CAP_INT_EN necessary
3) Captured value can be read from TMRn_CAP1_VALUE or TMRn_CAP2_VALUE

- Compare mode

1) Enable TMR_MAT_INT_EN necessary
2) Enable compare mode by setting TMR_COMP_EN

- PWM mode

1) Configure TMR_PWM_TRG, PWM_POL, and TMR_PULSE_TGL at TMR_CONTROL as approiate.
2) timer overflow event is used as PWM trigger, timer should work in user-defined mode and period of PWM wave solely depends on setting of TMR_LOAD_VALUE.
For example, 500MHz PWM is wanted and clocked by APB clock at 75MHz

When Timer in count-up mode, TMR1_LOAD_VALUE (0x4) should be calculated as
0xffffffff – (75000000 / (500*2)) + 1 = 0xfffedb08
When Timer in count-down mode, it should be
(75000000 / (500*2)) – 1 = 0x124F7

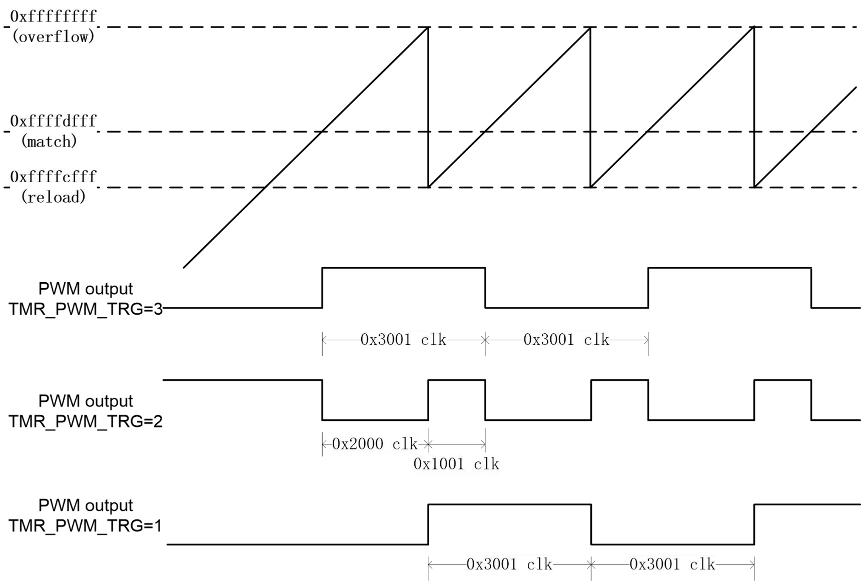


Figure 4- 6 Example of PWM output in toggle mode

66.4. WDT

66.4.1. Overview

One 32-bit watchdog timer in ARM subsystem is used to prevent system lock-up the software becomes trapped in a deadlock. This should be a down counter capable of generating a pulse on the reset pin and an interrupt to the device system modules, following an over flow condition. The timer should serve resets to the system reset module and watchdog interrupts to the host ARM. In normal operation, the Linux operating system reloads the watchdog at regular intervals before the timer overflow occurs. A timer overflow should generate an internal reset of the ARM system. An indication that a watchdog reset occurred should be available in the form of a register bit, which can be read and must be explicitly cleared by firmware.

66.4.2. Features

The Watchdog timer features should include the following:

- Free-running 32-bit down counter
- On-the-fly read/write register (while counting)
- Programmable prescale divisor for input clock
- The watchdog timers are reset either on power-on or after a warm reset before they start counting.
- Reset or interrupt actions when a timer overflow condition occurs
- Programmable Interrupt Time-out Period

66.4.3. Block Diagram

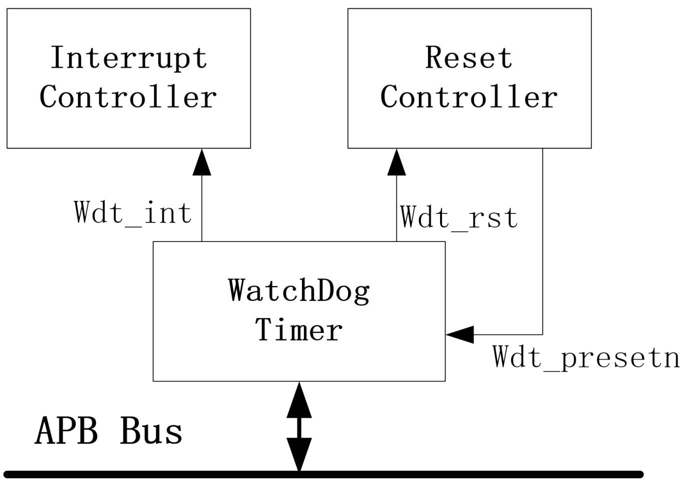


Figure 4.8.3- 1 WDT System Connection

66.4.4. Function Description

The Watchdog Timer has a 32-bit down counter which counts from a programmable load value to zero. When the counter reaches zero, an interrupt request or reset request is generated, depending on the Watchdog Timer Action setting. The counter will rewrap to FFFF_FFFFh and continue down counting when interrupt generated or on reset disable edge. The counter can be stopped by setting the Timer Enable bit to 0. The load value is loaded to the counter on the restart flow. To avoid the generation of interrupt, software may periodically restart the counter. After the interrupt request generate, it can be cleared by restarting counter or write "1" to the interrupt status register. As a safety feature to prevent accidental restarts, the value WDT_CNT_RESTART_VALUE must be written to the Watchdog Timer Counter Restart Register. The count value is readable by software anytime; it is set to FFFF_FFFFh by a hardware reset or setting the Timer Enable bit to 0.

The WDT prescale divisor select for the input clock (pclk) to control the timer range. The WDT program prescale divisor is 8-bit. The four higher are used to divide the input clock (pclk) with program prescale divisor to generate divide-clock signals; the other four lower are used to divide the divide-clock with program prescale divisor to generate the WDT Counter clock signals. It includes two 14- counters to generate the watchdog counter clock. The two counters are enabled when the watchdog is active.

There is a WDT controller block including the watchdog counter and watchdog program registers. It generates the watchdog interrupt signal or watchdog reset signal when the watchdog counter reaches zero.

66.4.5. Registers

66.4.5.1. Register Summary

Table 4.8- 1 WDT Register Summary

Addr offset	Name	Width	R/W	Default value	
0x0	WDT_LDR	32	RW	0x0	Watchdog Timer Load Register
0x4	WDT_CRR	32	W	0x0	Watchdog Timer Counter Restart Register
0x8	WDT_TCR	32	RW	0x1DE	Watchdog Timer Control Register
0xC	WDT_PLR	32	RW	0xFF	Watchdog Timer Reset Pulse Length Register
0x10	WDT_CNT	32	R	0xFFFF_FFFF	Watchdog Timer Counter Register
0x14	WDT_ISR	32	RW	0x0	Watchdog Timer Interrupt Status Register
0x18 ~ 0x3FF	Reserved				

66.4.5.2. **Register**

Table 4.8- 2 WDT_LDR

Bit Field	Name	Width	R/W	Default value	
31:0	LoadVal	32	RW	0x0	Watchdog Timer load value Note: write “0x76” to WDT_CRR, the Load value can restart the counter values.

Table 4.8- 3 WDT_CRR

Bit Field	Name	Width	R/W	Default value	
31:8	Reserved				
7:0	CounterRestart	8	W	0x0	This register is used to restart the WDT counter. As a safety feature to prevent accidental restarts. write “0x76” to WDT_CRR, then the LDR value can be load to Restart counter. Write other values of WDT_CRR, the LDR value cannot be load to Restart WDT counter.

Table 4.8- 4 WDT_TCR

Bit Field	Name	Width	R/W	Default value	
31:13	Reserved				
12:10	Time-out range.	3	RW	0x0	When a WDT Interrupt is generated, it is not cleared before the Time-out time, then it generates a system reset. 3’b000: 64K of WDT Counter clock cycles 3’b001: 32K of WDT Counter clock cycles 3’b010: 16K of WDT Counter clock cycles 3’b011: 8K of WDT Counter clock cycles 3’b100: 4K of WDT Counter clock cycles 3’b101: 2K of WDT Counter clock cycles 3’b110: 1K of WDT Counter clock cycles 3’b111: Reserved
9	WDT Enable	1	RW	0x0	1’b0: Timer is disabled (count value is set to FFFF_FFFFh) 1’b1: Timer is enabled
8	WDT Action	1	RW	0x1	1’b0: When the timer count reaches zero, a reset output is generated. 1’b1: When the timer count reaches zero, an

					interrupt request is generated.
7:4	Prescale Divisor (higher)	4	RW	0xD	4 Higher of the Pre scale Divisor The 4 (7:4) divide the system clock by following prescale divisors. 4'b0000: Divide by 2 4'b0001: Divide by 4 4'b0010: Divide by 8 4'b0011: Divide by 16 4'b0100: Divide by 32 4'b0101: Divide by 64 4'b0110: Divide by 128 4'b0111: Divide by 256 4'b1000: Divide by 512 4'b1001: Divide by 1024 4'b1010: Divide by 2048 4'b1011: Divide by 4096 4'b1100: Divide by 8192 4'b1101: Divide by 16384 4'b1110: Reserved 4'b1111: Reserved
3:0	Prescale Divisor (lower)	4	RW	0xE	The (3:0) further divide 4 higher (7:4) divided-clock by following prescale divisors . 4'b0000: No divide 4'b0001: Divide by 2 4'b0010: Divide by 4 4'b0011: Divide by 8 4'b0100: Divide by 16 4'b0101: Divide by 32 4'b0110: Divide by 64 4'b0111: Divide by 128 4'b1000: Divide by 256 4'b1001: Divide by 512 4'b1010: Divide by 1024 4'b1011: Divide by 2048 4'b1100: Divide by 4096 4'b1101: Divide by 8192 4'b1110: Divide by 16384 4'b1111: Reserved

Table 4.8- 5 WDT_CNT

Bit Field	Name	Width	R/W	Default value	

31:0	Counter Value	32	R	0xFFFF_FFFF	Current value of the timer counter

Table 4.8- 6 WDT_ISR

Bit Field	Name	Width	R/W	Default value	
31:1	Reserved				
0	IntStat	1	RW	0x0	This register shows the interrupt status of the WDT. 1’b1: Interrupt is active. 1’b0: Interrupt is inactive. When this register is written “1”, the WDT interrupt is cleared to 0 without restarting the watchdog counter.

66.4.6. Application Notes

WDT IP will not be reset by WDT reset of itself. But the WDT_TCR register will return to initial status after WDT reset. Below figure describes the WDT reset solution.

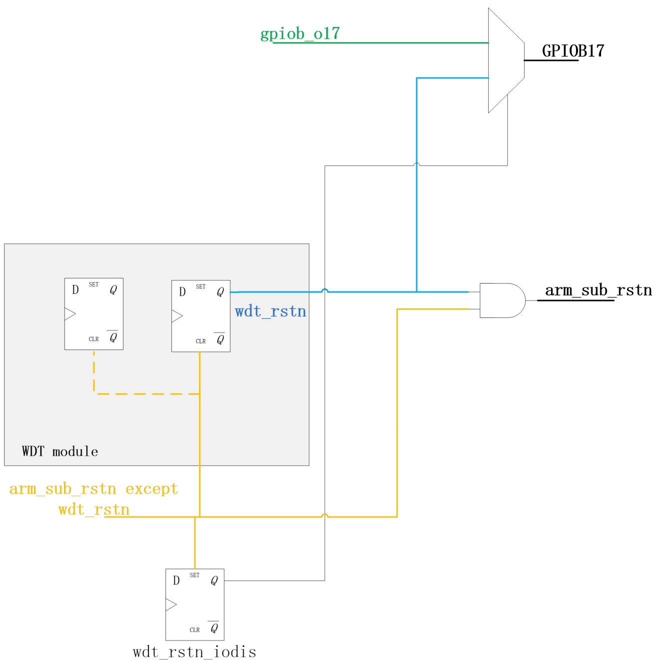


Figure 4.8.6- 1 WDT Reset Connection

66.4.7. Programming Guide

66.4.7.1. **Programming Considerations**

As an APB slave, the Watchdog_timer component is little-endian. As the largest register width can be 32-bits, all registers are aligned to 32-bit boundaries. Aligning to 32-bit boundaries keeps the same memory map for all bus widths. The APB bus reset, (presetn), resets all registers. The base address of Watchdog_timer is not fixed and is determined by the watchdog_timer in the generation of the psel signal for Watchdog_timer. Offset addresses from the base address are used for each register. When a register to be read is narrower than the data bus width, there is no need for coherency logic. There are coherency registers when the data bus width is less than the counter width. It is possible for the WDT_CCV

66.4.7.2. **Example Operational Flow**

Figure following illustrates the operational flow of a Watchdog_timer component configured with the following configuration parameters:

- Single timeout period
- Response mode not hard coded
- Reset pulse length not hard coded
- WDT not always enabled
- Generates interrupt, and then system reset

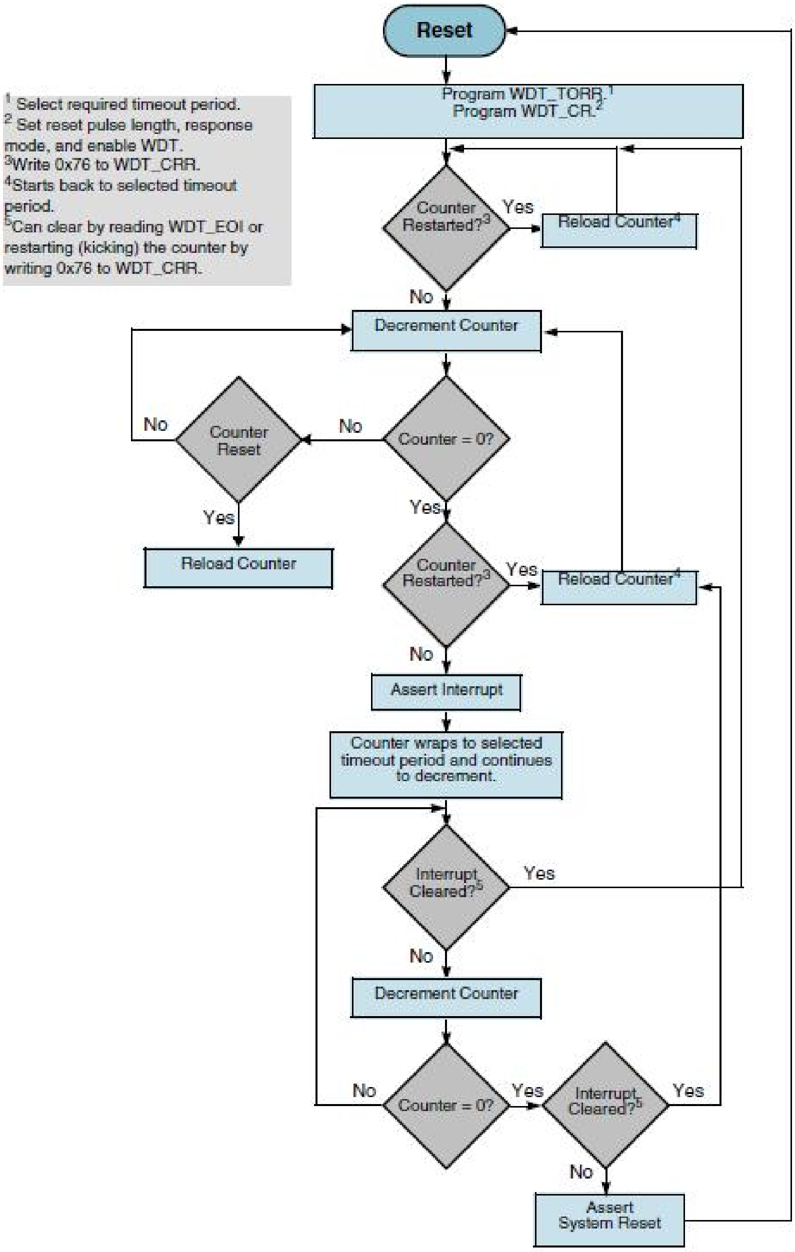


Figure 4.8.7- 1 Operation of an Example Watchdog_timer_Configuration

66.5. RTC

66.5.1. Time and Calendar Counter

The RTC module is able to provide real-time date and time information

Date and Time counter

- 100-th Second counter (100THSEC)
- Second counter (SEC)
- Minute counter (MIN)

- Hour counter (HOUR)
- Day-of-week counter (WEEK)
- Day counter (DAY)
- Month counter (MONTH)
- Year counter (YEAR and CEN)

YEAR represents last two decimal digit of year. CEN is 1900 or 2000 indication.
Each register is encoded in BCD (Binary Coded Decimal) code.

66.5.2.100TH Sec Counter (100THSEC)

This counter counts up from 00 to 99 for period of one second. It is incremented for every 326/327/328/329 OSC32K cycles depends on calibrated second period. The jitter within 1 second is limited to 1 HCLK cycle .

66.5.3.Time Counter (SEC, MIN and HOUR)

Time counter is defined as follows:
SEC : When this counter counts up from 59 to 00, MIN counter is incremented.
MIN : When this counter counts up from 59 to 00, HOUR counter is incremented.
HOUR : This counter consists of H20_PA (1bit) and HOUR19 (5bit).
In 12-hour clock mode, when [H20_PA, HOUR19] is counts up from [1,11] (11PM) to [0,12] (12AM), the DAY counter is incremented.
In 24-hour clock mode, when [H20_PA, HOUR19] is counts up from [1,3] (23H) to [0,0] (0H), the DAY counter is incremented. non-existent time has been written, any carry from lower digits may cause the time counters to malfunction. Therefore, such incorrect setting will be ignored.
Clock system is selectable from 12-hour clock mode or 24-hour clock mode by setting of 12_24 bit in HOUR register.
0 : 12-hour clock mode
1 : 24-hour clock mode
HOUR is represented by two registers, HOUR19 and H20_PA. The value of HOUR19 is encoded in BCD. Time format depends on clock system setting. In 12-hour clock mode, H20_PA means a.m. or p.m. In 24-hour clock mode, H20_PA represents the twenty's place.

Time	24Hours		12Hours	
	H20_PA	HOUR19	H20_PA	HOUR19
0 a.m.	0	0	0	12
1 a.m.	0	1	0	1
2 a.m.	0	2	0	2
3 a.m.	0	3	0	3
4 a.m.	0	4	0	4
5 a.m.	0	5	0	5

6 a.m.	0	6	0	6
7 a.m.	0	7	0	7
8 a.m.	0	8	0	8
9 a.m.	0	9	0	9
10 a.m.	0	10	0	10
11 a.m.	0	11	0	11
0 p.m.	0	12	1	12
1 p.m.	0	13	1	1
2 p.m.	0	14	1	2
3 p.m.	0	15	1	3
4 p.m.	0	16	1	4
5 p.m.	0	17	1	5
6 p.m.	0	18	1	6
7 p.m.	0	19	1	7
8 p.m.	1	0	1	8
9 p.m.	1	1	1	9
10 p.m.	1	2	1	10
11 p.m.	1	3	1	11

66.5.4. Date Counter

Day-of-Week counter (WEEK)

The value is binary code and range of this counter is 0 to 6. (7 is unused). The relation between WEEK counter and the actual day of week is user definable. (e.g. Sun=0, Mon=1...Sat=6)

Calendar counter (DAY, MONTH, YEAR, CEN)

The automatic calendar function provides the calendar digit displayed as follows.

DAY : Range of this counter is

01 to 31 : Jan, Mar, May, Jul, Aug, Oct, Dec

01 to 30 : Apr, Jun, Sep, Nov

01 to 29 : Feb in a leap year

01 to 28 : Feb in a common year

It carries to MONTH counter when wraps to 01.

MONTH : Range from 1 to 12 and carries to YEAR counter when wraps to 1.

YEAR: Range from 00 to 99 and carried to CEN counter when wraps to 0.

When CEN counter is 0, YEAR1=04, 08, ... 92, 96 are leap year.

When CEN counter is 1, YEAR1=00, 04, 08, ... 92, 96 are leap year.

Within the range of 1900 to 2099, leap year can be identied.

Any illegal input will be ignored.

66.5.5.Time and Date Setting

In order to prevent unexpected carry up while writing Time and Date counters, second timer is reset to 0 and frozen till end of write operation. The time/date setting operation should follow sequence as below

1)Write 1 to RTC_WRSTA_CPU, time counting is stopped
2)Confirm RTC_WRSTA is completed by checking RTC_ PEND_WR bit to be 0
3)Complete write operation as fast as possible
4)Write 1 to RTC_WRSTP_CPU, time starts counting from this point

Day of week value should be written after year and month has been correctly set, otherwise, the illegal day value may not be identied and ignored.
When 100th second counter needs to be configured, it is necessary to turn off trimming by setting TRIM_EN = 0 before step 1, and turn it on again after completion of step 4 desired.

66.5.6.Time and Date Reading

Before starting read operation to the time and date register, the RTC need to copy real time values of the time/date registers into shadow read registers. This isolation assures the integrity of read data of time/date values and not be subject to changing values from time updates. The time/date read operation should follow sequence as below

1)Write 1 to RTC_RDSTA_CPU
2) Confirm RTC_WRSTA is completed by checking RTC_ PEND_WR bit to be 0
3) Complete read operation as fast as possible
4) Write 1 to RTC_RDSTP_CPU

Value hold in shadow registers can be updated by RTC_RDSTA_* when no read sequence is ongoing.

66.5.7.ALARM Function

There are two alarms defined, ALARM1 and ALARM2.
Register ALM1_* are defined as alarm Date and Time configuration and Mask register:
Date and Time registers are
ALM1_100THSEC, ALM1_SEC, ALM1_MIN, ALM1_HOUR, ALM1_WEEK, ALM1_DAY, ALM1_MONTH,
ALM1_YEAR
Mask register is
ALM1_MASK

Mask register controls ALARM condition. When ALM1_MASK register is all 0, alarm is disabled.

For individual bit, 1 means alarm is on when the corresponding register matches, 0 means alarm is on regardless the corresponding register value. The alarm flag and interrupt can be generated when the time increments to match the enabled time set by ALM1_* registers.

Example of RTC ALARM1 Setting

ALM1 Time/Date register								ALM1_MASK Register								
ALM1_YEAR	ALM1_MONTH	ALM1_DAY	ALM1_WEEK	ALM1_HOUR	ALM1_MIN	ALM1_SEC	ALM1_100THSEC	A1MYEAR	A1MMON	A1MDAY	A1MWEEK	A1MHOUR	A1MMIN	A1MSEC	A1M100THSEC	Alarm Condition
2	3	4	5	6	7	8	9	0	0	0	0	0	0	0	0	Disabled
2	3	4	5	6	7	8	9	0	0	0	0	0	1	1	0	Every hour at 07:08.
2	3	4	5	6	7	8	9	1	1	1	0	1	1	1	0	06:07:08 on March 4, 2002
2	3	4	5	6	7	8	9	0	0	1	0	1	1	1	0	06:07:08 on every 4th day
2	3	4	5	6	7	8	9	0	0	0	1	1	1	1	0	06:07:08 on every Friday

ALARM2 is to generate periodical tick interrupt.
Interrupt period varies from every min, to every 1/128 second according to register setting.

The alarm status can be checked from register RTC_IRQ_RAW and RTC_IRQ_STA
The combined interrupt output to CPU is controlled by register of RTC_IRQ_MASK and RTC_IRQ_CLEAR, it maintains its value until cleared by AHB access.

66.5.8. Oscillation Adjustment

The module is able to calibrate OSC clock input frequency to be within +/- 5ppm of nominal 32.768KHz. The adjustment function changes the maximum value of clock counter from 32,768 at specied interval. Adjust step is +/-1cycle and max range is +/-128 cycles with 8bit trimming register (TRIM[7:0]) for each adjustment.
TRIM[7:0] register adjusts maximum counter value.
  The max value will be increased by (TRIM[6:0]) when TRIM[7] is set to 0.
  The max value will be decreased by (~TRIM[6:0] + 1) when TRIM[7] is set to 1.
The max value will not change when TRIM[7:0] are set to 0
TRIM_MODE register defines how frequent such adjustment is performed.
  0x0: Once every 60 seconds (when SEC=00)
  0x1: Once every 30 seconds (when SEC=00, 30)
  0x2: Once every 15 seconds (when SEC=00, 15, 30, 45)

0x3: Once every 6 seconds (when SEC=00, 06, 12, 18, 24, 30, 36, 42, 48, 54)

TRIM[7:0] value is pre-stored in NAND Flash which can be obtained by firmware.

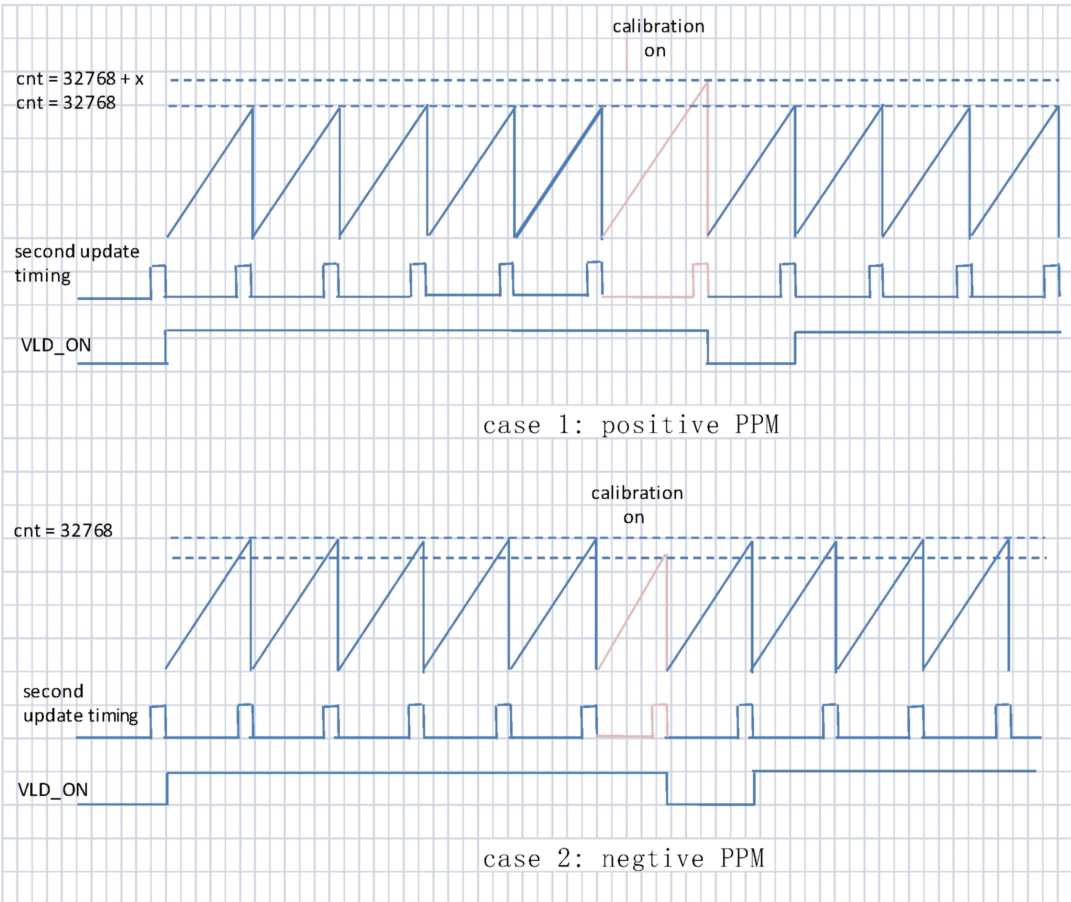


Figure 4.9- 1 Timing Diagram of Oscillation Adjustment

The compensation ensures the minute update timing is independent on oscillator frequency accuracy.

Example calculation of oscillation adjustment function

TRIM[7:0]		TRIM_MODE = 0x0		TRIM_MODE = 0x1	
bin	Compensation value	total clock cycles in 60sec	[ppm]	total clock cycles in 30sec	[ppm]
00000000	0	32768*60	0.0	32768*30	0.0
00000001	1	32768*59 + 32769*1	0.5	32768*29 + 32769*1	1
11111111	-1	32768*59 + 32767*1	-0.5	32768*29 + 32767*1	-1
00111111	63	32768*59 + 32831*1	32	32768*29 + 32831*1	64.6
11000000	-64	32768*59 + 32704*1	-32.2	32768*29 + 32704*1	-65.1
01111111	127	32768*59 + 32895*1	64.6	32768*29 + 32895*1	129.2
10000000	-128	32768*59 + 32640*1	-65.1	32768*29 + 32640*1	-130.2

TRIM[7:0]		TRIM_MODE = 0x2		TRIM_MODE = 0x3	
bin	Compensation value	total clock cycles in 15sec	[ppm]	total clock cycles in 6sec	[ppm]

00000000	0	32768*15	0.0	32768*6	0.0
00000001	1	32768*14 + 32769*1	1.9	32768*5 + 32769*1	5.1
11111111	-1	32768*14 + 32767*1	-1.9	32768*5 + 32767*1	-5.1
00111111	63	32768*14+ 32831*1	120.2	32768*5 + 32831*1	320.4
11000000	-64	32768*14+ 32704*1	-122	32768*5 + 32704*1	-325.5
01111111	127	32768*14 + 32895*1	242.2	32768*5 + 32895*1	646
10000000	-128	32768*14 + 32640*1	-244.2	32768*5 + 32640*1	-651

With the given target ppm value, TRIM and TRIM_MODE needs to be properly calculated to have actual ppm corrected as close as possible to target. It is also recommended to perform adjustment more frequently with smaller step by choosing higher TRIM_MODE setting whenever possible.

The output signal VLD_ON_O is asserted for calibrated clock validation. VLD_ON_O outputs positive pulse of 6/15/30/60 seconds based on VLD_MODE setting, which facilitates timing measurement using external reference clock.

Suggested trimming procedure:

1)32K oscillator's actual frequency is measured as fosc (Hz) at SOC's output port RTC_CLKOUT

2) Trimming value can be obtained by calculating the closest integer to
(fosc * N – 32768 * N) when N = 6, 15, 30, 60
Calibrated timing can be validated by measuring positive pulse width on SOC's output port of RTC_CAL_CLKOUT after properly configured registers of TRIM, TRIM_MODE, TRIM_EN and VLD_EN.

Please be noted VLD_ON and RTC_CLKOUT is not accessible in QFN128 package, therefore only limited trimming function is supported.

66.5.9. **Registers**

66.5.9.1. **100THSEC**

RTC Second Counter Register.

Address Offset: 0x0

	Name	R/W	Reset	
7:0	100THSEC		0	This is the 1/100 second counter. The value is encoded with BCD (Binary Coded Decimal). Range of this counter is 00 to 99. When it counts up from 99 to 00, SEC counter is incremented.

66.5.9.2. SEC

RTC Second Counter Register.
Address Offset: 0x4

	Name	R/W	Reset	
6:0	SEC	R/W	0	This is the second counter. The value is encoded with BCD (Binary Coded Decimal). Range of this counter is 00 to 59. When it counts up from 59 to 00, MIN counter is incremented. When this counter is written, the 16bit 32KHz clock counter value (less than one second) is reset.

66.5.9.3. MIN

RTC Minute Counter Register.
Address Offset: 0x8

	Name	R/W	Reset	
6:0	MIN	R/W	0	This is the minute counter. The value is encoded with BCD (Binary Coded Decimal). Range of this counter is 00 to 59. When it counts up from 59 to 00, HOUR counter is incremented.

66.5.9.4. HOUR

RTC Hour Counter Register.
Address Offset: 0xC

	Name	R/W	Reset	
7	HOUR12_24	R/W	0	Setting of the clock system. 0 : 12-hour clock mode 1 : 24-hour clock mode.
5	H20_PA	R/W	0	There are hour counters. The value of

	Name	R/W	Reset	
4:0	HOUR19	R/W	0	HOUR19 is encoded with BCD (Binary Coded Decimal). Time format depends on clock system. In 12-hour clock mode, H20_PA means AM or PM. In 24-hour clock mode, H20_PA means the 20's place. Refer 'Hour format' section to detail. In 12-hour clock mode, when [H20_PA, HOUR19] is counts up from [1,11] (11PM) to [0,12] (12AM), the DAY counter is incremented. In 24-hour clock mode, when [H20_PA, HOUR19] is counts up from [1,3] (23H) to [0,0] (0H), the DAY counter is incremented.

66.5.9.5. WEEK

RTC Week Counter Register.
Address Offset: 0x10

	Name	R/W	Reset	
2:0	WEEK	R/W	0	This is the DAY of WEEK counter. The value is binary code and range of this counter is 0 to 6. (7 is unused. It is prohibition to set 7 unless it does not use a day of week) The relation between WEEK counter and the days of week is user definable. (e.g. Sun=0, Mon=1...Sat=6)

66.5.9.6. DAY

RTC Day Counter Register.
Address Offset: 0x14

	Name	R/W	Reset	

	Name	R/W	Reset	
5:0	DAY	R/W	0	This is the day counter. The value is encoded with BCD (Binary Coded Decimal). Range of this counter is 01 to 31 : Jan, Mar, May, Jul, Aug, Oct, Dec 01 to 30 : Apr, Jun, Sep, Nov 01 to 29 : Feb in leap year 01 to 28 : Feb in no leap year When day counter counts up at end of month, MONTH counter is incremented.

66.5.9.7. CEN_MONTH

RTC Century and Month Counter Register.
Address Offset: 0x18

	Name	R/W	Reset	
7	CEN	R/W	0	Century counter. 0: 1900; 1: 2000
6:5	Reserved			
4:0	MONTH	R/W	0	This is the month counter. The value is encoded with BCD (Binary Coded Decimal). Range of this counter is 01 to 12. When it counts up from 12 to 01, YEAR counter is incremented.

66.5.9.8. YEAR

RTC Year Counter Register.
Address Offset: 0x1C

	Name	R/W	Reset	

	Name	R/W	Reset	
7:0	YEAR	R/W	0	It represents lower two digits of decimal year count. The value is encoded in BCD (Binary Coded Decimal) format. Range of this counter is 00 to 99. When it counts up from 99 to 00, CEN bit of CEN_MONTH may increment from 19 to 20. When CEN counter is 0, 04, 08, ... 92, 96 are leap year. When CEN counter is 1, 00, 04, 08, ... 92, 96 are leap year.

66.5.9.9. **ALM1_100THSEC**

RTC ALARM1 100th Second Configuration Register.
Address Offset: 0x30

	Name	R/W	Reset	
7:0	A1100THSEC	R/W	0	1/100 Second setting of ALARM1.

66.5.9.10. **ALM1_SEC**

RTC ALARM1 Second Configuration Register.
Address Offset: 0x34

	Name	R/W	Reset	
6:0	A1SEC	R/W	0	Second setting of ALARM1.

66.5.9.11. **ALM1_MIN**

RTC ALARM1 Minute Configuration Register.
Address Offset: 0x38

	Name	R/W	Reset	
6:0	A1MIN	R/W	0	Minute setting of ALARM1.

66.5.9.12. ALM1_HOUR

RTC ALARM1 Hour Configuration Register.

Address Offset: 0x3C

	Name	R/W	Reset	
7	A1H12_24	R/W	0	Hour setting of ALARM1. Set the value same as HOUR register. 0: HOUR12, 1: HOUR24
5	A1H20_PA	R/W	0	Hour setting of ALARM1.
4:0	A1HOUR19	R/W	0	

66.5.9.13. ALM1_WEEK

RTC ALARM1 Week Configuration Register.

Address Offset: 0x40

	Name	R/W	Reset	
2:0	A1WEEK	R/W	0	Day of week setting of ALARM1.

66.5.9.14. ALM1_DAY

RTC ALARM1 Day Configuration Register.

Address Offset: 0x44

	Name	R/W	Reset	
5:0	A1DAY	R/W	0	Day setting of ALARM1.

66.5.9.15. ALM1_MONTH

RTC ALARM1 Month Configuration Register.

Address Offset: 0x48

	Name	R/W	Reset	
4:0	A1MONTH	R/W	0	Month setting of ALARM1.

66.5.9.16. ALM1_YEAR

RTC ALARM1 Year Configuration Register.

Address Offset: 0x4C

	Name	R/W	Reset	
7:0	A1YEAR	R/W	0	Year setting of ALARM1.

66.5.9.17. **ALM1_MASK**

RTC ALARM1 Mask Register.

Address Offset: 0x50

	Name	R/W	Reset	
7	A1MYEAR	R/W	0	This is control register of A1YEAR condition. 0 :Condition not applied 1 :Condition applied
6	A1MMON	R/W	0	This is control register of A1MONTH condition. 0 :Condition not applied 1 :Condition applied
5	A1MDAY	R/W	0	This is control register of A1DAY condition. 0 :Condition not applied 1 :Condition applied
4	A1MWEEK	R/W	0	This is control register of A1WEEK condition. 0 :Condition not applied 1 :Condition applied
3	A1MHOUR	R/W	0	This is control register of A1HOUR condition. 0 :Condition not applied 1 :Condition applied
2	A1MMIN	R/W	0	This is control register of A1MIN condition. 0 :Condition not applied 1 :Condition applied
1	A1MSEC	R/W	0	This is control register of A1SEC condition. 0 :Condition not applied 1 :Condition applied
0	A1100THSEC	R/W	0	This is control register of A1100THSEC condition. 0 :Condition not applied 1 :Condition applied

66.5.9.18. ALM2_SET

RTC ALARM2 Set Register.
Address Offset: 0x54

	Name	R/W	Reset	
7:4	Reserved			
3:0	ALM2_PD	R/W	0	Period of ALARM2 interrupt output. 0000 :Not output 0001 :1 sec 0010 :1/2 sec 0011 :1/4 sec 0100 : 1/8 sec 0101 : 1/16 sec 0110 : 1/32 sec 0111 : 1/64 sec 1000 : 1/128 sec 1001 : 1min

66.5.9.19. ALM_EN

RTC ALARM Enable Control Register.
Address Offset: 0x58

	Name	R/W	Reset	
7:2	Reserved			
1	ALM2_EN	R/W	0	ALARM2 interrupt request enable 0: Disable; 1: Enable
0	ALM1_EN	R/W	0	ALARM 1 interrupt output enable 0: Disable; 1: Enable

66.5.9.20. RTC_IRQ_RAW

RTC ALARM IRQ Raw Status Register.
Address Offset: 0x5C

	Name	R/W	Reset	
7:2	Reserved			

1	ALM2_IRQ_RAW	R/W	0	ALARM2 RAW IRQ status enable 0: Inactive; 1: Active
0	ALM1_IRQ_RAW	R/W	0	ALARM 1 RAW IRQ status 0:Inactive; 1: Active

66.5.9.21. RTC_IRQ_MASK

RTC ALARM IRQ Mask Register.
Address Offset: 0x60

	Name	R/W	Reset	
7:2	Reserved			
1	ALM2_IRQ_MASK	R/W	0	ALARM2 interrupt request enable 1: Disable; 0: Enable
0	ALM1_IRQ_MASK	R/W	0	ALARM 1 interrupt output enable 1: Disable; 0: Enable

66.5.9.22. RTC_IRQ_STA

RTC ALARM IRQ Status Register.
Address Offset: 0x64

	Name	R/W	Reset	
7:2	Reserved			
1	ALM2_IRQ_STA	R	0	ALARM2 interrupt request status 0: Inactive; 1: Active
0	ALM1_IRQ_STA	R	0	ALARM 1 interrupt request status 0: Inactive; 1: Active

66.5.9.23. RTC_IRQ_CLR

RTC ALARM IRQ Clear Register.
Address Offset: 0x68

	Name	R/W	Reset	

7:2	Reserved			
1	ALM2_IRQ_CLR	W	0	ALARM2 interrupt request clear When write 0: No action; 1: Clear IRQ
0	ALM1_IRQ_CLR	W	0	ALARM 1 interrupt request clear When write 0: NO action; 1: Clear IRQ

66.5.9.24. TRIM

RTC Trimming Register.
Address Offset: 0x80

	Name	R/W	Reset	
7:0	TRIM	R/W	0x0	This is a control register of the clock error compensate function. TRIM[7:0] is signed compensation constants.

66.5.9.25. TRIM_EN

RTC TrimmingEnable Register
Address Offset: 0x84

	Name	R/W	Reset	
7:1	Resesrved			
0	TRIM_EN	R/W	0x0	Trimming enable control 0x0 : Disable 0x1 : Enable

66.5.9.26. TRIM_MODE

RTC Trimming Mode Register.
Address Offset: 0x88

	Name	R/W	Reset	
7:4	Resesrved			

	Name	R/W	Reset	
3:2	VLD_MODE	R/W	0x0	Setting for VLD_ON_O for trimming function 0x0: VLD_ON_O holds for 60 sec 0x1: VLD_ON_O holds for 30 sec 0x2: VLD_ON_O holds for 15 sec 0x3: VLD_ON_O holds for 6 sec
1:0	TRIM_MODE	R/W	0x0	This register controls the timing of clock adjustment. 0x0 : EVERY_60SEC 0x1 : EVERY_30SEC 0x2: EVERY_15SEC 0x3: EVERY_6SEC

66.5.9.27. **VLD_EN**

RTC Trimming ValidationEnable Register
Address Offset: 0x8C

	Name	R/W	Reset	
7:1	Resesrved			
0	VLD_EN	R/W	0x0	Validation enable control 0x0 : Disable 0x1 : Enable, VLD_ON pulse asserted periodically

66.5.9.28. **RTC_TEST**

RTC Test Register
This register is used for Software Debug or internal test. When 1 is written, corresponding carry up condition is triggered.
This function is valid when RTC_TEST_EN=1
Address Offset: 0x90

	Name	R/W	Reset	
4	MONTHCNTUP	R/W	0	0 : Normal operation 1 :Force MONTH counter increment for debug/test

	Name	R/W	Reset	
3	DAYCNTUP	R/W	0	0 : Normal operation 1 :Force DAY& WEEK counter increment for debug/test
2	HOURCNTUP	R/W	0	0 : Normal operation 1 :Force HOUR counter increment for debug/test
1	MINCNTUP	R/W	0	0 : Normal operation 1 :Force MIN counter increment for debug/test
0	SECCNTUP	R/W	0	0 : Normal operation 1 : Speedup SEC counter for debug/test When 1 is written to any bit, second carry up condition is triggered.

66.5.9.29. **RTC_TEST_EN**

RTC Test Enable Register
This register is used for Software Debug or internal test.
Address Offset: 0x94

	Name	R/W	Reset	
0	RTC_TEST_EN	R/W	0	This register enables RTC_TEST register function. 0 : Normal operation 1 : Enable test register

66.5.9.30. **RTC_PEND_WR**

RTC AHB write pending status register
Address Offset: 0xA0

	Name	R/W	Reset	
7:1	Reserved			
0	RTC_PEND_WR	R	0	AHB write pending status 0x0 :No AHB write pending 0x1 :AHB Write pending

66.5.9.31. RTC_WRSTA_CPU

RTC Write Start by A7 Core
Address Offset: 0xA4

	Name	R/W	Reset	
7:1	Reserved			
0	RTC_WRSTA_CPU	W	0x0	Write 1 to this register to indicate A7 starts to write Time and Date register

66.5.9.32. RTC_WRSTP_CPU

RTC Write Stop by A7 Core
Address Offset: 0xA8

	Name	R/W	Reset	
7:1	Reserved			
0	RTC_WRSTP_CPU	W	0x0	Write 1 to this register to indicate end of reading Time and Date registers by A7

66.5.9.33. RTC_RDSTA_CPU

RTC Read Start by A7 Core
Address Offset: 0xAC

	Name	R/W	Reset	
7:1	Reserved			
0	RTC_RDSTA_CPU	W	0x0	Write 1 to this register to indicate A7 starts to write Time and Date register

66.5.9.34. RTC_RDSTP_CPU

RTC Read Stop by RF A7 Core
Address Offset: 0xB0

	Name	R/W	Reset	
7:1	Reserved			
0	RTC_RDSTP_CPU	W	0x0	Write 1 to this register to indicate end

				of reading Time and Date registers by A7

66.6. TRNG

66.6.1. Overview

The TRNG is a True Random Number Generator (TRNG). It generates random numbers that are intended to be statistically equivalent to a uniformly distributed random data stream. The circuit includes a noise source, a conditioning component, and a Deterministic Random Bit Generator (DRBG). The noise source sends Independent and Identically Distributed (IID) noise stream to the conditioning component to produce full-entropy seed which is then fed into the DRBG to generate random numbers. A health test block is included to perform Known Answer Tests (KAT) and seven different statistical tests. The health test block is capable of performing start-up, on-demand, and continuous tests.

The TRNG can generate random seeds from the internal ring-oscillator based noise source or can be manually seeded through a host-provided nonce. Host-provided nonce are fed into the conditioning component to increase the entropy rate using a conditioning function. If the host's nonce has enough entropy, the nonce can also be directly loaded into the DRBG to be used as a seed.

66.6.2. Features

The following are the features of TRNG:

- Internal random seeding operation
- Host-driven nonce seeding option
- Start-up, continuous and on-demand health tests
- 128-bit random number generation
- 128- or 256-bits of security strength

66.6.3. Function Description

66.6.3.1. Architecture

The TRNG generates random data that is statistically equivalent to a uniformly distributed random data stream. The circuit includes:

- A live entropy source that provides full entropy upon request.
- A Deterministic Random Bit Generator (DRBG).
- Health test blocks that perform continuous statistical health tests and on-demand Known Answer

Tests (KAT).

The following figure shows the top level architecture of the TRNG:

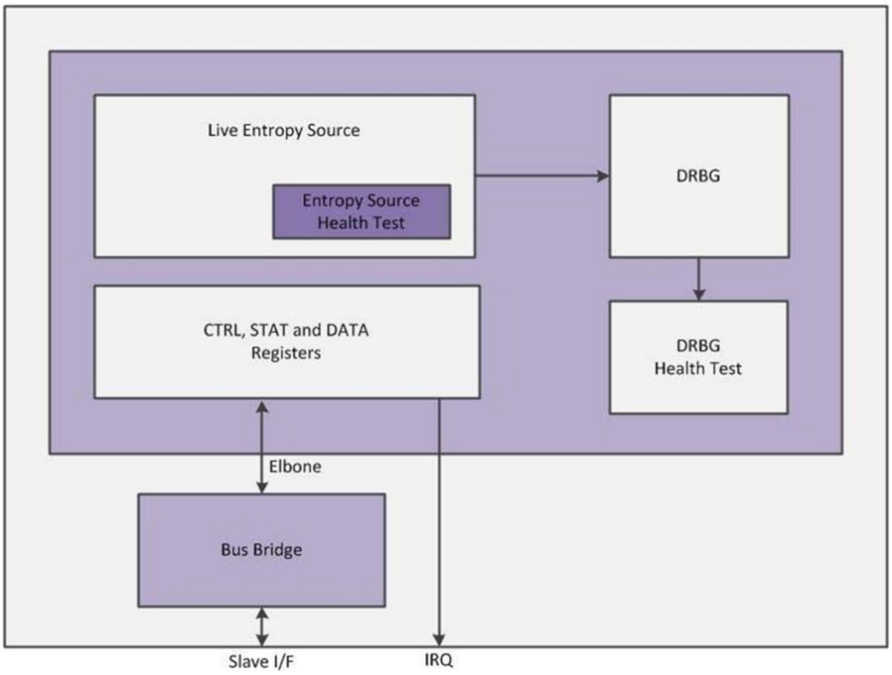


Figure 4- 2 TRNG Top Level Architecture

66.6.3.2. **Live Entropy Source**

There is a live entropy source in the TRNG to generate seeds for the DRBG. The source generates True Random Numbers upon user's request. It generates random numbers that are intended to be statistically equivalent to a uniformly distributed random data stream and contain full entropy. The Figure below shows the block diagram of the live entropy source.

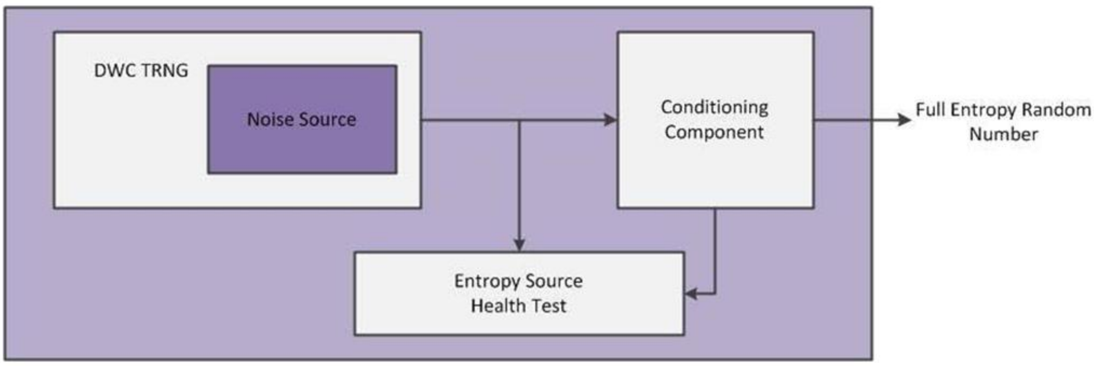


Figure 4- 3 Live Entropy Source Top Level Architecture

The circuit includes:

- Noise source that creates non-deterministic random bits.
- Block cipher Derivation Function (DF) as a conditioning component that is seeded either by the noise source or host-written nonce and provides 128-bit full entropy output.
- Health test component that is capable of performing continuous statistical health tests on the output

of the noise source as the non-deterministic part and known answer test on the conditioning component as the deterministic part. More details about the health tests are provided in “Health Tests”.

66.6.3.3. **Noise Source**

The ring-oscillator based seed generator front-end of the TRNG core is used as the noise source inside TRNG. The noise source provides non-deterministic random bits for the derivation function within the entropy source

66.6.3.4. **Conditioning Component**

The entropy source uses a block cipher derivation function as a conditioning component. This function is used to post-process the output of the noise source. The derivation function, reduces the bias of the data and increases the entropy. Hence, the resulting output contains full entropy in case of random self-seeding or a nonce seeding through a nonce with at least 0.5 entropy per bit (see “Seeding Options” for more information on seeding options).
The main component of the derivation function is an AES core. This module takes a 512-bit random stream and collects a 128-bit true random number stream into the RANDx registers. The random generate command can be run multiple times to produce as many random bits as required.

66.6.3.5. **DRBG**

The DRBG is a pseudo-random number generator. It produces a sequence of random bits from an initial value, that is, the seed, along with other possible inputs. The seed is provided by either the live entropy source or the host to initialize the internal state of the DRBG. As long as the seed remains secret, the output of the DRBG is unpredictable - up to the instantiated security strength.

Unlike the live entropy source (or other true random number generators) that usually takes an undefined amount of time to generate random numbers from probabilistic events, the major advantage of using a DRBG is that, once appropriately seeded, it can continuously generate data that appears random enough to pass statistical tests at low cost and rapid rate. It is also useful in testing since the process of generation can be easily replicated as long as the same seed is used again.

Periodic reseeding with sufficient entropy should be performed regularly to restore secrecy and reduce the likelihood of the seed being compromised over time. Using a fresh seed from the live entropy source can guarantee prediction resistance

66.6.3.6. **Health Tests**

The TRNG performs statistical health tests on the noise source output as the only non-deterministic part of the core and Known Answer Test (KAT) on the deterministic parts.

66.6.3.7. **Health Tests on Entropy Source**

The TRNG is capable of performing health tests on the live entropy source.
One round of KAT on the conditioning component plus a full round of statistical tests on the noise source are performed as the start-up test after reset. The number of noise bits that are generated and tested is configurable as a number of 256-bit blocks. All bits generated during the start-up test are discarded after the test regardless of the success or failure of the test.
Health test block receives bits coming out of the noise source and performs continuous statistical tests on those bits. The same tests are applied on any nonce provided by the host. The continuous health tests include:

- Repetition Count
- Adaptive Proportion
- Monobit
- Poker
- Run
- Long Run
- Auto-Correlation

These statistical tests can be individually turned on or off at build-time. This allows the customer to subscribe to the type of tests that they require and save area by disabling the rest. On-demand statistical test is also possible by rebooting the core. The KAT test is only applied at start-up and upon request through CTRL register.

66.6.3.8. **Health Tests on DRBG**

KAT tests are performed at start-up and upon request to check whether DRBG is operating correctly.

66.6.3.9. **Seeding Options**

The TRNG can operate in one of two modes: self-seeding mode or nonce mode.
In self-seeding mode, the live entropy source automatically produces a full-entropy seed that is fed directly into the DRBG.
Nonce mode is selected when the host chooses to use external entropy or noise sources other than the built-in modules. If the host-provided nonce contains enough entropy for the desired applications the nonce can be directly loaded into SEEDx register and be used as the DRBG seed. Another scenario of the nonce mode is to utilize the built-in conditioning component to increase the

entropy rate of a nonce before using it as a seed.
Note that a nonce directly loaded into SEEDx registers is not sent to the health test block, while on the other hand, any nonce value loaded into NPA_DATAx registers is sent to the health test block for statistical tests upon the assertion of the nonce seeding command.
In order to prevent backtracking attack, any transition between self-seeding mode and nonce mode causes the TRNG to erase all internal state information.

66.6.4. **Register Summary**

Table 4- 7 TRNG Register Summary

Address Offset	Register	Description
0x00	TRNG_CTRL	The CTRL register is used to cause the TRNG to execute one of a number of actions.
0x04	TRNG_MODE	The MODE register is used to enable or disable certain run-time features within the TRNG.
0x08	TRNG_SMODE	The SMODE register is used to enable or disable certain MISSION mode run-time features within the TRNG.
0x0C	TRNG_STAT	The STAT register allows the user to monitor the internal status of the TRNG.
0x10	TRNG_IE	The IE register is used to enable or disable interrupts within the TRNG.
0x14	TRNG_ISTAT	The ISTAT register allows the user to monitor the interrupt contributions of the TRNG.
0x18	TRNG_ALARMS	The ALARMS register allows the user to monitor the source of critical alarms.
0x1C	TRNG_COREKIT_REL	Contains the static coreKit release information.
0x20	TRNG_FEATURES	The FEATURES register returns the state of various build-time parameter values within the TRNG.
0x24	TRNG_RAND0	The RAND0 register is used by the host to read bits [31:0] of the newly generated 128-bit random data.
0x28	TRNG_RAND1	The RAND1 register is used by the host to read bits [63:32] of the newly generated 128-bit random data.
0x2C	TRNG_RAND2	The RAND2 register is used by the host to read bits [95:64] of the newly generated 128-bit random data.
0x30	TRNG_RAND3	The RAND3 register is used by the host to read bits [127:96] of the newly generated 128-bit random data
0x34	TRNG_NPA_DATA0	The NPA_DATA0 register holds Noise/Nonce/Personalization String/Additional Input - bits [31:0].
0x38	TRNG_NPA_DATA1	The NPA_DATA1 register holds Noise/Nonce/Personalization String/Additional Input - bits [63:32].

0x3C	TRNG_NPA_DATA2	The NPA_DATA2 register holds Noise/Nonce/Personalization String/Additional Input - bits [95:64].
0x40	TRNG_NPA_DATA3	The NPA_DATA3 register holds Noise/Nonce/Personalization String/Additional Input - bits [127:96].
0x44	TRNG_NPA_DATA4	The NPA_DATA4 register holds Noise/Nonce/Personalization String/Additional Input - bits [159:128].
0x48	TRNG_NPA_DATA5	The NPA_DATA5 register holds Noise/Nonce/Personalization String/Additional Input - bits [191:160].
0x4C	TRNG_NPA_DATA6	The NPA_DATA6 register holds Noise/Nonce/Personalization String/Additional Input - bits [223:192].
0x50	TRNG_NPA_DATA7	The NPA_DATA7 register holds Noise/Nonce/Personalization String/Additional Input - bits [255:224].
0x54	TRNG_NPA_DATA8	The NPA_DATA8 register holds Noise/Nonce/Personalization String/Additional Input - bits [287:256].
0x58	TRNG_NPA_DATA9	The NPA_DATA9 register holds Noise/Nonce/Personalization String/Additional Input - bits [319:288].
0x5C	TRNG_NPA_DATA10	The NPA_DATA10 register holds Noise/Nonce/Personalization String/Additional Input - bits [351:320].
0x60	TRNG_NPA_DATA11	The NPA_DATA11 register holds Noise/Nonce/Personalization String/Additional Input - bits [383:352].
0x64	TRNG_NPA_DATA12	The NPA_DATA12 register holds Noise/Nonce – bits [415:384].
0x68	TRNG_NPA_DATA13	The NPA_DATA13 register holds Noise/Nonce – bits [447:416].
0x6C	TRNG_NPA_DATA14	The NPA_DATA14 register holds Noise/Nonce – bits [479:448].
0x70	TRNG_NPA_DATA15	The NPA_DATA15 register holds Noise/Nonce – bits [511:480].
0x74	TRNG_SEED0	The SEED0 register holds seed value used in the DRBG – bits [31:0].
0x78	TRNG_SEED1	The SEED1 register holds seed value used in the DRBG - bits [63:32].
0x7C	TRNG_SEED2	The SEED2 register holds seed value used in the DRBG - bits [95:64].

0x80	TRNG_SEED3	The SEED3 register holds seed value used in the DRBG - bits [127:96].
0x84	TRNG_SEED4	The SEED4 register holds seed value used in the DRBG - bits [159:128].
0x88	TRNG_SEED5	The SEED5 register holds seed value used in the DRBG - bits [191:160].
0x8C	TRNG_SEED6	The SEED6 register holds seed value used in the DRBG - bits [223:192].
0x90	TRNG_SEED7	The SEED7 register holds seed value used in the DRBG - bits [255:224].
0x94	TRNG_SEED8	The SEED8 register holds seed value used in the DRBG - bits [287:256].
0x98	TRNG_SEED9	The SEED9 register holds seed value used in the DRBG - bits [319:288].
0x9C	TRNG_SEED10	The SEED10 register holds seed value used in the DRBG - bits [351:320].
0xA0	TRNG_SEED11	The SEED11 register holds seed value used in the DRBG - bits [383:352].
0xA4	TRNG_IA_RDATA	Data read from the indirectly accessible register.
0xA8	TRNG_IA_WDATA	Data write to the indirectly accessible register.
0xAC	TRNG_IA_ADDR	Address of the indirectly accessible register.
0xB0	TRNG_IA_CMD	The IA_CMD register is used to access many of the normally inaccessible regions within the TRNG.

66.6.5. Register Description

66.6.5.1. TRNG_CTRL

Description: The CTRL register is used to cause the TRNG to execute one of a number of actions. The are 9 active command operations available within the TRNG (in addition to 0, or NOP). CMD can only be written when the core is idle (see STAT.BUSY) unless the command is a ZEROIZE. Commands other than ZEROIZE during a non-idle condition is ignored.

Table 4- 8 TRNG_CTRL Register

Bits	Name	R/W	Description
31:4			**Reserved Field:** Yes
3:0	CMD	R/W	Execute a command. Enumerated values not listed are 'reserved'. **Values:**

			■ 0x0 (NOP): Execute a NOP ■ 0x1 (GEN_NOISE): Generate full-entropy seed from noise ■ 0x2 (GEN_NONCE): Generate seed from host-written nonce ■ 0x3 (CREATE_STATE): Move DRBG to create state ■ 0x4 (RENEW_STATE): Move DRBG to renew state ■ 0x5 (REFRESH_ADDIN): Move DRBG to refresh addin ■ 0x6 (GEN_RANDOM): Generate a random number ■ 0x7 (ADVANCE_STATE): Advance DRBG state ■ 0x8 (RUN_KAT): Run KAT on DRBG or entropy source ■ 0xf (ZEROIZE): Zeroize **Exists:** Always

66.6.5.2. TRNG_MODE

Description: The MODE register is used to enable or disable certain run-time features within the TRNG. This register can only be written when the core is not busy (see STAT.BUSY).
The SEC_ALG field indicates the security algorithm for the DF and DRBG, which determines the key size of the AES function. A SEC_ALG of 1 is only valid if the core is built with a 256-bit AES. If the AES module is an AES-128, the SEC_ALG never changes to one and a write of 1 to this field is ignored. This field must not change when the DRBG state is instantiated (that is, Create_State command has created an state for the DRBG). Any write to the SEC_ALG field while the DRBG state is instantiated is treated as a NOP.

Table 4- 9 TRNG_MODE Register

Bits	Name	R/W	Description
31:5			**Reserved Field:** Yes
4	ADDIN_PRESENT	R/W	Indicate availability of the Additional Input **Values:** ■ 0x0 (ADDIN_NOT_REQ): No Additional Input required from host ■ 0x1 (ADDIN_REQ): Additional input must be provided by host **Value After Reset:** 0x0 **Exists:** Always

3	PRED_RESIST	R/W	Enable/disable prediction resistance **Values:** ■ 0x0 (PRED_DISABLE): Prediction resistance is not required ■ 0x1 (PRED_ENABLED): Prediction resistance is required **Value After Reset:** 0x0 **Exists:** Always
2	KAT_SEL	R/W	Select test component for known-answer test **Values:** ■ 0x0 (KAT_DRBG): KAT on DRBG ■ 0x1 (KAT_DF): KAT on conditioning component (Derivation Function) **Value After Reset:** 0x0 **Exists:** Always
1	KAT_VEC	R/W	Select test vectors for known-answer test **Values:** ■ 0x0 (KAT_VEC0): KAT0 ■ 0x1 (KAT_VEC1): KAT1 **Value After Reset:** 0x0 **Exists:** Always
0	SEC_ALG	R/W	Select security strength in derivation function and DRBG **Values:** ■ 0x0 (SEC_128): AES-128 ■ 0x1 (SEC_256): AES-256 **Value After Reset:** 0x0 **Exists:** Always

66.6.5.3. **TRNG_SMODE**

Description: The SMODE register is used to enable or disable certain MISSION mode run-time features within the TRNG. This register should only be written when the core is not busy (see STAT.BUSY).

The MAX_REJECTS field dictates the maximum number of consecutive bit rejections that have to occur within a single bit generator before the ring feedback control system dynamically adjusted the frequency of the affected bit generator. There is usually no need to adjust this register. It is primarily for test purposes.

The NONCE field must be set to 1 and remain high as long as the state generated from the host-provided seed is being used in the DRBG.

In order to prevent certain attacks such as backtracking and spoofing, whenever the state of either of SECURE_EN or NONCE fields is changed, the DWC TRNG NIST SP800-90C automatically zeroizes all internal state information.

NOTE: Initiation of a self-seeding operation (a write of 1 to the CTRL.CMD register) when in the nonce mode, automatically clears the NONCE mode bit, performs a zeroize, and the command does not take effect. So, in order to execute a GEN_NOISE command when in the nonce mode, the host should change the mode back to self-seeding and then assert the command.

NOTE: It is not possible to both toggle the SECURE_EN bit and set other fields of the SMODE register in a single write operation since the toggling of SECURE_EN returns the state of the SMODE register to its reset state immediately after the write completes. These changes must be done in separate writes; for example, in order to set the NONCE while SECURE_EN is also changing, toggle the SECURE_EN bit first, then set NONCE mode in a subsequent write operation.

Table 4- 10 TRNG_SMODE Register

Bits	Name	R/W	Description
31:10			**Reserved Field:** Yes
9:2	MAX_REJ ECTS	R/W	Maximum number of consecutive bit rejections before issuing ring tweak. Defaults to 10. **Value After Reset:** 0xa **Exists:** Always
1	SECURE_ EN	R/W	Set the core in mission mode or test mode **Values:** ■ 0x0 (TEST_MODE): test mode ■ 0x1 (MISSION_MODE): mission mode **Exists:** Always
0	NONCE	R/W	Set the core in nonce seeding mode **Values:** ■ 0x0 (NONCE_DISABLED): Disable nonce mode ■ 0x1 (NONCE_ENABLED): Enable nonce mode **Value After Reset:** 0x0 **Exists:** Always

66.6.5.4. **TRNG_STAT**

Description: The STAT register allows the user to monitor the internal status of the TRNG.

NOTE: After the MODE.NONCE bit is set, the host needs to poll the NONCE_MODE in the STAT register until it is 1. Then, the host can load the nonce value into the core.

Table 4- 11 TRNG_STAT Register

Bits	Name	R/W	Description

31	BUSY	R	Indicates the state of the core. **Values:** ■ 0x0 (BUSY_NOT): Idle ■ 0x1 (BUSY_EXEC): Currently executing a command **Exists:** Always **Volatile:** true
30:9			**Reserved Field:** Yes
8:7	DRBG_STATE	R	Reflects how a DRBG state is instantiated. **Values:** ■ 0x0 (DRBG_NOT_INIT): State is not instantiated ■ 0x1 (DRBG_NS): State is instantiated using the built-in noise source ■ 0x2 (DRBG_HOST): State is instantiated using the host-provided nonce **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
6	SECURE	R	Reflects state of **SMODE.SECURE_EN**. **Exists:** Always **Volatile:** true
5	NONCE_MODE	R	Reflects state of **SMODE.NONCE**. **Values:** ■ 0x0 (NONCE_DISABLE): Nonce mode disabled ■ 0x1 (NONCE_ENABLE): Nonce mode enabled (allows**CTRL.CMD** value of 2) **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
4	SEC_ALG	R	Reflects state of **MODE.SEC_ALG**. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
0	NONCE	R/W	Last command **Values:** ■ 0x0 (NOP) ■ 0x1 (GEN_NOISE) ■ 0x2 (GEN_NONCE) ■ 0x3 (CREATE_STATE) ■ 0x4 (RENEW_STATE) ■ 0x5 (REFRESH_ADDIN) ■ 0x6 (GEN_RANDOM)

			■ 0x7 (ADVANCE_STATE) ■ 0x8 (RUN_KAT) ■ 0xf (ZEROIZE) **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

66.6.5.5. **TRNG_IE**

Description: The IE register is used to enable or disable interrupts within the TRNG.

Table 4- 12 TRNG_IE Register

Bits	Name	R/W	Description
31	GLBL	R/W	Global interrupt enable signal for the TRNG. **Values:** ■ 0x0 (GLBL_DIS): Disable **GLBL** interrupt contribution ■ 0x1 (GLBL_EN): Enable **GLBL** interrupt contribution **Exists:** Always
30:5			**Reserved Field:** Yes
4	DONE	R/W	Include or exclude **DONE** interrupt contribution **Values:** ■ 0x0 (DONE_DIS): Disable **DONE** interrupt contribution ■ 0x1 (DONE_EN): Enable **DONE** interrupt contribution **Exists:** Always
3	ALARMS	R/W	Include or exclude **ALARMS** interrupt contribution **Values:** ■ 0x0 (ALARMS_DIS): Disable **ALARMS** interrupt contribution ■ 0x1 (ALARMS_EN): Enable **ALARMS** interrupt contribution **Exists:** Always
2	NOISE_RDY	R/W	Include or exclude **NOISE_RDY** interrupt contribution **Values:** ■ 0x0 (NOISE_RDY_DIS): Disable **NOISE_RDY** interrupt contribution ■ 0x1 (NOISE_RDY_EN): Enable **NOISE_RDY**

			interrupt contribution **Exists:** Always
1	KAT_COMPLETED	R/W	Include or exclude **KAT_COMPLETED** interrupt contribution **Values:** ■ 0x0 (KAT_COMPLETED_DIS): Disable **KAT_COMPLETED** interrupt contribution ■ 0x1 (KAT_COMPLETED_EN): Enable **KAT_COMPLETED** interrupt contribution **Exists:** Always
0	ZEROIZED	R/W	Include or exclude **ZEROIZED** interrupt contribution. **Values:** ■ 0x0 (ZEROIZED_DIS): Disable **ZEROIZED** interruptcontribution ■ 0x1 (ZEROIZED_EN): Enable **ZEROIZED** interruptcontribution **Exists:** Always

66.6.5.6. **TRNG_ISTAT**

Description: The ISTAT register allows the user to monitor the interrupt contributions of the TRNG.

Note that these status signals are used for both polling mode and interrupt modes. Whether they assert the O_irq output depends on whether the IE.GLBL_EN and one or more of the other individual IE bits are set to 1.

ZEROIZED and KAT_COMPLETED bits are only set after a command has explicitly requested for a RUN_KAT or ZEROIZE operation. During an alarm event or a star-up test in which the core automatically performs a zeroize operation or runs a KAT, ZEROIZED and KAT_COMPLETED interrupts does not occur.

Table 4- 13 TRNG_ISTAT Register

Bits	Name	R/W	Description
30:5			**Reserved Field:** Yes
4	DONE	R/W1C	The **DONE** bit allows the user to poll the completion of all commands expect **RUN_KAT** and **ZEROIZE** which have their own interrupt. **Values:** ■ 0x0 (DONE_R0): R0: No unacknowledged command completion ■ 0x1 (DONE_R1): R1: Unacknowledged command

			completion ■ 0x0 (DONE_W0): W0: NOP ■ 0x1 (DONE_W1): W1: Clear DONE flag **Exists:** Always **Volatile:** true
3	ALARMS	R/W1C	The ALARMS bit allows the user to poll failures. When an alarm occurs, an automatic zeroize happens. Clearing this interrupt also clears the O_alarm pin. **Values:** ■ 0x0 (ALARMS_R0): R0: No unacknowledged ALARMS ■ 0x1 (ALARMS_R1): R1: Unacknowledged ALARMS ■ 0x0 (ALARMS_W0): W0: NOP ■ 0x1 (ALARMS_W1): W1: Clear ALARMS flag **Exists:** Always **Volatile:** true
2	NOISE_RDY	R/W1C	When DWC TRNG NIST SP800-90C is generating a full-entropy seed in the self-seeding mode and **SMODE.SECURE_EN** is 0 (TEST mode), the **NOISE_RDY** bit informs the user when 512 bits of noise have been generated. This interrupt never happens in the MISSION mode of operation. **Values:** ■ 0x0 (NOISE_RDY_R0): R0: No unacknowledged noise generation completion ■ 0x1 (NOISE_RDY_R1): R1: Unacknowledged noise generation completion ■ 0x0 (NOISE_RDY_W0): W0: NOP ■ 0x1 (NOISE_RDY_W1): W1: Clear NOISE_RDY flag **Exists:** Always **Volatile:** true
1	KAT_COMPLETED	R/W1C	Indicates the completion of the **RUN_KAT** command. **Values:** ■ 0x0 (KAT_COMPLETED_R0): R0: No unacknowledged KAT_COMPLETED command completion ■ 0x1 (KAT_COMPLETED_R1): R1: Unacknowledged KAT_COMPLETED command completion ■ 0x0 (KAT_COMPLETED_W0): W0: NOP ■ 0x1 (KAT_COMPLETED_W1): W1: Clear KAT_COMPLETED flag

			Exists: Always **Volatile:** true
0	ZEROIZED	R/W1C	Indicates the completion of the **ZEROIZE** command. **Values:** ■ 0x0 (ZEROIZED_R0): R0: No unacknowledged ZEROIZED command completion ■ 0x1 (ZEROIZED_R1): R1: Unacknowledged ZEROIZED command completion ■ 0x0 (ZEROIZED_W0): W0: NOP ■ 0x1 (ZEROIZED_W1): W1: Clear ZEROIZED flag **Exists:** Always **Volatile:** true

66.6.5.7. TRNG_ALARMS

Description: The ALARMS register allows the user to monitor the source of critical alarms. An alarm event appears at both the O_alarm I/O port (non-maskable) and in the ISTAT.ALARMS bit. The TRNG refuses to accept any command as long as any alarm is active.

Any alarm event causes an automatic zeroize of all internal state information. The fields of the ALARMS register can be cleared by writing a 1 to the ILLEGAL_CMD_SEQ field and 4'b1111 to the FAILED_TEST_ID field. Writing any other value to these fields has no effect.

Table 4- 14 TRNG_ALARMS Register

Bits	Name	R/W	Description
31:5			**Reserved Field:** Yes
4	ILLEGAL_CMD_SEQ	R/W	The ILLEGAL_CMD_SEQ field indicates that the S/W Driver has executed an illegal command sequence. **Exists:** Always
0	FAILED_TEST_ID	R/W	When an alarm is issued, the FAILED_TEST_ID field shows which test has failed. This fields only shows the first detected failed test and it should not be assumed that the remaining statistical tests are passed. **Values:** ■ 0x0 (FAILED_TEST_ID_0): no failure ■ 0x1 (FAILED_TEST_ID_1): failure in both KAT and statistical tests ■ 0x2 (FAILED_TEST_ID_2): KAT test failure ■ 0x3 (FAILED_TEST_ID_3): Monobit test failure ■ 0x4 (FAILED_TEST_ID_4): Run test failure ■ 0x5 (FAILED_TEST_ID_5): Long Run test failure

			■ 0x6 (FAILED_TEST_ID_6): Auto-correlation test failure ■ 0x7 (FAILED_TEST_ID_7): Poker test failure ■ 0x8 (FAILED_TEST_ID_8): Repetition Count test failure ■ 0x9 (FAILED_TEST_ID_9): Adaptive Proportion test failure **Exists:** Always

66.6.5.8. **TRNG_COREKIT_REL**

Description: Contains the static coreKit release information.

Table 4- 15 TRNG_COREKIT_REL Register

Bits	Name	R/W	Description
31:28	EXT_ENUM	R	Indicates the coreKit release extension type.For example, release '2.35e-lca04' is encoded as 0x1. **Values:** ■ 0x0 (EXT_ENUM_GA): GA release ■ 0x1 (EXT_ENUM_LCA): LCA release ■ 0x2 (EXT_ENUM_EA): EA release ■ 0x3 (EXT_ENUM_LP): LP release ■ 0x4 (EXT_ENUM_LPC): LPC release ■ 0x5 (EXT_ENUM_SOW): SOW release **Value After Reset:** ELP_NIST_TRNG_CONFIG_COREKIT_RELEASE_EXT_ENUM **Exists:** Always
27:24			**Reserved Field:** Yes
23:16	EXT_VER	R	Indicates the coreKit release extension version number. For example, release '2.35e-lp04' is encoded as 0x4. GA releases has a value of 0. **Value After Reset:** ELP_NIST_TRNG_CONFIG_COREKIT_RELEASE_EXT_VER **Exists:** Always
15:0	REL_NUM	R	Indicates the coreKit release version in pseudo-BCD. For example, release '2.35e-lca04' is encoded as 0x235e. **Value After Reset:**

			ELP_NIST_TRNG_CONFIG_COREKIT_RELEASE_NUM **Exists:** Always

66.6.5.9. TRNG_FEATURES

Description: The FEATURES register returns the state of various build-time parameter values within the TRNG.

Table 4- 16 TRNG_FEATURES Register

Bits	Name	R/W	Description
31:10			**Reserved Field:** Yes
9	AES_256	R	Indicates whether the instantiated AES is 128-bit or 256-bit. **Exists:** Always
8	PS_PRESENT	R	Determines whether the content of the **NPA_DATAx**registers is used as a Personalization String during the Create_State command. **Exists:** Always
7	DIAG_LEVEL_NS	R	Level of diagnostic circuitry for Noise Source output Registers. **Exists:** Always
6:4	DIAG_LEVEL_CLP800	R	Level of diagnostic circuitry for DWC_trng (Noise Source). **Exists:** Always
3:1	DIAG_LEVEL_ST_HLT	R	Level of diagnostic circuitry for the health test. **Exists:** Always
0	SECURE_RST_STATE	R	Indicates if the core resets to MISSION or TEST mode. **Exists:** Always

66.6.5.10. TRNG_RAND0

Description: The RAND0 register is used by the host to read bits [31:0] of the newly generated 128-bit random data.

Table 4- 17 TRNG_RAND0 Register

Bits	Name	R/W	Description

31:0	RAND	R	Random data word 0. **Exists:** Always **Volatile:** true

66.6.5.11. TRNG_RAND1

Description: The RAND1 register is used by the host to read bits [63:32] of the newly generated 128-bit random data.

Table 4- 18 TRNG_RAND1 Register

Bits	Name	R/W	Description
31:0	RAND	R	Random data word 1. **Exists:** Always **Volatile:** true

66.6.5.12. TRNG_RAND2

Description: The RAND2 register is used by the host to read bits [95:64] of the newly generated 128-bit random data.

Table 4- 19 TRNG_RAND2 Register

Bits	Name	R/W	Description
31:0	RAND	R	Random data word 2. **Exists:** Always **Volatile:** true

66.6.5.13. TRNG_RAND3

Description: The RAND3 register is used by the host to read bits [127:96] of the newly generated 128-bit random data.

Table 4- 20 TRNG_RAND3 Register

Bits	Name	R/W	Description
31:0	RAND	R	Random data word 3. **Exists:** Always **Volatile:** true

66.6.5.14. TRNG_NPA_DATA0

Description: The NPA_DATA0 register holds Noise/Nonce/Personalization String/Additional Input - bits [31:0].

Table 4- 21 TRNG_NPA_DATA0 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 0. **Exists:** Always **Volatile:** true

66.6.5.15. TRNG_NPA_DATA1

Description: The NPA_DATA1 register holds Noise/Nonce/Personalization String/Additional Input - bits [63:32].

Table 4- 22 TRNG_NPA_DATA1 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 1. **Exists:** Always **Volatile:** true

66.6.5.16. TRNG_NPA_DATA2

Description: The NPA_DATA2 register holds Noise/Nonce/Personalization String/Additional Input - bits [95:64].

Table 4- 23 TRNG_NPA_DATA2 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 2. **Exists:** Always **Volatile:** true

66.6.5.17. TRNG_NPA_DATA3

Description: The NPA_DATA3 register holds Noise/Nonce/Personalization String/Additional Input - bits [127:96].

Table 4- 24 TRNG_NPA_DATA3 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 3. **Exists:** Always **Volatile:** true

66.6.5.18. TRNG_NPA_DATA4

Description: The NPA_DATA4 register holds Noise/Nonce/Personalization String/Additional Input - bits [159:128].

Table 4- 25 TRNG_NPA_DATA4 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 4. **Exists:** Always **Volatile:** true

66.6.5.19. TRNG_NPA_DATA5

Description: The NPA_DATA5 register holds Noise/Nonce/Personalization String/Additional Input - bits [191:160].

Table 4- 26 TRNG_NPA_DATA5 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 5. **Exists:** Always **Volatile:** true

66.6.5.20. TRNG_NPA_DATA6

Description: The NPA_DATA6 register holds Noise/Nonce/Personalization String/Additional Input - bits [223:192].

Table 4- 27 TRNG_NPA_DATA6 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 6. **Exists:** Always **Volatile:** true

66.6.5.21. TRNG_NPA_DATA7

Description: The NPA_DATA7 register holds Noise/Nonce/Personalization String/Additional Input - bits [255:224].

Table 4- 28 TRNG_NPA_DATA7 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 7. **Exists:** Always **Volatile:** true

66.6.5.22. TRNG_NPA_DATA8

Description: The NPA_DATA8 register holds Noise/Nonce/Personalization String/Additional Input - bits [287:256].

Table 4- 29 TRNG_NPA_DATA8 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 8. **Exists:** Always **Volatile:** true

66.6.5.23. TRNG_NPA_DATA9

Description: The NPA_DATA1 register holds Noise/Nonce/Personalization String/Additional Input - bits [319:288].

Table 4- 30 TRNG_NPA_DATA9 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 9. **Exists:** Always **Volatile:** true

66.6.5.24. TRNG_NPA_DATA10

Description: The NPA_DATA10 register holds Noise/Nonce/Personalization String/Additional Input - bits [351:320].

Table 4- 31 TRNG_NPA_DATA10 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 10. **Exists:** Always **Volatile:** true

66.6.5.25. TRNG_NPA_DATA11

Description: The NPA_DATA11 register holds Noise/Nonce/Personalization String/Additional Input - bits [383:352].

Table 4- 32 TRNG_NPA_DATA11 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 11. **Exists:** Always **Volatile:** true

66.6.5.26. TRNG_NPA_DATA12

Description: The NPA_DATA12 register holds Noise/Nonce - bits [415:384].

Table 4- 33 TRNG_NPA_DATA12 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 12. **Exists:** Always **Volatile:** true

66.6.5.27. TRNG_NPA_DATA13

Description: The NPA_DATA13 register holds Noise/Nonce - bits [447:416].

Table 4- 34 TRNG_NPA_DATA13 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 13. **Exists:** Always **Volatile:** true

66.6.5.28. TRNG_NPA_DATA14

Description: The NPA_DATA14 register holds Noise/Nonce - bits [479:448].

Table 4-35 TRNG_NPA_DATA14 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 14. **Exists:** Always **Volatile:** true

66.6.5.29. TRNG_NPA_DATA15

Description: The NPA_DATA15 register holds Noise/Nonce - bits [511:480].

Table 4-36 TRNG_NPA_DATA15 Register

Bits	Name	R/W	Description
31:0	NPA_DATA	R/W	NPA data word 15. **Exists:** Always **Volatile:** true

66.6.5.30. TRNG_SEED0

Description: The SEED0 register holds seed value used in the DRBG - bits [31:0].

Table 4-37 TRNG_SEED0 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 0. **Exists:** Always **Volatile:** true

66.6.5.31. TRNG_SEED1

Description: The SEED1 register holds seed value used in the DRBG - bits [63:32].

Table 4-38 TRNG_SEED1 Register

Bits	Name	R/W	Description

31:0	SEED	R/W	SEED data word 1. **Exists:** Always **Volatile:** true

66.6.5.32. TRNG_SEED2

Description: The SEED2 register holds seed value used in the DRBG - bits [95:64].

Table 4-39 TRNG_SEED2 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 2. **Exists:** Always **Volatile:** true

66.6.5.33. TRNG_SEED3

Description: The SEED3 register holds seed value used in the DRBG - bits [127:96].

Table 4-40 TRNG_SEED3 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 3. **Exists:** Always **Volatile:** true

66.6.5.34. TRNG_SEED4

Description: The SEED4 register holds seed value used in the DRBG - bits [159:128].

Table 4-41 TRNG_SEED4 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 4. **Exists:** Always **Volatile:** true

66.6.5.35. TRNG_SEED5

Description: The SEED5 register holds seed value used in the DRBG - bits [191:160].

Table 4-42 TRNG_SEED5 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 5. **Exists:** Always **Volatile:** true

66.6.5.36. TRNG_SEED6

Description: The SEED6 register holds seed value used in the DRBG - bits [223:192].

Table 4-43 TRNG_SEED6 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 6. **Exists:** Always **Volatile:** true

66.6.5.37. TRNG_SEED7

Description: The SEED7 register holds seed value used in the DRBG - bits [255:224].

Table 4-44 TRNG_SEED7 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 7. **Exists:** Always **Volatile:** true

66.6.5.38. TRNG_SEED8

Description: The SEED8 register holds seed value used in the DRBG - bits [287:256].

Table 4-45 TRNG_SEED8 Register

Bits	Name	R/W	Description

31:0	SEED	R/W	SEED data word 8. **Exists:** Always **Volatile:** true

66.6.5.39. TRNG_SEED9

Description: The SEED9 register holds seed value used in the DRBG - bits [319:288].

Table 4- 46 TRNG_SEED9 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 9. **Exists:** Always **Volatile:** true

66.6.5.40. TRNG_SEED10

Description: The SEED10 register holds seed value used in the DRBG - bits [351:320].

Table 4- 47 TRNG_SEED10 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 10. **Exists:** Always **Volatile:** true

66.6.5.41. TRNG_SEED11

Description: The SEED11 register holds seed value used in the DRBG - bits [383:352].

Table 4- 48 TRNG_SEED11 Register

Bits	Name	R/W	Description
31:0	SEED	R/W	SEED data word 11. **Exists:** Always **Volatile:** true

66.6.5.42. TRNG_IA_RDATA

Description: Data read from the indirectly accessible register.

Table 4- 49 TRNG_IA_RDATA Register

Bits	Name	R/W	Description
31:0	RDATA	R	The **RDATA** register holds the data read from the IAP register pointed at by the IA_ADDR register. The actual read operation occurs when **IA_CMD.GO** is 1, **IA_CMD.W_nR**bits is 0 and the **SMODE.SECURE_EN** bit is 0. The returned data is available by the time the host reads the **IA_RDATA** register essentially immediately. **Exists:** Always **Volatile:** true

66.6.5.43. TRNG_IA_WDATA

Description: Data write to the indirectly accessible register.

Table 4- 50 TRNG_IA_WDATA Register

Bits	Name	R/W	Description
31:0	WDATA	R/W	The **WDATA** register holds the data to write to the IAP register pointed at by the IA_ADDR register. The actual write operation occurs when both the **IA_CMD.GO** and **IA_CMD.W_nR**bits are 1 and the **SMODE.SECURE_EN** bit is 0. **Exists:** Always **Volatile:** true

66.6.5.44. TRNG_IA_ADDR

Description: Address of the indirectly accessible register.

Table 4- 51 TRNG_IA_ADDR Register

Bits	Name	R/W	Description
31:10			**Reserved Field:** Yes

9:0	ADDR	R/W	The **ADDR** register holds the address of the IAP register to be accessed within the TRNG. **Exists:** Always **Volatile:** true

66.6.5.45. TRNG_IA_CMD

Description: The IA_CMD register is used to access many of the normally inaccessible regions within the TRNG. This register is not functional when SMODE.SECURE_EN is set 1.
Note that the GO field, as with all R/W1 bits, is auto-clearing and does not need to be written back to 0 by the host.
To use the IAP, the host must load the desired IAP address to the IA_ADDR register. If the operation is a write, the host must load the data to be written into the IA_WDATA register and load 32'h8000_0001 into the IA_CMD register. If the operation is a read, the host must write 32'h8000_0000 to the IA_CMD register and read the result from the IA_RDATA register.

Table 4- 52 TRNG_IA_CMD Register

Bits	Name	R/W	Description
31	GO	R/W	Execute W_nR command. **Exists:** Always **Volatile:** true
30:1			**Reserved Field:** Yes
9:0	W_nR	R/W	The **W_nR**register holds the write or read command to/from the IAP register. **Exists:** Always **Volatile:** true

66.7. Efuse Controller

66.7.1. Overview

This Efuse Controller is use to implement the operations that program and read the Efuse. and to enable and disable certain functions in the SOC.

66.7.2. Features

- One-time programmable nonvolatile EFUSE storage cells organized as 64x 1 bit
- 10us burning pulse width
- Control by Test mode or function mode

66.7.3. Efuse memory organization

S40NLLEFUSE_SISO64B_F2 is a serial-in/serial-out Electrical Fuse Macro IP which has 64B bits internal nonvolatile one-time programmable EFUSE storage. With a serial interface, 1-bit can be programmed each time in program mode and 1-bit can be read at one time in read mode.

66.7.4. Efuse Organization

Bit Address	Content	Comment
Bit0	boot_qspi_dio_mode[0]	Determin QSPI mode for mask ROM: 2’b00(Default): SPI mode 2’b01: Dual SPI mode 2’b10: Quad SPI mode 2’b11: Reserved
Bit1	boot_qspi_dio_mode[1]	Determin QSPI mode for mask ROM: Determin QSPIo mode for mask ROM: 2’b00(Default): SPI mode 2’b01: Dual SPI mode 2’b10: Quad SPI mode 2’b11: Reserved
Bit2	boot_qspi_clock_freq[0]	Determin QSPI clock frequency for mask ROM: 2’b00(default): 50MHz 2’b01: 37.5Mhz 2’b10: 30MHz 2’b11: 25MHz
Bit3	boot_qspi_clock_freq[1]	Determin QSPI clock frequency for mask ROM: 2’b00(default): 50MHz 2’b01: 37.5Mhz 2’b10: 30MHz 2’b11: 25MHz
Bit4	boot_uart_print_disable	Enable/Disable uart print in mask ROM: 1’b0(Default): Enable 1’b1: Disable
Bit5	boot_performance_test_enable	Enable print for boot performance testing. 1’b0(Default): Disable 1’b1: Enable

Bit6~Bit58	Reserved	Could be used by software after first-stage boot
59bit	jtag_disable	Jtag disable bit
60bit	Ldotrim[0]	Trim bit for LDO
61bit	Ldotrim[1]	Trim bit for LDO
62bit	Ldotrim[2]	Trim bit for LDO
63bit	Burn success indicator	Indicate whether efuse has been fused at production test

66.7.5. Functional Description

66.7.5.1. Block Diagram

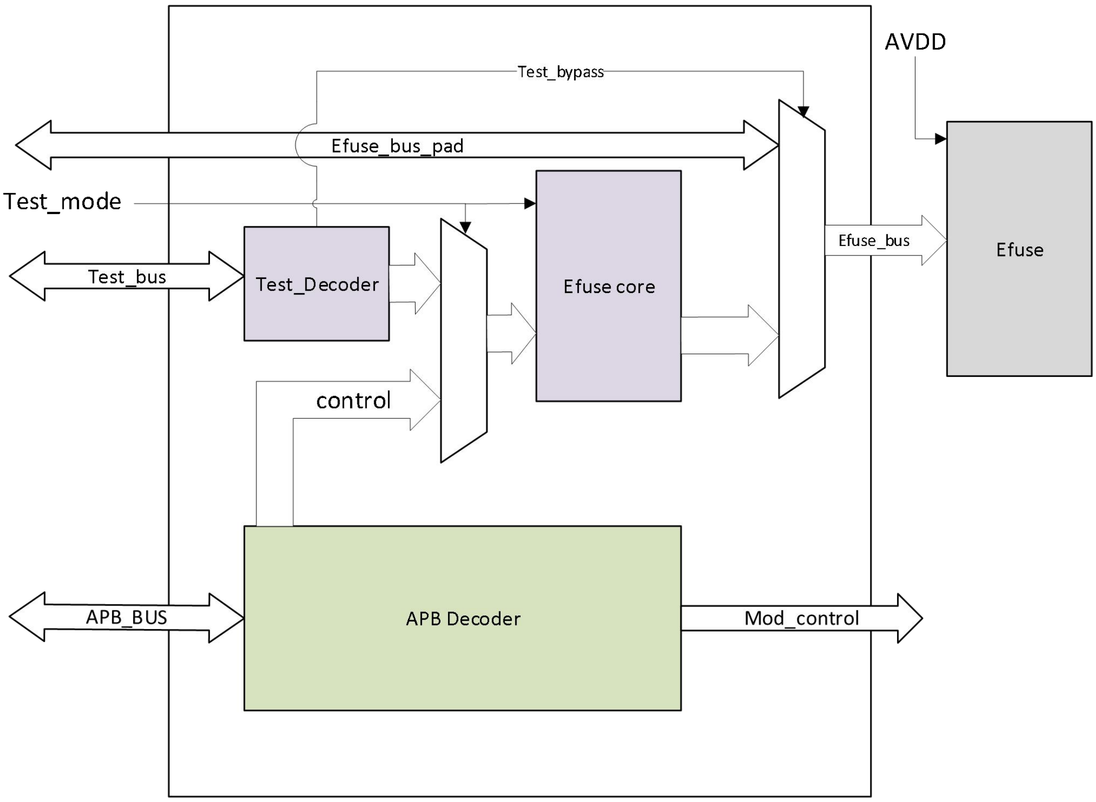


Figure 4- 4

66.7.5.2. APB Decoder

The Efuse controller provides a dedicated APB bus interface that is used to access the Efuse Core. The Efuse controller converts APB reads and writes into accesses to the Efuse Core. Finally, the control signal of the module is generated. The APB data width is fixed to 32 bits. The APB address width is 12 bits.

66.7.5.3. **Test Decoder**

efuse controller in test mode, the efuse controller can write efuse through Program mode on Test bus or Efuse Bus. Test bus consists of signal lines such as address, data, clock, etc. The test bus completes the burning of Efuse by configuring registers. The Efuse Bus is a straight-through mode that connects the control line of Efuse directly to the PAD, external logic to fully control the working sequence of Efuse.

66.7.5.4. **Efuse Core**

thus completing the burning. This module generates program/read commands to Efuse based on commands transmitted from AHB Decoder or Test Decoder.

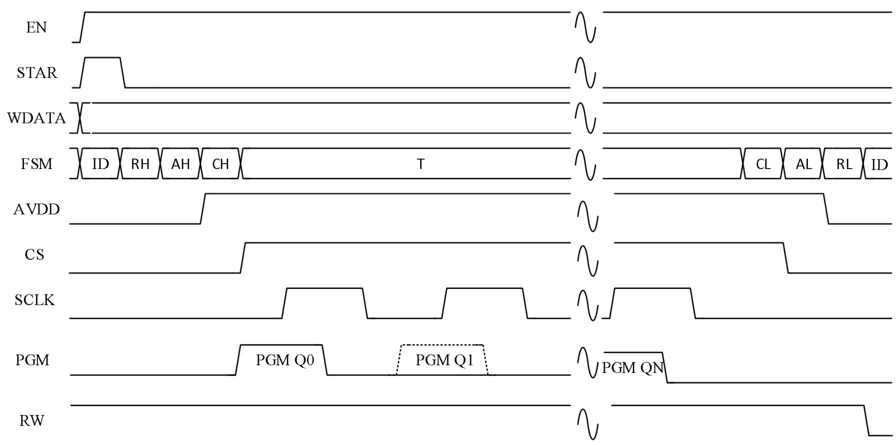


Figure 4- 5

66.7.6. **Register Summary**

66.7.6.1. **APB Register Summary**

Registers	Offset	Description
PRG_WDATA_L	0x00	Efuse program data register low
PRG_WDATA_H	0x04	Efuse program data register high
PRG_RDATA_L	0x08	Efuse read data low
PRG_RDATA_H	0x0c	Efuse read data ligh
PRG_EN	0x10	Efuse program enable register
PRG_START	0x14	Efuse program start trans mode register
PRG_WIRTE	0x18	Efuse program wirte or read sel
PRG_STATE	0x1c	Efuse program status register
PRG_PERIOD	0x20	Efuse program clk period config

66.7.6.2. TEST MODE Register Summary

Registers	Offset	Description
TMS_WDATA_0	0x00	Efuse program data register 0
TMS_WDATA_1	0x01	Efuse program data register 1
TMS_WDATA_2	0x02	Efuse program data register 2
TMS_WDATA_3	0x03	Efuse program data register 3
TMS_WDATA_4	0x04	Efuse program data register 4
TMS_WDATA_5	0x05	Efuse program data register 5
TMS_WDATA_6	0x06	Efuse program data register 6
TMS_WDATA_7	0x07	Efuse program data register 7
TMS_RDATA_0	0x08	Efuse read data register 0
TMS_RDATA_1	0x09	Efuse read data register 1
TMS_RDATA_2	0x0a	Efuse read data register 2
TMS_RDATA_3	0x0b	Efuse read data register 3
TMS_RDATA_4	0x0c	Efuse read data register 4
TMS_RDATA_5	0x0d	Efuse read data register 5
TMS_RDATA_6	0x0e	Efuse read data register 6
TMS_RDATA_7	0x0f	Efuse read data register 7
TMS_START	0x10	Efuse start trans mode register
TMS_WIRTE	0x11	Efuse trans wirte or read mode
TMS_STATUS	0x12	Efuse program status register
TMS_BYPASS	0x13	Test mode thought effuse to pin
TMS_APERIOD	0x14	Test mode clk period count
TMS_LPERIOD	0x15	Test mode clk low period count

66.7.7. Register Description

66.7.7.1. APB Register Description

Efuse register(PRG_WDATA,offset:0x00)

Bit field	Name	R/W	Reset value	Description
31:0	DATA_IN	R/W	0	Efuse program low write data

Efuse register(PRG_WDATA1,offset:0x04)

Bit field	Name	R/W	Reset value	Description
31:0	DATA_IN	R/W	0	Efuse program high write data

Efuse register(PRG_RDATA,offset:0x08)

Bit field	Name	R/W	Reset value	Description
31:0	DATA_OUT	R	0	Efuse program low read data

Efuse register(PRG_RDATA,offset:0x0C)

Bit field	Name	R/W	Reset value	Description
31:0	DATA_OUT	R	0	Efuse program high read data

Efuse program enable register (PRG_EN,offset:0x10)

Bit field	Name	R/W	Reset value	Description
31:1	Reserved			
0	PRG_START_EN	R/W	0	When trans: PRG_START is valid only if PRG_EN is valid 0x00: PRG_START disable 0x01: PRG_START enable

Efuse program enable register (PRG_STATR, offset:0x14)

Bit field	Name	R/W	Reset value	Description
31:1	Reserved			
0	PRG_TRANS START	R/W	0	Automatic zero clearing 0x00: write 0 is ignored 0x01: program data to Efuse

Efuse program enable register (PRG_ WIRTE, offset:0x18)

Bit field	Name	R/W	Reset value	Description
31:1	Reserved			
0	PRG_ WIRTE	R/W	0	0x00: effuse read mode 0x01: effuse write mode

Efuse program enable register (PRG_STATUS, offset:0x1c)

Bit field	Name	R/W	Reset value	Description
31:2	Reserved			
1	PRG_BUSY	R	0	0x00: Indicates that Efuse is idle 0x01: Indicates that Efuse is being operated and cannot perform other actions

0	PRG_COMP	R	0	0x00: indicates the EFUSE program is not completed 0x01: indicates the EFUSE program is completed. This state points to the 64th bit of effuse.

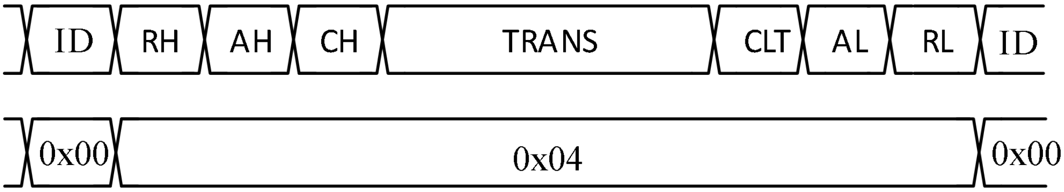


Figure 4- 6

Efuse register (PRG_PERIOD, offset:0x20)

Bit field	Name	R/W	Reset value	Description
31:26	Reserved			
25:16	PRG_LPERIOD	R/W	1	Clock low level counter. Prg_lperiod must be less than 1/2 Prg_aperiod PRG_LPERIOD must > 0
15:11	Reserved			
10:0	PRG_APERIOD	R/W	3	Clock cycle count SCLK = PRG_APERIOD*(1/P_CLK) PRG_APERIOD must > 2

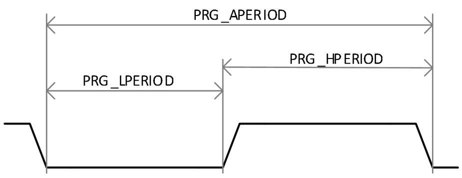


Figure 4- 7

Note: PRG HPERIOD must be between 9us and 11us.

66.7.7.2. **Test mode Register Description**

Efuse register(TMS_WDATA_0~ TMS_WDATA_8,offset:0x00~0x07)

Bit field	Name	R/W	Reset value	Description
7:0	TMS_PROG_DATA 0	R	0	Efuse program write data 0
7:0	TMS_PROG_DATA 1	R	0	Efuse program write data 1
7:0	TMS_PROG_DATA 2	R	0	Efuse program write data 2
7:0	TMS_PROG_DATA 3	R	0	Efuse program write data 3

7:0	TMS_PROG_DATA 4	R	0	Efuse program write data 4
7:0	TMS_PROG_DATA 5	R	0	Efuse program write data 5
7:0	TMS_PROG_DATA 6	R	0	Efuse program write data 6
7:0	TMS_PROG_DATA 7	R	0	Efuse program write data 7

Efuse register(TMS_WDATA_0~ TMS_WDATA_8,offset:0x08~0x0f)

Bit field	Name	R/W	Reset value	Description
7:0	TMS_READ_DATA 0	R/W	0	Efuse program write data 0
7:0	TMS_READ_DATA 1	R/W	0	Efuse program write data 1
7:0	TMS_READ_DATA 2	R/W	0	Efuse program write data 2
7:0	TMS_READ_DATA 3	R/W	0	Efuse program write data 3
7:0	TMS_READ_DATA 4	R/W	0	Efuse program write data 4
7:0	TMS_READ_DATA 5	R/W	0	Efuse program write data 5
7:0	TMS_READ_DATA 6	R/W	0	Efuse program write data 6
7:0	TMS_READ_DATA 7	R/W	0	Efuse program write data 7

Efuse program enable register (TMS_STATR, offset:0x10)

Bit field	Name	R/W	Reset value	Description
7:1	Reserved			
0	TMS_TRANS START	R/W	0	Automatic zero clearing 0x00: write 0 is ignored 0x01: program data to Efuse

Efuse program enable register (TMS_WIRTE, offset:0x11)

Bit field	Name	R/W	Reset value	Description
7:1	Reserved			
0	TMS_WIRTE	R/W	0	0x00: effuse read mode 0x01: effuse write mode

Efuse program enable register (TMS_STATUS, offset:0x12)

Bit field	Name	R/W	Reset value	Description
7:2	Reserved			
1	TMS_BUSY	R	0	0x00: Indicates that Efuse is idle 0x01: Indicates that Efuse is being operated and cannot perform other actions
0	TMS_COMP	R	0	0x00: indicates the EFUSE program is not completed 0x01: indicates the EFUSE program is

				completed. This state points to the 64th bit of effuse.

Efuse program enable register (TMS_APERIOD, offset:0x13)

Bit field	Name	R/W	Reset value	Description
7:4	Reserved			
3:0	TMS_APERIOD	R/W	3	Clock cycle count Example: SCLK = PRG_APERIOD*(1/P_CLK)

Efuse register (TMS_LPERIOD, offset:0x14)

Bit field	Name	R/W	Reset value	Description
7:4	Reserved			
3:0	TMS_LPERIOD	R/W	1	Clock low level counter. Prg_lperiod must be less than 1/2 Prg_aperiod

Efuse register (TMS_BYPASS, offset:0x15)

Bit field	Name	R/W	Reset value	Description
7:1	Reserved			
0	TMS_BYPASS	R/W	0	0x00: test mode control 0x01: pins of Efuse connected to PAD

66.8. LDO

66.8.1. Overview

There are two on-chip LDOs in the SOC to provide power for off-chip image sensor to reduce the total PCB BOM cost. The LDOA generates 1.2V-1.8V power supply. The LDOB generates 2.0V-3.3V power supply. Each output supports voltage configuration with 0.1V per-step.

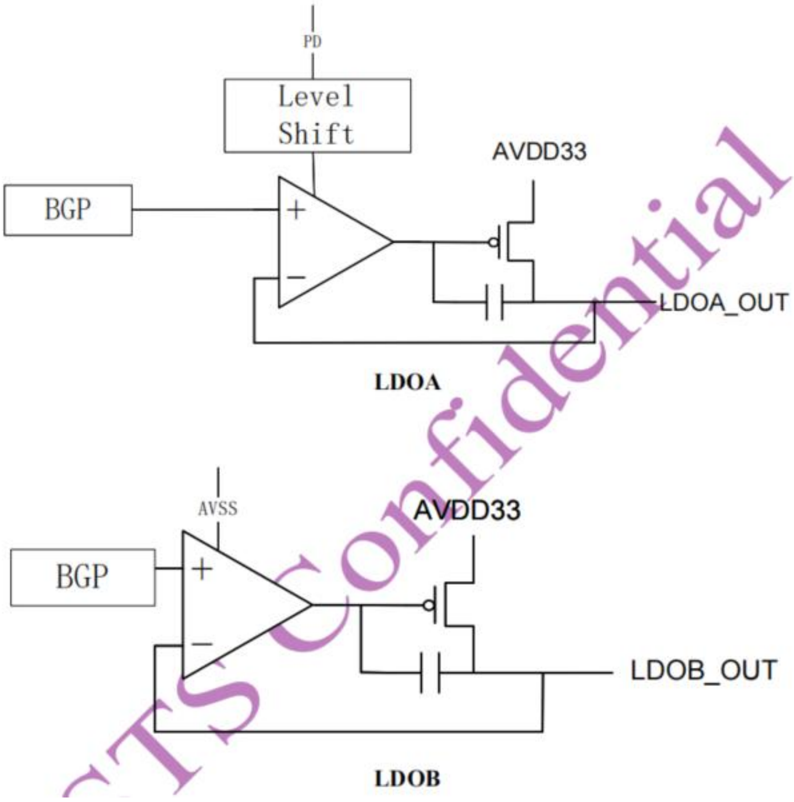


Figure 4- 8 Block Diagram of LDOA and LDOB

66.8.2. **Features**

- LDOA
  - 1.2V-1.8V voltage output
  - On/Off controlled by PD pin, which is driven by register bit
  - 100mA driving capability
  - 300mA voltage dropout
- LDOB
  - 2.0V-3.3V voltage output
  - On/Off controlled by PD pin, which is driven by register bit
  - 100mA driving capability
  - 300mA voltage dropout

66.8.3. **Functional Description**

The LDOA generates 1.2V-1.8V power supply. The LDOB generates 2.0V-3.3V power supply. Each output supports voltage configuration with 0.1V per-step.The power on/off, voltage output control and trim values are all controlled by registers.

66.8.4. **Register Description**

Please refer to 4.5.5.38 to 4.5.5.42 for LDO register description, which including LDOA_CTRL, LDOA_CFG, LDOB_CTRL, LDOB_CFG and LDO_BGTRIM.

66.8.5. **Programming Guide**

TBD.

67. Memory Interface

67.1. DDR

The Storm DDR controller and DDR PHY combine to create a complete solution for connecting an SoC application bus to the DDR memory devices. The combined solution enables the highest DDR performance and bandwidth, with the controller initiating DDR commands to transfer data with maximum efficiency.

67.1.1. Features

- Compatible with JEDEC standard DDR/LPDDR SDRAMs.
- Operating range of 400Mbps to 1066Mbps in DDR2/LPDDR2 mode
- 16-bit data-bus widths
- Supports 256, 512, 1024, and 2048-word page sizes
- Supports burst length of 4,8,16(sequential burst)
- Self-refresh and Power-Down modes for low power:
- Programmable I/O output and ODT impedance with dynamic PVT compensation
- Embedded Dynamic Drt Detection in the PHY to facilitate Dynamic Drt Compensation with the controller
- Utilizes Master and Slave DLLs for precise timing management

67.1.2. **Functional Description**

67.1.2.1. **Block Diagram**

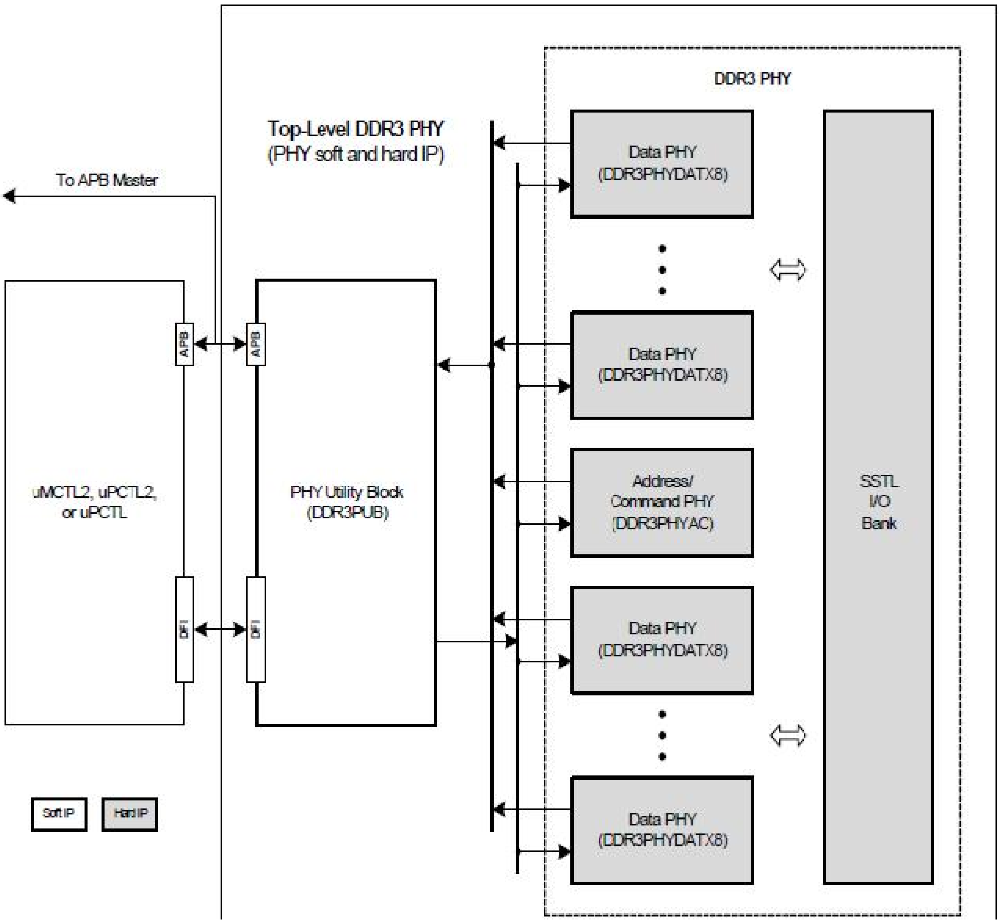


Figure 5.1.2- 1 DDR3/2 Subsystem Block Diagram

The DDR subsystem solution uses the PUB's (PHY Utility Block) DFI 2.1 interface to connect the DDR Controller to the PHY. The DDR PHYs utilize a PHY Utility Block (PUB) for managing operations such as data training, PHY calibration, BIST, etc.

67.1.2.2. **DDR Controller**

The DDR Controller controller converts system bus transactions into memory commands on the DFI interface that are compliant with the DDR protocols. Figure following is the block diagram of the DDR Controller.

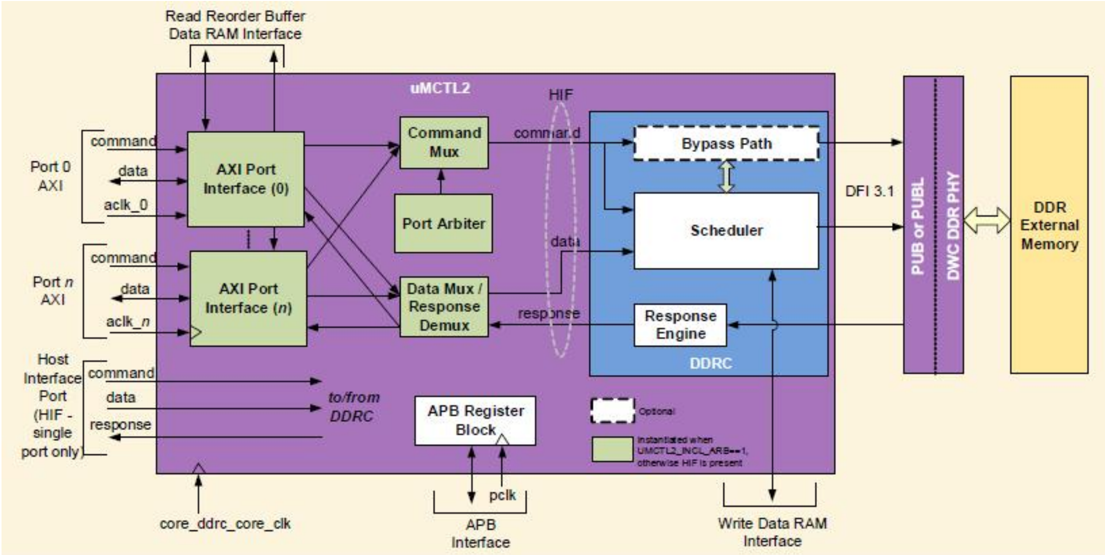


Figure 5.1.2- 2 DDR3/2 Controller Block Diagram

The DDR Controller controller performs the following functions:

Accepts requests from the SoC core with system addresses and associated data for writes. Performs address mapping from system addresses to SDRAM addresses (rank, bank, bank group, row).

- Prioritizes requests to minimize the latency of reads (especially high priority reads) and maximize page hits.
- Ensures that the SDRAM is properly initialized.
- Ensures that all requests made to the SDRAM are legal (accounting for associated SDRAM constraints).
- Ensures that refreshes and other SDRAM and PHY maintenance requests are inserted as required.
- Controls when the SDRAM enters and exits the various power-saving modes appropriately

67.1.2.3. DDR PHY Utility Block

The PUBL is a soft IP component to be used with the DDR PHY. It provides control features to ease the implementation of digitally controlled features of the PHY such as initialization, DQS gate training, and programmable configuration controls. The PUBL has built-in self test features to provide support for production testing of the DDR PHY. It also provides a DFI 2.1 compliant interface to the PHY.

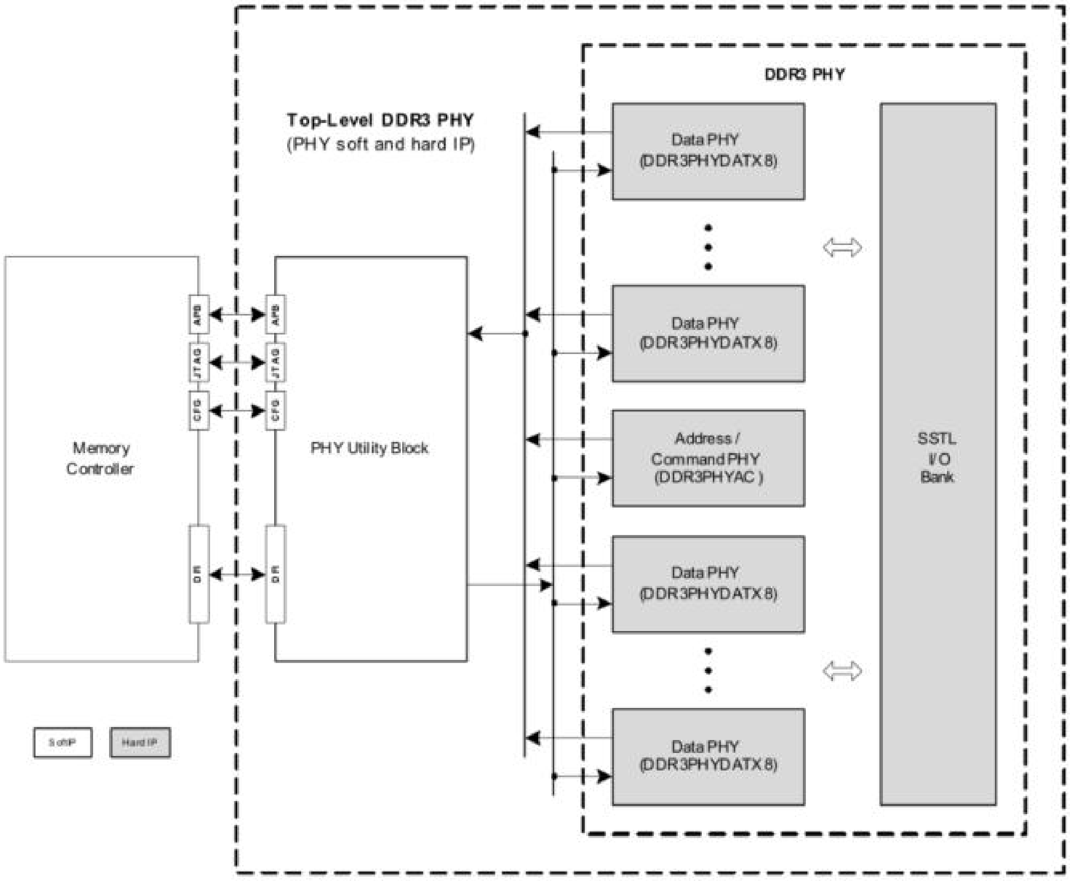


Figure 5.1.2- 3 DDR MulitPhy PUBL interface and connetion overview

The PUBL supports the following features:

- Scalable performance from 0 MHz (LPDDR) through 1066
- Maximum controller clock frequency of 533 MHz resulting in maximum SDRAM data rate of 1066Mbps
- Data path width scales in 8-bit increments
- Multiple memory rank support
- Complete PHY initialization, training and control
- Automatic DQS gate training and drift compensation
- At-speed built-in-self-test (BIST) loopback testing on both the address and data channels for DDR multiPHY
- PHY control and configuration registers
- APB or generic interfaces to configuration registers
- Additional JTAG interface to configure registers
- DFI 2.1 interface

67.1.2.4. **DDR PHY**

The DDR PHY includes the following features:

- Compatible with JEDEC standard DDR2/ DDR3/DDR3U/LPDDR (or Mobile DDR)/ LPDDR2 SDRAMs
- Operating range of 100MHz (200Mb/s) to 533MHz(1066Mb/s) in DDR2/DDR3/DDR3U/LPDDR2 modes
- PHY Utility Block (PUBL) component
- DFI 2.1 compliant interface to controller
- At-speed loopback testing
- Configurable external data bus widths in 8-bit increments
- Permits operating with SDRAMs using data widths narrower than the implemented data width
- Programmable output and ODT impedance with dynamic PVT compensation
- Embedded Dynamic Drift Detection in the PHY to facilitate Dynamic Drift Compensation with the controller
- Utilizes Master and Slave DLLs for precise timing management
- Lane-based architecture (Byte Lane, Command Lane)
- Test modes supporting IDDq and DLL characterization
- Library-based hard-IP PHY to permit maximum flexibility while ensuring high data rates

The PHY is implemented by using Library-based hard-IP approach. Both data and command lanes are composed of ITM, DLL and SSTL IO. And all components of the Byte Lane PHY are designed to permit connectivity by abutment. The ITM connects by abutment to the SSTL I/O, and the DLL connects by abutment to the ITM.

The data and command lane micro- architect are illustrated as below figures.

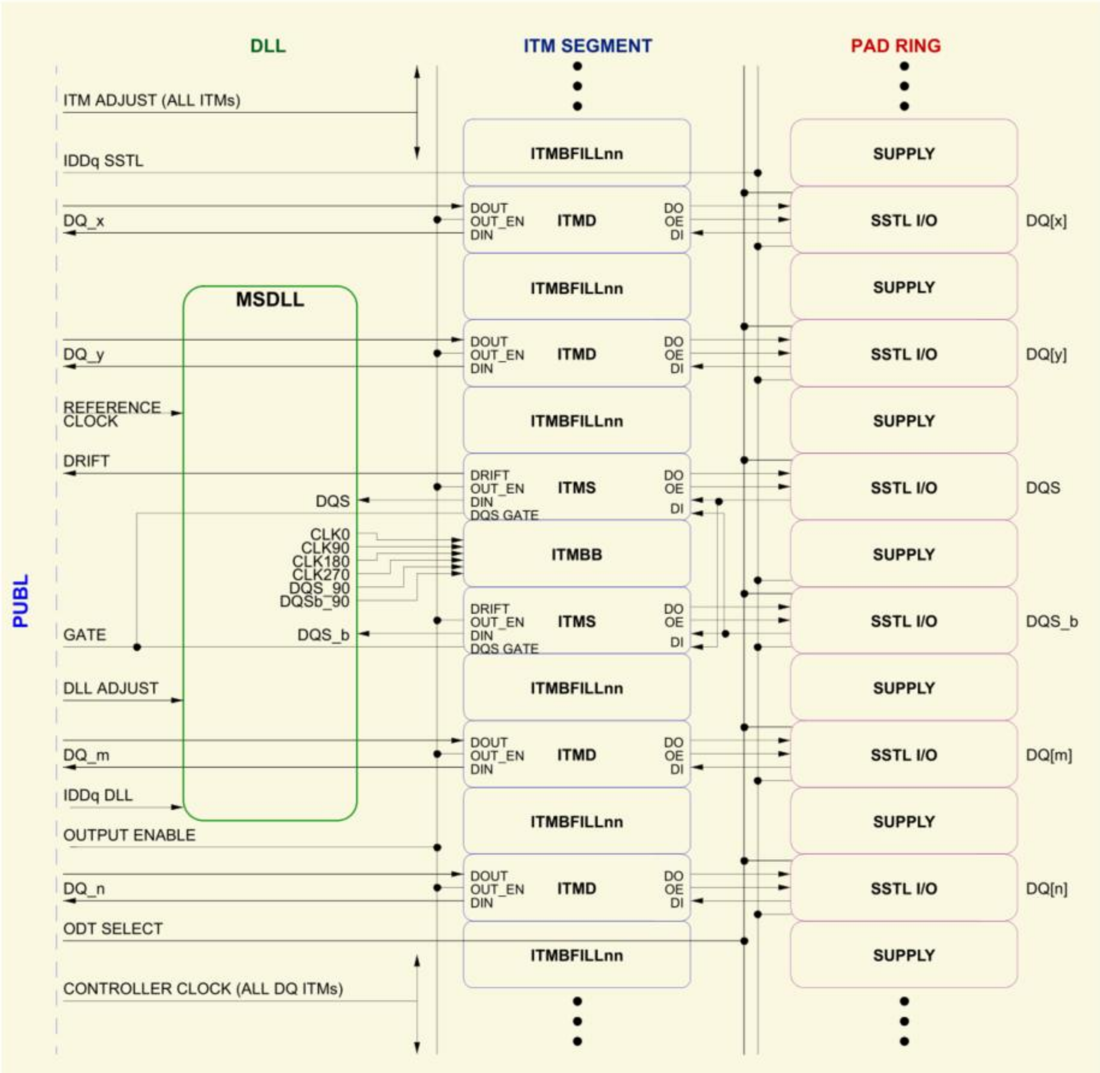


Figure 5.1.2- 4 DDR PHY Data Lane Overview

A Byte Lane consists of the following I/O slots:

- 8 data bits (DQ)
- data strobe bits (DQS / DQS_b)
- 1 data mask bit (DM)
- I/O power and ground cells
- Core power and ground cells

The number of required power cells is dependent on desired operating frequencies, packaging options, and core power requirements of the chip logic.

The ITMs provide a mechanism for monitoring read timing drift, which can be used to adjust timing to maintain optimum system margins. Drift analysis and compensation is performed by the controller on a per Byte Lane basis. The ITM components contain the functions to monitor DQS drift and permit timing adjustment, the controller provides the analysis and control for these functions. These functions operate dynamically for each data bit of every user-issued Read command.

A DLL microcell (MSDLL) consisting of a master DLL and 2 slave DLLs (mirror delay lines) is utilized at each Byte Lane to facilitate optimal PHY timing for drive and capture of DDR data streams, and allows the Lanes to be independent. The master DLL section provides outputs for DDR data stream creation to the SDRAMs and acts as a reference for the slave delay line sections. The slave delay line sections translate the incoming DQS/DQS_b into the center of the read data eye to maximize read system timing margins.

The DDR-specific SSTL I/Os include programmable ODT and output impedance selection. The ODT and output impedances can be dynamically calibrated to compensate for variations in voltage and temperature. The ODT feature can be disabled by the controller. When ODT is enabled by the controller, the SSTL I/O automatically enables its internal ODT circuitry when in input mode and disable this circuitry when in output mode, as determined by the output enable signal. The initial programming and subsequent calibration of the of the ODT and output impedance is achieved through the use of an impedance control loop that can be triggered to calibrate the ODT and output impedance values at the I/Os based on the desired impedance value when compared to an precision external resistor. All the necessary pieces of the impedance control loop are included in the SSTL I/O library. The SSTL I/Os include embedded boundary scan support logic to permit the insertion of boundary scan by the customer without interfering with the matched timing of the mission-mode paths.

DDR command and address lane is also composed the same as data lane of ITM, DLL and SSTL IO, which is illustrated as Figure 13-7.

A typical Command Lane consists of the following I/O slots:

- 1 system clock input (optional)
- Memory clocks (CK/CK_b)
- Command signals (RAS_b, CAS_b, WE_b)
- 1 or more clock enable (CKE)
- 1 or more on-die termination (ODT)
- 0 or more chip select (CS_b)
- 2 or 3 bank address (BA)
- Up to 16 row/column address (A)
- I/O power and ground cells
- Core power and ground cells

Note that for LPDDR2, the address/command address bus (CA) is 10-bit wide and there is no command signals (RAS_b, CAS_b, WE_b).

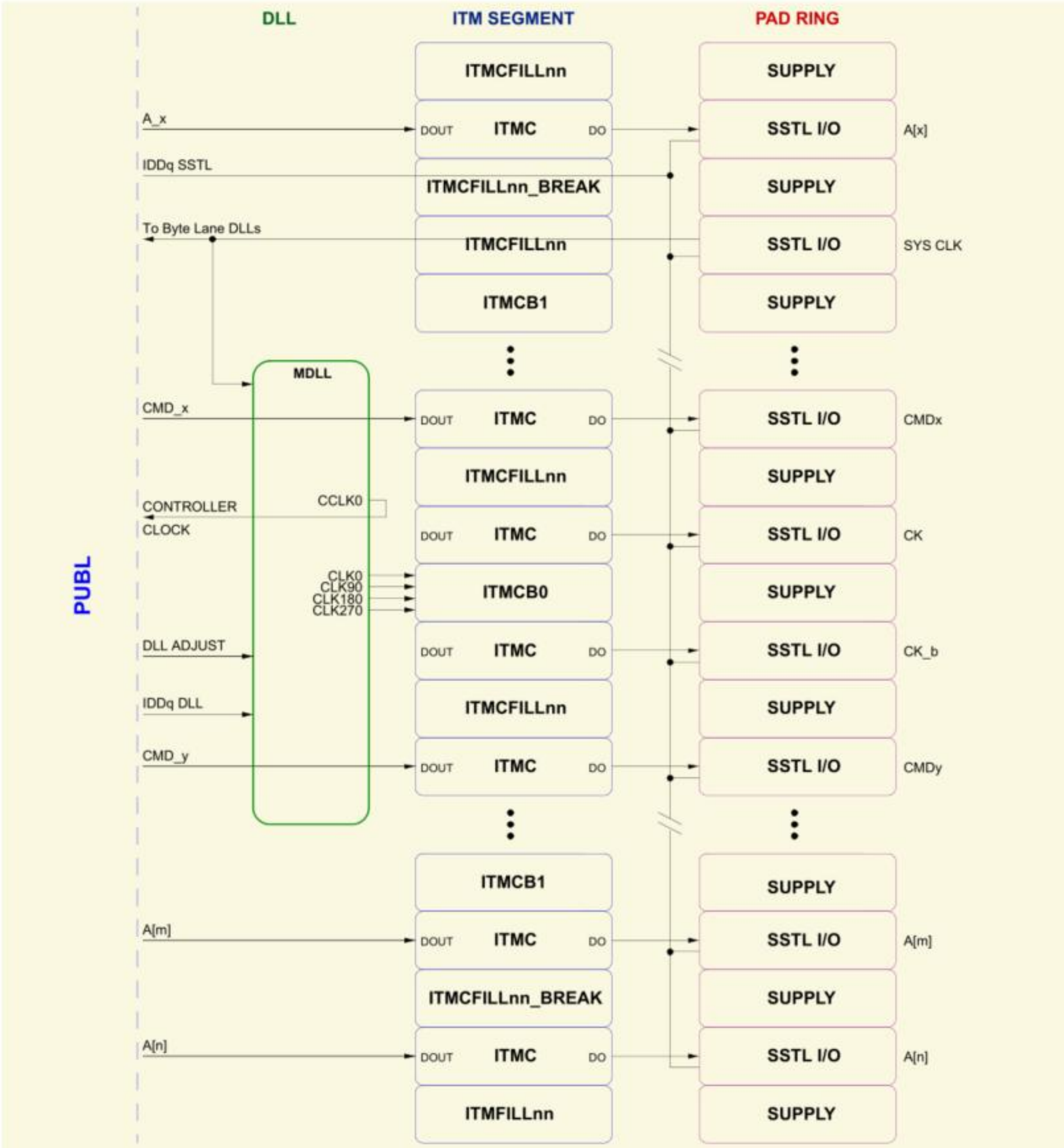


Figure 5.1.2- 5 DDR MulitPHY Command Lane Overview

67.1.3. Programming Guide

67.1.3.1. Initialization

This section describes the programming sequence that must be executed at power-up to bring the DDR controller, the PHY, and the memories into a state where HIF reads and writes can be performed.

67.1.3.2. **Clocks, Resets, and Registers**

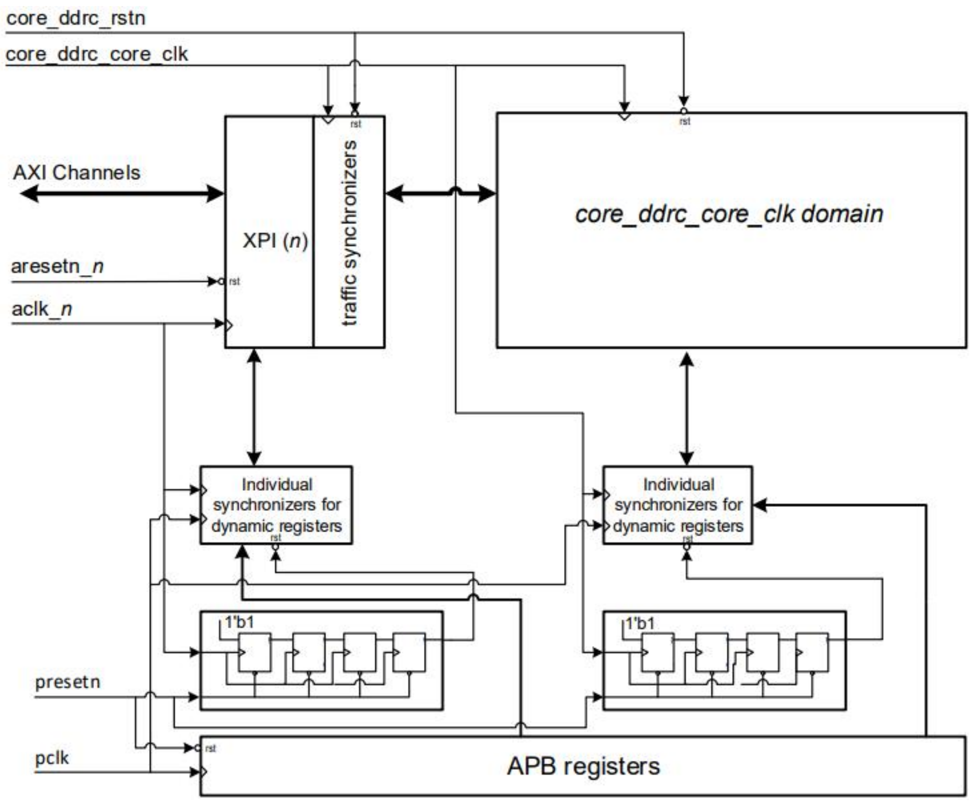


Figure 5.1.3- 1 Three Clock Domain in the DDR Solution

Initialize all the APB registers in reset mode (core_ddrc_rstn is asserted). The behavior of the DDR Controller is unpredictable if these registers contain illegal values. when the core_ddrc_rstn signal is deasserted, or any time thereafter.

Some APB registers can be changed at any time. They are defined as dynamic registers chapter. For more information about this, see Dynamic Registers. Other APB registers must not be changed after the core_ddrc_rstn is deasserted. Group 1: Registers that can be Written when No Read/Write Traffic is Present at the DFI, Group 2: Registers that can be Written in Self-refresh, DPD, and MPSM Modes, Refresh Related Registers are exceptions to this. The APB registers are reset by the signal presetn, which is deasserted synchronously with pclk.

Static registers are sampled directly in the destination domains (core_ddrc_core_clk/aclk domains) without synchronizing circuits. The deassertion of the signal presetn is synchronized to the destination domain clocks and used to deassert the reset on the static registers in their destination domains. Therefore, the static registers must be programmed when core_ddrc_core_clk/aclk_n and pclk are stable, presetn is deasserted (atleast for 4 cycles of clock in the destination domain) and core_ddrc_rstn/aresetn_n are asserted.

Synchronizing circuits are used to transfer the signals for dynamic registers to their destination domains. While these registers must be initialized with core_ddrc_rstn asserted, they can be changed thereafter with core_ddrc_rstn deasserted.

67.1.3.3. Controller Initialization

Figure following shows the relationship between clocks and resets for initialization of registers. Power and clocks must be in accordance with the requirements of full ASIC/SoC design requirements, and the SDRAM power and clocks must in accordance with the SDRAM specifications.

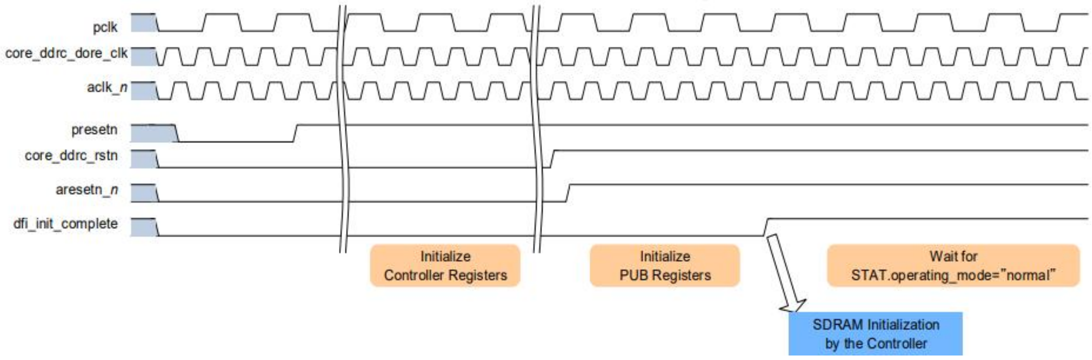


Figure 5.1.3- 2 Simplified Relationship Between Clocks and Resets for Register Initialization

After power-up:

1. Assert the core_ddrc_rstn and aresetn_n resets
2. Assert presetn
3. Start the clocks (pclk, core_ddrc_core_clk, aclk_n)
4. Deassert presetn once the clocks are active and stable
5. Allow 128 cycles for synchronization of presetn to core_ddrc_core_clk and aclk domains and to permit initialization of end logic
6. Initialize the registers
7. Deassert the resets (core_ddrc_rstn and aresetn_n)

Any subsequent access to the programming interface of the registers which may be accessed outside of reset must happen when both pclk and core_ddrc_core_clk are active and stable

67.2. QSPI

1 QSPI ports for NAND/NOR FLASH available.

67.2.1. Features

The main features of QSPI Flash Memory Controller:

- Memory mapped ‘direct’ mode of operation for performing FLASH data transfers and executing code from FLASH memory

- Software triggered ‘indirect’ mode of operation for performing low latency and non processor intensive FLASH data transfers
- Optional DMA Peripheral interface to communicate indirect mode status with external DMA
- Local SRAM of configurable size to reduce AHB overhead and buffer FLASH data during indirect transfers
- Set of software APB FLASH control registers to perform any FLASH command, including data transfers up to 8-bytes at a time

  Additional addressable Memory Bank to accommodate more than 8-bytes at a time
- Support for XIP (Execute in Place), sometimes referred to as continuous mode
- Support for DDR Mode and DTR Protocol
- Support for single, dual or quad I/O instructions
- Programmable device sizes
- Programmable write protected regions to block system writes from taking effect
- Programmable delays between transactions
- Legacy mode allowing software direct access to low level transmit and receive FOs, bypassing the higher layer processes
- An independent reference clock to decouple AHB clock from SPI clock – allows slow system clocks
- Programmable baud rate generator to generate divided clocks
- Features included to improve high-speed read data capture mechanism
- Option to use adapted clocks to further improve read data capturing
- Programmable interrupt generation
- Up to four external device selects

  Programmable AHB decoder, enables continuous addressing mode for each of connected devices and auto-detection of boundaries between devices
- Support BOOT mode

67.2.2. Functional

67.2.2.1. Block Diagram

The QSPI Flash Controller (QSPI_FLASH_CTRL) can be used to provide access to Serial Flash devices. Standard SerialPeripheral Interface (SPI) is supported along with high performance Dual and Quad SPI variants.

The QSPI_FLASH_CTRL connects to an SoC environment through its AMBA AHB bus and APB bus interfaces. The AHB interface is used to transfer data, either in a memory mapped direct fashion (for example a processor wishing to execute code directly from external FLASH memory), or in an indirect fashion where the controller is set up via configuration registers to silently perform some requested operation, signalling its completion via interrupts or status registers. For indirect operations, data is transferred between system memory and external FLASH memory via an internal SRAM which is loaded for writes and unloaded for reads by an AHB master within the SoC environment at low latency AHB system speeds. Interrupts or status registers are used to identy the specic times at which this SRAM should be accessed using user programmable configuration registers. The size of the SRAM is configurable. An optional DMA peripheral bus is also available to optimize the data transfers between an external master and QSPI_FLASH_CTRL during indirect transfers.

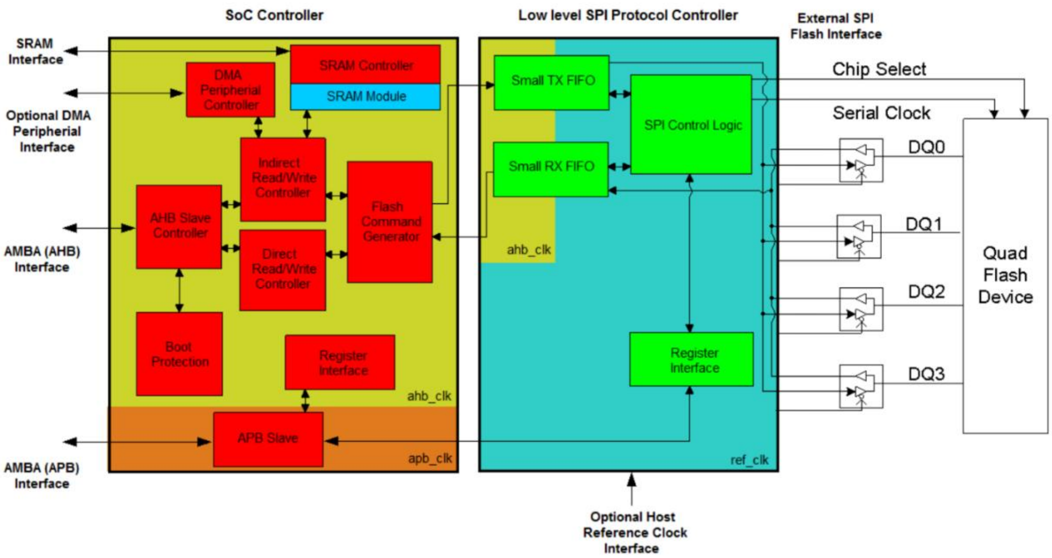


Figure 5.2- 3 QSPI Flash Memory Controller Block Diagram

67.2.2.2. **AHB Control Interface**

The AHB slave controller validates incoming AHB accesses, responds to invalid requests, performs any required byte and half-word reordering, blocks write that violate the programmed write protection rules (for direct access) and forwards the transfer request to either the direct access controller or the indirect access controller.

67.2.2.2.1. AHB Interface

The AHB interface conforms to the AMBA 3 AHB-Lite protocol specication from ARM. Locked transfers (HMASTLOCK) and transfers using the protection control signals (HPROT) are not supported. The AHB data-width is 32 wide. Therefore, byte, half-word and word accesses will be permitted. For writes, incrementing bursts are supported, wrapping write bursts received will generate an error i.e. the access completes with HRESP set to error. INCR16, INCR8, INCR4, INCR and SINGLE type bursts are permitted. For reads all burst types, including WRAPs, are supported. The slave will operate correctly a burst is terminated early due to interconnect burst termination or the slave errors an access within the burst.

67.2.2.2.2. Remapping the AHB address

The QSPI Flash Controller does not implement any specic address decoding to error incoming addresses that may lie outside the connected FLASH memory space but the AHB address decoder is implemented.   it is enabled, the QSPI Flash Controller is able to detect address range of individual device to give an option to auto-switching of active devices based on address from AHB. The incoming AHB address, by default, maps directly to the address sent serially to the FLASH device. the FLASH device has a 24-bit address, then the 24 LSB's of the AHB address will be forwarded. A remap feature is available to remap all incoming AHB addresses to ADDRESS+N, where N is the value store in the Remap Address Register (0x24). It is enabled via the QSPI Configuration Register. This feature could be used when software needs to move boot code to another FLASH region.

67.2.2.2.3. Write Protection

In order to protect the FLASH device, a write protection feature is implemented in hardware and controlled from software. Any AHB write detected (by using DAC controller), pointing to an area of FLASH that is protected, will generate an error, indicated on the HRESP error output. A programmable region of the FLASH device, defined as a number of FLASH "blocks" starting from a particular block number can be protected. Three programmable registers (Lower Write Protection Register, Upper Write Protection Register and Write Protection Register) are provided. The first register defines the FLASH block that is located at the bottom of the region to be protected. The second register defines the FLASH block that is located at the top of the region to be protected. The third register is a control register consisting of 2 . Bit 0 allows software to invert the region that is being protected, causing the programmed region to become the areas of FLASH memory that is not protected from writes. The bit 1 is the write protection enable bit. When low, the FLASH device is unprotected. For implementation, the AHB controller must map the incoming AHB address into its associated FLASH block. A block can be between 1 and 65K bytes, programmed via the Device Size Configuration Register.

67.2.2.2.4. Access Forwarding

For legal accesses, the AHB controller will forward all AHB accesses to one of two access controllers: the direct access and indirect access controllers. Assuming the direct access controller has been enabled via the QSPI Configuration Register (0x00), then by default all AHB accesses will be forwarded to the direct access controller. Before any accesses can be forwarded to the indirect access controller, it must first be configured by software. This process is fully explained in Section 11.1.3.3.3, "Indirect Access Controller (INDAC)". the direct access controller is disabled, any incoming AHB access that cannot be forwarded to the indirect access controller will be completed immediately with an HRESP error. DAC is enabled, the same access will be forwarded and serviced by it.

67.2.2.2.5. Sequential Access Detection and Burst Length

In order to maximize performance, the AHB interface signal "htrans" is not used by the controller to help identy sequential and non-sequential accesses. This is because "htrans" identies all accesses at the start of a new AHB burst as non-sequential, as well as all accesses after a 1K boundary even when the access may be sequential to the last. Instead non-sequential accesses are detected by comparing the address of the current access with the address of the previous access. A decision is then made as to whether the access is sequential or not. the direction of the access (write/read) has changed, then the access is deemed non-sequential. the size of the read access (byte/half-word/word) has changed, then the access is also non-sequential. AHB address decoder is enabled and transition between devices has been detected, then the new SPI transfer has to be generated because currently selected device needs to be initialized. This is the case when accesses are considered as non-sequential despite addresses are consecutive. Otherwise the access is sequential the address of the current transfer is sequential when compared the previous transfer. In addition, the "hburst" signal from AHB is not used by the controller to identy burst lengths. Instead a burst continues until a non-sequential access is received.

67.2.2.2.6. AHB Address Decoder

It is possible to connect up to four devices to the controller. Emerging memories reached size of 2Gb and they are still increasing over this parameter. To ensure high performance in sequential transfer of data greater than a few Gb, AHB Address Decoder can be enabled. Address range from AHB standpoint is merged to continuous form. It allows the software driver to manage of accesses without additional steps to detect address boundaries between devices. Switching is performed automatically by the controller based on actual address from AHB interface. Boundaries of addresses are calculated by hardware, based on programmable configuration from Device Size Configuration Register. Each connected device can have another size ranged from 512Mb to 4Gb. For example, each device was configured to have density of 512Mb, the inferred address range will be from address of 0x0000000 to 0x0FFFFFF what calculates into 4*512Mb = 2Gb of continuous address area. CS[0] is assigned to part from 0x0000000 to 0x03FFFFF, CS[1] to part from 0x0400000 to 0x07FFFFF and so on. AHB Address Decoder should be used for direct read transfers. Data to read should be stable to avoid necessity of polling each device after switching between them write access was issued before. To make sure about stability of the data, it is recommended to do first reads separately for each device without using of AHB decoder.

67.2.2.3. **Direct Access Controller (DAC)**

Direct access refers to the operation where AHB accesses directly trigger a read or write to FLASH memory. It is memory mapped and can be used to both access and directly execute code from external FLASH memory. Any incoming AHB access that is not recognized as being within the programmable indirect trigger region is assumed to be a direct accessand will be serviced by the direct access controller. Note that accesses that use the direct access controller do not use the embedded SRAM. The AHB will be throttled as the read or write burst is carried out, and the amount of wait states applied will be dependent on the latency through the controller. The latency has been carefully designed to be as small as possible. Latency is kept to a minimum when the use of XIP read instructions are enabled.

When servicing AHB reads, the DAC will always send one additional access downstream than was originally issuedby the AHB in any single AHB burst. This is invisible from the system interface. It is a predicted read and ensures the underlying SPI core can operate at maximum bandwidth. The AHB burst defined here is not an AHB burst as defined by AMBA. It is defined as follows:

1. The first access of the AHB burst is defined by:

a. An AHB access that is NON-SEQUENTIAL (Non-sequential based on address comparison, not based on htrans).
b. An AHB access that is NON-SEQUENTIAL or SEQUENTIAL when the downstream modules are IDLE.

2. The last access of the AHB burst is defined by an AHB access that is SEQUENTIAL and precedes a new burst as defined in 1.

3. The size of an AHB burst is the number of AHB accesses, starting from the first access and ending with the last one

4. The number of DAC requests per AHB burst is equal to the size of the AHB burst+1

When AHB clock is very slow comparing to the SPI data rate, the sequential read transfer might be interrupted on SPI side and then continued just after slow AHB clock had accepted the last data. The limiting value is getting narrower as SPI clock divider and AHB access size (hsize) go down and number of data IOs (single, dual or quad) go up. the system is not able to meet SPI data rate, it is recommended to use higher SPI clock divider what will decrease SPI data rate but makes the necessity of passing through opcode+address+dummy before each transfer part not to care what increases overall performance by keeping data transfer being continuous. There is not formal requirement covering marginal case value of AHB clock since phase relationship between this clock and reference one is unknown (they are asynchronous with respect to each other). The architecture of the controller was designed to be able to work on high data rates. Thepresence of predicted read can lead into the rubbish data during mixed AHB accesses on slow system data rate. Mixed accesses mean the Direct AHB sequence of single non-sequential access following a few sequential ones. The predicted read cannot be dropped because of SPI transfer interruption therefore it is sent as separate SPI transaction what creates rubbish on the interface for one access. To avoid this issue, it is recommended just to drop this data by system or just introduce wait states on AHB before issuing non-sequential direct AHB access. In spite of that, usage of Indirect Read Transfer is worth to consider instead for described working conditions.

For AHB writes, the direct access controller will trigger a sequence of write commands that mimic the way reads are processed, although the number of DAC requests for writes is equal to the number of AHB write requests received. For writes, the AHB controller will ensure FLASH bursts will not break a FLASH page boundary. When a page boundary is detected, the number of byte accesses up to that boundary will be forwarded. A sequential direct write request that crosses a page boundary must be detected as non-sequential to downstream modules, causing the downstream controller to force the Flash device to enter a self-timed page program cycle. The Controller supports splitting writes crossing page boundaries for word aligned addresses. The external AHB master supplying the data for writes should attempt to guarantee write data for a particular page can be provided to the controller in a timely fashion to avoid downstream starvation. there is too much delay in the system issuing a sequential write, then the FLASH write cycle may be initiated prematurely, reducing the effective le of the device. Note that this cannot be guaranteed and is of concern, then the indirect write operation can instead be used to avoid this issue.

FLASH erase operations, which may be required before a page write, are triggered by software using the documented programming interface Section, "*Software Triggered Instruction Generator (STIG)*" (detailed later). Once a page program cycle has been started, the QSPI Flash Controller will automatically poll for the write cycle to complete beforeallowing any further AHB accesses to complete. This is achieved by holding any subsequent AHB direct accesses in wait state.

67.2.2.4. **Indirect Access Controller (INDAC)**

67.2.2.4.1. Indirect Read Controller

The aim of the indirect mode of operation is to read signicant numbers of bytes from FLASH memory without requiring an AHB access to trigger it. Instead indirect operations are controlled and triggered by software via specic APB control/configuration registers (Indirect Read Transfer Control Register, Indirect Read Transfer Watermark Register, Indirect Read Transfer Start Address Register and Indirect Read Transfer Number Bytes Register). This block will communicate with an embedded low level SPI protocol state machine module to perform an efficient and optimized FLASH read burst, placing the read data into the local SRAM module ready for fast and low latency delivery to any external AHB master.

By default, the indirect read controller is disabled. Before enabling it, software must configure how much data is required and starting from what address. The starting address and total number of bytes to be fetched is defined in registers at addresses Indirect Read Transfer Start Address Register and Indirect Read Transfer Number Bytes Register respectively. Up to two indirect operations can be programmed at any one time. The second can be triggered while the first is in progress. Supporting two indirect operations allows a short turnaround time between the completion of one indirect operation and the start of the second. Refer to the Section "*Indirect Access Queuing*" for further details.

The total number of bytes to read in an indirect operation is not limited by the size of the SRAM. The size of SRAM will limit the size of requests made to the DMA (the DMA peripheral interface module is enabled). In the case of an SRAM overrun, the controller will back pressure FLASH reads until space becomes available in the SRAM. Back pressuring the reads on the SPI interface is handled by completing any current read burst, waiting until space in the SRAM becomes available and then issuing a new read burst at the address where the previous terminated burst ended.

An external master will be able to fetch the data the controller has read from external FLASH memory by issuing AHB reads to the controller. The AHB address of the incoming read access must be in the range AHB Indirect trigger address (programmed via the Indirect AHB Address Trigger Register) to AHB Indirect trigger address + 2**(Indirect AHB trigger address range) - 1. Default value of the range is equal to 16 locations. This allows a 16-beat burst to be applied starting from the AHB Indirect trigger address. The smaller bursts are possible to handle effectively as well with this approach. Furthermore, it is not strict requirement to push consecutive address sequence. Actual AHB address just has to be inside the Indirect Range to grant the SRAM as a source. Each valid AHB Indirect read will cause the internal SRAM to be popped, thereby decoupling the AHB address from the FLASH address – i.e. not direct mapped. Therefore, AHB Indirect trigger address does not have any relationship with FLASH address. It is just to indicate that data should take SRAM as source instead of FLASH Memory array after triggering of any valid Indirect Read. FLASH address for Indirect Read is taken from the Indirect Read Transfer Start Address Register (0x68). Assuming the requested data is present in the SRAM at the point the AHB access is received by the QSPI controller, then the data will be fetched from SRAM and the response to the read burst will be achieved with minimum latency. Once the data has been read from SRAM, the QSPI controller will free up the associated resource in the SRAM.

If an AHB read access is received whose address is not within the range described above then that access will not be completed using the indirect controller. It will instead be serviced by the direct access controller.

If an AHB read access is received whose address is within the range described above but the requested data is not immediately present in the SRAM then AHB wait states will be applied until the data has been read from FLASH and pushed to the SRAM.

If an AHB read burst is received whose access elements traverse the AHB Indirect trigger range, then the accesses within the Indirect trigger range will be processed by the indirect controller and the rest will be taken by the direct access controller. Note this is likely to be a software configuration error.

The external master is permitted to issue 32-bit AHB reads until the last word of an indirect transfer. This helps keep the SRAM control logic less complex. On the final read, the external master may issue a 16-bit (HalfWord) or byte accessto complete the transfer. It is also permitted for the external master to always issue a 32-bit Word read on the last indirect access. The controller will pad the upper of the response with zero.

The current expectation is that the SRAM will be kept fairly full while the read operation is carried out. The fill level of the SRAM is directly readable by software in SRAM Fill Level Register.   the DMA peripheral interface controller is enabled, this will automatically request an external DMA fetch the data from the SRAM via the AHB in efficient chunks of data, each chunk being a part of the overall indirect transfer.

An indirect operation may be cancelled at any time by setting Indirect Read Transfer Control Register bit[1].

Any bus master should be allowed to initiate an indirect access. The DMA bus peripheral interface has been included in the design to offload some of the necessary software overhead to efficiently manage data transfers when an external DMA supporting this interface is added. The alternative to this is for software to access the SRAM fill-level directly via APB registers and then decide for itself when the data should be fetched from the local SRAM. When the DMA peripheral interface is disabled, the fill level watermark register via the Indirect Read Transfer Watermark Register (0x64) is provided. When the SRAM fill level passes this watermark, an interrupt is generated. the watermark value is >0, the watermark interrupt is also generated when the final byte of data has been read by the QSPI controller and placed in the SRAM, even the actual SRAM fill level has not risen above the watermark. This last feature is useful to avoid software tracking how much data has been read and resetting the watermark value for the last few bytes of an indirect read transfer. Note that this watermark register is a dual-use register. When the DMA peripheral interface is enabled, it is used by hardware to control the rate at which the DMA requests are issued. When the DMA peripheral interface is disabled, the behaviour is as described above.

Two further interrupt sources are provided to help understand the status of an indirect operation. Firstly, an interrupt is generated when an indirect operation has completed. Secondly, an interrupt is generated an indirect read operation was requested but could not be accepted due to the fact 2 indirect operations have already been buffered by the QSPI controller.

Setting Indirect Read Transfer Control Register bit[0] starts an indirect read operation, and Indirect Read Transfer Control Register bit[2] will be available to check the status.

67.2.2.4.2. Indirect Write Controller

The aim of the indirect mode of operation is to perform bulk transfer of data from the processor or DMA into FLASH memory in the most efficient manner. The fewest possible write cycles inside the FLASH device will be carried out for the indirect transfer, thus maximizing the le of the device.

Indirect write operation can be thought of from a software perspective as the inverse of the indirect read. It is controlled and triggered by software via specic APB control/configuration registers (Indirect Write Transfer Control Register, Indirect Write Transfer Watermark Register, Indirect Write Transfer Start Address Register and Indirect Write Transfer Number Bytes Register). This block will await delivery of the write data via the external AHB master, placing it in the local SRAM before communicating with the existing legacy SPI IP core to perform an efficient and optimized FLASH write burst.

By default, the indirect write controller is disabled. Before enabling it, software must configure how much data is required and starting from what address. The starting address and total number of bytes to be written is defined in registers at addresses Indirect Write Transfer Start Address Register and Indirect Write Transfer Number Bytes Register respectively. Up to two indirect operations can be programmed at any one time. The second can be triggered while the first is in progress. Supporting two indirect operations allows a short turnaround time between the completion of one indirect operation and the start of the second. Indirect write queuing is very similar to indirect read queuing. Refer to the Section, "*Indirect Access Queuing*" for further details.

The total number of bytes to write in an indirect operation is not limited by the size of the SRAM. The size of SRAM will limit the amount of data that can be accepted from the external AHB master. When the external master is the DMA, the amount of data requested by the controller via the DMA peripheral interface will never exceed the current fill level of the SRAM. However, this does not guarantee in itself that the DMA, or other external master, the DMA is not used, will not attempt to push more data to the SRAM than the SRAM can accept. In the case of an SRAM overrun, the controller will back pressure the AHB with wait states. Note the fill level of the SRAM is readable via SRAM Fill Level Register and this can be used to avoid this situation.

An external master will provide the write data and will transfer this to the QSPI controller by issuing AHB writes. The AHB address of the incoming write access must be in the range AHB Indirect trigger address (programmed via the Indirect AHB Address Trigger Register) to AHB Indirect trigger address + 2**(Indirect AHB trigger address range) - 1. Default value of the range is equal to 16 locations. This allows a 16-beat burst to be applied starting from the AHB Indirect trigger address. The smaller bursts are possible to handle effectively as well with this approach. Furthermore, it is not strict requirement to push consecutive address sequence. Actual AHB address just has to be in the Indirect Range to grant SRAM as source. Each valid AHB Indirect write will cause the internal SRAM to be pushed, thereby decoupling the AHB address from the FLASH address – i.e. not direct mapped. Therefore AHB Indirect trigger address does not have any relationship with FLASH address. It is just to indicate that data should take SRAM as source instead of FLASH Memory array after triggering of any valid Indirect Write. FLASH address for Indirect Write is taken from the Indirect Write Transfer Start Address Register (0x78). Assuming the SRAM is not full at the point the AHB access is received by the QSPI controller, then the data will be pushed to the SRAM with minimum latency.

If an AHB write access is received whose address is not within the range described above then that access will not be completed using the indirect controller. It will instead be serviced by the direct access controller.

If an AHB write access is received whose address is within the range described above but the SRAM is full, then AHB wait states will be applied until some or all of the data has been pushed from SRAM onto the FLASH.

If an AHB write burst is received whose access elements traverse the AHB Indirect trigger range, then the accesses within the Indirect trigger range will be processed by the indirect controller, and the rest will be taken by the direct access controller. Note this is likely to be a software configuration error.

The external master is permitted to issue 32-bit AHB writes until the last word of an indirect transfer. This helps keep the SRAM control logic less complex. On the final write, the external master may issue a 32-bit word, 16-bit (half-word) or a byte access to complete the transfer. the number of bytes to write is less than 4 on the last transfer, the master is still permitted to issue a 32-bit transfer. In these cases, the extra bytes are discarded by the controller.

When the SRAM holds a number of bytes equal to or greater than the size of a FLASH page (which itself is programmable by the controller, with a default of 256 bytes) or when the SRAM holds all remaining bytes of the currently executing indirect transfer, the controller will initiate a write burst to the command generator.

An indirect operation may be cancelled at any time by setting Indirect Write Transfer Control Register bit[1].

Any bus master should be allowed to initiate an indirect access. The DMA bus peripheral interface has been included in the design to offload some of the necessary software overhead to efficiently manage data transfers when an external DMA supporting this interface is added. The alternative to this is for software to access the SRAM fill-level directly via APB registers and then decide for itself when the data should be written to the local SRAM. When the DMA peripheral interface is disabled, the fill level watermark register via the Indirect Write Transfer Watermark Register (0x74) is provided. When the SRAM fill level falls below this watermark, an interrupt is generated. Note that this watermark register is a dual-use register. When the DMA peripheral interface is enabled, it is used by hardware to control the rate at which the DMA requests are issued. In this mode, since the QSPI will not initiate a write operation unless there is at least one FLASH page worth of data in the local SRAM, or the remaining bytes of the indirect transfer are in the SRAM, then it is important the watermark in DMA mode is programmed to a value >= 1 flash page. Two further interrupt sources are provided to help understand the status of an indirect operation. Firstly, an interrupt is generated when an indirect operation has completed. Secondly, an interrupt is generated an indirect write operation was requested but could not be accepted due to the fact two indirect operations have already been buffered by the QSPI controller.

Two further interrupt sources are provided to help understand the status of an indirect operation. Firstly, an interrupt is generated when an indirect operation has completed. Secondly, an interrupt is generated an indirect read operation was requested but could not be accepted due to the fact 2 indirect operations have already been buffered by the QSPI controller.

Setting Indirect Write Transfer Control Register bit[0] starts an indirect write operation, and Indirect Write Transfer Control Register bit[2] will be available to check the status.

67.2.2.4.2.1. Indirect Access Queuing

Software is permitted to queue up to two indirect transfers for both the indirect write controller and the indirect read controller. Supporting two indirect operations allows a short turnaround time between the completion of one indirect operation and the start of the second. Any attempt to queue more than two operations will cause an interrupt to be generated. To take advantage of this feature, software should attempt to keep both indirect programming slots full at all times.

From the software perspective, indirect access queuing is achieved by triggering bit 0 of the indirect transfer control register (Indirect Read Transfer Control Register or Indirect Write Transfer Control Register) twice in short succession. The indirect number of bytes register and the indirect FLASH start address register must be setup with the relevant transfer data before bit 0 can be triggered for each transfer. Since these registers will change regularly, the hardware must keep sampled versions of these registers for the duration of the indirect transfer.

The internal register block will issue an indirect start trigger to the key underlying datapath blocks one at a time. There are 2 independent datapath blocks in the indirect access controller that will receive and independently sample this information. The first is the datapath block on the AHB side of the SRAM. For indirect reads, this is a read interface, for indirect writes, it is a write interface. The second is the datapath block on the FLASH side of the SRAM. For indirect reads, this is a write interface, for indirect writes, it is a read interface. Both blocks will process the indirect transfers at dferent times. For example, for an indirect read operation, the datapath block on the FLASH side of the SRAM will be able to start processing the second queued transfer as soon as the last byte of the first transfer has been written to SRAM. Before commencing the second transfer, this block must re-sample the number of bytes and the indirect FLASH start address register. Similarly, the datapath block on the AHB side will re-sample the same registers locally when it has forwarded all the FLASH data associated with the first indirect transfer from SRAM onto AHB.

67.2.2.4.2.2. Consecutive Writes and Reads using Indirect Transfers

It is permitted for software to trigger an indirect read operation while an indirect write operation is in progress. Similarly, it is permitted to trigger an indirect write while an indirect read operation is in progress. Indirect write operations will take overall precedence.

67.2.2.4.2.3. Accessing the SRAM

Physically, the SRAM will be a single port macro. The depth of this SRAM is a configuration choice at compile time. It will be segmented into two halves. The lower half is reserved for indirect read use. The upper is for indirect write use . The size of each half will be programmable via the SRAM Partition Configuration Register (0x18). This allows the programmer to allocate how many of the SRAM address bus are allocated to indirect read. By default, this is set so that exactly half of the SRAM is portioned for use by the indirect read controller. To ensure the AHB read data bus is not directly fed by the SRAM read data through combinatorial logic, an extra bank of holding registers is included in the indirect read data path. These registers act as an extra location to be added to the allocated number of SRAM locations for indirect read.

To illustrate how the SRAM (and the extra bank of holding registers) can be allocated between indirect read and write, the following example is provided. The depth of the SRAM in this example is configured to be 8 , equaling 256 locations...

- the SRAM Partition Register is set to 0x00, then 256 locations are allocated to indirect writes and 1 location to indirect reads.
- the SRAM Partition Register is set to 0x01, then 255 locations are allocated to indirect writes and 2 locations to indirect reads.
- the SRAM Partition Register is set to 0x02, then 254 locations are allocated to indirect writes and 3 locations to indirect reads.
- (...)
- the SRAM Partition Register is set to 0xfd, then 3 locations are allocated to indirect writes and 254 locations to indirect reads.

- the SRAM Partition Register is set to 0xfe, then 2 locations are allocated to indirect writes and 255 locations to indirect reads.
- the SRAM Partition Register is set to 0xff, then 1 locations are allocated to indirect writes and 256 location to indirect reads.

Note that a value of 0xFF or 0x00 in the SRAM Partition Register should be avoided by software, as the bottom 8 of the SRAM fill level are through software (i.e. up to 255 limit) via the SRAM Fill Level Register. I.e. the fill level reaches 256 on either the indirect read or write side, it will appear when reading the Fill Level to be 0.

There are four SRAM sources that are arbitrated and muxed onto the single SRAM port. Up to three sources may be accessing this port at any one time. The sources are described as follows:
1. Indirect Write, Write source. This is located on the AHB side of the SRAM.
2. Indirect Write, Read source. This is located on the FLASH side of the SRAM.
3. Indirect Read, Write source. This is located on the FLASH side of the SRAM.
4. Indirect Read, Read source. This is located on the AHB side of the SRAM.

A fixed priority arbitration scheme is implemented. The priority allocated to these sources is set as follows:

Table 5.2- 1 SRAM Access Priority

Transfer	Operation	Priority
Indirect Write	Write to SRAM (from System AHB)	3rd (exclusive with AHB read request)
Indirect Write	Read from SRAM (from QSPI Controller)	2nd
Indirect Read	Write to SRAM (from QSPI Controller)	1st
Indirect Read	Read from SRAM (from System AHB)	3rd (exclusive with AHB read request)

With the exception of the write port during an Indirect Read operation (on the FLASH side of the SRAM), the logic driving all four sources must not assume single cycle completion. Writes to the SRAM during an indirect read must be allowed to complete immediately to avoid data loss. Therefore, this port is given maximum priority.

67.2.2.5. DMA Peripheral Controller

The peripheral interface will be used to trigger an external DMA into performing low-latency AHB data burst accesses to the controller. The DMA the peripheral interface is directly meant for is the CoreLink DMA Controller DMA-330 core from ARM. The DMA peripheral interface is used for the Indirect modes of operation where data is buffered in local SRAM in order to quickly service incoming AHB requests and have the core perform the low level FLASH transfers over a longer period of time. There are two identical DMA interfaces, one for each indirect controller. For the indirect read controller, the QSPI controller will issue DMA requests after the data has been retrieved from FLASH and written to local SRAM. For the indirect write controller, the QSPI controller will issue DMA requests immediately after the transfer is triggered and will continue to do so until the entire indirect write transfer has been transferred. The rate at which these requests are made can be altered using the watermark registers.

67.2.2.5.1. DMA Interface

The following is true for both DMA interfaces.

The peripheral controller is responsible for driving the DR (peripheral to DMA) request bus and for acknowledging the DA (DMA to peripheral) acknowledge bus. The two buses are simple AXI like interfaces, which consist of a two-way VALID and READY handshake mechanism.

For the DR request bus, the controller drives the "drvalid" signal when valid information is available on "drtype" and "drlast" signals. This information will remain set until the external DMA drives the "drready" signal, indicating it can accept the request.

For the DA acknowledge bus, the DMA drives the "davalid" signal high when valid information is available on "datype" signal. This information will remain set until the DMA peripheral controller drives the "daready" signal high, indicating it has accepted the acknowledgement made by the DMA.

The controller will issue a request via the DR bus to the DMA when it requires the DMA (or other AHB master) to transfer data from the controller via its AHB slave port. The controller will issue two types of requests to achieve this, single or burst. Each of these request types will correspond to a programmable but fixed number of bytes. By default, both of these types will reset to a request size of just 1 byte. The DR bus is also used to acknowledge a Flush request which is described in more detail below. The DR request uses the "drtype" bus to indicate the request type.

The DMA uses the DA bus to indicate when the DMA has completed the data transfer. This is acknowledged by the controller by driving the "daready" signal, but no other action is taken. The DMA also uses the DA bus to request that the controller perform a flush operation. When this occurs, the controller will re-evaluate the FO levels and replay all DMArequests that may have been sent but have yet to be acknowledged. Note a request on the DR channel is valid at the point the flush is received, the controller must hold the "drvalid" and "drtype" signals steady until the DMA acknowledges the request via "drready". Once the flush operation has completed, the controller will acknowledge the flush by sending a unique request on the DR channel, setting drtype to 2’b10.

Table 5.2- 2 DMA Encoding

drtype/datype	Action
2’b00	Single Request/Acknowledge
2’b01	Burst Request/Acknowledge
2’b10	Flush Acknowledge/Request

67.2.2.5.2. DMA Order of operation

When an indirect operation is triggered, the total amount of data to be transferred from/to FLASH memory (in bytes) will be visible by the DMA peripheral controller. The controller will split this into a number of DMA burst and single requests by dividing the total number of bytes by the number of bytes programmed in the burst request, and then dividing the remainder by the number of bytes in a single request. It is for software to ensure there is no remainder following these divisions. To ensure the logic is kept simple, the divisions will always be a factor of 2, and the programmable registers defining them must be defined such. These programmable registers are defined in the DMA Peripheral Configuration Register. As an example, the total amount of data selected to be read from FLASH using the low-latency Indirect mode is 512 bytes, the size of the SRAM is a fixed 256 bytes and software configures the burst type transfer size to be 256 bytes, then the SPI controller will trigger a DMA burst request when the first 256 bytes have been locally buffered. It can then trigger the 2nd burst request when a further 256 bytes have been buffered in the local SRAM. Since the SRAM size was 256 bytes itself, this means the DMA will have had to fetch the entire contents of the SRAM before the next request can be issued.

The SRAM fill level will be made visible to the DMA peripheral controller. For indirect reads, requests will be issued to the external DMA based on the following state machine:

. Wait for START trigger

2. the next request to send is a BURST request, wait until both of the following is TRUE

a. SRAM fill level is greater than or equal to the value stored in the burst_size field of DMA Peripheral Configuration Register [11:8]

b. SRAM fill level is greater than or equal to the value stored in the Indirect Read Transfer Watermark Register ACTION when TRUE:

i. SEND BURST.

ii. Calculate local expected version of fill level: (SRAM fill level -num_bytes_requested_to_DMA)

iii. the next request to send is a BURST request, move to 4

iv. the next request to send is a SINGLE request, move to 5

v. Else move to 1

3. the next request to send is a SINGLE request, wait until both of the following is TRUE

a. SRAM fill level is greater than or equal to the value stored in the single_size field of DMA Peripheral Configuration Register [3:0]

ACTION when TRUE:

i. SEND SINGLE.

ii. Calculate local expected version of fill level: (SRAM fill level - num_bytes_requested_to_DMA)

iii. the next request to send is a SINGLE request, move to 5

iv. Else move to 1

4. The next request to send is a BURST request. the (SRAM fill level -num_bytes_requested_to_DMA) is less than the value stored in the burst_size field of DMA Peripheral Configuration Register [11:8] then goto 2, else:

a. SEND BURST.

b. Calculate local expected version of fill level: (SRAM fill level – num_bytes_requested_to_DMA)

c. the next request to send is a BURST request, move to 4

d. the next request to send is a SINGLE request, move to 5

e. Else move to 1

5. The next request to send is a SINGLE request. the (SRAM fill level - num_bytes_requested_to_DMA) is less than the value stored in the single_size field of DMA Peripheral Configuration Register [3:0] then goto 3, else:

a. SEND SINGLE.

b. Calculate local expected version of fill level: (SRAM fill level – num_bytes_requested_to_DMA)

c. the next request to send is a SINGLE request, move to 5

d. Else move to 1

For indirect writes, requests will be issued to the external DMA based on the following state machine:

1. Wait for START trigger

2. the next request to send is a BURST request, wait until both of the following is TRUE

a. Remaining space in SRAM is greater than or equal to the value stored in the burst_size field of DMA Peripheral Configuration Register [11:8]

b. SRAM fill level is less than or equal to the value stored in the Indirect Write Transfer Watermark Register

ACTION when TRUE:

i. SEND BURST.

ii. Calculate local expected version of fill level: (SRAM fill level + num_bytes_requested_to_DMA)

iii. the next request to send is a BURST request, move to 4

iv. the next request to send is a SINGLE request, move to 5

v. Else move to 1

3. the next request to send is a SINGLE request, wait until both of the following is TRUE

a. Remaining space in SRAM is greater than or equal to the value stored in the single_size field of DMA Peripheral Configuration Register [3:0]

ACTION when TRUE:

i. SEND SINGLE.

ii. Calculate local expected version of fill level: (SRAM fill level + num_bytes_requested_to_DMA)

iii. the next request to send is a SINGLE request, move to 5

iv. Else move to 1

4. The next request to send is a BURST request. the (SRAM fill level - num_bytes_requested_to_DMA) is less than the value stored in the burst_size field of DMA Peripheral Configuration Register [11:8] then goto 2, else:

a. SEND BURST.

b. Calculate local expected version of fill level: (SRAM fill level + num_bytes_requested_to_DMA)

c. the next request to send is a BURST request, move to 4

d. the next request to send is a SINGLE request, move to 5

e. Else move to 1

5. The next request to send is a SINGLE request. the (SRAM fill level - num_bytes_requested_to_DMA) is less than the value stored in the single_size field of DMA Peripheral Configuration Register [3:0] then goto 3, else:

a. SEND SINGLE.

b. Calculate local expected version of fill level: (SRAM fill level + num_bytes_requested_to_DMA)

c. the next request to send is a SINGLE request, move to 5

d. Else move to 1

For the indirect read operation, a programmable Indirect Read Transfer Watermark Register (0x64) will define the minimum fill level that the controller can issue the first DMA request – the higher this figure is, the more data that must be buffered in SRAM before the DMA will collect it. For the indirect write operation, a programmable Indirect Write Transfer Watermark Register (0x74) will define the maximum fill level that the controller can issue the first DMA burst/single request. The watermark registers allow the system to optionally concentrate AHB transfers over a short period of time. By default, the watermark registers are reset to zero, meaning the peripheral controller may issue DMA request as soon as possible. For the indirect read case, this means that there is sufficient data in the SRAM to perform a burst type request, or a single type request there less data left to transfer than the number of bytes programmed in the burst type request. Note the DMA peripheral controller must never send an indirect read DMA request the SRAM is empty, or an indirect write DMA request the SRAM is full.

The DMA peripheral controller will send a trigger event to the external DMA by setting drtype[1:0] to the required type of access (single, burst or flush) along with drvalid for a single hclk cycle to tell the DMA that there is READ data available. drlast is asserted at the same time as drtype to indicate the last request of the current transfer sequence. drready is driven by the DMA to indicate that it can accept requests from the Peripheral Controller.

The DMA will indicate completion of a previously requested data transfer which takes place on the AHB bus by asserting ACK on datype along with davalid. daready is used to indicate to the DMA that the Peripheral Controller is available to accept status information on datype. A series of requests will be sent on the DA bus until all data read/written from/to the FLASH device that is stored in the SRAM has been acknowledged as being transferred to/from the DMA.

The peripheral interface acts on the DA acknowledge channel the DMA issues an abort request. All other requests are simply accepted using daready as required. This means the controller is free to pipeline the DR requests as much as it likes, and will be limited by the pipelining requirements of the DMA.

The DMA peripheral interface may be disabled via software. When the interface is disabled, no DR channel request will ever be issued. An incoming flush request on the DA channel will be internally ignored, although the daready must still be set to acknowledge the flush request. Note an AHB master other than the DMA will perform the data transfer for indirect operations, the DMA peripheral interface must be disabled.

67.2.2.6. **Software Triggered Instruction Generator (STIG)**

The direct and indirect access controllers are used to transfer data. In order to access the volatile and non-volatile configuration registers, the legacy SPI Status register, other status/protection registers as well as to perform ERASE functions, a separate software controller is required. The software triggered instruction generator, or STIG, is controlled using the Flash Command Control Register (Using STIG) (0x90) by setting up the command to issue to the FLASH device. This is a generic controller and can be used to perform any instruction that the FLASH device supports from the extended SPI protocol. Configuring of instructions which are not compliant with the specication of FLASH Devices could cause unpredicted behaviour of the controller. Bit 0 is used to trigger the command, bit 1 used by software to poll the status of the command execution. For reads, when the command has been serviced (bit 1 toggles from "1" to "0"), up to 8 bytes of read data will be placed in the Flash Command Read Data Register (Flash Command Read Data Register (Lower) and Flash Command Read Data Register (Upper)). For writes, the write data should be placed in the Flash Command Write Data Register (Flash Command Write Data Register (Lower) and Flash Command Write Data Register (Upper)). The completion of the STIG request could be also checked by the corresponding interrupt. The occurrence of the interrupt indicates that the controller is ready for accepting a new STIG request. It is important to notice that completion of the STIG request is not equivalent to completion it on SPI side. E.g. STIG is configured to the command composed of data to transmit , the data is taken from the corresponding STIG register fields and put into TX FO. Since all bytes to write are known, another STIG can be queued before serialization of the current one is completed.

There are some commands which require more data to read than 8 bytes (for example READ ID command). The additional STIG Memory Bank was implemented in order to accommodate these data needed. The STIG Memory Bank is controlled by the Flash Command Control Register (Using STIG) Register bit[2]. enabled, the number of bytes to read in the STIG is extended to the value ranging from 16 to 512 as defined in Flash Command Control Memory Register (Using STIG) Register bit[18:16]. It should be noticed that there are very few commands (excluding Read Array ones which are not intended to handle effectively in STIG Mode but in Direct/Indirect Modes) which return more than 8 bytes to the controller. the maximum number of bytes to Read using STIG in target application is less than 512, the depth of the STIG Memory Bank can be set smaller what will result in saving noticeable part of the area (STIG Memory Bank is an internal component of the controller). The decision is made in pre-compilation time. The depth is configurable in hdl/hdl_src/cdns_qspi_flash_ctrl_defs_default.v by changing parameter called cdns_qspi_stig_mem_bank_depth. The default depth is set for the limited size of this parameter, namely 4. number of bytes to Read in the STIG as defined in Flash Command Control Memory Register (Using STIG) Register bit[18:16] exceeds the Memory Bank Depth, remaining data will overwrite the STIG Memory Bank locations starting from its first address. Flash Command Read Data Register (Lower) and Flash Command Read Data Register (Upper) keep the last 8 bytes read from the Flash Device by STIG when Memory Bank is enabled. Therefore, for example the user wants to get just a single byte from the last eight bytes from long continuous read SPI data chain, there is no need to access the STIG Memory Bank since data can be taken from suitable Flash Command Read Data Register. In order to access more data, STIG Memory Bank data request should be triggered. It is controlled by the Flash Command Control Memory Register (Using STIG) and works analogously for triggering STIG from the functional standpoint. Bit 0 is used to trigger the command, bit 1 used by software to poll the status of the command execution. When bit 1 toggles from "1" to "0", the byte of data (Flash Command Control Memory Register (Using STIG) Register bit[15:8]) from corresponding address (Flash Command Control Memory Register (Using STIG) Register bit[28:20]) is valid. The address should be set before triggering the STIG Memory Bank access. Each consecutive STIG access overwrites the previous one so that the data in the Bank always fit into byte index fetched by the last STIG access configured to use the Memory Bank (first incoming byte equals first address of the Memory Bank, second one equals the second address and so on).

67.2.2.6.1. Servicing a STIG request

A STIG request will cause the QSPI Flash Controller to interrogate the Flash Command Control Register (Using STIG) (0x90) to determine what and how many bytes it should send to the FLASH device. [31:24] of this register indicate the instruction to be sent and is always pushed first. there is an address to send, then the address (the size of which is also programmed in the same register) is sent next. The address itself is stored in the Flash Command Address Register (0x94). there are any dummy cycles to send (the size of which is also programmed in Flash Command Control Register (Using STIG)) then those are sent next. there is data to write or read (the size of which is also programmed in Flash Command Control Register (Using STIG)) then for the case of writes, up to 8 bytes can be sent (as stored in the Flash Command Write Data registers, Flash Command Write Data Register (Lower) and Flash Command Write Data Register (Upper)). In the read case, when the read data has been collected from the FLASH device, the QSPI Flash Controller stores that in the Flash Command Read Data Registers (Flash Command Read Data Register (Lower) and Flash Command Read Data Register (Upper)). Up to 8 bytes can be get Flash Command Control Register (Using STIG) Register bit[2] is disabled or up to 512 when enabled.

When the QSPI Flash Controller starts to service a STIG request, it sets Flash Command Control Register (Using STIG) Register bit[1] to indicate a command execution is in progress.

When the QSPI Flash Controller is in the auto-polling state, servicing a STIG request is slightly dferent. Most of devices are largely in after a program operation until the device has completed that write. Some group of them has a possibility to suspend programming page. It can be controlled by Polling Flash Status Register bit[8] which indicates active auto-polling phase. After requesting a STIG, the QSPI Flash Controller immediately issues appropriate OPCODE to Memory. During servicing a STIG (in auto-polling phase) the status bit of command execution remains steady and other parts of transfer such as ADDRESS or DUMMY etc. are disabled (to issued Program Suspend Command is needed OPCODE).

There is a programmable option to add delay between every repetitive poll operation (delay is defined by Write Completion Control Register [31:24]). This feature is implemented to free up SPI bandwidth needed.

67.2.2.7. **Arbitration between Direct/Indirect Access Controller and STIG**

When multiple controllers are active simultaneously, a simple fixed-priority arbitration scheme is used to arbitratebetween each interface and access the external FLASH. The fixed priority is defined as follows, highest priority first:

1. The Indirect Access Write
2. The Direct Access Write
3. The STIG
4. The Direct Access Read
5. The Indirect Access Read

Each controller is back pressured while waiting to be serviced.

67.2.2.8. **SPI Command Translation**

Requests issued by the direct access controller, the indirect access controller or the STIG will be translated into a sequence of byte transfers to send downstream (before serialization to the FLASH device). These sequences depend on the requested transfer but an example of a typical 1-byte non sequential READ is shown below:

INSTRUCTION OPCODE -> ADDRESS -> Mode Byte -> Dummy Cycles -> 1 byte of don’t care

For sequential accesses, an extra byte of data per read is pushed to the FLASH device on the back of the above sequence assuming it can be done so with no gap between each transferred byte.

The actual sequence sent to the FLASH device depends on the requested transfers, whether the transfer is non-sequential or sequential, whether the device has been configured in XIP mode and the state of the main Device Instruction Type programmable registers (at offsets Device Read Instruction Register and Device Write Instruction Register).

For writes, the write enable latch (or WEL) within the FLASH device itself must be high before a write sequence can be issued. The QSPI Flash Controller will automatically issue the write enable latch command before triggering a write command via the direct or indirect access controllers (DAC/INDAC) – i.e the user does not need to perform this operation. For increasing flexibility and performance user can turn off this feature by setting Device Write Instruction Register bit[8]. The opcode for WREN is typically 0x06 and is common between devices.

When write requests from the direct or indirect access controllers are no longer being received and all outstanding requests have been sent, the FLASH device will automatically start the page program write cycle. Any incoming request at this time will be held in wait states until the cycle has completed. The QSPI Flash Controller will automatically poll the FLASH device legacy SPI status register to identy when the write cycle has completed. This is achieved by sending the RDSR opcode to the FLASH device and waiting until the device itself has indicated the write cycle has completed (until the Write in Progress bit has cleared to zero and the write enable latch bit has also cleared to zero or device is ready bit has set to one). The WREN and the RDSR device instructions are the ones that are sent by the controller under the hood. For any other specic instruction that the user determines should be sent to the device (for example the device needs to be unprotected before a write command is issued), these should be handled separately by issuing FLASH commands via the STIG.

67.2.2.9. Selecting the Flash Instruction Type

In order to send the correct READ and WRITE opcodes, software should initialize the Device READ Instruction Type Register (Offset Device Read Instruction Register) and the Device WRITE Instruction Type Register (Offset Device Write Instruction Register). These registers include fields to setup the required instruction opcodes that is intended to be used to access the FLASH (default is basic READ and basic page program) as well as the instruction type, edge mode (DDR or SDR) and whether the instruction uses single, dual or quad pins for address and data transfer. Providing this level of control to the user provides a future proofed generic solution. To ensure the controller can operate from a reset state, the registers will be reset to an opcode compatible with SIO devices what can be modied using BOOT feature.

Despite being applicable for both READs and WRITEs, the Instruction Type field in the Device Read Instruction Register [9:8] appears once – it is not included in the Device Write Instruction Register. software sets this to anything other than "0", then the address transfer type and the data transfer type of both Device Read Instruction Register and Device Write Instruction Register Registers become don't care. It is made available to allow software to support the less common FLASH instructions where the opcode, address and data are sent on 2, 4 or 8 lanes (the opcode from most instructions are sent serially to the FLASH device, even for dual/quad instructions). Also note that for devices that support instructions that can send the OPCODE over multiple lanes, the name for these instructions are not consistently named in the FLASH datasheets. One of the devices that support these instructions is the Numonyx (Micron) N25Q128. The extra READ instructions are called DCFR and QCFR. The WRITE instructions are DCPP and QCPP.

There are group of devices (e.g. Numonyx (Micron) N25Q512A) capable to handling Read Operations in Dual Data Rate Mode (DDR) (it is also called Dual Transfer Rate Mode (DTR) and these two will be used interchangeably down the document). That means they can issue and capture the data on both rising and falling edges during working with dedicated command type. This enables the controller to maintain throughput at twice lower frequency of spi_clk. The Device Read

Instruction Register has DDR enable bit which inform QSPI Flash Controller that opcode written into Read Opcode field is capable with DDR command type. The another field defined in Read Data Capture Register [19:16] which enables the controller to sht the transmitted data in DDR mode. By default, data are shted by 1 clock cycle to ensure hold timing greater than 0 during DDR transactions. It may not be sufficient for high ref_clk frequency in accordance with the high dividers.

There is also MT25Q family of Micron devices where DTR protocol was implemented. It enables device to handle all commands in DTR Mode. DTR Read commands described above detect DTR Mode based on dedicated OPCODE. Therefore OPCODE has to be send as STR. When DTR protocol is enabled, device does not need OPCODE to detect edge Mode because can recognize it based on volatile or non-volatile bit from configuration register of itself.

For reference, below is a table illustrating how software should configure the controller for selected specic READ and WRITE instruction supported by the mentioned devices:

Table 5.2- 3 Read examples

Command	Opcode (how many lanes/edge mode)	Addr/Dummy/Mode (how many lanes/edge mode)	Data (how many lanes/edge mode)	Instruction Type (Device Read Instruction Register [9:8]	Address transfer type (Device Read Instruction Register [13:12]	Data transfer type (Device Read Instruction Register [17:16]	DDR bit enable (Device Read Instruction Register [10]
READ	1/SDR	1/SDR	1/SDR	0	0	0	0
FAST_READ	1/SDR	1/SDR	1/SDR	0	0	0	0
DTR_FAST_READ	1/SDR	1/SDR	1/SDR	0	0	0	1
DOFR (Dual O/p Fast Read)	1/SDR	1/SDR	2/SDR	0	0	1	0
DIOFR (Dual I/O Fast Read)	1/SDR	2/SDR	2/SDR	0	1	1	0
DDIOFR (DTR Dual I/O Fast Read)	1/SDR	2/SDR	2/SDR	0	1	1	1
QOFR (Quad O/p Fast Read)	1/SDR	1/SDR	4/SDR	0	0	2	0
QIOFR (Quad I/O Fast Read)	1/SDR	4/SDR	4/SDR	0	2	2	0
DQIOFR (DTR Quad I/O Fast Read)	1/SDR	4/SDR	4/SDR	0	2	2	1

DCFR (Dual Command Fast Read)	2/SDR	2/SDR	2/SDR	1	DON’T CARE	DON’T CARE	0
DDCFR (DTR Dual Command Fast Read)	2/SDR	2/SDR	2/SDR	1	DON’T CARE	DON’T CARE	1
QCFR (Quad Command Fast Read)	4/SDR	4/SDR	4/SDR	2	DON’T CARE	DON’T CARE	0
DQCFR (DTR Quad Command Fast Read)	4/SDR	4/SDR	4/SDR	2	DON’T CARE	DON’T CARE	1

NOTE1:
This data are applicable for both 3-byte or 4-byte address variants of the commands did not indicated otherwise.

NOTE2:

In DTR protocol all transfer phases (including opcode) take DDR edge mode independently on the command under execution. DTR protocol is to be enabled by QSPI Configuration Register bit[24] It has higher priority than DDR Mode enable bit from Device Read Instruction Register [10] (0x04).

Table 5.2- 4 Write examples

Command	Opcode (how many lanes)	Addr/Dummy/Mode (how many lanes)	Data (how many lanes)	Instruction Type (Device Read Instruction Register [9:8]	Address transfer type (Device Write Instruction Register [13:12]	Data transfer type (Device Write Instruction Register [17:16]
PP	1	1	1	0	0	0
DP (Dual Input Fast Program)	1	1	2	0	0	1
DIEFP (Dual Input Extended Fast Program)	1	2	2	0	1	1
QP (Quad Input Fast Program)	1	1	4	0	0	2

QIEFP (Quad Input Extended Fast Program)	1	4	4	0	2	2
DCPP (Dual Command Fast Program)	2	2	2	1	DON'T CARE	DON'T CARE
QCPP (Quad Command Fast Program)	4	4	4	2	DON'T CARE	DON'T CARE

NOTE1:
This data are applicable for both 3-byte or 4-byte address variants of the commands did not indicated otherwise.
NOTE2:
In DTR protocol all transfer phases (including opcode) take DDR edge mode independently on the command under execution. DTR protocol is to be enabled by QSPI Configuration Register.

67.2.2.10. APB Interface and Register Module

The APB interface conforms to AMBA v3.0. It is used to configure the core and perform software controlled FLASH accesses using the Flash Command Control register (refer to Section "*Software Triggered Instruction Generator(STIG)*"). The APB interface feeds a single register block containing the programmable register set. The register block is timed to the APB clock. All control or enable that are triggered by APB writes and cause an event to trigger, or an operation somewhere in the QSPI controller to be enabled are synchronized to the destination clock. Static configuration that have no effect while a separate enable bit is low do not require synchronization.

67.2.2.11. **Read Data Capturing**

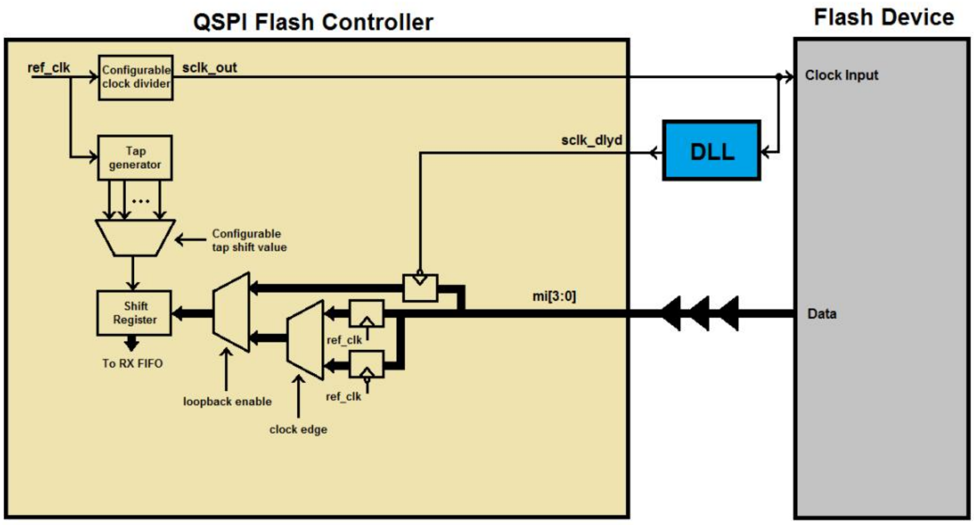


Figure 5.2- 4 Sampling data diagram

There are three mechanisms of data capturing in the controller. They can be combined in some parts to ensure reliable sampling solution independent on the system requirements and the controller configuration.

67.2.2.11.1.Mechanism using taps

After POR, the adapted loopback clock circuit and the ref_clk delay register line both wake in a disabled state. This should be valid for most applications, and can be used for device enumeration, where the device clock will be configured slowly(default SPI clock divider = 32). The Read Data Capture Register (0x10) provides the control for the mechanism using taps.

Bit 5'th selects the edge of the ref_clk, on which data outputs from flash memory are sampled. There is the compilation option, to turn off capturing the data on negedge of the ref_clk. Some systems are very limited over implementation of both negative and positive edges of flip-flops located in single domain. Furthermore necessity of increasing of sampling resolution two times is required for very high performance reads . When the system does not have such restrictive requirements, it is recommended to remove flip-flops (working on negedge of the ref_clk clock) from the project by change of the dedicated compilation parameter (CDNS_QSPI_BOTH_EDGES_SAMPLING) placed inside cdns_qspi_spi_ctrl_core_defs.v file.

4:1 controls the additional number of read data capture cycles (this is the fast ref_clk, running at least x4 of the device clock) that should be applied to the internal read data capture circuit. The large clock-to-out delay of the flash memory together with trace delays as well as other device delays may impose a maximum flash clock frequency which is less than the flash memory device itself can operate at. To compensate, software should set this register to a value that guarantees robust data captures.

Figure below shows an exemplary timing diagram of the capturing data with mechanism using taps in SDR edge mode:

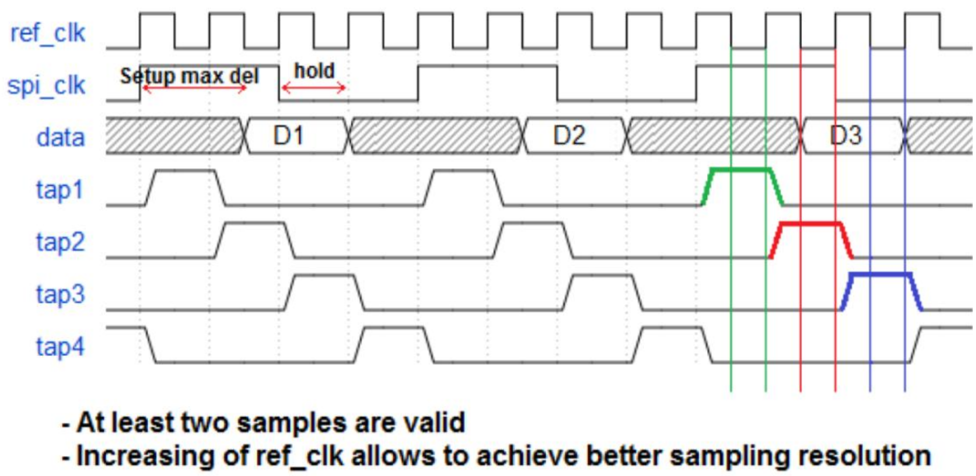


Figure 5.2- 5 Sampling mechanism using taps

67.2.2.11.2.Mechanism using external DLL

To further improve the input timing of the design, the master output clock (sclk_out) can be looped back in the logic that is instantiating the controller IP and then passed through an adjustable delay (not included in the QSPI IP core). This feature may be required when operating at high speeds with devices that have a very small guaranteed read data sampling window. For example, a device operating at 100MHz (ref_clk at 400MHz) where the clock to output valid time can be as high as 8ns or even 9ns, while the output HOLD time can be very low, even as low as just 1ns. This would give a valid sampling window of a few ns. While this would remain valid across all operating conditions, that window applies to the device . Extra delays associated with trace and pad in routing the read data back to the QSPI IP core must also be considered, and these will vary depending on operating conditions also. With such a small read data window, the ref_clk delay feature (allowing samples to be taken at 1.25ns (both edges) intervals when the ref_clk is 400MHz) may not be flexible enough for some purposes. Using the adapted delay line feature can provide a mechanism to allow the sampling point used by the QSPI IP to move beyond the ref_clk interval limitation and also may help compensate for operating condition variance. The adapted clock will be an input to the controller and, when enabled, will be used to clock the first register in the data-capture logic. Note that the read data window is large enough, the ref_clk delay register described earlier will be sufficient to capture all data and the adapted loopback circuit will not be required. Clearing bit 0 enables the adapted loopback clock circuit. Note that before the loopback clock is passed into the QSPI core, it must first be fed through a programmable delay line that is external to this IP. The user is responsible for implementing this delay line. The clock should be delayed such that the rising edge is in the center of the returned read data window.

Figure below shows a exemplary timing diagram of the capturing data with mechanism using external DLL:

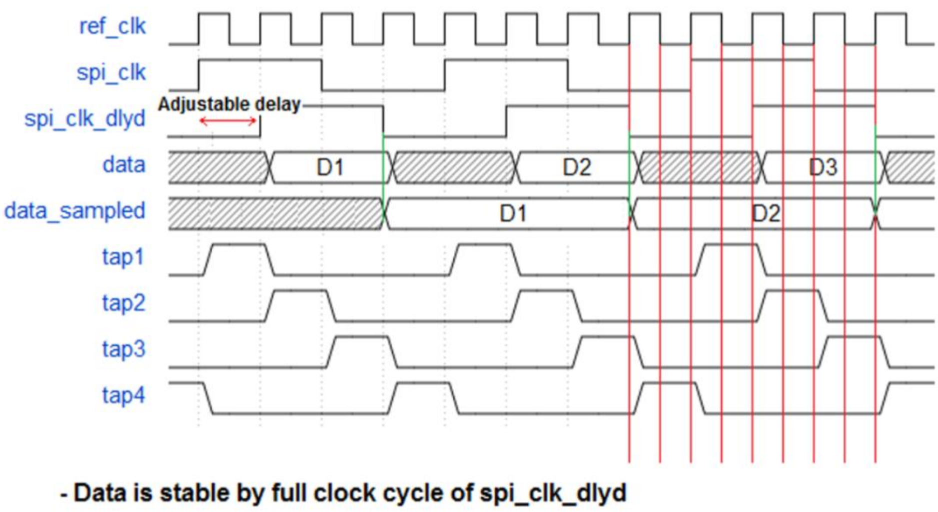


Figure 5.2- 6 Sampling mechanism using external DLL

67.2.2.12. BOOT feature

Default BOOT feature assumes following:

- Baud Rate Divisor = /32 (sclk = ref_clk/32)
- CS[0] is active (Default BOOT feature allows to fetch data from single Flash Device)
- Simple BOOT works in Extended SPI Protocol (Single Mode)
- Simple BOOT feature uses simple Read Data Command (opcode: 0x03)

To further improve the BOOT feature the user has an option to change of default values of some programmable configuration fields to make BOOT feature more flexible. It is left for user to provide configuration compliant with the specication of the controller and Flash Device/Devices being used. The controller supports BOOT operation in XIP Mode that allows to achieve the maximum performance. The mentioned parameters can be changed in cdns_qspi_flash_ctrl_defs_default.v file before compilation process. Correct setting of them requires checking of structure of individual commands for connected Flash Device. For reference, below is a table illustrating example of BOOT configuration for Fast Read instructions:

Table 5.2- 5 Write examples

BOOT Mode	Parameters to change
100 MHz QUAD SDR	cdns_qspi_boot_baud_rate = 4’h1 cdns_qspi_boot_read_dummy = 5’h08 cdns_qspi_boot_data_type = 2’h2 cdns_qspi_boot_addr_type = 2’h2 cdns_qspi_boot_read_opcode = 8’heb
100 MHz DUAL SDR	cdns_qspi_boot_baud_rate = 4’h1 cdns_qspi_boot_read_dummy = 5’h08

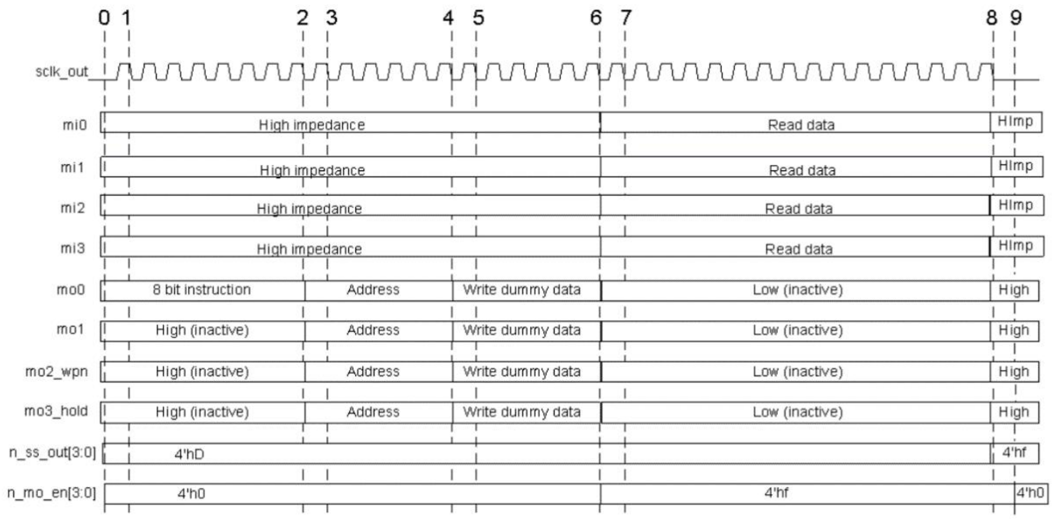
	cdns_qspi_boot_data_type = 2'h1 cdns_qspi_boot_addr_type = 2'h1 cdns_qspi_boot_read_opcode = 8'hbb
100 MHz SINGLE SDR	cdns_qspi_boot_baud_rate = 4'h1 cdns_qspi_boot_read_dummy = 5'h08 cdns_qspi_boot_data_type = 2'h0 cdns_qspi_boot_addr_type = 2'h0 cdns_qspi_boot_read_opcode = 8'h0b
50 MHz QUAD DDR	cdns_qspi_boot_baud_rate = 4'h3 cdns_qspi_boot_read_dummy = 5'h08 cdns_qspi_boot_data_type = 2'h2 cdns_qspi_boot_addr_type = 2'h2 cdns_qspi_boot_ddr_en = 1'h1 cdns_qspi_boot_read_opcode = 8'hed
100 MHz XIP QUAD SDR	cdns_qspi_boot_baud_rate = 4'h1 cdns_qspi_boot_xip = 1'h1 cdns_qspi_boot_read_dummy = 5'h08 cdns_qspi_boot_data_type = 2'h2 cdns_qspi_boot_addr_type = 2'h2 cdns_qspi_boot_read_opcode = 8'heb

The cdns_qspi_boot_baud_rate values presented in the table were calculated based on assumption that frequency of the ref_clk is 400 MHz. Presented values can be slightly dferent for various devices. Furthermore the user should be aware that to executing BOOT reads in one of Multiple I/O Modes, the device has to be configured to handle these modes by non-volatile manner. Similarly, it has to be configured to XIP Mode to handle BOOT feature in XIP.

Since all clock domains are asynchronous for one another, there might be an scenario when the Master launches booting before flops from every domain are reset. For example with fast hclk and slow pclk, the BOOT read request may be issued before the configuration (APB domain) is reset and propagated. Since the system is aware of the clock values from all domains it is able to calculate safe delay when it is ensured to have all clock domains reset.

67.2.2.13. **Example of an 8 byte Read Transfer**

To help understand the high level behaviour of the Multiple-SPI interface, the following diagram shows the primary stages in a read transfer. Note this is just one type of read instruction (QUAD I/O READ) so there are many variations on this. Refer to the specication of particular Flash Device to get more combinations.



0. Start of transaction, chip select (n_ss_out) activates slave 1 (transitions from 4'hf to 4'hd).
1. Start of instruction phase. The first of 8 bits are output only on mo0. Other mo pins unused at this stage.
2. End of instruction phase. The last of the 8 instruction bits is output.
3. Start of address phase. The example uses a 3 byte address output across all 4 mo pins. One address nibble is output per clock cycle so 3 bytes takes 6 cycles.
4. End of address phase. The last address nibble is output using all 4 mo pins.
5. The example read transaction requires 3 bytes of dummy write data. This edge is where the first of 6 nibbles are output across all 4 mo pins.
6. Last nibble of dummy write data. This is the last write phase of the read transfer therefore the output enable (n_mo_en) switches from 4'h0 to 4'hf.
7. First cycle of read data on input mi pins. Example is reading 8 bytes which will take 16 cycles using all 4 mi pins. While read data is being received, the mo signals are inactive and low.
8. Last cycle of read data input. Mi input pins switch back to being high impedance. Mo output pins switch to high. Chip enable is de-asserted for slave1. Transaction complete.
9. Output enable switches back to 4'h0 read for next transaction.

Figure 5.2- 7 SPI transfer example

67.2.3. **Programming Guide**

Software is responsible for configuring the QSPI controller before it can communicate with the FLASH device.

We assume that the controller's static configuration bits are setup before the QSPI controller is "enabled" via QSPI Configuration Register bit[0]. This is necessary in order to avoid any clock boundary issues in paths where metastability protection has not been implemented. If the user wishes to change the controller's configuration, we recommend that the QSPI enable bit is first set to "disabled" before reconfiguring.

67.2.3.1. Configuring the QSPI Controller for use after reset

The QSPI Controller has been designed to wake up in a state that is suitable for performing basic reads and writes using the direct access controller. The BASIC read (opcode 0x03) and BASIC write (opcode 0x02) instructions are operations supported by all target devices. The Controller also wakes up with a baud rate divider setting of divide-by-32. Assuming the reference clock is operating at 400MHz after reset, then this means the effective SPI clock is just 12.5 MHz. This should be slow enough to meet all timing requirements of all target devices without any further device programming.

If the target device does not use 3 address bytes, the device size configuration register must be modified to the appropriate size.

If software plans to write to the device, and the number of bytes per device page is NOT equal to 256, then the device size configuration register must also be modified. All devices tested on this project use a page size of 256 bytes, in which case no register update is required.

While not a requirement, it is prudent for software to enable the write protect feature prior to enabling the QSPI controller. This will block any AHB writes from taking effect. To do so, the protection registers (Lower Write Protection Register, Upper Write Protection Register and Write Protection Register) should be setup and the number of bytes per device block in the device size configuration register should also be setup.

After POR, software can read from and write to the FLASH device (albeit slowly). Enabling/Disabling the controller and DAC is achieved with just one write to corresponding fields of the QSPI Configuration Register (0x00). Take note to maintain the default values of the baud rate divisor and the default state of CPOL/CPHA which also are located in this register. A write data value of 0x00780081 is recommended.

67.2.3.2. Configuring the QSPI Controller for optimal use

To access the flash optimally, software must configure the controller accurately:

1. Wait until any pending STIG or INDAC operation has completed or poll QSPI Configuration Register bit[31]

2. Disable the DAC – QSPI Configuration Register bit[7]. It is permitted, but not necessary to also disable the QSPI controller completely here via QSPI Configuration Register bit[0].

3. Update the Device Read Instruction Register and Device Write Instruction Register for the

instruction type you wish to use for indirect and direct writes and reads.

4. Update the Mode Bit Configuration Register if mode bits have been enabled in the Device Read Instruction Register bit[20]

5. Update the device size configuration register if the contents are incorrect. Note parts or all of this register may have been updated after initialization. The number of address bytes is a key configuration setting required for performing reads and writes. The number of bytes per page is required for performing any write. The number of bytes per device block is only required if the write protect feature is used. If the default values are correct for the target device, or if some of the values (not including the number address bytes) were incorrect but device writes were not permitted.

6. Update the Device Delay Register This register allows the user to tweak how the chip select is driven after each FLASH access. This is required as each device may have different timing requirements. As the serial clock frequency is increased, these timing requirements become more important. Note the numbers programmed in this register are based on the period of ref_clk. Example: An ATMEL device needs 50ns minimum time before CS can be re-asserted after it has been de-asserted. By default, the controller will only provide a minimum of 1 SCLK period. When the device is operating at 100MHz, the SCLK period is only 10ns, so 40ns extra is required. Since the register defines the number of ref_clk cycles to add, and ref_clk is running at 400MHz (2.5ns period), then the user should program a value of at least 16 to the d_nss field of this register. This delay can be extended during auto-polling phase. There is possibility to define the polling repetition delay in the Write Completion Control Register bits[31:24]

7. Update the Remap Address Register, if required. Affects DAC path only. Refer to Section, "Remapping the AHB address" for further details.

8. Setup and enable write protection registers (Lower Write Protection Register, Upper Write Protection Register and Write Protection Register) if they are required and if they have not already been setup from post initialization.
9. Enable required interrupts via the Interrupt Mask Register.

10. Setup the baud rate divisor in the QSPI Configuration Register bit[22] to define the required clock frequency of the target device.

11. Update the Read Data Capture Register This register will delay when the read data is captured and can help when the read data path from the device to the controller is long and the device clock frequency is high. An update to this register may not be necessary.

12. Enable the QSPI Controller and the DAC via the QSPI Configuration Register

67.2.3.3. **Using the Flash Command Control Register (STIG Operation)**

The Flash Command Control register provides software means to access the FLASH device in a flexible and programmable manner. This is known as a STIG operation (Software Triggered Instruction Generator). The instruction opcode, number of address bytes (if any), the address itself, number of dummy cycles (if any), number of write data bytes (if any), the write data itself and the number of read data bytes (if any) can be programmed. Once these have been programmed, software can trigger the command via bit 0 and wait for its acceptance by polling bit 1. When this bit turns de-asserted, another STIG can be triggered. This method of accessing the FLASH is the typical mechanism that software would use to access the FLASH device's registers, as well as for performing ERASE operations. It can also be used to access the FLASH array itself, although the maximum of 8 data bytes may be read or written at any one time, defined in the Flash Command Write and Read Data registers (Flash Command Read Data Register (Lower), Flash Command Read Data Register (Upper), Flash Command Write Data Register (Lower) and Flash Command Write Data Register (Upper)). This number of bytes can be extended for Read Data commands using additional STIG Memory Bank controlled by Flash Command Control Register (Using STIG) Register bit[2] and Flash Command Control Memory Register (Using STIG).

Commands issued using this interface have a higher priority than all other READ accesses coming from AHB, and will therefore interrupt any READ commands being requested by the indirect or direct controllers.

67.2.3.4. **Using SPI Legacy Mode**

SPI legacy mode allows software to access the internal TX-FIFO and RX-FIFO directly, thus bypassing the direct, indirect and STIG controllers.

Legacy mode allows the user to issue any FLASH instruction to the device, but does place a heavy software overhead in order to manage the fill levels of the FIFO's effectively. This is because the legacy SPI core is bi-directional in nature, with data continuously being transferred in either direction while the chip select is enabled. Even if the driver only wishes
to read data from the FLASH device, dummy data must be written out to ensure the chip select stays active, and vice versa for write transactions. This means that to perform a basic READ of 4 bytes to a device that has 3 address bytes, software would have to write a total of 8 bytes to the TX FIFO. The first byte would be the instruction opcode, the next 3 bytes
are the address, and the final 4 bytes would be dummy data to ensure the chip select stays active while the read data is returned. Similarly, since 8 bytes were written to the TX-FIFO, software should expect 8 bytes to be returned in the RX- FIFO. The first 4 bytes of this would be discarded, leaving the final 4 bytes holding the required READ data.

This can place a lot of overhead on software. Interrupts are provided to indicate when the fill levels pass programmable watermarks, which are themselves programmable registers: TX Threshold Register and RX Threshold Register. The limited depth may impose the limitation over execution of some specific SPI commands in legacy mode. It should be

noticed that the controller interprets all transmitted bytes as valid. For example, if the Flash Device was configured to return valid data after many dummy cycles, the TX FIFO could become full before the controller sends all of dummy data.

67.2.3.5. Using XIP Mode

Since there is no consistent standard of servicing XIP read operations among FLASH Devices it is not ensured that described approaches cover all cases supported by FLASH Devices present on the market. Most common approaches have been listed.

67.2.3.5.1. 2.2.5.1. Entering XIP mode from POR

XIP is a mode that can be entered in a non-volatile way if the device has XIP enabled as a non-volatile configuration setting. Software will not be able to discover the state of XIP from POR via FLASH status register reads as the only operation a FLASH device will recognize when XIP mode is enabled is an XIP read operation.

If it is already known that the device will enter XIP from POR, then the Mode Bit Configuration Register and QSPI Configuration Register bit[18] should be set in initial boot.

If it is not already known that the device will enter XIP from POR, and XIP from POR may be supported by the attached FLASH device, then software can attempt to exit XIP mode by issuing an XIP exit command using a STIG command (via the Flash Command Control Register (Using STIG)). To do this, software must be aware of the mode bit requirements of that device, as XIP entry and exit changes per device. Devices that support XIP from POR from the list of tested devices are Micron N25Q and MT25Q devices. For these devices, exiting XIP mode can be performed by configuring the opcode to 8'h00, the number of address bytes to 3, the number of dummy cycles to 16 and the number of read bytes to 1. The address that should be programmed should be {8 mode bits for exiting XIP mode, 16'h0000}. The mode bits for the Micron device must be 8'b10000000.

Note the XIP from POR mode can be enabled by issuing a NVCR write command via the Flash Command Control

Register (Using STIG). As this does not take effect until the next POR sequence, the effect of this will be picked up during the next enumeration sequence.

67.2.3.5.2. Entering XIP mode otherwise

XIP mode is supported in most FLASH devices. However, FLASH manufacturers do not have a consistent standard approach for achieving this. Most use signature bits that are sent to the device immediately following the address bytes. Some, like the Micron devices, use signature bits and also require a FLASH device configuration register write to enable XIP. For the FLASH devices that must be compliant to this controller, the following steps can be taken by software to enter XIP mode:

67.2.3.5.3. Exiting XIP mode

To exit XIP mode, software should first disable the Direct Access Controller and Indirect Access Controller to ensure
no new AHB read accesses will be sent to the FLASH device. It should then set the mode bits to anything other than the mode bits specified in the specification of corresponding Flash Device. These are dependent on the FLASH device and manufacturer. Software should then reset QSPI Configuration Register bit[17].

Note the FLASH device must see a READ instruction before it can disable its internal XIP mode state, so this means XIP mode will internally stay active until the next READ instruction is serviced. Care must be taken to ensure that XIP mode is disabled before the end of any READ sequence.

67.2.3.6. **Using Indirect Data Transfer Mode**

The software sequence that should be followed when using the Indirect read controller, including how the DMA peripheral interface is setup is specified.

67.2.3.6.1. Indirect read transfer process

When the DMA peripheral controller is enabled, the following sequence can be followed:

1. Setup QSPI Configuration Register (0x00)

2. Setup the SRAM watermark in the Indirect Read Transfer Watermark Register (0x64)

3. Setup the indirect transfer's FLASH start address in the Indirect Read Transfer Start Address Register (0x68)

4. Setup the number of bytes to be transferred in the Indirect Read Transfer Number Bytes Register (0x6c)

5. Setup the indirect transfer’s AHB trigger address in the Indirect AHB Address Trigger Register (0x1c)

6. Setup the indirect transfer's AHB trigger address range in the Indirect Trigger Address Range Register (0x80)

7. Setup number of bytes for "single" and "burst" DMA peripheral burst type transfers in the DMA Peripheral Configuration Register (0x20)

8. Trigger Indirect Read access by setting Indirect Read Transfer Control Register bit[0]

9. The completion status of the indirect read operation can be polled via the Indirect Read Transfer Control Register bit[5]. Note this bit is write-to-clear. Alternatively, an Indirect Complete interrupt will be generated when the Indirect read operation has completed.

10. The number of indirect read operations completed will be readable in the same register - Indirect Read Transfer Control Register bits[7:6].

When the DMA peripheral controller is disabled, the following sequence can be followed:

1. Setup QSPI Configuration Register (0x00)

2. Setup the indirect transfer’s FLASH start address in the Indirect Read Transfer Start Address Register (0x68)

3. Setup the number of bytes to be transferred in the Indirect Read Transfer Number Bytes Register (0x6c)

4. Setup the indirect transfer’s AHB trigger address in the Indirect AHB Address Trigger Register (0x1c)

5. Setup the indirect transfer's AHB trigger address range in the Indirect Trigger Address Range Register (0x80)

6. If the watermark interrupt feature is to be used, set the Indirect Read Transfer Watermark Register (0x64) which will cause an interrupt to be generated when the fill level increases beyond the watermark level. Setting the watermark can be useful indication to software when to read the next part of the indirect read transfer. Note that if the watermark is set to a value other than zero, the watermark interrupt will always trigger once the final byte of indirect transfer has been fetched and placed in the embedded SRAM, even if the watermark value is higher than the actual

completed fill
level.

7. Trigger Indirect Read access by setting Indirect Read Transfer Control Register bit[0]

8. If the watermark interrupt feature is to be used, wait for watermark interrupt. Else poll the SRAM fill level to decide when sufficient data is in the SRAM to trigger AHB data fetches

9. Read the expected amount of data from SRAM. If there is still more data to fetch in order to complete the indirect read transfer, then loop back to 8. Otherwise continue to 10

10. The completion status of the indirect read operation can be polled via the Indirect Read Transfer Control Register bit[5].

11. An Indirect Complete interrupt will be generated when the Indirect read operation has completed

67.2.3.6.2. Indirect write transfer process

When the DMA peripheral controller is enabled, the following sequence can be followed:

1. Setup QSPI Configuration Register (0x00)

2. Setup the indirect transfer's FLASH start address in the Indirect Write Transfer Start Address Register (0x78)

3. Setup the number of bytes to be transferred in the Indirect Write Transfer Number Bytes Register (0x7c)

4. Setup the indirect transfer's AHB trigger address in the Indirect AHB Address Trigger Register (0x1c)

5. Setup the indirect transfer's AHB trigger address range in the Indirect Trigger Address Range Register (0x80)

6. Setup number of bytes for "single" and "burst" DMA peripheral burst type transfers in the DMA Peripheral Configuration Register (0x20)

7. Optional: Set the Indirect Write Transfer Watermark Register (0x74) to control the rate at which DMA requests will be issued.

8. Trigger Indirect Write access by setting Indirect Write Transfer Control Register bit[0].

9. The QSPI controller will request the DMA to transfer data using the DMA request interface.

10. The completion status of the indirect write operation can be polled via the Indirect Write Transfer Control Register bit[5]. This will be write-to-clear. The number of indirect write operations completed will be readable in the same register.

11. An Indirect Complete interrupt will be generated when the Indirect write operation has completed.

When the DMA peripheral controller is disabled, the following sequence can be followed:

1. Setup QSPI Configuration Register (0x00)

2. Setup the indirect transfer's FLASH start address in the Indirect Write Transfer Start Address Register (0x78)

3. Setup the number of bytes to be transferred in the Indirect Write Transfer Number Bytes Register (0x7c)

4. Setup the indirect transfer's AHB trigger address in the Indirect AHB Address Trigger Register (0x1c)

5. Setup the indirect transfer's AHB trigger address range in the Indirect Trigger Address Range Register (0x80)

6. It is functionally valid for software to simply write all the data to the SRAM in one block transfer. However, if the total number of bytes to write is greater than the size of the partitioned SRAM, then it is quite likely the SRAM will become full causing the QSPI to back-pressure the system AHB bus for a considerable time. This time is based on the FLASH data-rate and the page-write time of the device. To avoid sending all the write data in one block transfer, software can make use of the watermark interrupt to identify a convenient time to send data a page at a time to the SRAM module. Alternatively, software can poll the SRAM fill level register directly to identify how empty the SRAM is at any one time in order to make a judgement as to when the most practical time to send the next part of the transfer.

7. If the watermark interrupt feature is to be used, set the Indirect Write Transfer Watermark Register (0x74) which will cause an interrupt to be generated when the fill level falls below the watermark. The watermark should be set to a number between zero and a page size. I.e. if the page size is 256 bytes, then setting the watermark to a value between 10 and 250 is reasonable and will cause the interrupt to trigger when the fill level drops below the programmed number. Setting the watermark can be useful to provide an indication to software when to write the next page of data to the SRAM.

8. Trigger Indirect Write access by setting Indirect Write Transfer Control Register bit[0].

9. If the remaining number of bytes still to be transferred into the SRAM for the current indirect transfer is greater than a FLASH page, then write 1 FLASH page worth of data to the SRAM.

Otherwise send the remaining data from the indirect transfer to SRAM.

10. If all the data in the indirect transfer has now been sent to the SRAM, then go to 12. and await indirect complete status. Otherwise if there is more data still to be transferred then either:

a. If the watermark interrupt feature is being used, then wait for watermark interrupt.

b. Alternatively the SRAM fill level can be interrogated to identify a convenient time to send more data.

11. Loop back to 9.

12. Optional: The completion status of the Indirect write operation can be polled via Indirect Write Transfer Control Register bit[5].

13. An Indirect Complete interrupt will be generated when the Indirect write operation has completed.

67.2.3.7. Remapping AHB addresses

The remap feature of the QSPI allows software to define how incoming AHB addresses translate to FLASH addresses when the access passes through the direct access controller. By default there is a 1:1 mapping. When QSPI Configuration Register bit[16] is enabled, the incoming AHB address is remapped to address+N, where N is the value stored in the Mode Bit Configuration Register (0x24). It is recommended that the QSPI is disabled before configuring the remap registers.

67.2.3.8. Servicing Interrupts

The controller provides a single active high level sensitive interrupt pin. Refer to the Interrupt Status Register (0x40) for details of all the interrupt sources provided.

The interrupt source can be cleared when software reads the interrupt status register. Reported interrupts are cleaned by writing "1" (write-to-clear approach is implemented).

67.2.3.9. Using the AHB Protection Registers

On POR, the AHB protection mechanism wakes in a disabled state. Software can use the protection registers to block AHB writes to certain regions. Refer to Section, “Write Protection” for details. It is recommended that the QSPI is disabled before configuring the protection registers.

67.2.4. Register Summary

Offset	Function	R/W
0x00	QSPI Configuration Register	R/W
0x04	Device Read Instruction Register	R/W
0x08	Device Write Instruction Register	R/W
0x0C	QSPI Device Delay Register	R/W
0x10	Read Data Capture Register	R/W
0x14	Device Size Register	R/W
0x18	SRAM Partition Register	R/W
0x1C	Indirect AHB Address Trigger Register	R/W
0x20	DMA Peripheral Register	R/W
0x24	Remap Address Register	R/W
0x28	Mode Bit Register	R/W
0x2C	SRAM Fill Level Register	RO
0x30	TX Threshold Register	R/W
0x34	RX Threshold Register	R/W
0x38	Write Completion Control Register	R/W
0x3C	Polling Expiration Register	R/W
0x40	Interrupt Status Register	R/W
0x44	Interrupt Mask Register	R/W
0x50	Lower Write Protection Register	R/W
0x54	Upper Write Protection Register	R/W

67.2.5. Register Description

: This register contains basic configuration fields of the controller.

Table 5.2- 6 QSPI Configuration Register

Offset	Bit	R/W		Reset

0x00	31	RO	Serial Interface and QSPI pipeline is IDLE. This is a STATUS read- bit. Note this is a re-timed signal, so there will be some inherent delay on the generation of this status signal. **Note** It is recommended to wait at least 4 cycles of ref_clkbetween getting IDLE bit asserted and switching any configuration field. This latency ensures that all low level synchronization is done, FOs are empty and the controller can be safely introduced into another mode of operation	1'h1
0x00	30:25	RO	Unused . Read as zero.	6'h0
0x00	24	R/W	Enable DTR Protocol This bit should be set device is configured to work in DTR protocol. **Note** DTR protocol provides all commands in DTR Mode. There are DTR Read commands which can be handled in STR Protocol (but in DTR Mode). This bit should be equal "0" to executing them. DTR commands in STR protocol are controlled by 10th bit of Device Read Configuration Register.	1'h0
0x00	23	R/W	Enable AHB Decoder (Direct Access Mode) When set to "1", QSPI Configuration Register bit[13:10] are not relevant. Active slave is based on actual AHB address. The partition for each device is calculated with respect to Device Size Configuration Register [28:21]	1'h0
0x00	22:19	R/W	Master mode baud rate divisor (2 to 32), SPI baud rate = (master reference clock)/BD Where BD is: 4'b0000 = /2 4'b0001 = /4 4'b0010 = /6 4'b0011 = /8 4'b0100 = /10 4'b0101 = /12 4'b0110 = /14 4'b0111 = /16 4'b1000 = /18 ... 4'b1111 = /32 Set this register up before enabling the QSPI controller.	4'hf

0x00	18	R/W	Enter XIP Mode immediately 0 = XIP is enabled, then setting to "0" will cause the controller to exit XIP mode after the next READ instruction has completed. 1 = Operate the device in XIP mode immediately Use this register when the external device is introduced to XIP mode (as per the contents of its configuration register). The controller will assume the next READ instruction will be passed to the device as an XIP instruction, and therefore will not require the READ opcode to be transferred. **Note** To exit XIP mode, this bit should be set to "0". This will take effect in the attached device AFTER the next READ instruction is executed. Software should therefore ensure that at least one READ instruction is requested after resetting this bit before it can be sure XIP mode in the device is exited. This bit is synchronized in hardware.	1’h0
0x00	17	R/W	Enter XIP Mode on next READ 0 = XIP is enabled, then setting to "0" will cause the controller to exit XIP mode after the next READ instruction has completed. 1 = XIP is disabled, then setting to "1" will inform thecontroller that the device is ready to enter XIP on the next READ instruction. The controller will therefore send the appropriate command sequence, including mode to cause the device to enter XIP mode. Use this register after the controller has ensured the attached FLASH device has been configured to be ready to enter XIP mode. **Note** To exit XIP mode, this bit should be set to "0". This will take effect in the attached device AFTER the next READ instruction is executed. Software should therefore ensure that at least one READ instruction is requested after resetting this bit before it can be sure XIP mode in the device is exited. This bit is synchronized in hardware.	1’h0
0x00	16	R/W	Enable AHB Address Re-mapping (Direct Access Mode) When set to "1", the incoming AHB address will be adapted and sent to the FLASH device as (address + N), where N is the value stored in the remap address register. Use this register after the controller has ensured the attached	1’h0

			FLASH device has been configured to be ready to enter XIP mode. This bit is synchronized in hardware.	
0x00	15	R/W	Enable DMA Peripheral Interface Set to "1" to enable the DMA handshaking logic. When enabled the QSPI will trigger DMA transfer requests via the DMA peripheral interface. Set to '0' to disable DMA Peripheral Interface. This bit is synchronized in hardware.	1'h0
0x00	14	R/W	Set to drive the Write Protect pin of the FLASH device. This is resynchronized to the generated memory clock as necessary. Note that the WP pin is valid in SINGLE or DUAL transfer modes. During QUAD transfers, the WP pin is used for transferring data and therefore any setting of this register bit will be ignored. This bit is synchronized in hardware.	1'h0
0x00	13:10	R/W	Peripheral chip select lines bit 9 of this register = 0, ss[3:0] are output thus: ss[3:0] n_ss_out[3:0] 4'bxxx0 4'b1110 4'bxx01 4'b1101 4'bx011 4'b1011 4'b0111 4'b0111 4'b1111 4'b1111 (no peripheral selected) bit 9 of this register = 1, ss[3:0] directly drives n_ss_out[3:0]	4'h0
0x00	9	R/W	Peripheral select decode 0 = 1 of 4 selects n_ss_out[3:0] is active 1 = allow external 4-to-16 decode (n_ss_out=ss)	1'h0
0x00	8	R/W	Legacy IP Mode Enable 0 = Use Direct Access Controller/Indirect Access Controller or STIG interface for data transfers 1 = When set to "1", legacy mode is enabled. In this mode, any write to the controller via the AHB interface is serialized and sent to the FLASH device. Any valid AHB read will pop the internal RX-FO, retrieving data that was forwarded by the external FLASH device on the SPI lines. 4, 2 or 1 byte transfers are permitted and controlled via the HSIZE input. This bit is synchronized in hardware.	1'h0
0x00	7	R/W	Enable Direct Access Controller 0 = disable the Direct Access Controller once current transfer of the data word is completed. 1 = enable the Direct Access Controller When the Direct Access Controller and Indirect Access	1'h1

			Controller are both disabled, all AHB requests are completed with an error response. This bit is synchronized in hardware.	
0x00	6:3	RO	Unused . Read as zero.	4’h0
0x00	2	R/W	Clock phase - This maps to the standard SPI CPHA transfer format. **Note** Keep this bit low when operating in DDR Mode or DDRProtocol to ensure the falling edge of SPI clock as the last one.	1’h0
0x00	1	R/W	Clock polarity outside SPI word - This maps to the standard SPI CPOL transfer format. **Note** Keep this bit low when operating in DDR Mode or DDRProtocol to ensure the falling edge of SPI clock as the last one.	1’h0
0x00	0	R/W	QSPI Enable 0 = disable the Multiple-SPI interface once current transfer of the data word is completed. 1 = enable the Multiple-SPI interface When set to low, all output enables are inactive and all pins are set to input mode. This bit is synchronized in hardware.	1’h1

67.2.5.1. **Device Read Instruction Register**

: This register defines the configuration of Multiple-SPI READ instruction. This register should be setup
while the controller is idle.

Table 5.2- 7 Device Read Instruction Register

Offset	Bit	R/W		Reset
0x04	31:29	RO	Unused . Read as zero.	3’h0
0x04	28:24	R/W	Number of Dummy Clock Cycles required by device for Read Instruction.	5’h0

0x04	23:21	RO	Unused . Read as zero.	3'h0
0x04	20	R/W	Mode Bit Enable Set to "1" to ensure the mode as defined in the Mode Bit Configuration Register are sent following the address bytes	1'h0
0x04	19:18	RO	Unused . Read as zero.	2'h0
0x04	17:16	R/W	Data Transfer Type for Standard SPI modes (as defined by the instruction type in [9:8]) 2'b00: SIO mode - data is shted to the device on DQ0 and from the device on DQ1 2'b01: Used for Dual Input/Output instructions. For data transfers, DQ0 and DQ1 are used as both inputs and outputs. 2'b10: Used for Quad Input/Output instructions. For data transfers, DQ0, DQ1, DQ2 and DQ3 are used as both inputs and outputs.	2'h0
0x04	15:14	RO	Unused . Read as zero.	2'h0
0x04	13:12	R/W	Address Transfer Type for Standard SPI modes (as defined by the instruction type in [9:8]) 2'b00: Addresses can be shted to the device on DQ0 2'b01: Addresses can be shted to the device on DQ0 and DQ1 2'b10: Addresses can be shted to the device on DQ0, DQ1, DQ2 and DQ3	2'h0
0x04	11	RO	Unused . Read as zero.	1'h0
0x04	10	R/W	DDR Bit Enable Set to "1" when opcode from 7 to 0 is compliant with DDR command. Master output data is issued by the controller in DDR fashion starting of the address SPI transfer phase. It is not applicable for STIG Mode.	1'h0
0x04	9:8	R/W	Instruction Type 2'b00: Use Standard SPI mode (instruction always shted into the device on DQ0) 2'b01: Use DIO-SPI mode (Instructions, Address and Data always sent on DQ0 and DQ1) 2'b10: Use QIO-SPI mode (Instructions, Address and Data always sent on DQ0,DQ1,DQ2 and DQ3) **Note**	2'h0

			These are relevant not just to READ transfers. They are global settings and will affect READ's, WRITES via the DAC and INDAC as well as any transfer sent using the STIG.	
0x04	7:0	R/W	Read Opcode to use when not in XIP mode	8'h03

67.2.5.2. **Device Write Instruction Register**

: This register defines the configuration of Multiple-SPI WRITE (Program Page) instruction. This register should be setup while the controller is idle.

Table 5.2- 8 Device Write Instruction Register

Offset	Bit	R/W		Reset
0x08	31:29	RO	Unused . Read as zero.	3'h0
0x08	28:24	R/W	Number of Dummy Clock Cycles required by device for Write Instruction.	5'h0
0x08	23:18	RO	Unused . Read as zero.	6'h0
0x08	17:16	R/W	Data Transfer Type for Standard SPI modes (as defined by the instruction type in Device Read Instruction Register [9:8]) 2'b00: SIO mode - data is shted to the device on DQ0 and from the device on DQ1 2'b01: Used for Dual Input/Output instructions. For data transfers, DQ0 and DQ1 are used as both inputs and outputs. 2'b10: Used for Quad Input/Output instructions. For data transfers, DQ0, DQ1, DQ2 and DQ3 are used as both inputs and outputs.	2'h0
0x08	15:14	RO	Unused . Read as zero.	2'h0
0x08	13:12	R/W	Address Transfer Type for Standard SPI modes (as defined by the instruction type in Device Read Instruction Register [9:8]) 2'b00: Addresses can be shted to the device on DQ0 2'b01: Addresses can be shted to the device on DQ0 and	2'h0

			DQ1 2'b10: Addresses can be shted to the device on DQ0, DQ1, DQ2 and DQ3	
0x08	11:9	RO	Unused . Read as zero.	1’h0
0x08	8	R/W	WEL Disable This bit allows the user to turn off automatic issuing of WELCommand before write operation for DAC or INDAC.	1’h0
0x08	7:0	R/W	Write Opcode	8’h02

67.2.5.3. **Device Delay Register**

: This register is used to introduce relative delays into the generation of the master output signals. This allows the timings to be adjusted to meet the requirements of a specic flash device. All timings are defined in cycles of the SPI master ref clock. This register should be setup while the controller is idle.

For an illustration of the register fields, refer to the two diagrams below this table: Device Delay Register timing diagrams.

Table 5.2- 9 Device Delay Register

Offset	Bit	R/W		Reset
0x0C	31:24	R/W	CSDA - Chip Select De-Assert Added delay in master reference clocks (ref_clk) for the length that the master mode chip select outputs are de-asserted between transactions. The minimum delay for chip select to be de-asserted (CSDA=0) is: 1 sclk_out + 1 ref_clk to ensure the chip select is never reasserted within an sclk_out period. CSDA=N, then the chip select de-assert time will be: 1 sclk_out + 1 ref_clk + N ref_clks.	8’h0
0x0C	23:16	R/W	CSDADS – Chip Select De-Assert Dferent Slaves Delay in master reference clocks (ref_clk) between one chip select being de-activated and the activation of another. This is used to ensure a quiet period between the selection of two dferent slaves. CSDADS is relevant when switching between 2 dferent external flash devices. An example scenario might be you have 2 FLASH devices	8’h0

			named A and B, each connected to the controller on a dferent chip select. You might have a requirement to have chip select inactive for a particular period of time between each successive non-sequential transfer of the same device. You would control this using CSDA. There may be a dferent requirement to hold chip select inactive for a longer period of time between the end of one transfer on device A and the start of the next transfer on device B. You would control this using CSDADS. The minimum delay (CSDADS=0) is: 1 sclk_out + 3 ref_clk. CSDA=N, then the chip select de-assert time will be: 1 sclk_out + 1 ref_clk + N ref_clks.	
0x0C	15:8	R/W	CSEOT – Chip Select End Of Transfer Delay in master reference clocks between last bit of current transaction and de-asserting the device chip select (n_ss_out). By default (when CSEOT=0), the chip select will be de-asserted on the last falling edge of sclk_out in SPI Mode 0 or on the last rising edge + half spi clock cycle in SPI Mode 3 at the completion of the current transaction. CSEOT=N, then chip selected will de-assert N ref_clks after the last falling edge of sclk_out in SPI Mode 0 or on the last rising edge + half spi clock cycle in SPI Mode 3.	8’h0
0x0C	7:0	R/W	CSSOT – Chip Select Start Of Transfer Delay in master reference clocks between setting n_ss_out low and first bit transfer. By default (CSSOT=0), chip select will be asserted half a SCLK period before the first rising edge of sclk_out. CSSOT=N, chip select will be asserted half a sclk_out period before the first rising edge of sclk_out + N ref_clks.	8’h0

Figure below shows chip select behaviour between 2 consecutive transfers to the same slave

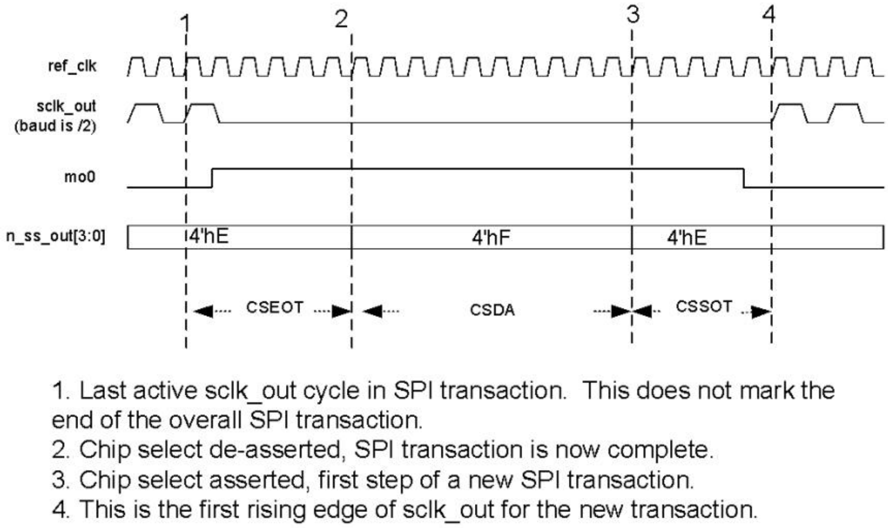


Figure 5.2- 8 Device Delay Register timing diagram (single slave)

Figure below shows chip select behaviour between 2 consecutive transfers to dferent slaves

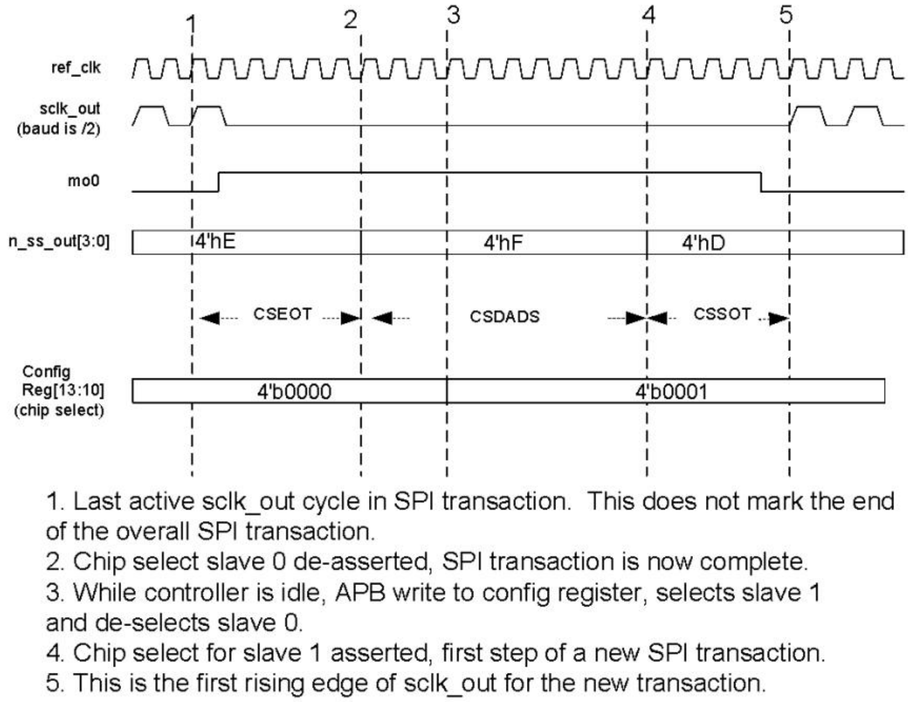


Figure 5.2- 9 Device Delay Register timing diagram (dferent slave)

67.2.5.4. **Read Data Capture Register**

: This register is used to adjust SPI transfer conditions in order to fetch and capture data reliably. This register should be setup while the controller is idle.

Table 5.2- 10 Read Data Capture Register

Offset	Bit	R/W		Reset
0x10	31:20	RO	Unused . Read as zero.	12'h0
0x10	19:16	R/W	Delay the transmitted data (Master Output) by the programmable number of ref_clk cycles (in order to improve the hold timing during transfers in DDR Mode). **Note** This field is relevant when DDR Read Command is executed. Otherwise can be ignored. delay_value = 1 x ref_clk + [field_value] x ref_clk	4'h0
0x10	15:6	RO	Unused . Read as zero.	10'h0
0x10	5	R/W	Sample edge selection (of the flash memory data outputs). 0 = data outputs from flash memory are sampled on falling edge of the ref_clk 1 = data outputs from flash memory are sampled on rising edge of the ref_clk	1'h0
0x10	4:1	R/W	Delay the read data capturing logic by the programmable number of ref_clk cycles	4'h0
0x10	0	R/W	Bypass of the adapted loopback clock circuit 0 = enable the adapted loopback clock circuit 1 = disable the adapted loopback clock circuit For detailed information please refer to Section, "*Mechanism using external DLL*"	1'h1

67.2.5.5. **Device Size Configuration Register**

: This register allows the user to define the memory organization of using Flash Devices. This register should be setup while the controller is idle.

Table 5.2- 11 Device Size Configuration Register

Offset	Bit	R/W		Reset
0x14	31:29	RO	Unused . Read as zero.	2'h0
0x14	28:27	R/W	Size of Flash Device connected to CS[3] pin. This field is valid when AHB Decoder is enabled. 2'b00 = size of 512Mb	2'h0

			2'b01 = size of 1Gb 2'b10 = size of 2Gb 2'b11 = size of 4Gb	
0x14	26:25	R/W	Size of Flash Device connected to CS[2] pin. This field is valid when AHB Decoder is enabled. 2'b00 = size of 512Mb 2'b01 = size of 1Gb 2'b10 = size of 2Gb 2'b11 = size of 4G	2’h0
0x14	24:23	R/W	Size of Flash Device connected to CS[1] pin. This field is valid when AHB Decoder is enabled. 2'b00 = size of 512Mb 2'b01 = size of 1Gb 2'b10 = size of 2Gb 2'b11 = size of 4Gb	2’h0
0x14	22:21	R/W	Size of Flash Device connected to CS[0] pin. This field is valid when AHB Decoder is enabled. 2'b00 = size of 512Mb 2'b01 = size of 1Gb 2'b10 = size of 2Gb 2'b11 = size of 4Gb	2’h0
0x14	20:16	R/W	Number of bytes per block This is required by the controller for performing the write protection logic. The number of bytes per block must be a power of 2 number. I.e. 0 = 1 byte 1 = 2 bytes 3 = 8 bytes (...) 16 = 65535 bytes etc.	5’h0
0x14	15:4	R/W	Number of bytes per device page This is required by the controller for performing FLASH writes up to and across page boundaries.	12’h100
0x14	3:0	R/W	Number of address bytes A value of 0 = 1 byte etc.	4’h2

67.2.5.6. **SRAM Partition Configuration Register**

: This register allows the user to allocate the SRAM area for indirect writes and reads. N in this register is the SRAM depth, configured in the cdns_qspi_flash_ctrl_defs_default.v. The default value of N is 8 what assumes that external SRAM hard macro has 2**8=256 memory locations with fixed data size of 32 . This register should be setup while the controller is idle.

Table 5.2- 12 SRAM Partition Configuration Register

Offset	Bit	R/W		Reset
0x18	31:N	RO	Unused . Read as zero.	N{1'b0}
0x18	N-1:0	R/W	Defines the size of the indirect read partition in the SRAM, in units of SRAM locations. By default, half of the SRAM is reserved for indirect read operation, and half for indirect write. The size of this register will scale with the depth of the SRAM. The number of locations allocated to indirect read = SRAM_PARTITION_REG+1 The number of locations allocated to indirect write = (2**N)- SRAM_PARTITION_REG Avoid setting SRAM_PARTITION_REG to N{1'b0} or (2**N)-1)	1'b1,{N-1{1'b0}}} Top bit is set, allother are zeroed

Indirect AHB Address Trigger Register

: This register allows the user to define the address distinguishing DAC access from triggered INDAC one. This register should be setup while the controller is idle.

Table 5.2- 13 Indirect AHB Address Trigger Register

Offset	Bit	R/W		Reset
0x1C	31:0	R/W	Indirect Trigger Address This is the base address that will be used by the AHB controller. When the incoming AHB read access address matches a range of addresses from this trigger address to the trigger address + [configured range in Indirect Trigger Address Range Register], then the AHB request will be completed by fetching data from the Indirect Controllers SRAM.	32'h0

67.2.5.7. DMA Peripheral Configuration Register

: This register allows the user to define the parameters of DMA peripheral controller. This register should be setup while the controller is idle.

Table 5.2- 14 Indirect AHB Address Trigger Register

Offset	Bit	R/W		Reset
0x20	31:12	RO	Unused . Read as zero.	20’h0
0x20	11:8	R/W	Number of bytes in a burst type request on the DMA peripheral request. A programmed value of 0 represents a single byte. This should be setup before starting the indirect read or write operation. The actual number of bytes used is 2**[value in this register] which will simply implementation.	4’h0
0x20	7:4	RO	Unused . Read as zero.	4’h0
0x20	3:0	R/W	Number of bytes in a single type request on the DMA peripheral request. A programmed value of 0 represents a single byte. This should be setup before starting the indirect read or write operation. The actual number of bytes used is 2**[value in this register] which will simply implementation.	4’h0

Remap Address Register

: This register allows the user to define the address offset for DAC accesses. This register should be setup while the controller is idle.

Table 5.2- 15 Remap Address Register

Offset	Bit	R/W		Reset
0x24	31:0	R/W	Remapping of incoming AHB address to a dferent address used by the FLASH device.	32’h0

Mode Bit Configuration Register

: This register allows the user to define the mode for corresponding Flash Device. This register should be setup while the controller is idle.

Table 5.2- 16 Mode Bit Configuration Register

Offset	Bit	R/W		Reset
0x28	31:8	RO	Unused . Read as zero.	24’h0

0x28	7:0	R/W	Mode These are the 8 that are sent to the device following the address bytes.	8'h0

67.2.5.8. **SRAM Fill Level Register**

: This register keeps the values of current fill levels of both SRAM partitions.

Table 5.2- 17 SRAM Fill Level Register

Offset	Bit	R/W		Reset
0x2C	31:16	RO	SRAM Fill Level (Indirect Write Partition) in units of SRAM Words (4 bytes).	16'h0
0x2C	15:0	RO	SRAM Fill Level (Indirect Read Partition) in units of SRAM Words (4 bytes).	16'h0

67.2.5.9. **Tx Threshold Register**

: This register allows the user to define the TX FO level arousing the corresponding interrupt. This register should be setup while the controller is idle.

Table 5.2- 18 Tx Threshold Register

Offset	Bit	R/W		Reset
0x30	31:5	RO	Unused . Read as zero.	27'h0
0x30	4:0	R/W	Defines the level at which the small TX FO not full interrupt is generated. This is relevant when accessing the FLASH device in legacy mode. It can be otherwise ignored.	5'h1

67.2.5.10. **Rx Threshold Register**

: This register allows the user to define the RX FO level arousing the corresponding interrupt. This register should be setup while the controller is idle.

Table 5.2- 19 Rx Threshold Register

Offset	Bit	R/W		Reset

0x34	31:5	RO	Unused . Read as zero.	27’h0
0x34	4:0	R/W	Defines the level at which the small RX FO not empty interrupt is generated. This is relevant when accessing the FLASH device in legacy mode. It can be otherwise ignored.	5’h1

67.2.5.11. **Write Completion Control Register**

: This register defines how the controller will poll the device following a write transfer. By default it is set to poll bit 0 of the device STATUS register (using opcode 0x05), which is common across all devices to indicate WRITE IN PROGRESS (or WIP). However, in some devices, notably the Micron devices that are >= 512MB in size, it is required that the controller polls a dferent bit of a dferent device register. This means the controller must issue a dferent command during this polling phase. For example the Micron N25Q devices that are >= 512MB in size, it is bit 7 of the FLAG STATUS register (opcode 0x70) instead of bit 0 of the normal STATUS register that must be polled. For this reason, the following register has been provided for software to determine the bit and the opcode to use to poll for write completion, and also the number of successive valid polls that should take place. In order not to killing the device by throttling it by continuously generated READ Status SPI transactions, it is possible to prolong the delay between them ([31:24]). Defining this delay is not always desired since the ready bit indication can come later what impacts the overall performance. This register should be setup while the controller is idle.

Table 5.2- 20 Write Completion Control Register

Offset	Bit	R/W		Reset
0x38	31:24	R/W	Polling repetition delay Defines additional delay for maintain Chip Select de-asserted during auto-polling phase.	8’h0
0x38	23:16	R/W	Polling count Defines the number of times the controller should expect to see a true result from the polling in successive reads of the device register. Adding additional cycles might be necessarily when the controller is working on high throughput.	8’h1
0x38	15	R/W	Enable polling expiration Set to "1" for enabling auto-polling expiration after number of auto-polling cycles defined in the Polling Expiration Register	1’h0
0x38	14	R/W	Disable polling This switches off the automatic polling function.	1’h0

0x38	13	R/W	Polling polarity Defines the polling polarity. 0 = The write transfer to the device will be complete the polled bit is equal to "0" 1 = The write transfer to the device will be complete the polled bit is equal to "1"	1’h0
0x38	12:11	RO	Unused . Read as zero.	2’h0
0x38	10:8	R/W	Polling bit index Defines the bit index that should be polled. 3'b000 = bit 0 of the returned data will be polled for. 3'b001 = bit 1 of the returned data will be polled for. 3'b010 = bit 2 of the returned data will be polled for. (...) 3'b101 = bit 5 of the returned data will be polled for. 3'b110 = bit 6 of the returned data will be polled for. 3'b111 = bit 7 of the returned data will be polled for.	3’h0
0x38	7:0	R/W	Polling opcode Defines the opcode that should be issued by the controller when it is automatically polling for device program completion. This command is issued followed all device write operations. By default, this will poll the standard device STATUS register using opcode 0x05.	8’h5

67.2.5.12. **Polling Expiration Register**

: This register defines maximum number of poll cycles. the expected value of the bit being polled is not gotten after number defined in this register, the auto-polling is done on the next phase. The value stored in this register matters Write Completion Control Register [15] is set to "1". auto-polling is not disabled by Write Completion Control Register [14], at least two auto-polling phases are queued before execution. This register should be setup while the controller is idle.

Table 5.2- 21 Polling Expiration Register

Offset	Bit	R/W		Reset
0x3C	31:0	R/W	Defines the numbers of poll cycles after which auto-polling phase terminates and polling expiration interrupt is generated (unmasked).	32’hffffffff

67.2.5.13. **Interrupt Status Register**

: The status fields in this register are set when the described event occurs and the interrupt is enabled in the mask register. When any of these bit fields are set, the interrupt output is asserted high. The fields are cleared using write-one-to-clear approach. Note that bit fields 7 through 11 are valid when legacy SPI mode is active.

Table 5.2- 22 Interrupt Status Register

Offset	Bit	R/W		Reset
0x40	31:15	R/W	Unused . Read as zero.	17’h0
0x40	14	RWC	The STIG request completion interrupt. The controller is ready for getting another STIG request.	1’h0
0x40	13	RWC	The maximum number of programmed polls cycles is expired.	1’h0
0x40	12	RWC	Indirect Read partition of SRAM is full and unable to immediately complete indirect operation.	1’h0
0x40	11	RWC	Small RX FO full (current FO status). Can be ignored in non-SPI legacy mode. 0 = FO is not full 1 = FO is full	1’h0
0x40	10	RWC	Small RX FO not empty (current FO status) Can be ignoredin non-SPI legacy mode. 0 = FO has less than RX THRESHOLD entries 1 = FO has at least RX THRESHOLD entries	1’h0
0x40	9	RWC	Small TX FO full (current FO status). Can be ignored innon-SPI legacy mode. 0 = FO is not full 1 = FO is full	1’h0
0x40	8	RWC	Small TX FO not full (current FO status). Can be ignored innon-SPI legacy mode. 0 = FO has more than TX THRESHOLD entries 1 = FO has at most TX THRESHOLD entries	1’h0
0x40	7	RWC	Receive Overflow This should occur in Legacy SPI mode. Set an attempt is made to push the RX FO when it is full. This bit is reset by a system reset and cleared when this register is read. a new push to the RX FO occurs coincident with a register read this flag will remain set.	1’h0

			0 = no overflow has been detected. 1 = an overflow has occurred.	
0x40	6	RWC	Indirect Transfer Watermark Level Breached To use this interrupt effectively, it is acceptable to mask/unmask it during the SPI transfer in order not to interrupt it what would impact the performance. It breaks the rule that Interrupt Mask Register (6th bit) should be modied when the controller is in the IDLE state. It may be useful to block the interrupt after it occurs and then unblock after AHB activity is completed. watermark level is set to value which ensures interrupt generation with at least a few AHB and APB clock cycles between, modying the Interrupt Mask Register (6th bit) is safe, because the information is synchronized and propagated on time.	1’h0
0x40	5	RWC	Illegal AHB Access Detected. AHB write wrapping bursts and the use of SPLIT/RETRY accesses will cause this interrupt to trigger. This interrupt is also triggered, AHB access is performed when the DAC controller is disabled.	1’h0
0x40	4	RWC	Write to protected area was attempted and rejected.	1’h0
0x40	3	RWC	Indirect operation was requested but could not be accepted. Two indirect operations already in storage.	1’h0
0x40	2	RWC	Controller has completed last triggered indirect operation.	1’h0
0x40	1	RWC	Underflow Detected 0 = no underflow has been detected 1 = underflow is detected and an attempt to transfer data is made when the small TX FO is empty. This may occur when AHB write data is being supplied too slowly to keep up with the requested write operation	1’h0
0x40	0	RWC	Mode fail Legacy interrupt which is not being used any more.	1’h0

67.2.5.14. **Interrupt Mask Register**

: This register allows the user to mask/unmask particular interrupt sources. This register should be setup while the controller is idle.

Table 5.2- 23 Interrupt Mask Register

Offset	Bit	R/W		Reset
0x44	31:15	R/W	Unused . Read as zero.	17’h0
0x44	14:0	RWC	0 = the interrupt for the corresponding interrupt status register bit is disabled. 1 = the interrupt for the corresponding interrupt status register bit is enabled.	15’h0

67.2.5.15. **Lower Write Protection Register**

: This register allows the user to define lower boundary of the write protection area. This register should be setup while the controller is idle.

Table 5.2- 24 Lower Write Protection Register

Offset	Bit	R/W		Reset
0x50	31:0	R/W	The block number that defines the lower block in the range of blocks that is to be locked from writing. The definition of a block in terms of number of bytes is programmable via the Device Size Configuration Register.Mody this bit when write protection feature (via bit 1 of the write protection register) is low.	32’h0

67.2.5.16. **Upper Write Protection Register**

: This register allows the user to define upper boundary of the write protection area. This register should be setup while the controller is idle.

Table 5.2- 25 Upper Write Protection Register

Offset	Bit	R/W		Reset
0x54	31:0	R/W	The block number that defines the upper block in the range of blocks that is to be locked from writing. The definition of a block in terms of number of bytes is programmable via the Device Size Configuration Register. Mody this bit when write protection feature (via bit 1 of the write protection register) is low.	32’h0

67.2.5.17. **Write Protection Register**

: This register allows the user to define the configuration of write protection settings. This register should be setup while the controller is idle.

Table 5.2- 26 Write Protection Register

Offset	Bit	R/W		Reset
0x58	31:2	RO	Unused . Read as zero.	30'h0
0x58	1	R/W	Write Protection Enable Bit When set to "1", any AHB write access with an address within the protection region defined in the lower and upper write protection registers is rejected. An AHB error response is generated and an interrupt source triggered unmasked. When set to "0", the protection region is disabled. This bit is internally synchronized in hardware	1'h0
0x58	0	RWC	Write Protection Inversion Bit When set to "1", the protection region defined in the lower and upper write protection registers is inverted meaning it is the region that the system is permitted to write to. When set to "0", the protection region defined in the lower and upper write protection registers is the region that the system is not permitted to write to. Mody this bit when write protection feature (via bit 1 of the write protection register) is low.	1'h0

67.2.5.18. **Indirect Read Transfer Control Register**

: This register allows the user to control of the Indirect Read Transfer logic.

Table 5.2- 27 Indirect Read Transfer Control Register

Offset	Bit	R/W		Reset
0x60	31:8	RO	Unused . Read as zero.	24'h0
0x60	7:6	RO	Number of indirect operations completed. This is used in conjunction with bit 5. Incremented by H/W when an indirect operation has completed, decremented when bit 5 is written with a "1"	2'h0

0x60	5	R/W	Indirect Completion Status (status) Set by H/W when an indirect operation has completed, cleared by writing a "1"	1’h0
0x60	4	RO	Two indirect read operations have been queued (status) This bit is set and cleared by H/W.	1’h0
0x60	3	R/W	SRAM full and unable to immediately complete indirect operation (status) Set by H/W, cleared by writing a "1"	1’h0
0x60	2	RO	Indirect read operation in progress (status) This bit can be cleared by H/W.	1’h0
0x60	1	WO	Cancel indirect read (control) Writing a "1" to this bit will cancel an ongoing Indirect READ operation. This will cancel all indirect operations currently in progress. This bit is internally synchronized in hardware.	1’h0
0x60	0	WO	Start indirect read (control) Writing a "1" to this bit will trigger an indirect READ operation. The assumption is that the indirect start address and Indirect number of bytes register is setup before triggering the indirect READ operation. This bit is internally synchronized in hardware.	1’h0

67.2.5.19. Indirect Read Transfer Watermark Register

: This register allows the user to define watermark level for Indirect read transfers. This register should be setup before an indirect read transfer is triggered.

Table 5.2- 28 Indirect Read Transfer Watermark Register

Offset	Bit	R/W		Reset
0x64	31:0	R/W	Watermark value This represents the minimum fill level of the SRAM before a DMA peripheral access is permitted. When the SRAM fill level passes the watermark, an interrupt source is also generated unmasked. This can be disabled by writing a value of all zeroes. In units of Bytes.	32’h0

Indirect Read Transfer Start Address Register

: This register allows the user to define start address of indirect read transfer which is about to be triggered. This register should be setup before an indirect read transfer is triggered.

Table 5.2- 29 Indirect Read Transfer Start Address Register

Offset	Bit	R/W		Reset
0x68	31:0	R/W	Start of Indirect Access This is the FLASH start address from which the indirect access will commence its READ operation.	32’h0

67.2.5.20. Indirect Read Transfer Number Bytes Register

: This register allows the user to define number of bytes to be read of indirect read transfer which is about to be triggered. This register should be setup before an indirect read transfer is triggered.

Table 5.2- 30 Indirect Read Transfer Number Bytes Register

Offset	Bit	R/W		Reset
0x6C	31:0	R/W	Indirect Number of Bytes This is the number of bytes that the indirect access will consume. This can be bigger than the configured size of SRAM.	32’h0

67.2.5.21. Indirect Write Transfer Control Register

: This register allows the user to control of the Indirect Write Transfer logic.

Table 5.2- 31 Indirect Write Transfer Control Register

Offset	Bit	R/W		Reset
0x70	31:8	RO	Unused . Read as zero.	24’h0
0x70	7:6	RO	Number of indirect operations completed. This is used in conjunction with bit 5. Incremented by H/W when an indirect operation has completed, decremented when bit 5 is written with a "1"	2’h0
0x70	5	R/W	Indirect Completion Status (status) Set by H/W when an indirect operation has completed, cleared by writing a "1"	1’h0
0x70	4	RO	Two indirect read operations have been queued (status) This bit is set and cleared by H/W.	1’h0

0x70	3	RO	Unused bit. Read as zero.	1'h0
0x70	2	RO	Indirect write operation in progress (status) This bit can be cleared by H/W.	1'h0
0x70	1	WO	Cancel indirect write (control) Writing a "1" to this bit will cancel an ongoing Indirect WRITE operation. This will cancel all indirect operations currently in progress. This bit is internally synchronized in hardware.	1'h0
0x70	0	WO	Start indirect write (control) Writing a "1" to this bit will trigger an indirect WRITE operation. The assumption is that the indirect start address and Indirect number of bytes register is setup before triggering the indirect WRITE operation. This bit is internally synchronized in hardware.	1'h0

67.2.5.22. Indirect Write Transfer Watermark Register

: This register allows the user to define watermark level for Indirect write transfers. This register should be setup before an indirect write transfer is triggered.

Table 5.2- 32 Indirect Write Transfer Watermark Register

Offset	Bit	R/W		Reset
0x74	31:0	R/W	Watermark value This represents the maximum fill level of the SRAM before a DMA peripheral access is permitted. When the SRAM fill level falls below the watermark, an interrupt source is also generated unmasked. This can be disabled by writing a value of all zeroes. In units of Bytes.	32'h0

Indirect Write Transfer Start Address Register

: This register allows the user to define start address of indirect write transfer which is about to be triggered. This register should be setup before an indirect write transfer is triggered.

Table 5.2- 33 Indirect Write Transfer Start Address Register

Offset	Bit	R/W		Reset

0x78	31:0	R/W	Start of Indirect Access This is the FLASH start address from which the indirect access will commence its READ operation.	32'h0

67.2.5.23. Indirect Write Transfer Number Bytes Register

: This register allows the user to define number of bytes to be written of indirect read transfer which is about to be triggered. This register should be setup before an indirect read transfer is triggered.

Table 5.2- 34 Indirect Write Transfer Number Bytes Register

Offset	Bit	R/W		Reset
0x7C	31:0	R/W	Indirect Number of Bytes This is the number of bytes that the indirect access will consume. This can be bigger than the configured size of SRAM.	32'h0

Indirect Trigger Address Range Register

: This register allows the user to define the indirect trigger address range. the configured range exceeds number of bytes programmed for particular indirect transfer, there is no need to detect indirect trigger address boundaries by software. This register should be setup before an indirect read transfer is triggered.

Table 5.2- 35 Indirect Trigger Address Range Register

Offset	Bit	R/W		Reset
0x80	31:4	RO	Unused . Read as zero.	28'h0
0x80	3:0	R/W	Indirect Range Width This is the address offset of the Indirect AHB Address Trigger Register (0x1C). When any valid Indirect Access is triggered and AHB address fits to the range, the request is forwarded into Indirect Write Controller or Indirect Read Controller depending on which one was requested by valid trigger. The value is given as power of 2. The default one is 2**4 = 16 what allows 16-byte bursts to be performed. This field reflects width of the range so number of locations of valid addresses (single location has 32) ranges from Indirect Trigger Address to Indirect Trigger Address + [Indirect Range With - 1]	4'h4

			(Indirect Trigger Address -> Indirect Trigger Address + 15 [by default]).	

67.2.5.24. Flash Command Control Memory Register (Using STIG)

: This register controls the Memory Bank accesses. It also defines the number of bytes intended to get by
STIG access configured to use the STIG Memory Bank.

Table 5.2- 36 Flash Command Control Memory Register (Using STIG)

Offset	Bit	R/W		Reset
0x8C	31:29	RO	Unused . Read as zero.	3'h0
0x8C	28:20	R/W	Memory Bank Address The address of the Memory Bank which data will be read from. It is equivalent to the index value of the byte read by the last STIG access configured to work with STIG Memory Bank. For example, setting this field for 255 (8'hff) will return (after triggering Memory Bank access) 255th byte read by the last STIG access configured to work with STIG Memory Bank. This should be setup before triggering the command via bit 0 of this register.	9'h0
0x8C	19	RO	Unused . Read as zero.	1'h0
0x8C	18:16	R/W	Number of STIG Memory Bank Read Bytes It defines the number of read bytes for the STIG configured to work with STIG Memory Bank as follows: 3'b000 = 16 bytes 3'b001 = 32 bytes 3'b010 = 64 bytes 3'b011 = 128 bytes 3'b100 = 256 bytes 3'b101 = 512 bytes 3'b11x = unused This should be setup before triggering the command via bit 0 of Flash Command Control Register (Using STIG)	3'h0
0x8C	15:8	RO	Memory Bank Read Data Last requested data from the STIG Memory Bank. It turns to be stable when bit 1 of this register toggles from "1" to "0".	8'h0

0x8C	7:2	RO	Unused . Read as zero.	6'h0
0x8C	1	RO	Memory Bank data request in progress	1'h0
0x8C	0	WO	Trigger the Memory Bank data request This bit is internally synchronized.	n/a

67.2.5.25. Flash Command Control Register (Using STIG)

: This register controls SPI transactions generated by STIG. It allows the user to define corresponding SPI frame to particular command, triggering the transfer and polling for its completion.

Table 5.2- 37 Flash Command Control Register (Using STIG)

Offset	Bit	R/W		Reset
0x90	31:24	R/W	Command Opcode The command opcode field should be setup before triggering the command. For example, 0x20 maps to SubSector Erase. Writing to the execute field (bit 0) of this register launches the command. **Note** Using this approach to issue commands to the device will make use of the instruction type of the device instruction configuration register. this field is set to 2'b00, then the command opcode, command address, command dummy cycles and command data will all be transferred in a serial fashion. this field is set to 2'b01, then the command opcode, command address, command dummy cycles and command data will all be transferred in parallel using DQ0 and DQ1 pins. this field is set to 2'b10, then the command opcode, command address, command dummy cycles and command data will all be transferred in parallel using DQ0, DQ1, DQ2 and DQ3 pins.	8'h0
0x90	23	R/W	Read Data Enable Set to "1" the command specied in 31:24 requires readdata bytes to be received from the device.	1'h0
0x90	22:20	R/W	Number of Read Data Bytes Up to 8 Data bytes may be read using this command (Set to 0 for 1 byte, 7 for 8 bytes).	3'h0

			In case of Flash Command Control Register (Using STIG) Register bit[2] is enabled, this field is not to care and Number of Read Data Bytes is as specied in Flash Command Control Memory Register (Using STIG) Register bit[15:8].	
0x90	19	R/W	Command Address Enable Set to "1" the command specied in 31:24 requires an address. This should be setup before triggering the command via bit 0 of this register.	1’h0
0x90	18	R/W	Mode Bit Enable Set to "1" to ensure the mode as defined in the Mode Bit Configuration Register are sent following the address bytes.	1’h0
0x90	17:16	R/W	Number of Address Bytes Set to the number of address bytes required (the address itself is programmed in the Flash Command Address Register). This should be setup before triggering the command via bit 0 of this register. 2'b00 = 1 address byte 2'b01 = 2 address bytes 2'b10 = 3 address bytes 2'b11 = 4 address bytes	2’h0
0x90	15	R/W	Write Data Enable Set to "1" the command specied in 31:24 requires write data bytes to be sent to the device.	1’h0
0x90	14:12	R/W	Number of Write Data Bytes Up to 8 Data bytes may be written using this command (Set to 0 for 1 byte, 7 for 8 bytes).	3’h0
0x90	11:7	R/W	Number of Dummy Cycles Set to the number of dummy cycles required for command specied in 31:24. This should be setup before triggering the command via bit 0 of this register.	5’h0
0x90	6:3	RO	Unused . Read as zero.	4’h0
0x90	2	R/W	STIG Memory Bank enable bit. This should be setup before triggering the command via bit 0 of this register.	1’h0
0x90	1	RO	STIG command execution in progress.	1’h0

0x90	0	WO	Execute the command This bit is internally synchronized.	n/a

Flash Command Address Register

: This register allows the user to define the address of the command using by the STIG controller. This
register should be setup before an indirect read transfer is triggered.

Table 5.2- 38 Flash Command Address Register

Offset	Bit	R/W		Reset
0x94	31:0	R/W	Command Address It is the address used by the command specied in Flash Command Control Register (Using STIG) Register [31:24].	32’h0

67.2.5.26. Flash Command Read Data Register (Lower)

: This register keeps the last 4 bytes read by STIG SPI access.

Table 5.2- 39 Flash Command Read Data Register (Lower)

Offset	Bit	R/W		Reset
0xA0	31:0	RO	Command Read Data (Lower) This is the data that is returned by the FLASH device for any status or configuration read operation carried out by triggeringthe event in the control register. The register will be valid when the Flash Command Control Register (Using STIG) Register bit[1] is low.	32’h0

67.2.5.27. Flash Command Read Data Register (Upper)

: This register keeps the last but 4 bytes read by STIG SPI access. This register in conjunction with the Flash Command Read Data Register (Lower) enables the controller to keep 8 last bytes read from the Flash Device using STIG.

Table 5.2- 40 Flash Command Read Data Register (Upper)

Offset	Bit	R/W		Reset

0xA4	31:0	RO	Command Read Data (Upper) This is the data that is returned by the FLASH device for any status or configuration read operation carried out by triggering the event in the control register. The register will be valid when the Flash Command Control Register (Using STIG) Register bit[1] is low.	32’h0

67.2.5.28. **Flash Command Write Data Register (Lower)**

: This register takes the first 4 bytes to be written by STIG.

Table 5.2- 41 Flash Command Write Data Register (Lower)

Offset	Bit	R/W		Reset
0xA8	31:0	RO	Command Write Data (Lower) This is the data that is to be written to the FLASH device for any status or configuration write operation carried out by triggering the event in the control register.	32’h0

67.2.5.29. **Flash Command Write Data Register (Upper)**

: This register takes the bytes ranging from 5 to 8 to be written by STIG.

Table 5.2- 42 Flash Command Write Data Register (Upper)

Offset	Bit	R/W		Reset
0xA8	31:0	RO	Command Write Data (Upper) This is the data that is to be written to the FLASH device for any status or configuration write operation carried out by triggering the event in the control register.	32’h0

67.2.5.30. **Polling Flash Status Register**

: This register provides auto-polling data. It acts as the extension for the Write Completion Control Register where full status is not available and any action can be taken relying on the indication of single bit being polled for.

Table 5.2- 43 Polling Flash Status Register

Offset	Bit	R/W		Reset
0xB0	31:20	RO	Unused . Read as zero.	12’h0
0xB0	19:16	R/W	Number of dummy cycles for auto-polling This field enables the user to define additional dummy cycles for read status during auto-polling state. Some cycles may be necessary to add external delay of read data path shts it outside the defined clock cycle. An option of adding dummy cycles may not force the target clock rate being degraded.	4’h0
0xB0	15:9	RO	Unused . Read as zero.	7’h0
0xB0	8	RO	Polling Status Valid This bit is set when value in from 7 to 0 is valid.	1’h0
0xB0	7:0	RO	Flash Status Defines the Status of Device returned by the last auto-polling access.	8’h0

67.2.5.31. **Module ID Register**

: This register provides the IP release number and the configuration data.

Table 5.2- 44 Module ID Register

Offset	Bit	R/W		Reset
0xFC	31:24	RO	Fix/patch number related to the Module/Revision ID	8’h1
0xFC	23:8	RO	Module/Revision ID	16’h1
0xFC	7:2	RO	Unused . Read as zero.	6’h0
0xFC	1:0	RO	Configuration ID number (QSPI)	2’h3

68. Video Subsystem

68.1. Overview

The Video Subsystem in Storm SOC is a high performance low power sub-system designed for various video applications such as smart IPC, smart doorbell system, smart home surveillance system etc. Its main feature and block diagram are as following.

- Multiple camera interfaces(1MIPI/1DVP)
  - Support MIPI RX
- 1 2-lane MIPI D-PHY interface , up to 900Mbit/s per lane
- RAW8,RAW10, RAW12 data type parsing
- YUV420, YUV420(Legacy), YUV420(CSPS), YUV422 of 8-bits and 10-bits
- Maximum input 1296P@60fps@2lane
  - Support DVP Interface
- Support BT.601,BT.656,BT.1120 protocol
- Support YUV422 (8bit) data format
  - Compatibility with mainstream HD CMOS sensors provided by Sony, OmniVision, SmartSense and Galaxycore etc.
- ISP
  - Maximum input 1296P@30fps
  - Test Pattern Generator(TPG)
  - Black Level Compensation (BLC)
  - Defect Pixel Correction (DPC)
  - Lens Shading Correction (De-Vignetting)
  - Digital Gain
  - 2 Stage Adaptive Noise Filter
  - Enhanced Color Interpolation (Demosaic)
  - Chromatic Aberration Correction (CAC)
  - Color Correction Matrix (CCM)
  - Color Space Conversion (CSM)
  - Gamma Correction
  - Auto Focus Measurement (AF)
  - Auto White Balancing (AWB)
  - Auto Exposure Measurement (AE)
  - Histogram Calculation
  - Anti-flicker
  - Cropping
  - Color Processing (Contrast, Saturation, Brightness, Hue) (CPROC)
  - Sharpening/Blurring Filter
  - Wide Dynamic Range Tone Mapping (WDR)
  - Multi-exposure HDR(DOL/Stagger, Stagger output)
  - Advanced 2DNR to remove spatial noise

- 3DNR to remove temporal noise
- H.264/JPEG encoding/decoding
- Up to 1080P@60fps image processing ability
- Support multi-stream encoding/decoding
- Efficient customized 2-D accelerator
- Up/Down Scale
- Support DPI interface
- YUV4:2:0/YUV 4:2:2 input
- OSD layer alpha-bending with scale layer
- YUV4:2:0/RGB888/RGBA output
- Fill and Copy
- 90,180,270 rotation
- Draw Rectangle
- LCD display controller
- Display interface
- Support MIPI DSI interface with 2lanes, up to 1.2Gbit/s per lane
- Support 16-bit, 18-bit, 24-bit RGB format, up to 1080x1920@30fps
- Support 8-bit 8080 mcu interface
- Support serial 8-bit RGB interface
- Display control
- Support up to 4 planes
- Layer0 & Layer1 & Layer2
- Support YUV420/YUV422 data format input
- Support configurable start pixel and start line
- OSD layer
- Support ARGB, RGB-24bit, RGB-16bit, ARGB-32bit, ARGB4444, ARGB1555, Monochrome
- OSD layer support configurable global alpha blending or pixel by pixel alpha blending

68.2. MIPI Rx

68.2.1. Overview

CSI (Camera Serial Interface) is an interface between a camera and a host processor for mobile applications.

Demand for increasingly higher image resolutions is pushing the bandwidth capacity of existing host processor-to-camera sensor interfaces. Common parallel interfaces are difficult to expand, require many interconnects and consume relatively large amounts of power. Emerging serial interfaces address many of the shortcomings of parallel interfaces while introducing their own problems. Incompatible, proprietary interfaces prevent devices from different manufacturers from working together. This can raise system costs and reduce system reliability by requiring “hacks” to force the devices to interoperate. The lack of a clear industry standard can slow innovation and inhibit new product market entry.

CSI-2 provides the mobile industry a standard, robust, scalable, low-power, high-speed, cost-effective interface that supports a wide range of imaging solutions for mobile devices.

68.2.2. Features

Support the following features:

- Compliant with the MIPI Alliance CSI-2 Interface specification, v1.2
- PPI interface to D-PHY with pipe-lined PPI enhanced support
- Configurable up to 2 data lanes
- Programmable multi-lane merging
- Supports all primary and secondary CSI-2 data formats
- Short and long packet formats
- Detection of low-power (LP) and ultra low-power (ULP) modes
- Error detection and correction with interrupt at PHY, packet, line and frame level
- AMBA-APB control and Configuration

68.2.3. Function Description

68.2.3.1. Function Overview

MIPI RX module including the following basic sub-modules, as shown in below figure:

- D-PHY Slave
- Image Pixel Interface (IPI)
- Image Date Interface (IDI)
- Control Logic

69.

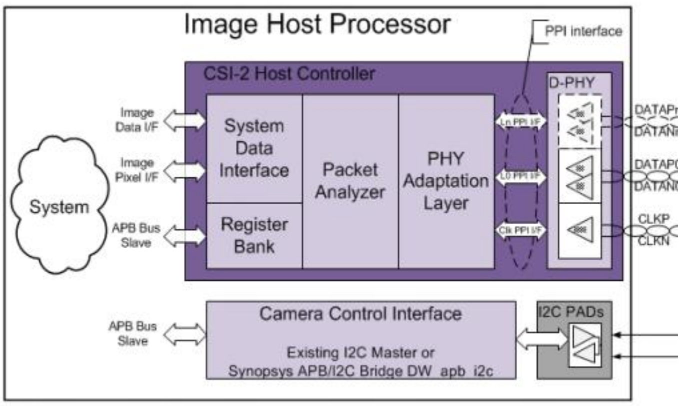


Figure 6- 1 MIPI_CSI Basic Sub-modules

69.1.1. Programming Guide

1. Trigger mipi shutdown and reset to initialize
2. Trigger IPI soft reset
3. Test mipi PHY by configure PHY_TEST_CTRL0/PHY_TEST_CTRL1
4. Configure IPI parameter setting like mode, data_type, HSA, VSA and so on
5. Enable mipi function

69.1.2. Register Summary

Table 6- 1 MIPI RX register summary

Name	Address Offset	R/W	Description
VERSION	0x0	RW	Contains the version of controller coded in 32-bit ASCII code
N_LANES	0x4	RW	Configures the number of active lanes
CSI2_RESETN	0x8	RW	Controls the logic reset state.
INT_ST_MAIN	0xC	RW	Current Status Register
DATA_IDS_1	0x10	RW	Programs the data Ids for matching line error reporting.
PHY_SHUTDOWNZ	0x40	RW	Controls the D-PHY Shutdown mode
DPHY_RSTZ	0x44	RW	Controls the D-PHY Reset mode
PHY_RX	0x48	RW	Contains the status of RX-related signals
PHY_STOPSTATE	0x4C	RW	Contains the STOPSTATE signal status
PHY_TEST_CTRL0	0x50	RW	Control for vendor specific interface in the PHY.
PHY_TEST_CTRL1	0x54	RW	Control for vendor specific interface in the PHY.
IPI_MODE	0x80	RW	This register selects how the IPI interface generates the video frame.
IPI_VCID	0x84	RW	This register selects the virtual channel processed by IPI.
IPI_DATA_TYPE	0X88	RW	This register selects the data type processed by IPI.
IPI_MEM_FLUSH	0X8C	RW	This register control the flush of IPI memory.
IPI_HSA_TIME	0X90	RW	This register configures the video Horizontal Synchronism Active (HSA) time.

IPI_HBP_TIME	0X94	RW	This register configures the video Horizontal Back Porch (HBP) time.
IPI_HSD_TIME	0X98	RW	This register configures the video Horizontal Sync Delay (HSD) time.
IPI_HLINE_TIME	0X9C	RW	This register configures the overall time for each video line.
IPI_SOFTRSTN	0XA0	RW	IPI soft reset
IPI_ADV_FEATURES	0XAC	RW	IPI advantage features
IPI_VSA_LINES	0XB0	RW	This register configures the vertical resolution of the video.
IPI_VBP_LINES	0XB4	RW	This register configures the Vertical Back Porch (VBP) period.
IPI_VFP_LINES	0XB8	RW	This register configures the Vertical Front Porch (VFP) period.
IPI_VACTIVE_LINES	0XBC	RW	This register configures the vertical resolution of the video.
PHY_CAL	0XCC	RW	Contains the CALIBRATION signal status from D-PHY
INT_ST_PHY_FATAL	0XE0	R	Groups the fatal interruptions caused by PHY Packet discarded
INT_MSK_PHY_FATAL	0XE4	RW	Fatal interruption masks
INT_FORCE_PHY_FATAL	0XE8	RW	Interrupt Force register for Packet
INT_ST_PKT_FATAL	0XF0	R	Groups the fatal interruption related with Packet construction
INT_MAK_PKT_FATAL	0XF4	RW	Interruption masks for Packet
INT_FORCE_PKT_FATAL	0XF8	RW	Interrupt Force register is used for test purposes
INT_ST_FRAME_FATAL	0X100	R	Groups the fatal interruption related with Frame
INT_MSK_FRAME_FATAL	0X104	RW	Interruption masks for Frame
INT_FORCE_FRAME_FATAL	0X108	RW	Interrupt Force register for Frame
INT_ST_PHY	0X110	R	Interruption caused by PHY
INT_MSK_PHY	0X114	RW	Interruption Mask for INT_ST_PHY
INT_FORCE_PHY	0X118	RW	Interrupt Force register for PHY
INT_ST_PKT	0X120	R	Interruption caused related Package construction
INT_MAK_PKT	0X124	RW	Interruption Mask for INT_ST_PKT
INT_FORCE_PKT	0X128	RW	Interrupt Force register for Packet
INT_ST_LINE	0X130	R	Interruption caused by Line construction
INT_MSK_LINE	0X134	RW	Interruption Mask for INT_ST_Line
INT_FORCE_LINE	0X138	RW	Interrupt Force register for Line
INT_ST_IPI	0X140	R	Fatal Interruption caused by IPI interface

INT_MSK_IPI	0X144	RW	Interruption Mask for INT_ST_IPI
INT_FORCE_IPI	0X148	RW	Interrupt Force register for IPI
INT_ST_IPI2	0X150	R	Fatal Interruption caused by IPI2 interface
INT_MSK_IPI2	0X154	RW	Interruption Mask for INT_ST_IPI2
INT_FORCE_IPI2	0X158	RW	Interrupt Force register for IPI2
IPI2_MODE	0X200	RW	This register selects how the IPI interface2 generates the video frame.
IPI2_VCID	0X204	RW	This register selects the virtual channel processed by IPI2.
IPI2_DATA_TYPE	0X208	RW	This register selects the data type processed by IPI2.
IPI2_MEM_FLUSH	0X20C	RW	This register control the flush of IPI2 memory.
IPI2_HSA_TIME	0X210	RW	This register configures the video Horizontal Synchronism Active (HSA) time for IPI2.
IPI2_HBP_TIME	0X214	RW	This register configures the video Horizontal Back Porch (HBP) time for IPI2.
IPI2_HSD_TIME	0X218	RW	This register configures the video Horizontal Sync Delay (HSD) time for IPI2.
IPI2_ADV_FEATURES	0X21C	RW	IPI2 advanced features

69.1.3. **Register Description**

69.1.3.1. **VERSION**

Table 6- 2 VERSION

Bits	Name	R/W	Reset value	Description
31:0	version	R	0x3131312a	Version information

69.1.3.2. **N_LANES**

Table 6- 3 N_LANES

Bits	Name	R/W	Reset value	Description
31:3	Reserved			Reserved
2:0	n_lanes	R/W	1	Number of active data lanes: - 000: 1 Data Lane (Lane 0) - 001: 2 Data Lanes (Lane 0 and 1) This can only be updated when the PHY lane is in stopstate

69.1.3.3. CSI2_RESETN

Table 6- 4 CSI2_RESETN

Bits	Name	R/W	Reset value	Description
31:1	Reserved			Reserved
0	csi2_resetn	R/W		CSI host reset output. Active Low.

69.1.3.4. INT_ST_MAIN

Table 6- 5 INT_ST_MAIN

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	status_int_ipi	R	0x0	Status of int_st_ipi
18	status_int_line	R	0x0	Status of int_st_line
17	status_int_pkt	R	0x0	Status of int_st_pkt
16	status_int_phy	R	0x0	Status of int_st_phy
15:3	Reserved			Reserved
2	status_int_frame_fatal	R	0x0	Status of int_st_frame_fatal
1	status_int_pkt_fatal	R	0x0	Status of int_st_pkt_fatal
0	status_int_phy_fatal	R	0x0	Status of int_st_phy_fatal

69.1.3.5. DATA_IDS_1

Table 6- 6 DATA_IDS_1

Bits	Name	R/W	Reset value	Description
31:30	di3_vc	RW	0x0	virtual channel for programmed data ID 3
29:24	di3_dt	RW	0x0	data type for programmed data ID 3
23:22	di2_vc	RW	0x0	virtual channel for programmed data ID 2
21:16	di2_dt	RW	0x0	data type for programmed data ID 2
15:14	di1_vc	RW	0x0	virtual channel for programmed data ID 1
13:8	di1_dt	RW	0x0	data type for programmed data ID 1
7:6	di0_vc	RW	0x0	virtual channel for programmed data ID 0
5:0	di0_dt	RW	0x0	data type for programmed data ID 0

69.1.3.6. **PHY_SHUTDOWNZ**

Table 6- 7 PHY_SHUTDOWNZ

Bits	Name	R/W	Reset value	Description
31:1	Reserved			Reserved
0	phy_shutdownz	R/W		Phy shutdown is used to place CSI phy in power down mode. All analog blocks are in power down mode and digital logic is cleared. Active Low.

69.1.3.7. **PHY_RX**

Table 6- 8 PHY_RX

Bits	Name	R/W	Reset value	Description
31:18	Reserved			Reserved
17	phy_rxclkactivehs	R	0x0	Indicates that D-PHY clock lane is actively receiving a DDR clock
16	phy_rxulpsclknot	R	0x0	Active Low. This signal indicates that D-PHY Clock Lane module has entered the Ultra Low Power state
15:2	Reserved			Reserved
1	phy_rxulpsesc_1			Lane module 1 has entered the Ultra Low Power mode
0	phy_rxulpsesc_0			Lane module 0 has entered the Ultra Low Power mode

69.1.3.8. **PHY_STOPSTATE**

Table 6- 9 PHY_STOPSTATE

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	phy_stopstateclk	R	0x0	D-PHY Clock lane in Stop state
15:2	Reserved			Reserved
1	phy_stopstatedata_1	R	0x0	Data lane 1 in Stop state
0	phy_stopstatedata_0	R	0x0	Data lane 0 in Stop state

69.1.3.9. **PHY_TEST_CTRL0**

Table 6- 10 PHY_TEST_CTRL0

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1	phy_testclk	RW	0x0	Clock to capture testdin bus contents into the PHY macro, with testen signal controlling the operation selection.
0	phy_testclr	RW	0x1	When active, performs vendor specific interface initialization.Active High. Note: This line needs an initial high pulse after power up for analog programmability default values to be preset.

69.1.3.10. **PHY_TEST_CTRL1**

Table 6- 11 PHY_TEST_CTRL1

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	phy_testen	RW	0x0	When asserted high, it configures an address write operation on the falling edge of testclk. When asserted low, it configures a data write operation on the rising edge of testclk.
15:8	phy_testdout	R	0x0	Vendor-specific 8-bit data output for reading data and other probing functionalities.
7:0	phy_testdin	RW	0x0	Test interface 8-bit data input for programming internal registers and accessing test functionalities.

69.1.3.11. **IPI_MODE**

Table 6- 12 IPI_MODE

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	ipi_enable	RW	0x0	This register enables the interface.
23:17	Reserved			Reserved
16	ipi_cut_through	RW	0x0	This field indicates cut-through mode state: -0: cut-through mode inactive -1: cut-through mode active
15:9	Reserved			Reserved
8	ipi_color_com	RW	0x0	This field indicates if color mode components

				are delivered as follows: -0: 48 bits interface -1: 16 bits interface
7:1	Reserved			Reserved
0	ipi_mode	RW	0x0	This field indicates the video mode transmission type as follows: -0: Camera timing -1: Controller timing

69.1.3.12. IPI_VCID

Table 6- 13 IPI_VCID

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1:0	ipi_vcid	RW	0x0	Virtual channel of data to be processed by pixel interface

69.1.3.13. IPI_DATA_TYPE

Table 6- 14 IPI_DATA_TYPE

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
8	embedded_data	RW	0x0	This bit enables embedded data processing on IPI interface.
7:6	Reserved			Reserved
5:0	ipi_data_type	RW	0x0	Data type of data to be processed by pixel interface.

69.1.3.14. IPI_MEM_FLUSH

Table 6- 15 IPI_MEM_FLUSH

Bits	Name	R/W	Reset value	Description
31:9	Reserved			Reserved
8	ipi_auto_flush	RW	0x0	Memory is automatically flushed at each vsync
7:1	Reserved			Reserved
0	ipi_flush	RW	0x0	Flush IPI memory. This bit is auto clear.

69.1.3.15. IPI_HSA_TIME

Table 6- 16 IPI_HSA_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hsa_time	RW	0x0	This field configures the HSA period in pixclk cycles

69.1.3.16. IPI_HBP_TIME

Table 6- 17 IPI_HBP_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hbp_time	RW	0x0	This field configures the HBP period in pixclk cycles

69.1.3.17. IPI_HSD_TIME

Table 6- 18 IPI_HSD_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hsd_time	RW	0x0	This field configures the HSD period in pixclk cycles

69.1.3.18. IPI_HLINE_TIME

Table 6- 19 IPI_HLINE_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hline_time	RW	0x0	This field configures the size of the line time counted in pixclk cycles.

69.1.3.19. IPI_SOFTRSTN

Table 6- 20 IPI_SOFTRSTN

Bits	Name	R/W	Reset value	Description
31:5	Reserved			Reserved
4	ipi2_softrstn	RW	0x0	This field resets IPI two. Active Low
3:1	Reserved			Reserved
0	ipi_softrstn	RW	0x0	This field resets IPI one. Active Low.

69.1.3.20. IPI_ADV_FEATURES

Table 6- 21 IPI_ADV_FEATURES

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	ipi_sync_event_mode	RW	0x0	For Camera Mode: -0: Frame Start dont trigger any sync event -1: Legacy mode. Frame start will trigger a sync event
23:22	Reserved			Reserved
21	en_embedded	RW	0x0	This register allows to use embedded packets for IPI synchronization events.
20	en_blanking	RW	0x0	This register allows to use blanking packets for IPI synchronization events.
19	en_null	RW	0x0	This register allows to use null packets for IPI synchronization events.
18	en_line_start	RW	0x0	This register allows to use line start packets for IPI synchronization events.
17	en_video	RW	0x0	This register allows to use video packets for IPI synchronization events.
16	line_event_selection	RW	0x0	For Camera Mode, this register allows to manualy select the Packet for line delimiter as follows: -0: controller select it automatically -1: select packets from list programmed in [17:21].
15:14	Reserved			Reserved
13:8	ipi_dt	RW	0x0	Datatype to overwrite.
7:1	Reserved			Reserved
0	ipi_dt_overwrite	RW	0x0	Ignore datatype of the header using the programed datatype for decoding.

69.1.3.21. IPI_VSA_LINES

Table 6- 22 IPI_VSA_LINES

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	ipi_vsa_lines	RW	0x0	This field configures the VSA period measured in number of horizontal lines

69.1.3.22. IPI_VBP_LINES

Table 6- 23 IPI_VBP_LINES

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	ipi_vbp_lines	RW	0x0	This field configures the VBP period measured in number of horizontal lines

69.1.3.23. IPI_VFP_LINES

Table 6- 24 IPI_VFP_LINES

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	ipi_vfp_lines	RW	0x0	This field configures the VFP period measured in number of horizontal lines

69.1.3.24. IPI_VACTIVE_LINES

Table 6- 25 IPI_VACTIVE_LINES

Bits	Name	R/W	Reset value	Description
31:14	Reserved			Reserved
13:0	ipi_vactive_line s	RW	0x0	This field configures the Vertical Active period measured in number of horizontal lines

69.1.3.25. PHY_CAL

Table 6- 26 PHY_CAL

Bits	Name	R/W	Reset value	Description
31:1	Reserved			Reserved
0	rxskewcalhs	RW	0x0	A low-to-high transition on rxskewcalhs signal means that the PHY has initiated the de-skew calibration.

69.1.3.26. INT_ST_PHY_FATAL

Table 6- 27 INT_ST_PHY_FATAL

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1	phy_errsotsynch	R	0x0	Start of transmission error on data lane 1 (no

	s_1			synchronization achieved)
0	phy_errsotsynchs_0	R	0x0	Start of transmission error on data lane 0 (no synchronization achieved)

69.1.3.27. INT_MSK_PHY_FATAL

Table 6- 28 INT_MSK_PHY_FATAL

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1	mask_phy_errsotsynchs_1	RW	0x0	Mask for phy_errsotsynchs_1
0	mask_phy_errsotsynchs_0	RW	0x0	Mask for phy_errsotsynchs_0

69.1.3.28. INT_FORCE_PHY_FATAL

Table 6- 29 INT_FORCE_PHY_FATAL

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1	force_phy_errsotsynchs_1	RW	0x0	Force phy_errsotsynchs_1
0	force_phy_errsotsynchs_0	RW	0x0	Force phy_errsotsynchs_0

69.1.3.29. INT_ST_PKT_FATAL

Table 6- 30 INT_ST_PKT_FATAL

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	err_ecc_double	R	0x0	Header ECC contains 2 errors, unrecoverable.
15:4	Reserved			Reserved
3	vc3_err_crc	R	0x0	Checksum error detected on virtual channel 3.
2	vc2_err_crc	R	0x0	Checksum error detected on virtual channel 2.
1	vc1_err_crc	R	0x0	Checksum error detected on virtual channel 1.
0	vc0_err_crc	R	0x0	Checksum error detected on virtual channel 0.

69.1.3.30. INT_MSK_PKT_FATAL

Table 6- 31 INT_MSK_PKT_FATAL

Bits	Name	R/W	Reset value	Description

31:17	Reserved			Reserved
16	mask_err_ecc_double	RW	0x0	Mask for err_ecc_double.
15:4	Reserved			Reserved
3	mask_vc3_err_crc	RW	0x0	Mask for vc3_err_crc
2	mask_2_err_crc	RW	0x0	Mask for vc2_err_crc
1	mask_1_err_crc	RW	0x0	Mask for vc1_err_crc
0	mask_0_err_crc	RW	0x0	Mask for vc0_err_crc

69.1.3.31. INT_FORCE_PKT_FATAL

Table 6- 32 INT_FORCE_PKT_FATAL

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	force_err_ecc_double	RW	0x0	Force err_ecc_double.
15:4	Reserved			Reserved
3	force_vc3_err_crc	RW	0x0	Force vc3_err_crc
2	force_2_err_crc	RW	0x0	Force vc2_err_crc
1	force_1_err_crc	RW	0x0	Force vc1_err_crc
0	force_0_err_crc	RW	0x0	Force vc0_err_crc

69.1.3.32. INT_ST_FRAME_FATAL

Table 6- 33 INT_ST_FRAME_FATAL

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	err_frame_data_vc3	R	0x0	Last received Frame in virtual channel 3, had at least one CRC error
18	err_frame_data_vc2	R	0x0	Last received Frame in virtual channel 2, had at least one CRC error
17	err_frame_data_vc1	R	0x0	Last received Frame in virtual channel 1, had at least one CRC error
16	err_frame_data_vc0	R	0x0	Last received Frame in virtual channel 0, had at least one CRC error
15:12	Reserved			Reserved
11	err_f_seq_vc3	R	0x0	Incorrect Frame sequence detected in Virtual Channel 3.
10	err_f_seq_vc2	R	0x0	Incorrect Frame sequence detected in Virtual

				Channel 2.
9	err_f_seq_vc1	R	0x0	Incorrect Frame sequence detected in Virtual Channel 1.
8	err_f_seq_vc0	R	0x0	Incorrect Frame sequence detected in Virtual Channel 0.
7:4	Reserved			Reserved
3	err_f_bndry_match_vc3	R	0x0	Error matching Frame Start with Frame End for virtual channel 3.
2	err_f_bndry_match_vc2	R	0x0	Error matching Frame Start with Frame End for virtual channel 2.
1	err_f_bndry_match_vc1	R	0x0	Error matching Frame Start with Frame End for virtual channel 1.
0	err_f_bndry_match_vc0	R	0x0	Error matching Frame Start with Frame End for virtual channel 0.

69.1.3.33. **INT_MSK_FRAME_FATAL**

Table 6- 34 INT_MSK_FRAME_FATAL

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	mask_err_frame_data_vc3	RW	0x0	Mask for err_frame_data_vc3
18	mask_err_frame_data_vc2	RW	0x0	Mask for err_frame_data_vc2
17	mask_err_frame_data_vc1	RW	0x0	Mask for err_frame_data_vc1
16	mask_err_frame_data_vc0	RW	0x0	Mask for err_frame_data_vc0
15:12	Reserved			Reserved
11	mask_err_f_seq_vc3	RW	0x0	Mask for err_f_seq_vc3
10	mask_err_f_seq_vc2	RW	0x0	Mask for err_f_seq_vc2
9	mask_err_f_seq_vc1	RW	0x0	Mask for err_f_seq_vc1
8	mask_err_f_seq_vc0	RW	0x0	Mask for err_f_seq_vc0
7:4	Reserved			Reserved
3	mask_err_f_bndry_match_vc3	RW	0x0	Mask for err_f_bndry_match_vc3
2	mask_err_f_bndry_match_vc2	RW	0x0	Mask for err_f_bndry_match_vc2

1	mask_err_f_bndry_match_vc1	RW	0x0	Mask for err_f_bndry_match_vc1
0	mask_err_f_bndry_match_vc0	RW	0x0	Mask for err_f_bndry_match_vc0

69.1.3.34. INT_FORCE_FRAME_FATAL

Table 6-35 INT_FORCE_FRAME_FATAL

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	force_err_frame_data_vc3	RW	0x0	Force err_frame_data_vc3
18	force_err_frame_data_vc2	RW	0x0	Force err_frame_data_vc2
17	force_err_frame_data_vc1	RW	0x0	Force err_frame_data_vc1
16	force_err_frame_data_vc0	RW	0x0	Force err_frame_data_vc0
15:12	Reserved			Reserved
11	force_err_f_seq_vc3	RW	0x0	Force err_f_seq_vc3
10	force_err_f_seq_vc2	RW	0x0	Force err_f_seq_vc2
9	force_err_f_seq_vc1	RW	0x0	Force err_f_seq_vc1
8	force_err_f_seq_vc0	RW	0x0	Force err_f_seq_vc0
7:4	Reserved			Reserved
3	force_err_f_seq_vc3	RW	0x0	Force err_f_bndry_match_vc3
2	force_err_f_seq_vc2	RW	0x0	Force err_f_bndry_match_vc2
1	force_err_f_seq_vc1	RW	0x0	Force err_f_bndry_match_vc1
0	force_err_f_seq_vc0	RW	0x0	Force err_f_bndry_match_vc0

69.1.3.35. INT_ST_PHY

Table 6-36 INT_ST_PHY

Bits	Name	R/W	Reset value	Description
31:18	Reserved			Reserved

17	phy_erresc_1	R	0x0	Start of Transmission Error on data lane 1 (no synchronization achieved)
16	phy_erresc_0	R	0x0	Start of Transmission Error on data lane 0 (no synchronization achieved)
15:2	Reserved			Reserved
1	phy_errsoths_1	R	0x0	Start of transmission error on data lane 1 (synchronization can still be achieved)
0	phy_errsoths_0	R	0x0	Start of transmission error on data lane 0 (synchronization can still be achieved)

69.1.3.36. INT_MSK_PHY

Table 6-37 INT_MSK_PHY

Bits	Name	R/W	Reset value	Description
31:18	Reserved			Reserved
17	mask_phy_erresc_1	RW	0x0	Mask for phy_erresc_1
16	mask_phy_erresc_0	RW	0x0	Mask for phy_erresc_0
15:2	Reserved			Reserved
1	mask_phy_errsoths_1	RW	0x0	Mask for phy_errsoths_1
0	mask_phy_errsoths_0	RW	0x0	Mask for phy_errsoths_0

69.1.3.37. INT_FORCE_PHY

Table 6-38 INT_FORCE_PHY

Bits	Name	R/W	Reset value	Description
31:18	Reserved			Reserved
17	force_phy_erresc_1	RW	0x0	Force phy_erresc_1
16	force_phy_erresc_0	RW	0x0	Force phy_erresc_0
15:2	Reserved			Reserved
1	force_phy_errsoths_1	RW	0x0	Force phy_errsoths_1
0	force_phy_errsoths_0	RW	0x0	Force phy_errsoths_0

69.1.3.38. INT_ST_PKT

Table 6- 39 INT_ST_PKT

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	vc3_err_ecc_corrected	R	0x0	Header error detected and corrected on virtual channel 3
18	vc2_err_ecc_corrected	R	0x0	Header error detected and corrected on virtual channel 2
17	vc1_err_ecc_corrected	R	0x0	Header error detected and corrected on virtual channel 1
16	vc0_err_ecc_corrected	R	0x0	Header error detected and corrected on virtual channel 0
15:4	Reserved			Reserved
3	err_id_vc3	R	0x0	Unrecognized or unimplemented data type detected in virtual channel 3.
2	err_id_vc2	R	0x0	Unrecognized or unimplemented data type detected in virtual channel 2.
1	err_id_vc1	R	0x0	Unrecognized or unimplemented data type detected in virtual channel 1.
0	err_id_vc0	R	0x0	Unrecognized or unimplemented data type detected in virtual channel 0.

69.1.3.39. INT_MSK_PKT

Table 6- 40 INT_MSK_PKT

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	mask_vc3_err_ecc_corrected	RW	0x0	Mask for vc3_err_ecc_corrected
18	mask_vc2_err_ecc_corrected	RW	0x0	Mask for vc2_err_ecc_corrected
17	mask_vc1_err_ecc_corrected	RW	0x0	Mask for vc1_err_ecc_corrected
16	mask_vc0_err_ecc_corrected	RW	0x0	Mask for vc0_err_ecc_corrected
15:4	Reserved			Reserved
3	mask_err_id_vc3	RW	0x0	Mask for err_id_vc3.
2	mask_err_id_vc2	RW	0x0	Mask for err_id_vc2.

1	mask_err_id_vc1	RW	0x0	Mask for err_id_vc1.
0	mask_err_id_vc0	RW	0x0	Mask for err_id_vc0.

69.1.3.40. **INT_FORCE_PKT**

Table 6- 41 INT_FORCE_PKT

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	force_vc3_err_ecc_corrected	RW	0x0	Force vc3_err_ecc_corrected
18	force_vc2_err_ecc_corrected	RW	0x0	Force vc2_err_ecc_corrected
17	force_vc1_err_ecc_corrected	RW	0x0	Force vc1_err_ecc_corrected
16	force_vc0_err_ecc_corrected	RW	0x0	Force vc0_err_ecc_corrected
15:4	Reserved			Reserved
3	force_err_id_vc3	RW	0x0	Force err_id_vc3.
2	force_err_id_vc2	RW	0x0	Force err_id_vc2.
1	force_err_id_vc1	RW	0x0	Force err_id_vc1.
0	force_err_id_vc0	RW	0x0	Force err_id_vc0.

69.1.3.41. **INT_ST_LINE**

Table 6- 42 INT_ST_LINE

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	err_l_seq_di3	R	0x0	Error in the sequence of lines for vc3 and dt3
18	err_l_seq_di2	R	0x0	Error in the sequence of lines for vc2 and dt2
17	err_l_seq_di1	R	0x0	Error in the sequence of lines for vc1 and dt1
16	err_l_seq_di0	R	0x0	Error in the sequence of lines for vc0 and dt0
15:4	Reserved			Reserved
3	err_l_bndry_match_di3	R	0x0	Error matching line start with line end for vc3 and dt3
2	err_l_bndry_match_di2	R	0x0	Error matching line start with line end for vc2 and dt2

1	err_l_bndry_match_di1	R	0x0	Error matching line start with line end for vc1 and dt1
0	err_l_bndry_match_di0	R	0x0	Error matching line start with line end for vc0 and dt0

69.1.3.42. INT_MSK_LINE

Table 6- 43 INT_MSK_LINE

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	mask_err_l_seq_di3	RW	0x0	Mask for err_l_seq_di3
18	mask_err_l_seq_di2	RW	0x0	Mask for err_l_seq_di2
17	mask_err_l_seq_di1	RW	0x0	Mask for err_l_seq_di1
16	mask_err_l_seq_di0	RW	0x0	Mask for err_l_seq_di0
15:4	Reserved			Reserved
3	mask_err_l_bndry_match_di3	RW	0x0	Mask for err_l_bndry_match_di3
2	mask_err_l_bndry_match_di2	RW	0x0	Mask for err_l_bndry_match_di2
1	mask_err_l_bndry_match_di1	RW	0x0	Mask for err_l_bndry_match_di1
0	mask_err_l_bndry_match_di0	RW	0x0	Mask for err_l_bndry_match_di0

69.1.3.43. INT_FORCE_LINE

Table 6- 44 INT_FORCE_LINE

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	force_err_l_seq_di3	RW	0x0	Force err_l_seq_di3
18	force_err_l_seq_di2	RW	0x0	Force err_l_seq_di2
17	force_err_l_seq_di1	RW	0x0	Force err_l_seq_di1
16	force_err_l_seq_di0	RW	0x0	Force err_l_seq_di0
15:4	Reserved			Reserved

3	force_err_l_bndry_match_di3	RW	0x0	Force err_l_bndry_match_di3
2	force_err_l_bndry_match_di2	RW	0x0	Force err_l_bndry_match_di2
1	force_err_l_bndry_match_di1	RW	0x0	Force err_l_bndry_match_di1
0	force_err_l_bndry_match_di0	RW	0x0	Force err_l_bndry_match_di0

69.1.3.44. INT_ST_IPI

Table 6- 45 INT_ST_IPI

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	int_event_fifo_overflow	R	0x0	Reporting internal fifo overflow.
4	pixel_if_hline_err	R	0x0	Horizontal line time error (only available in controller mode).
3	pixel_if_fifo_nempty_fs	R	0x0	The FIFO of pixel interface is not empty at the start of a new frame. If this is expected this interrupt should be masked.
2	pixel_if_frame_sync_err	R	0x0	New frame is received but previous has not been completed
1	pixel_if_fifo_overflow	R	0x0	The FIFO of pixel interface has lost information because some more data arrived when it was full.
0	pixel_if_fifo_underflow	R	0x0	The FIFO has become empty before the expected number of pixels (calculated from the packet's header) could be extracted to the pixel interface.

69.1.3.45. INT_MSK_IPI

Table 6- 46 INT_MSK_IPI

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	msk_int_event_fifo_overflow	R	0x0	Mask for int_event_fifo_overflow
4	msk_pixel_if_hline_err	R	0x0	Mask for pixel_if_hline_err
3	msk_pixel_if_fifo_nempty_err	R	0x0	Mask for pixel_if_fifo_nempty_fs

2	msk_pixel_if_frame_sync_err	R	0x0	Mask for pixel_if_frame_sync_err
1	msk_pixel_if_fifo_overflow	R	0x0	Mask for pixel_if_fifo_overflow.
0	msk_pixel_if_fifo_underflow	R	0x0	Mask for pixel_if_fifo_underflow

69.1.3.46. INT_FORCE_IPI

Table 6-47 INT_FORCE_IPI

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	force_int_event_fifo_overflow	R	0x0	Force int_event_fifo_overflow
4	force_pixel_if_hline_err	R	0x0	Force pixel_if_hline_err
3	force_pixel_if_fifo_nempty_err	R	0x0	Force pixel_if_fifo_nempty_fs
2	force_pixel_if_frame_sync_err	R	0x0	Force pixel_if_frame_sync_err
1	force_pixel_if_fifo_overflow	R	0x0	Force pixel_if_fifo_overflow.
0	force_pixel_if_fifo_underflow	R	0x0	Force pixel_if_fifo_underflow

69.1.3.47. INT_ST_IPI2

Table 6-48 INT_ST_IPI2

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	int_event_fifo_overflow	R	0x0	Reporting internal fifo overflow.
4	pixel_if_hline_err	R	0x0	Horizontal line time error (only available in controller mode).
3	pixel_if_fifo_nempty_fs	R	0x0	The FIFO of pixel interface is not empty at the start of a new frame. If this is expected this interrupt should be masked.
2	pixel_if_frame_sync_err	R	0x0	New frame is received but previous has not been completed
1	pixel_if_fifo_overflow	R	0x0	The FIFO of pixel interface has lost information because some more data arrived when it was full.

0	pixel_if_fifo_underflow	R	0x0	The FIFO has become empty before the expected number of pixels (calculated from the packet's header) could be extracted to the pixel interface.

69.1.3.48. INT_MSK_IPI2

Table 6- 49 INT_MSK_IPI2

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	msk_int_event_fifo_overflow	R	0x0	Mask for int_event_fifo_overflow
4	msk_pixel_if_hline_err	R	0x0	Mask for pixel_if_hline_err
3	msk_pixel_if_fifo_nempty_err	R	0x0	Mask for pixel_if_fifo_nempty_fs
2	msk_pixel_if_frame_sync_err	R	0x0	Mask for pixel_if_frame_sync_err
1	msk_pixel_if_fifo_overflow	R	0x0	Mask for pixel_if_fifo_overflow.
0	msk_pixel_if_fifo_underflow	R	0x0	Mask for pixel_if_fifo_underflow

69.1.3.49. INT_FORCE_IPI2

Table 6- 50 INT_FORCE_IPI2

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	force_int_event_fifo_overflow	R	0x0	Force int_event_fifo_overflow
4	force_pixel_if_hline_err	R	0x0	Force pixel_if_hline_err
3	force_pixel_if_fifo_nempty_err	R	0x0	Force pixel_if_fifo_nempty_fs
2	force_pixel_if_frame_sync_err	R	0x0	Force pixel_if_frame_sync_err
1	force_pixel_if_fifo_overflow	R	0x0	Force pixel_if_fifo_overflow.
0	force_pixel_if_fifo_underflow	R	0x0	Force pixel_if_fifo_underflow

69.1.3.50. IPI2_MODE

Table 6- 51 IPI2_MODE

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	ipi_enable	RW	0x0	This register enables the interface 2.
23:17	Reserved			Reserved
16	ipi_cut_through	RW	0x0	This field indicates cut-through mode state: -0: cut-through mode inactive -1: cut-through mode active
15:9	Reserved			Reserved
8	ipi_color_com	RW	0x0	This field indicates if color mode components are delivered as follows: -0: 48 bits interface -1: 16 bits interface
7:1	Reserved			Reserved
0	ipi_mode	RW	0x0	This field indicates the video mode transmission type as follows: -0: Camera timing (Pixel Interface 2 only supported in Camera Mode)

69.1.3.51. IPI2_VCID

Table 6- 52 IPI2_VCID

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1:0	ipi_vcid	RW	0x0	Virtual channel of data to be processed by pixel interface 2

69.1.3.52. IPI2_DATA_TYPE

Table 6- 53 IPI2_DATA_TYPE

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
8	embedded_data	RW	0x0	This bit enables embedded data processing on IPI 2 interface.
7:6	Reserved			Reserved
5:0	ipi_data_type	RW	0x0	Data type of data to be processed by pixel interface 2.

69.1.3.53. IPI2_MEM_FLUSH

Table 6- 54 IPI2_MEM_FLUSH

Bits	Name	R/W	Reset value	Description
31:9	Reserved			Reserved
8	ipi_auto_flush	RW	0x0	Memory is automatically flushed at each vsync
7:1	Reserved			Reserved
0	ipi_flush	RW	0x0	Flush IPI2 memory. This bit is auto clear.

69.1.3.54. IPI2_HSA_TIME

Table 6- 55 IPI2_HSA_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hsa_time	RW	0x0	This field configures the HSA period in pixclk cycles

69.1.3.55. IPI2_HBP_TIME

Table 6- 56 IPI2_HBP_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hbp_time	RW	0x0	This field configures the HBP period in pixclk cycles

69.1.3.56. IPI2_HSD_TIME

Table 6- 57 IPI2_HSD_TIME

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	ipi_hsd_time	RW	0x0	This field configures the HSD period in pixclk cycles

69.1.3.57. IPI2_ADV_FEATURES

Table 6- 58 IPI2_ADV_FEATURES

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	ipi_sync_event_	RW	0x0	For Camera Mode:

	mode			-0: Frame Start dont trigger any sync event -1: Legacy mode. Frame start will trigger a sync event
23:22	Reserved			Reserved
21	en_embedded	RW	0x0	This register allows to use embedded packets for IPI synchronization events.
20	en_blanking	RW	0x0	This register allows to use blanking packets for IPI synchronization events.
19	en_null	RW	0x0	This register allows to use null packets for IPI synchronization events.
18	en_line_start	RW	0x0	This register allows to use line start packets for IPI synchronization events.
17	en_video	RW	0x0	This register allows to use video packets for IPI synchronization events.
16	line_event_selection	RW	0x0	For Camera Mode, this register allows to manualy select the Packet for line delimiter as follows: -0: controller select it automatically -1: select packets from list programmed in [17:21].
15:14	Reserved			Reserved
13:8	ipi_dt	RW	0x0	Datatype to overwrite.
7:1	Reserved			Reserved
0	ipi_dt_overwrite	RW	0x0	Ignore datatype of the header using the programed datatype for decoding.

69.2. MIPI Tx

69.2.1. Overview

The Display Serial Interface (DSI) is part of a group of communication protocols defined by the MIPI Alliance. The MIPI DSI Host Controller is a digital core that implements all protocol functions defined in the MIPI DSI Specification, which provides an interface between the system and the MIPI D-PHY or Combo PHY (D-PHY and C-PHY), allowing the communication with a DSI-compliant display. There is a broad range of D-PHY IPs that includes bidirectional PHYs with two and four lanes for several technologies.

Typical applications built with the MIPI DSI Controller including handheld devices, smartphone, multimedia tablets, MID, navigation, gaming consoles, DVC, automotive, IoT and wearables.

69.2.2. Features

Support the following features:

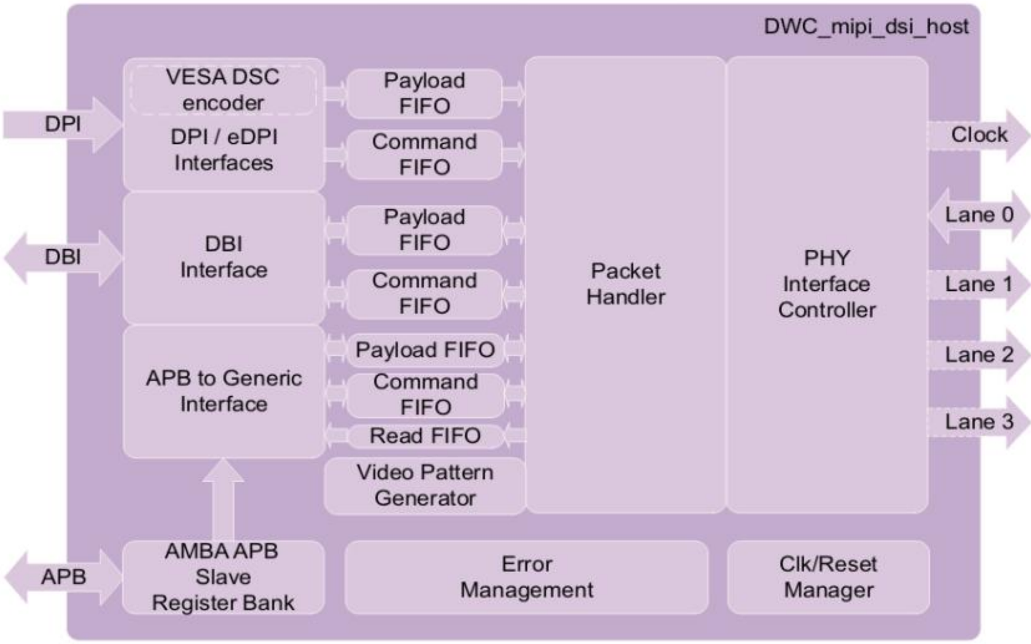
- Compliant with the Display Serial Interface (DSI) Version 1.2
- Video Format Supported: 16bitsRGB,18bitsRGB,24bitsRGB
- Interface with MIPI D-PHY following PHY Protocol Interface (PPI)
- Configurable up to 2 data lanes
- Supports all commands for Display Command Set (DCS)
- Bidirectional communication and escape mode support through data lane 0
- Supports continuous and non-continuous mode
- Supports Ultra Low-Power mode with PLL disabled
- ECC and Checksum capabilities
- Support for End of Transmission Packet (EoTp)
- Fault recovery schemes
- Video pattern generator
- Tear effect request by hardware
- AMBA-APB control and Configuration

69.2.3. Function Description

69.2.3.1. Function Overview

MIPI Tx module including the following basic sub-modules, as shown in below figure:

- DPI Interface
- Register Bank
- PHY Interface Control
- Packet Handler
- APB-to-Generic
- Error Management



69.2.4. Programming Guide

The following is the procedure and actions to program the DSI Host Controller.

1. Reset DSI Host using the global reset.
2. Configure n_lanes (PHY_IF_CFG) and interrupt trigger mask in INT_MSK_<group>.
3. Configure the DPI interface to define how the DPI interface interacts with the controller.
4. Select the video transmission mode to define how the processor requires the video line to be transported through the DSI link.
5. Define the DPI horizontal and vertical timing configuration.
6. Power-Up and reset the PHY using PHY_RSTZ register.
7. Configure the D-PHY PLL clock frequency through the Test interface
8. Setup TxClkEsc frequency between 2 to 20 MHz by writing into the tx_esc_clk_division field of the CLKMGR_CFG register. The tx_esc_clk_division field divides the byte clock, and generates a TX_ESC clock for the D-PHY.
9. Clear phy_shutdownz (PHY_RSTZ[0]) and phy_rstz (PHY_RSTZ[1]) to put the PHY in the reset state. Set phy_shutdownz (PHY_RSTZ[0]) and phy_rstz (PHY_RSTZ[1]) to remove the PHY from reset state.
10. Verify that D-PHY active lanes are in Stop state then power-up the controller.

69.2.5. Register Summary

Table 6- 59 MIPI TX register summary

Name	Offset	R/W	Description
VERSION	0x0	RW	Contains the version of controller coded in 32-bit ASCII code
PWR_UP	0x4	RW	Power up of the core.
CLKMGR_CFG	0x8	RW	Configures the factor for internal dividers to divide lanebyteclk for timeout purposes
DPI_VCID	0xC	RW	Configures the Virtual Channel ID for DPI traffic.
DPI_COLOR_CODING	0x10	RW	Configures DPI color coding.
DPI_CFG_POL	0x14	RW	Configures the polarity of DPI signals
DPI_LP_CMD_TIM	0x18	RW	Configures the timing for low-power commands sent while in video mode
PCKHDL_CFG	0x2C	RW	Configures how EoTp, BTA, CRC and ECC are to be used
GEN_VCID	0x30	RW	Configures the Virtual Channel ID of READ responses to store and return to Generic interface.
MODE_CFG	0x34	RW	Configures the mode of operation between Video or Command Mode.
VID_MODE_CFG	0x38	RW	Configures several aspects of Video mode operation, the transmission mode, switching to Low-Power in the middle of a frame, enabling acknowledge and whether to send commands in Low-Power.
VID_PKT_SIZE	0x3C	RW	Configures the video packet size.
VID_NUM_CHUNKS	0x40	RW	Configures the number of chunks to use. The data in each chunk has the size provided by VID_PKT_SIZE.
VID_NULL_SIZE	0X44	RW	Configures the size of null packets.
VID_HSA_TIME	0X48	RW	Configures the video HSA time.
VID_HBP_TIME	0X4C	RW	Configures the video HBP time.
VID_HLINE_TIME	0X50	RW	Configures the overall time for each video line.
VID_VSA_LINES	0X54	RW	Configures the VSA period.
VID_VBP_LINES	0X58	RW	Configures the VBP period.
VID_VFP_LINES	0X5C	RW	Configures the VFP period.
VID_VACTIVE_LINES	0X60	RW	Configures the vertical resolution of video.
CMD_MODE_CFG	0X68	RW	Configures several aspect of command mode operation, tearing effect, acknowledge for each packet and the speed mode to transmit each Data Type related to commands.
GEN_HDR	0X6C	RW	Sets the header for new packets sent using the Generic interface.

GEN_PLD_DATA	0X70	RW	Sets the payload for packets sent using the Generic interface and, when read, returns the contents of READ responses from the peripheral.
CMD_PKT_STATUS	0X74	R	Contains information about the status of FIFOs related to Generic interface.
TO_CNT_CFG	0X78	RW	Configures counters that trigger timeout errors.
HS_RD_TO_CNT	0X7C	RW	Configures the Peripheral Response timeout after High-Speed Read operations.
LP_RD_TO_CNT	0X80	RW	Configures the Peripheral Response timeout after Low-Power Read operations.
HS_WR_TO_CNT	0X84	RW	Configures the Peripheral Response timeout after High-Speed Write operations.
LP_WR_TO_CNT	0X88	RW	Configures the Peripheral Response timeout after Low-Power Write operations.
BTA_TO_CNT	0X8C	RW	Configures the Peripheral Response timeout after Bus Turnaround completion.
SDF_3D	0X90	RW	Stores 3D control information for VSS packets in video mode.
LPCLK_CTRL	0X94	RW	Configures the possibility for using non continuous clock in the clock lane.
PHY_TMR_LPCLK_CFG	0X98	RW	Sets the time used in calculations for the clock lane to switch between High-Speed and Low-Power.
PHY_TMR_CFG	0X9C	RW	Sets the time used in calculations for the data lanes to switch between High-Speed and Low-Power.
PHY_RSTZ	0XA0	RW	Controls resets and PLL settings for D-PHY.
PHY_IF_CFG	0XA4	RW	Configures the number of active lanes and the minimum time to remain in stop state.
PHY_ULPS_CTRL	0XA8	RW	Configures entering and leaving ULPS in the D-PHY.
PHY_TX_TRIGGERS	0XAC	RW	Configures the pins that activate triggers in the D-PHY.
PHY_STATUS	0XB0	R	Contains information about the status of the D-PHY.
PHY_TST_CTRL0	0XB4	RW	Controls clock and clear pins of the D-PHY vendor specific interface.
PHY_TST_CTRL1	0XB8	RW	Controls data and enable pins of the D-PHY vendor specific interface.
INT_ST0	0XBC	R	Contains information about the status of interrupt sources from acknowledge reports and the D-PHY.

INT_ST1	0XC0	R	Contains information about the status of interrupt sources related to timeouts, ECC, CRC, packet size, EoTp, Generic and DBI interfaces.
INT_MSK0	0XC4	RW	Configures masks for the sources of interrupts that affect the INT_ST0 register. Write 1 to Un-mask each error report.
INT_MSK1	0XC8	RW	Configures masks for the sources of interrupts that affect the INT_ST1 register. Write 1 to Un-mask each error report.
PHY_CAL	0XCC	RW	Controls the skew calibration of D-PHY.
INT_FORCE0	0XD8	RW	Forces trigger for interrupts that affect INT_ST0 register.
INT_FORCE1	0XDC	RW	Forces trigger for interrupts that affect INT_ST1 register.
DSC_PARAMETER	0XF0	RW	Configures Display Stream Compression
PHY_TMR_RD_CFG	0XF4	RW	Configures times related to PHY to perform some operations in lane byte clock cycles.
VID_SHADOW_CTRL	0X100	RW	Controls dpi shadow feature
DPI_VCID_ACT	0X10C	R	Value that controller is using for DPI_VCID
DPI_COLOR_CODING_ACT	0X110	R	Value that controller is using for DPI_COLOR_CODING.
DPI_LP_CMD_TIM_ACT	0X118	R	Value that controller is using for DPI_LP_CMD_TIM.
VID_MODE_CFG_ACT	0X138	R	Value that controller is using for VID_MODE_CFG.
VID_PKT_SIZE_ACT	0X13C	R	Value that controller is using for VID_PKT_SIZE.
VID_NUM_CHUNKS_ACT	0X140	R	Value that controller is using for VID_NUM_CHUNKS.
VID_NULL_SIZE_ACT	0X144	R	Value that controller is using for VID_NULL_SIZE.
VID_HSA_TIME_ACT	0X148	R	Value that controller is using for VID_HSA_TIME.
VID_HBP_TIME_ACT	0X14C	R	Value that controller is using for VID_HBP_TIME.
VID_HLINE_TIME_ACT	0X150	R	Value that controller is using for VID_HLINE_TIME.
VID_VSA_LINES_ACT	0X154	R	Value that controller is using for VID_VSA_LINES.
VID_VBP_LINES_ACT	0X158	R	Value that controller is using for VID_VBP_LINES.
VID_VFP_LINES_ACT	0X15C	R	Value that controller is using for VID_VFP_LINES.

VID_VACTIVE_LINES_ACT	0X160	R	Value that controller is using for VID_VACTIVE_LINES.
VID_PKT_STATUS	0X168	R	Contains information about the status of FIFOs related to DPI interfaces.
SDF_3D_ACT	0X190	R	Value that controller is using for SDF_3D.

69.2.6. Register Description

69.2.6.1. VERSION

Table 6- 60 VERSION

Bits	Name	R/W	Reset value	Description
31:0	version	R	0x3134302a	Version information

69.2.6.2. PWR_UP

Table 6- 61 PWR_UP

Bits	Name	R/W	Reset value	Description
31:1	Reserved			Reserved
0	shutdownz	R/W	0	Configures the core either to power up or to reset 0 (RESET): Reset the core 1 (POWERUP): Power up the core

69.2.6.3. CLKMGR_CFG

Table 6- 62 CLKMGR_CFG

Bits	Name	R/W	Reset value	Description
31:16	Reserved			Reserved
15:8	to_clk_division	R/W	0	This field indicates the division factor for the Time Out clock used as the timing unit in the configuration of HS to LP and LP to HS transition error.
7:0	tx_esc_clk_division	R/W	0	This field indicates the division factor for the TX Escape clock source (lanebyteclk). The values 0 and 1 stop the TX_ESC clock generation.

69.2.6.4. **DPI_VCID**

Table 6- 63 DPI_VCID

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1:0	dpi_vcid	R/W	0	This field configures the DPI virtual channel id that is indexed to the Video mode packets.

69.2.6.5. **DPI_COLOR_CODING**

Table 6- 64 DPI_COLOR_CODING

Bits	Name	R/W	Reset value	Description
31:9	Reserved			Reserved
8	loosely18_en	R/W	0	When set to 1, this bit activates loosely packed variant to 18-bit configurations.
7:4	Reserved			Reserved
3:0	dpi_color_coding	R/W	0	This field configures the DPI color for Video Mode Command Mode coding as follows: 0x0 (CC00): 16-bit configuration 1 0x1 (CC01): 16-bit configuration 2 0x2 (CC02): 16-bit configuration 3 0x3 (CC03): 18-bit configuration 1 0x4 (CC04): 18-bit configuration 2 0x5 (CC05): 24-bit 0x6 ~0xf: Reserved

69.2.6.6. **DPI_CFG_POL**

Table 6- 65 DPI_CFG_POL

Bits	Name	R/W	Reset value	Description
31:5	Reserved			Reserved
4	colorm_active_low	R/W	0	When set to 1, this bit configures the color mode pin (dpicolorm) as active low.
3	shutd_active_low	R/W	0	When set to 1, this bit configures the shutdown pin (dpishutdn) as active low.
2	hsync_active_low	R/W	0	When set to 1, this bit configures the horizontal synchronism pin (dpihsync) as active low.
1	vsync_active_low	R/W	0	When set to 1, this bit configures the vertical synchronism pin (dpivsync) as active low.
0	dataen_active_low	R/W	0	When set to 1, this bit configures the data enable pin (dpidataen) as active low.

69.2.6.7. DPI_LP_CMD_TIM

Table 6- 66 DPI_LP_CMD_TIM

Bits	Name	R/W	Reset value	Description
31:24	Reserved			Reserved
23:16	outvact_lpcmd_time	R/W	0	This field is used for the transmission of commands in low-power mode. It defines the size, in bytes, of the largest packet that can fit in a line during the VSA, VBP, and VFP regions.
15:8	Reserved			Reserved
7:0	invact_lpcmd_time	R/W	0	This field is used for the transmission of commands in low-power mode. It defines the size, in bytes, of the largest packet that can fit in a line during the VACT region.

69.2.6.8. PCKHDL_CFG

Table 6- 67 PCKHDL_CFG

Bits	Name	R/W	Reset value	Description
31:6	Reserved			Reserved
5	eotp_tx_lp_en	R/W	0	When set to 1, this bit enables the EoTp transmission in Low-Power.
4	crc_rx_en	R/W	0	When set to 1, this bit enables the CRC reception and error reporting.
3	ecc_rx_en	R/W	0	When set to 1, this bit enables the ECC reception, error correction, and reporting.
2	bta_en	R/W	0	When set to 1, this bit enables the Bus Turn-Around (BTA) request.
1	eotp_rx_en	R/W	0	When set to 1, this bit enables the EoTp reception.
0	eotp_tx_en	R/W	0	When set to 1, this bit enables the EoTp transmission in High-Speed.

69.2.6.9. GEN_VCID

Table 6- 68 GEN_VCID

Bits	Name	R/W	Reset value	Description
31:18	Reserved			Reserved
17:16	gen_vcid_tx_auto	R/W	0	This field indicates the Generic interface virtual channel identification where generic packet is

				automatically generated & transmitted.
15:10	Reserved			Reserved
9:8	gen_vcid_tear_auto	R/W	0	This field indicates the virtual channel identification for tear effect by hardware
7:2	Reserved			Reserved
1:0	gen_vcid_rx	R/W	0	This field indicates the Generic interface read-back virtual channel identification.

69.2.6.10. **MODE_CFG**

Table 6- 69 MODE_CFG

Bits	Name	R/W	Reset value	Description
31:1	Reserved			Reserved
0	cmd_video_mode	R/W	0	This bit configuers the operation mode: 0 (VIDMODE): video mode 1 (CMDMODE): command mode

69.2.6.11. **VID_MODE_CFG**

Table 6- 70 VID_MODE_CFG

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	vpg_orientation	R/W	0	This field indicates the color bar orientation as follows: 0 (VPGORENT0): Vertical mode 1 (VPGORENT1): Horizontal mode
23:21	Reserved			Reserved
20	vpg_mode	R/W	0	This field is to select the pattern: 0 (COLORBAR): horizontal or vertical 1 (BERPATTERN): vertical only
19:17	Reserved			Reserved
16	vpg_en	R/W	0	When set to 1, this bit enables the video mode pattern generator.
15	lp_cmd_en	R/W	0	When set to 1, this bit enables the command transmission only in low-power mode.
14	frame_bta_ack_en	R/W	0	When set to 1, this bit enables the request for an acknowledge response at the end of a frame.
13	lp_hfp_en	R/W	0	When set to 1, this bit enables the return to low-power inside the HFP period when timing allows
12	lp_hbp_en	R/W	0	When set to 1, this bit enables the return to

				low-power inside the HBP period when timing allows.
11	lp_vact_en	R/W	0	When set to 1, this bit enables the return to low-power inside the VACT period when timing allows.
10	lp_vfp_en	R/W	0	When set to 1, this bit enables the return to low-power inside the VFP period when timing allows.
9	lp_vbp_en	R/W	0	When set to 1, this bit enables the return to low-power inside the VBP period when timing allows.
8	lp_vsa_en	R/W	0	When set to 1, this bit enables the return to low-power inside the VSA period when timing allows.
7:2	Reserved			Reserved
1:0	vid_mode_type	R/W	0	This field indicates the video mode transmission type as follows:Values: 0x0 (VIDMODE0): Non-burst with sync pulses 0x1 (VIDMODE1): Non-burst with sync events 0x2 (VIDMODE2): Burst mode 0x3 (VIDMODE3): Burst mode

69.2.6.12. VID_PKT_SIZE

Table 6- 71 VID_PKT_SIZE

Bits	Name	R/W	Reset value	Description
31:14	Reserved			Reserved
13:0	vid_pkt_size	R/W	0	This field configures the number of pixels in a single video packet.

69.2.6.13. VID_NUM_CHUNKS

Table 6- 72 VID_NUM_CHUNKS

Bits	Name	R/W	Reset value	Description
31:13	Reserved			Reserved
12:0	vid_num_chunks	R/W	0	This register configures the number of chunks to be transmitted during a Line period (a chunk is pair made of a video packet and a null packet). If set to 0 or 1, video line is still transmitted in a single packet. If set to 1 that packet is part of a chunk, meaning that a null packet follows it (if vid_null_size>0).

				Otherwise, multiple chunks are used to transmit each video line.

69.2.6.14. VID_NULL_SIZE

Table 6- 73 VID_NULL_SIZE

Bits	Name	R/W	Reset value	Description
31:13	Reserved			Reserved
12:0	vid_null_size	R/W	0	This register configures the number of bytes inside a null packet. Setting to 0 disables null packets.

69.2.6.15. VID_HSA_SIZE

Table 6- 74 VID_HSA_SIZE

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	vid_hsa_time	R/W	0	This field configures the Horizontal Synchronism Active period in lane byte clock cycles.

69.2.6.16. VID_HBP_SIZE

Table 6- 75 VID_HBP_SIZE

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	vid_hbp_time	R/W	0	This field configures the Horizontal Back Porch period in lane byte clock cycles.

69.2.6.17. VID_HLINE_TIME

Table 6- 76 VID_HLINE_TIME

Bits	Name	R/W	Reset value	Description
31:15	Reserved			Reserved
14:0	vid_hline_time	R/W	0	This field configures the size of the total line time (HSA+HBP+HACT+HFP) counted in lane byte clock cycles.

69.2.6.18. VID_VSA_LINES

Table 6- 77 VID_VSA_LINES

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	vsa_lines	R/W	0	This field configures the Vertical Synchronism Active period measured in number of horizontal lines.

69.2.6.19. VID_VBP_LINES

Table 6- 78 VID_VBP_LINES

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	vbp_lines	R/W	0	This field configures the Vertical Back Porch period measured in number of horizontal lines.

69.2.6.20. VID_VFP_LINES

Table 6- 79 VID_VFP_LINES

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	vfp_lines	R/W	0	This field configures the Vertical Front Porch period measured in number of horizontal lines.

69.2.6.21. VID_VACTIVE_LINES

Table 6- 80 VID_VACTIVE_LINES

Bits	Name	R/W	Reset value	Description
31:14	Reserved			Reserved
13:0	v_active_lines	R/W	0	This field configures the Vertical Active period measured in number of horizontal lines.

69.2.6.22. CMD_MODE_CFG

Table 6- 81 CMD_MODE_CFG

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	max_rd_pkt_siz	R/W	0	This bit configures the maximum read packet

	e			size command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
23:20	Reserved			Reserved
19	dcs_lw_tx	R/W	0	This bit configures the DCS long write packet command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
18	dcs_sr_0p_tx	R/W	0	This bit configures the DCS short read packet with zero parameter command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
17	dcs_sw_1p_tx	R/W	0	This bit configures the DCS short write packet with one parameter command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
16	dcs_sw_0p_tx	R/W	0	This bit configures the DCS short write packet with zero parameter command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
15	Reserved			Reserved
14	gen_lw_tx	R/W	0	This bit configures the Generic long write packet command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
13	gen_sr_2p_tx	R/W	0	This bit configures the Generic short read packet with two parameters command transmission type: 0x0 (HIGHSPEED): Transition type is High

				Speed 0x1 (LOWPOWER): Transition type is Low Power
12	gen_sr_1p_tx	R/W	0	This bit configures the Generic short read packet with one parameter command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
11	gen_sr_0p_tx	R/W	0	This bit configures the Generic short read packet with zero parameter command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
10	gen_sw_2p_tx	R/W	0	This bit configures the Generic short write packet with two parameters command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
9	gen_sw_1p_tx	R/W	0	This bit configures the Generic short write packet with one parameters command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
8	gen_sw_0p_tx	R/W	0	This bit configures the Generic short write packet with zero parameters command transmission type: 0x0 (HIGHSPEED): Transition type is High Speed 0x1 (LOWPOWER): Transition type is Low Power
7:2	Reserved			Reserved
1	ack_rqst_en	R/W	0	When set to 1, this bit enables the acknowledge request after each packet transmission.
0	tear_fx_en	R/W	0	When set to 1, this bit enables the tearing effect acknowledge request.

69.2.6.23. GEN_HDR

Table 6- 82 GEN_HDR

Bits	Name	R/W	Reset value	Description
31:24	Reserved			Reserved
23:16	gen_wc_msbyte	R/W	0	This field configures the most significant byte of the header packet's word count for long packets or data 1 for short packets.
15:8	gen_wc_lsbyte	R/W	0	This field configures the least significant byte of the header packet's Word count for long packets or data 0 for short packets.
7:6	gen_vc	R/W	0	This field configures the Virtual Channel ID of the header packet.
5:0	gen_dt	R/W	0	This field configures the packet Data Type of the header packet.

69.2.6.24. GEN_PLD_DATA

Table 6- 83 GEN_PLD_DATA

Bits	Name	R/W	Reset value	Description
31:24	gen_pld_b4	R/W	0	This field indicates byte 4 of the packet payload.
23:16	gen_pld_b3	R/W	0	This field indicates byte 3 of the packet payload.
15:8	gen_pld_b2	R/W	0	This field indicates byte 2 of the packet payload.
7:0	gen_pld_b1	R/W	0	This field indicates byte 1 of the packet payload.

69.2.6.25. CMD_PKT_STATUS

Table 6- 84 CMD_PKT_STATUS

Bits	Name	R/W	Reset value	Description
31:20	Reserved			Reserved
19	gen_buff_pld_full	R	0	This bit indicates the full status of the generic payload internal buffer.
18	gen_buff_pld_empty	R	1	This bit indicates the empty status of the generic payload internal buffer.
17	gen_buff_cmd_full	R	0	This bit indicates the full status of the generic command internal buffer.
16	gen_buff_cmd_	R	1	This bit indicates the empty status of the

	empty			generic command internal buffer.
15:7	Reserved			Reserved
6	gen_rd_cmd_busy	R	0	This bit is set when a read command is issued and cleared when the entire response is stored in the FIFO for GENERIC interface.
5	gen_pld_r_full	R	0	This bit indicates the full status of the generic read payload FIFO.
4	gen_pld_r_empty	R	1	This bit indicates the empty status of the generic read payload FIFO.
3	gen_pld_w_full	R	0	This bit indicates the full status of the generic write payload FIFO.
2	gen_pld_w_empty	R	1	This bit indicates the empty status of the generic write payload FIFO.
1	gen_cmd_full	R	0	This bit indicates the full status of the generic command FIFO.
0	gen_cmd_empty	R	1	This bit indicates the empty status of the generic command FIFO.

69.2.6.26. TO_CNT_CFG

Table 6- 85 TO_CNT_CFG

Bits	Name	R/W	Reset value	Description
31:16	hstx_to_cnt	R/W	0	This field configures the timeout counter that triggers a high-speed transmission timeout contention detection (measured in TO_CLK_DIVISION cycles). If using non-burst mode and there is not sufficient time to switch from HS to LP and back in the period which is from one line data finishing to the next line sync start, the DSI link will return LP state once per frame, then you should configure the TO_CLK_DIVISION and hstx_to_cnt to satisfy the formula below: hstx_to_cnt * lanebyteclkperiod * TO_CLK_DIVISION >= the time of one FRAME data transmission * (1 + 10%) In burst mode, RGB pixel packets are time-compressed, leaving more time during a scan line. So if in burst mode and there is sufficient time to switch from HS to LP and back in the period of time from one line data finishing to the next line sync start, the DSI link can return LP mode and back in this time

				interval to save power. If you choose so, you should configure the TO_CLK_DIVISION and hstx_to_cnt to satisfy the formula below: hstx_to_cnt * lanebyteclkperiod * TO_CLK_DIVISION >= the time of one LINE data transmission * (1 + 10%)
15:0	lprx_to_cnt	R/W	0	This field configures the timeout counter that triggers a low-power reception timeout contention detection (measured in TO_CLK_DIVISION cycles).

69.2.6.27. **HS_RD_TO_CNT**

Table 6- 86 HS_RD_TO_CNT

Bits	Name	R/W	Reset value	Description
31:16	Reserved			Reserved
15:0	hs_rd_to_cnt	R/W	0	This field sets a period for which DSI host keeps the link still, after sending a High-Speed Read operation. This period is measured in cycles of lanebyteclk, starts to count when the D-PHY enters stop state and causes no interrupts.

69.2.6.28. **LP_RD_TO_CNT**

Table 6- 87 LP_RD_TO_CNT

Bits	Name	R/W	Reset value	Description
31:16	Reserved			Reserved
15:0	lp_rd_to_cnt	R/W	0	This field sets a period for which DSI host keeps the link still, after sending a High-Speed Read operation. This period is measured in cycles of lanebyteclk, starts to count when the D-PHY enters stop state and causes no interrupts.

69.2.6.29. **HS_WR_TO_CNT**

Table 6- 88 HS_WR_TO_CNT

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	presp_to_mode	R/W	0	When set to 1, this bit causes that peripheral

				response timeout caused by hs_wr_to_cnt is used only once per eDPI frame, after the following conditions: - dpivsync_edpiwms has risen and fallen. - Packets originated from eDPI have been transmitted and its FIFO is empty again. In this scenario no non-eDPI requests are sent to the D-PHY, even if there is traffic from generic or DBI ready to be sent, making it return to stop state. When it does so, PRESP_TO counter is activated and only when it finishes does the controller send any other traffic that is ready.
23:16	Reserved			Reserved
15:0	hs_wr_to_cnt	R/W	0	This field sets a period for which DSI host keeps the link still, after sending a High-Speed Write operation. This period is measured in cycles of lanebyteclk, starts to count when the D-PHY enters stop state and causes no interrupts.

69.2.6.30. **LP_WR_TO_CNT**

Table 6- 89 LP_WR_TO_CNT

Bits	Name	R/W	Reset value	Description
31:16	Reserved			Reserved
15:0	lp_wr_to_cnt	R/W	0	This field sets a period for which DSI host keeps the link still, after sending a Low-Power Write operation. This period is measured in cycles of lanebyteclk, starts to count when the D-PHY enters stop state and causes no interrupts.

69.2.6.31. **BTA_TO_CNT**

Table 6- 90 BTA_TO_CNT

Bits	Name	R/W	Reset value	Description
31:16	Reserved			Reserved
15:0	bta_to_cnt	R/W	0	This field sets a period for which DSI host keeps the link still, after completing a Bus Turnaround. This period is measured in cycles of lanebyteclk, starts to count when the D-PHY enters stop state and causes no interrupts.

69.2.6.32. SDF_3D

Table 6- 91 SDF_3D

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	send_3d_cfg	R/W	0	When set, causes the next VSS packet to include 3D control payload in every VSS packet.
15:6	Reserved			Reserved
5	right_first	R/W	0	This bit defines the left/right order: 0x0 (LFIRST): left eye is sent first, then right eye 0x1 (RFIRST): right eye data is sent first, then left eye
4	second_vsync	R/W	0	This field defines whether there is a second VSYNC pulse between Left and Right Images, when 3D Image Format is Frame-based: 0x0 (NOSYNC): No sync pulses between left and right data 0x1 (SYNC): Sync pulseHSYNC, VSYNC, blanking) between left and right data
3:2	format_3d	R/W	0	This field defines 3D Image Format: 0x0 (LINE): Alternating lines of left and right data 0x1 (FRAME): Alternating frames of left and right data 0x2 (PIXEL): Alternating pixels of left and right data 0x3 (RESERVED): Reserved, not used
1:0	mode_3d	R/W	0	This field defines 3D Mode On/Off and Display Orientation: 0x0 (MODE0): 3D Mode Off , 2D Mode On 0x1 (MODE1): 3D Mode On, Portrait Orientation 0x2 (MODE2): 3D Mode On, Landscape Orientation 0x3 (MODE3): Reserved, not used

69.2.6.33. LPCLK_CTRL

Table 6- 92 LPCLK_CTRL

Bits	Name	R/W	Reset value	Description

31:16	Reserved			Reserved
1	auto_clklane_ctrl	R/W	0	This bit enables the automatic mechanism to stop providing clock in the clock lane when time allows.
0	phy_txrequestclkhs	R/W	0	This bit controls the D-PHY PPI txrequestclkhs signal.

69.2.6.34. **PHY_TMR_LPCLK_CFG**

Table 6- 93 PHY_TMR_LPCLK_CFG

Bits	Name	R/W	Reset value	Description
31:26	Reserved			Reserved
25:16	phy_clkhs2lp_time	R/W	0	This field configures the maximum time that the D-PHY clock lane takes to go from high-speed to low-power transmission measured in lane byte clock cycles.
15:10	Reserved			Reserved
9:0	phy_clklp2hs_time	R/W	0	This field configures the maximum time that the D-PHY clock lane takes to go from low-power to high-speed transmission measured in lane byte clock cycles.

69.2.6.35. **PHY_TMR_CFG**

Table 6- 94 PHY_TMR_CFG

Bits	Name	R/W	Reset value	Description
31:26	Reserved			Reserved
25:16	phy_hs2lp_time	R/W	0	This field configures the maximum time that the D-PHY data lanes take to go from high-speed to low-power transmission measured in lane byte clock cycles.
15:10	Reserved			Reserved
9:0	phy_lp2hs_time	R/W	0	This field configures the maximum time that the D-PHY data lanes take to go from low-power to high-speed transmission measured in lane byte clock cycles.

69.2.6.36. **PHY_RSTZ**

Table 6- 95 PHY_RSTZ

Bits	Name	R/W	Reset value	Description

31:4	Reserved			Reserved
3	phy_forcepll	R/W	0	When the D-PHY is in ULPS, this bit enables the D-PHY PLL.
2	phy_enableclk	R/W	0	When set to 1, this bit enables the D-PHY Clock Lane Module.
1	phy_rstz	R/W	0	When set to 0, this bit places the digital section of the D-PHY in the reset state.
0	phy_shutdownz	R/W	0	When set to 0, this bit places the complete D-PHY macro in power-down state.

69.2.6.37. PHY_IF_CFG

Table 6- 96 PHY_IF_CFG

Bits	Name	R/W	Reset value	Description
31:16	Reserved			Reserved
15:8	phy_stop_wait_time	R/W	0	This field configures the minimum time PHY needs to stay in StopState before requesting an highspeed transmission
7:2	Reserved			Reserved
1:0	n_lanes	R/W	1	This field configures the number of active data lanes: 0x0 (ONELANES): lane 0 0x1 (TWOLANES): lanes 0 and 1

69.2.6.38. PHY_ULPS_CTRL

Table 6- 97 PHY_ULPS_CTRL

Bits	Name	R/W	Reset value	Description
31:4	Reserved			Reserved
3	phy_txexitulpslan	R/W	0	ULPS mode Exit on all active data lanes.
2	phy_txrequlpslan	R/W	0	ULPS mode Request on all active data lanes.
1	phy_txexitulpsclk	R/W	0	ULPS mode Exit on clock lane.
0	phy_txrequlpsclk	R/W	0	ULPS mode Request on clock lane.

69.2.6.39. PHY_TX_TRIGGERS

Table 6- 98 PHY_TX_TRIGGERS

Bits	Name	R/W	Reset value	Description
31:4	Reserved			Reserved
3:0	phy_tx_triggers	R/W	0	This field controls the trigger transmissions.

69.2.6.40. PHY_STATUS

Table 6- 99 PHY_STATUS

Bits	Name	R/W	Reset value	Description
31:9	Reserved			Reserved
8	phy_ulpsactivenot1lane	R		This bit indicates the status of ulpsactivenot1lane D-PHY signal.
7	phy_stopstate1lane	R		This bit indicates the status of phystopstate1lane D-PHY signal.
6	phy_rxulpsesc0lane	R		This bit indicates the status of rxulpsesc0lane D-PHY signal.
5	phy_ulpsactivenot0lane	R		This bit indicates the status of ulpsactivenot0lane D-PHY signal.
4	phy_stopstate0lane	R		This bit indicates the status of phystopstate0lane D-PHY signal.
3	phy_ulpsactivenotclk	R		This bit indicates the status of phyulpsactivenotclk D-PHY signal.
2	phy_stopstateclklane	R		This bit indicates the status of phystopstateclklane D-PHY signal.
1	phy_direction	R		This bit indicates the status of phydirection D-PHY signal.
0	phy_lock	R		This bit indicates the status of phylock D-PHY signal.

69.2.6.41. PHY_TST_CTRL0

Table 6- 100 PHY_TST_CTRL0

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1	phy_testclk	R/W	0	This bit is used to clock the TESTDIN bus into the D-PHY.
0	phy_testclr	R/W	0	PHY test interface clear (active high).

69.2.6.42. PHY_TST_CTRL1

Table 6- 101 PHY_TST_CTRL1

Bits	Name	R/W	Reset value	Description

31:17	Reserved			Reserved
16	phy_testen	R/W	0	PHY test interface operation selector: 0x1 (WRITEADDR): the address write operation is set on the falling edge of the testclk signal. 0x0 (WRITEDATA): the data write operation is set on the rising edge of the testclk signal.
15:8	phy_testdout	R	0	PHY output 8-bit data bus for read-back and internal probing functionalities.
7:0	phy_testdin	R/W	0	PHY test interface input 8-bit data bus for internal register programming and test functionalities access.

69.2.6.43. **INT_ST0**

Table 6- 102 INT_ST0

Bits	Name	R/W	Reset value	Description
31:21	Reserved			Reserved
20	dphy_errors_4	R	0	This bit indicates LP1 contention error ErrContentionLP1 from Lane 0.
19	dphy_errors_3	R	0	This bit indicates LP0 contention error ErrContentionLP0 from Lane 0.
18	dphy_errors_2	R	0	This bit indicates control error ErrControl from Lane 0.
17	dphy_errors_1	R	0	This bit indicates ErrSyncEsc low-power data transmission synchronization error from Lane 0.
16	dphy_errors_0	R	0	This bit indicates ErrEsc escape entry error from Lane 0.
15	ack_with_err_15	R	0	This bit retrieves the DSI protocol violation from the Acknowledge error report.
14	ack_with_err_14	R	0	This bit retrieves the reserved (specific to device) from the Acknowledge error report.
13	ack_with_err_13	R	0	This bit retrieves the invalid transmission length from the Acknowledge error report.
12	ack_with_err_12	R	0	This bit retrieves the DSI VC ID Invalid from the Acknowledge error report.
11	ack_with_err_11	R	0	This bit retrieves the not recognized DSI data type from the Acknowledge error report.
10	ack_with_err_10	R	0	This bit retrieves the checksum error (long packet only) from the Acknowledge error report.
9	ack_with_err_9	R	0	This bit retrieves the ECC error, multi-bit

				(detected, not corrected) from the Acknowledge error report.
8	ack_with_err_8	R	0	This bit retrieves the ECC error, single-bit (detected and corrected) from the Acknowledge error report.
7	ack_with_err_7	R	0	This bit retrieves the reserved (specific to device) from the acknowledge error report.
6	ack_with_err_6	R	0	This bit retrieves the False Control error from the Acknowledge error report.
5	ack_with_err_5	R	0	This bit retrieves the Peripheral Timeout error from the Acknowledge error report.
4	ack_with_err_4	R	0	This bit retrieves the LP Transmit Sync error from the Acknowledge error report.
3	ack_with_err_3	R	0	This bit retrieves the Escape Mode Entry Command error from the Acknowledge error report.
2	ack_with_err_2	R	0	This bit retrieves the EoT Sync error from the Acknowledge error report.
1	ack_with_err_1	R	0	This bit retrieves the SoT Sync error from the Acknowledge error report.
0	ack_with_err_0	R	0	This bit retrieves the SoT error from the Acknowledge error report.

69.2.6.44. INT_ST1

Table 6- 103 INT_ST1

Bits	Name	R/W	Reset value	Description
31:21	Reserved			Reserved
20	tear_request_err	R	0	This bit indicates that tear_request has occurred but tear effect is not active in DSI host and device.
19	dpi_buff_pld_under	R	0	This bit indicates that an underflow has occurred when reading payload to build DSI packet for video mode
18:13	Reserved			Reserved
12	gen_pld_recev_err	R	0	This bit indicates that during a generic interface packet read back, the payload FIFO becomes full and the received data is corrupted.
11	gen_pld_rd_err	R	0	This bit indicates that during a DCS read data, the payload FIFO becomes empty and the data sent to the interface is corrupted.
10	gen_pld_send_err	R	0	This bit indicates that during a Generic interface packet build, the payload FIFO

				becomes empty and corrupt data is sent.
9	gen_pld_wr_err	R	0	This bit indicates that the system tried to write a payload data through the Generic interface and the FIFO is full. Therefore, the payload is not written.
8	gen_cmd_wr_err	R	0	This bit indicates that the system tried to write a command through the Generic interface and the FIFO is full. Therefore, the command is not written.
7	dpi_pld_wr_err	R	0	This bit indicates that during a DPI pixel line storage, the payload FIFO becomes full and the data stored is corrupted.
6	eopt_err	R	0	This bit indicates that the EoTp packet has not been received at the end of the incoming peripheral transmission.
5	pkt_size_err	R	0	This bit indicates that the packet size error has been detected during the packet reception.
4	crc_err	R	0	This bit indicates that the CRC error has been detected in the received packet payload.
3	ecc_milti_err	R	0	This bit indicates that the ECC multiple error has been detected in a received packet.
2	ecc_single_err	R	0	This bit indicates that the ECC single error has been detected and corrected in a received packet.
1	to_lp_rx	R	0	This bit indicates that the low-power reception timeout counter reached the end and contention has been detected.
0	to_hs_tx	R	0	This bit indicates that the high-speed transmission timeout counter reached the end and contention has been detected.

69.2.6.45. **INT_MSK0**

Table 6- 104 INT_MSK0

Bits	Name	R/W	Reset value	Description
31:21	Reserved			Reserved
20	mask_dphy_errors_4	R/W	0	Mask for dphy_errors_4
19	mask_dphy_errors_3	R/W	0	Mask for dphy_errors_3
18	mask_dphy_errors_2	R/W	0	Mask for dphy_errors_2
17	mask_dphy_erro	R/W	0	Mask for dphy_errors_1

	rs_1			
16	mask_dphy_errors_0	R/W	0	Mask for dphy_errors_0
15	mask_ack_with_err_15	R/W	0	Mask for ack_with_err_15
14	mask_ack_with_err_14	R/W	0	Mask for ack_with_err_14
13	mask_ack_with_err_13	R/W	0	Mask for ack_with_err_13
12	mask_ack_with_err_12	R/W	0	Mask for ack_with_err_12
11	mask_ack_with_err_11	R/W	0	Mask for ack_with_err_11
10	mask_ack_with_err_10	R/W	0	Mask for ack_with_err_10
9	mask_ack_with_err_9	R/W	0	Mask for ack_with_err_9
8	mask_ack_with_err_8	R/W	0	Mask for ack_with_err_8
7	mask_ack_with_err_7	R/W	0	Mask for ack_with_err_7
6	mask_ack_with_err_6	R/W	0	Mask for ack_with_err_6
5	mask_ack_with_err_5	R/W	0	Mask for ack_with_err_5
4	mask_ack_with_err_4	R/W	0	Mask for ack_with_err_4
3	mask_ack_with_err_3	R/W	0	Mask for ack_with_err_3
2	mask_ack_with_err_2	R/W	0	Mask for ack_with_err_2
1	mask_ack_with_err_1	R/W	0	Mask for ack_with_err_1
0	mask_ack_with_err_0	R/W	0	Mask for ack_with_err_0

69.2.6.46. **INT_MSK1**

Table 6- 105 INT_MSK1

Bits	Name	R/W	Reset value	Description
31:21	Reserved			Reserved
20	mask_tear_requ	R/W	0	Mask for tear_request_err

	est_err			
19	mask_dpi_buff_pld_under	R/W	0	Mask for dpi_buff_pld_under
18:13	Reserved			Reserved
12	mask_gen_pld_recev_err	R/W	0	Mask for gen_pld_recev_err
11	mask_gen_pld_rd_err	R/W	0	Mask for gen_pld_rd_err
10	mask_gen_pld_send_err	R/W	0	Mask for gen_pld_send_err
9	mask_gen_pld_wr_err	R/W	0	Mask for gen_pld_wr_err
8	mask_gen_cmd_wr_err	R/W	0	Mask for gen_cmd_wr_err
7	mask_dpi_pld_wr_err	R/W	0	Mask for dpi_pld_wr_err
6	mask_eopt_err	R/W	0	Mask for eopt_err
5	mask_pkt_size_err	R/W	0	Mask for pkt_size_err
4	mask_crc_err	R/W	0	Mask for crc_err
3	mask_ecc_milti_err	R/W	0	Mask for ecc_milti_err
2	mask_ecc_single_err	R/W	0	Mask for ecc_single_err
1	mask_to_lp_rx	R/W	0	Mask for to_lp_rx
0	mask_to_hs_tx	R/W	0	Mask for to_hs_tx

69.2.6.47. **PHY_CAL**

Table 6- 106 PHY_CAL

Bits	Name	R/W	Reset value	Description
31:1	Reserved			Reserved
0	txskewcalhs	R/W	0	High-speed skew calibration is started when txskewcalhs is set high (assuming that PHY is in Stop state).

69.2.6.48. **INT_FORCE0**

Table 6- 107 INT_FORCE0

Bits	Name	R/W	Reset value	Description
31:21	Reserved			Reserved
20	force_dphy_erro	R/W	0	Force for dphy_errors_4

	rs_4			
19	force_dphy_errors_3	R/W	0	Force for dphy_errors_3
18	force_dphy_errors_2	R/W	0	Force for dphy_errors_2
17	force_dphy_errors_1	R/W	0	Force for dphy_errors_1
16	force_dphy_errors_0	R/W	0	Force for dphy_errors_0
15	force_ack_with_err_15	R/W	0	Force for ack_with_err_15
14	force_ack_with_err_14	R/W	0	Force for ack_with_err_14
13	force_ack_with_err_13	R/W	0	Force for ack_with_err_13
12	force_ack_with_err_12	R/W	0	Force for ack_with_err_12
11	force_ack_with_err_11	R/W	0	Force for ack_with_err_11
10	force_ack_with_err_10	R/W	0	Force for ack_with_err_10
9	force_ack_with_err_9	R/W	0	Force for ack_with_err_9
8	force_ack_with_err_8	R/W	0	Force for ack_with_err_8
7	force_ack_with_err_7	R/W	0	Force for ack_with_err_7
6	force_ack_with_err_6	R/W	0	Force for ack_with_err_6
5	force_ack_with_err_5	R/W	0	Force for ack_with_err_5
4	force_ack_with_err_4	R/W	0	Force for ack_with_err_4
3	force_ack_with_err_3	R/W	0	Force for ack_with_err_3
2	force_ack_with_err_2	R/W	0	Force for ack_with_err_2
1	force_ack_with_err_1	R/W	0	Force for ack_with_err_1
0	force_ack_with_err_0	R/W	0	Force for ack_with_err_0

69.2.6.49. INT_FORCE1

Table 6- 108 INT_FORCE1

Bits	Name	R/W	Reset value	Description
31:21	Reserved			Reserved
20	force_tear_request_err	R/W	0	Force for tear_request_err
19	force_dpi_buff_pld_under	R/W	0	Force for dpi_buff_pld_under
18:13	Reserved			Reserved
12	force_gen_pld_recev_err	R/W	0	Force for gen_pld_recev_err
11	force_gen_pld_rd_err	R/W	0	Force for gen_pld_rd_err
10	force_gen_pld_send_err	R/W	0	Force for gen_pld_send_err
9	force_gen_pld_wr_err	R/W	0	Force for gen_pld_wr_err
8	force_gen_cmd_wr_err	R/W	0	Force for gen_cmd_wr_err
7	force_dpi_pld_wr_err	R/W	0	Force for dpi_pld_wr_err
6	force_eopt_err	R/W	0	Force for eopt_err
5	force_pkt_size_err	R/W	0	Force for pkt_size_err
4	force_crc_err	R/W	0	Force for crc_err
3	force_ecc_milti_err	R/W	0	Force for ecc_milti_err
2	force_ecc_single_err	R/W	0	Force for ecc_single_err
1	force_to_lp_rx	R/W	0	Force for to_lp_rx
0	force_to_hs_tx	R/W	0	Force for to_hs_tx

69.2.6.50. PHY_TMR_RD_CFG

Table 6- 109 PHY_TMR_RD_CFG

Bits	Name	R/W	Reset value	Description
31:15	Reserved			Reserved
14:0	max_rd_time	R/W	0	This field configures the maximum time required to perform a read command in lane byte clock cycles. This register can only be

				modified when no read command is in progress.

69.2.6.51. VID_SHADOW_CTRL

Table 6- 110 VID_SHADOW_CTRL

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	vid_shadow_pin_req			When set to 1, the video request is done by external pin. In this mode vid_shadow_req is ignored.
15:9	Reserved			Reserved
8	vid_shadow_req			When set to 1, this bit request that the dpi registers from regbank are copied to the auxiliary registers. When the request is completed this bit is auto clear.
7:1	Reserved			Reserved
0	vid_shadow_en	R/W	0	When set to 1, DPI receives the active configuration from the auxiliary registers. When the feature is set at the same time with vid_shadow_req then the auxiliary registers are automatically updated.

69.2.6.52. DPI_VCID_ACT

Table 6- 111 DPI_VCID_ACT

Bits	Name	R/W	Reset value	Description
31:2	Reserved			Reserved
1:0	dpi_vcid	R	0	This field configures the DPI virtual channel id that is indexed to the Video mode packets.

69.2.6.53. DPI_COLOR_CODING_ACT

Table 6- 112 DPI_COLOR_CODING_ACT

Bits	Name	R/W	Reset value	Description
31:9	Reserved			Reserved
8	loosely18_en	R	0	When set to 1, this bit activates loosely packed variant to 18-bit configurations.
7:4	Reserved			Reserved
3:0	dpi_color_coding	R	0	This field configures the DPI color for Video Mode Command Mode coding as follows: 0x0 (CC00): 16-bit configuration 1

				0x1 (CC01): 16-bit configuration 2 0x2 (CC02): 16-bit configuration 3 0x3 (CC03): 18-bit configuration 1 0x4 (CC04): 18-bit configuration 2 0x5 (CC05): 24-bit 0x6 ~0xf: Reserved

69.2.6.54. DPI_LP_CMD_TIM_ACT

Table 6- 113 DPI_LP_CMD_TIM_ACT

Bits	Name	R/W	Reset value	Description
31:24	Reserved			Reserved
23:16	outvact_lpcmd_time	R	0	This field is used for the transmission of commands in low-power mode. It defines the size, in bytes, of the largest packet that can fit in a line during the VSA, VBP, and VFP regions.
15:8	Reserved			Reserved
7:0	invact_lpcmd_time	R	0	This field is used for the transmission of commands in low-power mode. It defines the size, in bytes, of the largest packet that can fit in a line during the VACT region.

69.2.6.55. VID_MODE_CFG_ACT

Table 6- 114 VID_MODE_CFG_ACT

Bits	Name	R/W	Reset value	Description
31:25	Reserved			Reserved
24	vpg_orientation	R	0	This field indicates the color bar orientation as follows: 0 (VPGORENT0): Vertical mode 1 (VPGORENT1): Horizontal mode
23:21	Reserved			Reserved
20	vpg_mode	R	0	This field is to select the pattern: 0 (COLORBAR): horizontal or vertical 1 (BERPATTERN): vertical only
19:17	Reserved			Reserved
16	vpg_en	R	0	When set to 1, this bit enables the video mode pattern generator.
15	lp_cmd_en	R	0	When set to 1, this bit enables the command transmission only in low-power mode.
14	frame_bta_ack_	R	0	When set to 1, this bit enables the request for an

	en			acknowledge response at the end of a frame.
13	lp_hfp_en	R	0	When set to 1, this bit enables the return to low-power inside the HFP period when timing allows
12	lp_hbp_en	R	0	When set to 1, this bit enables the return to low-power inside the HBP period when timing allows.
11	lp_vact_en	R	0	When set to 1, this bit enables the return to low-power inside the VACT period when timing allows.
10	lp_vfp_en	R	0	When set to 1, this bit enables the return to low-power inside the VFP period when timing allows.
9	lp_vbp_en	R	0	When set to 1, this bit enables the return to low-power inside the VBP period when timing allows.
8	lp_vsa_en	R	0	When set to 1, this bit enables the return to low-power inside the VSA period when timing allows.
7:2	Reserved			Reserved
1:0	vid_mode_type	R	0	This field indicates the video mode transmission type as follows:Values: 0x0 (VIDMODE0): Non-burst with sync pulses 0x1 (VIDMODE1): Non-burst with sync events 0x2 (VIDMODE2): Burst mode 0x3 (VIDMODE3): Burst mode

69.2.6.56. **VID_PKT_SIZE_ACT**

Table 6- 115 VID_PKT_SIZE_ACT

Bits	Name	R/W	Reset value	Description
31:14	Reserved			Reserved
13:0	vid_pkt_size	R	0	This field configures the number of pixels in a single video packet.

69.2.6.57. **VID_NUM_CHUNKS_ACT**

Table 6- 116 VID_NUM_CHUNKS_ACT

Bits	Name	R/W	Reset value	Description
31:13	Reserved			Reserved
12:0	vid_num_chunks	R	0	This register configures the number of chunks to be transmitted during a Line period (a chunk

				is pair made of a video packet and a null packet). If set to 0 or 1, video line is still transmitted in a single packet. If set to 1 that packet is part of a chunk, meaning that a null packet follows it (if vid_null_size>0). Otherwise, multiple chunks are used to transmit each video line.

69.2.6.58. VID_NULL_SIZE_ACT

Table 6- 117 VID_NULL_SIZE_ACT

Bits	Name	R/W	Reset value	Description
31:13	Reserved			Reserved
12:0	vid_null_size	R	0	This register configures the number of bytes inside a null packet. Setting to 0 disables null packets.

69.2.6.59. VID_HSA_SIZE_ACT

Table 6- 118 VID_HSA_SIZE_ACT

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	vid_hsa_time	R	0	This field configures the Horizontal Synchronism Active period in lane byte clock cycles.

69.2.6.60. VID_HBP_SIZE_ACT

Table 6- 119 VID_HBP_SIZE_ACT

Bits	Name	R/W	Reset value	Description
31:12	Reserved			Reserved
11:0	vid_hbp_time	R	0	This field configures the Horizontal Back Porch period in lane byte clock cycles.

69.2.6.61. VID_HLINE_TIME_ACT

Table 6- 120 VID_HLINE_TIME_ACT

Bits	Name	R/W	Reset value	Description
31:15	Reserved			Reserved

14:0	vid_hline_time	R	0	This field configures the size of the total line time (HSA+HBP+HACT+HFP) counted in lane byte clock cycles.

69.2.6.62. **VID_VSA_LINES_ACT**

Table 6- 121 VID_VSA_LINES_ACT

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	vsa_lines	R	0	This field configures the Vertical Synchronism Active period measured in number of horizontal lines.

69.2.6.63. **VID_VBP_LINES_ACT**

Table 6- 122 VID_VBP_LINES_ACT

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	vbp_lines	R	0	This field configures the Vertical Back Porch period measured in number of horizontal lines.

69.2.6.64. **VID_VFP_LINES_ACT**

Table 6- 123 VID_VFP_LINES_ACT

Bits	Name	R/W	Reset value	Description
31:10	Reserved			Reserved
9:0	vfp_lines	R	0	This field configures the Vertical Front Porch period measured in number of horizontal lines.

69.2.6.65. **VID_VACTIVE_LINES_ACT**

Table 6- 124 VID_VACTIVE_LINES_ACT

Bits	Name	R/W	Reset value	Description
31:14	Reserved			Reserved
13:0	v_active_lines	R	0	This field configures the Vertical Active period measured in number of horizontal lines.

69.2.6.66. VID_PKT_STATUS

Table 6- 125 VID_PKT_STATUS

Bits	Name	R/W	Reset value	Description
31:18	Reserved			Reserved
17	dpi_buff_pld_full	R	0	This bit indicates the full status of the payload internal buffer for video Mode. This bit is set to 0 for command Mode.
16	dpi_buff_pld_empty	R	1	This bit indicates the empty status of the payload internal buffer for video Mode. This bit is set to 0 for command Mode.
15:4	Reserved			Reserved
3	dpi_pld_w_full	R	0	This bit indicates the full status of write payload FIFO for video Mode. This bit is set to 0 for command Mode.
2	dpi_pld_w_empty	R	1	This bit indicates the empty status of write payload FIFO for video Mode. This bit is set to 0 for command Mode.
1	dpi_cmd_w_full	R	0	This bit indicates the full status of write command FIFO for video Mode. This bit is set to 0 for command Mode.
0	dpi_cmd_w_empty	R	1	This bit indicates the empty status of write command FIFO for video Mode. This bit is set to 0 for command Mode.

69.2.6.67. SDF_3D_ACT

Table 6- 126 SDF_3D_ACT

Bits	Name	R/W	Reset value	Description
31:17	Reserved			Reserved
16	send_3d_cfg	R	0	When set, causes the next VSS packet to include 3D control payload in every VSS packet.
15:6	Reserved			Reserved
5	right_first	R	0	This bit defines the left/right order: 0x0 (LFIRST): left eye is sent first, then right eye 0x1 (RFIRST): right eye data is sent first, then left eye
4	second_vsync	R	0	This field defines whether there is a second VSYNC pulse between Left and Right Images, when 3D Image Format is Frame-based:

				0x0 (NOSYNC): No sync pulses between left and right data 0x1 (SYNC): Sync pulseHSYNC, VSYNC, blanking) between left and right data
3:2	format_3d	R	0	This field defines 3D Image Format: 0x0 (LINE): Alternating lines of left and right data 0x1 (FRAME): Alternating frames of left and right data 0x2 (PIXEL): Alternating pixels of left and right data 0x3 (RESERVED): Reserved, not used
1:0	mode_3d	R	0	This field defines 3D Mode On/Off and Display Orientation: 0x0 (MODE0): 3D Mode Off , 2D Mode On 0x1 (MODE1): 3D Mode On, Portrait Orientation 0x2 (MODE2): 3D Mode On, Landscape Orientation 0x3 (MODE3): Reserved, not used

69.3. Video In Controller

69.3.1. Overview

Video_in is the system of video input, including MIPI interface, DVP interface.

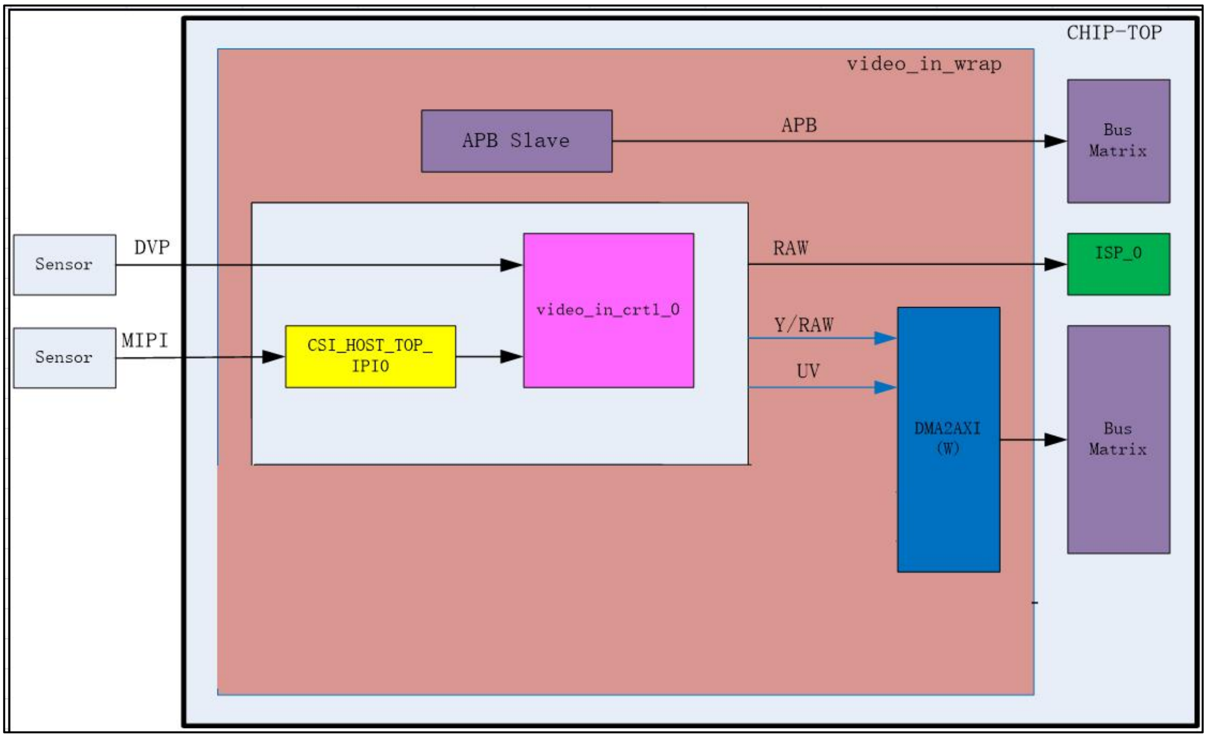


Figure 6- 2 Video_in module program

69.3.2. Features

Video In has the following features:

- Support one dual lane MIPI @ 900MHz bps
- Support one DVP interface for video in
- Support 8/10/12bit RGB Bayer DC for timing video input
- Support BT601、BT.656、BT.1120 video input interface
- Compliant with Mainstream CMOS sensor , such as SONY, OmniVision, SmartSensTech, GalaxyCore
- Compliant with multiple sensor with parallel/differential electrical characters
- Provide programmable sensor clock out
- Max 2M resolution ratio for one lane

69.3.3. Function Description

69.3.3.1. Function Overview

The VIDEO_IN system includes two subsystems. The two subsystems are DVP system and MIPI system.

DVP system supports most common standard and nonstandard video input formats. The input video format includes YUV422 8bit color depth, RAW8, RAW10, RAW12. Synchronization

attributes refer to how the horizontal and vertical sync signals are
configured. The DVP system supports both separate synchronization and embedded sync code synchronization. Separate synchronization involves placing the horizontal sync, vertical sync, and data enable signals on separate input pins. The embedded sync code synchronization standard is defined in ITU-R BT.656 and ITU-R BT1120.
The MIPI system includes these modules.

69.3.4. Programming Guide

1. Set CTRL reg
2. Set PIXEL_FORMAT and related registers
3. Set LINE_CTRL and INTERRUPT
4. SET dma related reg

69.3.5. Register Summary

Table 6- 127 Video_in register summary

Name	Address Offset	R/W	Description
HREG00	0x0	RW	Control register
HREG01	0x4	RW	Pixel format control
HREG02	0x8	RW	Set hsync and vsync
HREG03	0xC	RC	Set v_size and h_size total
HREG04	0x10	RW	Set v_size and h_size active
HREG05	0x14	RW	Set v_start and h_start
HREG06	0x18	RW	Interrupt signal
HREG10	0x40	RW	Ctrl_frame base address0 y
HREG11	0X44	RW	Ctrl_frame base address1 y
HREG12	0X48	RW	Ctrl_frame base address0 uv
HREG13	0X4C	RW	Ctrl_frame base address1 uv
HREG14	0X50	RW	Line stride
HREG18	0X60	RW	DMA arbiter mode
HREG19	0X64	RW	DMA weight
HREG1A	0X68	RW	DMA priority
HREG1B	0X6C	RW	DMA ID

69.3.6. Register Description

69.3.6.1. HREG00

Table 6- 128 HREG00

Bit Field	Name	Width	R/W	Default value	Description
0	dma_soft_reset	1	RW	0x1	dma/axi soft reset
1	pixel_clk_in	1	RW	0x1	module in pixel clock domain gate
3:2	reserved_3_2	2	R	0x3	
4	video_in	1	RW	0x1	Video_/In module configuration done
5	auto_flush	1	RW	0x1	Auto flush DMA fifo at each frame start
7:6	reserved_7_6	2	R	0x3	
8	dma_y_enable	1	RW	0x0	dma Y/AXI module enable
9	dma_uv_enable	1	RW	0x0	dma UV module enable
10	win_enable	1	RW	0x0	win module enable(blank pixel remove)
11	timing_enable	1	RW	0x0	timing module enable(sync code decode,emb)
12	sync_output_enable	1	RW	0x0	vsync/hsync output enable(RGB to ISP)
13	hsync_mask_enable	1	RW	0x0	hsync_mask enable
14	axi_8bit_swap	1	RW	0x0	axi write 8 bit swap
15	reserved_15	1	RW	0x0	1: 1 bit swap (also low of rawdata[x:0] first push into fifo, use in ISP mode); 0: 8 bit swap
16	mipi_clk_pol	1	RW	0x1	sensor/mipi clock polarity, 0:active low; 1: active high
17	mipi_vsync_pol	1	RW	0x1	sensor/mipi vsync polarity, 0:active low; 1: active high

18	mipi_hsync_pol	1	RW	0x1	sensor/mipi hsync polarity, 0:active low; 1: active high
19	mipi_field_pol	1	RW	0x1	sensor/mipi field polarity, 0:active low; 1: active high
20	ISP_clk_pol	1	RW	0x1	ISP clock polarity, 0:active low; 1: active high
21	ISP_vsync_pol	1	RW	0x1	ISP vsync polarity, 0:active low; 1: active high
22	ISP_hsync_pol	1	RW	0x1	ISP hsync polarity, 0:active low; 1: active hight
23	ISP_field_pol	1	RW	0x1	ISP field polarity, 0:active low; 1: active high
24	sync_select	1	RW	0x0	sync select; 0:external sync ; 1:internal sync
25	input_chnl	1	RW	0x0	input channel; 0: parallel; 1:mipi
26	chnl_num	1	RW	0x0	channel number 1x/2x; 0:BT656 or 2 clocks for pixel(DVP422in should choose this); 1: BT1120,or 1 clock for one pixel
27	emb_enable	1	RW	0x0	1'b0: emb disable; 1'b1: emb enable
28	emb_mode	1	RW	0x0	emb mode; 1'b0:8bit 656 ; 1'b1:full range 656
29	emb_ecc_enable	1	RW	0x0	1'b0: emb ecc disable; 1'b1:emb ecc enable
31:30	embed_position	2	RW	0x0	embed position; 2'b00: lower; 2'b01:upper; 2'b10:both

69.3.6.2. **HREG01**

Table 6- 129 HREG01

Bit Field	Name	Width	R/W	Default value	Description
0	sensor_input_format	1	RW	0x0	1'b0:RGB; 1'b1:YUV
1	YUV_format_input	1	RW	0x0	1'b0:YUV422; 1'b1:YUV420
2	YUV_format_output	1	RW	0x0	1'b0:YUV422; 1'b1:YUV420
4:3	pixel_width	2	RW	0x0	2'b00:8bit; 2'b01:10bit; 2'b10:12bit;2'b11:14bit
7:5	pixel_form	3	RW	0x0	[7:5]:YUV444/REG888

	at				3`b000:YUVYUV/GBRGBR  3`b001:YVUYVU/GRBGRB 3`b010:UYVUYV/BGRBGR  3`b011:VYUVYU/RGBRGB 3`b100:UVYUVY/BRGBRG  3`b101:VUYVUY/RBGRBG 3`b111:UYVUYV/BGRBGR other: not allowed [7:5]:YUV422 3`b000:YUYV 3`b001:YVYU 3`b010:UYVY 3`b011:VYUY 3`b111:UYVY other: not allowed [7:5]:YUV420 3`b000:YYYY/YUYV 3`b001:YYYY/YVYU 3`b010:YYYY/UYVY 3`b011:YYYY/VYUY 3`b100:YUYV/YYYY 3`b101:YVYU/YYYY 3`b110:UYVY/YYYY 3`b111:VYUY/YYYY 3`b111:YYYY/UYVY other: not allowed
9:8	dvp_remap	2	RW	0x0	input remap: 00:sensor[11:0],4'b0 01:sensor[9:0],6'b0 10:sensor[7:0],8'b0 11:sensor[11:6],2'b0,sensor[5:0],2'b0
11:10	reserved	2	RW	0x0	
14:12	win_ctrl	3	RW	0x0	000:vertical sync/horizontal valid 001:vertical sync/horizontal

					sync 010:vertical valid/horizontal valid 011:vertical valid/horizontal sync 100:vertical sync/data enable 101:vertical valid/data enable 110:vertical valid 111:vertical sync
31:15	reserved	17	R	0x0	

69.3.6.3. **HREG02**

Table 6- 130 HREG02

Bit Field	Name	Width	R/W	Default value	Description
13:0	vsync_width	14	RW	0x0	vsync width,(actual-1)
15:14	reserved_15_14	2	R	0x0	
29:16	hsync_width	14	RW	0x0	hsync width,(actual-1)
31:30	reserved_15_14	2	R	0x0	reset output

69.3.6.4. **HREG03**

Table 6- 131 HREG03

Bit Field	Name	Width	R/W	Default value	Description
13:0	vsync_total	14	RW	0x0	vsync total size,(actual-1)
15:14	reserved_15_14	2	R	0x0	
29:16	hsync_total	14	RW	0x0	hsync total size,(actual-1)
31:30	reserved_31_30	2	R	0x0	

69.3.6.5. HREG04

Table 6- 132 HREG04

Bit Field	Name	Width	R/W	Default value	Description
13:0	vsync_active	14	RW	0x0	active vsync size,(actual-1)
15:14	reserved_15_14	2	R	0x0	
29:16	hsync_active	14	RW	0x0	active hsync size,(actual-1)
31:30	reserved_31_30	2	R	0x0	

69.3.6.6. HREG05

Table 6- 133 HREG05

Bit Field	Name	Width	R/W	Default value	Description
13:0	vsync_start	14	RW	0x0	active vsync start position,(actual-1)
15:14	reserved_15_14	2	R	0x0	
29:16	hsync_start	14	RW	0x0	active hsync start position,(actual-1)
31:30	reserved_31_30	2	R	0x0	

69.3.6.7. HREG06

Table 6- 134 HREG06

R	Name	Width	R/W	Default value	Description
1:0	int_sts_clr	2	RW	0x0	Interrupt status, write "1" to clear bit1: dma fifo overflow bit0: dma done
7:6	reserved	6	R	0x0	
15:8	int_mask	8	RW	0x0	Interrupt mask, 1:mask;0:enable
15:10	reserved	6	R	0x0	

16	int_pol	1	RW	0x0	Interrupt polarity, 1:high active;0:low active
31:17	reserved	15	R	0x0	

69.3.6.8. HREG10

Table 6- 135 HREG10

Bit Field	Name	Width	R/W	Default value	Description
31:0	ctrl_frame_base_addr0_y	32	RW	0x0	frame base address Y buffer0 , byte unit, at least 8pixel align

69.3.6.9. HREG11

Table 6- 136 HREG11

Bit Field	Name	Width	R/W	Default value	Description
31:0	ctrl_frame_base_addr1_y	32	RW	0x0	frame base address Y buffer1 , byte unit, at least 8pixel align

69.3.6.10. HREG12

Table 6- 137 HREG12

Bit Field	Name	Width	R/W	Default value	Description
31:0	ctrl_frame_base_addr0_uv	32	RW	0x0	frame base address UV buffer0 , byte unit, at least 8pixel align

69.3.6.11. HREG13

Table 6- 138 HREG13

Bit Field	Name	Width	R/W	Default value	Description
31:0	ctrl_frame_base_addr1_uv	32	RW	0x0	frame base address UV buffer1 , byte unit, at least 8pixel align

69.3.6.12. **HREG14**

Table 6- 139 HREG14

Bit Field	Name	Width	R/W	Default value	Description
12:0	ctrl_hstride_y	13	RW	0x0	line stride Y,byte unit, at least 8pixel align,=width*Ybits per pix/64
15:13	reserved_15_13	3	R	0x0	Reserved
28:16	ctrl_hstride_uv	13	RW	0x0	line stride UV,byte unit, at least 8pixel align
31:29	reserved_31:29	3	R	0x0	Reserved

69.3.6.13. **HREG18**

Table 6- 140 HREG18

Bit Field	Name	Width	R/W	Default value	Description
1:0	dma_arb_mode	2	RW	0x0	read channel arbiter mode. 2`b0:priority configured by cpu 2`b1:weight+round-robin
31:2	reserved	30	R	0x0	

69.3.6.14. **HREG19**

Table 6- 141 HREG19

Bit Field	Name	Width	R/W	Default value	Description
7:0	chnl_0_weight	8	RW	0x1	Chanel 0 write weight
15:8	chnl_1_weight	8	RW	0x0	Chanel 1 write weight
23:16	chnl_2_weight	8	RW	0x1	Chanel 2 write weight
31:24	chnl_3_weight	8	RW	0x0	Chanel 3 write weight

69.3.6.15. HREG1A

Table 6- 142 HREG1A

Bit Field	Name	Width	R/W	Default value	Description
3:0	chnl_0_priority	4	RW	0x7	Chanel 0 priority
7:4	chnl_1_priority	4	RW	0x6	Chanel 1 priority
11:8	chnl_2_priority	4	RW	0x5	Chanel 2 priority
15:12	chnl_3_priority	4	RW	0x4	Chanel 3 priority
19:16	chnl_4_priority	4	RW	0x3	Chanel 4 priority
23:20	chnl_5_priority	4	RW	0x2	Chanel 5 priority
27:24	chnl_6_priority	4	RW	0x1	Chanel 6 priority
31:28	chnl_7_priority	4	RW	0x0	Chanel 7 priority

69.3.6.16. HREG1B

Table 6- 143 HREG1B

Bit Field	Name	Width	R/W	Default value	Description
3:0	chnl_0_ID	4	RW	0x0	Chanel 0 id
7:4	chnl_1_ID	4	RW	0x1	Chanel 1 id
11:8	chnl_2_ID	4	RW	0x2	Chanel 2 id
15:12	chnl_3_ID	4	RW	0x3	Chanel 3 id
19:16	chnl_4_ID	4	RW	0x4	Chanel 4 id
23:20	chnl_5_ID	4	RW	0x5	Chanel 5 id
27:24	chnl_6_ID	4	RW	0x6	Chanel 6 id
31:28	chnl_7_ID	4	RW	0x7	Chanel 7 id

69.4. ISP

69.4.1. Overview

The ISP processor is a high-performance image signal processor. The typical application scenario is to process 1296P30 BAYER image stream @148.5MHz in a single pixel per clock pipeline. The ISP processor is intended for surveillance cameras, smart home cameras, AI-based entrance guard cameras and some automotive related camera products.

69.4.2. Features

- 2:1 DOL/stagger HDR.
- Raw data linearization.
- Defect pixel correction.
- Green equalization.
- Chroma aberration.
- CFA denoising.
- Luma denoising.
- Chroma denoising and suppression.
- 3D denoising.
- Local tone mapping.
- Bayer CFA demosaic.
- Color matrix correction.
- De-purple fringe.
- Gamma correction.
- Contrast enhancement.
- Edge sharpening.
- RAW and YUV scaling.
- Horizontal distortion.
- Special digital effect.
- Multi-stream in and multi-stream out.
- Zero shuttle lag.
- Super resolution image.
- AE/AWB/AF.

69.4.3. Function Description

The following figure shows the component blocks of the ISP processor.

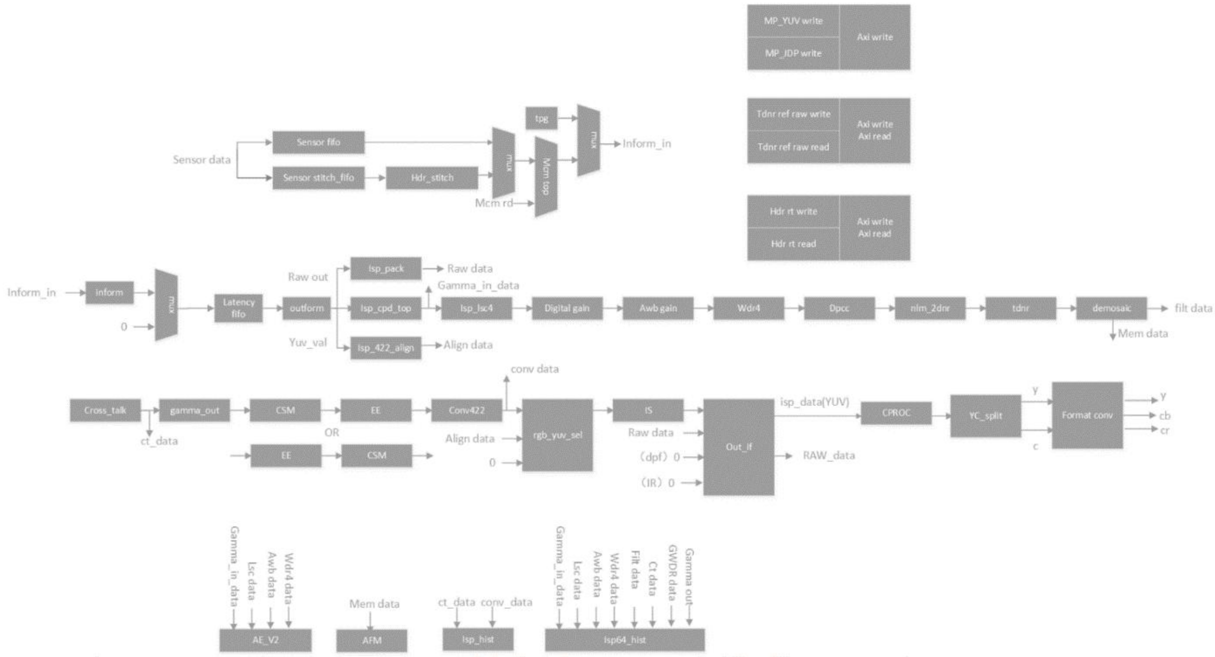


Figure 6- 2 component blocks of the ISP

The entire ISP pipe is simply divided into three domains. Bayer CFA raw data is processed in the RAW domain, red/green/blue data after the de-mosaic block is processed in the RGB domain, and Y/U/V data after the color space convertor block is processed in the YUV domain.

69.4.3.1. **ISP Architecture**

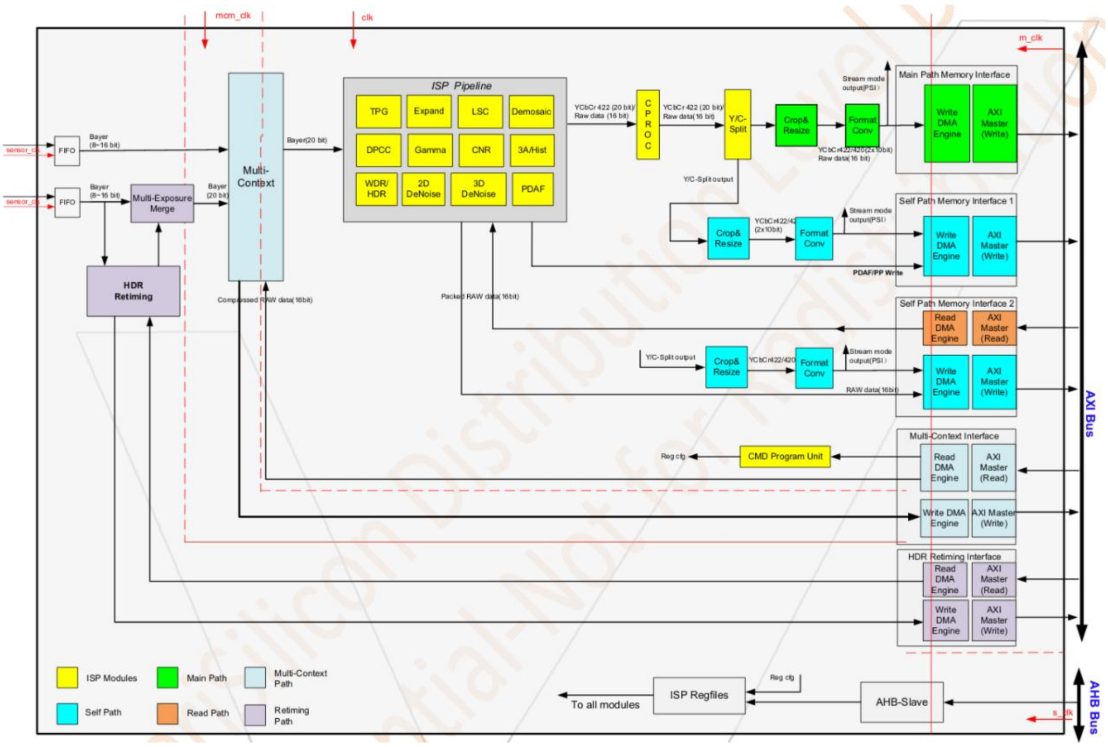


Figure 6- 3 ISP architecture

Abbreviation	Meaning
VIN	Video Input
TPG	Test Pattern Generation
DPCC	Defect Pixel Correction
CAC	Chroma Aberration Correction
BLC	Black Level Correction
LSC	Lens Shading
CCM	Color Correction Matrix
CSC	Color Space Conversion
WDR/HDR	Wide Dynamic Range/ High Dynamic Range
3A	Auto Exposure/ Auto White Balance/ Auto Focus

69.5. VPU

69.5.1. Overview

VPU (Video Process Unit) is hardware supporting encoder/decoder, which is compliant with H264/MJPEG video standard. VPU has the features like low CPU usage, low bandwidth usage, low latency, low pow consumption and so on.

69.5.2. Features

VPU support the following features:

- H.264 encoder: BP/MP/HP， Main Profile
- H.264 decoder
- JPEG encoder and decoder
- Real time code ability:4K@15fps/1080P@60fps
- Real time decode ability:1080P@60fps

69.5.3. Function Description

69.5.3.1. Function Overview

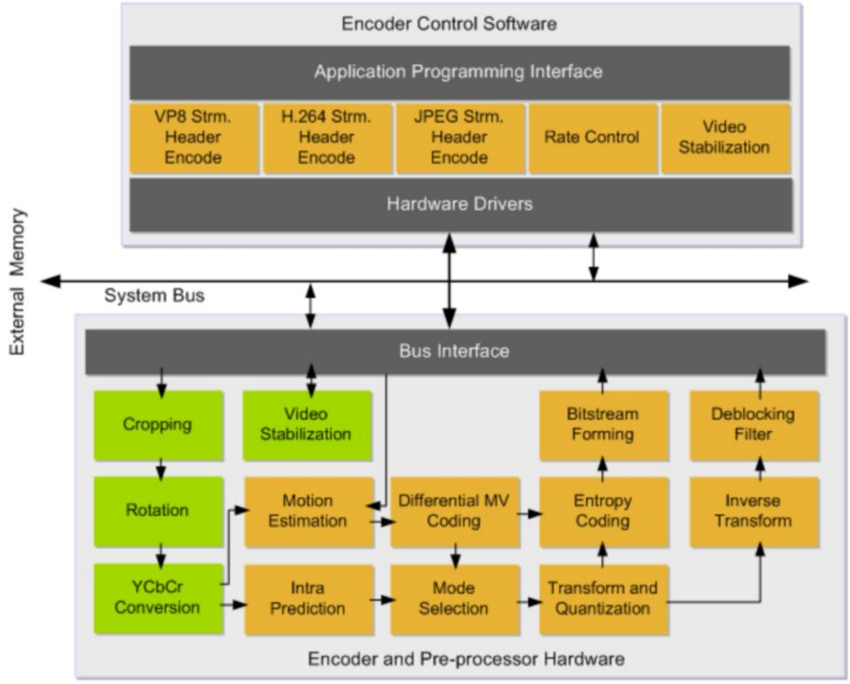


Figure 6- 3 VPU Encoder application

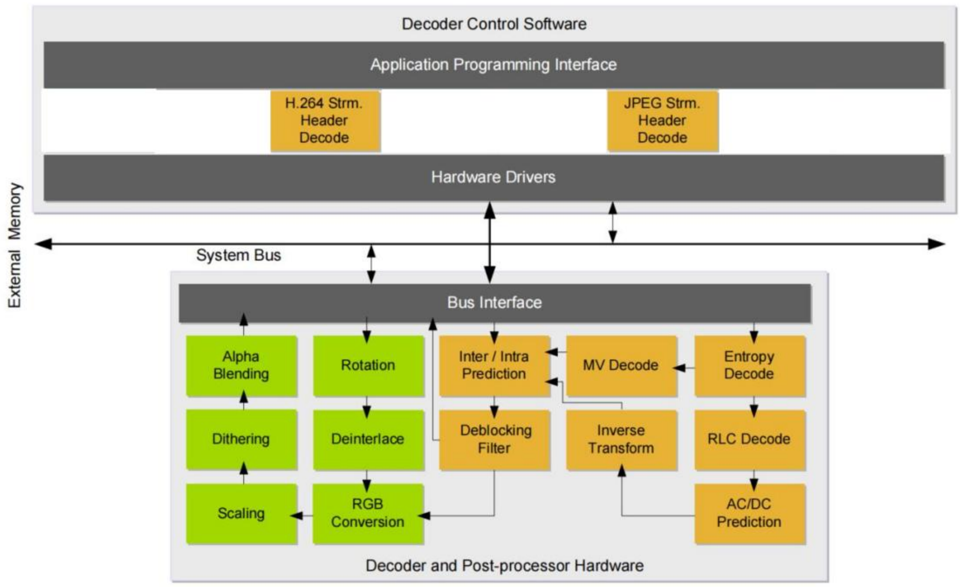


Figure 6- 4 VPU Decoder application

69.5.4. Register Summary

69.5.4.1. Encoder

Register	Address	Description	Default	R/W

name	Offset		value	
swreg0	0x00	ID register (read only)	0x8290 0760	R
swreg1	0x04	Interrupt register encoder	0	R/W
swreg2	0x08	Device configuration register	0	R/W
swreg3	0x0C	Test control register	0	R/W
swreg4	0x10	Reserved	0	R
swreg5	0x14	Base address for output stream data	0	R/W
swreg6	0x18	Base address for output control data	0	R/W
swreg7	0x1C	Base address for reference luma	0	R/W
swreg8	0x20	Base address for reference chroma	0	R/W
swreg9	0x24	Base address for reconstructed luma	0	R/W
swreg10	0x28	Base address for reconstructed chroma	0	R/W
swreg11	0x2C	Base address for input picture luma	0	R/W
swreg12	0x30	Base address for input picture cb	0	R/W
swreg13	0x34	Base address for input picture cr	0	R/W
swreg14	0x38	Encoder control register 0	0	R/W
swreg15	0x3C	Encoder control register 1	0	R/W
swreg16	0x40	H.264 contrl register 0	0	R/W
swreg17	0x44	H.264 contrl register 1	0	R/W
swreg18	0x48	H.264 contrl register 2	0	R/W
swreg19	0x4C	H.264 contrl register 3	0	R/W
swreg20	0x50	JPEG contrl register 0	0	R/W
swreg21	0x54	H.264 contrl register 4	0	R/W
swreg22	0x58	Stream header remainder bits MSB (MSB aligned)	0	R/W
swreg23	0x5C	Stream header remainder bits LSB (MSB aligned)	0	R/W
swreg24	0x60	Stream buffer limit (64bit addresses) output stream size (bits)	0	R/W
swreg25	0x64	MAD adjustment/QP Sum output div2	0	R/W
swreg26	0x68	Intra slice bitmap for slices 64..95	0	R/W
swreg27	0x6C	QP control/Checkpoint distance	0	R/W
swreg28	0x70	Checkpoint 1/2 word target/usage div32	0	R/W
swreg29	0x74	Checkpoint 3/4 word target/usage div32	0	R/W
swreg30	0x78	Checkpoint 5/6 word target/usage div32	0	R/W
swreg31	0x7C	Checkpoint 7/8 word target/usage div32	0	R/W
swreg32	0x80	Checkpoint 9/10 word target/usage div32	0	R/W
swreg33	0x84	Checkpoint word error 1/2 div4	0	R/W
swreg34	0x88	Checkpoint word error 3/4 div4	0	R/W
swreg35	0x8C	Checkpoint word error 5/6 div4	0	R/W
swreg36	0x90	Checkpoint delta QP	0	R/W
swreg37	0x94	Stream start offset/RLC codeword count div4	0	R/W
swreg38	0x98	Macroblock count output/MAD undercount output	0	R/W
swreg39	0x9C	Base address for next pic luminance	0	R/W
swreg40	0xA0	Stabilization contrl	0	R/W

swreg41	0xA4	Stabilization motion sum	0	R
swreg42	0xA8	Stabilization matrix output	0	R
swreg43	0xAC	Stabilization matrix output	0	R
swreg44	0xB0	Stabilization matrix output	0	R
swreg45	0xB4	Stabilization matrix output	0	R
swreg46	0xB8	Stabilization matrix output	0	R
swreg47	0xBC	Stabilization matrix output	0	R
swreg48	0xC0	Stabilization matrix output	0	R
swreg49	0xC4	Stabilization matrix output	0	R
swreg50	0xC8	Stabilization matrix output	0	R
swreg51	0xCC	Base address for cabac context tables	0	R/W
swreg52	0xD0	Base address for MV output writing	0	R/W
swreg53	0xD4	Reserved	0	R/W
swreg54	0xD8	Reserved	0	R/W
swreg55	0xDC	Reserved	0	R/W
swreg56	0xE0	Intra area control	0	R/W
swreg57	0xE4	CIR first intra mb/ CIR intra mb interval	0	R/W
swreg58	0xE8	Intra slice bitmap for slices 0..31	0	R/W
swreg59	0xEC	Intra slice bitmap for slices 32..63	0	R/W
swreg60	0xF0	1st ROI area	0	R/W
swreg61	0xF4	2nd ROI area	0	R/W
swreg62	0xF8	MVC control/ ROI delta QP	0	R/W
swreg63	0xFC	Synthesis configuration register (read only)	NA	R
swreg64 ~swreg79	0x100~0x13C	JPEG luma quantization table	0	R/W
swreg80 ~swreg95	0x140~0x17C	JPEG chroma quantization table	0	R/W

69.5.4.2. **Decoder**

Register name	Address Offset	Desctription	Default value	R/W
swreg0	0x00	ID register (read only)	0x67312398	R
swreg1	0x04	Interrupt register	0	R/W
swreg2	0x08	Device configuration register	0x400	R/W
swreg3	0x0C	Decoder control register 0 (decmode,picture type etc)	0	R/W
swreg4	0x10	Decoder control register 1 (picture parameters)	0	R/W
swreg5	0x14	Decoder control register 2 (stream decoding table selects)	0	R/W
swreg6	0x18	Decoder control register 3 (stream buffer information)	0	R/W
swreg7	0x1C	Decoder control register 4 (H264/progressive JPEG control)	0	R/W

swreg8	0x20	Decoder control register 5(H264/progressive JPEG control)	0	R/W
swreg9	0x24	Decoder control register 6	0	R/W
swreg10	0x28	H264 P initial fwd ref pic list register (4-9)	0	R/W
swreg11	0x2C	Base address for H.264 intra prediction 4x4 H264 P initial fwd ref pic list register (10-15)	0	R/W
swreg12	0x30	Reserved	0	R/W
swreg13	0x34	Base address for decoded picture Base address for JPEG decoder output Y	0	R/W
swreg14	0x38	Base address for reference picture index 0 Base address for JPEG decoder output UV	0	R/W
swreg15	0x3C	Base address for reference picture index 1 JPEG control	0	R/W
swreg16	0x40	Base address for reference picture index 2 List of VLC code lengths in first JPEG AC table	0	R/W
swreg17	0x44	Base address for reference picture index 3 List of VLC code lengths in first JPEG AC table	0	R/W
swreg18	0x48	Base address for reference picture index 4 List of VLC code lengths in first JPEG AC table	0	R/W
swreg19	0x4C	Base address for reference picture index 5 List of VLC code lengths in first/second JPEG AC table	0	R/W
swreg20	0x50	Base address for reference picture index 6 List of VLC code lengths in second JPEG AC table	0	R/W
swreg21	0x54	Base address for reference picture index 7 List of VLC code lengths in second JPEG AC table	0	R/W
swreg22	0x58	Base address for reference picture index 8 List of VLC code lengths in second JPEG AC table	0	R/W
swreg23	0x5C	Base address for reference picture index 9 List of VLC code lengths in first JPEG DC table	0	R/W
swreg24	0x60	Base address for reference picture index 10 List of VLC code lengths in first JPEG DC table	0	R/W
swreg25	0x64	Base address for reference picture index 11 List of VLC code lengths in second JPEG DC table	0	R/W
swreg26	0x68	Base address for reference picture index 12 List of VLC code lengths in second JPEG DC table	0	R/W
swreg27	0x6C	Base address for reference picture index 13 Progressive JPEG DC table	0	R/W
swreg28	0x70	Base address for reference picture index 14 Progressive JPEG DC table	0	R/W
swreg29	0x74	Base address for reference picture index 15	0	R/W
swreg30	0x78	Reference picture numbers for index 0 and 1 (H264 VLC)	0	R/W
swreg31	0x7C	Reference picture numbers for index 2 and 3 (H264 VLC)	0	R/W
swreg32	0x80	Reference picture numbers for index 4 and 5 (H264 VLC)	0	R/W
swreg33	0x84	Reference picture numbers for index 6 and 7 (H264 VLC)	0	R/W
swreg34	0x88	Reference picture numbers for index 8 and 9 (H264 VLC)	0	R/W

swreg35	0x8C	Reference picture numbers for index 10 and 11 (H264 VLC)	0	R/W
swreg36	0x90	Reference picture numbers for index 12 and 13 (H264 VLC)	0	R/W
swreg37	0x94	Reference picture numbers for index 14 and 15 (H264 VLC)	0	R/W
swreg38	0x98	Reference picture long term flags (H264 VLC)	0	R/W
swreg39	0x9C	Reference picture valid flags (H264 VLC)	0	R/W
swreg40	0xA0	Base address for standard dependent tables	0	R/W
swreg41	0xA4	Base address for direct mode motion vectors Progressive JPEG AC/DC coeffient read/write base	0	R/W
swreg42	0xA8	bi_dir initial ref pic list register (0-2) Progressive JPEG Cb ACDC coefficient base	0	R/W
swreg43	0xAC	bi-dir initial ref pic list register (3-5) Progressive JPEG Cr ACDC coefficient base	0	R/W
swreg44	0xB0	bi-dir initial ref pic list register (6-8)	0	R/W
swreg45	0xB4	bi-dir initial ref pic list register (9-11)	0	R/W
swreg46	0xB8	bi-dir initial ref pic list register (12-14)	0	R/W
swreg47	0xBC	bi-dir and P fwd initial ref pic list register (15 and P 0-3)	0	R/W
swreg48	0xC0	Error concealment register	0	R/W
swreg49	0xC4	Prediction filter tap register for H264	0	R/W
swreg50	0xC8	Synthesis configuration register decoder 0 (read only)	NA	R
swreg51	0xCC	Reference picture buffer control register	0	R/W
swreg52	0xD0	Reference picture buffer information register 1 (read only)	0	R
swreg53	0xD4	Reference picture buffer information register 2 (read only)	0	R
swreg54	0xD8	Synthesis configuration register decoder 1 (read only)	NA	R
swreg55	0xDC	Reference picture buffer 2 Advanced prefetch control register 0	0	R/W
swreg56	0xE0	Reference buffer information register 3 (read only)	0	R
swreg57	0xE4	Decoder fuse register (read only)	0	R
swreg58	0xE8	Decoder debug register 0 (read only)	0	R
swreg59	0xEC	H264 Chrominance 8 pixel interleaved data base	0	R/W

69.6. 2D Graphics

69.6.1. Overview

This module mainly performs picture scaling and alpha-blending or some other 2D Graphics processing. Result will be written back to external memory.

69.6.2. Features

- 64 bits DMA interface to access external memory

- Video layer scaler and can zoom out to 1/8, zoom in to 16
- Scaler support YUV input
- Support YUV420 output to save bandwidth
- Support one OSD layer alpha-blending with one video layer
- Support Color Space Conversion
- Support Memory Copy mechanism
- Support Memory Fill with the same configured value
- Support Rotation(90,180,270,360)
- Limitation:Base address should be 8 bytes aligned due to 64 bits BUS interfaces.

69.6.3. Function Description

The TWOD engine contains following sub-blocks: AXI DMA master, AHB bus slave interface, register configuration block, CSC module, Scaler module, OSD alpha-blending module, TD copy,TD fill,TD rotate and some internal buffer.

69.6.3.1. Block Diagram

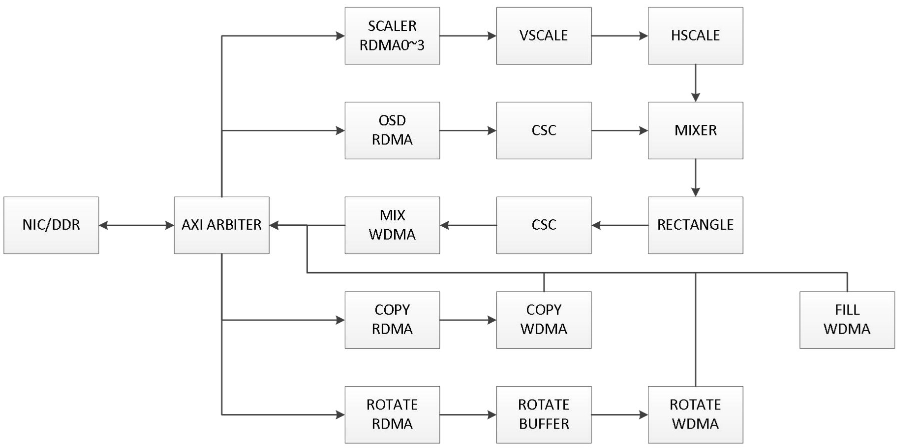


Figure 6- 5 Twod module top diagram

69.6.3.2. AXI DMA master interface

This logic convert 2D data request to DMA interface and send to external AXI logic. DMA burst length is configurable by software.

69.6.3.3. AHB Bus slave interface

This logic implements bidirectional data convertion between 32-bit AHB 2.0 slave interface and 32-bit local control register interface.

The AHB address should be 4 bytes aligned due to 32-bit AHB 2.0 slave interface.

69.6.3.4. Register Configuration

The register logic block contains all memory mapped register. The registers definitions are shown in Table 6- 1.

69.6.3.5. CSC logic

Do YUV2RGB for scaling and alpha-blending or RGB2YUV to send out YUV420 data in order to save bandwidth.

69.6.3.6. TD Scaler

4-tap 64-phase polyphase scaler, support downscaling/upscaling ratio from 1/8 to 16.

69.6.3.7. OSD Alpha-blending

Support multiple type of alpha-blending on OSD layer.

69.6.3.8. TD copy

Copy image data from one memory address space to another memory address space.

69.6.3.9. TD Fill

Fill specified Memory address space with fixed pixel value.

69.6.3.10. Data endian

LSB or msb option.

69.6.3.11. TD ROTATE

Support 90°/180°/270° image rotation.

69.6.4. **Programming Guide**

69.6.4.1. **TD Scaling**

1. Configure TWOD_LAYER0_SCALE_BLENTH register to setburst length.
2. Configure TD_VS_COE_BASE, TD_HS_COE_BASE, to set zoom and calculate parameter.
3. Configure LAYER0_H_NUM_CTL, LAYER0_V_NUM_CTL, to set the scaling ratio for the horizontal and vertical directions.
4. Configure TWOD_LAYER0_SCALE_CTL, to set upscale or downscale.
5. Configure TWOD_LAYER0_SHIFT, to set scaling factor.
6. Configure TWOD_LAYER0_IMG_IN_DAT_MODE register, to set input picture format.
7. Configure TWOD_LAYER0_IMG_IN_PIX_SIZE register, to define picture size.
((vsize-1)<<16 | (div_hsize -1)) Note: The h_size required is multiplied by 8.
8. Set TWOD_LAYER0_BADDR_Y, to define read segmented image Y data address;
9. Set TWOD_LAYER0_BADDR_UV, to define read segmented image UV data address.
10. Configure TWOD_LAYER0_HSTRIDE, to set the dma stride.
(hsize/8 -1)
11. Configure TWOD_LAYER0_IMG_OUT_PIX_SIZE,register to define picture size.
(vsize-1 <<16 | (hsize -1))
12. Configure TWOD_LAYER0_PAD_NUM_CTL, to set zoom padding parameter.
(layer0_t_pad_num<<24 | layer0_b_pad_num<<16 | div_pad_l<<8 | div_pad_r).
The leftmost segmented image padl is layer0_l_pad_num, the rightmost segmented image padr is layer0_r_pad_num, and the padl in the middle needs to be read more in the front, as well as its padr is 0.
Notice that starting address of segmented image must be aligned.
13. Configure TWOD_LAYER0_H_CUR_OFFSET, to set resumable zoom parameter.
(pad_signbit<<31 | ccur<<16 | ycur<<8 | pcur).
14. Configure TWOD_LAYER0_CTL, set 0x1 to enable zoom.
15. Configure TWOD_DISP_MIX_LAYER_GLB_EN,enable 2d mix layer.
16. Configure TWOD_DISP_MIX_LAYER_GLB_ALPHA0,TWOD_DISP_MIX_LAYER_GLB_ALPHA1 and TWOD_DISP_BACKGROUND.
17. Configure ADDR_TWOD_DISP_LAYER0_XCTL, ADDR_TWOD_DISP_LAYER0_YCTL.
(l0_hsize_out-1)<<16) + ((l0_hloc<<16)|l0_hloc)+(hporch<<16 | hporch) and (l0_vsize_out-1)<<16) + ((l0_vloc<<16)|l0_vloc) +(vporch<<16 | vporch)(hporch&vporch=0)
18. Configure TWOD_DISP_SIZE, to define write picture size. Note: The out_size required is multiplied by 8.
19. Configure TWOD_DISP_YUV2RGB_CTL, to set csc logic.
20. Configure TWOD_MIX_OUT_CTRL, to set output burst length and picture format(YUV or RGB).
21. Configure TWOD_MIX_OUT_ADDR0 and TWOD_MIX_OUT_ADDR1, to define write Y address and UV address of segmented image.

22. Configure TWOD_MIX_OUT_STRIDE, to set the dma stride (Ostride_yuv<<16)|Ostride_yuv) or Ostride_rgb.
23. Configure TWOD_MIX_OUT_DMA_CTRL0,TWOD_MIX_OUT_DMA_CTRL1. ((background_vsize/2+background_vsize%2-1)<<16)|(background_hsize-1)) and (((background_vsize-1) << 16) | (background_hsize - 1))
24. Configure TWOD_DISP_ENABLE, to set 0x1 to start this operation.
25. Configure TWOD_DISP_IRQ0_CTL, (0x100000 | background_vsize), and TWOD_DISP_IRQ1_CTL,(0x0), and TWOD_DISP_IRQ2_CTL(0x0) to write interrupt indicate that this frame has done.
26. Configure TWOD_DISP_ENABLE, to set 0x80 to stop this operation.
Set 0x1 to clear TWOD_DISP_IRQ, to wait for next segment input.

69.6.4.2. TD fill

1. Configure ADDR_TWOD_FIL_P0_SIZE,register to define rectangle size.
2. Set FIL_P0_ADDR to define Upper left corner vertex address.
3. Configure TWOD_FIL_P0_STRIDE, to set the dma stride(hsize*3).
4. Configure ADDR_TWOD_FIL_P1_SIZE,register to define rectangle size.
5. Set FIL_P1_ADDR to define Upper right corner vertex address.
6. Configure TWOD_FIL_P1_STRIDE, to set the dma stride(hsize*3).
7. Configure ADDR_TWOD_FIL_P2_SIZE,register to define rectangle size.
8. Set FIL_P2_ADDR to define bottom left corner vertex address.
9. Configure TWOD_FIL_P2_STRIDE, to set the dma stride(hsize*3).
10. Configure ADDR_TWOD_FIL_P3_SIZE,register to define rectangle size.
11. Set FIL_P3_ADDR to define bottom right corner vertex address.
12. Configure TWOD_FIL_P3_STRIDE, to set the dma stride(hsize*3).
13. Configure TWOD_FIL_DATA to set rectangle color.
14. Configure TWOD_FIL_CTRL to 0x10031to start this operation

69.6.4.3. TD copy

1. Configure TWOD_CPY_DMA_SIZE,register to define picture size.
   (vsize-1 <<16 | (hsize*3/8 -1))
2. Set TWOD_CPY_RD_ADDR to define read picture address.
3. Configure CPY_RD_STRIDE, to set the dma stride(hsize*3 >>3).
4. ConfigureCPY_WR_ADDR, to define write picture address.
5. ConfigureCPY_WR_STRIDE to set the dma stride(hsize*3).
6. Configure CPY_CTRL,set 0x10001 to start this opration.
7. Wite interrupt .
8. Configure CPY_CTRL,set 0x0 to stop this opration.

69.6.4.4. **TD OSD operation.**

1. Configure TD_R_GAMMEA_BASE, TD_G_GAMMEA_BASE and TD_B_GAMMEA_BASE to set zoom and calculate parameter.
2. Configure TWOD_LAYER0_LINE0_BD_CTL,TWOD_LAYER0_LINE1_BD_CTL,TWOD_LAYER0_LINE3_BD_CTL,TWOD_LAYER3_LINE0_BD_CTL and TWOD_LAYER0_SCALE_BLENTH register to set band width.(0x301)
3. Configure ADDR_TWOD_DISP_LAYER3_XCT ,ADDR_TWOD_DISP_LAYER3_YCTL register to set OSD picture location.
4. Configure TWOD_DISP_OSD_INFO,register to define picture format.
5. Configure ADDR_TWOD_DISP_OSD_SIZE,register to define size.
   (vsize-1 <<16 | (hsize -1))
6. Configure ADDR_TWOD_DISP_OSD_VLU_ADDR,to set OSD RGB base address.
7. Configure TWOD_DISP_OSD_ALP_ADDR,to set OSD alpha base address.
8. Configure TWOD_DISP_OSD_DMA_CTRL,to set OSD DMA contorl.(0x3)
9. Configure TWOD_DISP_OSD_STRIDE to set hsize_in*3 >> 3
10. Configure TWOD_DISP_MIX_LAYER_GLB_EN,enable 2d mix layer .
11. Configure TWOD_DISP_MIX_LAYER_GLB_ALPHA0,TWOD_DISP_MIX_LAYER_GLB_ALPHA1 and TWOD_DISP_BACKGROUND.
12. Configure ADDR_TWOD_DISP_LAYER0_XCTL,ADDR_TWOD_DISP_LAYER0_YCTL, (l0_hsize_out-1) << 16) + ((l0_hloc<<16)|l0_hloc)+(hporch<<16 | hporch) and (l0_vsize_out-1) << 16) + ((l0_vloc<<16)|l0_vloc) +(vporch<<16 | vporch)
13. Configure TWOD_DISP_SIZE, to define write picture size.
14. Configure TWOD_MIX_OUT_CTRL.to set output picture format(YUV or RGB)
15. Configure TWOD_MIX_OUT_ADDR0 and TWOD_MIX_OUT_ADDR1, to define write picture Y/RGB address and UV address
16. Configure TWOD_MIX_OUT_STRIDE, to set the dma stride (Ostride_yuv<<16)|Ostride_yuv) or Ostride_rgb;
17. Configure TWOD_MIX_OUT_DMA_CTRL0,TWOD_MIX_OUT_DMA_CTRL1,(background_vsize-1) << 16) | (background_hsize/8-1) and (background_vsize/2-1) << 16) | (background_hsize/8-1)
18. Configure ADDR_TWOD_DISP_ENABLE,set 0x9 to start this opration.
19. Wite interrupt .
20. Configure ADDR_TWOD_DISP_ENABLE,set 0x0 to stop this opration.

69.6.5. **Register Summary**

Table 6- 144 2D register and memory summary

ADDR	70. Name	**Description**

0x0	TWOD_SOFT_RST_CTL	Sub system soft reset
0x4	TWOD_TABLE_CLK_CTL	Coeff table clk select
0x8	TWOD_DMA_ARB_MODE	AXI master arbiter mode
0xC	TWOD_DMA_WEIGHT_WR0	0~3 channel write priority weight
0x10	TWOD_DMA_WEIGHT_WR1	4~7 channel write priority weight
0x14	TWOD_DMA_WEIGHT_RD0	0~3 channel read priority weight
0x18	TWOD_DMA_WEIGHT_RD1	4~7 channel read priority weight
0x1C	TWOD_DMA_PRIORITY_WR	0~7 channel write priority Channel mapping: Channel 0: Scaler+OSD Y/RGB output Channel 1: Scaler+OSD UV Channel 2: Copy Channel 3: Fill Channel 4: Rotate Channel 5~7: reserved
0x20	TWOD_DMA_PRIORITY_RD	0~7 channel read priority Channel mapping: Channel 0: Scaler Channel 1: Scaler Channel 2: Scaler Channel 3: Scaler Channel 4: Copy Channel 5: reserved Channel 6: OSD Channel 7: Rotate
0x24	TWOD_DMA_ID_RD	DMA ID select
0x28	TWOD_DMA_ID_WR	DMA ID select
0x4C	TWOD_MIX_LAYER0_XCTL	Scaler layer horizontal start & stop
0x50	TWOD_MIX_LAYER0_YCTL	Scaler layer vertical start & stop
0x54	TWOD_MIX_LAYER3_XCTL	OSD layer horizontal start & stop
0x58	TWOD_MIX_LAYER3_YCTL	OSD layer vertical start & stop
0x5C~0x70		Reserved
0x74	TWOD_MIX_ENABLE	Mixer Layer enable
0x78	TWOD_MIX_SIZE	Mixer output total size
0x7C~0xAC		Reserved

0xB0	TWOD_LAYER0_CTL	Layer0 enable
0xB4	TWOD_LAYER0_SCALE_BLENTH	Layer0 line memory burst length
0xB8	TWOD_LAYER0_IMG_IN_DAT_MODE	Layer0 input image data format: YUV420, YUV422
0xBC	TWOD_LAYER0_BADDR_Y	Layer0 in image y base address
0xC0	TWOD_LAYER0_BADDR_UV	Layer0 in image uv base address
0xC4	TWOD_LAYER0_IMG_IN_PIX_SIZE	Layer0 in image size in pixel
0xC8	TWOD_LAYER0_IMG_OUT_PIX_SIZE	Layer0 out image size in pixel
0xCC	TWOD_LAYER0_IMG_IN_SIZE	Layer0 in image size in 8 pixel
0xD0	TWOD_LAYER0_IMG_IN_MEM_HSIZE	
0xD4	TWOD_LAYER0_IMG_IN_OFFSET	Layer0 in image horizontal offset(in 8 pixel) & vertical offset(in line)
0xD8	TWOD_LAYER0_SCALE_CTL	Scaler mode select
0xDC	TWOD_LAYER0_HSTRIDE	Layer0 in image stride
0xE0	TWOD_LAYER0_H_NUM_CTL	Scaler horizontal ratio
0xE4	TWOD_LAYER0_V_NUM_CTL	Scaler vertical ratio
0xE8	TWOD_LAYER0_PAD_NUM_CTL	Scaler padding pixel size
0xEC	TWOD_LAYER0_SHIFT	Scaler factor
0xF0	TWOD_LAYER0_H_CUR_OFFSET	Slice H offset
0xF4	TWOD_LAYER0_V_CUR_OFFSET	Slice V offset
0xF8~0x124		
0x128	TWOD_MIX_YUV2RGB_CTL	YUV2RGB conversion enable
0x12C	TWOD_MIX_YUV2RGB_COEF00	YUV2RGB coefficient
0x130	TWOD_MIX_YUV2RGB_COEF01	YUV2RGB coefficient
0x134	TWOD_MIX_YUV2RGB_COEF02	YUV2RGB coefficient
0x138	TWOD_MIX_YUV2RGB_COEF03	YUV2RGB coefficient
0x13C	TWOD_MIX_YUV2RGB_COEF10	YUV2RGB coefficient
0x140	TWOD_MIX_YUV2RGB_COEF11	YUV2RGB coefficient
0x144	TWOD_MIX_YUV2RGB_COEF12	YUV2RGB coefficient
0x148	TWOD_MIX_YUV2RGB_COEF13	YUV2RGB coefficient
0x14C	TWOD_MIX_YUV2RGB_COEF20	YUV2RGB coefficient
0x150	TWOD_MIX_YUV2RGB_COEF21	YUV2RGB coefficient
0x154	TWOD_MIX_YUV2RGB_COEF22	YUV2RGB coefficient
0x158	TWOD_MIX_YUV2RGB_COEF23	YUV2RGB coefficient
0x160	TWOD_MIX_LAYER_GLB_EN	Display mix alpha source selection
0x164	TWOD_MIX_LAYER_GLB_ALPHA0	Display mix global alpha value
0x16C	TWOD_MIX_SRC_SEL	Display mix source select
0x170	TWOD_MIX_BACKGROUND	Background RGB color value
0x178	TWOD_IRQ0_CTL	Display irq0 control
0x17C	TWOD_IRQ1_CTL	Display irq1 control
0x180	TWOD_IRQ2_CTL	Display irq2 control
0x184	TWOD_IRQ	Dislay irq clear & status
0x188	TWOD_OSD_INFO	OSD image data format

0x18C	TWOD_OSD_SIZE	OSD image size
0x190	TWOD_OSD_VLU_ADDR	OSD RGB base address
0x194	TWOD_OSD_ALP_ADDR	OSD alpha base address
0x198	TWOD_OSD_DMA_CTRL	OSD DMA contorl
0x19C	TWOD_OSD_STRIDE	OSD frame address space stride
0x1A0	TWOD_RECTANGLE_EN	Rectangle enable
0x1A4	TWOD_RECTANGLE_DATA	Rectangle pixel color
0x1A8	TWOD_RECTANGLE0_POSITION	Rectangle start position
0x1AC	TWOD_RECTANGLE0_LENGTH	Rectangle Length and width
0x1B0	TWOD_RECTANGLE1_POSITION	Rectangle start position
0x1B4	TWOD_RECTANGLE1_LENGTH	Rectangle Length and width
0x1B8	TWOD_RECTANGLE2_POSITION	Rectangle start position
0x1BC	TWOD_RECTANGLE2_LENGTH	Rectangle Length and width
0x1C0	TWOD_RECTANGLE3_POSITION	Rectangle start position
0x1C4	TWOD_RECTANGLE3_LENGTH	Rectangle Length and width
0x1C8	TWOD_RECTANGLE4_POSITION	Rectangle start position
0x1CC	TWOD_RECTANGLE4_LENGTH	Rectangle Length and width
0x1D0	TWOD_RECTANGLE5_POSITION	Rectangle start position
0x1D4	TWOD_RECTANGLE5_LENGTH	Rectangle Length and width
0x1D8	TWOD_RECTANGLE6_POSITION	Rectangle start position
0x1DC	TWOD_RECTANGLE6_LENGTH	Rectangle Length and width
0x1E0	TWOD_RECTANGLE7_POSITION	Rectangle start position
0x1E4	TWOD_RECTANGLE7_LENGTH	Rectangle Length and width
0x1E8	TWOD_RECTANGLE8_POSITION	Rectangle start position
0x1EC	TWOD_RECTANGLE8_LENGTH	Rectangle Length and width
0x1F0	TWOD_RECTANGLE9_POSITION	Rectangle start position
0x1F4	TWOD_RECTANGLE9_LENGTH	Rectangle Length and width
0x1F8~0x1FC		
0x200	TWOD_OSD_YUV2RGB_CTL	Osd layer RGB2YUV enable
0x204	TWOD_OSD_YUV2RGB_COEFF00	RGB2YUV coefficient
0x208	TWOD_OSD_YUV2RGB_COEFF01	RGB2YUV coefficient
0x20C	TWOD_OSD_YUV2RGB_COEFF02	RGB2YUV coefficient
0x210	TWOD_OSD_YUV2RGB_COEFF03	RGB2YUV coefficient
0x214	TWOD_OSD_YUV2RGB_COEFF10	RGB2YUV coefficient
0x218	TWOD_OSD_YUV2RGB_COEFF11	RGB2YUV coefficient
0x21C	TWOD_OSD_YUV2RGB_COEFF12	RGB2YUV coefficient
0x220	TWOD_OSD_YUV2RGB_COEFF13	RGB2YUV coefficient
0x224	TWOD_OSD_YUV2RGB_COEFF20	RGB2YUV coefficient
0x228	TWOD_OSD_YUV2RGB_COEFF21	RGB2YUV coefficient
0x22C	TWOD_OSD_YUV2RGB_COEFF22	RGB2YUV coefficient
0x230	TWOD_OSD_YUV2RGB_COEFF23	RGB2YUV coefficient
0x234	TWOD_OSD_RGBA_CTL	
0x238~0x23C		

0x240	TWOD_AXI_RST_CTL	Reset AXI bus
0x244	TWOD_AXI_RST_STATUS	AXI bus status
0x248~0x24C		
0x250	TWOD_MIX_OUT_CTRL	MIX output ctrl
0x254	TWOD_MIX_OUT_ADDR0	Mix output start address0
0x258	TWOD_MIX_OUT_ADDR1	Mix output start address1
0x25C	TWOD_MIX_OUT_STRIDE	Mix output stride
0x260	TWOD_MIX_OUT_DMA_CTRL0	Mix output dma ctrl
0x264	TWOD_MIX_OUT_DMA_CTRL1	Mix output dma ctrl
0x268	TWOD_MIX_OUT_DMAY_LAST_ADDR	
0x26C	TWOD_MIX_OUT_DMAUV_LAST_ADDR	
0x270~0x27C		
0x280	TWOD_CPY_DMA_SIZE	Copy image size
0x284	TWOD_CPY_RD_ADDR	Copy read address
0x288	TWOD_CPY_RD_STRIDE	Copy read stride
0x28C	TWOD_CPY_WR_ADDR	Copy write address
0x290	TWOD_CPY_WR_STRIDE	Copy write stride
0x294	TWOD_CPY_CTRL	Copy ctrl
0x298	TWOD_CPY_IRQ	Copy interrupt ctrl
0x29C~0x2AC		
0x2B0	TWOD_FIL_P0_SIZE	Fill point0 size
0x2B4	TWOD_FIL_P0_STRIDE	Fill point0 stride
0x2B8	TWOD_FIL_P0_ADDR	Fill point0 address
0x2BC	TWOD_FIL_P1_SIZE	Fill point1 size
0x2C0	TWOD_FIL_P1_STRIDE	Fill point1 stride
0x2C4	TWOD_FIL_P1_ADDR	Fill point1 address
0x2C8	TWOD_FIL_P2_SIZE	Fill point2 size
0x2CC	TWOD_FIL_P2_STRIDE	Fill point2 stride
0x2D0	TWOD_FIL_P2_ADDR	Fill point2 address
0x2D4	TWOD_FIL_P3_SIZE	Fill point3 size
0x2D8	TWOD_FIL_P3_STRIDE	Fill point3 stride
0x2DC	TWOD_FIL_P3_ADDR	Fill point3 address
0x2E0	TWOD_FIL_DATA	Fill data
0x2E4	TWOD_FIL_CTRL	Fill ctrl
0x2E8	TWOD_FIL_IRQ	Fill interrupt ctrl
0x2EC	TWOD_ENDIAN_SWAP_CTRL	Endia ctrl
0x2F0	TWOD_TRANS_DMA_SIZE	Rotate size
0x2F4	TWOD_TRANS_RD_ADDR	Rotate address
0x2F8	TWOD_TRANS_RD_STRIDE	Rotate stride
0x2FC	TWOD_TRANS_WR_ADDR	Rotate write address
0x300	TWOD_TRANS_WR_STRIDE	Rotate write stride
0x304	TWOD_TRANS_CTRL	Rotate ctrl
0x308	TWOD_TRANS_IRQ	Rotate interrupt ctrl

0x30C	TWOD_TRANS_ANGLE	Rotate angle
0x310	TWOD_TRANS_TOTAL_SIZE	Rotate image total size

70.1.1. Register Description

70.1.1.1. TWOD_SOFT_RST_CTL

Table 6-145 .TWOD_SOFT_RST_CTL

Address: BASE + 0x000				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_SF_RST_AHB	R/W	b0	When '1', reset ahb clock domain logic
1	TWOD_SF_RST_AXI	R/W	b0	When '1', reset axi clock domain logic
31-2	--	--	--	Reserved

70.1.1.2. TWOD_TABLE_CLK_CTL

Table 6-146 .TWOD_TABLE_CLK_CTL

Address: BASE + 0x004				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	VO_VSCALE_CONFIG	R/W	b0	Vertical Coeff Table Config Clk
1	VO_HSCALE_CONFIG	R/W	b0	Horizontal Coeff Table Config Clk
31-2	--	--	--	Reserved

70.1.1.3. TWOD_DMA_ARB_MODE

Table 6-147 .TWOD_DMA_ARB_MODE

Address: BASE + 0x008				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	RD_ARB_MODE	R/W	b0	Read channel arbiter mode. 0:use configured weight 1:use fixed round-robin
1	WR_ARB_MODE	R/W	b0	Write channel arbiter mode. 0:use configured weight 1:use fixed round-robin
31-2	--	--	--	Reserved

70.1.1.4. TWOD_DMA_WEIGHT_WR0

Table 6-148 .TWOD_DMA_WEIGHT_WR0

Address: BASE + 0x00C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
7-0	TWOD_DMA_WEIGHT_WRITE0	R/W	H01	Chanel 0 weight
15-8		R/W	H01	Chanel 1 weight
23-16		R/W	H01	Chanel 2 weight
31-24		R/W	H01	Chanel 3 weight

70.1.1.5. TWOD_DMA_WEIGHT_WR1

Table 6-149 .TWOD_DMA_WEIGHT_WR1

Address: BASE + 0x010				
Bit	**Name**	**Access**	**Reset Value**	**Description**
7-0	TWOD_DMA_WEIGHT_WRITE1	R/W	H01	Chanel 4 weight
15-8		R/W	H01	Chanel 5 weight
23-16		R/W	H01	Chanel 6 weight
31-24		R/W	H01	Chanel 7 weight

70.1.1.6. TWOD_DMA_WEIGHT_RD0

Table 6-150 .TWOD_DMA_WEIGHT_RD0

Address: BASE + 0x014				
Bit	**Name**	**Access**	**Reset Value**	**Description**
7-0	TWOD_DMA_WEIGHT_READ0	R/W	H01	Chanel 0 read weight
15-8		R/W	H01	Chanel 1 read weight
23-16		R/W	H01	Chanel 2 read weight
31-24		R/W	H01	Chanel 3 read weight

70.1.1.7. TWOD_DMA_WEIGHT_RD1

Table 6-151 .TWOD_DMA_WEIGHT_RD1

Address: BASE + 0x018				
Bit	**Name**	**Access**	**Reset Value**	**Description**
7-0	TWOD_DMA_WEIGHT_READ1	R/W	H01	Chanel 4 read weight
15-8		R/W	H01	Chanel 5 read weight
23-16		R/W	H01	Chanel 6 read weight
31-24		R/W	H01	Chanel 7 read weight

70.1.1.8. TWOD_DMA_PRIORITY_WR

Table 6-152 .TWOD_DMA_PRIORITY_WR

Address: BASE + 0x01C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
3-0	TWOD_DMA_PRIORITY_WR	R/W	h0	Chanel 0 read priority
7-4		R/W	h1	Chanel 1 read priority
11-8		R/W	h2	Chanel 2 read priority
15-12		R/W	h3	Chanel 3 read priority
19-16		R/W	h4	Chanel 4 read priority
23-20		R/W	h5	Chanel 5 read priority
27-24		R/W	h6	Chanel 6 read priority
31-28		R/W	h7	Chanel 7 read priority

70.1.1.9. TWOD_DMA_PRIORITY_RD

Table 6-153 .TWOD_DMA_PRIORITY_RD

Address: BASE + 0x020				
Bit	**Name**	**Access**	**Reset Value**	**Description**
3-0	TWOD_DMA_PRIORITY_RD	R/W	h0	Chanel 0 read priority
7-4		R/W	h1	Chanel 1 read priority
11-8		R/W	h2	Chanel 2 read priority
15-12		R/W	h3	Chanel 3 read priority
19-16		R/W	h4	Chanel 4 read priority
23-20		R/W	h5	Chanel 5 read priority
27-24		R/W	h6	Chanel 6 read priority
31-28		R/W	h7	Chanel 7 read priority

70.1.1.10. TWOD_DMA_ID_RD

Table 6-154 .TWOD_DMA_ID_RD

Address: BASE + 0x024				
Bit	**Name**	**Access**	**Reset Value**	**Description**
3-0	RID_CH0	R/W	h7	Chanel 0 read ID
7-4	RID_CH1	R/W	h6	Chanel 1 read ID
11-8	RID_CH2	R/W	h5	Chanel 2 read ID
15-12	RID_CH3	R/W	h4	Chanel 3 read ID
19-16	RID_CH4	R/W	h3	Chanel 4 read ID
23-20	RID_CH5	R/W	h2	Chanel 5 read ID
27-24	RID_CH6	R/W	h1	Chanel 6 read ID
31-28	RID_CH7	R/W	h0	Chanel 7 read ID

70.1.1.11. TWOD_DMA_ID_WR

Table 6-155 .TWOD_DMA_ID_WR

Address: BASE + 0x028				
Bit	**Name**	**Access**	**Reset Value**	**Description**
3-0	WID_CH0	R/W	h7	Chanel 0 read ID
7-4	WID_CH1	R/W	h6	Chanel 1 read ID
11-8	WID_CH2	R/W	h5	Chanel 2 read ID
15-12	WID_CH3	R/W	h4	Chanel 3 read ID
19-16	WID_CH4	R/W	h3	Chanel 4 read ID
23-20	WID_CH5	R/W	h2	Chanel 5 read ID
27-24	WID_CH6	R/W	h1	Chanel 6 read ID
31-28	WID_CH7	R/W	h0	Chanel 7 read ID

70.1.1.12. TWOD_LAYER0_LINE0_BD_CTL

Table 6-156 .TWOD_LAYER0_LINE0_BD_CTL

Address: BASE + 0x02C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER0_LINE0_LIMIT_EN	R/W	b1	Scaler Line 0 DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER0_LINE0_OUTSTAND	R/W	h7	Scaler Line 0 DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.13. TWOD_LAYER0_LINE1_BD_CTL

Table 6-157 .TWOD_LAYER0_LINE1_BD_CTL

Address: BASE + 0x030				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER0_LINE1_LIMIT_EN	R/W	b1	Scaler Line 1 DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER0_LINE1_OUTSTAND	R/W	h7	Scaler Line 1 DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.14. TWOD_LAYER0_LINE2_BD_CTL

Table 6-158 .TWOD_LAYER0_LINE2_BD_CTL

Address: BASE + 0x034				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER0_LINE2_LIMIT_EN	R/W	b1	Scaler Line 2 DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER0_LINE2_OUTSTAND	R/W	h7	Scaler Line 2 DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.15. TWOD_LAYER0_LINE3_BD_CTL

Table 6-159 .TWOD_LAYER0_LINE3_BD_CTL

Address: BASE + 0x038				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER0_LINE3_LIMIT_EN	R/W	b1	Scaler Line 3 DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER0_LINE3_OUTSTAND	R/W	h7	Scaler Line 3 DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.16. TWOD_LAYER1_BD_CTL

Table 6-160 .TWOD_LAYER1_BD_CTL

Address: BASE + 0x03C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER1_LIMIT_EN	R/W	b1	TD Copy DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER1_OUTSTAND	R/W	h7	TD Copy DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.17. TWOD_LAYER2_BD_CTL

Table 6-161 .TWOD_LAYER1_BD_CTL

Address: BASE + 0x040				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER2_LIMIT_EN	R/W	b1	TD Copy DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER2_OUTSTAND	R/W	h7	TD Copy DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.18. TWOD_LAYER3_BD_CTL

Table 6-162 .TWOD_LAYER3_BD_CTL

Address: BASE + 0x044				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER3_LIMIT_EN	R/W	b1	OSD DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER3_OUTSTAND	R/W	h7	OSD DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.19. TWOD_LAYER4_BD_CTL

Table 6-163 .TWOD_LAYER4_BD_CTL

Address: BASE + 0x048				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_LAYER4_LIMIT_EN	R/W	b1	TD Copy DMA outstanding limit enable
7-1	--	--	--	Reserved
12-8	TWOD_LAYER4_OUTSTAND	R/W	h7	TD Copy DMA maximum outstanding (actual outstand value – 1)
31-13	--	--	--	reserved

70.1.1.20. TWOD_MIX_LAYER0_XCTL

Table 6-164 .TWOD_MIX_LAYER0_XCTL

Address: BASE + 0x04c				
Bit	Name	Access	Reset Value	Description
11:0	71. TWOD_MIX_LAYER0_XSTART	R/W	h0	72. Display layer0 horizontal start

				pixel address (min pixel address is 1)
15-12	73. --	74. --	75. --	76. Reserved
27-16	77. TWOD_MIX_LAYER0_XSTOP	R/W	h0	78. Display layer0 horizontal stop pixel address (min pixel address is 1)
32-28	--	--	--	reserved

78.1.1.1. TWOD_MIX_LAYER0_YCTL

Table 6-165 .TWOD_MIX_LAYER0_YCTL

Address: BASE + 0x050				
Bit	Name	Access	Reset Value	Description
11:0	79. TWOD_MIX_LAYER0_YSTART	R/W	h0	80. Display layer0 vertical start pixel address (min pixel address is 1)
15-12	81. --	82. --	83. --	84. Reserved
27-16	85. TWOD_MIX_LAYER0_YSTOP	R/W	h0	86. Display layer0 vertical stop pixel address (min pixel address is 1)
32-28	--	--	--	reserved

86.1.1.1. TWOD_MIX_LAYER3_XCTL

Table 6-166 .TWOD_MIX_LAYER3_XCTL

Address: BASE + 0x054				
Bit	Name	Access	Reset Value	Description

87. 11:0	88. TWOD_MIX_LAYER3_XSTART	89. R/W	90. h0	91. Display layer3 horizontal start pixel address (min pixel address is 1)
92. 15-12	93. --	94. --	95. --	96. Reserved
27-16	97. TWOD_MIX_LAYER3_XSTOP	R/W	h0	98. Display layer3 horizontal stop pixel address (min pixel address is 1)
32-28	--	--	--	reserved

98.1.1.1. TWOD_MIX_LAYER3_YCTL

Table 6-167 .TWOD_MIX_LAYER3_YCTL

Address: BASE + 0x058				
Bit	Name	Access	Reset Value	Description
11:0	99. TWOD_MIX_LAYER3_YSTART	R/W	h0	100. Display layer3 vertical start pixel address (min pixel address is 1)
15-12	101. --	102. --	103. --	104. Reserved
27-16	105. TWOD_MIX_LAYER3_YSTOP	R/W	h0	106. Display layer3 vertical stop pixel address (min pixel address is 1)
32-28	--	--	--	reserved

106.1.1.1. TWOD_MIX_ENABLE

Table 6-168 .TWOD_MIX_ENABLE

Address: BASE + 0x0A0				
Bit	Name	Access	Reset Value	Description
0	TWOD_MIX_SCALER_EN	R/W	b1	Scaler Layer output enable
1	TWOD_OSD_EN	R/W	b0	OSD Layer output enable
6-2				Reserved
7	TWOD_MIX_RESTART	R/W	B1	When ‘1’, Mixer is disabled.
32-8				Reserved

106.1.1.2. TWOD_MIX_SIZE

Table 6-169 .TWOD_MIX_SIZE

Address: BASE + 0x0A4				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_MIX_HSIZE	R/W	h0	Horizontal size of display frame. for image of 800x600，it is 800
15-12	--	--	--	Reserved
27-16	TWOD_MIX_VSIZE	R/W	h0	Vertical size of display frame. . for image of 800x600，it is 600
31-28	--	--	--	Reserved

106.1.1.3. TWOD_LAYER0_CTL

Table 6-170 .TWOD_LAYER0_CTL

Address: BASE + 0x0B0				
Bit	Name	Access	Reset Value	Description
0	TWOD_LAYER0_ENABLE	R/W	b0	When '1', layer0 is enable When '0', layer0 is disable
31-1	--	--	--	Reserved

106.1.1.4. TWOD_LAYER0_SCALE_BLENTH

Table 6-171 .TWOD_LAYER0_SCALE_BLENTH

Address: BASE + 0x0B4				
Bit	Name	Access	Reset Value	Description
3-0	TWOD_LAYER0_SCALE_BLENTH	R/W	h3	DMA burst length of layer0 scale line memory 0~4(min length is 0)
31-4	--	--	--	Reserved

106.1.1.5. TWOD_LAYER0_IMG_IN_DAT_MODE

Table 6-172 .TWOD_LAYER0_IMG_IN_DAT_MODE

Address: BASE + 0x0B8

Bit	Name	Access	Reset Value	Description
4-0	TWOD_LAYER0_IMG_IN_DAT_MODE	R/W	H8	When 5'h8, layer0 input image is YUV4:2:0

106.1.1.6. TWOD_LAYER0_BADDR_Y

Table 6-173 .TWOD_LAYER0_BADDR_Y

Address: BASE + 0x0BC				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_LAYER0_BADDR_Y	R/W	h0	Layer0 Y frame base address of input imiage.

106.1.1.7. TWOD_LAYER0_BADDR_UV

Table 6-174 .TWOD_LAYER0_BADDR_UV

Address: BASE + 0x0C0				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_LAYER0_BADDR_UV	R/W	h0	Layer0 UV frame base address of input imiage.

106.1.1.8. TWOD_LAYER0_IMG_IN_PIX_SIZE

Table 6-175 .TWOD_LAYER0_IMG_IN_PIX_SIZE

Address: BASE + 0x0C4				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_LAYER0_IMG_IN_PIX_HSIZE	R/W	h0	Actually reading Horizontal size of layer0, in pixel unit. As example, for image of 800x600, is 799.
15-12	--	--	--	Reserved
27-16	TWOD_LAYER0_IMG_IN_PIX_VSIZE	R/W	h0	Actually reading Vertical line size of layer0, in line unit. As example, for image of 800x600, is 599.
31-28	--	--	--	Reserved

106.1.1.9. TWOD_LAYER0_IMG_OUT_PIX_SIZE

Table 6-176 .TWOD_LAYER0_IMG_OUT_PIX_SIZE

Address: BASE + 0x0C8				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_LAYER0_IMG_OUT_PIX_HSIZE	R/W	h0	Horizontal size in pixel of layer0 scaler output. As example, for image of 800x600, is 799.
15-12	--	--	--	Reserved
27-16	TWOD_LAYER0_IMG_OUT_PIX_VSIZE	R/W	h0	Vertical line size of layer0 scaler output. As example, for image of 800x600, is 599.
31-28	--	--	--	Reserved

106.1.1.10. TWOD_LAYER0_IMG_IN_SIZE

Table 6-177 .TWOD_LAYER0_IMG_IN_SIZE

Address: BASE + 0x0CC				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_LAYER0_IMG_IN_PIX_HSIZE	R/W	h0	Horizontal size of layer 0 image in DDR, in double word(8-pixel). of layer0 input image. As example, for image of 800x600, is 99.
15-12	--	--	--	Reserved
27-16	TWOD_LAYER0_IMG_IN_PIX_VSIZE	R/W	h0	Vertical line size of layer0 image in DDR. input image. As example, for image of 800x600, is 599.
31-28	--	--	--	Reserved

106.1.1.11. TWOD_LAYER0_IMG_IN_OFFSET

Table 6-178 .TWOD_LAYER0_IMG_IN_OFFSET

Address: BASE + 0x0D4

Bit	Name	Access	Reset Value	Description
s				
15-12	--	--	--	Reserved
27-16	TWOD_LAYER0_IMG_IN_VOFFSET	R/W	h0	Vertical offset line number relative to start line(0) of layer0 input image.
31-28	--	--	--	Reserved

106.1.1.12. **TWOD_LAYER0_SCALE_CTL**

Table 6-179 .TWOD_LAYER0_SCALE_CTL

Address: BASE + 0x0D8				
Bit	Name	Access	Reset Value	Description
0	TWOD_LAYER0_HSCALE_MODE	R/W	h0	Hscale mode. 0: downscale 1: upscale
1	TWOD_LAYER0_VSCALE_MODE	R/W	h0	Vscale mode. 0: downscale 1: upscale
31-2	--	--	--	Reserved

106.1.1.13. **TWOD_LAYER0_HSTRIDE**

Table 6-180 .TWOD_LAYER0_HSTRIDE

Address: BASE + 0XDC				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_LAYER0_HSTRIDE	R/W	h0	Horizontal stride (minus 1) of layer 0 image in DDR, in double word(8-pixel). for example if stride is 100*8byte, this field will be configured as 99.
31-12	--	--	--	Reserved

106.1.1.14. **TWOD_LAYER0_H_NUM_CTL**

Table 6-181 .TWOD_LAYER0_H_NUM_CTL

Address: BASE + 0x0E0				
Bit	Name	Access	Reset	Description

			Value	
11-0	TWOD_LAYER0_HSCALE_M	R/W	h0	Horizontal Scaling ratio denominator/numerator (whichever is larger)
15-12	--	--	--	Reserved
27-16	TWOD_LAYER0_HSCALE_N	R/W	h0	Horizontal Scaling ratio denominator/numerator (whichever is smaller)
31-28	--	--	--	Reserved

106.1.1.15. TWOD_LAYER0_V_NUM_CTL

Table 6-182 .TWOD_LAYER0_V_NUM_CTL

Address: BASE + 0x0E4				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_LAYER0_VSCALE_M	R/W	h0	Vertical Scaling ratio denominator/numerator (whichever is larger)
15-12	--	--	--	Reserved
27-16	TWOD_LAYER0_VSCALE_N	R/W	h0	Vertical Scaling ratio denominator/numerator (whichever is smaller)
31-28	--	--	--	Reserved

106.1.1.16. TWOD_LAYER0_PAD_NUM_CTL

Table 6-183 .TWOD_LAYER0_PAD_NUM_CTL

Address: BASE + 0x0E8				
Bit	Name	Access	Reset Value	Description
7-0	TWOD_LAYER0_PAD_R	R/W	h0	Scaler input right padding
15-8	TWOD_LAYER0_PAD_L	R/W	h0	Scaler input left padding
23-16	TWOD_LAYER0_PAD_B	R/W	h0	Scaler input bottom padding
31-24	TWOD_LAYER0_PAD_T	R/W	h0	Scaler input top padding

106.1.1.17. TWOD_LAYER0_SHIFT

Table 6-184 .TWOD_LAYER0_SHIFT

Address: BASE + 0x0EC				
Bit	Name	Access	Reset Value	Description
7-0	TWOD_LAYER0_SHIFT_V	R/W	h0	Scaler input right padding
15-8	TWOD_LAYER0_SHIFT_H	R/W	h0	Scaler input left padding
31-16	--	R/W	h0	Reserved

106.1.1.18. TWOD_LAYER0_H_CUR_OFFSET

Table 6-185 .TWOD_LAYER0_H_CUR_OFFSET

Address: BASE + 0x0F0				
Bit	Name	Access	Reset Value	Description
7-0	TWOD_LAYER0_H_PCUR	R/W	h0	H direction pcur offset
15-8	TWOD_LAYER0_H_YCUR	R/W	h0	H directionycur offset
23-16	TWOD_LAYER0_H_CCUR	R/W	h0	H directionccur offset
31	TWOD_LAYER0_PAD_SIGNBIT	R/W	h0	remain pad 0 :positive, 1: negative

106.1.1.19. TWOD_MIX_OUT_YUV2RGB_CTL

Table 6-186 .TWOD_MIX_OUT_YUV2RGB_CTL

Address: BASE + 0x128				
Bit	**Name**	**Access**	**Reset Value**	**Description**
0	TWOD_MIX_OUT_YUV2RGB_EN	R/W	b1	When '1', YUV444 -> RGB after display mixer is enabled. When '0', YUV444 -> RGB after display mixer is disabled.
31-1	--	--	--	Reserved

106.1.1.20. TWOD_MIX_OUT_YUV2RGB_COEFF00

Table 6-187 .TWOD_MIX_OUT_YUV2RGB_COEFF00

Address: BASE + 0x12C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_Y	R/W	h400	YUV444 -> RGB coefficient

	UV2RGB_COEFF00			
31-12	--	--	--	Reserved

106.1.1.21. TWOD_MIX_OUT_YUV2RGB_COEFF01

Table 6-188 .TWOD_MIX_OUT_YUV2RGB_COEFF01

Address: BASE + 0x130				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF01	R/W	h0	YUV444 -> RGB coefficient
31-12	--	--	--	Reserved

106.1.1.22. TWOD_MIX_OUT_YUV2RGB_COEFF02

Table 6-189 .TWOD_MIX_OUT_YUV2RGB_COEFF02

Address: BASE + 0x134				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF02	R/W	h5a1	YUV444 -> RGB coefficient
31-12	--	--	--	Reserved

106.1.1.23. TWOD_MIX_OUT_YUV2RGB_COEFF03

Table 6-190 .TWOD_MIX_OUT_YUV2RGB_COEFF03

Address: BASE + 0x138				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF03	R/W	hf4c	YUV444 -> RGB offset value
31-12	--	--	--	Reserved

106.1.1.24. TWOD_MIX_OUT_YUV2RGB_COEFF10

Table 6-191 .TWOD_MIX_OUT_YUV2RGB_COEFF10

Address: BASE + 0x13C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF10	R/W	h400	YUV444 -> RGB coefficient

31-12	--	--	--	Reserved

106.1.1.25. TWOD_MIX_OUT_YUV2RGB_COEFF11

Table 6-192 .TWOD_MIX_OUT_YUV2RGB_COEFF11

Address: BASE + 0x140				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF11	R/W	he9e	YUV444 -> RGB coefficient
31-12	--	--	--	Reserved

106.1.1.26. TWOD_MIX_OUT_YUV2RGB_COEFF12

Table 6-193 .TWOD_MIX_OUT_YUV2RGB_COEFF12

Address: BASE + 0x144				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF12	R/W	hd22	YUV444 -> RGB coefficient
31-12	--	--	--	Reserved

106.1.1.27. TWOD_MIX_OUT_YUV2RGB_COEFF13

Table 6-194 .TWOD_MIX_OUT_YUV2RGB_COEFF13

Address: BASE + 0x148				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF13	R/W	h88	YUV444 -> RGB offset value
31-12	--	--	--	Reserved

106.1.1.28. TWOD_MIX_OUT_YUV2RGB_COEFF20

Table 6-195 .TWOD_MIX_OUT_YUV2RGB_COEFF20

Address: BASE + 0x14C				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF20	R/W	h400	YUV444 -> RGB coefficient

31-12	--	--	--	Reserved

106.1.1.29. TWOD_MIX_OUT_YUV2RGB_COEFF21

Table 6-196 .TWOD_MIX_OUT_YUV2RGB_COEFF21

Address: BASE + 0x150				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF21	R/W	h71e	YUV444 -> RGB coefficient
31-12	--	--	--	Reserved

106.1.1.30. TWOD_MIX_OUT_YUV2RGB_COEFF22

Table 6-197 .TWOD_MIX_OUT_YUV2RGB_COEFF22

Address: BASE + 0x154				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	TWOD_MIX_OUT_YUV2RGB_COEFF22	R/W	h0	YUV444 -> RGB coefficient
31-12	--	--	--	Reserved

106.1.1.31. TWOD_MIX_OUT_YUV2RGB_COEFF23

Table 6-198 .TWOD_MIX_OUT_YUV2RGB_COEFF23

Address: BASE + 0x158				
Bit	**Name**	**Access**	**Reset Value**	**Description**
11-0	VO_DISP_YUV2RGB_COEFF23	R/W	hf1c	YUV444 -> RGB offset value
31-12	--	--	--	Reserved

106.1.1.32. TWOD_MIX_LAYER_GLB_EN

Table 6-199 .TWOD_MIX_LAYER_GLB_EN

Address: BASE + 0x160				
Bit	Name	Access	Reset Value	Description
0	TWOD_LAYER0_GLB_EN	R/W	b1	when ‘1’, layer0 is to be mixed use global alpha blending value. In current

				design, pixel by pixel alpha blending is not supported.
1	TWOD_LAYER3_GLB_EN	R/W	b1	when ‘1’, layer3(OSD layer) is to be mixed use global alpha blending value. when ‘0’, layer3(OSD layer) is to be mixed use pixel-by-pixel alpha blending.
31-4	--	--	--	Reserved

106.1.1.33. **TWOD_MIX_LAYER_GLB_ALPHA0**

Table 6-200 .TWOD_MIX_LAYER_GLB_ALPHA0

Address: BASE + 0x164				
Bit	Name	Access	Reset Value	Description
7-0	TWOD_LAYER0_GLB_ALPHA	R/W	hff	layer0 global alpha value
15-8	TWOD_LAYER3_GLB_ALPHA	R/W	hff	layer3(OSD layer) global alpha value
23-16				**Reserved**
31-24	TWOD_MIX_OUT_ALPHA	R/W	hff	Output alpha value in ARGB mode

106.1.1.34. **TWOD_MIX_SRC_SEL**

Table 6-201 .TWOD_MIX_SRC_SEL

Address: BASE + 0x16C				
Bit	Name	Access	Reset Value	Description
2-0	TWOD_PIPE0_SRC_SEL	R/W	0	mixer pipe0 source select 0: Scaler 1:OSD
3				**Reserved**
6-4	TWOD_PIPE1_SRC_SEL	R/W	1	mixer pipe1 source select 0: Scaler 1:OSD
31-7				**Reserved**

106.1.1.35. **TWOD_MIX_BACKGROUND**

Table 6-202 .TWOD_MIX_BACKGROUND

Address: BASE + 0x170				
Bit	Name	Access	Reset Value	Description
7-0	TWOD_MIX_BACKGROUND_Y	R/W	hff	R component color of background
15-8	TWOD_MIX_BACKGROUND_U	R/W	h0	G component color of background
23-16	TWOD_MIX_BACKGROUND_V	R/W	h0	B component color of background
31-24	--	--	--	reserved

106.1.1.36. TWOD_IRQ0_CTL

Table 6-203 .TWOD_IRQ0_CTL

Address: BASE + 0x178				
Bit	Name	Access	Reset Value	Description
15-12	--	--	--	reserved
19-18	--	--	--	reserved
20	TWOD_IRQ0_EN	R/W	b0	when ‘1’, irq0 is enabled. when ‘0’, irq0 is disabled. Note: this IRQ0 is high, not make sure AXI master have drain all data to fabric, only make sure DMA write buffer is empty.
31-21	--	--	--	reserved

106.1.1.37. TWOD_IRQ1_CTL

Table 6-204 .TWOD_IRQ1_CTL

Address: BASE + 0x17C				
Bit	Name	Access	Reset Value	Description
11-0				
15-12	--	--	--	reserved
19-18	--	--	--	reserved

20	TWOD_SCALE_ERR_IRQ1_EN	R/W	b0	when ‘1’, irq1 is enabled. when ‘0’, irq1 is disabled.
31-21	--	--	--	reserved

106.1.1.38. **TWOD_IRQ2_CTL**

Table 6-205 .TWOD_IRQ2_CTL

Address: BASE + 0x180				
Bit	Name	Access	Reset Value	Description
11-0				
15-12	--	--	--	reserved
19-18	--	--	--	reserved
20	TWOD_SCALE_ERR_IRQ2_EN	R/W	b0	when ‘1’, irq2 is enabled. when ‘0’, irq2 is disabled.
21	TWOD_SCALE_ERR_IRQ3_EN	R/W	b0	when ‘1’, irq3 is enabled. when ‘0’, irq3 is disabled
31-22				

106.1.1.39. **TWOD_MIX_IRQ**

Table 6-206 .TWOD_MIX_IRQ

Address: BASE + 0x184				
Bit	Name	Access	Reset Value	Description
0	TWOD_MIX_IRQ0	R	b0	TWOD_MIX_IRQ0 status, read only.
0	TWOD_MIX_IRQ0_CLR	W	b0	Write ‘1’ to clear TWOD_MIX_IRQ0, write only.
1	TWOD_MIX_IRQ1	R	b0	TWOD_MIX_IRQ1 status, read only.
1	TWOD_MIX_IRQ1_CLR	W	b0	Write ‘1’ to clear TWOD_MIX_IRQ1, write only.
2	TWOD_MIX_IRQ2	R	b0	TWOD_MIX_IRQ2 status, read only.
2	TWOD_MIX_IRQ2_CLR	W	b0	Write ‘1’ to clear TWOD_MIX_IRQ2, write only.

31-3	--	--	--	reserved

106.1.1.40. **TWOD_OSD_INFO**

Table 6-207 .TWOD_OSD_INFO

Address: BASE + 0x188				
Bit	Name	Access	Reset Value	Description
3-0	OSD_TYPE	R/W	h0	when 0: RGB-24bit when 1: Monochrome-8bit when 2: RGB-16bit (565) when 3: RGB-32bit
6-4	ALPHA_TYPE	R/W	h0	ALPHA mode: 0: Fix value defined by Register 1: Load from OSD_ALP_ADDR. 2: ALPHA on R channel 3: ALPHA on G channel 4: ALPHA on B channel 5: ALPHA on A channel 6: Pixel value as ALPHA(Only available when pixel format is set to MONO-8bit) 7: ALPHA_RGB as transparent pixel.
7	--	--	--	reserved
31-8	ALPHA_RGB	--	--	transparent pixel value in RGB24 Format

106.1.1.41. **TWOD_OSD_SIZE**

Table 6-208 .TWOD_OSD_SIZE

Address: BASE + 0x18c				
Bit	Name	Access	Reset Value	Description
15-0	OSD_WIDTH	R/W	h0	OSD image width (in unit of pixel)
31-16	OSD_HEIGHT	R/W	h0	OSD image height (in unit of pixel) Address: BASE + 0x1C8

106.1.1.42. TWOD_OSD_VLU_ADDR

Table 6-209 .TWOD_OSD_VLU_ADDR

Address: BASE + 0x190				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_OSD_VLU_ADDR	R/W	h0	OSD RGB image base address in DDR (should be 8byte aligned and must be start address of new pixel).

106.1.1.43. TWOD_OSD_ALP_ADDR

Table 6-210 .TWOD_OSD_ALP_ADDR

Address: BASE + 0x194				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_OSD_ALP_ADDR	R/W	h0	OSD ALPHA image base address in DDR (should be 8byte aligned and must be start address of new pixel).

106.1.1.44. TWOD_OSD_DMA_CTRL

Table 6-211 .TWOD_OSD_DMA_CTRL

Address: BASE + 0x198				
Bit	Name	Access	Reset Value	Description
3-0	DMA_REQUEST_LENGHT	R/W	h7	Beat number for DMA read request is [DMA request length] + 1.

5-4	DMA_MAP	R/W	h0	DMA data map manually in a d-word (32bit) by byte (8bit). 00: original order 10: 1-2-3-4 to 2-1-4-3 11: 1-2-3-4 to 4-3-2-1
6	RGB_REV	R/W	b0	0: mem[OSD_ADDR] is R 1: mem[OSD_ADDR] is B
31-7	--	--	--	reserved

106.1.1.45. **TWOD_OSD_STRIDE**

Table 6-212 .TWOD_OSD_STRIDE

Address: BASE + 0x19C				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_OSD_STRIDE	R/W	b0	OSD frame stride in pixel 8-byte unit for each pixel line
31-16	TWOD_MIX_ALP_STRIDE --	R/W--	B0--	ALPHA stride in unit of 8-byte for each pixel line

106.1.1.46. **TWOD_RECTANGLE_EN**

Table 6-213 .TWOD_RECTANGLE_EN

Address: BASE + 0x1A0				
Bit	Name	Access	Reset Value	Description
9-0	RECTANGLE_EN	R/W	h0	Rectangle 0~9 mix enable
31-10	--	--	--	reserved

106.1.1.47. **TWOD_RECTANGLE_DATA**

Table 6-214 .TWOD_RECTANGLE_DATA

Address: BASE + 0x1A4

Bit	Name	Access	Reset Value	Description
23-0	RECTANGLE_DATA	R/W	h0	Rectangle pixel color in YUV domain.
31-24	--	--	--	reserved

106.1.1.48. **TWOD_RECTANGLE0_POSITION**

Table 6-215 .TWOD_RECTANGLE0_POSITION

Address: BASE + 0x1A8				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.49. **TWOD_RECTANGLE0_LENGTH**

Table 6-216 .TWOD_RECTANGLE0_LENGTH

Address: BASE + 0x1AC				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.50. **TWOD_RECTANGLE1_POSITION**

Table 6-217 .TWOD_RECTANGLE1_POSITION

Address: BASE + 0x1B0				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.51. TWOD_RECTANGLE1_LENGTH

Table 6-218 .TWOD_RECTANGLE1_LENGTH

Address: BASE + 0x1B4				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.52. TWOD_RECTANGLE2_POSITION

Table 6-219 .TWOD_RECTANGLE2_POSITION

Address: BASE + 0x1B8				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.53. TWOD_RECTANGLE2_LENGTH

Table 6-220 .TWOD_RECTANGLE2_LENGTH

Address: BASE + 0x1BC				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.54. TWOD_RECTANGLE3_POSITION

Table 6-221 .TWOD_RECTANGLE3_POSITION

Address: BASE + 0x1C0

Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.55. TWOD_RECTANGLE3_LENGTH

Table 6-222 .TWOD_RECTANGLE3_LENGTH

Address: BASE + 0x1C4				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.56. TWOD_RECTANGLE4_POSITION

Table 6-223 .TWOD_RECTANGLE4_POSITION

Address: BASE + 0x1C8				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.57. TWOD_RECTANGLE4_LENGTH

Table 6-224 .TWOD_RECTANGLE4_LENGTH

Address: BASE + 0x1CC				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.

31-28	--	--	--	reserved

106.1.1.58. TWOD_RECTANGLE5_POSITION

Table 6-225 .TWOD_RECTANGLE5_POSITION

Address: BASE + 0x1D0				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.59. TWOD_RECTANGLE5_LENGTH

Table 6-226 .TWOD_RECTANGLE5_LENGTH

Address: BASE + 0x1D4				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.60. TWOD_RECTANGLE6_POSITION

Table 6-227 .TWOD_RECTANGLE6_POSITION

Address: BASE + 0x1D8				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.61. TWOD_RECTANGLE6_LENGTH

Table 6-228 .TWOD_RECTANGLE6_LENGTH

Address: BASE + 0x1DC				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.62. TWOD_RECTANGLE7_POSITION

Table 6-229 .TWOD_RECTANGLE7_POSITION

Address: BASE + 0x1E0				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.63. TWOD_RECTANGLE7_LENGTH

Table 6-230 .TWOD_RECTANGLE7_LENGTH

Address: BASE + 0x1E4				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.64. TWOD_RECTANGLE8_POSITION

Table 6-231 .TWOD_RECTANGLE8_POSITION

Address: BASE + 0x1E8

Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.65. **TWOD_RECTANGLE8_LENGTH**

Table 6-232 .TWOD_RECTANGLE8_LENGTH

Address: BASE + 0x1EC				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.
31-28	--	--	--	reserved

106.1.1.66. **TWOD_RECTANGLE9_POSITION**

Table 6-233 .TWOD_RECTANGLE9_POSITION

Address: BASE + 0x1F0				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE START X POSITION	R/W	h0	Rectangle start x position.
15-12	--	--	--	reserved
27-16	RECTANGLE START Y POSITION	R/W	h0	Rectangle start y position
30-28	--	--	--	reserved

106.1.1.67. **TWOD_RECTANGLE9_LENGTH**

Table 6-234 .TWOD_RECTANGLE9_LENGTH

Address: BASE + 0x1F4				
Bit	Name	Access	Reset Value	Description
11-0	RECTANGLE WIDTH	R/W	h0	Rectangle width.
15-12	--	--	--	reserved
27-16	RECTANGLE HEIGHT	R/W	h0	Rectangle height.

31-28	--	--	--	reserved

106.1.1.68. TWOD_OSD_RGB2YUV_CTL

Table 6-235 .TWOD_OSD_RGB2YUV_CTL

Address: BASE + 0x200				
Bit	Name	Access	Reset Value	Description
0	TWOD_OSD_RGB2YUV_EN	R/W	b1	set ‘1’, RGB -> YUV444 in OSD layer is enabled. set ‘0’, RGB -> YUV444 in OSD layer is disabled.
31-1	--	--	--	reserved

106.1.1.69. TWOD_OSD_RGB2YUV_COEFF00

Table 6-236 .TWOD_OSD_RGB2YUV_COEFF00

Address: BASE + 0x204				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF00	R/W	h132	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.70. TWOD_OSD_RGB2YUV_COEFF01

Table 6-237 .TWOD_OSD_RGB2YUV_COEFF01

Address: BASE + 0x208				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF01	R/W	h259	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.71. TWOD_OSD_RGB2YUV_COEFF02

Table 6-238 .TWOD_OSD_RGB2YUV_COEFF02

Address: BASE + 0x020C				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF02	R/W	h75	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.72. TWOD_OSD_RGB2YUV_COEFF03

Table 6-239 .TWOD_OSD_RGB2YUV_COEFF03

Address: BASE + 0x210				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF03	R/W	h0	YUV444 -> RGB offset value.
31-12	--	--	--	reserved

106.1.1.73. TWOD_OSD_RGB2YUV_COEFF10

Table 6-240 .TWOD_OSD_RGB2YUV_COEFF10

Address: BASE + 0x214				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF10	R/W	hf50	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.74. TWOD_OSD_RGB2YUV_COEFF11

Table 6-241 .TWOD_OSD_RGB2YUV_COEFF11

Address: BASE + 0x218				
Bit	Name	Access	Reset	Description

			Value	
11-0	TWOD_OSD_RGB2YUV_COEFF11	R/W	hea5	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.75. TWOD_OSD_RGB2YUV_COEFF12

Table 6-242 .TWOD_OSD_RGB2YUV_COEFF12

Address: BASE + 0x21C				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF12	R/W	h20b	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.76. TWOD_OSD_RGB2YUV_COEFF13

Table 6-243 .TWOD_OSD_RGB2YUV_COEFF13

Address: BASE + 0x220				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF13	R/W	h80	YUV444 -> RGB offset value.
31-12	--	--	--	reserved

106.1.1.77. TWOD_OSD_RGB2YUV_COEFF20

Table 6-244 .TWOD_OSD_RGB2YUV_COEFF20

Address: BASE + 0x224				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF20	R/W	h20b	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.78. TWOD_OSD_RGB2YUV_COEFF21

Table 6-245 .TWOD_OSD_RGB2YUV_COEFF21

Address: BASE + 0x228				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF21	R/W	he4a	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.79. TWOD_OSD_RGB2YUV_COEFF22

Table 6-246 .TWOD_OSD_RGB2YUV_COEFF22

Address: BASE + 0x22C				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF22	R/W	hfab	YUV444 -> RGB coefficient.
31-12	--	--	--	reserved

106.1.1.80. TWOD_OSD_RGB2YUV_COEFF23

Table 6-247 .TWOD_OSD_RGB2YUV_COEFF23

Address: BASE + 0x230				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_OSD_RGB2YUV_COEFF23	R/W	h80	YUV444 -> RGB offset value.
31-12	--	--	--	reserved

106.1.1.81. TWOD_OSD_RGBA_CTL

Table 6-248 .TWOD_OSD_RGBA_CTL

Address: BASE + 0x234				
Bit	Name	Access	Reset	Description

			Value	
0	vo_osd_rgba_en	R/W	h0	Ouput RGBA
31-12	--	--	--	reserved

106.1.1.82. TWOD_AXI_RST_CTL

Table 6-249 .TWOD_AXI_RST_CTL

Address: BASE + 0x240				
Bit	Name	Access	Reset Value	Description
1:0	TWOD_AXI_RST_CTL	R/W	h0	
31-12	--	--	--	reserved

106.1.1.83. TWOD_AXI_RST_STATUS

Table 6-250 .TWOD_AXI_RST_STATUS

Address: BASE + 0x244				
Bit	Name	Access	Reset Value	Description
7:0	TWOD_AXI_RST_STATUS	R/W	h0	
31-12	--	--	--	reserved

106.1.1.84. TWOD_MIX_OUT_CTRL

Table 6-251 .TWOD_MIX_OUT_CTRL

Address: BASE + 0x250				
Bit	Name	Access	Reset Value	Description
7-0	TWOD_MIX_OUT_FORMAT	R/W	H1	1: RGB24 2：YUV420 – UV 4：YUV420 - VU
15-8	--	--	--	reserved
20-16	twod_mix_out_uvblen_cfg	R/W	H8	
23-21	--	--	--	reserved
28-24	twod_mix_out_yblen_cfg	R/W	H8	
29	--	--	--	reserved
30	TWOD_MIX_OUT_DMAY_BU	RO	H0	1: Indicate DMA Y

	F_EMPTY			write buffer is empty
31	TWOD_MIX_OUT_DMAUV_BUF_EMPTY	RO	H0	1: Indicate DMA UV write buffer is empty

106.1.1.85. TWOD_MIX_OUT_START_ADDR0

Table 6-252 .TWOD_MIX_OUT_START_ADDR0

Address: BASE + 0x254				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_MIX_OUT_START_ADDR0	R/W	h0	RGB24/Y output address

106.1.1.86. TWOD_MIX_OUT_START_ADDR1

Table 6-253 .TWOD_MIX_OUT_START_ADDR1

Address: BASE + 0x258				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_MIX_OUT_START_ADDR1	R/W	h0	UV output address

106.1.1.87. TWOD_MIX_OUT_STRIDE

Table 6-254 .TWOD_MIX_OUT_STRIDE

Address: BASE + 0x25C				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_MIX_ADDR0_STRIDE	R/W	b0	TWOD_MIX_OUT_START_ADDR0's Stride, and unit is byte, and should be 8-byte aligned
31-16	TWOD_MIX_ADDR1_STRIDE	R/W	B0	TWOD_MIX_OUT_START_ADDR1's Stride, and unit is byte, and should be 8-byte aligned

106.1.1.88. TWOD_MIX_OUT_DMA_CTRL0

Table 6-255 .TWOD_MIX_OUT_DMA_CTRL0

Address: BASE + 0x260				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_MIX_DMA0_HSIZE	R/W	h0	Horizontal size in DDR, in double word (8-byte, actual value -1). As example, for image of 532x160, Y output, it is Ceiling[532/8]-1=66.
27-16	TWOD_MIX_DMA0_VSIZE	R/W	h0	Vertical line size (image line number - 1). for image of 532x160, Y output, it is 159
31-28	--	--	--	Reserved

106.1.1.89. TWOD_MIX_OUT_DMA_CTRL1

Table 6-256 .TWOD_MIX_OUT_DMA_CTRL1

Address: BASE + 0x264				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_MIX_DMA1_HSIZE	R/W	h0	Horizontal size in DDR, in double word (8-byte, actual value -1). As example, for image of 532x160, YUV420-UV output, it is Ceiling[532/8]-1=66.
27-16	TWOD_MIX_DMA1_VSIZE	R/W	h0	Vertical line size (image line number - 1). for image of 532x160, YUV420-UV output, it is 160/2-1=79
31-28	--	--	--	Reserved

106.1.1.90. TWOD_MIX_OUT_DMAY_LAST_ADDR

Table 6-257 .TWOD_MIX_OUT_DMAY_LAST_ADDR

Address: BASE + 0x268

Bit	Name	Access	Reset Value	Description
31-0	TWOD_MIX_OUT_DMAY_LAST_ADDR0	RO	h0	Last write address of the frame

106.1.1.91. **TWOD_MIX_OUT_DMAUV_LAST_ADDR**

Table 6-258 .TWOD_MIX_OUT_DMAUV_LAST_ADDR

Address: BASE + 0x26C				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_MIX_OUT_DMAUV_LADT_ADDR1	RO	h0	Last write address of the frame

106.1.1.92. **TWOD_CPY_DMA_SIZE**

Table 6-259 .TWOD_CPY_DMA_SIZE

Address: BASE + 0x280				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_CPY_DMA_HSIZE	R/W	h0	Horizontal size in DDR, in double word (8-byte, actual value -1). As example, for image of 532x160, Y output, it is Ceiling[532/8]-1=66.
27-16	TWOD_CPY_DMA_VSIZE	R/W	h0	Vertical line size (image line number - 1). for image of 532x160, Y output, it is 159
31-28	--	--	--	Reserved

106.1.1.93. **TWOD_CPY_RD_ADDR**

Table 6-260 .TWOD_CPY_RD_ADDR

Address: BASE + 0x284				
Bit	Name	Access	Reset Value	Description

31-0	TWOD_CPY_RD_ADDR	R/W	h0	Base address for data to be copied

106.1.1.94. **TWOD_CPY_RD_ STRIDE**

Table 6-261 .TWOD_CPY_RD_ STRIDE

Address: BASE + 0x288				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_CPY_RD_STRIDE	R/W	b0	unit is 8-byte, and should be 8-byte aligned

106.1.1.95. **TWOD_CPY_WR_ADDR**

Table 6-262 .TWOD_CPY_WR_ADDR

Address: BASE + 0x28C				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_CPY_WR_ADDR	R/W	h0	Base address for data to be stored

106.1.1.96. **TWOD_CPY_WR_ STRIDE**

Table 6-263 .TWOD_CPY_WR_ STRIDE

Address: BASE + 0x290				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_CPY_WR_STRIDE	R/W	b0	unit is byte, and should be 8-byte aligned

106.1.1.97. **TWOD_CPY_CTRL**

Table 6-264 .TWOD_CPY_CTRL

Address: BASE + 0x294				
Bit	Name	Access	Reset Value	Description
0	TWOD_CPY_EN	R/W	b0	Write 1 to enable TWOD copy;

15-12	--	--	--	reserved
16	TWOD_CPY_IRQ_EN	R/W	b0	when ‘1’, irq is enabled. when ‘0’, irq is disabled.
31-17	--	--	--	reserved

106.1.1.98. **TWOD_CPY_IRQ**

Table 6-265 .TWOD_CPY_IRQ

Address: BASE + 0x298				
Bit	Name	Access	Reset Value	Description
0	TWOD_CPY_IRQ	R	b0	TWOD_CPY_IRQ status, read only. Note: if this IRQ is high, not make sure AXI master have drain all data to fabric, only make sure internal DMA write buffer is empty.
0	TWOD_CPY_IRQ_CLR	W	b0	Write ‘1’ to clear TWOD_CPY_IRQ, write only.

106.1.1.99. **TWOD_FILL_P0_SIZE**

Table 6-266 .TWOD_FILL_P0_SIZE

Address: BASE + 0x2B0				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_P0_X_SIZE	R/W	h0	Horizontal size of fill. for size of 800x600， it is 800
15-12	--	--	--	Reserved
27-16	TWOD_FIL_P0_Y_SIZE	R/W	h0	Vertical size of fill. . for size of 800x600， it is 600
31-28	--	--	--	Reserved

106.1.1.100. **TWOD_FIL_P0_ STRIDE**

Table 6-267 .TWOD_FIL_P0_ STRIDE

Address: BASE + 0x2B4				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_FIL_P0_STRIDE	R/W	b0	unit is byte, and should be 8-byte aligned

106.1.1.101. **TWOD_FIL_P0_ADDR**

Table 6-268 .TWOD_FIL_P0_ADDR

Address: BASE + 0x2B8				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_START_P0_X_ADDR	R/W	h0	X coordinate of starting point
27-16	TWOD_FIL_START_P0_Y_ADDR	R/W	h0	Y coordinate of starting point

106.1.1.102. **TWOD_FILL_P1_SIZE**

Table 6-269 .TWOD_FILL_P1_SIZE

Address: BASE + 0x2BC				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_P1_X_SIZE	R/W	h0	Horizontal size of fill. for size of 800x600，it is 800
15-12	--	--	--	Reserved
27-16	TWOD_FIL_P1_Y_SIZE	R/W	h0	Vertical size of fill. . for size of 800x600，it is 600
31-28	--	--	--	Reserved

106.1.1.103. **TWOD_FIL_P1_ STRIDE**

Table 6-270 .TWOD_FIL_P1_ STRIDE

Address: BASE + 0x2C0

Bit	Name	Access	Reset Value	Description
15-0	TWOD_FIL_P1_STRIDE	R/W	b0	unit is byte, and should be 8-byte aligned

106.1.1.104. **TWOD_FIL_P1_ADDR**

Table 6-271 .TWOD_FIL_P1_ADDR

Address: BASE + 0x2C4				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_START_P1_X_ADDR	R/W	h0	X coordinate of starting point
27-16	TWOD_FIL_START_P1_Y_ADDR	R/W	h0	Y coordinate of starting point

106.1.1.105. **TWOD_FILL_P2_SIZE**

Table 6-272 .TWOD_FIL_P2_SIZE

Address: BASE + 0x2C8				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_P2_X_SIZE	R/W	h0	Horizontal size of fill. for size of 800x600，it is 800
15-12	--	--	--	Reserved
27-16	TWOD_FIL_P2_Y_SIZE	R/W	h0	Vertical size of fill. . for size of 800x600，it is 600
31-28	--	--	--	Reserved

106.1.1.106. **TWOD_FIL_P2_ STRIDE**

Table 6-273 .TWOD_FIL_P2_ STRIDE

Address: BASE + 0x2CC				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_FIL_P2_STRIDE	R/W	b0	unit is byte, and should be 8-byte aligned

106.1.1.107. TWOD_FIL_P2_ADDR

Table 6-274 .TWOD_FIL_P2_ADDR

Address: BASE + 0x2D0				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_START_P2_X_ADDR	R/W	h0	X coordinate of starting point
27-16	TWOD_FIL_START_P2_Y_ADDR	R/W	h0	Y coordinate of starting point

106.1.1.108. TWOD_FILL_P3_SIZE

Table 6-275 .TWOD_FIL_P3_SIZE

Address: BASE + 0x2BD4				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_P3_X_SIZE	R/W	h0	Horizontal size of fill. for size of 800x600， it is 800
15-12	--	--	--	Reserved
27-16	TWOD_FIL_P3_Y_SIZE	R/W	h0	Vertical size of fill. . for size of 800x600， it is 600
31-28	--	--	--	Reserved

106.1.1.109. TWOD_FIL_P3_ STRIDE

Table 6-276 .TWOD_FIL_P3_ STRIDE

Address: BASE + 0x2D8				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_FIL_P0_STRIDE	R/W	b0	unit is byte, and should be 8-byte aligned

106.1.1.110. TWOD_FIL_P3_ADDR

Table 6-277 .TWOD_FIL_P3_ADDR

Address: BASE + 0x2DC

Bit	Name	Access	Reset Value	Description
11-0	TWOD_FIL_START_P3_X_ADDR	R/W	h0	X coordinate of starting point
27-16	TWOD_FIL_START_P3_Y_ADDR	R/W	h0	Y coordinate of starting point

106.1.1.111. **TWOD_FIL_DATA**

Table 6-278 .TWOD_FIL_P3_ADDR

Address: BASE + 0x2E0				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_FIL_DATA	R/W	b0	Data for each pixel to fill, bit7-0 is low address.

106.1.1.112. **TWOD_FIL_CTRL**

Table 6-279 .TWOD_FIL_CTRL

Address: BASE + 0x2E4				
Bit	Name	Access	Reset Value	Description
0	TWOD_FIL_EN	R/W	b0	Write 1 to enable TWOD fill; and Need software to clean this bit to 0 if it is 1 before enable fill task.
3-1	--	--	--	reserved
7-4	TWOD_FIL_PSIZE	R/W	B0	Byte number for each pixel, for example, if format is RGB24, this value should be 4'd3, and fill value is TWOD_FIL_DATA[23:0]
15-8	--	--	--	reserved
16	TWOD_FIL_IRQ_EN	R/W	b0	when '1', irq is enabled. when '0', irq is disabled.
31-17	--	--	--	reserved

106.1.1.113. **TWOD_FIL_IRQ**

Table 6-280 .TWOD_FIL_IRQ

Address: BASE + 0x2E8				
Bit	Name	Access	Reset Value	Description
0	TWOD_FIL_IRQ	R	b0	TWOD_FIL_IRQ status, read only. Note: if this IRQ is high, not make sure AXI master have drain all data to fabric, only make sure internal DMA write buffer is empty.
0	TWOD_FIL_IRQ_CLR	W	b0	Write '1' to clear TWOD_FIL_IRQ, write only.

106.1.1.114. TWOD_ENDIAN_SWAP_CTRL

Table 6-281 .TWOD_ENDIAN_SWAP_CTRL

Address: BASE + 0x2EC				
Bit	Name	Access	Reset Value	Description
0	TWOD_16BITS_ENDIAN_SWAP	R/W	b0	When '0' disable endian swap. When '1' enable endian swap.
1	TWOD_8BITS_ENDIAN_SWAP	R/W	b0	When '0' disable endian swap. When '1' enable endian swap.
31-2	--	--	--	reserved

106.1.1.115. TWOD_ROTATE_DMA_SIZE

Table 6-282 .TWOD_ROTATE_DMA_SIZE

Address: BASE + 0x2F0				
Bit	Name	Access	Reset Value	Description
11-0	TWOD_ROTATE_DMA_HSIZE	R/W	h0	Horizontal size in DDR, in double word (8-byte, actual

				value -1). As example, for image of 532x160, Y output, it is Ceiling[532/8]-1=66.
27-16	TWOD_ROTATE_DMA_VSIZE	R/W	h0	Vertical line size (image line number - 1). for image of 532x160, Y output, it is 159
31-28	--	--	--	reserved

106.1.1.116. **TWOD_ROTATE_RD_ADDR**

Table 6-283 .TWOD_ROTATE_RD_ADDR

Address: BASE + 0x2F4				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_TRANS_RD_ADDR	R/W	h0	Base address for data to be rotated.

106.1.1.117. **TWOD_ROTATE_RD_STRIDE**

Table 6-284 .TWOD_ROTATE_RD_STRIDE

Address: BASE + 0x2F0				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_ROTATE_RD_STRIDE	R/W	b0	unit is 8-byte, and should be 8-byte aligned
31-16	--	--	--	reserved

106.1.1.118. **TWOD_ROTATE_WR_ADDR**

Table 6-285 .TWOD_ROTATE_WR_ADDR

Address: BASE + 0x2F0				
Bit	Name	Access	Reset Value	Description

31-0	TWOD_ROTATE_WR_ADDR	R/W	h0	Base address for data to be stored

106.1.1.119. **TWOD_ROTATE_WR_STRIDE**

Table 6-286 .TWOD_ROTATE_WR_STRIDE

Address: BASE + 0x2F0				
Bit	Name	Access	Reset Value	Description
15-0	TWOD_ROTATE_WR_STRIDE	R/W	b0	unit is byte, and should be 8-byte aligned
31-2	--	--	--	reserved

106.1.1.120. **TWOD_ROTATE_CTRL**

Table 6-287 .TWOD_ROTATE_CTRL

Address: BASE + 0x2F0				
Bit	Name	Access	Reset Value	Description
0	TWOD_ROTATE_EN	R/W	b0	Write 1 to enable TWOD rotate;
15-12	--	--	--	reserved
9-8	trans_yuv_sel	R/W	b0	when ‘0’, Y channel is enabled. when ‘1’, UV channel is enabled.
15-12	--	--	--	reserved
16	TWOD_ROTATE_IRQ_EN	R/W	b0	when ‘1’, irq is enabled. when ‘0’, irq is disabled.

106.1.1.121. **TWOD_ROTATE_ANGLE**

Table 6-288 .TWOD_ROTATE_ANGLE

Address: BASE + 0x2F4				
Bit	Name	Access	Reset Value	Description
1-0	TWOD_ROTATE_ANGLE	R/W	b0	When ‘0’ not rotate.

				When‘1’ 90 degree. When ‘2’ 180 degree. When ‘3’ 270 degree.
31-2	--	--	--	reserved

106.1.1.122. **TWOD_ROTATE_TOTAL_SIZE**

Table 6-289 .TWOD_ROTATE_TOTAL_SIZE

Address: BASE + 0x2F4				
Bit	Name	Access	Reset Value	Description
31-0	TWOD_ROTATE_TOTAL_SIZE	R/W	b0	‘Y’ is H*V ‘UV’ is H*V/2

106.2. Video Out Controller

106.2.1. Overview

The video output unit generates the picture that will be displayed, which are from the video layer and the OSD layer. The unit support up to 4 display plane, which are back-ground layer, 3 video layer and one OSD layer. The size and position of planes could be configurable. The layer 0 video layer could scale up or down input image at horizontal and vertical direction, independently.

The video output unit reads image data from DDR by AXI interface. After mixed with each other, the mixed picture are fed out through DPI interface.

106.2.2. Features

- Support Up to 4 planes
- Support Supports all common picture formats: 1080p, 1080i, 720p, 576i, 480i.
- Layer 0 & Layer 1 & Layer 2:
  - Support 2-plane, YUV420 & YUV422 data format input
  - Support configurable start pixel & start line
- OSD layer support 1 or 2 plane ARGB format:

■ RGB-24bit with separate alpha channel
■ RGB-16bit (565) with separate alpha channel
■ ARGB-32bit
■ ARGB4444
■ ARGB1555
■ Monochrome(8bit)

- Display mix:

■ OSD layer support configurable global alpha blending or pixel by pixel alpha blending
■ Video layer only support configurable global alpha blending

- Up to 3 interrupt, support configurable line position to generate interrupt
- Support configurable color space conversion
- Support gamma correction on RGB data before output

106.2.3. **Function Description**

106.2.3.1. **Block Diagram**

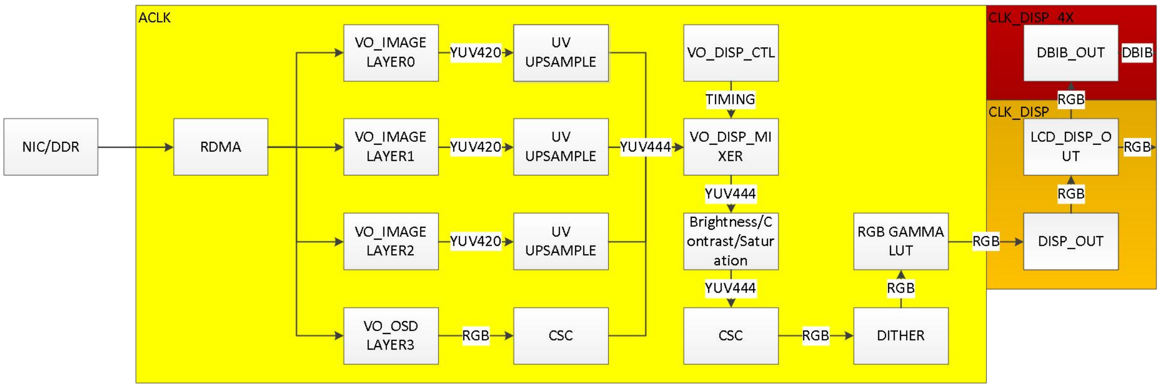


Figure 6- 6 Video output system top diagram

106.2.3.2. Display layer

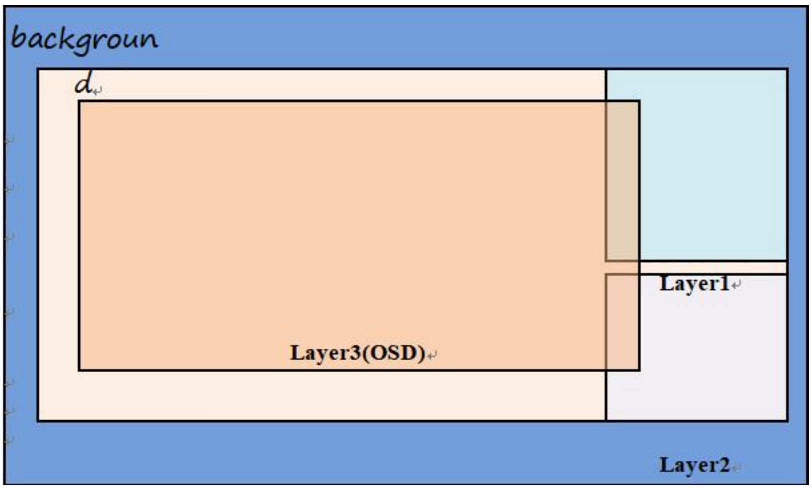


Figure 6- 7 Display layer overlap

106.2.3.3. Layer 0&1&2

- Supports all common picture formats (i.e. 1080p, 1080i, 720p, 576i, 480i)
- Support YUV 4:2:0 & 4:2:2 input format
- 4:2:0->4:4:4 & 4:2:2->4:4:4 up-sampling conversion
- 8-bit YUV 4:4:4 output
- Programmable display size
- Programmable display position
- Global alpha blending

106.2.3.4. OSD Layer

OSD layer will get OSD data from external memory and send to mix block to do merge with other layer. OSD data include the value of Alpha-channel and RGB channel; you can refer register description of OSD to know more about OSD data type and configuration.

Support below features

- Supports all common OSD data formats:
  - RGB-24bit with separate alpha channel
  - RGB-16bit (565) with separate alpha channel
  - ARGB-32bit
  - ARGB4444
  - ARGB1555
  - Monochrome(8bit)
- Programmable display size

- Programmable display position
- Support OSD frame crop by program OSD start-address (should be 8byte aligned and must be start address of new pixel), OSD stride (in unit of pixel) and display size (OSD size in unit of pixel).
- 8-bit output for each ARGB to mix with other layer
- Global or pixel-by-pixel alpha blending

106.2.3.5. **Display Mixer**

The display mixer mixes all layers to a final picture. It allows also a layer reordering which gives a maximum flexibility to the display generation. Each mixing input has also a programmable alpha blending value.
The display controller is signaling the window of each plane. Once the plane becomes active, the pixel is mixed with the input value of the mixer and the alpha value that's coming together with the pixel information.
Display mixer has serveral internal layers 0~3. Internal layer0 is the first to do blending, and then internal layer1, and so on. Each internal mixer layer source could be selected from 4 external layer0~3, such that blending sequence could be configurable.
Each mixing input has also a programmable alpha blending value.

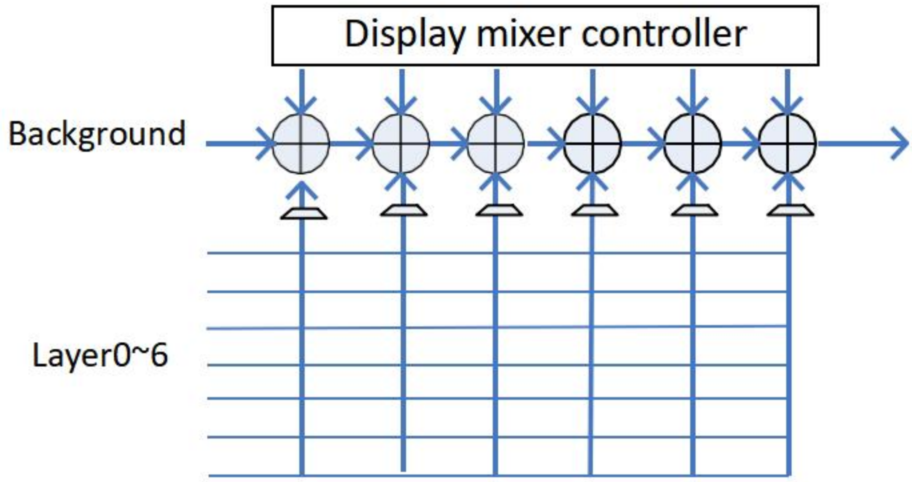


Figure 6- 8 Display mixer structure

106.2.3.6. **Color Space Conversion**

The CSC(color space conversion) has one conversion matrix-es plus look up tables, which could function as RGB -> YUV or YUV -> RGB by configure corresponding set conversion matrix.

$$\begin{pmatrix} P_{1out} \\ P_{2out} \\ P_{3out} \end{pmatrix} = \left(\begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \bullet \begin{pmatrix} P_{1in} \\ P_{2in} \\ P_{3in} \end{pmatrix} + \begin{pmatrix} 128 \\ 128 \\ 128 \end{pmatrix} \right) \bullet \frac{1}{256} + \begin{pmatrix} M_{14} \\ M_{24} \\ M_{34} \end{pmatrix}$$

106.2.3.1. BCS module

The BCS (Brightness Contrast Saturation) module allow some picture improvements like color saturation, contrast or brightness adjustment.

106.2.3.1. RGB Gamma module

The RGB Gamma module use 3 separate 256 entry LUT to allow Gamma adjustment before display output.

106.2.3.2. Display Controller

The display controller is the timing generator for all signals required within the video output unit. It has a programmable horizontal and vertical counter to be able to support all standards which are required. It’s main function is to generate the Hsync, Vsync and DE signals for the DPI interface as well the horizontal and vertical position of all layers.

Below features are supported:

- Timing generator for all video output relevant features
- Synchronization signal output
- IRQ generation
- Support interlaced output

When interlaced output, the last line length of either field or only the second field, could be shortened. This mechanism can be used to adapt the frame rate to the encoder frame rate, when horizontal and vertical size do not match exactly with the input rate. A possible application would be if an LCD panel is connected directly to the chip. It cannot be used for the HDMI IF, because all lines need to have the exact same length, here.

Up to 3 interrupts are supported, each interrupt is of configurable features:

- Interrupt line position is programmable
- Interrupt field is programmable when interlaced output
- Frame interrupts is supported
- interrupt is level triggered

106.2.4. Programming Guide

106.2.5. Register Summary

There are 4 address space block for VO_sys sub-sytem.

- register block: BASE + 0000 ~ BASE + 7FFF
- R gamma table block: BASE + C000 ~ BASE + C3FF
- G gamma table block: BASE + C400 ~ BASE + C7FF
- B gamma table block: BASE + C800 ~ BASE + CBFF

Table 6- 290 Video_out Register Summary

me	fset address	scription
)_SOFT_RST_CTL	)000	b-system soft reset
)_TABLE_CLK_CTL	)004	b-system clock selection
)_DMA_ARB_MODE	)008	I master arbiter mode
)_DMA_WEIGHT_WR0	)00C	served
)_DMA_WEIGHT_WR1	)010	served
)_DMA_WEIGHT_RD0	)014	3 channel read priority weight
)_DMA_WEIGHT_RD1	)018	7 channel read priority weight
)_DMA_PRIORITY_WR	)01C	served
)_DMA_PRIORITY_RD	)020	8 channel read priority
)_DMA_ID_RD	)024	
)_DMA_ID_WR	)028	served
)_LAYER0_BD_CTL	)038	yer 0 bandwidth control
)_LAYER1_BD_CTL	)03C	yer 1 bandwidth control
)_LAYER2_BD_CTL	)040	yer 2 bandwidth control
)_LAYER3_BD_CTL	)044	yer3(osd layer) bandwidth control
)_DISP_XZONE_CTL	)048	ckground layer horizontal start & stop
)_DISP_YZONE_CTL	)04C	ckground layer vertical start & stop
)_DISP_LAYER0_XCTL	)050	yer0 horizontal start & stop
)_DISP_LAYER0_YCTL	)054	yer0 vertical start & stop
)_DISP_LAYER1_XCTL	)058	yer1 horizontal start & stop
)_DISP_LAYER1_YCTL	)05C	yer1 vertical start & stop
)_DISP_LAYER2_XCTL	)060	yer2 horizontal start & stop
)_DISP_LAYER2_YCTL	)064	yer2 vertical start & stop
)_DISP_LAYER3_XCTL	)068	yer3(osd layer) horizontal start & stop
)_DISP_LAYER3_YCTL	)06C	yer3(osd layer) vertical start & stop
)_DISP_HSYNC_CTL	)088	splay hsync start & stop position
)_DISP_HSYNC1_CTL	)08C	splay horizontal position of VSYN1

)_DISP_VSYNC1_CTL	)090	splay vertical position of VSYNC1
)_DISP_HSYNC2_CTL	)094	splay horizontal position of VSYN2
)_DISP_VSYNC2_CTL	)098	splay vertical position of VSYNC2
)_DISP_CTL	)09C	splay control
)_DISP_ENABLE	)0A0	splay output enable
)_DISP_SIZE	)0A4	splay frame horizontal & vertical size
)_LAYER0_CTL	)0C8	yer0 enable
)_LAYER0_IMG_IN_BADDR_Y	)0CC	yer0 in image y base address
)_LAYER0_IMG_IN_BADDR_UV	)0D0	yer0 in image uv base address
)_LAYER0_IMG_IN_HOFFSET	)0D4	yer0 in image horizontal offset(in 8 pixel)
)_LAYER0_IMG_IN_VOFFSET	)0D8	yer0 in image vertical offset (in 8 pixel)
)_LAYER0_IMG_IN_BLENTH	)0DC	yer0 burst length of DMA access
)_LAYER0_IMG_IN_SIZE	)0E0	yer0 in image horizontal size(in 8 pixel) & vertical size(in line
)_LAYER0_IMG_IN_PIX_SIZE	)0E4	yer0 in image horizontal size(in 1 pixel) & vertical size(in line
)_LAYER1_CTL	)0E8	yer1 enable
)_LAYER1_IMG_IN_BADDR_Y	)0EC	yer1 in image y base address
)_LAYER1_IMG_IN_BADDR_UV	)0F0	yer1 in image uv base address
)_LAYER1_IMG_IN_HOFFSET	)0F4	yer1 in image horizontal offset(in 8 pixel)
)_LAYER1_IMG_IN_HOFFSET	)0F8	yer1 in image vertical offset(in 8 pixel)
)_LAYER1_IMG_IN_BLENTH	)0FC	yer1 burst length of DMA access
)_LAYER1_IMG_IN_SIZE	)100	yer1 in image horizontal size(in 8 pixel) & vertical size(in line
)_LAYER1_IMG_IN_PIX_SIZE	)104	yer1 in image horizontal size(in 1 pixel) & vertical size(in line
)_LAYER2_CTL	)108	yer2 enable
)_LAYER2_IMG_IN_BADDR_Y	)10C	yer2 in image y base address
)_LAYER2_IMG_IN_BADDR_UV	)110	yer2 in image uv base address
)_LAYER2_IMG_IN_OFFSET	)114	yer2 in image horizontal offset(in 8 pixel)
)_LAYER2_IMG_IN_OFFSET	)118	yer2 in image vertical offset(in 8 pixel)
)_LAYER2_IMG_IN_BLENTH	)11C	yer2 burst length of DMA access
)_LAYER2_IMG_IN_SIZE	)120	yer2 in image horizontal size(in 8 pixel) & vertical size(in line
)_LAYER2_IMG_IN_PIX_SIZE	)124	yer2 in image horizontal size(in 1 pixel) & vertical size(in line
)_DISP_YUV2RGB_CTL	)128	V2RGB conversion enable
)_DISP_YUV2RGB_COEFF00	)12C	V2RGB coefficient
)_DISP_YUV2RGB_COEFF01	)130	V2RGB coefficient
)_DISP_YUV2RGB_COEFF02	)134	V2RGB coefficient
)_DISP_YUV2RGB_COEFF03	)138	V2RGB coefficient
)_DISP_YUV2RGB_COEFF10	)13C	V2RGB coefficient
)_DISP_YUV2RGB_COEFF11	)140	V2RGB coefficient
)_DISP_YUV2RGB_COEFF12	)144	V2RGB coefficient
)_DISP_YUV2RGB_COEFF13	)148	V2RGB coefficient
)_DISP_YUV2RGB_COEFF20	)14C	V2RGB coefficient

)_DISP_YUV2RGB_COEFF21	)150	JV2RGB coefficient
)_DISP_YUV2RGB_COEFF22	)154	JV2RGB coefficient
)_DISP_YUV2RGB_COEFF23	)158	JV2RGB coefficient
)_DISP_DITH_CTL	)15C	splay dither enable
)_DISP_MIX_LAYER_GLB_EN	)160	splay mix alpha source selection
)_DISP_MIX_LAYER_GLB_ALPHA0	)164	yer0~2 & osd layer global alpha
)_DISP_MIX_LAYER_GLB_ALPHA1	)168	served
)_DISP_MIX_SEL	)16C	splay mixing layer sequence selecton
)_DISP_BACKGROUND	)170	ckground RGB color value
)_DISP_CLUT_CTL	)174	mma correction enable
)_DISP_IRQ0_CTL	)178	splay irq0 control
)_DISP_IRQ1_CTL	)17C	splay irq1 control
)_DISP_IRQ2_CTL	)180	splay irq2 control
)_DISP_IRQ	)184	splay irq clear & status
)_DISP_OSD_INFO	)1C0	SD image data format
)_DISP_OSD_SIZE	)1C4	SD image size
)_DISP_OSD_VLU_ADDR	)1C8	SD RGB base address
)_DISP_OSD_ALP_ADDR	)1CC	SD alpha base address
)_DISP_OSD_DMA_CTRL	)1D0	SD DMA control
)_DISP_OSD_STRIDE	)1D4	SD frame address space stride
)_OSD_YUV2RGB_CTL	)200	SD layer RGB2YUV enable
)_OSD_YUV2RGB_COEFF00	)204	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF01	)208	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF02	)20C	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF03	)210	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF10	)214	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF11	)218	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF12	)21C	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF13	)220	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF20	)224	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF21	)228	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF22	)22C	GB2YUV coefficient
)_OSD_YUV2RGB_COEFF23	)230	GB2YUV coefficient
)_TEST_MODE	)234	stpattern set
)_DATA_ENDIAN_SWAP	)238	dian swap
)_TRANS_MODEL	)23C	ans angle
)DR_VO_AXI_RST_CTL	)240	(I bus reset
)DR_VO_AXI_RST_STATUS	0244	(I bus read channel status
)_LCD_CTL	0254	
)DR_VO_DBIB_CTL	0258	

DR_VO_DBIB_CFG	025C	
DR_VO_DBIB_CMD	0260	
DR_VO_DBIB_STATUS	0264	
DR_VO_DBIB_RDATA	0268	
DR_VO_RECTANGLE_EN	0280	
DR_VO_RECTANGLE_DATA	0284	
_RECTANGLE0_POSITION	0288	ctangle start position
_RECTANGLE0_LENGTH	028C	ctangle Length and width
_RECTANGLE1_POSITION	0290	ctangle start position
_RECTANGLE1_LENGTH	0294	ctangle Length and width
_RECTANGLE2_POSITION	0298	ctangle start position
_RECTANGLE2_LENGTH	029C	ctangle Length and width
_RECTANGLE3_POSITION	02A0	ctangle start position
_RECTANGLE3_LENGTH	02A4	ctangle Length and width
_RECTANGLE4_POSITION	02A8	ctangle start position
_RECTANGLE4_LENGTH	02AC	ctangle Length and width
_RECTANGLE5_POSITION	02B0	ctangle start position
_RECTANGLE5_LENGTH	02B4	ctangle Length and width
_RECTANGLE6_POSITION	02B8	ctangle start position
_RECTANGLE6_LENGTH	02BC	ctangle Length and width
_RECTANGLE7_POSITION	02C0	ctangle start position
_RECTANGLE7_LENGTH	02C4	ctangle Length and width
_RECTANGLE8_POSITION	02C8	ctangle start position
_RECTANGLE8_LENGTH	02CC	ctangle Length and width
_RECTANGLE9_POSITION	02D0	ctangle start position
_RECTANGLE9_LENGTH	02D4	ctangle Length and width

106.2.6. Register Description

106.2.6.1. VO_SOFT_RST_CTL

dress: BASE + 0x000				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_SF_RST_DISP	R/W	b0	en ‘1’, reset display clock domain logic
	_SF_RST_VO	R/W	b0	en ‘1’, reset axi clock domain logic
	_SF_RST_AHB	R/W	b0	en ‘1’, reset ahb clock domain logic
31-	--	--	--	erved

106.2.6.2. VO_TABLE_CLK_CTL

dress: BASE + 0x004				
Bit	Name	Access	Reset Value	cription
	_GAMMA_CLK_ G	R/W	b0	en '1', clock of gamma table select clk_ahb, else select clk_axi
	_VSCALE_CLK_ G	R/W	b0	en '1', clock of gamma table select clk_ahb, else select clk_axi
	_HSCALE_CLK_ G	R/W	b0	en '1', clock of gamma table select clk_ahb, else select clk_axi
31-	--	--	--	erved

106.2.6.3. VO_DMA_ARB_MODE

dress: BASE + 0x008				
Bit	Name	Access	Reset Value	cription
1-	_DMA_RAB_MO DE	R/W	b0	: read channel arbiter mode. en '0', use priority configured by cpu When '1', use weight+round-robin : write channel arbiter mode. en '0', use priority configured by cpu When '1', use weight+round-robin
31-		--	--	erved

106.2.6.4. VO_DMA_WEIGHT_RD0

dress: BASE + 0x014				
Bit	Name	Access	Reset Value	cription
7-	_DMA_WEIGHT	R/W	hff	nel 0 read weight
15-	:AD0	R/W	hff	nel 1 read weight
23-1		R/W	hff	nel 2 read weight
31-2		R/W	hff	nel 3 read weight

106.2.6.5. VO_DMA_WEIGHT_RD1

dress: BASE + 0x018				
Bit	Name	Access	Reset Value	cription
7-	_DMA_WEIGHT	R/W	hff	nel 4 read weight

15-	:AD1	R/W	hff	nel 5 read weight
23-1		R/W	hff	nel 6 read weight
31-2		R/W	hff	nel 7 read weight

106.2.6.6. VO_DMA_PRIORITY_RD

dress: BASE + 0x020				
Bit	**Name**	**Access**	**Reset Value**	**cription**
3-	_DMA_PRIORIT Y_RD	R/W	h0	nel 0 read priority
7-		R/W	h1	nel 1 read priority
11-		R/W	h2	nel 2 read priority
15-1		R/W	h3	nel 3 read priority
19-1		R/W	h4	nel 4 read priority
23-2		R/W	h5	nel 5 read priority
27-2		R/W	h6	nel 6 read priority
31-2		R/W	h7	nel 7 read priority

106.2.6.7. VO_DMA_ID_RD

dress: BASE + 0x024				
Bit	**Name**	**Access**	**Reset Value**	**cription**
3-	_DMA_ID_RD	R/W	h7	nel 0 read ID
7-		R/W	h6	nel 1 read ID
11-		R/W	h5	nel 2 read ID
15-1		R/W	h4	nel 3 read ID
19-1		R/W	h3	nel 4 read ID
23-20		R/W	h2	nel 5 read ID
27-2		R/W	h1	nel 6 read ID
31-2		R/W	h0	nel 7 read ID

106.2.6.8. VO_LAYER0_BD_CTL

dress: BASE + 0x038				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_LAYER0_LIMIT_EN	R/W	b1	en ‘1’, layer0 DMA access limit enable
7-		--	--	erved
12-	_LAYER0_OUTSTAND	R/W	h7	er0 DMA access outstanding number(actual outstand value – 1)
31-1	--	--	--	rved

106.2.6.9. VO_LAYER1_BD_CTL

dress: BASE + 0x03C				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_LAYER1_LIMIT I	R/W	b1	en '1', layer1 DMA access limit enable
7-		--	--	erved
12-	_LAYER1_OUTS ID	R/W	h7	er1 DMA access outstanding number(actual outstand value – 1)
31-1	--	--	--	rved

106.2.6.10. VO_LAYER2_BD_CTL

dress: BASE + 0x040				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_LAYER2_LIMIT I	R/W	b1	en '1', layer2 DMA access limit enable
7-		--	--	erved
12-	_LAYER2_OUTS ID	R/W	h7	er2 DMA access outstanding number(actual outstand value – 1)
31-1	--	--	--	rved

106.2.6.11. VO_LAYER3_BD_CTL

dress: BASE + 0x044				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_LAYER3_LIMIT I	R/W	b1	en '1', layer3 DMA access limit enable
7-		--	--	erved
12-	_LAYER3_OUTS ID	R/W	h7	er3 DMA access outstanding number(actual outstand value – 1)
31-1	--	--	--	rved

106.2.6.12. VO_DISP_XZONE_CTL

dress: BASE + 0x048

Bit	Name	ess	et Value	cription
11:	_DISP_ZONE_X ɅRT	/	h0	ɔlay active zone horizontal start pixel address(min pixel address)
15-1				ɛrved
27-1	_DISP_ZONE_X ɔP	/	h0	ɔlay active zone horizontal stop pixel address(min pixel address)
32-2				ɛrved

106.2.6.13. VO_DISP_YZONE_CTL

ɖress: BASE + 0x04C				
Bit	**Name**	**ess**	**et Value**	**cription**
11:	_DISP_ZONE_Y ɅRT	/	h0	ɔlay active zone vertical start pixel address(min pixel address is
15-1				ɛrved
27-1	_DISP_ZONE_Y ɔP	/	h0	ɔlay active zone vertical stop pixel address(min pixel address is
32-2				ɛrved

106.2.6.14. VO_DISP_LAYER0_XCTL

ɖress: BASE + 0x050				
Bit	**Name**	**ess**	**et Value**	**cription**
11:	_DISP_LAYER0_ ΓART	/	h0	ɔlay layer0 horizontal start pixel address(min pixel address is 1)
15-1				ɛrved
27-1	_DISP_LAYER0_ ΓOP	/	h0	ɔlay layer0 horizontal stop pixel address(min pixel address is 1)
32-2				ɛrved

106.2.6.15. VO_DISP_LAYER0_YCTL

ɖress: BASE + 0x054				
Bit	**Name**	**ess**	**et Value**	**cription**
11:	_DISP_LAYER0_ ΓART	/	h0	ɔlay layer0 vertical start pixel address(min pixel address is 1)
15-1				ɛrved

27-1	_DISP_LAYER0_ 'OP	/	h0	play layer0 vertical stop pixel address(min pixel address is 1)
32-2				erved

106.2.6.16. VO_DISP_LAYER1_XCTL

dress: BASE + 0x058				
Bit	**Name**	**ess**	**et Value**	**cription**
11:	_DISP_LAYER1_ 'ART	/	h0	play layer1 horizontal start pixel address(min pixel address is 1)
15-1				erved
27-1	_DISP_LAYER1_ 'OP	/	h0	play layer1 horizontal stop pixel address(min pixel address is 1)
32-2				erved

106.2.6.17. VO_DISP_LAYER1_YCTL

dress: BASE + 0x05C				
Bit	**Name**	**ess**	**et Value**	**cription**
11:	_DISP_LAYER1_ 'ART	/	h0	play layer1 vertical start pixel address(min pixel address is 1)
15-1				erved
27-1	_DISP_LAYER1_ 'OP	/	h0	play layer1 vertical stop pixel address(min pixel address is 1)
32-2				erved

106.2.6.18. VO_DISP_LAYER2_XCTL

dress: BASE + 0x060				
Bit	**Name**	**ess**	**et Value**	**cription**
11:	_DISP_LAYER2_ 'ART	/	h0	play layer2 horizontal start pixel address(min pixel address is 1)
15-1				erved
27-1	_DISP_LAYER2_ 'OP	/	h0	play layer2 horizontal stop pixel address(min pixel address is 1)
32-2				erved

106.2.6.19. VO_DISP_LAYER2_YCTL

dress: BASE + 0x064				
Bit	Name	ess	et Value	cription
11:	_DISP_LAYER2_ 'ART	/	h0	play layer2 vertical start pixel address(min pixel address is 1)
15-1				erved
27-1	_DISP_LAYER2_ 'OP	/	h0	play layer2 vertical stop pixel address(min pixel address is 1)
32-2				erved

106.2.6.20. VO_DISP_LAYER3_XCTL

dress: BASE + 0x068				
Bit	Name	ess	et Value	cription
11:	_DISP_LAYER3_ 'ART	/	h0	play layer3(OSD layer) horizontal start pixel address(min pixel ress is 1)
15-1				erved
27-1	_DISP_LAYER3_ 'OP	/	h0	play layer3(OSD layer) horizontal stop pixel address(min pixel ress is 1)
32-2				erved

106.2.6.21. VO_DISP_LAYER3_YCTL

dress: BASE + 0x06C				
Bit	Name	ess	et Value	cription
11:	_DISP_LAYER3_ 'ART	/	h0	play layer3(OSD layer) vertical start pixel address(min pixel ress is 1)
15-1				erved
27-1	_DISP_LAYER3_ 'OP	/	h0	play layer3(OSD layer) vertical stop pixel address(min pixel ress is 1)
32-2				erved

106.2.6.22. VO_DISP_HSYNC_CTL

dress: BASE + 0x088				
Bit	Name	ess	et Value	cription

11-	_DISP_HSYNC_ ART	/	h1	zontal start pixel position of Hsync(min pixel address is 1)
15-1				erved
27-1	_DISP_HSYNC_ OP	/	h8	zontal stop pixel position of Hsync(min pixel address is 1)
31-2				erved

106.2.6.23. VO_DISP_HSYNC1_CTL

dress: BASE + 0x08C				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	_DISP_HSYNC1 ART	/	h0	zontal start pixel position of Vsync1(min pixel address is 1)
15-1				erved
27-1	_DISP_HSYNC1 OP	/	h0	zontal stop pixel position of Vsync1(min pixel address is 1)
31-2				rved

106.2.6.24. VO_DISP_VSYNC1_CTL

dress: BASE + 0x090				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	_DISP_VSYNC1 ART	/	h0	ical start line position of Vsync1(min line address is)
15-1				erved
27-1	_DISP_VSYNC1 OP	/	h0	ical stop line position of Vsync1(min line address is)
31-2				erved

106.2.6.25. VO_DISP_HSYNC2_CTL

dress: BASE + 0x094				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	_DISP_HSYNC2 ART	/	h0	zontal start pixel position of Vsync2(min pixel address is 1)
15-1				erved
27-1	_DISP_HSYNC2	/	h0	zontal stop pixel position of Vsync2(min pixel address is 1)

	OP			
31-2				rved

106.2.6.26. VO_DISP_VSYNC2_CTL

dress: BASE + 0x098				
Bit	Name	ess	et Value	cription
11-	DISP_VSYNC2 ART	/	h0	ical start line position of Vsync2(min line address is DISP_VCNT_BIT0)
15-1				erved
27-1	DISP_VSYNC2 OP	/	h0	ical stop line position of Vsync2(min line address is DISP_VCNT_BIT0)
31-2				erved

106.2.6.27. VO_DISP_CTL

dress: BASE + 0x09C				
Bit	Name	ess	et Value	cription
	DISP_VCNT_BI	/	h1	en '1', Display vcnt start line number is 1. en '0', Display vcnt start line number is 0.
7-				erved
	DISP_2NDF_O ET	/	h2	et line number of second field, when display in interlaced output le.
15-				erved
23-1	VO_FRAME_SYNC _DLY	/	h2	et line number of frame reset for sync between two display trol, when two display output are supported.
31-2				erved

106.2.6.28. VO_DISP_ENABLE

dress: BASE + 0x0A0				
Bit	Name	Access	Reset Value	cription
	DISP_LAYER0_	R/W	b1	en '1', layer 0 display enable.
	DISP_LAYER1_	R/W	b0	en '1', layer 1 display enable.
	DISP_LAYER2_	R/W	b0	en '1', layer 2 display enable.

	VO_DISP_LAYER3_ EN	R/V	b	en '1', OSD layer display enable.
6-				erved
			b	
31-				erved

106.2.6.29. VO_DISP_SIZE

dress: BASE + 0x0A4

Bit	Name	ess	et Value	cription
11-	DISP_HSIZE	/	h0	zontal size of display frame.
15-1				erved
27-1	DISP_VSIZE	/	h0	ical size of display frame.
31-2				erved

106.2.6.30. VO_LAYER_MODE

dress: BASE + 0x0B4

Bit	Name	ess	et Value	cription
4:	LAYER2_MODE	/	b0	or mode for layer 2. 00 8-bit color palette look-up table (LUT8) 01 16-bit RGBA5551 color format 02 32-bit ABGR8888 color format 04 8-bit RGB332 color format 05 16-bit RGB565 color format 06 32-bit BGRA8888 color format 07 L8 Grayscale/Palette format 0A Reserved 0B 24-bit RGB color format 0C Reserved 0D 32-bit RGBA8888 color format 0E 32-bit ARGB8888 color format 10 Reserved 12 TSc2 13 TSc4 14 Reserved 15 16-bit RGBA4444 color format 18 16-bit ARGB4444 color format

9:	LAYER3_MODE	/		r mode for layer 3. 00 8-bit color palette look-up table (LUT8) 01 16-bit RGBA5551 color format 02 32-bit ABGR8888 color format 04 8-bit RGB332 color format 05 16-bit RGB565 color format 06 32-bit BGRA8888 color format 07 L8 Grayscale/Palette format 0A Reserved 0B 24-bit RGB color format 0C Reserved 0D 32-bit RGBA8888 color format 0E 32-bit ARGB8888 color format 10 Reserved 12 TSc2 13 TSc4 14 Reserved 15 16-bit RGBA4444 color format 18 16-bit ARGB4444 color format
31-				erved

106.2.6.31. VO_LAYER0_CTL

dress: BASE + 0x0C8				
Bit	**Name**	**ess**	**et Value**	**cription**
	LAYER0_EN	/	b0	n '1', layer0 is enabled. n '0', layer0 is disabled.
3-				erved
	LAYER0_YUV4 EN	/		'1', when layer0 image is YUV4:2:2 format.
7-				erved
	LAYER0_YUV4 EN	/		'1', when layer0 image is YUV4:2:0 format.
31-				erved

106.2.6.32. VO_LAYER0_IMG_IN_BADDR_Y

dress: BASE + 0xCC

	Name	Access	Reset Value	cription
)	_LAER0_IMG_IN DDR_Y	R/W	h0	er0 Y frame base address of input imiage.

106.2.6.33. VO_LAYER0_IMG_IN_BADDR_UV

dress: BASE + 0xD0				
	Name	Access	Reset Value	cription
)	_LAER0_IMG_IN DDR_UV	R/W	h0	er0 UV frame base address of input imiage.

106.2.6.34. VO_LAYER0_IMG_IN_HOFFSET

dress: BASE + 0xD4				
Bit	Name	ess	et Value	cription
11-	_LAYER0_IMG_I HOFFSET	/	h0	zontal offset in double word(8-pixel) relative to start address(0) yer0 input image.
31-1				erved

106.2.6.35. VO_LAYER0_IMG_IN_VOFFSET

dress: BASE + 0xD8				
Bit	Name	ess	et Value	cription
23-	_LAYER0_IMG_I /OFFSET	/	h0	ical offset in double word(8-pixel) relative to start address(0) yer0 input image.
31-2				erved

106.2.6.36. VO_LAYER0_IMG_IN_BLENTH

dress: BASE + 0xDC				
Bit	Name	ess	et Value	cription
3-	_LAYER0_IMG_I BLENTH	/	h3	er0 DMA burst length
31-				erved

106.2.6.37. VO_LAYER0_IMG_IN_SIZE

dress: BASE + 0xE0				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	LAYER0_IMG_I HSIZE	/	h0	zontal size of layer0 image in DDR, in double word(8-pixel) As example, for image of 800x600, is 99.
15-1				erved
27-1	LAYER0_IMG_I VSIZE	/	h0	ical line size of layer0 input image in DDR, in line unit. As mple, for image of 800x600, is 599.
31-2				erved

106.2.6.38. VO_LAYER0_IMG_IN_PIX_SIZE

dress: BASE + 0xE4				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	LAYER0_IMG_I PIX_HSIZE	/	h0	ually reading horizontal size of layer0 image, in pixel unit. f r1 input image. As example, for image of 800x600, is 799.
15-1				erved
27-16	LAYER0_IMG_I PIX_HSIZE	/		ually reading vertical size of layer0 image, in line mple, for image of 800x600, is 599.
31-1				erved

106.2.6.39. VO_LAYER1_CTL

dress: BASE + 0xE8				
Bit	**Name**	**ess**	**et Value**	**cription**
	LAYER1_EN	/	b0	en '1', layer1 is enabled. en '0', layer1 is disabled.
3-				erved
	LAYER1_YUV4 EN	/	b1	'1', when layer1 image is YUV4:2:2 format.
7-				erved
	LAYER1_YUV4 EN	/	b0	'1', when layer1 image is YUV4:2:0 format.
31-				erved

106.2.6.40. VO_LAYER1_IMG_IN_BADDR_Y

dress: BASE + 0xEC				
	Name	Access	Reset Value	cription
)	_LAER1_IMG_IN DDR_Y	R/W	h0	er1 Y frame base address of input imiage.

106.2.6.41. VO_LAYER1_IMG_IN_BADDR_UV

dress: BASE + 0x0F0				
	Name	Access	Reset Value	cription
)	_LAER1_IMG_IN DDR_UV	R/W	h0	er1 UV frame base address of input imiage.

106.2.6.42. VO_LAYER1_IMG_IN_HOFFSET

dress: BASE + 0xF4				
Bit	Name	ess	et Value	cription
11-	_LAYER1_IMG_I IOFFSET	/	h0	zontal offset in double word(8-pixel) relative to start address(0) yer1 input image.
31-1				erved

106.2.6.43. VO_LAYER1_IMG_IN_VOFFSET

dress: BASE + 0xF8				
Bit	Name	ess	et Value	cription
23-	_LAYER1_IMG_I /OFFSET	/	h0	ical offset in double word(8-pixel) relative to start address(0) yer1 input image.
31-2				erved

106.2.6.44. VO_LAYER1_IMG_IN_BLENTH

dress: BASE + 0xFC				
Bit	Name	ess	et Value	cription
3-		/	h3	er1 DMA burst length
31-				erved

106.2.6.45. VO_LAYER1_IMG_IN_SIZE

dress: BASE + 0x100				
Bit	Name	ess	et Value	cription
11-	LAYER1_IMG_I HSIZE	/	h0	zontal size of layer1 image in DDR, in double word(8-pixel) . As example, for image of 800x600, is 99.
15-1				erved
27-1	LAYER1_IMG_I VSIZE	/	h0	ical line size of layer1 input image in DDR, in line unit. As mple, for image of 800x600, is 599.
31-2				erved

106.2.6.46. VO_LAYER1_IMG_IN_PIX_SIZE

dress: BASE + 0x104				
Bit	Name	ess	et Value	cription
11-	LAYER1_IMG_I PIX_HSIZE	/	h0	ually reading horizontal size of layer1 image, in pixel unit. f r1 input image. As example, for image of 800x600, is 799.
15-1				erved
27-16	LAYER1_IMG_I PIX_HSIZE	/		ually reading vertical size of layer1 image, in line mple, for image of 800x600, is 599.
31-1				erved

106.2.6.47. VO_LAYER2_CTL

dress: BASE + 0x108				
Bit	Name	ess	et Value	cription
	_LAYER2_EN	/	b0	en '1', layer2 is enabled. en '0', layer2 is disabled.
3-				erved
	_LAYER2_YUV4 EN	/	b1	'1', when layer2 image is YUV4:2:2 format.
7-				erved
	_LAYER2_YUV4 EN	/	b0	'1', when layer2 image is YUV4:2:0 format.
31-				erved

106.2.6.48. VO_LAYER2_IMG_IN_BADDR_Y

dress: BASE + 0x10C				
	Name	Access	Reset Value	cription
)	_LAYER2_IMG_IN DDR_Y	R/W	h0	er2 Y frame base address of input imiage.

106.2.6.49. VO_LAYER2_IMG_IN_BADDR_UV

dress: BASE + 0x110				
	Name	Access	Reset Value	cription
)	_LAYER2_IMG_IN DDR_UV	R/W	h0	er2 UV frame base address of input imiage.

106.2.6.50. VO_LAYER2_IMG_IN_HOFFSET

dress: BASE + 0x114				
Bit	Name	ess	et Value	cription
11-	_LAYER2_IMG_I HOFFSET	/	h0	zontal offset in double word(8-pixel) relative to start address(0) yer2 input image.

31-1				erved

106.2.6.51. VO_LAYER2_IMG_IN_VOFFSET

dress: BASE + 0x118				
Bit	**Name**	**ess**	**et Value**	**cription**
23-	LAYER2_IMG_I /OFFSET	/	h0	ical offset in double word(8-pixel) relative to start address(0) ıyer2 input image.
31-2				erved

106.2.6.52. VO_LAYER2_IMG_IN_BLENTH

dress: BASE + 0x11C				
Bit	**Name**	**ess**	**et Value**	**cription**
3-	LAYER2_IMG_IN_BLE H	/	h3	er2 DMA burst length
31-				erved

106.2.6.53. VO_LAYER2_IMG_IN_SIZE

dress: BASE + 0x120				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	LAYER2_IMG_I HSIZE	/	h0	zontal size of layer2 image in DDR, in double word(8-pixel) . As example, for image of 800x600, is 99.
15-1				erved
27-1	LAYER1_IMG_I /SIZE	/	h0	ical line size of layer2 input image in DDR, in line unit. As mple, for image of 800x600, is 599.
31-2				erved

106.2.6.54. VO_LAYER1_IMG_IN_PIX_SIZE

dress: BASE + 0x124				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	LAYER2_IMG_I PIX_HSIZE	/	h0	ually reading horizontal size of layer2 image, in pixel unit. f r1 input image. As example, for image of 800x600, is 799.
15-1				erved

27-16	LAYER2_IMG_I IX_HSIZE	/		ually reading vertical size of layer2 image, in line mple, for image of 800x600, is 599.
31-1				erved

106.2.6.55. VO_DISP_YUV2RGB_CTL

dress: BASE + 0x128				
Bit	**Name**	**ess**	**et Value**	**cription**
	DISP_YUV2RG N	/	b1	en '1', YUV444 -> RGB after display mixer is enabled. en '0', YUV444 -> RGB after display mixer is disabled.
31-		R	all '0'	erved

106.2.6.56. VO_DISP_YUV2RGB_COEFF00

dress: BASE + 0x12C				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	DISP_YUV2RG OEFF00	/	h400	/444 -> RGB coefficient
31-1				erved

106.2.6.57. VO_DISP_YUV2RGB_COEFF01

dress: BASE + 0x130				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	DISP_YUV2RG OEFF01	/	h0	/444 -> RGB coefficient
31-1				erved

106.2.6.58. VO_DISP_YUV2RGB_COEFF02

dress: BASE + 0x134				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	DISP_YUV2RG OEFF02	/	h5a1	/444 -> RGB coefficient
31-1				erved

106.2.6.59. VO_DISP_YUV2RGB_COEFF10

dress: BASE + 0x13C				
Bit	Name	ess	et Value	cription
11-|	DISP_YUV2RG :OEF10	/	h400	/444 -> RGB coefficient
31-1				erved

106.2.6.60. VO_DISP_YUV2RGB_COEFF11

dress: BASE + 0x140				
Bit	Name	ess	et Value	cription
11-|	DISP_YUV2RG :OEFF11	/	he9e	/444 -> RGB coefficient
31-1				erved

106.2.6.61. VO_DISP_YUV2RGB_COEFF12

dress: BASE + 0x144				
Bit	Name	ess	et Value	cription
11-|	DISP_YUV2RG :OEFF12	/	hd22	/444 -> RGB coefficient
31-1				erved

106.2.6.62. VO_DISP_YUV2RGB_COEFF13

dress: BASE + 0x148				
Bit	Name	ess	et Value	cription
11-|	DISP_YUV2RG :OEFF13	/	h88	/444 -> RGB offset value
31-1				erved

106.2.6.63. VO_DISP_YUV2RGB_COEFF20

dress: BASE + 0x14C				
Bit	Name	ess	et Value	cription

11-	DISP_YUV2RG :OEFF20	/	h400	/444 -> RGB coefficient
31-1				erved

106.2.6.64. VO_DISP_YUV2RGB_COEFF21

dress: BASE + 0x150				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	DISP_YUV2RG :OEFF21	/	h71e	/444 -> RGB coefficient
31-1				erved

106.2.6.65. VO_DISP_YUV2RGB_COEFF22

dress: BASE + 0x154				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	DISP_YUV2RG :OEFF22	/	h0	/444 -> RGB coefficient
31-1				erved

106.2.6.66. VO_DISP_YUV2RGB_COEFF23

dress: BASE + 0x158				
Bit	**Name**	**ess**	**et Value**	**cription**
11-	DISP_YUV2RG :OEFF23	/	hf1c	/444 -> RGB offset value
31-1				erved

106.2.6.67. VO_DISP_DITH_CTL

dress: BASE + 0x15C				
Bit	**Name**	**ess**	**et Value**	**cription**
	DISP_DITH_EN	/	h0	n '1', 10-8-bit dither after display mixer is enabled. n '0', 10-8-bit dither after display mixer is disabled.
31-				erved

106.2.6.68. VO_DISP_MIX_LAYER_GLB_EN

dress: BASE + 0x160				
Bit	Name	ess	et Value	cription
	_DISP_LAYER0_ ʒ_EN	/	b1	n '1', layer0 is to be mixed use global alpha blending value. In ent design, pixel by pixel alpha blending is not supported.
	_DISP_LAYER0_ ʒ_EN	/	b1	n '1', layer1 is to be mixed use global alpha blending value. In ent design, pixel by pixel alpha blending is not supported.
	_DISP_LAYER0_ ʒ_EN	/	b1	n '1', layer2 is to be mixed use global alpha blending value. In ent design, pixel by pixel alpha blending is not supported.
	_DISP_LAYER0_ ʒ_EN	/	b1	n '1', layer3(OSD layer) is to be mixed use global alpha blendin e. n '0', layer3(OSD layer) is to be mixed use pixel-by-pixel alpha ding.
31-				erved

106.2.6.69. VO_DISP_MIX_LAYER_GLB_ALPHA0

dress: BASE + 0x164				
Bit	Name	Access	Reset Value	cription
7-	_DISP_LAYER0_ ʒ_ALPHA	R/W	hff	r0 global alpha value
15-	_DISP_LAYER1_ ʒ_ALPHA	R/W	hff	r1 global alpha value
23-1	_DISP_LAYER2_ ʒ_ALPHA	R/W	hff	r2 global alpha value
31-2	_DISP_LAYER3_ ʒ_ALPHA	R/W	hff	r3(OSD layer) global alpha value

106.2.6.70. VO_DISP_MIX_SEL

dress: BASE + 0x16C				
Bit	Name	ess	et Value	cription
2-	_DISP_LAYER0_SEL	/	h0	er layer0 source selection: 0: layer0; 3'h1: layer1; 3'h2: layer2; 3'h3: layer3(osd layer)

				rved
6-	DISP_LAYER1_SEL	/	h1	r layer1 source selection:: : layer0; 3'h1: layer1; 3'h2: layer2; 3'h3: layer3(osd layer)
				rved
10-	DISP_LAYER2_SEL	/	h2	r layer2 source selection:: : layer0; 3'h1: layer1; 3'h2: layer2; 3'h3: layer3(osd layer)
1				rved
14-1	DISP_LAYER3_SEL	/	h3	r layer3 source selection:: : layer0; 3'h1: layer1; 3'h2: layer2; 3'h3: layer3(osd layer)
31-1				rved

106.2.6.71. VO_DISP_BACKGROUND

dress: BASE + 0x170				
Bit	**Name**	**Access**	**Reset Value**	**cription**
7-	DISP_BACKGR ND_Y	R/W	hff	mponent color of background
15-	DISP_BACKGR ND_U	R/W	h0	mponent color of background
23-1	DISP_BACKGR ND_V	R/W	h0	mponent color of background
31-2	--	--	--	rved

106.2.6.72. VO_DISP_CLUT_CTL

dress: BASE + 0x174				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	DISP_CLUT_EN	R/W	b0	n '1', gamma correction is enabled. n '0', gamma correction is disabled.
	VO_DISP_BCS_EN	R/W	b0	n '1', bright/contrast/saturation correction is enabled. n '0', bright/contrast/saturation correction is disabled.
7-	--	--	--	rved
15-	VO_DISP_BRI	R/W	b0	htness correction in Y level (-127~+127)
23-1	VO_DISP_CON	R/W	b0	trast correction in 1.7 fixed point format (0~1.992x)
31-2	VO_DISP_SAT	R/W	b0	ration correction in 1.7 fixed point format (0~1.992x)

106.2.6.73. VO_DISP_IRQ0_CTL

dress: BASE + 0x178				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_DISP_IRQ0_VP	R/W	b0	number to trigger irq0. And, irq0 is high active, level.
15-1	--	--	--	rved
1	_DISP_IRQ0_FL	R/W	b0	n '0', irq0 will be triggered in field 0. n '1', irq0 will be triggered in field 1.
1	--	--	--	--
19-1	--	--	--	rved
2	VO_DISP_IRQ0_EN	R/W	b0	n '1', irq0 is enabled. n '0', irq0 is disabled.
31-2	--	--	--	rved

106.2.6.74. VO_DISP_IRQ1_CTL

dress: BASE + 0x17C				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_DISP_IRQ1_VP	R/W	b0	number to trigger irq1. And, irq1 is high active, level.
15-1	--	--	--	rved
1	_DISP_IRQ1_FL	R/W	b0	n '0', irq1 will be triggered in field 0. n '1', irq1 will be triggered in field 1.
1	--	--	--	--
19-1	--	--	--	rved
2	VO_DISP_IRQ1_EN	R/W	b0	n '1', irq1 is enabled. n '0', irq1 is disabled.
31-2	--	--	--	rved

106.2.6.75. VO_DISP_IRQ2_CTL

dress: BASE + 0x180				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_DISP_IRQ2_VP	R/W	b0	number to trigger irq2. And, irq2 is high active, level.
15-1	--	--	--	rved
1	_DISP_IRQ2_FL	R/W	b0	n '0', irq2 will be triggered in field 0. n '1', irq2 will be triggered in field 1.
1	--	--	--	--
19-1	--	--	--	rved
2	VO_DISP_IRQ2_EN	R/W	b0	n '1', irq2 is enabled. n '0', irq2 is disabled.
31-2	--	--	--	rved

106.2.6.76. VO_DISP_IRQ

dress: BASE + 0x184				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_DISP_IRQ0	R	b0	_DISP_IRQ0 status, read only.
	_DISP_IRQ0_CL	W	b0	e '1' to clear VO_DISP_IRQ0, write only.
	_DISP_IRQ1	R	b0	_DISP_IRQ1 status, read only.
	_DISP_IRQ1_CL	W	b0	e '1' to clear VO_DISP_IRQ1, write only.
	_DISP_IRQ2	R	b0	_DISP_IRQ2 status, read only.
	_DISP_IRQ2_CL	W	b0	e '1' to clear VO_DISP_IRQ2, write only.
	_DBIB_TE_IRQ	R	b0	_DBIB_TE_IRQ status, read only.
	_DBIB_TE_IRQ_CLR	W	b0	e '1' to clear VO_DBIB_TE_IRQ, write only.
31-	--	--	--	rved

106.2.6.77. VO_LAYER3_IMG_IN_BADDR

dress: BASE + 0x188				
	Name	Access	Reset Value	cription
)	_LAER3_IMG_IN DDR	R/W	h0	er3 Y frame base address of input imiage.

106.2.6.78. VO_LAYER3_IMG_IN_BLENTH

dress: BASE + 0x18c				
Bit	Name	ess	et Value	cription
3-		/	h3	er3 DMA burst length
31-				erved

106.2.6.79. VO_LAYER3_IMG_IN_SIZE

dress: BASE + 0x190				
Bit	Name	ess	et Value	cription
11-	_LAYER3_IMG_I ISIZE	/	h0	zontal size of layer1 image in DDR, in double word(8-pixel) . As example, for image of 800x600, is 99.
15-1				erved
27-1	_LAYER3_IMG_I /SIZE	/	h0	ical line size of layer1 input image in DDR, in line unit. As mple, for image of 800x600, is 599.
31-2				erved

106.2.6.80. VO_LAYER3_IMG_IN_PIX_SIZE

dress: BASE + 0x194				
Bit	Name	ess	et Value	cription
11-	_LAYER3_IMG_I PIX_HSIZE	/	h0	ially reading horizontal size of layer1 image, in pixel unit. f r1 input image. As example, for image of 800x600, is 799.

15-1				erved
27-16	LAYER1_IMG_I IX_HSIZE	/		ually reading vertical size of layer1 image, in line mple, for image of 800x600, is 599.
31-1				erved

106.2.6.81. VO_LAYER3_IMG_IN_STRIDE

dress: BASE + 0x194				
Bit	**Name**	**Access**	**Reset Value**	**cription**
15-	DISP_LAYER3_ST E	R/W	b0	) stride in 8-byte unit for each pixel line
31-1	VO_DISP_ALP_ST RIDE	R/W	B0	HA stride in unit of 8-byte for each pixel line

106.2.6.82. VO_DISP_OSD_INFO

dress: BASE + 0x1C0				
Bit	**Name**	**Access**	**Reset Value**	**cription**
3-	)_TYPE	R/W	b0	GB24 1: Mono 8 2: RGB565 3: RGB32
6-	HA_TYPE	R/W	b0	HA mode: ix value defined by Register oad from OSD_ALP_ADDR. LPHA on R channel LPHA on G channel LPHA on B channel LPHA on A channel ixel value as ALPHA(Only available in MONO-8bit mode) LPHA_RGB as transparent pixel.
	--	--	--	rved
31-	ALPHA_OSD	R/W	b0	sparent pixel value in RGB24 Format

106.2.6.83. VO_DISP_OSD_SIZE

dress: BASE + 0x1C4				
Bit	**Name**	**Access**	**Reset Value**	**cription**
15-	D_WIDTH	R/W	h0	D image width (in unit of pixel)
31-1	OSD_HEIGHT	R/W	h0	D image height (in unit of pixel)

106.2.6.84. VO_DISP_OSD_VLU_ADDR

dress: BASE + 0x1C8				
Bit	**Name**	**Access**	**Reset Value**	**cription**
31-	D_VLU_ADDR	R/W	h0	D RGB image base address in DDR (should be 8byte aligned and mus start address of new pixel).

106.2.6.85. VO_DISP_OSD_ALP_ADDR

dress: BASE + 0x1CC				
Bit	**Name**	**Access**	**Reset Value**	**cription**
31-	D_ALP_ADDR	R/W	h0	D ALPHA image base address in DDR (should be 8byte aligned and mus start address of new pixel).

106.2.6.86. VO_DISP_OSD_DMA_CTRL

dress: BASE + 0x1D0				
Bit	**Name**	**Access**	**Reset Value**	**cription**
3-	DMA_LEN	R/W	h7	burst length per DMA read request
5-	DMA_MAP	R/W	h0	A data map manually in a d-word (32bit) by byte (8bit). 00: original orde 1-2-3-4 to 2-1-4-3 11: 1-2-3-4 to 4-3-2-1
	RGB_REV	R/W	b0	SB is R channel SB is B channel
31-	--	--	--	erved

106.2.6.87. VO_DISP_OSD_STRIDE

dress: BASE + 0x1D4				
Bit	**Name**	**Access**	**Reset Value**	**cription**

15-	OSD_STRIDE	R/W	b0	) frame stride in pixel 8-byte unit for each pixel line
31-1	ALP_STRIDE --	R/W--	B0--	'HA stride in unit of 8-byte for each pixel line

106.2.6.88. VO_OSD_YUV2RGB_CTL

dress: BASE + 0x200				
Bit	**Name**	**Access**	**Reset Value**	**cription**
	_OSD_YUV2RG :N	R/W	b10	'1', RGB -> YUV444 in OSD layer is enabled. 0', RGB -> YUV444 in OSD layer is disabled.
31-	--	--	--	erved

106.2.6.89. VO_OSD_YUV2RGB_COEFF00

dress: BASE + 0x204				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_OSD_YUV2RG :OEFF00	R/W	h132	/444 -> RGB coefficient.
31-1	--	--	--	erved

106.2.6.90. VO_OSD_YUV2RGB_COEFF01

dress: BASE + 0x208				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_OSD_YUV2RG :OEFF01	R/W	h259	/444 -> RGB coefficient.
31-1	--	--	--	erved

106.2.6.91. VO_OSD_YUV2RGB_COEFF02

dress: BASE + 0x020C				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_OSD_YUV2RG :OEFF02	R/W	h75	/444 -> RGB coefficient.
31-1	--	--	--	erved

106.2.6.92. VO_OSD_YUV2RGB_COEFF03

dress: BASE + 0x210				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_OSD_YUV2RG :OEFF03	R/W	h0	/444 -> RGB offset value.
31-1	--	--	--	erved

106.2.6.93. VO_OSD_YUV2RGB_COEFF10

dress: BASE + 0x214				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_OSD_YUV2RG :OEFF10	R/W	hf50	/444 -> RGB coefficient.
31-1	--	--	--	erved

106.2.6.94. VO_OSD_YUV2RGB_COEFF11

dress: BASE + 0x218				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	_OSD_YUV2RG :OEFF11	R/W	hea5	/444 -> RGB coefficient.
31-1	--	--	--	erved

106.2.6.95. VO_OSD_YUV2RGB_COEFF12

dress: BASE + 0x21C				
Bit	Name	Access	Reset Value	cription
11-	_OSD_YUV2RG :OEFF12	R/W	h20b	/444 -> RGB coefficient.
31-1	--	--	--	rved

106.2.6.96. VO_OSD_YUV2RGB_COEFF13

dress: BASE + 0x220				
Bit	Name	Access	Reset Value	cription
11-	_OSD_YUV2RG :OEFF13	R/W	h80	/444 -> RGB offset value.
31-1	--	--	--	rved

106.2.6.97. VO_OSD_YUV2RGB_COEFF20

dress: BASE + 0x224				
Bit	Name	Access	Reset Value	cription
11-	_OSD_YUV2RG :OEFF20	R/W	h20b	/444 -> RGB coefficient.
31-1	--	--	--	rved

106.2.6.98. VO_OSD_YUV2RGB_COEFF21

dress: BASE + 0x228				
Bit	Name	Access	Reset Value	cription
11-	_OSD_YUV2RG :OEFF21	R/W	he4a	/444 -> RGB coefficient.
31-1	--	--	--	rved

106.2.6.99. VO_OSD_YUV2RGB_COEFF22

dress: BASE + 0x22C				
Bit	Name	Access	Reset Value	cription
11-(	OSD_YUV2RG COEFF22	R/W	hfab	/444 -> RGB coefficient.
31-1	--	--	--	erved

106.2.6.100. VO_OSD_YUV2RGB_COEFF23

dress: BASE + 0x230				
Bit	Name	Access	Reset Value	cription
11-(	OSD_YUV2RG COEFF23	R/W	h80	/444 -> RGB offset value.
31-1	--	--	--	erved

106.2.6.101. VO_TEST_MODE

dress: BASE + 0x234				
Bit	Name	Access	Reset Value	cription
	T MODEL ENABLE	R/W	h0	0', testmodel is disabled. 1', testmodel is enabled.
7-	--	--	--	erved
11-8	T_MODEL CHOICE		h0	0', red model. 1', green model. 2', blue model. 3', gradient color model.
31-1	--	--	--	erved

106.2.6.102. VO_DATA_ENDIAN_SWAP

dress: BASE + 0x238				
Bit	Name	Access	Reset Value	cription
	ENDIAN_SWAP	R/W	h0	0', endian swap is disabled. 1', endian swap is enabled.
7-	--	--	--	rved
	16bit ENDIAN_SWAP		h0	0', endian swap is disabled. 1', endian swap is enabled.
31-	--	--	--	rved

106.2.6.103. VO_AXI_RST_CTL

dress: BASE + 0x240				
Bit	Name	Access	Reset Value	cription
	AXI_RST_CTL	R/W	h0	0', reset vo axi read channel disable. 1', reset vo axi read channel enable.
	AXI_RST_CTL	R/W	h0	0', reset vo axi write channel disable. 1', reset vo axi write channel enable.
31-	--	--	--	rved

106.2.6.104. VO_AXI_RST_STATUS

dress: BASE + 0x244				
Bit	Name	Access	Reset Value	cription
3-	--	--	--	rved
6-	ts AXI dma read status	RO	h0	d '0', IDLE. d '1', read request. d '2', wait ack .
	bit axi dma read ready	RO	h0	d '0', dma is busy. d '1', dma is ready.
31-	--	--	--	rved

106.2.6.105. **VO_LCD_CTL**

dress: BASE + 0x254				
Bit	**Name**	**Access**	**Reset Value**	**cription**
7-	_SYNC_POLARITY	R/W	h0	image encoding or not. ormat coding disable 1 : format coding enable reversal HSYNC enable. when '0', normal HSYNC. when '1', reversal HSYNC. reversal VSYNC enable. when '0', normal VSYNC. when '1', reversal VSYNC. reversal DE enable. when '0', normal DE . when '1', reversal DE .
15-	_VIDEO_OUT_FORMAT	R/W	h0	0', RGB8b. 1', RGB8b delta. 2', RGB8b stride. 5', Bt656. 6', MCU mode.
23-1	VO_BTMODE_CTL	R/W	h0	Bt encoding enable when '0', disable . when '1', enable . output reorder when '0', not reorder . when '1', reorder UV . mode choice when '0', separate mode. when '1', Embedded mode .(not support) Bt1120 input reorder when '0', not reorder. when '1', reorder UV.
31-2	--	--	--	rved

106.2.6.106. **VO_DBIB_CTL**

dress: BASE + 0x258				
Bit	Name	Access	Reset Value	cription
	DBIB enable	R/W	h0	disable dbib module . enable dibi module.
30-	--	--	--	erved

106.2.6.107. VO_DBIB_CFG

dress: BASE + 0x25c				
Bit	Name	Access	Reset Value	cription
31-	DBIB config	R/W	h0	:0] TLCDCFG_FORMAT TLCDCFG_OPT2 TLCDCFG_SINGLE_RD 1] TLCDCFG_LP_EN 1] TLCDCFG_EN_STALL 2] TLCDCFG_SPI4 3] TLCDCFG_SPI3 4] TLCDCFG_DMA 5] TLCDCFG_RESX 7] TLCDCFG_TE_IRQ_EDGE_SEL 8] TLCDCFG_TE_IRQ_EN 9] TLCDCFG_FRC_CSX_V 0] TLCDCFG_FRC_CSX 1] TLCDCFG_DIS_TE

106.2.6.108. VO_DBIB_STATUS

dress: BASE + 0x264				
Bit	Name	Access	Reset Value	cription
2-	DBIB status	R/W	h0	CMD pend No CMD ack Cmd go or rw pending or write side not empty
15-	--	--	--	erved

1	DBIB TE status	R	'h0	B TE pin status
31-1	--	--	--	erved

106.2.6.109. VO_DBIB_RDATA

dress: BASE + 0x264				
Bit	**Name**	**Access**	**Reset Value**	**cription**
31-	DBIB CMD	R/W	h0	Read data from lcd

106.2.6.110. VO_DBIB_CMD

dress: BASE + 0x260				
Bit	**Name**	**Access**	**Reset Value**	**cription**
31-	DBIB CMD	R/W	h0	

106.2.6.111. VO_RECTANGLE_EN

dress: BASE + 0x280				
Bit	**Name**	**Access**	**Reset Value**	**cription**
4-	CTANGLE_EN	R/W	h0	tangle en
30-	--	--	--	erved

106.2.6.112. VO_RECTANGLE_DATA

dress: BASE + 0x284				
Bit	**Name**	**Access**	**Reset Value**	**cription**
23-	CTANGLE DATA	R/W	h0	tangle data

31-2	--	--	--	rved

106.2.6.113. VO_RECTANGLE0_POSITION

dress: BASE + 0x288				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE START SITION	R/W	h0	tangle start x position.
15-1	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.114. VO_RECTANGLE0_LENGTH

dress: BASE + 0x28c				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	rved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	rved

106.2.6.115. VO_RECTANGLE1_POSITION

dress: BASE + 0x290				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE START SITION	R/W	h0	tangle start x position.
15-1	--	--	--	rved

27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.116. VO_RECTANGLE1_LENGTH

dress: BASE + 0x294				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	rved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	rved

106.2.6.117. VO_RECTANGLE2_POSITION

dress: BASE + 0x298				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE START SITION	R/W	h0	tangle start x position.
15-1	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.118. VO_RECTANGLE2_LENGTH

dress: BASE + 0x29c				
Bit	**Name**	**Access**	**Reset Value**	**cription**

11-	:TANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	rved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	rved

106.2.6.119. VO_RECTANGLE3_POSITION

dress: BASE + 0x2a0				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	:TANGLE START) SITION	R/W	h0	tangle start x position.
15-1	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.120. VO_RECTANGLE3_LENGTH

dress: BASE + 0x2a4				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	:TANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	rved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	rved

106.2.6.121. VO_RECTANGLE4_POSITION

dress: BASE + 0x2a8				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	:TANGLE START)	R/W	h0	tangle start x position.

	SITION			
15-1	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.122. VO_RECTANGLE4_LENGTH

dress: BASE + 0x2ac				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	rved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	rved

106.2.6.123. VO_RECTANGLE5_POSITION

dress: BASE + 0x2b0				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE START SITION	R/W	h0	tangle start x position.
15-1	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.124. VO_RECTANGLE5_LENGTH

dress: BASE + 0x2b4				
Bit	**Name**	**Access**	**Reset Value**	**cription**

11-ꞆTANGLE WIDTH		R/W	h0	tangle width.
15-1	--	--	--	ɛrved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	ɛrved

106.2.6.125. VO_RECTANGLE6_POSITION

ꞇress: BASE + 0x2b8				
Bit	Name	Access	Reset Value	cription
11-	ꞆTANGLE START ꞁ ꞩITION	R/W	h0	tangle start x position.
15-1	--	--	--	ɛrved
27-1	ꞆECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	ɛrved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.126. VO_RECTANGLE6_LENGTH

ꞇress: BASE + 0x2bc				
Bit	Name	Access	Reset Value	cription
11-	ꞆTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	ɛrved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	ɛrved

106.2.6.127. VO_RECTANGLE7_POSITION

ꞇress: BASE + 0x2c0				
Bit	Name	Access	Reset Value	cription

11-	CTANGLE START] SITION	R/W	h0	tangle start x position.
15-1|	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.128. VO_RECTANGLE7_LENGTH

dress: BASE + 0x2c4				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	rved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	rved

106.2.6.129. VO_RECTANGLE8_POSITION

dress: BASE + 0x2c8				
Bit	**Name**	**Access**	**Reset Value**	**cription**
11-	CTANGLE START] SITION	R/W	h0	tangle start x position.
15-1	--	--	--	rved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	rved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.130. VO_RECTANGLE8_LENGTH

dress: BASE + 0x2cc				
Bit	Name	Access	Reset Value	cription
11-	CTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	erved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	erved

106.2.6.131. VO_RECTANGLE9_POSITION

dress: BASE + 0x2d0				
Bit	Name	Access	Reset Value	cription
11-	CTANGLE START SITION	R/W	h0	tangle start x position.
15-1	--	--	--	erved
27-1	ECTANGLE START Y POSITION	R/W	h0	tangle start y position
30-2	--	--	--	erved
3	VO_RECTANGLE ENABLE		h0	0', retangle is disabled. 1', retangle is enabled.

106.2.6.132. VO_RECTANGLE9_LENGTH

dress: BASE + 0x2d4				
Bit	Name	Access	Reset Value	cription
11-	CTANGLE WIDTH	R/W	h0	tangle width.
15-1	--	--	--	erved
27-1	RECTANGLE HIGH	R/W	h0	tangle high
31-2	--	--	--	erved

106.3. VSSC

106.3.1. Overview

VSSC stands for Video Sub-System Configure block, it serves 2 main purposes:

1. Provide configurable features in the video subsystem such as QOS priority and video data-path selection. See 6.8.2 for details
2. Provide hardware part of the low-latency video data-path solution. See 6.8.3 for details.

106.3.2. Video Subsystem System Control Register

Below is the details of VSSC registers, whose base address is 0x9300_F000.

Table 6- 291 VSSC Registers

Offset	Name	R/W	Description
0x0	VSS_BUS_SW_RST	RW	[0] : 0: VSSC is not in reset status. 1:VSSC is in reset status. [31:1] Reserved
0x4	VSS_QOS_VIN0	RW	[31:8] Reserved [7:4]AXI Write master QOS field of the IP [3:0]AXI Read master QOS field of the IP
0x8	VSS_QOS_VIN1	RW	[31:8] Reserved [7:4]AXI Write master QOS field of the IP [3:0]AXI Read master QOS field of the IP
0xC	VSS_QOS_ISP	RW	[31:24] Reserved [23:20]AXI Write master2 QOS field of the IP [19:16]AXI Read master2 QOS field of the IP [15:12]AXI Write master1 QOS field of the IP [11:8]AXI Read master1 QOS field of the IP [7:4]AXI Write master0 QOS field of the IP [3:0]AXI Read master0 QOS field of the IP
0x10	Reserved	RW	
0x14	VSS_QOS_VPU	RW	[31:8] Reserved [7:4]AXI Write master QOS field of the IP [3:0]AXI Read master QOS field of the IP
0x18	VSS_QOS_DMA	RW	[31:8] Reserved [7:4]AXI Write master QOS field of the IP [3:0]AXI Read master QOS field of the IP
0x1C	VSS_QOS_LCD	RW	[31:8] Reserved [7:4]AXI Write master QOS field of the IP

Offset	Name	R/W	Description
			[3:0]AXI Read master QOS field of the IP
0x20	VSS_CACHE_VIN0	RW	[31:8] Reserved [7:4]AXI Write master CACHE field of the IP [3:0]AXI Read master CACHE field of the IP
0x24	VSS_CACHE_VIN1	RW	[31:8] Reserved [7:4]AXI Write master CACHE field of the IP [3:0]AXI Read master CACHE field of the IP
0x28	VSS_CACHE_ISP	RW	[31:24] Reserved [23:20]AXI Write master2 CACHE field of the IP [19:16]AXI Read master2 CACHE field of the IP [15:12]AXI Write master1 CACHE field of the IP [11:8]AXI Read master1 CACHE field of the IP [7:4]AXI Write master0 CACHE field of the IP [3:0]AXI Read master0 CACHE field of the IP
0x2C	Reserved	RW	
0x30	VSS_CACHE_VPU	RW	[31:8] Reserved [7:4]AXI Write master CACHE field of the IP [3:0]AXI Read master CACHE field of the IP
0x34	VSS_CACHE_DMA	RW	[31:8] Reserved [7:4]AXI Write master CACHE field of the IP [3:0]AXI Read master CACHE field of the IP
0x38	VSS_CACHE_LCD	RW	[31:8] Reserved [7:4]AXI Write master CACHE field of the IP [3:0]AXI Read master CACHE field of the IP
0x3C	EXT_CTL	RW	Legacy Power/Reset gpio control for Sensor. Could be replaced by GPIOs.
0x40	Reserved	RW	
0x44	ISP_CH_SEL	RW	[1:0] ISP Source Select(Depends on ISP_IN_SEL): 0: DVP0/MIPI IPI0 1: DVP1/MIPI IPI1 2: Reserved 3: MIPI-HDR [3:2] MIPI-HDR IPI0 VCID remap value for ISP [5:4] MIPI-HDR IPI1 VCID remap value for ISP
0x48	VSS_MODE_SEL	RW	[31:2] Reserved. [1]VPU mode Select: 0: Enable Encoder; 1: Enable Decoder [0]ISP Disable 0: Enable ISP; 1: Disable ISP
0x4C	DSI_INF_CTRL	RW	DSI Interface Ctrl [31:3] Reserved.

Offset	Name	R/W	Description
			[2]DPI_UPDATECFG [1]DPI_COLORM [0]DPI_SHUTDN
0x50	ISP_SNOOP_CTRL	RW	[31:2] Reserved. [1]Interrupt enable. Interrupt will be triggered when ISP_HCNT_BUFFER=ISP_HCNT_THR [0]Enable ISP output snooping which triggers Hsync counting.
0x54	ISP_HCNT_FRAME	RO	[31:0] Hsync counter snooping from ISP DVP output interface. Cleared by SOF.
0x58	ISP_HCNT_BUFFER	RO	[31:0] Hsync counter snooping from ISP DVP output interface. Cleared by .ISP_HCNT_BUFFER=ISP_HCNT_THR
0x5C	ISP_HCNT_THR	RW	[31:0]. Buffer size(line number) to trigger interrupt.
0x60	ISP_VSSC_INT	RW	Read:[31:0] reprensents ready buffer number not cleared Write: Write 0x5aa5beef clears latest buffer ready and reduces ready buffer number by 1. If current buffer number not cleared is 0, writing 0x5aa5beef has no effect.
0x48	ISP_IN_SEL	RW	[31:1] Reserved. [0]ISP Source type select 0: MIPI 1: DVP

106.3.3. **Video Subsystem Low-latency Encoding Solution**

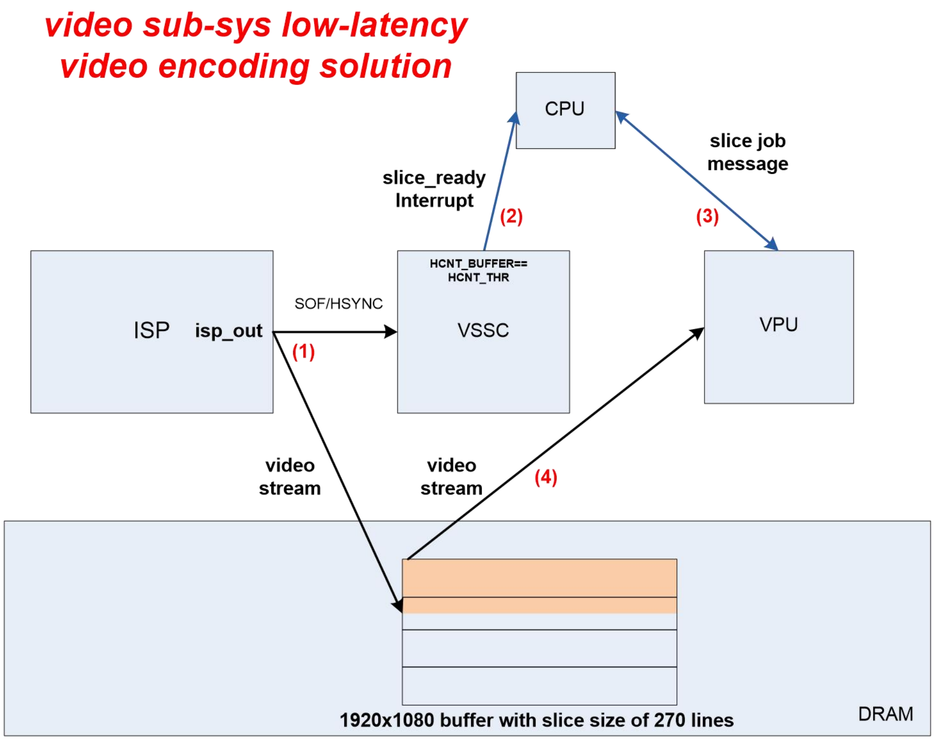


Figure 6- 9 Video Subsystem Low-Latency Video Encoding Solution

In order to achieve low-latency video encoding system solution, VPU needs to start encoding as early as possible. In VPU design, video frame could be encoded on slice-bases. For example, if the input resolution is 1920x1080 and we divide the input frame into 4 slices, VPU could start encoding when there is 270 lines of the frame data rather than a full frame. As described in above diagram, it works like following:

1. ISP allocate only 1 frame for output. VSSC logic in our system snoops the ISP output SOF/HSYNC information, couting the line number.
2. When there is slice of data ready(HCNT_BUFFER==HCNT_THR) VSSC generates an interrupt to inform CPU a slice for VPU input is ready. The VSSC interrupt number is 50
3. CPU informs VPU to start encoding.
4. VPU start working reading from the same buffer.

107. Audio Subsystem

107.1. Audio CODEC

107.1.1. Overview

The Audio Codec includes 1-Channel Stereo ADC with Differential Mic-Phone Inputs and 2-Channel Stereo DAC with Single-Ended Line-Out Support.

107.1.2. Features

- 1-Channel Differential Mic-In or Line-In ADCs with 1-channel Mic-Bias
- 2-Channel DAC with Single-Ended Line Outputs
- APB Bus for ADC and DAC Control
- ADC Master Clock is 128*fs
- ADC Support 8kHz~96KHz Sample Rates
- Internal Fractional PLL
- DAC Master clock is 256*fs
- DAC Support 8kHz~192kHz Sample rates
- Power Supply, Digital Core Supply 0.99V-1.21V, Analog Supply 2.97V-3.63V
- SMIC 40nm LL 1P7M 1TM Process

107.1.3. **Function Description**

107.1.3.1. **Block Diagram**

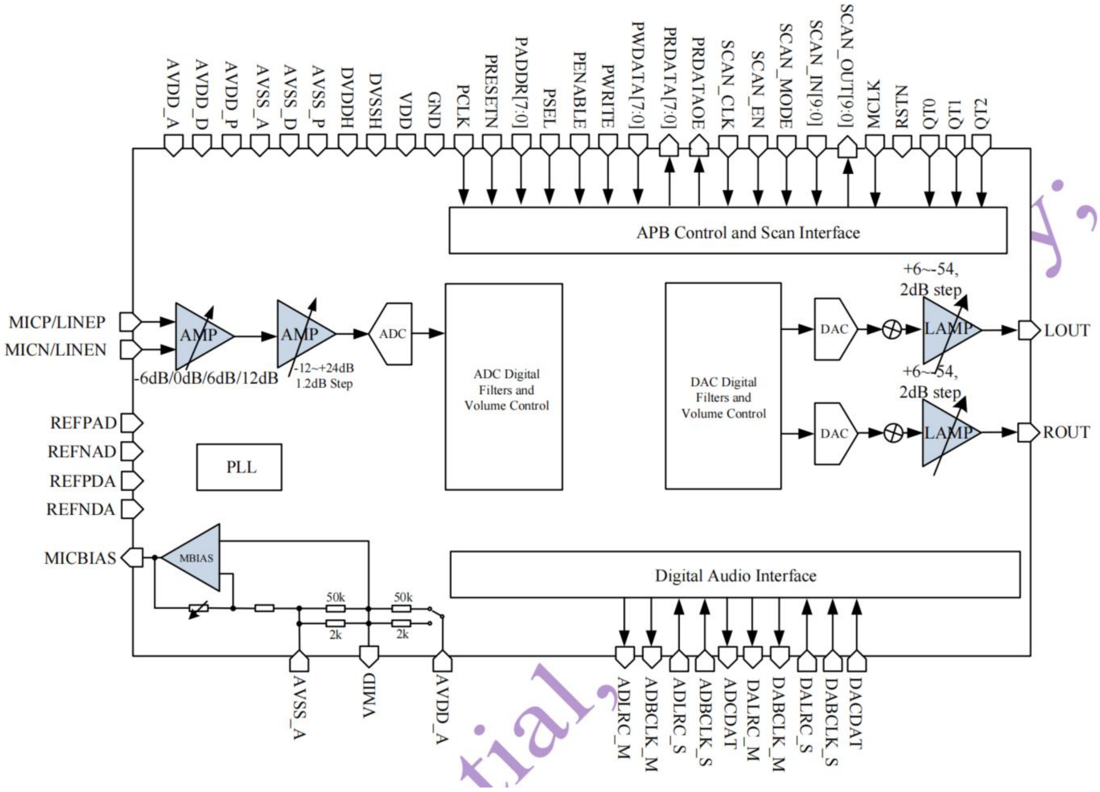


Figure 7- 1 Function Block Diagram

107.1.4. **Register Summary**

Table 7- 1 Register List for Codec

Register name	Address Offset	Description
ADC Analog On/Off Control Register	0x00	ADC Analog On/Off Control
ADC Analog Volume Control Register	0x04	ADC Analog Volume Control
ADC Digital Volume Control Register	0x0C	ADC Digital Volume Control
CODEC Digital Block Control Register	0x14	CODEC Digital Block Control
Left Channel DAC Output and Volume Control Register	0x18	Left Channel DAC Output and Volume Control

Register name	Address Offset	Description
Right Channel DAC Output and Volume Control Register	0x20	Right Channel DAC Output and Volume Control
DAC Analog and PLL Block Control Register	0x28	DAC Analog and PLL Block Control
Fractional PLL Configuration Control Register:DPADC[7:0]	0x2C	Fractional PLL Configuration Control:DPADC[7:0]
Fractional PLL Configuration Control Register:DPDAC[7:0]	0x30	Fractional PLL Configuration Control:DPDAC[7:0]
Fractional PLL Configuration Control Register:FRAC[23:16]	0x34	Fractional PLL Configuration Control:FRAC[23:16]
Fractional PLL Configuration Control Register:FRAC[15:8]	0x38	Fractional PLL Configuration Control:FRAC[15:8]
Fractional PLL Configuration Control Register:FRAC[7:0]	0x3C	Fractional PLL Configuration Control: FRAC[7:0]
Fractional PLL Configuration Control Register:DM[7:0]	0x40	Fractional PLL Configuration Control: DM[7:0]
Fractional PLL Configuration Control Register:DN[7:0]	0x44	Fractional PLL Configuration Control: DN[7:0]
CODEC Common Analog Block Control Register	0x48	CODEC Common Analog Block Control
CODEC Digital Control Register	0x4C	CODEC Digital Control
ADC Analog Input Control Register	0x50	ADC Analog Input Control

107.1.5. Register Description

107.1.5.1. ADC Analog On/Off Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION

dress:00H fault:FFH tus: R/W	MIC		C-Buffer Power Down Control: 'ower-On 'ower-Down
	erved		erved
	SDM		C SDM Power Down Control: DM Power-On DM Power-Down
	PGA		A Power Down Control: 'GA Power-On 'GA Power-Down
	CTRL		-Bias Voltage Control: .8V .5V
	MBIAS		CBIAS Power-Down Control: 1ICBIAS Power-On 1ICBIAS Power-Down
	erved		erved
	erved		erved

107.1.5.2. ADC Analog Volume Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:01H fault:19H tus: R/W	erved		
	INEVOL[4:0]	10	A Gain Control:)00: -12dB(1.2dB Step))01:······-10.8dB 10: 24dB 11: Mute

	ICVOL[1:0]		:-Phone PGA Gain Control: -6dB 0dB 6dB 12dB

107.1.5.3. ADC Digital Volume Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:03H fault:7AH tus: R/W	served		
	DDVOL[6:0]	_1010	M Digital Gain Control. The value can be grammed from 1Bh to 7Fh, stepped by 1.2dB +6dB +4.8dB : +3.6dB +2.4dB +1.2dB : 0 dB (Default) -1.2dB -2.4dB -3.6dB : -96dB

107.1.5.4. CODEC Digital Block Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:05H	SERVED		

ault:00H tus: R/W	CEN		C Digital Part Control Signal: igital Part Power-Down igital Part Power-On
	FEN		C High Pass Filter Control Signal: ypass HPF(Default) nable HPF(3.75Hz@8KSPS)
	CEN		C Digital Part Control Signal: igital Part Power-Down igital Part Power-On
	CMONO		C Digital Input Control Signal: Stereo Data Input L Mono R Mono (L+R)/2 Mono
	CMUTEL		t Channel DIGITAL Volume Control: igital Volume not Set to Mute: igital Volume Set to Mute

107.1.5.5. **Left Channel DAC Output and Volume Control**

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:06H ault:23H tus: R/W	N		t Channel DAC Line-Out Signal Control: 0: From Left Channel DAC 1: From External Signal
	erved		
	CAVL[4:0]	011	C Left Channel Line PGA Buffer Gain Control: I: +6dB (-2dB Step) I: +4dB …… I: -54dB I: Mute

dress:07H fault:80H tus: R/W	ACDVOL[7:0]	)0_0000	arse Gain Step Control: 128: 0dB 64: -6dB 32: -12dB 16: -18dB 08: -24dB 04: -30dB 02: -36dB 01: -42dB

DAC Volume Control: Coarse Gain + Fine Gain, Total Range: +6dB ~ -96dB

107.1.5.6. Right Channel DAC Output and Volume Control

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:08H fault:23H tus: R/W	IN		ht Channel DAC Line-Out Signal Control: 1'b0: From ht Channel DAC 1: From External Signal
	erved		
	CAVR[4:0]	)011	C Right Channel Line PGA Buffer Gain Control: 00H IB (-2dB Step) I: +4dB…… I: -54dB I: Mute
dress:09H fault:80H tus: R/W	ACDVOL[7:0]	)0_0000	arse Gain Step Control: 128: 0dB 64: -6dB 32: -12dB 16: -18dB

			08: -24dB 04: -30dB 02: -36dB 01: -42dB

DAC Volume Control: Coarse Gain + Fine Gain, Total Range: +6dB ~ -96dB

107.1.5.7. DAC Analog and PLL Block Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:0AH fault: FFH tus: R/W	MUTE		C Analog Filter Mute Control: Mute (Default) Normal Work
	ANAFL		t-Channel DAC Analog Filter Power Down 1: Power Down 0: Power On
	ANAFR		ht-Channel DAC Analog Filter Power Down 1: Power Down 0: Power On
	LBUF		t-Channel DAC Line-Out Buffer Power Down 1: Power Down 0: Power On
	RBUF		ht-Channel DAC Line-Out Buffer Power Down 1'b1: Powe wn 0: Power On
	FRA		L Working Mode Control: nteger Mode (Reserved) ractional Mode
	PLL		ver Down or Bypass PLL
	LBYP		L Bypass Mode Control: PLL Mode

			ʾLL Bypass

107.1.5.8. Fractional PLL Configuration Control Register

DRESS	MBOL	ſS[7:0]	SCRIPTION
dress:0BH fault:0FH tus: R/W	ADC[7:0]	)0_1111	Output Divider For ADC Master Clock. *Fs Output
dress:0CH fault:07H tus: R/W	DAC[7:0]	)0_0111	Output Divider For DAC Master Clock. *Fs Output.
dress:0DH fault:31H tus: R/W	AC[23:16]	1_0001	ctional Part Input
dress:0EH fault:26H tus: R/W	AC[15:8]	0_0110	ctional Part Input
dress:0FH fault:E9H tus: R/W	AC[7:0]	0_1001	ctional Part Input
dress:10H fault:00H tus: R/W	I[7:0]	)0_0000	Input Divider
dress:11H fault:07H tus: R/W	[7:0]	)0_0111	Feedback Divider

For audio CODEC, the main clock is generated by a 24-bit accuracy fractional PLL. The configuration of the internal dividers are:

M= DM [7:0] +1: Input divider

N= DN [6:0] +1: Feedback divider

P= DP [7:0] +1: Output divider

In Fractional Mode, Via setting FRAC [23:0], The feedback divider equals to N.X, where:

X=FRAC [23:0]/ 2^{24}: Fractional number

N.X=N+X: Fractional divider

The frequency relationship of the PLL blocks and the output clock can be calculated by the following equations:

$F_{pfd} = F_{refin}/M = F_{clkvco}/N.X$

$F_{clkoutput} = F_{clkvco} / P$

The ADC and DAC can be configured with different sample rates. When PLL VCO output frequency equals to 98.304MHz, the ADC sample rate can be configured to 8K, 12K, 16K, 24K, 32K, 48KSPS, and the DAC sample rate can be configured to 8K, 12K, 16K, 24K, 32K, 48K,96KSPS. When PLL VCO output frequency equals to 90.3168MHz, the ADC sample rate can be configured to 11.025K, 22.05K, 44.1KSPS, and DAC sample rate can be configured to 11.025K, 22.05K, 44.1K and 88.2KSPS.

107.1.5.9. CODEC Common Analog Block Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:12H fault:A2H tus: R/W	BIAS		DEC Bias Power Down Control: Power On Power Down
	OI		IID to analog Output resistance High Z K Ohm
	MID		IID Resistor Divider Control: 5K Ohm K Ohm (Just For Fast Start Up)
	RL	010	DEC Low Power Mode Control: 01: High Power 10: Middle Power ers: Reserved

107.1.5.10. CODEC Digital Control Register

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:13H fault:3CH tus: R/W	RTEST		R TEST, Reserved
	FRSTN		C Logic Part Digital Filter Soft Reset Control: 0: Soft Rese Filter

			Iormal Work
	FRSTN		C Logic Part Digital Filter Soft Reset Control: 0: Soft Rese Filter Iormal Work
	LRSTN		L Logic Part Modulator Soft Reset Control: 0: Soft Reset th er Iormal Work
	I2SMODE		C I2S Interface Mode Selection: Aaster Mode Slave Mode
	BW		Bit Width Mode Selection: 32 Bit Mode 16 Bit Mode
	I2SMODE		C I2S Interface Mode Selection: Aaster Mode Slave Mode

107.1.5.11. **ADC Analog Input Control Register**

DRESS	MBOL	TS[7:0]	SCRIPTION
dress:14H fault: 11H tus: R/W	INSELL[3:0]	)1	t Channel ADC Input Control:)1: LMICP Input 0: LMICN Input)0: LMIC Input with MIC Buffer ers: No Use
	INSELR[3:0]		ht Channel ADC Input Control:)1: RMICP Input 0: RMICN Input)0: RMIC Input with MIC Buffer ers: No Use

107.2. I2S

107.2.1. Overview

The I2S component which is a serial link designed for digital audio systems. It is an AMBA 2.0-compliant Advanced Peripheral Bus (APB) slave device.

The I2S is designed to be used in systems that process digital audio signals, such as:

- A/D and D/A converters
- digital signal processors
- error correction for compact disc and digital recording
- digital filters
- digital input/output interfaces

There are 4 I2S block in the chip, namely I2S_ADC, I2S_DAC which bridge Audio Codec and MCU system and I2S0, I2S1, which serve as chip's audio signal interface.

Table 7- 2 I2S list

	Base Addr	Channels	Max Resolution	Master/Slave
I2S_ADC	0x9201_1000	1 RX	16bit	Slave
I2S_DAC	0x9201_2000	1 TX	16ibt	Slave

It should be noted only one of I2S0 and I2S1 module can be in operation at anytime due to limited package pin. I2S0 is selected when I2S master is needed in application and I2S1 is selected when slave is desired.

107.2.2. Features

The I2S has the following features:

- APB data bus widths of 32 bits

- I2S transmitter and/or receiver based on the Philips I2S serial protocol
- Full duplex communication due to the independence of transmitter and receiver
- Asynchronous clocking of APB bus and I2S sclk
- Master or slave mode of operation
- External sclk gating and enable signals
- Configurable FIFO depth is16 words
- Configurable support for programmable DMA registers
- Programmable FIFO thresholds
- Programmable DMA data mode (L only/R only/Stereo)
- Component parameters for configurable software driver

107.2.3. Function Description

107.2.3.1. Block Description

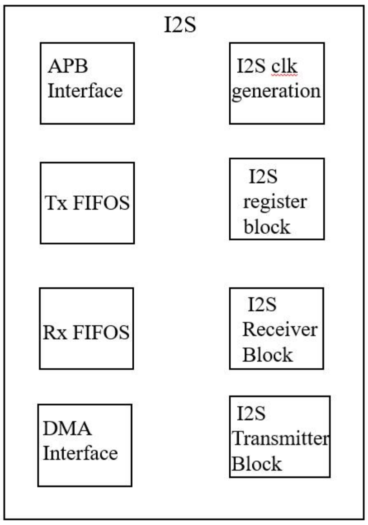


Figure 7- 2 Block Diagram of I2S

The bus consists of a serial data line (sd), a word select line (ws), and a serial clock (sclk). The serial data line is time multiplexed to allow the transfer of two data streams (such as, left and right stereo data).

The I2S bus can only handle audio data transmissions; subcoding and controls are handled by another device, such as an I2C. The I2S protocol requires a minimum of three wires—data (sd), word select (ws), and serial clock (sclk)—keeping the design simple and the pin count minimal.

Storm have two I2S: I2S_ADC/I2S_DAC:

I2S_ADC as slave,just have one recive channel; I2S_DAC as slave,just have one transmitter channel;

107.2.3.2. Timing Description

The i2s as a master, the number of sclk cycles during which ws_out is held high or low can be reprogrammed during operation of the component by setting the Word Select Size (WSS) bits [4:3] of the Clock Configuration Register (CCR). You must disable the Clock Generation block (CER[0] = 0) before you can change the word select size.

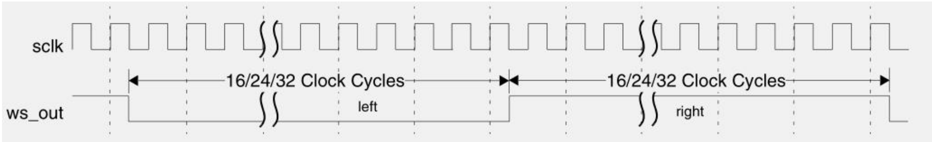


Figure 7- 3 Number of SCLK Cycles Per Left/Right WS Cycle

The i2s as a master and the audio data resolution of the receive and transmit channels is less than the current word select size, the sclk can be gated off for the remainder of the left/right cycle, can be reprogrammed during operation of the component by setting the SCLK Gating (SCLKG) bits [2:0] of the Clock Configuration Register (CCR).

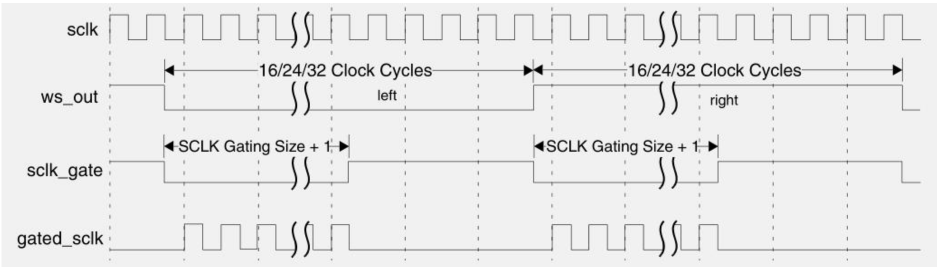


Figure 7- 4 Gating of SCLK

107.2.4. **Programming Guide**

107.2.4.1. **I2S as Transmitter and Slave Mode**

This section describes how to program the i2s when it is configured as a transmitter in either slave mode with one stereo channel.

The following subsections describe normal and TX DMA sequences when i2s is a transmitter in slave mode.

To program i2s when it is a transmitter in slave mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Fill the TX-FIFO by writing data to the Left Transmit Holding Register (LTHR) and the Right Transmit Holding Register (RTHR), respectively. Keep writing in this order—left and then right—until the FIFO is filled with data.

   The i2s starts transmitting stereo data on the first left cycle when the ws signal goes low.
3. Enable the I2S Transmitter block by writing 1 in bit 0 (TXEN) of the I2S Transmitter Block EnableRegister (ITER).

To program i2s when it is a transmitter in slave mode, and when TX DMA mode is enabled, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Transmitter block by writing a 1 in bit 0 (TXEN) of the I2S Transmitter Block Enable Register (ITER).
3. Fill the TX-FIFO by writing to the Transmitter Block DMA Register (TXDMA).

107.2.4.2. I2S as Transmitter and Master Mode

To program i2s when it is a transmitter in master mode, complete the following steps:

1. Complete steps 1 to 3 of the previous procedure (transmitter, slave mode).
2. Enable the I2S Clock Generation block by writing 1 in bit 0 (CLKEN) of the Clock Enable Register(CER).

107.2.4.3. I2S as Receiver and Slave Mode

To program i2s when it is a receiver in slave mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Receiver block by writing 1 in bit 0 (RXEN) of the I2S Receiver Block Enable Register (IRER).
3. Read bit 0 (RXDA) of the Interrupt Status Register (ISR). When bit 0 of the goes high, the defaulttrigger has been reached.
4. Read the contents of the Left Receive Buffer Register (LRBR) and Right Receive Buffer Register (RRBR).

The bit is dependent on how you have configured (or programmed) the trigger level.

To program i2s when RX DMA mode is enabled, When RX DMA is enabled, the data contents is read from RXDMA instead of the Left Receive BufferRegister (LRBR) and the Right Receive Buffer Register (RRBR), as defined in step 4 of the previousprocedure (receiver, slave mode).

107.2.4.4. I2S as Receiver and Master Mode

To program i2s when it is a receiver in master mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Receiver block by writing 1 in bit 0 (RXEN) of the I2S Receiver Block Enable Register(IRER).
3. Enable the I2S Clock Generation block by writing 1 in bit 0 (CLKEN) of the Clock Enable Register(CER).
4. Read bit 0 (RXDA) of the Interrupt Status Register (ISR). When bit 0 of the goes high, the default trigger has been reached.
5. Read the contents of the Left Receive Buffer Register (LRBR) and Right Receive Buffer Register(RRBR).

The bit is dependent on how you have configured (or programmed) the trigger level.

When RX DMA is enabled, the data contents is read from RXDMA instead of the Left Receive Buffer Register (LRBR) and the Right Receive Buffer Register (RRBR), as defined in step 5 of the previous procedure (receiver, master mode).

107.2.4.5. I2S as Transmitter with DMA Handshake Interface

To program i2s when it is a transmitter in slave mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Transmitter block by writing 1 in bit 0 (TXEN) of the I2S Transmitter Block Enable Register (ITER).
3. Enable DMA handshaking by writing 1 in bit 8 (DMAEN_TXCH_0) of the DMA Control Register (DMACR).
4. Enable the Transmit channel by writing 1 in bit 0 (TXCHEN0) of the Transmit Enable Register 0 (TER0).
5. Fill the TX-FIFO by writing to the Transmitter Block DMA Register (TXDMA).

To program i2s when it is a transmitter in master mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Transmitter block by writing 1 in bit 0 (TXEN) of the I2S Transmitter Block Enable Register (ITER).
3. Enable the I2S Clock Generation block by writing 1 in bit 0 (CLKEN) of the Clock Enable Register (CER).
4. Enable DMA handshaking by writing 1 in bit 8 (DMAEN_TXCH_0) of the DMA Control Register (DMACR).
5. Enable the Transmit channel by writing 1 in bit 0 (TXCHEN0) of the Transmit Enable Register 0 (TER0).
6. Fill the TX-FIFO by writing to the Transmitter Block DMA Register (TXDMA).

107.2.4.6. I2S as Receiver with DMA Handshake Interface

To program i2s when it is a receiver in slave mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Receiver block by writing 1 in bit 0 (RXEN) of the I2S Receiver Block Enable Register (IRER).
3. Enable DMA handshaking by writing 1 in bit 8 (DMAEN_RXCH_0) of the DMA Control Register (DMACR).
4. Enable the Receiver channel by writing 1 in bit 0 (RXCHEN0) of the Receiver Enable Register 0 (RER0).
5. Empty the RX-FIFO by reading from the Receiver Block DMA Register (RXDMA).

To program i2s when it is a receiver in master mode, complete the following steps:

1. Enable the i2s by setting bit 0 of the i2s Enable Register (IER) to 1.
2. Enable the I2S Receiver block by writing 1 in bit 0 (RXEN) of the I2S Receiver Block Enable Register (IRER).
3. Enable the I2S Clock Generation block by writing 1 in bit 0 (CLKEN) of the Clock Enable Register (CER).
4. Enable DMA handshaking by writing 1 in bit 8 (DMAEN_RXCH_0) of the DMA Control Register (DMACR).
5. Enable the Receiver channel by writing 1 in bit 0 (RXCHEN0) of the Receiver Enable Register 0

(RER0).
6. Empty the RX-FIFO by reading from the Receiver Block DMA Register (RXDMA).

107.2.5. **Register Summary**

Table 7- 3 Register List for I2S

Address Offset	Register	Description
0x0	I2S_IER	This register acts as a global enable/disable for I2S.
0x04	I2S_IRER	This register acts as an enable/disable for the I2S Receiver block.
0x08	I2S_ITER	This register acts as an enable/disable for the I2S Transmitter block.
0x0C	I2S_CER	This register acts as an enable/disable for the I2S Clock Generation block.
0x10	I2S_CCR	This register configures the ws_out and sclk_gate signals when I2S is a master.
0x14	I2S_RXFFR	This register specifies the Receiver Block FIFO Reset Register.
0x18	I2S_TXFFR	This register specifies the Transmitter Block FIFO Reset Register.
0x20+0x40*x	I2S_LRBRx	This specifies the Left Receive Buffer Register.
0x20+0x40*x	I2S_LTHRx	This specifies the Left Transmit Holding Register.
0x24+0x40*x	I2S_RRBRx	This specifies the Right Receive Buffer Register.
0x24+0x40*x	I2S_RTHRx	This specifies the Right Transmit Holding Register.
0x28+0x40*x	I2S_RERx	This specifies the Receive Enable Register.
0x2C+0x40*x	I2S_TERx	This specifies the Transmit Enable Register.
0x30+0x40*x	I2S_RCRx	This specifies the Receive Configuration Register.
0x34+0x40*x	I2S_TCRx	This specifies the Transmit Configuration Register.
0x38+0x40*x	I2S_ISRx	This specifies the Interrupt Status Register.
0x3C+0x40*x	I2S_IMRx	This specifies the Interrupt Mask Register.
0x40+0x40*x	I2S_RORx	This specifies the Receive Overrun Register.
0x44+0x40*x	I2S_TORx	This specifies the Transmit Overrun Register.
0x48+0x40*x	I2S_RFCRx	This specifies the Receive FIFO Configuration Register.
0x4C+0x40*x	I2S_TFCRx	This specifies the Transmit FIFO

		Configuration Register.
0x50+0x40*x	I2S_RFFx	This specifies the Receive FIFO Flush Register.
0x54+0x40*x	I2S_TFFx	This specifies the Transmit FIFO Flush Register.
0x1C0	I2S_RXDMA	The RXDMA register allows access to all enabled Receive channels via a single point rather than through the LRBRx and RRBRx registers. The Receive channels are targeted in a cyclical fashion (starting at the lowest numbered enabled channel) and takes two reads (left and right stereo data) before the component points to the next channel.
0x1C4	I2S_RRXDMA	The RXDMA can be reset to the lowest enabled Channel via the RRXDMA register. The RRXDMA register can be written to at any stage of the RXDMA's read cycle, however, it has no effect when the component is in the middle of a stereo pair read. The following example describes the operation of this register for a system with four Receive channels, where channels 0, 1, 2, and 3 are enabled.
0x1C8	I2S_TXDMA	The TXDMA register functions similar to the RXDMA register and allows write accesses to all of the enabled Transmit channels via a single point rather than through the LTHRx and RTHRx registers.
0x1CC	I2S_RTXDMA	This register provides the same functionality as the RRXDMA register but targets TXDMA instead.
0x1F0	I2S_COMP_PARAM_2	This specifies bits for Component Parameter Register 2.
0x1F4	I2S_COMP_PARAM_1	This specifies bits for Component Parameter Register 1.
0x1F8	I2S_COMP_VERSION	This register specifies the I2S Component Version.
0x1FC	I2S_COMP_TYPE	This register specifies the I2S Component Type.
0x200	I2S_DMACR	This register is only valid when I2S is configured with a set of DMA Controller interface signals (I2S_HAS_DMA_INTERFACE = 1). When I2S is not configured

		with DMA handshake interface, this register will not exist and writing to the registers address will have no effect and reading from this register address will return zero. The register is used to enable the DMA Controller interface operation.

The x value for all registers at receiver endian assigned as (for x=0; x<=I2S_RX_CHANNELS-1) and its value for all registers at transmitter endian assigned as (for x=0; x<=I2S_TX_CHANNELS-1).

107.2.6. Register Description

107.2.6.1. I2S_IER

Table 7- 4 Fields for Register: IER

Bits	Name	R/W	Reset	Description
31:1	RSVD_IER	R	0x00	RSVD_IER Reserved bits - Read Only
0	IEN	R/W	0x00	i2s enable. This bit enables or disables i2s. A disable on this bit overrides any other block or channel enables and flushes all FIFOs. Values: ■0x0 (DISABLED): i2s disabled. ■0x1 (ENABLED): i2s enabled

107.2.6.2. I2S_IRER

Table 7- 5 Fields for Register: IRER

Bits	Name	R/W	Reset	Description
31:1	RSVD_IRER	R	0x00	RSVD_IER Reserved bits - Read Only
0	RXEN	R/W	0x00	Receiver block enable. This bit enables or disables the receiver. A disable on this bit overrides any individual receive channel enables. Values: ■0x0 (DISABLED): i2s disabled. ■0x1 (ENABLED): i2s enabled

107.2.6.3. I2S_ITER

Table 7- 6 Fields for Register: ITER

Bits	Name	R/W	Reset	Description
31:1	RSVD_IRER	R	0x00	RSVD_ITER Reserved bits - Read Only
0	TXEN	R/W	0x00	Transmitter block enable. This bit enables or disables the transmitter. A disable on thisbit overrides any individual transmit channel enables. Values: ■0x0 (DISABLED): i2s disabled. ■0x1 (ENABLED): i2s enabled

107.2.6.4. I2S_CER

Table 7- 7 Fields for Register: CER

Bits	Name	R/W	Reset	Description
31:1	RSVD_CER	R	0x00	RSVD_CER Reserved bits - Read Only
0	CLKEN	R/W	0x00	Clock generation enable/disable. This bit enables/disables the clock generation signals when i2s is a master: sclk_en, ws_out, and sclk_gate. Values: ■0x0 (DISABLED): i2s disabled. ■0x1 (ENABLED): i2s enabled

107.2.6.5. I2S_CCR

Table 7- 8 Fields for Register: CCR

Bits	Name	R/W	Reset	Description
31:5	RSVD_CCR	R	0x00	RSVD_CCR Reserved bits - Read Only

Bits	Name	R/W	Reset	Description
4:3	WSS	R/W	0x00	These bits are used to program the number of sclk cycles for which the word select line (ws_out) stays in the left or right sample mode. The I2S Clock Generation block must be disabled (CER[0] = 0) prior to any changes in this value. Values: ■0x0 (CLOCK_CYCLES_16): 16 sclk cycles ■0x1 (CLOCK_CYCLES_24): 24 sclk cycles ■0x2 (CLOCK_CYCLES_32): 32 sclk cycles
2:0	SCLKG	R/W	0x00	These bits are used to program the gating of sclk. The programmed gating value must be less than or equal to the largest configured/programmed audio resolution to prevent the truncating of RX/TX data. The I2S Clock Generation block must be disabled (CER[0] = 0) before making any changes in this value. Values: ■0x0 (NO_CLOCK_GATING): Clock gating is disabled ■ 0x1 (CLOCK_CYCLES_12): Gating after 12 sclk cycles ■ 0x2 (CLOCK_CYCLES_16): Gating after 16 sclk cycles ■ 0x3 (CLOCK_CYCLES_20): Gating after 20 sclk cycles ■ 0x4 (CLOCK_CYCLES_24): Gating after 24 sclk cycles

107.2.6.6. **I2S_RXFFR**

Table 7- 9 Fields for Register: RXFFR

Bits	Name	R/W	Reset	Description
31:1	RSVD_RXFFR	W	0x00	RSVD_RXFFR Reserved bits - Write Only

Bits	Name	R/W	Reset	Description
0	RXFFR	W	0x00	Receiver FIFO Reset. Writing a 1 to this register flushes all the RX FIFOs (this is a self clearing bit). The Receiver Block must be disabled efore writing to this bit. Values: ■ 0x0 (NO_FLUSH): Does not flush the RX FIFO ■0x1 (FLUSH): Flushes the RX FIFO

107.2.6.7. **I2S_TXFFR**

Table 7- 10 Fields for Register: TXFFR

Bits	Name	R/W	Reset	Description
31:1	RSVD_TXFFR	W	0x00	RSVD_TXFFR Reserved bits - Write Only
0	TXFFR	W	0x00	Transmitter FIFO Reset. Writing a 1 to this register flushes all the TX FIFOs (this is a self clearing bit). The Transmitter Block must be disabled prior to writing this bit. Values: ■ 0x0 (NO_FLUSH): Does not flush the TX FIFO ■0x1 (FLUSH): Flushes the TX FIFO

107.2.6.8. **I2S_LRBRx**

Table 7- 11 Fields for Register: LRBRx

Bits	Name	R/W	Reset	Description
32-I2S_RX_WORDSIZE_x	RSVD_LRBx	R	0x00	RSVD_LRBRx Reserved bits - Read Only

Bits	Name	R/W	Reset	Description
I2S_RX_WORDSIZE_x	LRBRx	R	0x00	The left stereo data received serially from the receive channel input (sdix). If the RX FIFO is full and the two-stage read operation (for instance, a read from LRBRx followed by a read from RRBRx) is not performed before the start of the next stereo pair, then the new data is lost and an overrun interrupt occurs. (data already in the RX FIFO is preserved.) Note: Before reading this register again, the right stereo data must be read from RRBRx or the status/interrupts will not be valid.

107.2.6.9. **I2S_LTHRx**

Table 7- 12 Fields for Register: LTHRx

Bits	Name	R/W	Reset	Description
32-I2S_TX_WORDSIZE_x	RSVD_LTHRx	W	0x00	RSVD_LRBRx Reserved bits - Read Only
I2S_TX_WORDSIZE_x	LTHRx	R	0x00	The left stereo data to be transmitted serially through the transmit channel output (sdox) is written through this register. Writing is a two-stage process: 1. A write to this register passes the left stereo sample to the transmitter. 2. This MUST be followed by writing the right stereo sample to the RTHRx register. Data must only be written to the FIFO when it is not full. Any attempt to write to a full FIFO results in that data being lost and an overrun interrupt being generated.

107.2.6.10. **I2S_RRBRx**

Table 7- 13 Fields for Register: RRBRx

Bits	Name	R/W	Reset	Description
32-I2S_RX_WORDSIZE_x	RSVD_RRBRx	R	0x00	RSVD_RRBRx Reserved bits - Read Only
I2S_RX_WORDSIZE_x	RRBRx	R	0x00	The right stereo data received serially from the receive channel input (sdix) is read through this register. If the RX FIFO is full and the two-stage read operation (for instance,read from LRBRx followed by a read from RRBRx) is not performed before the start of the next stereo pair, then the new data is lost and an overrun interrupt occurs. (Data already in the RX FIFO is preserved.) Note: Prior to reading this register, the left stereo data MUST be read from LRBRx, or the status/interrupts will not be valid.

107.2.6.11. **I2S_RTHRx**

Table 7- 14 Fields for Register: RTHRx

Bits	Name	R/W	Reset	Description
32-I2S_TX_WORDSIZE_x	RSVD_RTHRx	W	0x00	RSVD_RTHRx Reserved bits - Write Only

Bits	Name	R/W	Reset	Description
I2S_TX_WORDSIZE_x	RTHRx	W	0x00	The right stereo data to be transmitted serially through the transmit channel output (sdox) is written through this register.Writing is a two-stage process: 1. A left stereo sample MUST be written to the LTHRx register. 2. A write to this register passes the right stereo sample to the transmitter. Data should only be written to the FIFO when it is not full. Any attempt to write to a full FIFO results in that data being lost and an overrun interrupt being generated.

107.2.6.12. **I2S_RERx**

Table 7- 15 Fields for Register: RERx

Bits	Name	R/W	Reset	Description
31:1	RSVD_RERx	R	0x00	RSVD_RERx Reserved bits - Read Only
0	RXCHENx	R/W	0x1	Receive channel enable. This bit enables/disables a receive channel, independently of all other channels. On enable, the channel begins receiving on the next left stereo cycle. A global disable of i2s (IER[0] = 0) or the Receiver block (IRER[0] = 0) overrides this value. Values: ■ 0x0 (DISABLED): Receive Channel Disable ■0x1 (ENABLED): Receive Channel Enable

107.2.6.13. **I2S_TERx**

Table 7- 16 Fields for Register: TERx

Bits	Name	R/W	Reset	Description
31:1	RSVD_TERx	R	0x00	RSVD_TERx Reserved bits - Read Only

Bits	Name	R/W	Reset	Description
0	TXCHENx	R/W	0x1	Transmit channel enable. This bit enables/disables a transmit channel, independently of all other channels. On enable, the channel begins transmitting on the next left stereo cycle. A global disable of i2s (IER[0] = 0) or Transmitter block (ITER[0] = 0) overrides this value. Values: ■ 0x0 (DISABLED): Transmit Channel Disable ■ 0x1 (ENABLED): Transmit Channel Enable

107.2.6.14. **I2S_RCRx**

Table 7- 17 Fields for Register: RCRx

Bits	Name	R/W	Reset	Description
31:3	RSVD_RCRx	R	0x00	RSVD_RCRx Reserved bits - Read Only
2:0	WLEN	R/W	0x4	These bits are used to program the desired data resolution of the receiver and enables the LSB of the incoming left (or right) word to be placed in the LSB of the LRBRx (or RRBRx) register. The channel must be disabled prior to any changes in this value (RER0[0] = 0). Values: ■ 0x0 (IGNORE_WORD_LENGTH): Ignore the word length ■ 0x1 (RESOLUTION_12_BIT): 12-bit data resolution of the receiver. ■ 0x2 (RESOLUTION_16_BIT): 16-bit data resolution of the receiver. ■ 0x3 (RESOLUTION_20_BIT): 20-bit data resolution of the receiver. ■ 0x4 (RESOLUTION_24_BIT): 24-bit data resolution of the receiver. ■ 0x5 (RESOLUTION_32_BIT): 32-bit data resolution of the receiver

107.2.6.15. **I2S_TCRx**

Table 7- 18 Fields for Register: TCRx

Bits	Name	R/W	Reset	Description
31:3	RSVD_TCRx	R	0x00	RSVD_TCRx Reserved bits - Read Only
2:0	WLEN	R/W	0x4	These bits are used to program the data resolution of the transmitter and ensures the MSB of the data is transmitted The channel must be disabled prior to any changes in this value (TER0[0] = 0). Values: ■ 0x0 (IGNORE_WORD_LENGTH): Ignore the word length ■ 0x1 (RESOLUTION_12_BIT): 12-bit data resolution of the transmitter. ■ 0x2 (RESOLUTION_16_BIT): 16-bit data resolution of the transmitter. ■ 0x3 (RESOLUTION_20_BIT): 20-bit data resolution of the transmitter. ■ 0x4 (RESOLUTION_24_BIT): 24-bit data resolution of the transmitter. ■ 0x5 (RESOLUTION_32_BIT): 32-bit data resolution of the transmitter.

107.2.6.16. **I2S_ISRx**

Table 7- 19 Fields for Register: ISRx

Bits	Name	R/W	Reset	Description
31:6	RSVD31_6	R	0x00	RSVD31_6 Reserved bits - Read Only
5	TXFO	R	0x1	Status of Data Overrun interrupt for the TX channel.This bit specifies whether the TX FIFO write is valid or an overrun. Attempt to write to full TX FIFO. Values: ■0x0 (WRITE_VALID): TX FIFO write valid ■0x1 (WRITE_OVERRUN): TX FIFO write overrun

Bits	Name	R/W	Reset	Description
4	TXFE	R	0x1	Status of Transmit Empty Trigger interrupt. This bit specifies whether the TX FIFO trigger level has reached or not. TX FIFO is empty. Values: ■ 0x0 (REACHED_TRIGGER_LEVEL): TX FIFO trigger level is reached ■ 0x1 (NOT_REACHED): TX FIFO trigger level is not reached
3:2	RSVD3_2	R	0x00	RSVD3_2 Reserved bits - Read Only
1	RXFO	R	0x00	Status of Data Overrun interrupt for the RX channel.Incoming data lost due to a full RX FIFO. Values: ■ 0x0 (WRITE_VALID): RX FIFO write valid ■ 0x1 (WRITE_OVERRUN): RX FIFO write overrun
0	RXDA	R	0x00	Status of Receive Data Available interrupt. This bit denotes the status of the RX FIFO trigger level. Values: ■ 0x1 (REACHED_TRIGGER_LEVEL): RX FIFO triggerlevel is reached ■ 0x0 (NOT_REACHED): RX FIFO trigger level is not reached

107.2.6.17. **I2S_IMRx**

Table 7- 20 Fields for Register: IMRx

Bits	Name	R/W	Reset	Description
31:6	RSVD_IMR0_6_31	R	0x00	RSVD_IMR0_6_31 Reserved bits - Read Only
5	TXFOM	R/W	0x1	Mask TX FIFO Overrun interrupt. This bit masks or unmasks a TX FIFO overrun interrupt. Values: ■ 0x1 (MASK_INTERRUPT): Masks TX FIFO Overrun interrupt ■ 0x0 (UNMASK_INTERRUPT): Unmasks TX FIFO Overrun interrup

Bits	Name	R/W	Reset	Description
4	TXFEM	R/W	0x1	Mask TX FIFO Empty interrupt. This bit masks or unmasks a TX FIFO Empty interrupt. Values: ■ 0x1 (MASK_INTERRUPT): Masks TX FIFO Empty interrupt ■ 0x0 (UNMASK_INTERRUPT): Unmasks TX FIFO Empty interrup
3:2	RSVD_IMR0_2_3	R	0x00	RSVD_IMRO_2_3 Reserved bits - Read Only
1	RXFOM	R	0x00	Mask RX FIFO Overrun interrupt. This bit masks or unmasks an RX FIFO Overrun interrupt. Values: ■ 0x1 (MASK_INTERRUPT): Masks RX FIFO Overrun interrupt ■ 0x0 (UNMASK_INTERRUPT): Unmasks RX FIFO Overrun interrupt
0	RXDAM	R	0x00	Mask RX FIFO Data Available interrupt. This bit masks or unmasks an RX FIFO Data Available interrupt. Values: ■ 0x1 (MASK_INTERRUPT): Masks RX FIFO data available interrupt ■ 0x0 (UNMASK_INTERRUPT): Unmasks RX FIFO data available interrupt

107.2.6.18. **I2S_RORx**

Table 7-21 Fields for Register: RORx

Bits	Name	R/W	Reset	Description
31:1	RSVD_RORx	R	0x00	RSVD_RORx Reserved bits - Read Only
0	RXCHO	R	0x00	Read this bit to clear the RX FIFO Data Overrun interrupt. Values: ■ 0x0 (WRITE_VALID): RX FIFO write valid ■ 0x1 (WRITE_OVERRUN): RX FIFO write overrun

107.2.6.19. **I2S_TORx**

Table 7- 22 Fields for Register: TORx

Bits	Name	R/W	Reset	Description
31:1	RSVD_TORx	R	0x00	RSVD_TORx Reserved bits - Read Only
0	TXCHO	R	0x00	Read this bit to clear the TX FIFO Data Overrun interrupt. Values: ■0x0 (WRITE_VALID): TX FIFO write valid ■0x1 (WRITE_OVERRUN): TX FIFO write overrun

107.2.6.20. **I2S_RFCRx**

Table 7- 23 Fields for Register: RFCRx

Bits	Name	R/W	Reset	Description
31:4	RSVD_RFCRx	R	0x00	RSVD_RFCRx Reserved bits - Read Only

Bits	Name	R/W	Reset	Description
3:0	RXCHDT	R/W	0x00	These bits program the trigger level in the RX FIFO at whichthe Received Data Available interrupt and DMA request is generated. Trigger Level = Programmed Value + 1 Values: ■ 0x0 (TRIGGER_LEVEL_1): Interrupt trigger and DMA request asserted when FIFO level is 1. ■ 0x1 (TRIGGER_LEVEL_2): Interrupt trigger and DMA request asserted when FIFO level is 2. ■ 0x2 (TRIGGER_LEVEL_3): Interrupt trigger and DMA request asserted when FIFO level is 3. ■ 0x3 (TRIGGER_LEVEL_4): Interrupt trigger and DMA request asserted when FIFO level is 4. ■ 0x4 (TRIGGER_LEVEL_5): Interrupt trigger and DMA request asserted when FIFO level is 5. ■ 0x5 (TRIGGER_LEVEL_6): Interrupt trigger and DMA request asserted when FIFO level is 6. ■ 0x6 (TRIGGER_LEVEL_7): Interrupt trigger and DMA request asserted when FIFO level is 7. ■ 0x7 (TRIGGER_LEVEL_8): Interrupt trigger and DMA request asserted when FIFO level is 8. ■ 0x8 (TRIGGER_LEVEL_9): Interrupt trigger and DMA request asserted when FIFO level is 9. ■ 0x9 (TRIGGER_LEVEL_10): Interrupt trigger and DMA request asserted when FIFO level is 10. ■ 0xa (TRIGGER_LEVEL_11): Interrupt trigger and DMA request asserted when FIFO level is 11. ■ 0xb (TRIGGER_LEVEL_12): Interrupt trigger and DMA request asserted when FIFO level is 12. ■ 0xc (TRIGGER_LEVEL_13): Interrupt trigger and DMA request asserted when FIFO level is 13. ■ 0xd (TRIGGER_LEVEL_14): Interrupt trigger and DMA request asserted when FIFO level is 14. ■ 0xe (TRIGGER_LEVEL_15): Interrupt

107.2.6.21. **I2S_TFCRx**

Table 7- 24 Fields for Register: TFCRx

Bits	Name	R/W	Reset	Description
31:4	RSVD_TFCRx	R	0x00	RSVD_TFCRx Reserved bits - Read Only

Bits	Name	R/W	Reset	Description
3:0	TXCHET	R/W	0x00	These bits program the trigger level in the TX FIFO at which the Empty Threshold Reached Interrupt and DMA request is generated. Trigger Level = TXCHET Values: ■ 0x0 (TRIGGER_LEVEL_1): Interrupt trigger and DMA request asserted when FIFO level is 1. ■ 0x1 (TRIGGER_LEVEL_2): Interrupt trigger and DMA request asserted when FIFO level is 2. ■ 0x2 (TRIGGER_LEVEL_3): Interrupt trigger and DMA request asserted when FIFO level is 3. ■ 0x3 (TRIGGER_LEVEL_4): Interrupt trigger and DMA request asserted when FIFO level is 4. ■ 0x4 (TRIGGER_LEVEL_5): Interrupt trigger and DMA request asserted when FIFO level is 5. ■ 0x5 (TRIGGER_LEVEL_6): Interrupt trigger and DMA request asserted when FIFO level is 6. ■ 0x6 (TRIGGER_LEVEL_7): Interrupt trigger and DMA request asserted when FIFO level is 7. ■ 0x7 (TRIGGER_LEVEL_8): Interrupt trigger and DMA request asserted when FIFO level is 8. ■ 0x8 (TRIGGER_LEVEL_9): Interrupt trigger and DMA request asserted when FIFO level is 9. ■ 0x9 (TRIGGER_LEVEL_10): Interrupt trigger and DMA request asserted when FIFO level is 10. ■ 0xa (TRIGGER_LEVEL_11): Interrupt trigger and DMA request asserted when FIFO level is 11. ■ 0xb (TRIGGER_LEVEL_12): Interrupt trigger and DMA request asserted when FIFO level is 12. ■ 0xc (TRIGGER_LEVEL_13): Interrupt trigger and DMA request asserted when FIFO level is 13. ■ 0xd (TRIGGER_LEVEL_14): Interrupt trigger and DMA request asserted when FIFO level is 14. ■ 0xe (TRIGGER_LEVEL_15): Interrupt

107.2.6.22. I2S_RFFx

Table 7- 25 Fields for Register: RFFx

Bits	Name	R/W	Reset	Description
31:1	RSVD_RFFx	W	0x00	RSVD_RFFx Reserved bits - Write Only
0	RXCHFR	W	0x00	Receive Channel FIFO Reset. Writing a 1 to this register flushes an individual RX FIFO (This is a self clearing bit.). A Rx channel or block must be disabled prior to writing to this bit. Values: ■ 0x1 (FLUSH): Flushes an individual RX FIFO

107.2.6.23. I2S_TFFx

Table 7- 26 Fields for Register: TFFx

Bits	Name	R/W	Reset	Description
31:1	RSVD_TFFx	W	0x00	RSVD_TFFx Reserved bits - Write Only
0	TXCHFR	W	0x00	Transmit Channel FIFO Reset. Writing a 1 to this register flushes channel's TX FIFO (This is a self clearing bit.). The TX channel or block must be disabled prior to writing to this bit. Values: ■ 0x1 (FLUSH): Flushes an individual TX FIFO

107.2.6.24. I2S_RXDMA

Table 7- 27 Fields for Register: RXDMA

Bits	Name	R/W	Reset	Description
31:0	RXDMA	R	0x00	Receiver Block DMA Register. These bits are used to cycle repeatedly through the enabled receive channels (from lowest numbered to highest), reading stereo data pairs.

107.2.6.25. I2S_RRXDMA

Table 7- 28 Fields for Register: RRXDMA

Bits	Name	R/W	Reset	Description
31:1	RSVD_RRXDMA	W	0x00	RSVD_RRXDMA Reserved bits - Write Only
0	RRXDMA	W	0x00	Reset Receiver Block DMA Register. Writing a 1 to this self-clearing register resets the RXDMA register mid-cycle to point to the lowest enabled Receive channel. Note: Writing to this register has no effect if the component is performing a stereo pair read (such as, when left stereo data has been read but not right stereo data). Values: ■ 0x1 (RESET): Reset Receiver Block DMA Register

107.2.6.26. I2S_TXDMA

Table 7- 29 Fields for Register: TXDMA

Bits	Name	R/W	Reset	Description
31:0	TXDMA	W	0x00	Transmitter Block DMA Register. The register bits can be used to cycle repeatedly through the enabled Transmit channels (from lowest numbered to highest) to allow writing of stereo data pairs

107.2.6.27. I2S_RTXDMA

Table 7- 30 Fields for Register: RTXDMA

Bits	Name	R/W	Reset	Description
31:1	RSVD_RTXDMA	W	0x00	RSVD_RTXDMA Reserved bits - Write Only

Bits	Name	R/W	Reset	Description
0	RTXDMA	W	0x00	Reset Transmitter Block DMA Register. Writing a 1 to this self-clearing register resets the TXDMA register mid-cycle to point to the lowest enabled Transmit channel. Note: This register has no effect in the middle of a stereo pair write (such as, when left stereo data has been written but not right stereo data). Values: ■ 0x1 (RESET): Reset Receiver Block DMA Register

107.2.6.28. **I2S_COMP_PARAM_2**

Table 7-31 Fields for Register: I2S_COMP_PARAM_2

Bits	Name	R/W	Reset	Description
31:13	Reserved	R	0x00	Reserved
12:10	I2S_RX_WORDSIZE_3	R		These bits specify the RX resolution for WORDSIZE_3. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution
9:7	I2S_RX_WORDSIZE_2	R		These bits specify the RX resolution for WORDSIZE_2. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution

Bits	Name	R/W	Reset	Description
6	Reserved	R	0x00	Reserved
5:3	I2S_RX_WORDSIZE_1	R		These bits specify the RX resolution for WORDSIZE_1. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution
2:0	I2S_RX_WORDSIZE_0	R		These bits specify the RX resolution for WORDSIZE_0. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution

107.2.6.29. **I2S_COMP_PARAM_1**

Table 7- 32 Fields for Register: I2S_COMP_PARAM_1

Bits	Name	R/W	Reset	Description
31:28	Reserved	R	0x00	Reserved

Bits	Name	R/W	Reset	Description
27:25	I2S_TX_WORDSIZE_3	R		These bits specify the TX resolution for WORDSIZE_3. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution
24:22	I2S_TX_WORDSIZE_2	R		These bits specify the TX resolution for WORDSIZE_2. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution
21:19	I2S_TX_WORDSIZE_1	R		These bits specify the TX resolution for WORDSIZE_1. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution

Bits	Name	R/W	Reset	Description
18:16	I2S_TX_WORDSIZE_0	R		These bits specify the TX resolution for WORDSIZE_0. Values: ■ 0x0 (RESOLUTION_12_BIT): 12-bit Resolution ■ 0x1 (RESOLUTION_16_BIT): 16-bit Resolution ■ 0x2 (RESOLUTION_20_BIT): 20-bit Resolution ■ 0x3 (RESOLUTION_24_BIT): 24-bit Resolution ■ 0x4 (RESOLUTION_32_BIT): 32-bit Resolution
15:11	Reserved	R	0x00	Reserved
10:9	I2S_TX_CHANNELS	R		These bits specify the number of TX channels. Values: ■ 0x0 (TX_CHANNEL_1): 1 Transmit Channel ■ 0x1 (TX_CHANNEL_2): 2 Transmit Channels ■ 0x2 (TX_CHANNEL_3): 3 Transmit Channels ■ 0x3 (TX_CHANNEL_4): 4 Transmit Channels
8:7	I2S_RX_CHANNELS	R		These bits specify the number of RX channels. Values: ■ 0x0 (RX_CHANNEL_1): 1 Receive Channel ■ 0x1 (RX_CHANNEL_2): 2 Receive Channels ■ 0x2 (RX_CHANNEL_3): 3 Receive Channels ■ 0x3 (RX_CHANNEL_4): 4 Receive Channels
6	I2S_RECEIVER_BLOCK	R		This bit specifies whether the receiver block is enabled or not. Values: ■0x0 (FALSE): Receiver block is disabled ■0x1 (TRUE): Receiver block is enabled

Bits	Name	R/W	Reset	Description
5	I2S_TRANSMITTER_BLOCK	R		This bit specifies whether the transmitter block is enabled or not. Values: ■0x0 (FALSE): Transmitter block is disabled ■0x1 (TRUE): Transmitter block is enabled
4	I2S_MODE_EN	R		This bit specifies whether the master mode is enabled or not. Values: ■0x0 (FALSE): Master mode is disabled ■0x1 (TRUE): Master mode is enabled
3:2	I2S_FIFO_DEPTH_GLOBAL	R		These bits specify the FIFO depth for TX and RX channels. Values: ■ 0x0 (FIFO_DEPTH_2): FIFO depth is equals to 2 for TX and RX channels ■ 0x1 (FIFO_DEPTH_4): FIFO depth is equals to 4 for TX and RX channels ■ 0x2 (FIFO_DEPTH_8): FIFO depth is equals to 8 for TX and RX channels ■ 0x3 (FIFO_DEPTH_16): FIFO depth is equals to 16 for TX and RX channels
1:0	APB_DATA_WIDTH	R		These bits specify the APB data width. Values: ■0x0 (BITS_8): 8 bits APB data width ■0x1 (BITS_16): 16 bits APB data width ■0x2 (BITS_32): 32 bits APB data width

107.2.6.30. **I2S_COMP_VERSION**

Table 7- 33 Fields for Register: I2S_COMP_VERSION

Bits	Name	R/W	Reset	Description
31:0	I2S_COMP_VERSION	R	0x00	These bits specify the I2S component version.

107.2.6.31. **I2S_COMP_TYPE**

Table 7- 34 Fields for Register: I2S_COMP_TYPE

Bits	Name	R/W	Reset	Description

Bits	Name	R/W	Reset	Description
31:20	I2S_COMP_TYPE	R	0x00	DesignWare Component Type number = 0x445701a0. This unique hexadecimal value is constant and is derived from the two ASCII letters 'DW' followed by a 16-bit unsigned number.

107.2.6.32. **I2S_DMACR**

Table 7- 35 Fields for Register: DMACR

Bits	Name	R/W	Reset	Description
31:0	Reserved	R	0x00	Reserved
19:18	DMA_MODE	RW	0x00	DMA data mode (Storm I2S only): Values: ■ 0x0 (STEREO): DMA FIFO cycle through both left and right channel. ■ 0x1 (RD_L): DMA data from left channel only. Sending DMA data to both channel. ■ 0x2 (RD_R): DMA data from right channel only. Sending DMA data to both channel.
17	DMAEN_TXBLOCK	RW	0x00	DMA Enable for transmit block. The corresponding bits of this field enables/disables the DMA handshake logic for transmitter block. Values: ■ 0x0 (DISABLED): DMA disabled for transmit block ■ 0x1 (ENABLED): DMA enabled for transmit block
16	DMAEN_RXBLOCK	RW	0x00	DMA Enable for receive block. The corresponding bits of this field enables/disables the DMA handshake logic for receiver block. Values: ■ 0x0 (DISABLED): DMA disabled for receiver block ■ 0x1 (ENABLED): DMA enabled for receiver block
15:0	Reserved	R	0x00	Reserved

108. Power Management Unit

108.1. Power Domains

There are 2 power domains defined at chip level to save powers as maximum as, including:

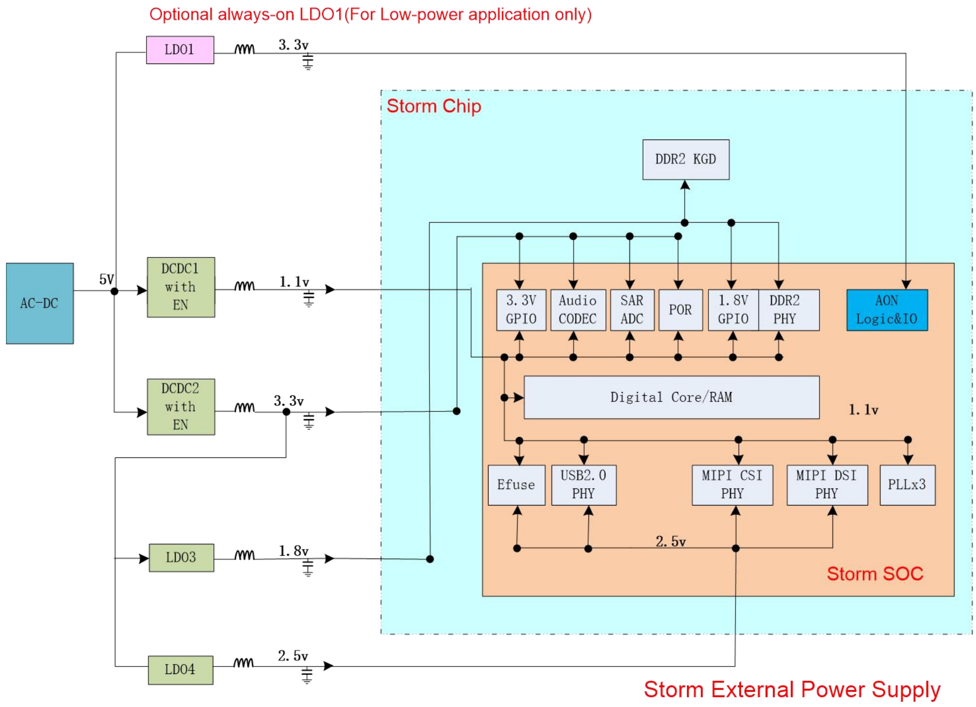
- Always-on(AON) Domain, which mainly consists of the PMU, RTC, and always on GPIO to communicate with off-chip LDO/PMIC to enter/exit deep sleep mode.
- TOP Power Domain, which includes rest of the design

108.2. Low Power Solution

Blow figure describes the low-power solution of Storm SOC. There are 3 power modes:

1) Full speed mode with all 2 power domains alive
2) Deep sleep mode with only AON Domain alive

To enter the deep sleep mode, software needs to write to PMU AON GPIO(LDO_EN) to shut down the power from outside the chip.TOP power domain can be waked up by GPIO(UART0_RX/UART0_TX/PWM0/PMW1 and RESETN assert) in AON domain from board level and PMU hardware will handle the power on sequence.



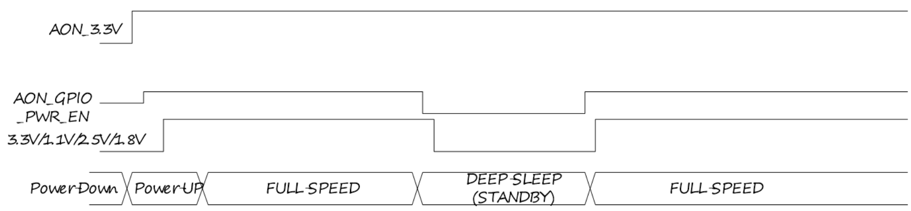


Figure 8- 1 Storm External Power Supply Solution

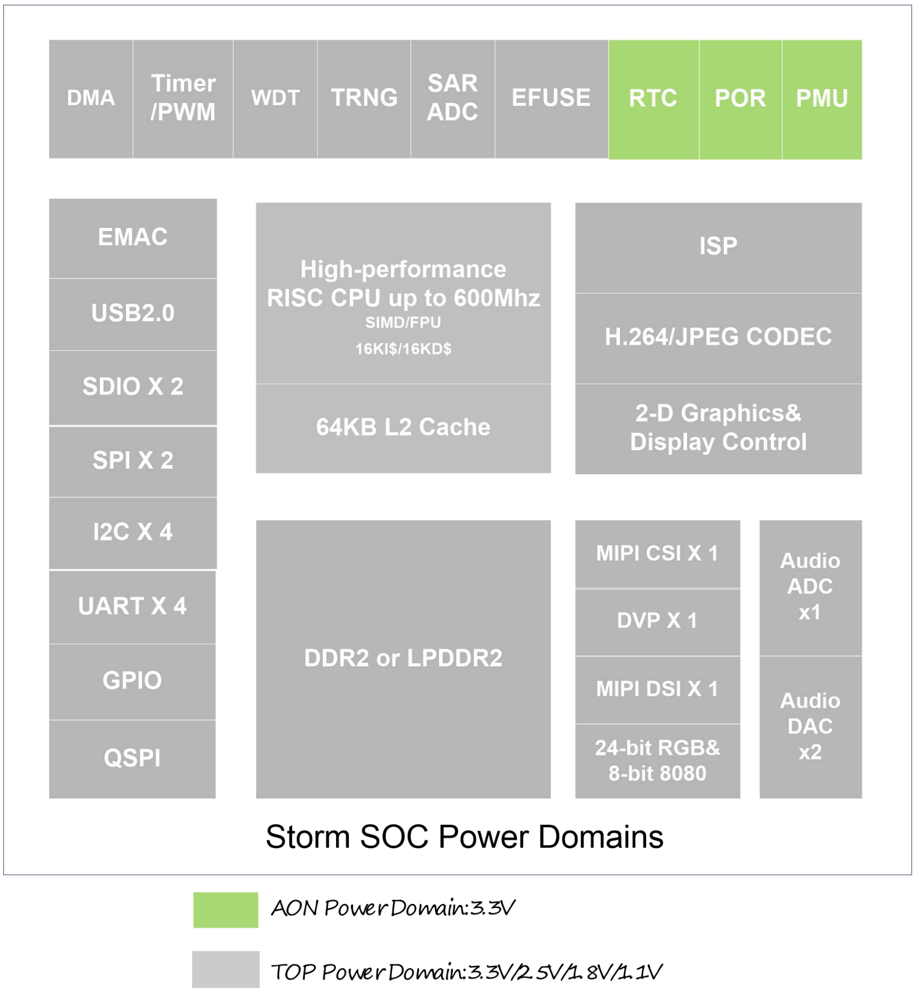


Figure 8- 2 Storm Power Domains

Table 8- 1 Storm SOC Power State

Power Mode	AON Domain	TOP Domain
Full Speed Mode	ON	ON
Deep Sleep Mode	ON	OFF

108.3. PMU State Machine

Power management is achieved through state machine in AON power domain. TOP power domain has its own power state machine. The states of power state machine are defined as following:

- Sleep State: Reset state after power-on-reset asserted and transit to power up state after POR system initialization time.
- Power Up State: Transit to Power Active state after power-up counter expires and release system reset and isolation is disabled for power switch subsequently
- Power Active State: logic in this domain can function well and stays in this state until accept power

down request before transiting to Power Down state. Isolation enables for power switch and later system reset is asserted

- Power Down State: It is a intermediate state for top power domain for housekeeping work to be done, after the top domain can goes to deep sleep state.

Deep Sleep(Standby) State: All logic in this state is in deep sleep mode and power shuts down to save power exception the AON logic. When power-up request detected, it prepares to wake up the TOP domain. GPIO in AON domain act as wake-up resource.

Following diagram illustrates power state machine and timing for power-down/isolation/power domain reset signals during power-up and power-down sequence.

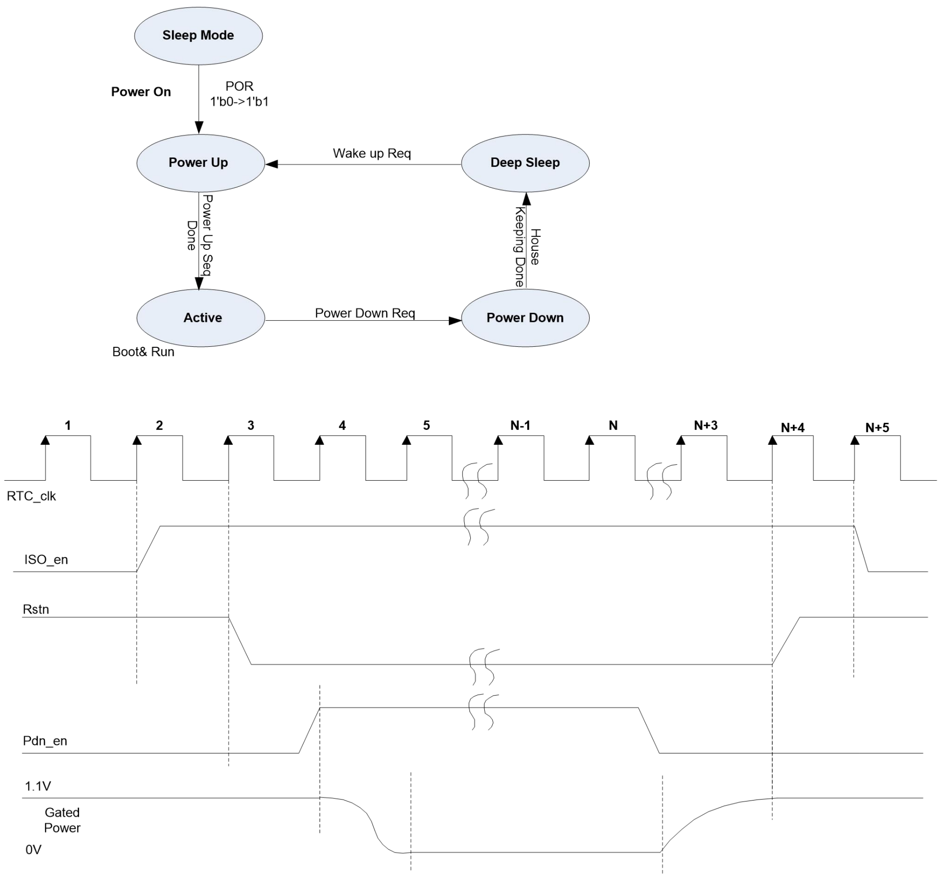


Figure 8- 3 Top Domain Power State Machine and Timing Diagram

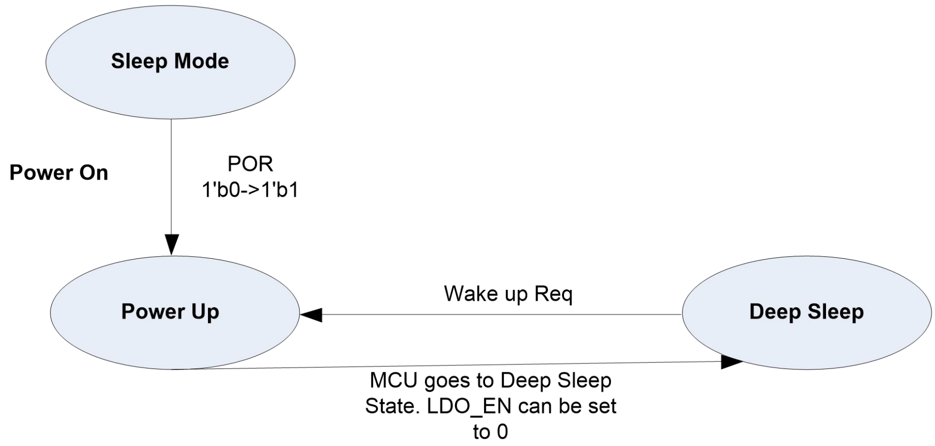


Figure 8- 4 AON Domain Power State Machine and Timing Diagram

Below also illustrates power-up sequence for whole chip starting from POR to timing window that PLL outputs stable clock and chip works in normal state.

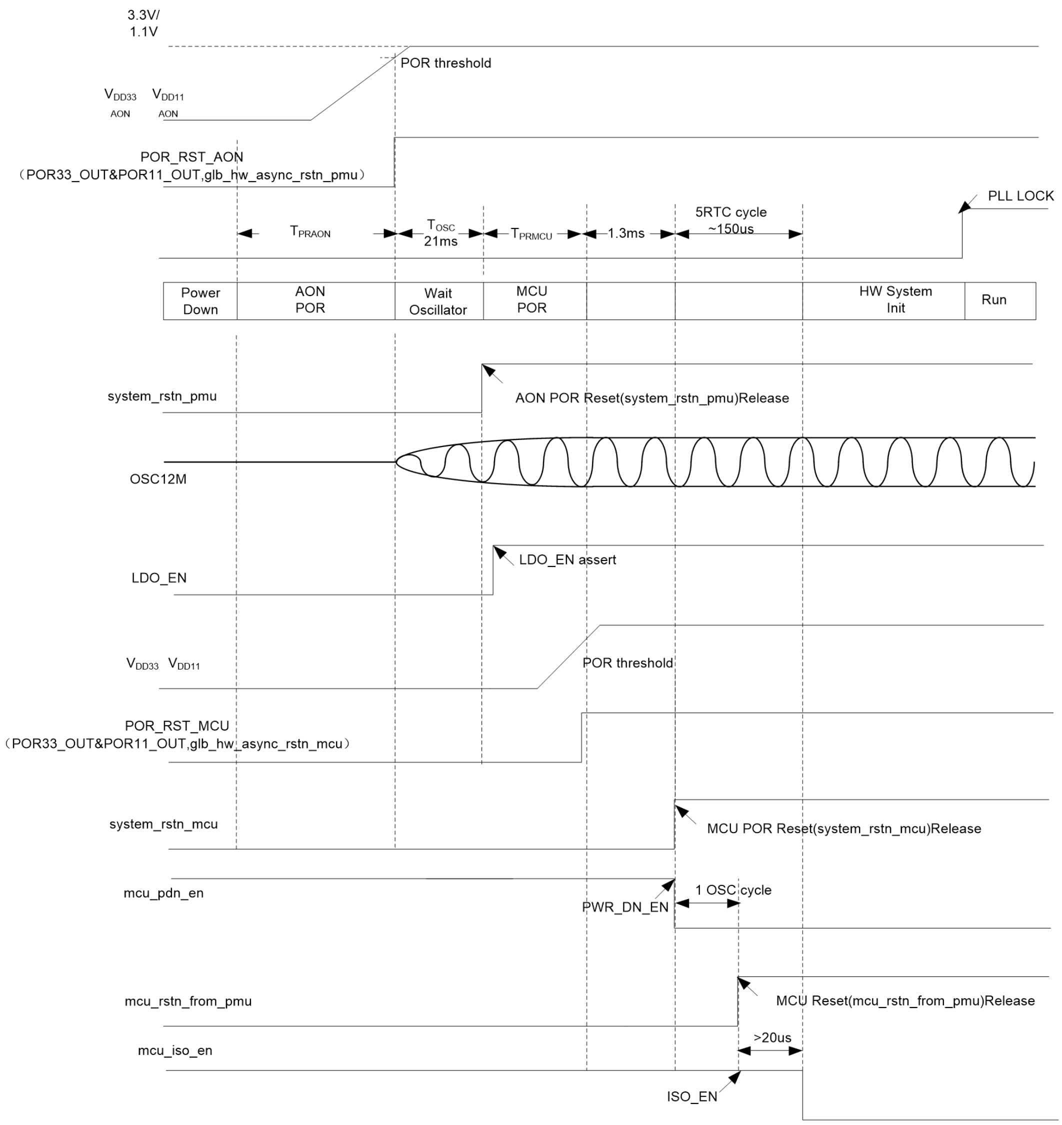
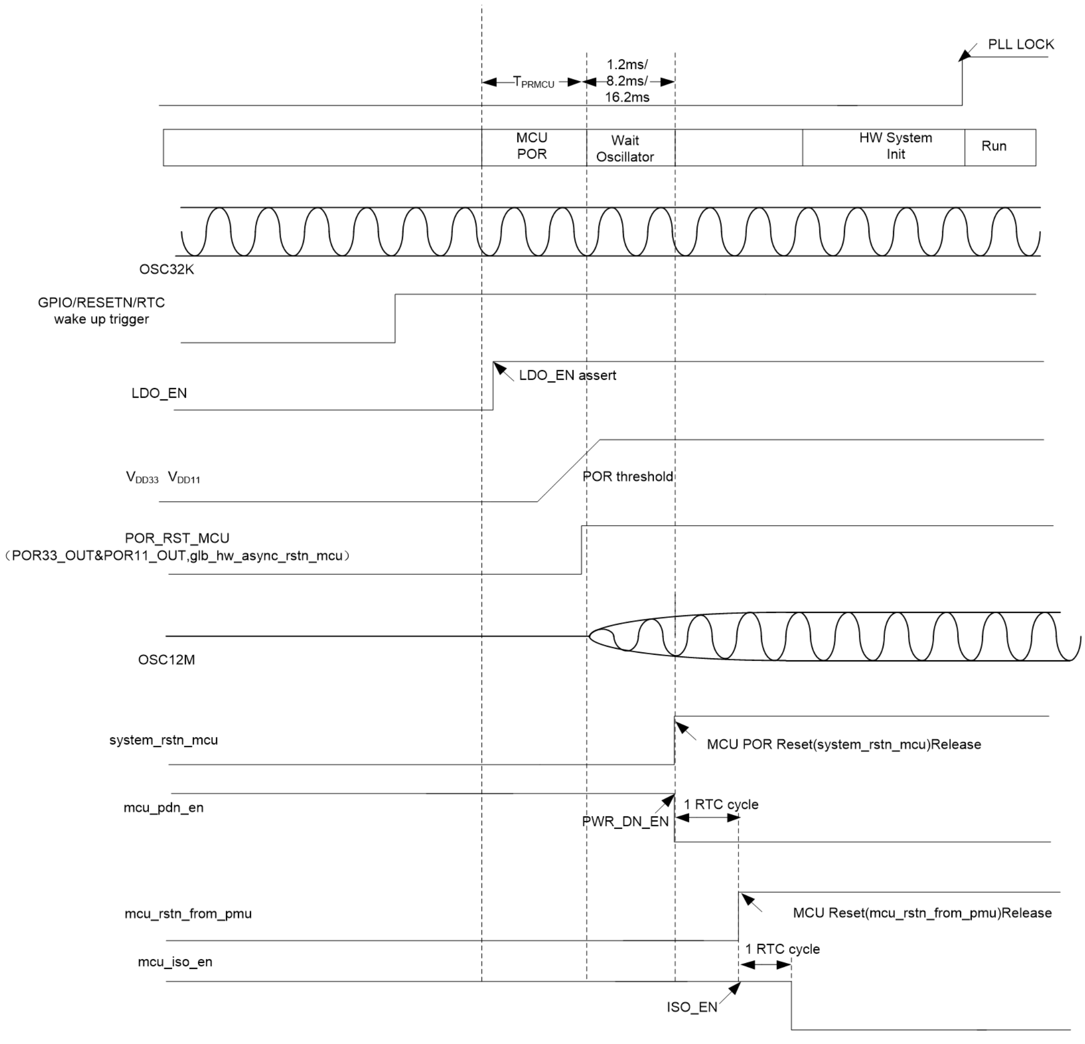


Figure 8- 5 Power-up Sequence Timing Diagram

Below describes how the SOC wake up from deep-sleep state. For go into deep-sleep state, see Figure 8-3 for reference.



108.4. PMU Registers

PMU register address starts from 0x9100_0100. For example, the GPIO_DATA address offset is 12’h100 where its system address is 0x9100_0100.

108.4.1. GPIO_DATA

Address Offset: 12’h100

	Access	Default	
[3:0]	RW	4’h0	Each bit driving output data of corresponding gpio_out in software mode. Bit 0 for pin PWM1 Bit 1 for pin PWM0 Bit 2 for pin UART0_RXD Bit 3 for pin UART0_TXD
[31:4]	Reserved		

108.4.2. GPIO_DIR

Address Offset: 12’h104

	Access	Default	
[3:0]	RW	4’h4	Each bit mapping to gpio_en_n and control direction of corresponding gpio_xp in software mode: "0" for input from gpio_in, "1" for output to gpio_out.
[31:4]	Reserved		

108.4.3. GPIO_CTRL

Address Offset: 12’h108

	Access	Default	
[3:0]	RW	4’h0	"0" for software mode, each bit of gpio_dir is used to setup direction for corresponding pin; "1" for hardware mode, auxiliary hardware signal, gpio_hw_en_n, is used to configure direction for corresponding pin.
[31:4]	Reserved		

108.4.4. GPIO_EXT

Address Offset: 12’h10C

	Access	Default	
[3:0]	RO	4’h0	For each bit, value is corresponding pin of gpio_in in input direction; or same as corresponding bit of gpio_data in output direction.

[31:4]	Reserved		

108.4.5. GPIO_IEN

Address Offset: 12'h110

	Access	Default	
[3:0]	RW	4'h0	For each bit, "0" is disable interrupt event on corresponding pin, while "1" to enable corresponding pins trigger interrupt.
[31:4]	Reserved		

108.4.6. GPIO_IS

Address Offset: 12'h114

	Access	Default	
[3:0]	RW	4'h0	For each bit, "0" is edge-detecting and "1" is level-sensitive on corresponding pin.

108.4.7. GPIO_IBE

Address Offset: 12'h118

	Access	Default	
[3:0]	RW	4'h0	For each bit, "0" is single edge-sensitive (eir falling edge or rising edge), decided by bit of gpio_iev. "1" is both edges on corresponding pin trigger an interrupt.
[31:4]	Reserved		

108.4.8. GPIO_IEV

Address Offset: 12'h11C

	Access	Default	
[3:0]	RW	4'h0	For each bit, "0" means that interrupt is triggered in falling edge of corresponding pins, "1" means interrupt is triggered in rising edge of corresponding pins.
[31:4]	Reserved		

108.4.9. GPIO_RIS

Address Offset: 12’h120

	Access	Default	
[3:0]	RO	4’h0	Raw Interrupt status register before masked. “1”: Raw Interrupt “0”: No raw Interrupt
[31:4]	Reserved		

108.4.10. GPIO_IM

Address Offset: 12’h124

	Access	Default	
[3:0]	RW	4’h0	For each bit, "0" means that interrupt on corresponding gpio_ext bit is unmasked while "1" is masked.
[31:4]	Reserved		

108.4.11. GPIO_MIS

Address Offset: 12’h128

	Access	Default	
[3:0]	RO	4’h0	For each bit, this is masked interrupt status on corresponding gpio_ext bit.
[31:4]	Reserved		

108.4.12. GPIO_IC

Address Offset: 12’h12C

	Access	Default	
[3:0]	WO	4’h0	For each bit, write "1" to clear edge-sensitive interrupt status on corresponding pin, write "0" to a bit has no effects on corresponding interrupt.
[31:4]	Reserved		

108.4.13. GPIO_DB

Address Offset: 12’h130

	Access	Default	
[3:0]	RW	4’h0	For each bit, write "1" to enable function of debounce for each line, write "0" to disable debounce for each line.
[31:4]	Reserved		

108.4.14. GPIO_DFG

Address Offset: 12’h134

	Access	Default	
[3:0]	RW	4’h0	value is 32'h0000_0000, db_clk, which is used to define glitch, will be pclk. value is 32'hFFFF_FFFF, db_clk will be x_clk. Orwise db_clk will be gpio_dfg+1. Anyway, it has no effect on de-bounce when gpio_db is 0.
[31:4]	Reserved		

108.4.15. GPIO_IG

Address Offset: 12’h138

	Access	Default	
[3:0]	RW	4’hc	Determines interrupt group for 32 GPIO pins, each bit represents one GPIO pin. It will be associated with intr[1] when bit is programmed to 1, and associated with intr[0] when bit is programmed to 0.
[31:4]	Reserved		

108.4.16. CHIP_PD_REQ_ADDR

Address Offset: 12’h140

+Field	RW	Default Value	
0	RW	0x0	Chip Power Down Request (one-hot mode orwise won’t take effect) 1’b1: Request to enter into Deep Sleep Mode, that is TOP OFF ON

31:4	RO	0x0	Reserved Field

108.4.17. MCU_WAKEUP_SRC_ADDR

Address Offset: 12’h148

Field	RW	Default Value	
2:0	RO	0x0	MCU Wake Up Source from Deep sleep mode 3’b01:RTC Wake up 3'b10: PMU External Wakeup GPIO0 Int 3'b11: PMU External Wakeup GPIO1 Int 3’b100: Reset pin wakeup
3	RO	0x0	Reserved Field
4	RO	0x0	MCU Reset identity for boot code 1’b0: Reset from POR reset 1’b1: Reset due to waking up from deep sleep mode
31:5	RO	0x0	Reserved Field

108.4.18. CHIP_PWR_STATE_ADDR

Address Offset: 12’h14C

Field	RW	Default Value	
2:0	RO	0x0	TOP Power State: 3’b000: TOP_SLEEP_MODE 3'b001: TOP_POWERUP_MODE 3'b011: TOP_ACTIVE_MODE 3'b010: TOP_POWERDOWN_STATE 3'b100: TOP_STANDBY_STATE
3	RO	0x0	Reserved Field
7	RO	0x0	Reserved Field
10:8	RO	0x0	Reserved Field

11	RO	0x0	Reserved Field
31:12	RO	0x0	Reserved Field

108.4.19. **PMU_SW_AUX0_ADDR**

Address Offset: 12’h150

Field	RW	Default Value	
7:0 [31:8]	RW	0x5a	SW Auxiliary 0 register for boot usage Reserved Field

108.4.20. **PMU_SW_AUX1_ADDR**

Address Offset: 12’h154

Field	RW	Default Value	
7:0 [31:8]	RW	0x5a	SW Auxiliary 1 register for boot usage Reserved Field

108.4.21. **PMU_SW_AUX2_ADDR**

Address Offset: 12’h158

Field	RW	Default Value	
7:0 [31:8]	RW	0x5a	SW Auxiliary 2 register for boot usage Reserved Field

108.4.22. **PMU_MISC_CFG_ADDR**

Address Offset: 12’h160

Field	RW	Default Value	
0	RW	0x0	0: Select PMU OSC12M as system 12M clock source 1: Select back-up OSC12M as system 12M clock source(PMU OSC12M will not be changed).
1	RO	0x1	PMU clock selection status. 1: OSC12M is selected as pmu logic clock 0: OSC32K is selected as pmu logic clock. After OSC32K stable, software should write to PMU_MISC_CTRL.OSC32K_STATBLE to switch the pmu clock source from OSC12M to OSC32K. The switching process is not finished until software poll this bit and got

			value 0.
3:2	RW	0x0	Wake up time configuration: 2’bx0: 16.2ms wake up time for OSC12M stable 2’b01: 8.2ms wake up time for OSC12M stable 2’b11: 1.2ms wake up time for OSC12M stable
31:4	R	0x0	Reserved

108.4.23. PMU_GPIO_CFG_ADDR

Each four bit from low to high configures the control signals for PMW1, PMW0, UART0_RX, UART0_TX
Address Offset: 12’h164

Field	RW	Default Value	
3:0	RW	0xF	Four PMU GPIO Pull-Selection(PS) config
7:4	RW	0x0	Four PMU GPIO Pull-Enable(PE) config
31:8	RO	0x0	Reserved

108.4.24. PMU_GPIO_CFG2_ADDR

Address Offset: 12’h168

Field	RW	Default Value	
3:0	RW	0x0	Four PMU GPIO Driving strenghtg bit0(DR0) config
7:4	RW	0x0	Four PMU GPIO Driving strenghtg bit1(DR1) config
31:8	RO	0x0	Reserved

108.4.25. PMU_GPIO_CFG3_ADDR

Address Offset: 12’h16C

Field	RW	Default Value	
3:0	RW	0xF	Four PMU GPIO Input-Enabled config
7:4	RW	0x0	Reserved
31:8	RO	0x0	Reserved

108.4.26. **PMU_GPIO_CFG4_ADDR**

Address Offset: 12’h170

Field	RW	Default Value	
7:0	RW	0x7C	[7:6] AGC gain [5] enable internal internal RF resistor. When set 0, need 10Mohm resistor on PCB [4] RF resistor walue when [5] is 1: 1-10Mohm 0-15Mohm [3] 0: dont bypass agc function 1 bypass agc function [2:0] operation current:
31:8	RO	0x0	Reserved

108.4.27. **PMU_GPIO_CFG5_ADDR**

Address Offset: 12’h174

Field	RW	Default Value	
6:0	RW	0x19	[6:5] agc gain, default 2'b00 [4] 0-dont bypass agc 1-bypass agc [3:0] 60uA 16M config 1001
7	RW	0x1	osc12m_nopd: 0 allow osc12m to be powered down when in deepstandby mode. 1 osc12m is always on
31:6	RO	0x0	Reserved

109. Peripherals

109.1. Ethernet

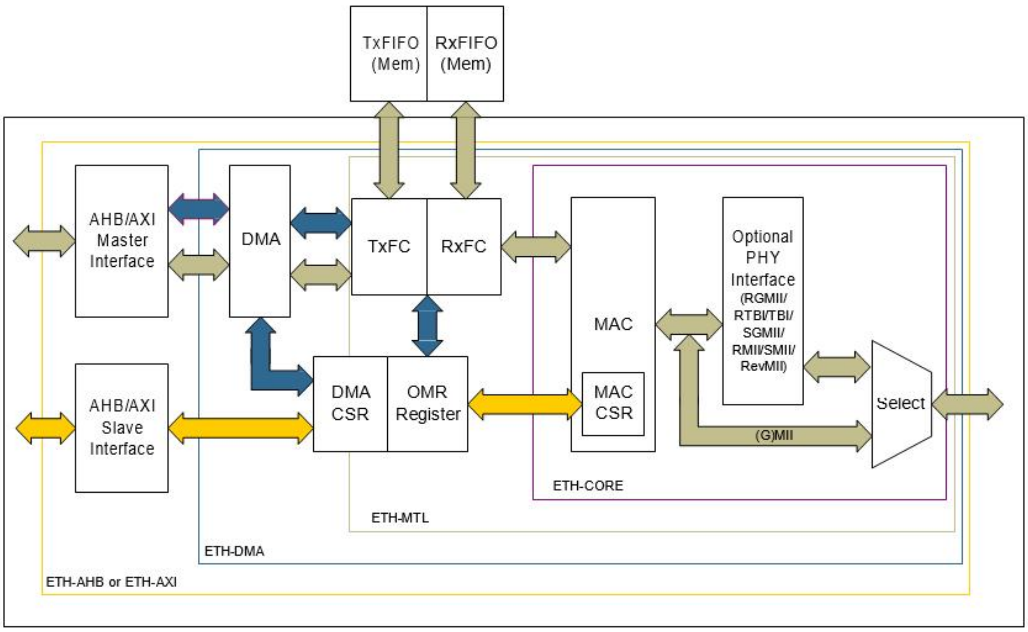
109.1.1. Overview

The ETH enables a host to transmit and receive data over Ethernet in compliance with the IEEE 802.3-2008 standard. The ETH is a configurable product that can be optimized for gate count and latency to meet the application requirements. You can use the ETH in number of applications such as switches and network interface cards.

109.1.2. Features

109.1.2.1. Block Diagram

Figure 9.1- 1 System-Level Block Diagram



109.1.2.2. Standard Compliance

In addition to the default interfaces defined in the IEEE 802.3 specifications, the ETH supports several industry standard interfaces to the PHY. The ETH is compliant with the following standards:

- IEEE 802.3-2008 for Ethernet MAC, Media Independent Interface (MII), Ten Bit Interface

(TBI)

- IEEE 1588-2008 standard for precision networked clock synchronization
- IEEE 802.1AS-2011and 802.1-Qav-2009 for Audio Video (AV) traffic
- IEEE 802.3az-2010, for Energy Efficient Ethernet (EEE)
- AMBA 2.0 for AHB master and slave ports
- AMBA 3.0 for AXI master and slave and APB3 slave ports
- RMII specification version 1.2 from RMII consortium

109.1.2.3. MAC

The MAC supports the following features:

- 10, 100Mbps data transfer rates with the following PHY interfaces:
  - ❑ IEEE 802.3-compliant MII interface to communicate with an external Gigabit/Fast Ethernet PHY
  - ❑ RMII interface to communicate with an external Fast Ethernet PHY
- Full-duplex operation:
  - ❑ IEEE 802.3x flow control automatic transmission of zero-quanta Pause frame on flow control input de-assertion
- Half-duplex operation:
  - ❑ CSMA/CD Protocol support
  - ❑ Flow control using backpressure support (based on implementation-specific white papers and UNH Ethernet Clause 4 MAC Test Suite - Annex D)
  - ❑ Frame bursting and frame extension in 1000 Mbps half-duplex operation
- Preamble and start of frame data (SFD) insertion in Transmit path
- Preamble and SFD deletion in the Receive path
- Automatic CRC and pad generation controllable on a per-frame basis
- Automatic Pad and CRC Stripping options for receive frames
- Flexible address filtering modes, such as:
  - ❑ Up to 31 additional 48-bit perfect (DA) address filters with masks for each byte
  - ❑ Up to 96 additional 48-bit perfect (DA) address filters that can be selected in blocks of 32 and 64 registers
  - ❑ Up to 31 48-bit SA address comparison check with masks for each byte
  - ❑ 64-bit, 128-bit, or 256-bit Hash filter (optional) for multicast and unicast (DA) addresses
  - ❑ Option to pass all multicast addressed frames
  - ❑ Promiscuous mode to pass all frames without any filtering for network monitoring
  - ❑ Pass all incoming packets (as per filter) with a status report
- Programmable frame length to support Standard or Jumbo Ethernet frames with up to 16 KB of size
- Programmable Interframe Gap (IFG) (40-96 bit times in steps of 8)
- Option to transmit frames with reduced preamble size

- Separate 32-bit status for transmit and receive packets
- IEEE 802.1Q VLAN tag detection for reception frames
- Additional frame filtering:
  - VLAN tag-based: Perfect match and Hash-based (optional) filtering
  - Layer 3 and Layer 4-based: TCP or UDP over IPv4 or IPv6
- Separate transmission, reception, and control interfaces to the application
- Configurable big-endian and little-endian for Transmit and Receive paths
- 32-bit, 64-bit, or 128-bit data transfer interface on system-side
- Network statistics (optional) with RMON/MIB Counters (RFC2819/RFC2665)
- Optional module to detect remote wake-up frames and AMD magic packets
- Optional Receive module for checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame (Type 1)
- Optional Enhanced Receive module for checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams (Type 2)
- Optional module to support Ethernet frame timestamping as described in IEEE 1588-2002 and IEEE 1588-2008. The 64-bit timestamps are given in the transmit or receive status of each frame
- MDIO master interface (optional) for PHY device configuration and management
- Standard IEEE 802.3az-2010 for Energy Efficient Ethernet
- Flexibility to control the Pulse-Per-Second (PPS) output signal (ptp_pps_o)
- CRC replacement, Source Address field insertion or replacement, and VLAN insertion, replacement, and deletion in transmitted frames with per-frame control
- Programmable watchdog timeout limit in the receive path
- Option to select up to 4 general purpose inputs and outputs

The general purpose inputs are programmable to generate interrupts on rising or falling edges.

109.1.2.4. Transaction Layer (MTL)

The MTL block consists of two sets of FIFOs: a Transmit FIFO with programmable threshold capability, and a Receive FIFO with a programmable threshold (default of 64 bytes).

The MTL block supports the following features:

- 32-bit, 64-bit, or 128-bit Transaction Layer block (bridges the application and the ETH-CORE)
- Single-channel Transmit and Receive engines
- Data transfers executed using simple FIFO-protocol
- Synchronization for all clocks in the design (Transmit, Receive, and system clocks)
- Optimization for packet-oriented transfers with frame delimiters
- Four separate ports for system-side and ETH-CORE-side transmission and reception
- Two 2-port RAM-based asynchronous FIFOs with synchronous/asynchronous Read and

Write operation with respect to the Read and Write clocks (one for transmission and one for reception)

- FIFO instantiation outside the top-level module to facilitate memory testing/instantiation
- 128-byte, 256-byte, or 512-byte, or 1-KB, 2-KB, 4-KB, 8-KB, 16-KB, or 32-KB receive FIFO sizes on reception
- Optional interface to indicate the length of a received frame at the top of the MTL Rx FIFO in the ETH-MTL configuration
- Programmable burst length for starting a burst up to half the size of the MTL Rx and Tx FIFO in the ETH-MTL configuration
- Insertion of Receive Status vectors into the Receive FIFO after the EOF transfer

This enables multiple-frame storage in the Receive FIFO without requiring another FIFO to store those frames' Receive Status.

- Configurable Receive FIFO threshold (default fixed at 64 bytes) in Cut-Through or Threshold mode
- Option to filter all error frames on reception and not forward them to the application in the store- and-forward mode
- Option to forward under-sized good frames
- Supports statistics by generating pulses for frames dropped or corrupted (due to overflow) in the Receive FIFO
- 256-byte or 512-byte, or 1-KB, 2-KB, 4-KB, 8-KB, or 16-KB FIFO sizes on transmission
- Store and Forward mechanism for transmission to the MAC
- Threshold control for transmit buffer management
- Configurable number of frames to be stored in FIFO at any time

The default is two frames (fixed) with internal DMA, and up to eight frames in ETH-MTL configuration

- Automatic generation of Pause frame control or backpressure signal to the MAC based on Receive FIFO-fill (threshold configurable) level
- Automatic retransmission of Collision frames for transmission
- Discard frames on late collision, excessive collisions, excessive deferral, and under-run conditions
- Software control to flush Tx FIFO
- Disabling of Data FIFO RAM chip-select when inactive to reduce power consumption
- Optional module to calculate and insert IPv4 header checksum and TCP, UDP, or ICMP checksum in frames transmitted in the store-and-forward mode

109.1.2.5. DMA Block

The DMA block exchanges data between the MTL block and host memory. The host can use a set of registers (DMA CSR) to control the DMA operations.

The DMA block supports the following features:

- 32-bit, 64-bit, and 128-bit data transfers
- Single-channel Transmit and Receive engines
- Fully synchronous design operating on a single system clock (except for CSR module, when a separate CSR clock is configured)
- Optimization for packet-oriented DMA transfers with frame delimiters
- Byte-aligned addressing for data buffer support
- Dual-buffer (ring) or linked-list (chained) descriptor chaining
- Descriptor architecture to allow large blocks of data transfer with minimum CPU intervention (each descriptor can transfer up to 8 KB of data)
- Comprehensive status reporting for normal operation and transfers with errors
- Individual programmable burst size for Transmit and Receive DMA Engines for optimal host bus utilization
- Programmable interrupt options for different operational conditions
- Per-frame Transmit or Receive complete interrupt control
- Round-robin or fixed-priority arbitration between Receive and Transmit engines
- Start and Stop modes
- Separate ports for host CSR access and host data interface

109.1.2.6. **AMBA AHB Interface**

The AMBA AHB interface features can be divided into the following categories:

- AHB Master Interface
- AHB Slave Interface

109.1.2.7. **AMBA AXI Interface**

The AMBA AXI interface features are divided into the AXI Master Interface and AXI Slave Interface categories.

109.1.2.8. **AMBA APB Slave Interface**

The ETH supports the APB slave interface for all configurations. It provides the following features:

- 32-bit access without any byte-enable
- APB3 interface with pready handshake signal

The AMBA APB slave interface does not generate Error response in the APB3 interface.

109.1.2.9. Audio Video Support

The ETH supports the following Audio Video (AV) features:

- Separate channels or queues for AV data transfer in 100 Mbps and 1000 Mbps modes
- Configuring up to two additional channels (Channel 1 and Channel 2) on the transmit and receive paths for AV traffic

The Channel 0 is available by default and carries the legacy best-effort Ethernet traffic on the transmit side.

- IEEE 802.1-Qav specified credit-based shaper (CBS) algorithm for the additional transmit channels
- Separate DMA, Tx FIFO, and Rx FIFO (MTL) for each additional channel (system-side interface [AHB, AXI, or native] remains the same)
- Programmable control to route received VLAN tagged non-AV packets to channels or queues.

109.1.2.10. Monitoring, Testing, and Debugging Support

The ETH supports the following features that you can use for monitoring, testing, and debugging:

- Internal loopback on the GMII or MII for debugging
- DMA states (Tx and Rx) as status bits
- Debug status register that gives status of FSMs in Transmit and Receive data-paths and FIFO fill- levels
- Application Abort status bits
- MMC (RMON) module
- Current Tx/Rx Buffer pointer as status registers
- Current Tx/Rx Descriptor pointer as status registers
- Statistical counters to calculate the bandwidth served by each transmit channel when AV support is enabled

109.1.3. Programming Guide

109.1.3.1. Initializing DMA

The initialization sequence in this section is used for ETH configurations without Audio Video

(AV) feature. For information about DMA initialization steps for configurations with AV feature, see "Initializing the DMA"

Complete the following steps to initialize the DMA:

1. Provide a software reset. This resets all of the ETH internal registers and logic. (DMA Register 0 (Bus Mode Register) – bit 0).

2. Wait for the completion of the reset process (poll bit 0 of the DMA Register 0 (Bus Mode Register), which is only cleared after the reset operation is completed).

3. (Only for ETH-AHB and ETH-AXI configurations) Poll the bits of Register 11 (AHB or AXI Status Register) to confirm that all previously initiated (before soft reset) or ongoing AHB or AXI transactions are complete.

4. Program the following fields to initialize the Bus Mode Register by setting values in DMA Register 0 (Bus Mode Register):

a. Mixed Burst and AAL (only if ETH-AHB/ETH-AXI configuration is selected)
b. Fixed burst or undefined burst
c. Burst length values and burst mode values.
d. Descriptor Length (only valid if Ring Mode is used)
e. Tx and Rx DMA Arbitration scheme

5. Program the AXI Interface options in Register 10 (AXI Bus Mode Register). If fixed burst length is enabled, then select the maximum burst length possible on the AXI bus (bits[7:1]).

6. Create a proper descriptor chain for transmit and receive. In addition, ensure that the receive descriptors are owned by DMA (bit 31 of descriptor should be set). When OSF mode is used, at least two descriptors are required.

7. Make sure that your software creates three or more different transmit or receive descriptors in the chain before reusing any of the descriptors.

8. Initialize receive and transmit descriptor list address with the base address of the transmit and receive descriptor (Register 3 (Receive Descriptor List Address Register) and Register 4 (Transmit Descriptor List Address Register) respectively).

9. Program the following fields to initialize the mode of operation in Register 6 (Operation Mode Register)

a. Receive and Transmit Store And Forward
b. Receive and Transmit Threshold Control (RTC and TTC)
c. Hardware Flow Control enable
d. Flow Control Activation and De-activation thresholds for MTL Receive and Transmit FIFO (RFA and RFD)
e. Error Frame and undersized good frame forwarding enable
f. OSF Mode

10. Enable the interrupts by programming Register 7 (Interrupt Enable Register).

11. (Only for ETH-AHB and ETH-AXI configurations) Read Register 11 (AHB or AXI Status Register) to confirm that all previous AHB or AXI transactions are complete.

12. Start the Receive and Transmit DMA by setting SR (bit 1) and ST (bit 13) of the control register (DMA Register 6 (Operation Mode Register)).

109.1.3.2. **Initializing MAC**

The following MAC Initialization operations can be performed after DMA initialization. If the MAC initialization is done before the DMA is set-up, then enable the MAC receiver (last step in following sequence) only after the DMA is active. Otherwise, received frames fills the Rx FIFO and overflow. Steps (1) and (2) are to be followed if the TBI, SGMII, or RTBI PHY interface is enabled, otherwise follow steps (3) and (4).

1. Program the ETH Register 48 (AN Control Register) to enable Auto-negotiation ANE (bit-12). Setting ELE (bit-14) of this register enables the PHY to loop back the transmit data and RAN (bit-9) can be set to restart auto-negotiation (only for TBI, SGMII, and RTBI PHY interfaces).

2. Check the ETH Register 49 (AN Status Register) for completion of the Auto-negotiation process. ANC (bit-5) should be set. The link status (bit-2), when set, indicates that the link is up (only for TBI, SGMII, and RTBI PHY interfaces).

3. Program the ETH Register 4 (GMII Address Register) for controlling the management cycles for external PHY. For example, Physical Layer Address PA (bits 15-11). In addition, set bit 0 (GMII Busy) for writing into PHY and reading from PHY.

4. Read the 16-bit data of Register 5 (GMII Data Register) from the PHY for link up, speed of operation, and mode of operation, by specifying the appropriate address value in bits 15-11 of Register 4 (GMII Address Register).

5. Provide the MAC address registers (Register 16 (MAC Address0 High Register) and Register 17 (MAC Address0 Low Register)). If more than one MAC address is enabled in your configuration (during configuration in coreConsultant), program the MAC addresses appropriately.

6. If Hash filtering is enabled in your configuration, program Register 2 (Hash Table High Register) and Register 3 (Hash Table Low Register).

7. Program the following fields to set the appropriate filters for the incoming frames in Register 1 (MAC Frame Filter):

a. Receive All
b. Promiscuous mode
c. Hash or Perfect Filter
d. Unicast, multicast, broadcast, and control frames filter settings

8. Program the following fields for proper flow control in Register 6 (Flow Control Register):
a. Pause time and other Pause frame control bits
b. Receive and Transmit Flow control bits
c. Flow Control Busy/Backpressure Activate

9. Program the Interrupt Mask register bits, as required, and if applicable, for your configuration.

10. Program the appropriate fields in Register 0 (MAC Configuration Register). For example, Inter-frame gap while transmission and jabber disable. Based on the Auto-negotiation you can set the Duplex mode (bit 11) or port select (bit 15).

11. Set Bit 3 (TE) and Bit 2 (RE) in Register 0 (MAC Configuration Register).

109.1.3.3. Performing Normal Receive and Transmit Operation

For normal operation, complete the following steps:

1. For normal transmit and receive interrupts, read the interrupt status. Then, poll the descriptors, reading the status of the descriptor owned by the Host (either transmit or receive).

2. Set appropriate values for the descriptors, ensuring that transmit and receive descriptors are owned by the DMA to resume the transmission and reception of data.

3. If the descriptors are not owned by the DMA (or no descriptor is available), the DMA goes into SUSPEND state. The transmission or reception can be resumed by freeing the descriptors and issuing a poll demand by writing 0 into the Tx/Rx poll demand register (Register 1 (Transmit Poll Demand Register) and Register 2 (Receive Poll Demand Register)).

4. The values of the current host transmitter or receiver descriptor address pointer can be read for the debug process (Register 18 (Current Host Transmit Descriptor Register) and Register 19 (Current Host Receive Descriptor Register)).

5. The values of the current host transmit buffer address pointer and receive buffer address pointer can be read for the debug process (Register 20 (Current Host Transmit Buffer Address Register) and Register 21 (Current Host Receive Buffer Address Register)).

109.1.3.4. Stopping and Starting Transmission

Complete the following steps to pause the transmission for some time:

1. Disable the Transmit DMA (if applicable) by clearing Bit 13 (ST) of Register 6 (Operation Mode Register).
2. Wait for any previous frame transmissions to complete. You can check this by reading the appropriate bits of Register 9 (Debug Register).
3. Disable the MAC transmitter and MAC receiver by clearing Bit 3 (TE) and Bit 2 (RE) in Register 0 (MAC Configuration Register).
4. Disable the Receive DMA (if applicable), after making sure that the data in the Rx FIFO is transferred to the system memory (by reading Register 9 (Debug Register)).
5. Make sure that both Tx FIFO and Rx FIFO are empty.
6. To restart the operation, first start the DMAs, and then enable the MAC Transmitter and Receiver.

109.1.3.5. Programming Guidelines for GMII Link Transitions

109.1.3.5.1. Transmit and Receive Clocks are Running when the Link is Down

Complete the following steps when the link is down but the Transmit and Receive clocks are running:

1. Disable the Transmit DMA (if applicable), by clearing bit 13 (ST) of Register 6 (Operation Mode Register).

2. Disable the MAC receiver by clearing Bit 2 (RE) of Register 0 (MAC Configuration Register).

3. Wait for any previous frame transmissions to complete from the Tx FIFO. You can do this by reading the appropriate bits of Register 9 (Debug Register) or-Flush the Tx FIFO for faster empty operation.

4. Disable the MAC transmitter by clearing bit 3 (TE) in Register 0 (MAC Configuration Register).

5. After the link is up, read the PHY registers to know the latest configuration and accordingly program the MAC registers.

6. Restart the operation by starting the Tx DMA, and then enabling the MAC Transmitter and Receiver.

You do not need to disable the Rx DMA. As the Receiver is disabled, the FIFO does not get any data in the Rx FIFO.

109.1.3.5.2. Transmit and Receive Clocks are Stopped when the Link is Down

Complete the following steps when the link is down and the Transmit and Receive clocks are stopped:

1. Wait till the link is up and the Transmit and Receive clocks are active. When the Transmit and Receive clocks are stopped, then disabling the transmit or receive operations does not have any effect. Therefore, the software must wait till the link is up again.

2. Disable the Transmit DMA (if applicable), by clearing Bit 13 (ST) of the Register 6 (Operation Mode Register).

3. Disable the MAC receiver by clearing Bit 2 (RE) of Register 0 (MAC Configuration Register).

4. Wait for any previous frame transmissions to complete from the Tx FIFO. You can do this by reading the appropriate bits of Register 9 (Debug Register) or Flush the Tx FIFO for faster empty operation.

5. Disable the MAC transmitter by clearing Bit 3 (TE) in Register 0 (MAC Configuration Register).

6. After the link is up, read the PHY registers to know the latest configuration and accordingly program the MAC registers.

7. Restart the operation by starting the Tx DMA, and then enabling the MAC Transmitter and Receiver

109.1.3.6. **Programming Guidelines for IEEE 1588 Timestamping**

109.1.3.6.1. Initialization Guideline for System Time Generation

You can enable the timestamp feature by setting Bit 0 of the Timestamp control register. However, it is essential that the timestamp counter should be initialized after this bit is set. Complete the following steps during ETH initialization:

1. Mask the Timestamp Trigger interrupt by setting Bit 9 of Register 15 (Interrupt Mask Register).
2. Program Bit 0 in Register 448 (Timestamp Control Register) to enable timestamping.
3. Program Register 449 (Sub-Second Increment Register) based on the PTP clock frequency.
4. If you are using the Fine Correction approach, program Register 454 (Timestamp Addend Register) and set Bit 5 of Register 448 (Timestamp Control Register).
5. Poll the Timestamp Control register until Bit 5 is cleared.
6. Program Bit 1 of Register 448 (Timestamp Control Register) to select the Fine Update method (if required).
7. Program Register 452 (System Time – Seconds Update Register) and Register 453 (System Time – Nanoseconds Update Register) with the appropriate time value.
8. Set Bit 2 in Register 448 (Timestamp Control Register). The timestamp counter starts operation as soon as it is initialized with the value written in the Timestamp Update registers.
9. Enable the MAC receiver and transmitter for proper timestamping.

109.1.3.6.2. System Time Correction

To synchronize or update the system time in one process (coarse correction method), complete the following steps:

1. Set the offset (positive or negative) in the Timestamp Update registers (Register 452 (System Time – Seconds Update Register) and Register 453 (System Time – Nanoseconds Update Register)).
2. Set Bit 3 (TSUPDT) of the Register 448 (Timestamp Control Register).

The value in the Timestamp Update registers is added to or subtracted from the system time when the TSUPDT bit is cleared.

To synchronize or update the system time to reduce system-time jitter (fine correction method), complete the following steps:

1. With the help of the algorithm explained in “System Time Register Module”, calculate the rate by which you want to make the system time increments slower or faster.
2. Update the Register 454 (Timestamp Addend Register) with the new value and set Bit 5 of the Register 448 (Timestamp Control Register).
3. Wait for the time for which you want the new value of the Addend register to be active. You can do this by enabling the Timestamp Trigger interrupt after the system time reaches the target value.
4. Program the required target time in Register 455 (Target Time Seconds Register) and Register 456 (Target Time Nanoseconds Register).

5. Unmask the Timestamp interrupt by clearing bit 9 of Register 15 (Interrupt Mask Register).

6. Set bit 4 in Register 448 (Timestamp Control Register).

7. When this trigger causes an interrupt, read Register 14 (Interrupt Status Register).

8. Reprogram Register 454 (Timestamp Addend Register) with the old value and set bit 5 again.

109.1.3.7. Programming Guidelines for AV Feature

109.1.3.7.1. Initializing the DMA

This initialization sequence is used only for ETH-AHB configuration with AV feature. Complete the following steps to initialize the DMA:

1. Provide a software reset to reset all ETH internal registers and logic (bit 0 in Register 0 (Bus Mode Register)).

2. Wait for the completion of the reset process. Poll bit 0 of the Register 0 (Bus Mode Register), which is cleared only after the reset operation is completed.

3. Program the following fields to initialize the Bus Mode Register by setting the values in Register 0 (Bus Mode Register)

a. Mixed Burst and AAL (only if ETH-AHB/ETH-AXI configuration is selected)
b. Fixed burst or undefined burst
c. Burst length values and burst mode values
d. Descriptor Length (only valid if Ring Mode is used)
e. Tx and Rx DMA Arbitration scheme and two level priority weight for the channel

4. Create a proper descriptor chain for transmit and receive. In addition, ensure that the DMA owns the receive descriptors by setting Bit 31 of the descriptor. When OSF mode is used, at least two descriptors are required.

5. Make sure that your software creates three or more different transmit or receive descriptors in the chain before reusing any of the descriptors.

6. Initialize receive and transmit descriptor list address with the base address of the transmit and receive descriptor (Register 3 (Receive Descriptor List Address Register) and Register 4 (Transmit Descriptor List Address Register), respectively).

7. Program the following fields to initialize the mode of operation by setting the values in DMA Register 6 (Operation Mode Register):

a. Receive and Transmit Store And Forward
b. Receive and Transmit Threshold Control (RTC and TTC)
c. Error Frame and undersized good frame forwarding enable d. OSF Mode

8. Clear the interrupt requests, by writing to those bits of the status register (interrupt bits only) that are set. For example, writing 1 into bit 16 (normal interrupt summary) clears this bit (Register 5 (Status Register).

9. Enable the interrupts by programming the Register 7 (Interrupt Enable Register).

10. Repeat steps 3 through 9 for all additional channels of AV feature.

11. Program the CBS control register, idleSlope, sendSlope, hiCredit, and loCredit registers of Channel 1 and Channel 2.

12. Start the Receive and Transmit DMA by setting SR (bit 1) and ST (bit 13) of the control registers
(DMA Register 6, Register 70 and Register 134 operation mode register) for all the channels.

109.1.3.7.2. Enabling Slot Number Checking

You can use the slot number check feature to specify the intervals at which the Channel 1 or Channel 2 DMA fetches the frames from the AHB/AXI system bus. This feature is useful for a uniform and periodic transfer of the AV traffic from the host memory. The feature is available only when you enable timestamping and program the Register 449 (Sub-Second Increment Register). Complete the following steps to enable the slot number checking:

1. Enable timestamping by following the steps described in "Initialization Guideline for System Time Generation".

2. Make sure that the SLOTNUM field (bits 6:3) of Transmit Descriptor Word 0 (TDES0) contains a valid slot number. You can read the current reference slot number from the Slot Function Control and Status register.

3. Set Bit 0 (ESC) of the Slot Function Control and Status register of a channel to enable the slot number checking.

109.1.3.7.3. Enabling Average Bits Per Slot Reporting

The CBS Status register of the additional AV channels (Channel 1 and Channel 2) provides information about the average bits that are transmitted in a slot. The software can asynchronously read this register to retrieve information about the average bits transmitted per slot. Complete the following steps to enable average bits per slot reporting:

1. Enable timestamping by following the steps described in "Initialization Guideline for System Time Generation".
2. Program Bits [6:4], SLC, of the CBS Control register of a channel with number of slots over which the average transmitted bits per slot need to be computed.
3. Enable Bit 17 (ABPSSIE) of the CBS Control register of a channel to generate the average bits per slot interrupt.
4. Read Bits [16:0], ABS, from the CBS Status register of a channel on each interrupt.

109.1.3.8. **Programming Guidelines for Energy Efficient Ethernet**

109.1.3.8.1. Entering and Exiting the Tx LPI Mode

You can configure the Energy Efficient Ethernet (EEE) feature in coreConsultant. Complete the following steps during ETH core initialization:

1. Read the PHY register through the MDIO interface, check if the remote end has the EEE capability, and then negotiate the timer values.

2. Program the PHY registers through the MDIO interface (including the RX_CLK_stoppable bit that indicates to the PHY whether to stop Rx clock in LPI mode.)

3. Program Bits[26:15] and Bits[15:0] in Register 13 (LPI Timers Control Register).

4. Read the link status of the PHY chip by using the MDIO interface and update Bit 17 of Register 12 (LPI Control and Status Register) accordingly. This update should be done

whenever the link status in the PHY chip changes.

5. Set Bit 16 of Register 12 (LPI Control and Status Register) to make the MAC enter the LPI state.

The MAC enters the LPI mode after completing the transmission in progress and sets Bit 0.

6. Reset Bit 16 of Register 12 (LPI Control and Status Register) to bring the MAC out of the LPI state.

The MAC waits for the time programmed in Bits [15:0] before setting the TLPIEX interrupt status bit and resuming the transmission.

109.1.3.8.2. Gating Off the CSR Clock in the LPI Mode

You can gate off the CSR clock to save the power when the MAC is in the Low-Power Idle (LPI) mode.

109.1.3.8.2.1. Gating Off the CSR Clock in the Rx LPI Mode

The following operations are performed when the MAC receives the LPI pattern from the PHY.

1. The MAC RX enters the LPI mode and the Rx LPI entry interrupt status [RLPIEN interrupt of
Register 12 (LPI Control and Status Register)] is set.

2. The interrupt pin (sbd_intr_o) is asserted. The sbd_intr_o interrupt is cleared when the host reads the Register 12 (LPI Control and Status Register).

After the sbd_intr_o interrupt is asserted and the MAC Tx is also in the LPI mode, you can gate-off the CSR clock. If the MAC TX is not in the LPI mode when you gate off the CSR clock, the events on the MAC transmitter do not get reported or updated in the CSR.

For restoring the CSR clock, wait for the LPI exit indication from the PHY after which the MAC asserts the LPI exit interrupt on lpi_intr_o (synchronous to clk_rx_i). The lpi_intr_o interrupt is cleared when Register 12 is read.

109.1.3.8.2.2. Gating Off the CSR Clock in the Tx LPI Mode

The following operations are performed when Bit 16 (LPIEN) of Register 12 (LPI Control and Status Register) is set:

1. The Transmit LPI Entry interrupt (TLPIEN bit of Register 12) is set.

2. The interrupt pin (sbd_intr_o) is asserted. The sbd_intr_o interrupt is cleared when the host reads the Register 12.

After the sbd_intr_o interrupt is asserted and the MAC RX is also in the LPI mode, you can gate off the CSR clock. If the MAC RX is not in the LPI mode when you gate off the CSR clock, the events on the MAC receiver do not get reported or updated in the CSR.

For restoring the CSR clock, switch on the CSR clock when the MAC has to come out of the TX LPI mode. After the CSR clock is resumed, reset Bit 16 (LPIEN) of Register 12 (LPI Control and Status Register) to bring the MAC out of the LPI mode.

109.1.3.9. Programming Guidelines for Flexible Pulse-Per-Second (PPS) Output

109.1.3.9.1. Generating Single Pulse on PPS

To generate single Pulse on PPS:

1. Program 11 or 10 (for interrupt) in Bits [6:5], TRGTMODSEL, of Register 459 (PPS Control Register).
This instructs the MAC to use the Target Time registers (register 455 and 456) for start time of PPS signal output.

2. Program the start time value in the Target Time registers (register 455 and 456).

3. Program the width of the PPS signal output in Register 473 (PPS0 Width Register).

4. Program Bits [3:0], PPSCMD, of Register 459 (PPS Control Register) to 0001. This instructs the MAC to generate single pulse on the PPS signal output at the time programmed in the Target Time registers (register 455 and 456).

When the PPSCMD is executed (PPSCMD bits = 0), you can cancel the pulse generation by giving the Cancel Start Command (PPSCMD=0011) before the programmed start time elapses. You can also program the behavior of the next pulse in advance. To program the next pulse:

1. Program the start time for the next pulse in the Target Time registers (register 455 and 456). This time should be more than the time at which the falling edge occurs for the previous pulse.

2. Program the width of the next PPS signal output in Register 473 (PPS0 Width Register).

3. Program Bits [3:0], PPSCMD, of Register 459 (PPS Control Register) to generate a single pulse after the time at which the previous pulse is de-asserted. This instructs the MAC to generate single pulse on the PPS signal output, at the time programmed in Target Time registers.

If you give this command before the previous pulse becomes low, then the new command overwrites the previous command and the ETH may generate only 1 extended pulse.

109.1.3.9.2. Generating a Pulse Train on PPS

To generate a pulse train on PPS:

1. Program 11 or 10 (for interrupt) in Bits [6:5], TRGTMODSEL, of Register 459 (PPS Control Register).
This instructs the MAC to use the Target Time registers (register 455 and 456) for start time of the PPS signal output.

2. Program the start time value in the Target Time registers (register 455 and 456).

3. Program the interval value between the train of pulses on the PPS signal output in Register 473 (PPS0 Width Register).

4. Program the width of the PPS signal output in Register 473 (PPS0 Width Register).

5. Program Bits[3:0], PPSCMD, of Register 459 (PPS Control Register) to 0010. This instructs the MAC to generate train of pulses on the PPS signal output with start time programmed in Target Time registers (register 455 and 456).

By default, the PPS pulse train is free-running unless stopped by 'STOP Pulse train at time' or 'STOP Pulse Train immediately' commands.

6. Program the stop value in the Target Time registers (register 455 and 456). Ensure that Bit 31 (TSTRBUSY) of Register 456 (Target Time Nanoseconds Register) is reset before programming the Target Time registers (register 455 and 456) again.
7. Program the PPSCMD field (bit 3:0) of Register 459 (PPS Control Register) to 0100. This stops the train of pulses on PPS signal output after the programmed stop time specified in Step 6 elapses.

You can stop the pulse train at any time by programming 0101 in the PPSCMD field. Similarly, you can cancel the Stop Pulse train command (given in Step 7) by programming 0110 in the PPSCMD field before the time (programmed in Step 6) elapses. You can cancel the pulse train generation by programming 0011 in the PPSCMD field before the programmed start time (in Step 2) elapses.

109.1.3.9.3. Generating an Interrupt without Affecting the PPS

The Bits [6:5], TRGTMODSEL, of the Register 459 (PPS Control Register) enable you to program the Target
Time registers (register 455 and 456) to do any one of the following:

- Generate only interrupts.
- Generate interrupts and the PPS start and stop time.
- Generate only PPS start and stop time.

To program the Target Time registers (register 455 and 456) to generate only interrupt event:

1. Program 00 (for interrupt) in Bits [6:5], TRGTMODSEL, of Register 459 (PPS Control Register). This instructs the MAC to use the Target Time registers (register 455 and 456) for target time interrupt.
2. Program a target time value in the Target Time registers (register 455 and 456). This instructs the
MAC to generate an interrupt when the target time elapses.

If Bits [6:5], TRGTMODSEL, are changed (for example, to control the PPS), then the interrupt generation is over-written with the new mode and new programmed Target Time register value.

109.1.3.10. **Programming Guidelines for Switching the AHB or AXI Clock Frequency**

To dynamically change the AHB or AXI clock frequency (without applying soft rest or hard reset) in ETH-AHB or ETH-AXI configuration, complete the following:

1. Disable the Transmit DMA (if applicable) and wait for any previous frame transmissions to complete.

When the frame transmissions is complete, the Tx FIFO becomes empty and the Tx DMA enters the STOP state. The Tx FIFO status is given in Register 9 (Debug Register) and the status of DMA is given in Register 5 (Status Register).

2. Disable the MAC transmitter and the MAC receiver by clearing the appropriate bits in

Register 0 (MAC Configuration Register).

3. Disable the Receive DMA (if applicable) after making sure that the data in the Rx FIFO is transferred to the system memory.

The Rx FIFO empty status is given in Register 9 (Debug Register).

4. Ensure that the application does not perform any register read or write operation.

5. Change the frequency of the AHB or AXI clock.

6. Enable the MAC Transmitter or the MAC Receiver and the Transmit or Receive DMA.

These steps ensure that no valid data is present in the Tx FIFO or Rx FIFO at the time of clock frequency switching and prevent any data corruption.

109.1.4. Register Summary

109.1.4.1. DMA Register Map

The register maps in this section provide high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions in "Register Descriptions". To view the detailed description of a register, double-click the register name.

109.1.4.1.1. DMA Register Map

Table below provides the address map of the DMA registers. When you enable the AV feature, the information in Table becomes specific to Channel 0.

Table 9.1-1 DMA Register Map

Register No.	Offset Address	Register Name and Description
0	0x1000	Register 0 (Bus Mode Register) Controls the Host Interface Mode.
1	0x1004	Register 1 (Transmit Poll Demand Register) Used by the host to instruct the DMA to poll the Transmit Descriptor list.
2	0x1008	Register 2 (Receive Poll Demand Register) Used by the host to instruct the DMA to poll the Receive Descriptor list.
3	0x100C	Register 3 (Receive Descriptor List Address Register) Points the DMA to the start of the Receive Descriptor list.
4	0x1010	Register 4 (Transmit Descriptor List Address Register) Points the DMA to the start of the Transmit Descriptor list.
5	0x1014	Register 5 (Status Register) The Software driver (application) reads this register during interrupt service routine or polling to determine the status of the DMA.
6	0x1018	Register 6 (Operation Mode Register) Establishes the Receive and Transmit operating modes and command. Note: This register is valid and present in the ETH-MTL configuration.

7	0x101C	Register 7 (Interrupt Enable Register) Enables the interrupts reported by the Status Register.
8	0x1020	Register 8 (Missed Frame and Buffer Overflow Counter Register) Contains the counters for discarded frames because no host Receive Descriptor was available or because of Receive FIFO Overflow.
9	0x1024	Register 9 (Receive Interrupt Watchdog Timer Register) Watchdog timeout for Receive Interrupt (RI) from DMA.
10	0x1028	Register 10 (AXI Bus Mode Register) Controls AXI master behavior (mainly controls burst splitting and number of outstanding requests).
11	0x102C	Register 11 (AHB or AXI Status Register) Gives the idle status of the AHB master interface in the ETH-AHB configuration. Gives the idle status of the AXI master's read or write channel in the ETH-AXI configuration.
12–17	0x1030–0x1044	Reserved Note: When Register 76 (Channel 1 Slot Function Control and Status Register) is present, the register at offset address 0x1030 is a shadow register (with same bit access attributes and reset values) of Register 76.
18	0x1048	Register 18 (Current Host Transmit Descriptor Register) Points to the start of current Transmit Descriptor read by the DMA.
19	0x104C	Register 19 (Current Host Receive Descriptor Register) Points to the start of current Receive Descriptor read by the DMA.
20	0x1050	Register 20 (Current Host Transmit Buffer Address Register) Points to the current Transmit Buffer address read by the DMA.
21	0x1054	Register 21 (Current Host Receive Buffer Address Register) Points to the current Receive Buffer address read by the DMA.
22	0x1058	Register 22 (HW Feature Register) Indicates the presence of the optional features of the core.

Table below provides the address map of the DMA registers for Channel 1. The address map in table is applicable only when you enable the AV Feature and select one or more additional Transmit channels.

In the ETH-MTL configuration, only registers from address 0x1160 to 0x1174 are present. In other configurations, all registers described in table are present.

Table 9.1-2 Channel 1 DMA Register Map

Register No.	Offset Address	Register Name and Description
64	0x1100	Register 64 (Channel 1 Bus Mode Register) Controls the Host Interface mode for Channel 1. For information about this register, see Register 0 (Bus Mode Register).
65	0x1104	Register 65 (Channel 1 Transmit Poll Demand Register) Used by the host to instruct the DMA to poll the Transmit Descriptor list. For information about this register, see Register 1 (Transmit Poll Demand Register).

66	0x1108	Register 66 (Channel 1 Receive Poll Demand Register) Used by the Host to instruct the DMA to poll the Receive Descriptor list. For information about this register, see Register 2 (Receive Poll Demand Register).
67	0x110C	Register 67 (Channel 1 Receive Descriptor List Address Register) Points the DMA to the start of the Receive Descriptor list. For information about this register, see Register 3 (Receive Descriptor List Address Register).
68	0x1110	Register 68 (Channel 1 Transmit Descriptor List Address Register) Points the DMA to the start of the Transmit Descriptor list. For information about this register, see Register 4 (Transmit Descriptor List Address Register).
69	0x1114	Register 69 (Channel 1 Status Register) The Software driver (application) reads this register during interrupt service routine or polling to determine the status of the DMA. Bits 29:26 are reserved for the Channel 1 Status Register. For information about this register, see Register 5 (Status Register).
70	0x1118	Register 70 (Channel 1 Operation Mode Register) Establishes the Receive and Transmit operating modes and command. For information about this register, see Register 6 (Operation Mode Register).
71	0x111C	Register 71 (Channel 1 Interrupt Enable Register) Enables the interrupts reported by the Status Register. For information about this register, see Register 7 (Interrupt Enable Register).
72	0x1120	Register 72 (Channel 1 Missed Frame and Buffer Overflow Counter Register) Contains the counters for discarded frames because no host Receive Descriptor was available, and discarded frames because of Receive FIFO Overflow. For information about this register, see Register 8 (Missed Frame and Buffer Overflow Counter Register).
73	0x1124	Register 73 (Channel 1 Receive Interrupt Watchdog Timer Register) Watchdog timeout for Receive Interrupt (RI) from DMA. For information about this register, see Register 9 (Receive Interrupt Watchdog Timer Register).
74–75	0x1128–0x112C	Reserved for this configuration. Note: When Register 10 (AXI Bus Mode Register) is present, the register at offset address 0x1128 is a shadow register (with same bit access attributes and reset values) of Register 10 (AXI Bus Mode Register).
76	0x1130	Register 76 (Channel 1 Slot Function Control and Status Register) Contains the control bits for slot function and its status for Channel 1 transmit path.
77–81	0x1134–0x1144	Reserved
82	0x1148	Register 82 (Channel 1 Current Host Transmit Descriptor Register) Points to the start of current Transmit Descriptor read by the DMA. For information about this register, see Register 18 (Current Host Transmit Descriptor Register).

83	0x114C	Register 83 (Channel 1 Current Host Receive Descriptor Register) Points to the start of current Receive Descriptor read by the DMA. For information about this register, see Register 19 (Current Host Receive Descriptor Register).
84	0x1150	Register 84 (Channel 1 Current Host Transmit Buffer Address Register) Points to the current Transmit Buffer address read by the DMA. For information about this register, see Register 20 (Current Host Transmit Buffer Address Register).
85	0x1154	Register 85 (Channel 1 Current Host Receive Buffer Address Register) Points to the current Receive Buffer address read by the DMA. For information about this register, see Register 21 (Current Host Receive Buffer Address Register).
86–87	0x1158–0x115C	Reserved The register at 0x1158 is a shadow register of Register 22 (HW Feature Register) at 0x1058 and it gives the same value on a read access.
88	0x1160	Register 88 (Channel 1 CBS Control Register) Controls the Channel 1 credit shaping operation on the transmit path.
89	0x1164	Register 89 (Channel 1 CBS Status Register) Provides the average traffic transmitted in Channel 1.
90	0x1168	Register 90 (Channel 1 idleSlopeCredit Register) Contains the idleSlope credit value required for the credit-based shaper algorithm for Channel 1.
91	0x116C	Register 91 (Channel 1 sendSlopeCredit Register) Contains the sendSlope credit value required for the credit-based shaper algorithm for Channel 1.
92	0x1170	Register 92 (Channel 1 hiCredit Register) Contains the hiCredit value required for the credit-based shaper algorithm for Channel 1.
93	0x1174	Register 93 (Channel 1 loCredit Register) Contains the loCredit value required for the credit-based shaper algorithm for Channel 1.

Table below provides the address map of the DMA registers for Channel 2. The address map in table is applicable only when you enable the AV feature and select two additional Transmit channels.

In the ETH-MTL configuration, only registers from address 0x1260 to 0x1274 are present. In other configurations, all registers mentioned in table are present.

Table 9.1-3 Channel 2 DMA Register Map

Register No.	Offset Address	Register Name and Description
128	0x1200	Register 128 (Channel 2 Bus Mode Register) Controls the Host Interface mode for Channel 2. For information about this register, see Register 0 (Bus Mode Register).

129	0x1204	Register 129 (Channel 2 Transmit Poll Demand Register) Used by the host to instruct the DMA to poll the Transmit Descriptor list. For information about this register, see Register 1 (Transmit Poll Demand Register).
130	0x1208	Register 130 (Channel 2 Receive Poll Demand Register) Used by the Host to instruct the DMA to poll the Receive Descriptor list. For information about this register, see Register 2 (Receive Poll Demand Register).
131	0x120C	Register 131 (Channel 2 Receive Descriptor List Address Register) Points the DMA to the start of the Receive Descriptor list. For information about this register, see Register 3 (Receive Descriptor List Address Register).
132	0x1210	Register 132 (Channel 2 Transmit Descriptor List Address Register) Points the DMA to the start of the Transmit Descriptor List. For information about this register, see Register 4 (Transmit Descriptor List Address Register).
133	0x1214	Register 133 (Channel 2 Status Register) The software driver (application) reads this register during interrupt service routine or polling to determine the status of the DMA. Bits [29:26] are reserved for the Channel 2 Status Register. For information about this register, see Register 5 (Status Register).
134	0x1218	Register 134 (Channel 2 Operation Mode Register) Establishes the Receive and Transmit operating modes and command. For information about this register, see Register 6 (Operation Mode Register).
135	0x121C	Register 135 (Channel 2 Interrupt Enable Register) Enables the interrupts reported by the Status Register. For information about this register, see Register 7 (Interrupt Enable Register).
136	0x1220	Register 136 (Channel 2 Missed Frame and Buffer Overflow Counter Register) For information about this register, see Register 8 (Missed Frame and Buffer Overflow Counter Register).
137	0x1224	Register 137 (Channel 2 Receive Interrupt Watchdog Timer Register) Watchdog timeout for Receive Interrupt (RI) from DMA. For information about this register, see Register 9 (Receive Interrupt Watchdog Timer Register).
138–139	0x1228–0x122C	Reserved for this configuration. Note: When Register 10 (AXI Bus Mode Register) is present, the register at offset address 0x1228 is a shadow register (with same bit access attributes and reset values) of Register 10 (AXI Bus Mode Register).
140	0x1230	Register 140 (Channel 2 Slot Function Control and Status Register) Contains the control bits for slot function and its status for Channel 2 transmit path. For information about this register, see Register 76 (Channel 1 Slot Function Control and Status Register).
141–145	0x1234–0x1244	Reserved

146	0x1248	Register 146 (Channel 2 Current Host Transmit Descriptor Register) Points to the start of current Transmit Descriptor read by the DMA. For information about this register, see Register 18 (Current Host Transmit Descriptor Register).
147	0x124C	Register 147 (Channel 2 Current Host Receive Descriptor Register) Points to the start of current Receive Descriptor read by the DMA. For information about this register, see Register 19 (Current Host Receive Descriptor Register).
148	0x1250	Register 148 (Channel 2 Current Host Transmit Buffer Address Register) Points to the current Transmit Buffer address read by the DMA. For information about this register, see Register 20 (Current Host Transmit Buffer Address Register).
149	0x1254	Register 149 (Channel 2 Current Host Receive Buffer Address Register) Points to the current Receive Buffer address read by the DMA. For information about this register, see Register 21 (Current Host Receive Buffer Address Register).
150–151	0x1258–0x125C	Reserved The register at 0x1258 is a shadow register of Register 22 (HW Feature Register) at 0x1058 and it gives the same value on a read access.
152	0x1260	Register 152 (Channel 2 CBS Control Register) Controls the Channel 2 credit shaping operation on the transmit path. For information about this register, see Register 88 (Channel 1 CBS Control Register).
153	0x1264	Register 153 (Channel 2 CBS Status Register) Provides the average traffic transmitted in Channel 2. For information about this register, see Register 89 (Channel 1 CBS Status Register).
154	0x1268	Register 154 (Channel 2 idleSlopeCredit Register) Contains the idleSlope credit value required for the credit-based shaper algorithm for Channel 2. For information about this register, see Register 90 (Channel 1 idleSlopeCredit Register).
155	0x126C	Register 155 (Channel 2 sendSlopeCredit Register) Contains the sendSlope credit value required for the credit-based shaper algorithm for Channel 2. For information about this register, see Register 91 (Channel 1 sendSlopeCredit Register).
156	0x1270	Register 156 (Channel 2 hiCredit Register) Contains the hiCredit value required for the credit-based shaper algorithm for Channel 2. For information about this register, see Register 92 (Channel 1 hiCredit Register).
157	0x1274	Register 157 (Channel 2 loCredit Register) Contains the loCredit value required for the credit-based shaper algorithm for Channel 2. For information about this register, see Register 93 (Channel 1 loCredit Register).

109.1.4.2. ETH Register Map

Table below provides the address map of the ETH registers. Most of the registers are optional and present in the source code only if selected during configuration in coreConsultant. If a register is not configured, then that address is reserved.

Table 9.1-4 ETH Register Map

Register No.	Offset Address	Register Name and Description
0	0x0000	Register 0 (MAC Configuration Register) This is the operation mode register for the MAC.
1	0x0004	Register 1 (MAC Frame Filter) Contains the frame filtering controls.
2	0x0008	Register 2 (Hash Table High Register) Contains the higher 32 bits of the Multicast Hash table. This register is presen only when you select the 64-bit Hash filter function in coreConsultant.
3	0x000C	Register 3 (Hash Table Low Register) Contains the lower 32 bits of the Multicast Hash table. This register is present only when you select the Hash filter function in coreConsultant.
4	0x0010	Register 4 (GMII Address Register) Controls the management cycles to an external PHY. This register is present only when you select the Station Management (MDIO) feature in coreConsultant.
5	0x0014	Register 5 (GMII Data Register) Contains the data to be written to or read from the PHY register. This register is present only when you select the Station Management (MDIO) feature in coreConsultant.
6	0x0018	Register 6 (Flow Control Register) Controls the generation of control frames.
7	0x001C	Register 7 (VLAN Tag Register) Identifies IEEE 802.1Q VLAN type frames.
8	0x0020	Register 8 (Version Register) Identifies the version of the Core.
9	0x0024	Register 9 (Debug Register) Gives the status of various internal blocks for debugging.
10	0x0028	Remote Wake-Up Frame Filter Register This is the address through which the application writes or reads the remote wake-up frame filter registers (wkupfmfilter_reg). The wkupfmfilter_reg register is a pointer to eight wkupfmfilter_reg registers. The wkupfmfilter_re register is loaded by sequentially loading the eight register values. Eight sequential writes to this address (0x0028) write all wkupfmfilter_reg registers. Similarly, eight sequential reads from this address (0x0028) read all wkupfmfilter_reg registers. This register is present only when you select the PMT module Remote
11	0x002C	PMT Control and Status Register This register is present only when you select the PMT module in coreConsultant.

12	0x0030	Register 12 (LPI Control and Status Register) Controls the Low Power Idle (LPI) operations and provides the LPI status of the core. This register is present only when you select the Energy Efficient Ethernet feature in coreConsultant.
13	0x0034	Register 13 (LPI Timers Control Register) Controls the timeout values in LPI states. This register is present only when you select the Energy Efficient Ethernet feature in coreConsultant.
14	0x0038	Register 14 (Interrupt Status Register) Contains the interrupt status.
15	0x003C	Register 15 (Interrupt Mask Register) Contains the masks for generating the interrupts.
16	0x0040	Register 16 (MAC Address0 High Register) Contains the higher 16 bits of the first MAC address.
17	0x0044	Register 17 (MAC Address0 Low Register) Contains the lower 32 bits of the first MAC address.
18	0x0048	Register 18 (MAC Address1 High Register) Contains the higher 16 bits of the second MAC address. This register is present only when Enable MAC Address1 is selected in coreConsultant.
19	0x004C	Register 19 (MAC Address1 Low Register) Contains the lower 32 bits of the second MAC address. This register is present only when Enable MAC Address1 is selected in coreConsultant.
20	0x0050	MAC Address2 High Register Contains the higher 16 bits of the third MAC address. This register is present only when Enable MAC Address2 is selected in coreConsultant.
21	0x0054	MAC Address2 Low Register Contains the lower 32 bits of the third MAC address. This register is present only when Enable MAC Address2 is selected in coreConsultant.
22	0x0058	MAC Address3 High Register Contains the higher 16 bits of the fourth MAC address. This register is present only when Enable MAC Address3 is selected in coreConsultant.
23	0x005C	MAC Address3 Low Register Contains the lower 32 bits of the fourth MAC address. This register is present only when Enable MAC Address3 is selected in coreConsultant.
24	0x0060	MAC Address4 High Register Contains the higher 16 bits of the fifth MAC address. This register is present only when Enable MAC Address4 is selected in coreConsultant.
25	0x0064	MAC Address4 Low Register Contains the lower 32 bits of the fifth MAC address. This register is present only when Enable MAC Address4 is selected in coreConsultant.
26	0x0068	MAC Address5 High Register Contains the higher 16 bits of the sixth MAC address. This register is present only when Enable MAC Address5 is selected in coreConsultant.
27	0x006C	MAC Address5 Low Register Contains the lower 32 bits of the sixth MAC address. This register is present only when Enable MAC Address5 is selected in coreConsultant.
28	0x0070	MAC Address6 High Register Contains the higher 16 bits of the seventh MAC address.This register is present only when Enable MAC Address6 is selected in coreConsultant.
29	0x0074	MAC Address6 Low Register Contains the lower 32 bits of the seventh MAC address. This register is present only when Enable MAC Address6 is selected in coreConsultant.
30	0x0078	MAC Address7 High Register

		Contains the higher 16 bits of the eighth MAC address. This register is present only when Enable MAC Address7 is selected in coreConsultant.
31	0x007C	MAC Address7 Low Register Contains the lower 32 bits of the eighth MAC address. This register is present only when Enable MAC Address7 is selected in coreConsultant.
32	0x0080	MAC Address8 High Register Contains the higher 16 bits of the ninth MAC address. This register is present only when Enable MAC Address8 is selected in coreConsultant.
33	0x0084	MAC Address8 Low Register Contains the lower 32 bits of the ninth MAC address. This register is present only when Enable MAC Address8 is selected in coreConsultant.
34	0x0088	MAC Address9 High Register Contains the higher 16 bits of the 10th MAC address. This register is present only when Enable MAC Address9 is selected in coreConsultant.
35	0x008C	MAC Address9 Low Register Contains the lower 32 bits of the 10th MAC address. This register is present only when Enable MAC Address9 is selected in coreConsultant.
36	0x0090	MAC Address10 High Register Contains the higher 16 bits of the 11th MAC address. This register is present only when Enable MAC Address10 is selected in coreConsultant.
37	0x0094	MAC Address10 Low Register Contains the lower 32 bits of the 11th MAC address. This register is present only when Enable MAC Address10 is selected in coreConsultant.
38	0x0098	MAC Address11 High Register This register contains the higher 16 bits of the 12th MAC address. This register is present only when Enable MAC Address11 is selected in coreConsultant.
39	0x009C	MAC Address11 Low Register Contains the lower 32 bits of the 12th MAC address. This register is present only when Enable MAC Address11 is selected in coreConsultant.
40	0x00A0	MAC Address12 High Register Contains the higher 16 bits of the 13th MAC address. This register is present only when Enable MAC Address12 is selected in coreConsultant.
41	0x00A4	MAC Address12 Low Register Contains the lower 32 bits of the 13th MAC address. This register is present only when Enable MAC Address12 is selected in coreConsultant.
42	0x00A8	MAC Address13 High Register Contains the higher 16 bits of the 14th MAC address. This register is present only when Enable MAC Address13 is selected in coreConsultant.
43	0x00AC	MAC Address13 Low Register Contains the lower 32 bits of the 14th MAC address. This register is present only when Enable MAC Address13 is selected in coreConsultant.
44	0x00B0	MAC Address14 High Register Contains the higher 16 bits of the 15th MAC address. This register is present only when Enable MAC Address14 is selected in coreConsultant.
45	0x00B4	MAC Address14 Low Register Contains the lower 32 bits of the 15th MAC address. This register is present only when Enable MAC Address14 is selected in coreConsultant.
46	0x00B8	MAC Address15 High Register Contains the higher 16 bits of the 16th MAC address. This register is present only when Enable MAC Address15 is selected in coreConsultant.
47	0x00BC	MAC Address15 Low Register Contains the lower 32 bits of the 16th MAC address. This register is present

		only when Enable MAC Address15 is selected in coreConsultant.
48	0x00C0	Register 48 (AN Control Register) Enables and/or restarts auto-negotiation. This register also enables the Physical Coding Sublayer (PCS) loopback. This register is present only when you select the TBI, RTBI, or SGMII interface in coreConsultant.
49	0x00C4	Register 49 (AN Status Register) Indicates the link and auto-negotiation status. This register is present only when you select the TBI, RTBI, or SGMII interface in coreConsultant.
50	0x00C8	Register 50 (Auto-Negotiation Advertisement Register) This register is configured before auto-negotiation begins. It contains the advertised ability of the MAC. This register is present only when you select the TBI or RTBI interface in coreConsultant.
51	0x00CC	Register 51 (Auto-Negotiation Link Partner Ability Register) Contains the advertised ability of the link partner. Its value is valid after successful completion of auto-negotiation or when a new base page has been received (indicated in the Auto-Negotiation Expansion Register). This register is present only when you select the TBI or RTBI interface in coreConsultant.
52	0x00D0	Register 52 (Auto-Negotiation Expansion Register) Indicates whether a new base page has been received from the link partner. This register is present only when you select the TBI or RTBI interface in coreConsultant.
53	0x00D4	Register 53 (TBI Extended Status Register) Indicates all modes of operation of the MAC. This register is present only when you select the TBI or RTBI interface in coreConsultant.
54	0x00D8	Register 54 (SGMII/RGMII/SMII Control and Status Register) Indicates the status signals received from the PHY through the SGMII, RGMII, or SMII interface. This register is present only when you select the SGMII, RGMII, or SMII interface in coreConsultant.
55	0x00DC	Register 55 (Watchdog Timeout Register) Controls the watchdog timeout for received frames.
56	0x00E0	Register 56 (General Purpose IO Register) Provides the control to drive up to 4 bits of output ports (GPO) and also provides the status of up to 4 input ports (GPIS).
57–63	0x00E4–0x00FC	Reserved
64–191	0x0100–0x02FC	MMC Register Map
192–255	0x0300–0x03FC	Reserved
256	0x0400	Register 256 (Layer 3 and Layer 4 Control Register 0) Controls the operations of the Layer 3 and Layer 4 frame filtering.
257	0x0404	Register 257 (Layer 4 Address Register 0) Layer 4 Port number field. It contains the 16-bit Source and Destination Port numbers of the TCP or UDP frame.
258–259	0x408–0x40C	Reserved
260	0x0410	Register 260 (Layer 3 Address 0 Register 0) Layer 3 Address field. For IPv4 frames, it contains the 32-bit IP Source Address field. For IPv6 frames, it contains Bits [31:0] of the 128-bit IP Source Address or Destination Address field.
261	0x0414	Register 261 (Layer 3 Address 1 Register 0) Layer 3 Address 1 field. For IPv4 frames, it contains the 32-bit IP Destination Address field. For IPv6 frames, it contains Bits [63:32] of the 128-bit IP Source Address or Destination Address field.

262	0x0418	Register 262 (Layer 3 Address 2 Register 0) Layer 3 Address 2 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [95:64] of the 128-bit IP Source Address or Destination Address field.
263	0x041C	Register 263 (Layer 3 Address 3 Register 0) Layer 3 Address 3 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [127:96] of the 128-bit IP Source Address or Destination Address field.
264–267	0x0420–0x42C	Reserved
268	0x0430	Register 268 (Layer 3 and Layer 4 Control Register 1) Controls the operations of the Layer 3 and Layer 4 frame filtering. This register is similar to Register 256 (Layer 3 and Layer 4 Control Register 0).
269	0x0434	Register 269 (Layer 4 Address Register 1) Layer 4 Port number field. It contains the 16-bit Source and Destination Port numbers of TCP or UDP frame. This register is similar to Register 257 (Layer 4 Address Register 0).
270-271	0x438-0x43C	Reserved
272	0x0440	Register 272 (Layer 3 Address 0 Register 1) Layer 3 Address field. For IPv4 frames, it contains the 32-bit IP Source Address field. For IPv6 frames, it contains Bits [31:0] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 260 (Layer 3 Address 0 Register 0).
273	0x0444	Register 273 (Layer 3 Address 1 Register 1) Layer 3 Address 1 field. For IPv4 frames, it contains the 32-bit IP Destination Address field. For IPv6 frames, it contains Bits [63:32] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 261 (Layer 3 Address 1 Register 0).
274	0x0448	Register 274 (Layer 3 Address 2 Register 1) Layer 3 Address 2 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [95:64] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 262 (Layer 3 Address 2 Register 0).
275	0x044C	Register 275 (Layer 3 Address 3 Register 1) Layer 3 Address 3 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [127:96] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 263 (Layer 3 Address 3 Register 0).
276–279	0x0450–0x45C	Reserved
280	0x0460	Register 280 (Layer 3 and Layer 4 Control Register 2) Controls the operations of the Layer 3 and Layer 4 frame filtering. This register is similar to Register 256 (Layer 3 and Layer 4 Control Register 0).
281	0x0464	Register 281 (Layer 4 Address Register 2) Layer 4 Port number field. It contains the 16-bit Source and Destination Port numbers of TCP or UDP frame. This register is similar to Register 257 (Layer 4 Address Register 0).
282–283	0x468–0x46C	Reserved
284	0x0470	Register 284 (Layer 3 Address 0 Register 2) Layer 3 Address field. For IPv4 frames, it contains the 32-bit IP Source Address field. For IPv6 frames, it contains Bits [31:0] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 260 (Layer 3 Address 0 Register 0).
285	0x0474	Register 285 (Layer 3 Address 1 Register 2) Layer 3 Address 1 field. For IPv4 frames, it contains the 32-bit IP Destination Address field. For IPv6 frames, it contains Bits [63:32] of the 128-bit IP

		Source Address or Destination Address field. This register is similar to Register 261 (Layer 3 Address 1 Register 0).
286	0x0478	Register 286 (Layer 3 Address 2 Register 2) Layer 3 Address 2 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [95:64] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 262 (Layer 3 Address 2 Register 0).
287	0x047C	Register 287 (Layer 3 Address 3 Register 2) Layer 3 Address 3 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [127:96] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 263 (Layer 3 Address 3 Register 0).
288–291	0x0480–0x48C	Reserved
292	0x0490	Register 292 (Layer 3 and Layer 4 Control Register 3) Controls the operations of the Layer 3 and Layer 4 frame filtering. This register is similar to Register 256 (Layer 3 and Layer 4 Control Register 0).
293	0x0494	Register 293 (Layer 4 Address Register 3) Layer 4 Port number field. It contains the 16-bit Source and Destination Port numbers of TCP or UDP frame. This register is similar to Register 257 (Layer 4 Address Register 0).
294–295	0x0498–0x049C	Reserved
296	0x04A0	Register 296 (Layer 3 Address 0 Register 3) Layer 3 Address field. For IPv4 frames, it contains the 32-bit IP Source Address field. For IPv6 frames, it contains Bits [31:0] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 260 (Layer 3 Address 0 Register 0).
297	0x04A4	Register 285 (Layer 3 Address 1 Register 3) Layer 3 Address 1 field. For IPv4 frames, it contains the 32-bit IP Destination Address field. For IPv6 frames, it contains Bits [63:32] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 261 (Layer 3 Address 1 Register 0).
298	0x04A8	Register 286 (Layer 3 Address 2 Register 3) Layer 3 Address 2 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [95:64] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 262 (Layer 3 Address 2 Register 0).
299	0x04AC	Register 287 (Layer 3 Address 3 Register 3) Layer 3 Address 3 field. This register is reserved for IPv4 frames. For IPv6 frames, it contains Bits [127:96] of the 128-bit IP Source Address or Destination Address field. This register is similar to Register 263 (Layer 3 Address 3 Register 0).
300–319	0x04B0–0x4FC	Reserved
320	0x0500	Register 320 (Hash Table Register 0) This register contains the first 32 bits of the hash table when the width of the Hash table is 128 bits or 256 bits.
321	0x0504	Register 321 (Hash Table Register 1) This register contains the second 32 bits of the hash table when the width of the Hash table is 128 bits or 256 bits. This register is similar to Register 320 (Hash Table Register 0).
322	0x0508	Register 322 (Hash Table Register 2) This register contains the third 32 bits of the hash table when the width of the Hash table is 128 bits or 256 bits. This register is similar to Register 320 (Hash Table Register 0).

323	0x050C	Register 323 (Hash Table Register 3) This register contains the fourth 32 bits of the hash table when the width of the Hash table is 128 bits or 256 bits. This register is similar to Register 320 (Hash Table Register 0).
324	0x0510	Register 324 (Hash Table Register 4) This register contains the fifth 32 bits of the hash table when the width of the Hash table is 256 bits. This register is similar to Register 320 (Hash Table Register 0).
325	0x0514	Register 325 (Hash Table Register 5) This register contains the sixth 32 bits of the hash table when the width of the Hash table is 256 bits. This register is similar to Register 320 (Hash Table Register 0).
326	0x0518	Register 326 (Hash Table Register 6) This register contains the seventh 32 bits of the hash table when the width of the Hash table is 256 bits. This register is similar to Register 320 (Hash Table Register 0).
327	0x051C	Register 327 (Hash Table Register 7) This register contains the eighth 32 bits of the hash table when the width of the Hash table is 256 bits. This register is similar to Register 320 (Hash Table Register 0).
328–352	0x0520–0x0580	Reserved
353	0x0584	Register 353 (VLAN Tag Inclusion or Replacement Register) This register contains the VLAN tag for insertion into or replacement in the transmit frames.
354	0x0588	Register 354 (VLAN Hash Table Register) This register contains the VLAN hash table.

109.1.5. **Register Description**

109.1.5.1. **DMA Register Description**

This section defines the bits for each DMA register. The write data inputs to the DMA registers are qualified with the corresponding mci_be_i signal inputs (MCI interface). Thus, non-32 bit accesses are allowed as long as the address is Word-aligned. For the APB interface, only 32-bit accesses are possible, while for AHB slave interfaces, byte, half-word, or word accesses are possible.

109.1.5.1.1. Register 0 (Bus Mode Register)

The Bus Mode register establishes the bus operating modes for the DMA.

Table 9.1-5 Register 0 (Bus Mode Register)

Field	Field Name	Description	Reset	Access

31	RIB	Rebuild INCRx Burst When this bit is set high and the AHB master gets an EBT (Retry, Split, or Losing bus grant), the AHB master interface rebuilds the pending beats of any burst transfer initiated with INCRx. The AHB master interface rebuilds the beats with a combination of specified bursts with INCRx and SINGLE. By default, the AHB master interface rebuilds pending beats of an EBT with an unspecified (INCR) burst. This bit is valid only in the ETH-AHB configuration. It is reserved in all other configuration.	0	R_W
30		Reserved	0	RO
29:28	PRWG	Channel Priority Weights This field sets the priority weights for Channel 0 during the round-robin arbitration between the DMA channels for the system bus. ■ 00: The priority weight is 1. ■ 01: The priority weight is 2. ■ 10: The priority weight is 3. ■ 11: The priority weight is 4. This field is present in all ETH configurations except ETH-AXI when you select the AV feature. Otherwise, this field is reserved and read-only (RO). For more information about the priority weights of DMA channels, see “DMA Arbiter Functions”	00	R_W
27	TXPR	Transmit Priority When set, this bit indicates that the transmit DMA has higher priority than the receive DMA during arbitration for the system-side bus. In the ETH-AXI configuration, this bit is reserved and read-only (RO).	0	R_W
26	MB	Mixed Burst When this bit is set high and the FB bit is low, the AHB master interface starts all bursts of length more than 16 with INCR (undefined burst), whereas it reverts to fixed burst transfers (INCRx and SINGLE) for burst length of 16 and less. This bit is valid only in the ETH-AHB configuration and reserved in all other configuration.	0	R_W
25	AAL	Address-Aligned Beats When this bit is set high and the FB bit is equal to 1, the AHB or AXI interface generates all bursts aligned to the start address LS bits. If the FB bit is equal to 0, the first burst (accessing the start address of data buffer) is not aligned, but subsequent bursts are aligned to the address. This bit is valid only in the ETH-AHB and ETH-AXI configurations and is reserved (RO with default value 0) in all other configurations.	0	R_W
24	PBLx8	PBLx8 Mode When set high, this bit multiplies the programmed PBL value (Bits [22:17] and Bits[13:8]) eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Note: This bit function is not backward compatible. Before release 3.50a, this bit was 4xPBL.	0	R_W
23	USP	Use Separate PBL When set high, this bit configures the Rx DMA to use the value configured in Bits [22:17] as PBL. The PBL value in Bits [13:8] is applicable only to the Tx DMA operations. When reset to low, the PBL value in Bits [13:8] is applicable for both DMA engines.	0	R_W

22:17	RPBL	Rx DMA PBL This field indicates the maximum number of beats to be transferred in one Rx DMA transaction. This is the maximum value that is used in a single block Read or Write. The Rx DMA always attempts to burst as specified in the RPBL bit each time it starts a Burst transfer on the host bus. You can program RPBL with values of 1, 2, 4, 8, 16, and 32. Any other value results in undefined behavior. This field is valid and applicable only when USP is set high.	01H	R_W
16	FB	Fixed Burst This bit controls whether the AHB or AXI master interface performs fixed burst transfers or not. When set, the AHB interface uses only SINGLE, INCR4, INCR8, or INCR16 during start of the normal burst transfers. When reset, the AHB or AXI interface uses SINGLE and INCR burst transfer operations. For more information, see Bit 0 (UNDEF) of the AXI Bus Mode register in the ETH-AXI configuration.	0	R_W
15:14	PR	Priority Ratio These bits control the priority ratio in the weighted round-robin arbitration between the Rx DMA and Tx DMA. These bits are valid only when Bit 1 (DA) is reset. The priority ratio is Rx:Tx or Tx:Rx depending on whether Bit 27 (TXPR) is reset or set. ■ 00: The Priority Ratio is 1:1. ■ 01: The Priority Ratio is 2:1. ■ 10: The Priority Ratio is 3:1. ■ 11: The Priority Ratio is 4:1. In the ETH-AXI configuration, these bits are reserved and read-only (RO).	00	R_W
13:8	PBL	Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA transaction. This is the maximum value that is used in a single block Read or Write. The DMA always attempts to burst as specified in PBL each time it starts a Burst transfer on the host bus. PBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value results in undefined behavior. When USP is set high, this PBL value is applicable only for Tx DMA transactions. If the number of beats to be transferred is more than 32, then perform the following steps 1. Set the PBLx8 mode. 2. Set the PBL. For example, if the maximum number of beats to be transferred is 64, then first set PBLx8 to 1 and then set PBL to 8. The PBL values have the following limitation: The maximum number of possible beats (PBL) is limited by the size of the Tx FIFO and Rx FIFO in the MTL layer and the data bus width on the DMA. The FIFO has a constraint that the maximum beat supported is half the depth of the FIFO, except when specified. For different data bus widths and FIFO sizes, the valid PBL range (including x8 mode) is provided.	01H	R_W
7	ATDS	Alternate Descriptor Size When set, the size of the alternate descriptor increases to 32 bytes (8 DWORDS). This is required when the Advanced Timestamp feature or the IPC Full Checksum Offload Engine (Type 2) is enabled in the receiver. The enhanced descriptor is not required if the Advanced Timestamp and IPC Full Checksum Offload Engine (Type 2) features are not enabled. In such case, you can use the 16 bytes descriptor to save 4 bytes of memory. This bit is present only when you select the Alternate Descriptor feature and any one of the following features during core configuration: ■ Advanced Timestamp feature ■ IPC Full Checksum Offload Engine (Type 2) feature Otherwise, this bit is reserved and is read-only. When reset, the descriptor size reverts back to 4 DWORDs (16 bytes). This bit preserves the backward compatibility for the descriptor size. In versions prior to 3.50a, the descriptor size is 16 bytes for both normal and enhanced descriptors. In version 3.50a, descriptor size is increased to 32 bytes because of the Advanced Timestamp and IPC Full Checksum Offload Engine (Type 2) features.	0	R_W

6:2	DSL	Descriptor Skip Length This bit specifies the number of Word, Dword, or Lword (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of current descriptor to the start of next descriptor. When the DSL value is equal to zero, the descriptor table is taken as contiguous by the DMA in Ring mode.	00H	R_W
1	DA	DMA Arbitration Scheme This bit specifies the arbitration scheme between the transmit and receive paths of Channel 0. ■ 0: Weighted round-robin with Rx:Tx or Tx:Rx The priority between the paths is according to the priority specified in Bits [15:14] (PR) and priority weights specified in Bit 27 (TXPR). ■ 1: Fixed priority The transmit path has priority over receive path when Bit 27 (TXPR) is set. Otherwise, receive path has priority over the transmit path. In the ETH-AXI configuration, these bits are reserved and are read-only (RO).	0	R_W
0	SWR	Software Reset When this bit is set, the MAC DMA Controller resets the logic and all internal registers o the MAC. It is cleared automatically after the reset operation is complete in all of the ETH clock domains. Before reprogramming any register of the ETH, you should read a zero (0) value in this bit. Note: ■ The Software reset function is driven only by this bit. Bit 0 of Register 64 (Channel 1 Bus Mode Register) or Register 128 (Channel 2 Bus Mode Register) has no impact on the Software reset function. ■ The reset operation is completed only when all resets in all active clock domains are de-asserted. Therefore, it is essential that all PHY inputs clocks (applicable for the selected PHY interface) are present for the software reset completion. The time to complete the software reset operation depends on the frequency of the slowest active clock.	1	R_WS_SC

Table below provides the valid PBL range (including x8 mode) for different data bus widths and FIFO sizes.

Table 9.1-6 Valid PBL Range

Data Bus Width	FIFO Depth	Valid PBL Range in Full Duplex Mode	Valid PBL Range in Half Duplex Mode (Only for Tx FIFO)
32	128 Bytes	16 or less	8 or less (10/100 Mbps mode only)
	256 Bytes	32 or less	32 or less (10/100 Mbps mode only)
	512 Bytes	64 or less	64 or less (10/100 Mbps mode only)
	1 KB	128 or less	▪ 128 or less in 10/100 Mbps mode ▪ 64 or less in 1000 Mbps mode
	2 KB and higher	All	All
64	128 Bytes	8 or less	4 or less (10/100 Mbps mode only)
	256 Bytes	16 or less	16 or less (10/100 Mbps mode only)
	512 Bytes	32 or less	32 or less (10/100 Mbps mode only)
	1 KB	64 or less	▪ 64 or less in 10/100 Mbps mode ▪ 32 or less in 1000 Mbps mode
	2 KB	128 or less	128 or less

	4 KB and higher	All	All
128	128 Bytes	4 or less	2 or less (10/100 Mbps mode only)
	256 Bytes	8 or less	8 or less (10/100 Mbps mode only)
	512 Bytes	16 or less	16 or less (10/100 Mbps mode only)
	1 KB	32 or less	▪ 32 or less in 10/100 Mbps mode only ▪ 16 or less in 1000 Mbps mode only
	2 KB	64 or less	64 or less
	4 KB	128 or less	128 or less
	8 KB and higher	All	All

109.1.5.1.2. Register 1 (Transmit Poll Demand Register)

The Transmit Poll Demand register enables the Tx DMA to check whether or not the DMA owns the current descriptor. The Transmit Poll Demand command is given to wake up the Tx DMA if it is in the Suspend mode. The Tx DMA can go into the Suspend mode because of an Underflow error in a transmitted frame or the unavailability of descriptors owned by it. You can give this command anytime, and the Tx DMA resets this command when it again starts fetching the current descriptor from host memory. When this register is read, it always returns zero.

Table 9.1-7 Register 1 (Transmit Poll Demand Register)

Field	Field Name	Description	Reset	Access
31:0	TPD	Transmit Poll Demand When these bits are written with any value, the DMA reads the current descriptor to which the Register 18 (Current Host Transmit Descriptor Register) is pointing. If that descriptor is not available (owned by the Host), the transmission returns to the Suspend state and Bit 2 (TU) of Register 5 (Status Register) is asserted. If the descriptor is available, the transmission resumes.	0000_0000H	RO_WT

109.1.5.1.3. Register 2 (Receive Poll Demand Register)

The Receive Poll Demand register enables the Rx DMA to check for new descriptors. This command is given to wake up the Rx DMA from the SUSPEND state. The Rx DMA can go into the SUSPEND state only because of the unavailability of descriptors it owns. When this register is read, it always returns zero.

Table 9.1-8 Register 2 (Receive Poll Demand Register)

Field	Field Name	Description	Reset	Access
31:0	RPD	Receive Poll Demand When these bits are written with any value, the DMA reads the current descriptor to which the Register 19 (Current Host Receive Descriptor Register) is pointing. If that descriptor is not available (owned by the Host), the reception returns to the Suspended state and Bit 7 (RU) of Register 5 (Status Register) is asserted. If the descriptor is available, the Rx DMA returns to the active state.	0000_0000H	RO_WT

109.1.5.1.4. Register 3 (Receive Descriptor List Address Register)

The Receive Descriptor List Address register points to the start of the Receive Descriptor List. The descriptor lists reside in the host's physical memory space and must be Word, Dword, or Lword-aligned (for 32-bit, 64- bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given.

You can write to this register only when Rx DMA has stopped, that is, Bit 1 (SR) is set to zero in Register 6 (Operation Mode Register). When stopped, this register can be written with a new descriptor list address. When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

If this register is not changed when the SR bit is set to 0, then the DMA takes the descriptor address where it was stopped earlier.

Table 9.1-9 Register 3 (Receive Descriptor List Address Register)

Field	Field Name	Description	Reset	Access
31:0	RDESLA	Start of Receive List This field contains the base address of the first descriptor in the Receive Descriptor list. The LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width are ignored and internally taken as all-zero by the DMA. Therefore, these LSB bits are read-only (RO).	0000_0000H	R_W

109.1.5.1.5. Register 4 (Transmit Descriptor List Address Register)

The Transmit Descriptor List Address register points to the start of the Transmit Descriptor List. The descriptor lists reside in the host's physical memory space and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to this register only when the Tx DMA has stopped, that is, Bit 13 (ST) is set to zero in Register6 (Operation Mode Register). When stopped, this register can be written with a new descriptor list address.
When you set the ST bit to 1, the DMA takes the newly programmed descriptor base address.

If this register is not changed when the ST bit is set to 0, then the DMA takes the descriptor address where it was stopped earlier.

Table 9.1-10 Register 4 (Transmit Descriptor List Address Register)

Field	Field Name	Description	Reset	Access
31:0	TDESLA	Start of Transmit List This field contains the base address of the first descriptor in the Transmit Descriptor list. The LSB bits (1:0, 2:0, 3:0) for 32-bit, 64-bit, or 128-bit bus width are ignored and are internally taken as all-zero by the DMA. Therefore, these LSB bits are read-only (RO).	0000_0000H	R_W

109.1.5.1.6. Register 5 (Status Register)

The Status register contains all status bits that the DMA reports to the host. The Software driver reads this register during an interrupt service routine or polling. Most of the fields in this register cause the host to be interrupted. The bits of this register are not cleared when read. Writing 1’b1 to (unreserved) Bits [16:0] of this register clears these bits and writing 1’b0 has no effect. Each field (Bits[16:0]) can be masked by masking the appropriate bit in Register 7 (Interrupt Enable Register).

Table 9.1-11 Register 5 (Status Register)

Field	Field Name	Description	Reset	Access
31	-	Reserved	0	RO
30	GLPII -or- GTMSI	GLPII: ETH LPI Interrupt (for Channel 0) This bit indicates an interrupt event in the LPI logic of the MAC. To reset this bit to 1'b0, the software must read the corresponding registers in the ETH to get the exact cause of the interrupt and clear its source. **Note**: GLPII status is given only in Channel 0 DMA register and is applicable only when the Energy Efficient Ethernet feature is enabled. Otherwise, this bit is reserved. When this bit is high, the interrupt signal from the MAC (sbd_intr_o) is high. -or- GTMSI: ETH TMS Interrupt (for Channel 1 and Channel 2) This bit indicates an interrupt event in the traffic manager and scheduler logic of ETH. To reset this bit, the software must read the corresponding registers (Channel Status Register to get the exact cause of the interrupt and clear its source. **Note**: GTMSI status is given only in Channel 1 and Channel 2 DMA register when the AV feature is enabled and corresponding additional transmit channels are present. Otherwise, this bit is reserved. When this bit is high, the interrupt signal from the MAC (sbd_intr_o) is high.	0	RO
29	TTI	Timestamp Trigger Interrupt This bit indicates an interrupt event in the Timestamp Generator block of the ETH. The software must read the corresponding registers in the ETH to get the exact cause of the interrupt and clear its source to reset this bit to 1'b0. When this bit is high, the interrupt signal from the ETH subsystem (sbd_intr_o) is high. This bit is applicable only when the IEEE 1588 Timestamp feature is enabled. Otherwise, this bit is reserved.	0	RO
28	GPI	ETH PMT Interrupt This bit indicates an interrupt event in the PMT module of the ETH. The software must read the PMT Control and Status Register in the MAC to get the exact cause of interrupt and clear its source to reset this bit to 1’b0. The interrupt signal from the ETH subsystem (sbd_intr_o) is high when this bit is high. This bit is applicable only when the Power Management feature is enabled. Otherwise, this bit is reserved. Note: The GPI and pmt_intr_o interrupts are generated in different clock domains.	0	RO
27	GMI	ETH MMC Interrupt This bit reflects an interrupt event in the MMC module of the ETH. The software must read the corresponding registers in the ETH to get the exact cause of the interrupt and clear the source of interrupt to make this bit as 1’b0. The interrupt signal from the ETH subsystem (sbd_intr_o) is high when this bit is high. This bit is applicable only when the MAC Management Counters (MMC) are enabled. Otherwise, this bit is reserved.	0	RO

26	GLI	ETH Line Interface Interrupt When set, this bit reflects any of the following interrupt events in the ETH interfaces (if present and enabled in your configuration): ■ PCS (TBI, RTBI, or SGMII): Link change or auto-negotiation complete event ■ SMII or RGMII: Link change event ■ General Purpose Input Status (GPIS): Any LL or LH event on the gpi_i input ports To identify the exact cause of the interrupt, the software must first read Bit 11 and Bits[2:0] of Register 14 (Interrupt Status Register) and then to clear the source of interrupt (which also clears the GLI interrupt), read any of the following corresponding registers: ■ PCS (TBI, RTBI, or SGMII): Register 49 (AN Status Register) ■ SMII or RGMII: Register 54 (SGMII/RGMII/SMII Control and Status Register) ■ General Purpose Input (GPI): Register 56 (General Purpose IO Register) The interrupt signal from the ETH subsystem (sbd_intr_o) is high when this bit is high.	0	RO
25:23	EB	Error Bits This field indicates the type of error that caused a Bus Error, for example, error response on the AHB or AXI interface. This field is valid only when Bit 13 (FBI) is set. This field does not generate an interrupt. ■ 0 0 0: Error during Rx DMA Write Data Transfer ■ 0 1 1: Error during Tx DMA Read Data Transfer ■ 1 0 0: Error during Rx DMA Descriptor Write Access ■ 1 0 1: Error during Tx DMA Descriptor Write Access ■ 1 1 0: Error during Rx DMA Descriptor Read Access ■ 1 1 1: Error during Tx DMA Descriptor Read Access Note: 001 and 010 are reserved.	000	RO
22:20	TS	Transmit Process State This field indicates the Transmit DMA FSM state. This field does not generate an interrupt. ■ 3’b000: Stopped; Reset or Stop Transmit Command issued ■ 3’b001: Running; Fetching Transmit Transfer Descriptor ■ 3’b010: Running; Waiting for status ■ 3’b011: Running; Reading Data from host memory buffer and queuing it to transmit buffer (Tx FIFO) ■ 3’b100: TIME_STAMP write state ■ 3’b101: Reserved for future use ■ 3’b110: Suspended; Transmit Descriptor Unavailable or Transmit Buffer Underflow ■ 3’b111: Running; Closing Transmit Descriptor	000	RO
19:17	RS	Receive Process State This field indicates the Receive DMA FSM state. This field does not generate an interrupt. ■ 3’b000: Stopped: Reset or Stop Receive Command issued ■ 3’b001: Running: Fetching Receive Transfer Descriptor ■ 3’b010: Reserved for future use ■ 3’b011: Running: Waiting for receive packet ■ 3’b100: Suspended: Receive Descriptor Unavailable ■ 3’b101: Running: Closing Receive Descriptor ■ 3’b110: TIME_STAMP write state ■ 3’b111: Running: Transferring the receive packet data from receive buffer to host memory	000	RO

16	NIS	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in Register 7 (Interrupt Enable Register): ■ Register 5[0]: Transmit Interrupt ■ Register 5[2]: Transmit Buffer Unavailable ■ Register 5[6]: Receive Interrupt ■ Register 5[14]: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in Register 7) affect the Normal Interrupt Summary bit. This is a sticky bit and must be cleared (by writing 1 to this bit) each time a corresponding bit, which causes NIS to be set, is cleared.	0	R_SS_WC
15	AIS	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in Register 7 (Interrupt Enable Register): ■ Register 5[1]: Transmit Process Stopped ■ Register 5[3]: Transmit Jabber Timeout ■ Register 5[4]: Receive FIFO Overflow ■ Register 5[5]: Transmit Underflow ■ Register 5[7]: Receive Buffer Unavailable ■ Register 5[8]: Receive Process Stopped ■ Register 5[9]: Receive Watchdog Timeout ■ Register 5[10]: Early Transmit Interrupt ■ Register 5[13]: Fatal Bus Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit and must be cleared (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared.	0	R_SS_WC
14	ERI	Early Receive Interrupt This bit indicates that the DMA filled the first data buffer of the packet. This bit is cleared when the software writes 1 to this bit or Bit 6 (RI) of this register is set (whichever occurs earlier).	0	R_SS_WC
13	FBI	Fatal Bus Error Interrupt This bit indicates that a bus error occurred, as described in Bits [25:23]. When this bit is set, the corresponding DMA engine disables all of its bus accesses.	0	R_SS_WC
12:11	-	Reserved	00	RO
10	ETI	Early Transmit Interrupt This bit indicates that the frame to be transmitted is fully transferred to the MTL Transmit FIFO.	0	R_SS_WC
9	RWT	Receive Watchdog Timeout When set, this bit indicates that the Receive Watchdog Timer expired while receiving the current frame and the current frame is truncated after the watchdog timeout.	0	R_SS_WC
8	RPS	Receive Process Stopped This bit is asserted when the Receive Process enters the Stopped state.	0	R_SS_WC
7	RU	Receive Buffer Unavailable This bit indicates that the host owns the Next Descriptor in the Receive List and the DMA cannot acquire it. The Receive Process is suspended. To resume processing Receive descriptors, the host should change the ownership of the descriptor and issue a Receive Poll Demand command. If no Receive Poll Demand is issued, the Receive Process resumes when the next recognized incoming frame is received. This bit is set only when the previous Receive Descriptor is owned by the DMA.	0	R_SS_WC

6	RI	Receive Interrupt This bit indicates that the frame reception is complete. When reception is complete, the Bit 31 of RDES1 (Disable Interrupt on Completion) is reset in the last Descriptor, and the specific frame status information is updated in the descriptor. The reception remains in the Running state.	0	R_SS_WC
5	UNF	Transmit Underflow This bit indicates that the Transmit Buffer had an Underflow during frame transmission. Transmission is suspended and an Underflow Error TDES0[1] is set.	0	R_SS_WC
4	OVF	Receive Overflow This bit indicates that the Receive Buffer had an Overflow during frame reception. If the partial frame is transferred to the application, the overflow status is set in RDES0[11].	0	R_SS_WC
3	TJT	Transmit Jabber Timeout This bit indicates that the Transmit Jabber Timer expired, which happens when the fram size exceeds 2,048 (10,240 bytes when the Jumbo frame is enabled). When the Jabber Timeout occurs, the transmission process is aborted and placed in the Stopped state. Thi causes the Transmit Jabber Timeout TDES0[14] flag to assert.	0	R_SS_WC
2	TU	Transmit Buffer Unavailable This bit indicates that the host owns the Next Descriptor in the Transmit List and the DMA cannot acquire it. Transmission is suspended. Bits[22:20] explain the Transmit Process state transitions. To resume processing Transmit descriptors, the host should change the ownership of the descriptor by setting TDES0[31] and then issue a Transmit Poll Demand command.	0	R_SS_WC
1	TPS	Transmit Process Stopped This bit is set when the transmission is stopped.	0	R_SS_WC
0	TI	Transmit Interrupt This bit indicates that the frame transmission is complete. When transmission is complete, Bit 31 (OWN) of TDES0 is reset, and the specific frame status information is updated in the descriptor.	0	R_SS_WC

109.1.5.1.7. Register 6 (Operation Mode Register)

The Operation Mode register establishes the Transmit and Receive operating modes and commands. This register should be the last CSR to be written as part of the DMA initialization. This register is also present in the ETH-MTL configuration with bits unused and reserved 24, 13, 2, and 1.

Table 9.1-12 Register 6 (Operation Mode Register)

Field	Field Name	Description	Reset	Access
31:27	-	Reserved	0H	RO
26	DT	Disable Dropping of TCP/IP Checksum Error Frames When this bit is set, the MAC does not drop the frames which only have errors detected by the Receive Checksum Offload engine. Such frames do not have any errors (including FCS error) in the Ethernet frame received by the MAC but have errors only in the encapsulated payload. When this bit is reset, all error frames are dropped if the FEF bit is reset. If the IPC Full Checksum Offload Engine (Type 2) is disabled, this bit is reserved (RO with value 1'b0).	0	R_W

25	RSF	Receive Store and Forward When this bit is set, the MTL reads a frame from the Rx FIFO only after the complete frame has been written to it, ignoring the RTC bits. When this bit is reset, the Rx FIFO operates in the cut-through mode, subject to the threshold specified by the RTC bits.	0	R_W
24	DFF	Disable Flushing of Received Frames When this bit is set, the Rx DMA does not flush any frames because of the unavailability of receive descriptors or buffers as it does normally when this bit is reset. (See “Receive Process Suspended”.) This bit is reserved (and RO) in the ETH-MTL configuration.	0	R_W
23	RFA_2	MSB of Threshold for Activating Flow Control If the ETH is configured for an Rx FIFO size of 8 KB or more, this bit (when set) provides additional threshold levels for activating the flow control in both half- duplex and full-duplex modes. This bit (as Most Significant Bit), along with the RFA (Bits [10:9]), gives the following thresholds for activating flow control: ■ 100: Full minus 5 KB, that is, FULL — 5 KB ■ 101: Full minus 6 KB, that is, FULL — 6 KB ■ 110: Full minus 7 KB, that is, FULL — 7 KB ■ 111: Reserved This bit is reserved (and RO) if the Rx FIFO is 4 KB or less deep.	0	R_W
22	RFD_2	MSB of Threshold for Deactivating Flow Control If the ETH is configured for Rx FIFO size of 8 KB or more, this bit (when set) provides additional threshold levels for deactivating the flow control in both half-duplex and full-duplex modes. This bit (as Most Significant Bit) along with the RFD (Bits [12:11]) gives the following thresholds for deactivating flow control: ■ 100: Full minus 5 KB, that is, FULL — 5 KB ■ 101: Full minus 6 KB, that is, FULL — 6 KB ■ 110: Full minus 7 KB, that is, FULL — 7 KB ■ 111: Reserved This bit is reserved (and RO) if the Rx FIFO is 4 KB or less deep.	0	R_W
21	TSF	Transmit Store and Forward When this bit is set, transmission starts when a full frame resides in the MTL Transmit FIFO. When this bit is set, the TTC values specified in Bits [16:14] are ignored. This bit should be changed only when the transmission is stopped.	0	R_W
20	FTF	Flush Transmit FIFO When this bit is set, the transmit FIFO controller logic is reset to its default values and thus all data in the Tx FIFO is lost or flushed. This bit is cleared internally when the flushing operation is complete. The Operation Mode register should not be written to until this bit is cleared. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt frame transmission. Note: The flush operation is complete only when the Tx FIFO is emptied of its contents and all the pending Transmit Status of the transmitted frames are accepted by the host. In order to complete this flush operation, the PHY transmit clock (clk_tx_i) is required to be active.	0	R_WS_ SC
19:17	-	Reserved	000	RO

16:14	TTC	Transmit Threshold Control These bits control the threshold level of the MTL Transmit FIFO. Transmission starts when the frame size within the MTL Transmit FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are also transmitted. These bits are used only when Bit 21 (TSF) is reset. ■ 000: 64 ■ 001: 128 ■ 010: 192 ■ 011: 256 ■ 100: 40 ■ 101: 32 ■ 110: 24 ■ 111: 16	000	R_W
13	ST	Start or Stop Transmission Command When this bit is set, transmission is placed in the Running state, and the DMA checks the Transmit List at the current position for a frame to be transmitted. Descriptor acquisition is attempted either from the current position in the list, which is the Transmit List Base Address set by Register 4 (Transmit Descriptor List Address Register), or from the position retained when transmission was stopped previously. If the DMA does not own the current descriptor, transmission enters the Suspended state and Bit 2 (Transmit Buffer Unavailable) of Register 5 (Status Register) is set. The Start Transmission command is effective only when transmission is stopped. If the command is issued before setting Register 4 (Transmit Descriptor List Address Register), then the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current frame. The Next Descriptor position in the Transmit List is saved, and it becomes the current position when transmission is restarted. To change the list address, you need to program Register 4 (Transmit Descriptor List Address Register) with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current frame is complete or the transmission is in the Suspended state. Note: For information about how to pause the transmission, see “Stopping and Starting Transmission”.	0	R_W
12:11	RFD	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (Fill-level of Rx FIFO) at which the flow control is de-asserted after activation. ■ 00: Full minus 1 KB, that is, FULL — 1 KB ■ 01: Full minus 2 KB, that is, FULL — 2 KB ■ 10: Full minus 3 KB, that is, FULL — 3 KB ■ 11: Full minus 4 KB, that is, FULL — 4 KB The de-assertion is effective only after flow control is asserted. If the Rx FIFO is 8 KB or more, an additional Bit (RFD_2) is used for more threshold levels as described in Bit 22. These bits are reserved and read-only when the Rx FIFO depth is less than 4 KB. Note: For proper flow control, the value programmed in the “RFD_2, RFD” fields should be equal to or more than the value programmed in the “RFA_2, RFA” fields.	00	R_W

10:9	RFA	Threshold for Activating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (Fill level of Rx FIFO) at which the flow control is activated. ■ 00: Full minus 1 KB, that is, FULL—1KB. ■ 01: Full minus 2 KB, that is, FULL—2KB. ■ 10: Full minus 3 KB, that is, FULL—3KB. ■ 11: Full minus 4 KB, that is, FULL—4KB. These values are applicable only to Rx FIFOs of 4 KB or more and when Bit 8 (EFC) is set high. If the Rx FIFO is 8 KB or more, an additional Bit (RFA_2) is used for more threshold levels as described in Bit 23. These bits are reserved and read-only when the depth of Rx FIFO is less than 4 KB. Note: When FIFO size is exactly 4 KB, although the ETH allows you to program the value of these bits to 11, the software should not program these bits to 2'b11. The value 2'b11 means flow control on FIFO empty condition.	00	R_W
8	EFC	Enable HW Flow Control When this bit is set, the flow control signal operation based on the fill-level of Rx FIFO is enabled. When reset, the flow control operation is disabled. This bit is not used (reserved and always reset) when the Rx FIFO is less than 4 KB.	0	R_W
7	FEF	Forward Error Frames When this bit is reset, the Rx FIFO drops frames with error status (CRC error, collision error, GMII_ER, giant frame, watchdog timeout, or overflow). However, if the start byte (write) pointer of a frame is already transferred to the read controller side (in Threshold mode), then the frame is not dropped. In the ETH-MTL configuration in which the Frame Length FIFO is also enabled during core configuration, the Rx FIFO drops the error frames if that frame's start byte is not transferred (output) on the ARI bus. When the FEF bit is set, all frames except runt error frames are forwarded to the DMA. If the Bit 25 (RSF) is set and the Rx FIFO overflows when a partial frame is written, then the frame is dropped irrespective of the FEF bit setting. However, if the Bit 25 (RSF) is reset and the Rx FIFO overflows when a partial frame is written, then a partial frame may be forwarded to the DMA. Note: When FEF bit is reset, the giant frames are dropped if the giant frame status is given in Rx Status in the following configurations: ■ The IP checksum engine (Type 1) and full checksum offload engine (Type 2) are not selected. ■ The advanced timestamp feature is not selected but the extended status is selected. The extended status is available with the following features: - L3-L4 filter in ETH-CORE or ETH-MTL configurations - Full checksum offload engine (Type 2) with enhanced descriptor format in the ETH-DMA, ETH-AHB, or ETH-AXI configurations.	0	R_W
6	FUF	Forward Undersized Good Frames When set, the Rx FIFO forwards Undersized frames (that is, frames with no Error and length less than 64 bytes) including pad-bytes and CRC. When reset, the Rx FIFO drops all frames of less than 64 bytes, unless a frame is already transferred because of the lower value of Receive Threshold, for example, RTC = 01.	0	R_W

5	DGF	Drop Giant Frames When set, the MAC drops the received giant frames in the Rx FIFO, that is, frames that are larger than the computed giant frame limit. When reset, the MAC does not drop the giant frames in the Rx FIFO. Note: This bit is available in the following configurations in which the giant frame status is not provided in Rx status and giant frames are not dropped by default: ■ Configurations in which IP Checksum Offload (Type 1) is selected in Rx ■ Configurations in which the IPC Full Checksum Offload Engine (Type 2) is selected in Rx with normal descriptor format ■ Configurations in which the Advanced Timestamp feature is selected In all other configurations, this bit is not used (reserved and always reset).	0	R_W
4:3	RTC	Receive Threshold Control These two bits control the threshold level of the MTL Receive FIFO. Transfer (request) to DMA starts when the frame size within the MTL Receive FIFO is larger than the threshold. In addition, full frames with length less than the threshold are automatically transferred. The value of 11 is not applicable if the configured Receive FIFO size is 128 bytes. These bits are valid only when the RSF bit is zero, and are ignored when the RSF bit is set to 1. ■ 00: 64 ■ 01: 32 ■ 10: 96 ■ 11: 128	00	R_W
2	OSF	Operate on Second Frame When this bit is set, it instructs the DMA to process the second frame of the Transmit data even before the status for the first frame is obtained.	0	R_W
1	SR	Start or Stop Receive When this bit is set, the Receive process is placed in the Running state. The DMA attempts to acquire the descriptor from the Receive list and processes the incoming frames. The descriptor acquisition is attempted from the current position in the list, which is the address set by the Register 3 (Receive Descriptor List Address Register) or the position retained when the Receive process was previously stopped. If the DMA does not own the descriptor, reception is suspended and Bit 7 (Receive Buffer Unavailable) of Register 5 (Status Register) is set. The Start Receive command is effective only when the reception has stopped. If the command is issued before setting Register 3 (Receive Descriptor List Address Register), the DMA behavior is unpredictable. When this bit is cleared, the Rx DMA operation is stopped after the transfer of the current frame. The next descriptor position in the Receive list is saved and becomes the current position after the Receive process is restarted. The Stop Receive command is effective only when the Receive process is in either the Running (waiting for receive packet) or in the Suspended state. Note: For information about how to pause the transmission, see "Stopping and Starting Transmission".	0	R_W
0	-	Reserved	0	RO

109.1.5.1.8. Register 7 (Interrupt Enable Register)

The Interrupt Enable register enables the interrupts reported by Register 5 (Status Register). Setting a bit to 1'b1 enables a corresponding interrupt. After a hardware reset or software reset, all interrupts are disabled.

Table 9.1-13 Register 7 (Interrupt Enable Register)

Field	Field Name	Description	Reset	Access
31:17	-	Reserved	0000H	RO

16	NIE	Normal Interrupt Summary Enable When this bit is set, normal interrupt summary is enabled. When this bit is reset, normal interrupt summary is disabled. This bit enables the following interrupts in Register 5 (Status Register): ▪ Register 5[0]: Transmit Interrupt ▪ Register 5[2]: Transmit Buffer Unavailable ▪ Register 5[6]: Receive Interrupt ▪ Register 5[14]: Early Receive Interrupt	0	R_W
15	AIE	Abnormal Interrupt Summary Enable When this bit is set, abnormal interrupt summary is enabled. When this bit is reset, the abnormal interrupt summary is disabled. This bit enables the following interrupts in Register 5 (Status Register): ■ Register 5[1]: Transmit Process Stopped ■ Register 5[3]: Transmit Jabber Timeout ■ Register 5[4]: Receive Overflow ■ Register 5[5]: Transmit Underflow ■ Register 5[7]: Receive Buffer Unavailable ■ Register 5[8]: Receive Process Stopped ■ Register 5[9]: Receive Watchdog Timeout ■ Register 5[10]: Early Transmit Interrupt ■ Register 5[13]: Fatal Bus Error	0	R_W
14	ERE	Early Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (Bit 16), the Early Receive Interrupt is enabled. When this bit is reset, the Early Receive Interrupt is disabled.	0	R_W
13	FBE	Fatal Bus Error Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Fatal Bus Error Interrupt is enabled. When this bit is reset, the Fatal Bus Error Enable Interrupt is disabled.	0	R_W
12:11	-	Reserved	00	RO
10	ETE	Early Transmit Interrupt Enable When this bit is set with an Abnormal Interrupt Summary Enable (Bit 15), the Early Transmit Interrupt is enabled. When this bit is reset, the Early Transmit Interrupt is disabled.	0	R_W
9	RWE	Receive Watchdog Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Receive Watchdog Timeout Interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout Interrupt is disabled.	0	R_W
8	RSE	Receive Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped Interrupt is disabled.	0	R_W
7	RUE	Receive Buffer Unavailable Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Receive Buffe Unavailable Interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable Interrupt is disabled.	0	R_W
6	RIE	Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (Bit 16), the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.	0	R_W

5	UNE	Underflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Transmit Underflow Interrupt is enabled. When this bit is reset, the Underflow Interrupt is disabled.	0	R_W
4	OVE	Overflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Receive Overflow Interrupt is enabled. When this bit is reset, the Overflow Interrupt is disabled.	0	R_W
3	TJE	Transmit Jabber Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Transmit Jabber Timeout Interrupt is enabled. When this bit is reset, the Transmit Jabber Timeout Interrupt is disabled.	0	R_W
2	TUE	Transmit Buffer Unavailable Enable When this bit is set with Normal Interrupt Summary Enable (Bit 16), the Transmit Buffer Unavailable Interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable Interrupt is disabled.	0	R_W
1	TSE	Transmit Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (Bit 15), the Transmission Stopped Interrupt is enabled. When this bit is reset, the Transmission Stopped Interrupt is disabled.	0	R_W
0	TIE	Transmit Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (Bit 16), the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.	0	R_W

The sbd_intr_o interrupt is generated as shown in Figure below. It is asserted only when the TTI, GPI, GMI, GLI, or GLPII bit of the DMA Status register is asserted, or when the NIS or AIS Status bit is asserted and the corresponding Interrupt Enable bits (NIE or AIE) are enabled.

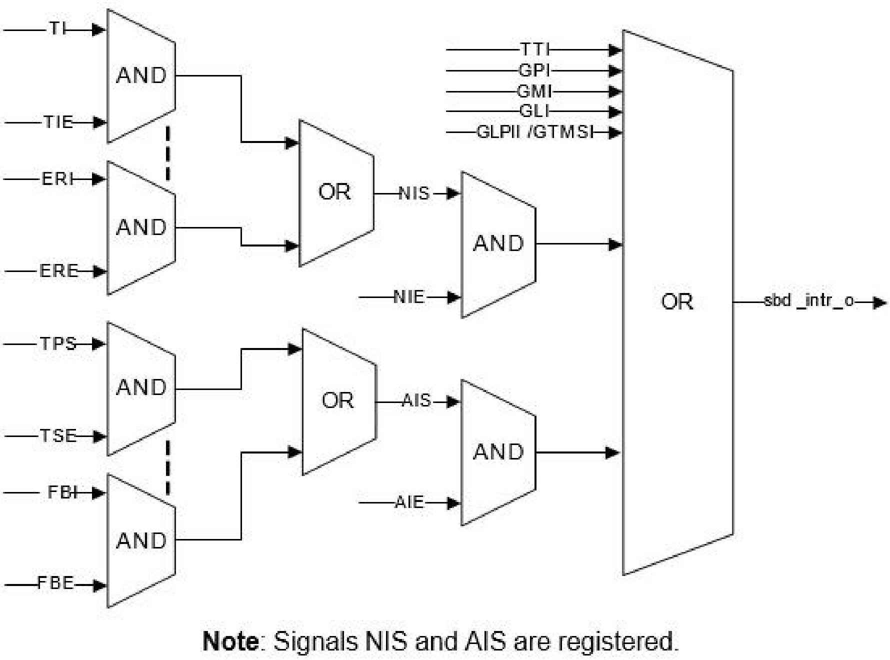


Figure 9.1- 2 Sbd_intr_o Generation

109.1.5.1.9. Register 8 (Missed Frame and Buffer Overflow Counter Register)

The DMA maintains two counters to track the number of frames missed during reception. This register reports the current value of the counter. The counter is used for diagnostic purposes. Bits[15:0] indicate missed frames because of the host buffer being unavailable. Bits[27:17] indicate missed frames because of buffer overflow conditions (MTL and MAC) and runt frames (good frames of less than 64 bytes) dropped by the MTL.

Table 9.1-14 Register 8 (Missed Frame and Buffer Overflow Counter Register)

Field	Field Name	Description	Reset	Access
31:29	-	Reserved	000	RO
28	OVFCNTOVF	Overflow Bit for FIFO Overflow Counter This bit is set every time the Overflow Frame Counter (Bits[27:17]) overflows, that is, the Rx FIFO overflows with the overflow frame counter at maximum value. In such a scenario, the overflow frame counter is reset to all-zeros and this bit indicates that the rollover happened.	0	R_SS_RC
27:17	OVFFRMCNT	Overflow Frame Counter This field indicates the number of frames missed by the application. This counter is incremented each time the MTL FIFO overflows. The counter is cleared when this register is read with mci_be_i[2] at 1'b1.	000H	R_SS_RC
16	MISCNTOVF	Overflow Bit for Missed Frame Counter This bit is set every time Missed Frame Counter (Bits[15:0]) overflows, that is, the DMA discards an incoming frame because of the Host Receive Buffer being unavailable with the missed frame counter at maximum value. In such a scenario, the Missed frame counter is reset to all-zeros and this bit indicates that the rollover happened.	0	R_SS_RC
15:0	MISFRMCNT	Missed Frame Counter This field indicates the number of frames missed by the controller because of the Host Receive Buffer being unavailable. This counter is incremented each time the DMA discards an incoming frame. The counter is cleared when this register is read with mci_be_i[0] at 1'b1.	0000H	R_SS_RC

109.1.5.1.10. Register 9 (Receive Interrupt Watchdog Timer Register)

This register, when written with a non-zero value, enables the watchdog timer for the Receive Interrupt (Bit 6) of Register 5 (Status Register).

Table 9.1-15 Register 9 (Receive Interrupt Watchdog Timer Register)

Field	Field Name	Description	Reset	Access
31:8	-	Reserved	000000H	RO

7:0	RIWT	RI Watchdog Timer Count This bit indicates the number of system clock cycles multiplied by 256 for which the watchdog timer is set. The watchdog timer gets triggered with the programmed value after the Rx DMA completes the transfer of a frame for which the RI status bit is not set because of the setting in the corresponding descriptor RDES1[31]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per RDES1[31] of any received frame.	00H	R_W

109.1.5.1.11. Register 10 (AXI Bus Mode Register)

The AXI Bus Mode Register controls the behavior of the AXI master. It is mainly used to control the burst splitting and the number of outstanding requests. This register is present and valid only in the ETH-AXI configuration. In addition, this register is valid only in the Channel 0 DMA when multiple channels are present in the AV mode.

Table 9.1-16 Register 10 (AXI Bus Mode Register)

Field	Field Name	Description	Reset	Access
31	EN_LPI	Enable Low Power Interface (LPI) When set to 1, this bit enables the LPI mode supported by the ETH-AXI configuration and accepts the LPI request from the AXI System Clock controller. When set to 0, this bit disables the LPI mode and always denies the LPI request from the AXI System Clock controller.	0	R_W
30	LPI_XIT_FRM	Unlock on Magic Packet or Remote Wake-Up Frame When set to 1, this bit enables the ETH-AXI to come out of the LPI mode only when the magic packet or remote wake-up frame is received. When set to 0, this bit enables the ETH-AXI to come out of LPI mode when any frame is received.	0	R_W
29:24	-	Reserved	00H	RO
23:20	WR_OSR_LMT	AXI Maximum Write Outstanding Request Limit This value limits the maximum outstanding request on the AXI write interface. Maximum outstanding requests = WR_OSR_LMT+1 **Note:** ▪ Bit 22 is reserved if AXI_GM_MAX_WR_REQUESTS = 4. ▪ Bit 23 bit is reserved if AXI_GM_MAX_WR_REQUESTS != 16.	‘h1	R_W
19:16	RD_OSR_LMT	AXI Maximum Read Outstanding Request Limit This value limits the maximum outstanding request on the AXI read interface. Maximum outstanding requests = RD_OSR_LMT+1 **Note**: ▪ Bit 18 is reserved if AXI_GM_MAX_RD_REQUESTS = 4. ▪ Bit 19 is reserved if AXI_GM_MAX_RD_REQUESTS != 16.	‘h1	R_W
15:14	-	Reserved	00	RO
13	ONEKBBE	1 KB Boundary Crossing Enable for the ETH-AXI Master When set, the ETH-AXI master performs burst transfers that do not cross 1 KB boundary. When reset, the ETH-AXI master performs burst transfers that do not cross 4 KB boundary.	0	R_W

12	AXI_AAL	Address-Aligned Beats This bit is read-only bit and reflects the Bit 25 (AAL) of Register 0 (Bus Mode Register). When this bit is set to 1, the ETH-AXI performs address-aligned burst transfers on both read and write channels.	0	RO
11:8	-	Reserved	0H	RO
7	BLEN256	AXI Burst Length 256 When this bit is set to 1, the ETH-AXI is allowed to select a burst length of 256 on the AXI master interface. This bit is present only when the configuration parameter AXI_BL is set to 256. Otherwise, this bit is reserved and is read-only (RO).	0	R_W
6	BLEN128	AXI Burst Length 128 When this bit is set to 1, the ETH-AXI is allowed to select a burst length of 128 on the AXI master interface. This bit is present only when the configuration parameter AXI_BL is set to 128 or more. Otherwise, this bit is reserved and is read-only (RO).	0	R_W
5	BLEN64	AXI Burst Length 64 When this bit is set to 1, the ETH-AXI is allowed to select a burst length of 64 on the AXI master interface. This bit is present only when the configuration parameter AXI_BL is set to 64 or more. Otherwise, this bit is reserved and is read-only (RO).	0	R_W
4	BLEN32	AXI Burst Length 32 When this bit is set to 1, the ETH-AXI is allowed to select a burst length of 32 on the AXI master interface. This bit is present only when the configuration parameter AXI_BL is set to 32 or more. Otherwise, this bit is reserved and is read-only (RO).	0	R_W
3	BLEN16	AXI Burst Length 16 When this bit is set to 1 or UNDEF is set to 1, the ETH-AXI is allowed to select a burst length of 16 on the AXI master interface.	0	R_W
2	BLEN8	AXI Burst Length 8 When this bit is set to 1, the ETH-AXI is allowed to select a burst length of 8 on the AXI master interface. Setting this bit has no effect when UNDEF is set to 1.	0	R_W
1	BLEN4	AXI Burst Length 4 When this bit is set to 1, the ETH-AXI is allowed to select a burst length of 4 on the AXI master interface. Setting this bit has no effect when UNDEF is set to 1.	0	R_W
0	UNDEF	AXI Undefined Burst Length This bit is read-only bit and indicates the complement (invert) value of Bit 16 (FB) in Register 0 (Bus Mode Register). ■ When this bit is set to 1, the ETH-AXI is allowed to perform any burst length equal to or below the maximum allowed burst length programmed in Bits[7:3]. ■ When this bit is set to 0, the ETH-AXI is allowed to perform only fixed burst lengths as indicated by BLEN256, BLEN128, BLEN64, BLEN32, BLEN16, BLEN8, or BLEN4, or a burst length of 1. If UNDEF is set and none of the BLEN bits is set, then ETH-AXI is allowed to perform a burst length of 16.	1	RO

109.1.5.1.12. Register 11 (AHB or AXI Status Register)

This register provides the active status of the read and write channels of the AHB master interface or AXI interface. This register is present and valid only in the ETH-AHB and

ETH-AXI configurations. This register is useful for debugging purposes. In addition, this register is valid only in the Channel 0 DMA when multiple channels are present in the AV mode.

Table 9.1-17 Register 11 (AHB or AXI Status Register)

Field	Field Name	Description	Reset	Access
31:2	-	Reserved	0000_0000H	RO
1	AXIRDSTS	AXI Master Read Channel Status When high, it indicates that AXI master's read channel is active and transferring data.	0	RO
0	AXWHSTS	AXI Master Write Channel or AHB Master Status When high, it indicates that AXI master's write channel is active and transferring data in the ETH-AXI configuration. In the ETH-AHB configuration, it indicates that the AHB master interface FSMs are in the non-idle state.	0	RO

109.1.5.1.13. Register 18 (Current Host Transmit Descriptor Register)

The Current Host Transmit Descriptor register points to the start address of the current Transmit Descriptor read by the DMA.

Table 9.1-18 Register 18 (Current Host Transmit Descriptor Register)

Field	Field Name	Description	Reset	Access
31:0	CURTDESAPTR	Host Transmit Descriptor Address Pointer Cleared on Reset. Pointer updated by the DMA during operation.	0000_0000H	RO

109.1.5.1.14. Register 19 (Current Host Receive Descriptor Register)

The Current Host Receive Descriptor register points to the start address of the current Receive Descriptor read by the DMA.

Table 9.1-19 Register 19 (Current Host Receive Descriptor Register)

Field	Field Name	Description	Reset	Access
31:0	CURRDESAPTR	Host Receive Descriptor Address Pointer Cleared on Reset. Pointer updated by the DMA during operation.	0000_0000H	RO

109.1.5.1.15. Register 20 (Current Host Transmit Buffer Address Register)

The Current Host Transmit Buffer Address register points to the current Transmit Buffer Address being read by the DMA.

Table 9.1-20 Register 20 (Current Host Transmit Buffer Address Register)

Field	Field Name	Description	Reset	Access
31:0	CURTBUFAPTR	Host Transmit Buffer Address Pointer Cleared on Reset. Pointer updated by the DMA during operation.	0000_0000H	RO

109.1.5.1.16. Register 21 (Current Host Receive Buffer Address Register)

The Current Host Receive Buffer Address register points to the current Receive Buffer address being read by the DMA.

Table 9.1-21 Register 21 (Current Host Receive Buffer Address Register)

Field	Field Name	Description	Reset	Access
31:0	CURRBUFAPTR	Host Receive Buffer Address Pointer Cleared on Reset. Pointer updated by the DMA during operation.	0000_0000H	RO

109.1.5.1.17. Register 22 (HW Feature Register)

This register indicates the presence of the optional features or functions of the ETH. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.

Table 9.1-22 Register 22 (HW Feature Register)

Field	Field Name	Description	Access
Note: All bits are set or reset as per the selection of features during the core configuration in coreConsultant.			
31	-	Reserved	RO
30:28	ACTPHYIF	Active or selected PHY interface When you have multiple PHY interfaces in your configuration, this field indicates the sampled value of phy_intf_sel_i during reset de-assertion. ■ 000: GMII or MII ■ 001: RGMII ■ 010: SGMII ■ 011: TBI ■ 100: RMII ■ 101: RTBI ■ 110: SMII ■ 111: RevMII ■ All Others: Reserved	RO
27	SAVLANINS	Source Address or VLAN Insertion	RO
26	FLEXIPPSEN	Flexible Pulse-Per-Second Output	RO
25	INTTSEN	Timestamping with Internal System Time	RO
24	ENHDESSEL	Alternate (Enhanced Descriptor)	RO
23:22	TXCHCNT	Number of additional Tx Channels	RO
21:20	RXCHCNT	Number of additional Rx Channels	RO
19	RXFIFOSIZE	Rx FIFO > 2,048 Bytes	RO
18	RXTYP2COE	IP Checksum Offload (Type 2) in Rx	RO
17	RXTYP1COE	IP Checksum Offload (Type 1) in Rx Note: If IPCHKSUM_EN = Enabled and IPC_FULL_OFFLOAD = Enabled, then RXTYP1COE = 0 and RXTYP2COE = 1.	RO
16	TXCOESEL	Checksum Offload in Tx	RO
15	AVSEL	AV feature	RO

14	EEESEL	Energy Efficient Ethernet	RO
13	TSVER2SEL	IEEE 1588-2008 Advanced timestamp	RO
12	TSVER1SEL	Only IEEE 1588-2002 timestamp	RO
11	MMCSEL	RMON module	RO
10	MGKSEL	PMT magic packet	RO
9	RWKSEL	PMT remote wake-up frame	RO
8	SMASEL	SMA (MDIO) Interface	RO
7	L3L4FLTREN	Layer 3 and Layer 4 feature	RO
6	PCSSEL	PCS registers (TBI, SGMII, or RTBI PHY interface)	RO
5	ADDMACADRSEL	Multiple MAC Address registers	RO
4	HASHSEL	HASH filter	RO
3	EXTHASHEN	Expanded DA Hash filter	RO
2	HDSEL	Half-duplex support	RO
1	GMIISEL	1000 Mbps support	RO
0	MIISEL	10 or 100 Mbps support	RO

109.1.5.1.18. Register 76 (Channel 1 Slot Function Control and Status Register)

This register controls the slot comparison feature that the Channel 1 transmit DMA uses to fetch the buffer data from system memory.

Table 9.1-23 Register 76 (Channel 1 Slot Function Control and Status Register)

Field	Field Name	Description	Reset	Access
31:20	-	Reserved	00H	RO
19:16	RSN	Reference Slot Number This field gives the current value of the reference slot number in DMA used for comparison checking.	0H	RO
15:2	-	Reserved	000H	RO
1	ASC	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the transmit descriptor is: ▪ equal to the reference slot number given in Bits [19:16] -or- ▪ ahead of the reference slot number by up to two slots This bit is applicable only when Bit 0 (ESC) is set.	0	R_W

0	ESC	Enable Slot Comparison When set, this bit enables the checking of the slot numbers, programmed in the transmit descriptor, with the current reference given in Bits [19:16]. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or is ahead of the reference slot number by one slot. When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed.	0	R_W

109.1.5.1.19. Register 88 (Channel 1 CBS Control Register)

This register controls the credit-based shaper algorithm in the Traffic Manager for scheduling the frames for transmission. This register is present only when you select the Transmit Channel 1 in the AV mode.

Table 9.1-24 Register 88 (Channel 1 CBS Control Register)

Field	Field Name	Description	Reset	Access
31:18	-	Reserved	00H	RO
17	ABPSSIE	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts an interrupt (sbd_intr_o or mci_intr_o) when the average bits per slot status is updated (Bit 17 (ABSU) in Register 89) for Channel 1. When this bit is cleared, interrupt is not asserted for such an event.	0	R_W
16:7	-	Reserved	00H	RO
6:4	SLC	Slot Count The software can program the number of slots (of duration 125 micro-sec) over which the average transmitted bits per slot (provided in the CBS Status register) need to be computed for Channel 1 when the credit-based shaper algorithm is enabled. The encoding is as follows: ▪ 3'b000: 1 Slot ▪ 3'b001: 2 Slots ▪ 3'b010: 4 Slots ▪ 3'b011: 8 Slots ▪ 3'b100: 16 Slots ▪ 3'b101-3'b111: Reserved	00	R_W
3:2	-	Reserved	00	RO
1	CC	Credit Control When reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no frame to transmit in Channel 1. When there is no frame waiting in Channel 1 and other channel is transmitting, no credit is accumulated. When set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no frame to transmit in Channel 1. The credit accumulates even when there is no frame waiting in Channel 1 and another channel is transmitting.	0	R_W

0	CBSD	Credit-Based Shaper Disable When set, the MAC disables the credit-based shaper algorithm for Channel 1 traffic and makes the traffic management algorithm to strict priority for Channel 1 over Channel 0 When reset, the credit-based shaper algorithm schedules the traffic in Channel 1 for transmission.	0	R_W

109.1.5.1.20. Register 89 (Channel 1 CBS Status Register)

This register provides the average traffic transmitted in Channel 1. This register is present only when you select the Transmit Channel 1 in the AV mode.

Table 9.1-25 Register 89 (Channel 1 CBS Status Register)

Field	Field Name	Description	Reset	Access
31:18	-	Reserved	000H	RO
17	ABSU	ABS Updated When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application reads the ABS value.	0	R_SS_SC
16:0	ABS	Average Bits per Slot This field contains the average transmitted bits per slot. This field is computed over programmed number of slots (SLC bits in the CBS Control Register) for Channel 1 traffic. The maximum value is 0x30D4 for 100 Mbps and 0x1E848 for 1000 Mbps.	000H	RO

109.1.5.1.21. Register 90 (Channel 1 idleSlopeCredit Register)

This register provides the bandwidth allocated for the AV traffic on Channel 1. This register is present only when you select the Transmit Channel 1 in the AV mode.

Table 9.1-26 Register 90 (Channel 1 idleSlopeCredit Register)

Field	Field Name	Description	Reset	Access
31:14	-	Reserved	000H	RO
13:0	ISC	idleSlopeCredit This field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Channel 1. This is the rate of change of credit in bits per cycle (40ns and 8ns for 100 Mbps and 1000 Mbps respectively) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1024. The maximum value is portTransmitRate, that is, 0x2000 in 1000 Mbps mode and 0x1000 in 100 Mbps mode.	000H	R_W

109.1.5.1.22. Register 91 (Channel 1 sendSlopeCredit Register)

This register provides the bandwidth that is available for the AV traffic on other channels. This register is present only when you select the Transmit Channel 1 in the AV mode.

Table 9.1-27 Register 91 (Channel 1 sendSlopeCredit Register)

Field	Field Name	Description	Reset	Access
31:14	-	Reserved	000H	RO
13:0	SSC	sendSlopeCredit This field contains the sendSlopeCredit value required for credit-based shaper algorithm for Channel 1. This is the rate of change of credit in bits per cycle (40ns and 8ns for 100 Mbps and 1000 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1024. The maximum value is portTransmitRate, that is, 0x2000 in 1000 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Channel 1 is selected for transmission.	000H	R_W

109.1.5.1.23. Register 92 (Channel 1 hiCredit Register)

This register provides the maximum value that can be accumulated for Channel 1 in the credit parameter of the credit-based shaper algorithm. This register is present only when you select the Transmit Channel 1 in the AV mode.

Table 9.1-28 Register 92 (Channel 1 hiCredit Register)

Field	Field Name	Description	Reset	Access
31:29	-	Reserved	0H	RO
28:0	HC	hiCredit This field contains the hiCredit value required for the credit-based shaper algorithm for Channel 1. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum frame size which is 16,384 bytes or 131,072 bits. The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000.	0000000H	R_W

109.1.5.1.24. Register 93 (Channel 1 loCredit Register)

This register provides the minimum value that can be accumulated for Channel 1 in the credit parameter of the credit-based shaper algorithm. This register is present only when you select Transmit Channel 1 in the AV mode.

Table 9.1-29 Register 93 (Channel 1 loCredit Register)

Field	Field Name	Description	Reset	Acces s
31:29	-	Reserved	0x7	RO
28:0	LC	loCredit This field contains the loCredit value required for the credit-based shaper algorithm for Channel 1. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum frame size which is 16,384 bytes or 131,072 bits. The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. The programmed value is 2's complement (negative number), that is, 0xF800_0000.	0x1FFF_FFFF	R_W

109.1.5.2. ETH Register Description

109.1.5.2.1. Register 0 (MAC Configuration Register)

The MAC Configuration register establishes receive and transmit operating modes.

Table 9.1-30 Register 0 (MAC Configuration Register)

Field	Field Name	Description	Reset	Access
31	-	Reserved	0	RO
30:28	SARC	Source Address Insertion or Replacement Control This field controls the source address insertion or replacement for all transmitted frames. Bit 30 specifies which MAC Address register (0 or 1) is used for source address insertion or replacement based on the values of Bits [29:28]: ■ 2'b0x: The input signals mti_sa_ctrl_i and ati_sa_ctrl_i control the SA field generation. ■ 2'b10: - If Bit 30 is set to 0, the MAC inserts the content of the MAC Address 0 registers (registers 16 and 17) in the SA field of all transmitted frames. - If Bit 30 is set to 1 and the Enable MAC Address Register 1 option is selected during core configuration, the MAC inserts the content of the MAC Address 1 registers (registers 18 and 19) in the SA field of all transmitted frames. ■ 2'b11: - If Bit 30 is set to 0, the MAC replaces the content of the MAC Address 0 registers (registers 16 and 17) in the SA field of all transmitted frames. - If Bit 30 is set to 1 and the Enable MAC Address Register 1 option is selected during core configuration, the MAC replaces the content of the MAC Address 1 registers (registers 18 and 19) in the SA field of all transmitted frames. Note: ■ Changes to this field take effect only on the start of a frame. If you write this register field when a frame is being transmitted, only the subsequent frame can use the updated value, that is, the current frame does not use the updated value. ■ These bits are reserved and RO when the Enable SA, VLAN, and CRC Insertion on TX feature is not selected during core configuration.	0	R_W
27	TWOKPE	IEEE 802.3as Support for 2K Packets When set, the MAC considers all frames, with up to 2,000 bytes length, as normal packets. When Bit 20 (JE) is not set, the MAC considers all received frames of size more than 2K bytes as Giant frames. When this bit is reset and Bit 20 (JE) is not set, the MAC considers all received frames of size more than 1,518 bytes (1,522 bytes for tagged) as Giant frames. When Bit 20 is set, setting this bit has no effect on Giant Frame status. For more information about how the setting of this bit and Bit 20 impact the Giant frame status, see Table 6-31.	00	R_W
26	SFTERR	SMII Force Transmit Error When set, this bit indicates to the PHY to force a transmit error in the SMII frame being transmitted. This bit is reserved if the SMII PHY port is not selected during core configuration.	0	R_W

25	CST	CRC Stripping for Type Frames When this bit is set, the last 4 bytes (FCS) of all frames of Ether type (Length/Type field greater than or equal to 1,536) are stripped and dropped before forwarding the frame to the application. This function is not valid when the IP Checksum Engine (Type 1) is enabled in the MAC receiver. This function is valid when Type 2 Checksum Offload Engine is enabled.	0	R_W
24	TC	Transmit Configuration in RGMII, SGMII, or SMII When set, this bit enables the transmission of duplex mode, link speed, and link up or down information to the PHY in the RGMII, SMII, or SGMII port. When this bit is reset, no such information is driven to the PHY. This bit is reserved (and RO) if the RGMII, SMII, or SGMII PHY port is not selected during core configuration. The details of this feature are explained in the following sections: ■ "Reduced Gigabit Media Independent Interface" ■ "Serial Media Independent Interface" ■ "Serial Gigabit Media Independent Interface"	0	R_W
23	WD	Watchdog Disable When this bit is set, the MAC disables the watchdog timer on the receiver. The MAC can receive frames of up to 16,383 bytes. When this bit is reset, the MAC does not allow a receive frame which more than 2,048 bytes (10,240 if JE is set high) or the value programmed in Register 55 (Watchdog Timeout Register). The MAC cuts off any bytes received after the watchdog limit number of bytes.	0	R_W
22	JD	Jabber Disable When this bit is set, the MAC disables the jabber timer on the transmitter. The MAC can transfer frames of up to 16,383 bytes. When this bit is reset, the MAC cuts off the transmitter if the application sends out more than 2,048 bytes of data (10,240 if JE is set high) during transmission.	0	R_W
21	BE	Frame Burst Enable When this bit is set, the MAC allows frame bursting during transmission in the GMII half-duplex mode. This bit is reserved (and RO) in the 10/100 Mbps only or full-duplex-only configurations.	0	R_W
20	JE	Jumbo Frame Enable When this bit is set, the MAC allows Jumbo frames of 9,018 bytes (9,022 bytes for VLAN tagged frames) without reporting a giant frame error in the receive frame status.	0	R_W
19:17	IFG	Inter-Frame Gap These bits control the minimum IFG between frames during transmission. ■ 000: 96 bit times ■ 001: 88 bit times ■ 010: 80 bit times ■ ... ■ 111: 40 bit times In the half-duplex mode, the minimum IFG can be configured only for 64 bit times (IFG = 100). Lower values are not considered. In the 1000-Mbps mode, the minimum IFG supported is 64 bit times (and above) in the ETH-CORE configuration and 80 bit times (and above) in other configurations. When a JAM pattern is being transmitted because of backpressure activation, the MAC does not consider the minimum IFG.	000	R_W

16	DCRS	Disable Carrier Sense During Transmission When set high, this bit makes the MAC transmitter ignore the (G)MII CRS signal during frame transmission in the half-duplex mode. This request results in no errors generated because of Loss of Carrier or No Carrier during such transmission. When this bit is low, the MAC transmitter generates such errors because of Carrier Sense and can even abort the transmissions. This bit is reserved (and RO) in the full-duplex-only configurations.	0	R_W
15	PS	Port Select This bit selects the Ethernet line speed. ■ 0: For 1000 Mbps operations ■ 1: For 10 or 100 Mbps operations In 10 or 100 Mbps operations, this bit, along with FES bit, selects the exact line speed In the 10/100 Mbps-only (always 1) or 1000 Mbps-only (always 0) configurations, this bit is read-only with the appropriate value. In default 10/100/1000 Mbps configuration, this bit is R_W. The mac_portselect_o or mac_speed_o[1] signal reflects the value of this bit.	0	R_W
14	FES	Speed This bit selects the speed in the MII, RMII, SMII, RGMII, SGMII, or RevMII interface: ■ 0: 10 Mbps ■ 1: 100 Mbps This bit is reserved (RO) by default and is enabled only when the parameter SPEED_SELECT = Enabled. This bit generates link speed encoding when Bit 24 (TC) is set in the RGMII, SMII, or SGMII mode. This bit is always enabled for RGMII, SGMII, SMII, or RevMII interface. In configurations with RGMII, SGMII, SMII, or RevMII interface, this bit is driven as an output signal (mac_speed_o[0]) to reflect the value of this bit in the mac_speed_o signal. In configurations with RMII, MII, or GMII interface, you can optionally drive this bit as an output signal (mac_speed_o[0]) to reflect its value in the mac_speed_o signal.	0	R_W
13	DO	Disable Receive Own When this bit is set, the MAC disables the reception of frames when the phy_txen_o is asserted in the half-duplex mode. When this bit is reset, the MAC receives all packets that are given by the PHY while transmitting. This bit is not applicable if the MAC is operating in the full-duplex mode. This bit is reserved (RO with default value) if the MAC is configured for the full-duplex-only operation.	0	R_W
12	LM	Loopback Mode When this bit is set, the MAC operates in the loopback mode at GMII or MII. The (G)MII Receive clock input (clk_rx_i) is required for the loopback to work properly, because the Transmit clock is not looped-back internally.	0	R_W
11	DM	Duplex Mode When this bit is set, the MAC operates in the full-duplex mode where it can transmit and receive simultaneously. This bit is RO with default value of 1'b1 in the full-duplex-only configuration.	0	R_W

10	IPC	Checksum Offload When this bit is set, the MAC calculates the 16-bit one's complement of the one's complement sum of all received Ethernet frame payloads. It also checks whether the IPv4 Header checksum (assumed to be bytes 25–26 or 29–30 (VLAN- tagged) of the received Ethernet frame) is correct for the received frame and gives the status in the receive status word. The MAC also appends the 16-bit checksum calculated for the IP header datagram payload (bytes after the IPv4 header) and appends it to the Ethernet frame transferred to the application (when Type 2 COE is deselected). When this bit is reset, this function is disabled. When Type 2 COE is selected, this bit, when set, enables the IPv4 header checksum checking and IPv4 or IPv6 TCP, UDP, or ICMP payload checksum checking. When this bit is reset, the COE function in the receiver is disabled and the corresponding PCE and IP HCE status bits are always cleared. If the IP Checksum Offload feature is not enabled during core configuration, this bit is reserved (RO with default value).	0	R_W
9	DR	Disable Retry When this bit is set, the MAC attempts only one transmission. When a collision occurs on the GMII or MII interface, the MAC ignores the current frame transmission and reports a Frame Abort with excessive collision error in the transmit frame status. When this bit is reset, the MAC attempts retries based on the settings of the BL field (Bits [6:5]). This bit is applicable only in the half-duplex mode and is reserved (RO with default value) in the full-duplex-only configuration.	0	R_W
8	LUD	Link Up or Down This bit indicates whether the link is up or down during the transmission of configuration in the RGMII, SGMII, or SMII interface: ■ 0: Link Down ■ 1: Link Up This bit is reserved (RO with default value) and is enabled when the RGMII, SGMII, or SMII interface is enabled during core configuration.	0	R_W
7	ACS	Automatic Pad or CRC Stripping When this bit is set, the MAC strips the Pad or FCS field on the incoming frames only if the value of the length field is less than 1,536 bytes. All received frames with length field greater than or equal to 1,536 bytes are passed to the application without stripping the Pad or FCS field. When this bit is reset, the MAC passes all incoming frames, without modifying them, to the Host.	0	R_W
6:5	BL	Back-Off Limit The Back-Off limit determines the random integer number (r) of slot time delays (4,096 bit times for 1000 Mbps and 512 bit times for 10/100 Mbps) for which the MAC waits before rescheduling a transmission attempt during retries after a collision. This bit is applicable only in the half-duplex mode and is reserved (RO) in the full-duplex-only configuration. ■ 00: k= min (n, 10) ■ 01: k = min (n, 8) ■ 10: k = min (n, 4) ■ 11: k = min (n, 1) where n = retransmission attempt. The random integer r takes the value in the range $0 \leq r < 2k$	00	R_W

4	DC	Deferral Check When this bit is set, the deferral check function is enabled in the MAC. The MAC issues a Frame Abort status, along with the excessive deferral error bit set in the transmit frame status, when the transmit state machine is deferred for more than 24,288 bit times in the 10 or 100 Mbps mode. If the MAC is configured for 1000 Mbps operation or if the Jumbo frame mode is enabled in the 10 or 100 Mbps mode, the threshold for deferral is 155,680 bits times. Deferral begins when the transmitter is ready to transmit, but it is prevented because o an active carrier sense signal (CRS) on GMII or MII. The defer time is not cumulative. For example, if the transmitter defers for 10,000 bit times because the CRS signal is active and then the CRS signal becomes inactive, the transmitter transmits and collision happens. Because of collision, the transmitter needs to back off and then defer again after back off completion. In such a scenario, the deferral timer is reset to 0 and it is restarted. When this bit is reset, the deferral check function is disabled and the MAC defers unti the CRS signal goes inactive. This bit is applicable only in the half-duplex mode and i reserved (RO) in the full-duplex-only configuration.	0	R_W
3	TE	Transmitter Enable When this bit is set, the transmit state machine of the MAC is enabled for transmissior on the GMII or MII. When this bit is reset, the MAC transmit state machine is disable after the completion of the transmission of the current frame, and does not transmit any further frames.	0	R_W
2	RE	Receiver Enable When this bit is set, the receiver state machine of the MAC is enabled for receiving frames from the GMII or MII. When this bit is reset, the MAC receive state machine i disabled after the completion of the reception of the current frame, and does not receive any further frames from the GMII or MII.	0	R_W
1:0	PRELEN	Preamble Length for Transmit frames These bits control the number of preamble bytes that are added to the beginning of every Transmit frame. The preamble reduction occurs only when the MAC is operating in the full-duplex mode. ■ 2'b00: 7 bytes of preamble ■ 2'b01: 5 bytes of preamble ■ 2'b10: 3 bytes of preamble ■ 2'b11: Reserved	00	R_W

Table below shows how the settings of Bit 27 and Bit 20 of Register 0 (MAC Configuration Register) impact the giant frame status.

Table 9.1-31 Giant Frame Status based on Bit 27 and Bit 20

Length/Type Field	Received Frame Length	Bit 27 (TWOKPE)	Bit 20 (JE)	Giant Frame Status
Untagged packet	> 1,518	0	0	1
	> 2,000	1	0	1
	> 9,018	x	1	1
VLAN tagged packet	> 1,522	0	0	1
	> 2,000	1	0	1
	> 9,022	x	1	1

Note: For all other combinations, the Giant Frame status is 0.

Table below shows how the settings of Bit 7 and Bit 25 of Register 0 (MAC Configuration Register) impact whether CRC length is included in the frame length.

Table 9.1-32 Frame Length based on Bit 7 and Bit 25

Receive Checksum Offload Engine	Received Frame Length	Bit 7 (ACS)	Bit 25 (CST)	FCS Stripping Done
IPCHKSUM_EN = 0 and IPC_FULL_OFFLOAD = 0 or IPCHKSUM_EN = 1 and IPC_FULL_OFFLOAD = 1	< 1,536	0	x	No
		1	x	Yes (for Ethernet Frames)
	>= 1,536	x	0	No
		x	1	Yes (for Type Frames)
IPCHKSUM_EN = 1 and IPC_FULL_OFFLOAD = 0	< 1,536	0	x	No
		1	x	Yes (for Ethernet Frames)
	>= 1,536	x	x	No

109.1.5.2.2. Register 1 (MAC Frame Filter)

The MAC Frame Filter register contains the filter controls for receiving frames. Some of the controls from this register go to the address check block of the MAC, which performs the first level of address filtering. The second level of filtering is performed on the incoming frame, based on other controls such as Pass Bad Frames and Pass Control Frames.

Table 9.1-33 Register 1 (MAC Frame Filter)

Field	Field Name	Description	Reset	Access
31	RA	Receive All When this bit is set, the MAC Receiver module passes all received frames, irrespective of whether they pass the address filter or not, to the Application. The result of the SA or DA filtering is updated (pass or fail) in the corresponding bits in the Receive Status Word. When this bit is reset, the Receiver module passes only those frames to the Application that pass the SA or DA address filter.	0	R_W
30:22	-	Reserved	00H	RO
21	DNTU	Drop non-TCP/UDP over IP Frames When set, this bit enables the MAC to drop the non-TCP or UDP over IP frames. The MAC forward only those frames that are processed by the Layer 4 filter. When reset, this bit enables the MAC to forward all non-TCP or UDP over IP frames. If the Layer 3 and Layer 4 Filtering feature is not selected during core configuration, thi bit is reserved (RO with default value).	0	R_W

20	IPFE	Layer 3 and Layer 4 Filter Enable When set, this bit enables the MAC to drop frames that do not match the enabled Layer 3 and Layer 4 filters. If Layer 3 or Layer 4 filters are not enabled for matching, this bit does not have any effect. When reset, the MAC forwards all frames irrespective of the match status of the Layer 3 and Layer 4 fields. If the Layer 3 and Layer 4 Filtering feature is not selected during core configuration, thi bit is reserved (RO with default value).	0	R_W
19:17	-	Reserved	000	RO
16	VTFE	VLAN Tag Filter Enable When set, this bit enables the MAC to drop VLAN tagged frames that do not match the VLAN Tag comparison. When reset, the MAC forwards all frames irrespective of the match status of the VLAN Tag.	0	R_W
15:11	-	Reserved	00	R_W
10	HPF	Hash or Perfect Filter When this bit is set, it configures the address filter to pass a frame if it matches either the perfect filtering or the hash filtering as set by the HMC or HUC bits. When this bit i low and the HUC or HMC bit is set, the frame is passed only if it matches the Hash filter. This bit is reserved (and RO) if the Hash filter is not selected during core configuration.	0	R_W
9	SAF	Source Address Filter Enable When this bit is set, the MAC compares the SA field of the received frames with the values programmed in the enabled SA registers. If the comparison fails, the MAC drops the frame. When this bit is reset, the MAC forwards the received frame to the application with updated SAF bit of the Rx Status depending on the SA address comparison. Note: According to the IEEE specification, Bit 47 of the SA is reserved and set to 0. However, in ETH, the MAC compares all 48 bits. The software driver should take this into consideration while programming the MAC address registers for SA.	0	R_W
8	SAIF	SA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the SA address comparison. The frames whose SA matches the SA registers are marked as failing the SA Address filter. When this bit is reset, frames whose SA does not match the SA registers are marked as failing the SA Address filter.	0	R_W

7:6	PCF	Pass Control Frames These bits control the forwarding of all control frames (including unicast and multicast Pause frames). ■ 00: MAC filters all control frames from reaching the application. ■ 01: MAC forwards all control frames except Pause frames to application even if they fail the Address filter. ■ 10: MAC forwards all control frames to application even if they fail the Address Filter. ■ 11: MAC forwards control frames that pass the Address Filter. The following conditions should be true for the Pause frames processing: ■ Condition 1: The MAC is in the full-duplex mode and flow control is enabled by setting Bit 2 (RFE) of Register 6 (Flow Control Register) to 1. ■ Condition 2: The destination address (DA) of the received frame matches the special multicast address or the MAC Address 0 when Bit 3 (UP) of the Register 6 (Flow Control Register) is set. ■ Condition 3: The Type field of the received frame is 0x8808 and the OPCODE field is 0x0001. Note: This field should be set to 01 only when the Condition 1 is true, that is, the MAC is programmed to operate in the full-duplex mode and the RFE bit is enabled. Otherwise, the Pause frame filtering may be inconsistent. When Condition 1 is false, the Pause frames are considered as generic control frames. Therefore, to pass all control frames (including Pause frames) when the full-duplex mode and flow control is not enabled, you should set the PCF field to 10 or 11 (as required by the application).	00	R_W
5	DBF	Disable Broadcast Frames When this bit is set, the AFM module blocks all incoming broadcast frames. In addition, it overrides all other filter settings. When this bit is reset, the AFM module passes all received broadcast frames.	0	R_W
4	PM	Pass All Multicast When set, this bit indicates that all received frames with a multicast destination address (first bit in the destination address field is '1') are passed. When reset, filtering of multicast frame depends on HMC bit.	0	R_W
3	DAIF	DA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the DA address comparison for both unicast and multicast frames. When reset, normal filtering of frames is performed.	0	R_W
2	HMC	Hash Multicast When set, the MAC performs destination address filtering of received multicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for multicast frames, that is, it compares the DA field with the values programmed in DA registers. If Hash Filter is not selected during core configuration, this bit is reserved (and RO).	0	R_W
1	HUC	Hash Unicast When set, the MAC performs destination address filtering of unicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for unicast frames that is, it compares the DA field with the values programmed in DA registers. If Hash Filter is not selected during core configuration, this bit is reserved (and RO).	0	R_W

0	PR	Promiscuous Mode When this bit is set, the Address Filter module passes all incoming frames irrespective of the destination or source address. The SA or DA Filter Fails status bits of the Receive Status Word are always cleared when PR is set.	0	R_W

109.1.5.2.3. Register 2 (Hash Table High Register)

The 64-bit Hash table is used for group address filtering. For hash filtering, the contents of the destination address in the incoming frame is passed through the CRC logic, and the upper 6 bits of the CRC register are used to index the contents of the Hash table. The most significant bit determines the register to be used (Hash Table High or Hash Table Low), and the other 5 bits determine which bit within the register. A hash value of 5b'00000 selects Bit 0 of the selected register, and a value of 5b'11111 selects Bit 31 of the selected register.

The hash value of the destination address is calculated in the following way:

1. Calculate the 32-bit CRC for the DA (See IEEE 802.3, *Section 3.2.8* for the steps to calculate CRC32).

2. Perform bitwise reversal for the value obtained in Step 1.

3. Take the upper 6 bits from the value obtained in Step 2.

For example, if the DA of the incoming frame is received as 0x1F52419CB6AF (0x1F is the first byte received on GMII interface), then the internally calculated 6-bit Hash value is 0x2C and Bit 12 of Hash Table High register is checked for filtering. If the DA of the incoming frame is received as 0xA00A98000045, then the calculated 6-bit Hash value is 0x07 and Bit 7 of Hash Table Low register is checked for filtering.

If the corresponding bit value of the register is 1'b1, the frame is accepted. Otherwise, it is rejected. If the PM (Pass All Multicast) bit is set in Register 1, then all multicast frames are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table High or Low registers are written. Consecutive writes to these register should be performed only after at least 4 clock cycles in the destination clock domain when double-synchronization is enabled.

The Hash Table High register contains the higher 32 bits of the Hash table.

Table 9.1-34 Register 2 (Hash Table High Register)

Field	Field Name	Description	Reset	Access
31:0	HTH	Hash Table High This field contains the upper 32 bits of the Hash table.	0000_0000H	R_W

109.1.5.2.4. Register 3 (Hash Table Low Register)

The Hash Table Low register contains the lower 32 bits of the Hash table. Both Register 2 and Register 3 are reserved if the Hash Filter Function is disabled or the 128-bit or 256-bit Hash Table is selected during core configuration.

Table 9.1-35 Register 3 (Hash Table Low Register)

Field	Field Name	Description	Reset	Access
31:0	HTL	Hash Table Low This field contains the lower 32 bits of the Hash table.	0000_0000H	R_W

109.1.5.2.5. Register 4 (GMII Address Register)

The GMII Address register controls the management cycles to the external PHY through the management interface.

Table 9.1-36 Register 4 (GMII Address Register)

Field	Field Name	Description	Reset	Access
31:16	-	Reserved	0000H	RO
15:11	PA	Physical Layer Address This field indicates which of the 32 possible PHY devices are being accessed. For RevMII, this field gives the PHY Address of the RevMII module.	00H	R_W
10:6	GR	GMII Register These bits select the desired GMII register in the selected PHY device. For RevMII, these bits select the desired CSR register in the RevMII Registers set.	00H	R_W

5:2	CR	CSR Clock Range The CSR Clock Range selection determines the frequency of the MDC clock according to the CSR clock frequency used in your design. The suggested range of CSR clock frequency applicable for each value (when Bit[5] = 0) ensures that the MDC clock is approximately between the frequency range 1.0 MHz–2.5 MHz. ■ 0000: The CSR clock frequency is 60–100 MHz and the MDC clock frequency is CSR clock/42. ■ 0001: The CSR clock frequency is 100–150 MHz and the MDC clock frequency is CSR clock/62. ■ 0010: The CSR clock frequency is 20–35 MHz and the MDC clock frequency is CSR clock/16. ■ 0011: The CSR clock frequency is 35–60 MHz and the MDC clock frequency is CSR clock/26. ■ 0100: The CSR clock frequency is 150–250 MHz and the MDC clock frequency is CSR clock/102. ■ 0101: The CSR clock frequency is 250–300 MHz and the MDC clock is CSR clock/124. ■ 0110, 0111: Reserved When Bit 5 is set, you can achieve higher frequency of the MDC clock than the frequency limit of 2.5 MHz (specified in the IEEE Std 802.3) and program a clock divider of lower value. For example, when CSR clock is of 100 MHz frequency and you program these bits as 1010, then the resultant MDC clock is of 12.5 MHz which is outside the limit of IEEE 802.3 specified range. Program the following values only if the interfacing chips support faster MDC clocks. ■ 1000: CSR clock/4 ■ 1001: CSR clock/6 ■ 1010: CSR clock/8 ■ 1011: CSR clock/10 ■ 1100: CSR clock/12 ■ 1101: CSR clock/14 ■ 1110: CSR clock/16 ■ 1111: CSR clock/18 These bits are not used for accessing RevMII. These bits are read-only if the RevMII interface is selected as single PHY interface.	0000	R_W
1	GW	GMII Write When set, this bit indicates to the PHY or RevMII that this is a Write operation using the GMII Data register. If this bit is not set, it indicates that this is a Read operation, that is, placing the data in the GMII Data register.	0	R_W
0	GB	GMII Busy This bit should read logic 0 before writing to Register 4 and Register 5. During a PHY or RevMII register access, the software sets this bit to 1'b1 to indicate that a Read or Write access is in progress. Register 5 is invalid until this bit is cleared by the MAC. Therefore, Register 5 (GMII Data) should be kept valid until the MAC clears this bit during a PHY Write operation. Similarly for a read operation, the contents of Register 5 are not valid until this bit is cleared. The subsequent read or write operation should happen only after the previous operation is complete. Because there is no acknowledgment from the PHY to MAC after a read or write operation is completed, there is no change in the functionality of this bit even when the PHY is not present.	0	R_WS_SC

109.1.5.2.6. Register 5 (GMII Data Register)

The GMII Data register stores Write data to be written to the PHY register located at the address specified in Register 4 (GMII Address Register). This register also stores the Read data from the PHY register located at the address specified by Register 4.

Table 9.1-37 Register 5 (GMII Data Register)

Field	Field Name	Description	Reset	Access
31:16	-	Reserved	0000H	RO
15:0	GD	GMII Data This field contains the 16-bit data value read from the PHY or RevMII after a Management Read operation or the 16-bit data value to be written to the PHY or RevMII before a Management Write operation.	0000H	R_W

109.1.5.2.7. Register 6 (Flow Control Register)

The Flow Control register controls the generation and reception of the Control (Pause Command) frames by the MAC's Flow control module. A Write to a register with the Busy bit set to '1' triggers the Flow Control block to generate a Pause frame. The fields of the control frame are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control frame. The Busy bit remains set until the control frame is transferred onto the cable. The Host must make sure that the Busy bit is cleared before writing to the register.

Table 9.1-38 Register 6 (Flow Control Register)

Field	Field Name	Description	Reset	Access
31:16	PT	Pause Time This field holds the value to be used in the Pause Time field in the transmit control frame. If the Pause Time bits is configured to be double-synchronized to the (G)MII clock domain, then consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain.	0000H	R_W
15:8	-	Reserved	00H	RO
7	DZPQ	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the Zero-Quanta Pause frames on the de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i/mti_flowctrl_i). When this bit is reset, normal operation with automatic Zero-Quanta Pause frame generation is enabled.	0	R_W
6	-	Reserved	0	RO

5:4	PLT	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause frame. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot-times), and PLT = 01, then a second Pause frame is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256 – 28) slot times after the first Pause frame is transmitted. The following list provides the threshold values for different values: ■ 00: The threshold is Pause time minus 4 slot times (PT – 4 slot times). ■ 01: The threshold is Pause time minus 28 slot times (PT – 28 slot times). ■ 10: The threshold is Pause time minus 144 slot times (PT – 144 slot times). ■ 11: The threshold is Pause time minus 256 slot times (PT – 256 slot times). The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface.	00	R_W
3	UP	Unicast Pause Frame Detect A pause frame is processed when it has the unique multicast address specified in the IEEE Std 802.3. When this bit is set, the MAC can also detect Pause frames with unicast address of the station. This unicast address should be as specified in the MAC Address0 High Register and MAC Address0 Low Register. When this bit is reset, the MAC only detects Pause frames with unique multicast address. Note: The MAC does not process a Pause frame if the multicast address of received frame is different from the unique multicast address.	0	R_W
2	RFE	Receive Flow Control Enable When this bit is set, the MAC decodes the received Pause frame and disables its transmitter for a specified (Pause) time. When this bit is reset, the decode function of the Pause frame is disabled.	0	R_W
1	TFE	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to transmit Pause frames. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause frames. In the half-duplex mode, when this bit is set, the MAC enables the backpressure operation. When this bit is reset, the backpressure feature is disabled.	0	R_W
0	FCB_ BPA	Flow Control Busy or Backpressure Activate This bit initiates a Pause frame in the full-duplex mode and activates the backpressure function in the half-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to the Flow Control register. To initiate a Pause frame, the Application must set this bit to 1'b1. During a transfer of the Control Frame, this bit continues to be set to signify that a frame transmission is in progress. After the completion of Pause frame transmission, the MAC resets this bit to 1'b0. The Flow Control register should not be written to until this bit is cleared. In the half-duplex mode, when this bit is set (and TFE is set), then backpressure is asserted by the MAC. During backpressure, when the MAC receives a new frame, the transmitter starts sending a JAM pattern resulting in a collision. This control register bit is logically ORed with the mti_flowctrl_i input signal for the backpressure function. When the MAC is configured for the full-duplex mode, the BPA is automatically disabled.	0	R_WS_SC for FCB R_W for BPA

109.1.5.2.8. Register 7 (VLAN Tag Register)

The VLAN Tag register contains the IEEE 802.1Q VLAN Tag to identify the VLAN frames. The MAC compares the 13th and 14th bytes of the receiving frame (Length/Type) with 16’h8100, and the following two bytes are compared with the VLAN tag. If a match occurs, the MAC sets the received VLAN bit in the receive frame status. The legal length of the frame is increased from 1,518 bytes to 1,522 bytes.

If the VLAN Tag register is configured to be double-synchronized to the (G)MII clock domain, then consecutive writes to these register should be performed only after at least four clock cycles in the destination clock domain.

Table 9.1-39 Register 7 (VLAN Tag Register)

Field	Field Name	Description	Reset	Access
31:20	-	Reserved	000H	RO
19	VTHM	VLAN Tag Hash Table Match Enable When set, the most significant four bits of the VLAN tag’s CRC are used to index the content of Register 354 (VLAN Hash Table Register). A value of 1 in the VLAN Hash Table register, corresponding to the index, indicates that the frame matched the VLAN hash table. When Bit 16 (ETV) is set, the CRC of the 12-bit VLAN Identifier (VID) is used for comparison whereas when ETV is reset, the CRC of the 16-bit VLAN tag is used for comparison. When reset, the VLAN Hash Match operation is not performed. If the VLAN Hash feature is not enabled during core configuration, this bit is reserved (RO with default value).	0	R_W
18	ESVL	Enable S-VLAN When this bit is set, the MAC transmitter and receiver also consider the S-VLAN (Type = 0x88A8) frames as valid VLAN tagged frames.	0	R_W
17	VTIM	VLAN Tag Inverse Match Enable When set, this bit enables the VLAN Tag inverse matching. The frames that do not have matching VLAN Tag are marked as matched. When reset, this bit enables the VLAN Tag perfect matching. The frames with matched VLAN Tag are marked as matched.	0	R_W
16	ETV	Enable 12-Bit VLAN Tag Comparison When this bit is set, a 12-bit VLAN identifier is used for comparing and filtering instead of the complete 16-bit VLAN tag. Bits [11:0] of VLAN tag are compared with the corresponding field in the received VLAN-tagged frame. Similarly, when enabled, only 12 bits of the VLAN tag in the received frame are used for hash-based VLAN filtering. When this bit is reset, all 16 bits of the 15th and 16th bytes of the received VLAN frame are used for comparison and VLAN hash filtering.	0	R_W

15:0	VL	VLAN Tag Identifier for Receive Frames This field contains the 802.1Q VLAN tag to identify the VLAN frames and is compared to the 15th and 16th bytes of the frames being received for VLAN frames. The following list describes the bits of this field: ■ Bits [15:13]: User Priority ■ Bit 12: Canonical Format Indicator (CFI) or Drop Eligible Indicator (DEI) ■ Bits[11:0]: VLAN tag's VLAN Identifier (VID) field When the ETV bit is set, only the VID (Bits[11:0]) is used for comparison. If VL (VL[11:0] if ETV is set) is all zeros, the MAC does not check the fifteenth and 16th bytes for VLAN tag comparison, and declares all frames with a Type field value of 0x8100 or 0x88a8 as VLAN frames.	0000H	R_W

109.1.5.2.9. Register 8 (Version Register)

The Version registers identifies the version of the ETH. This register contains two bytes: one that Synopsys uses to identify the core release number, and the other that you set during core configuration.

Table 9.1-40 Register 8 (Version Register)

Field	Field Name	Description	Reset	Access
31:16	-	Reserved	0000H	RO
15:8	USERVER	User-defined Version (configured with coreConsultant)	xxH	RO
7:0	SNPSVER	Synopsys-defined Version (3.7)	37H	RO

109.1.5.2.10. Register 9 (Debug Register)

The Debug register gives the status of all main modules of the transmit and receive data-paths and the FIFOs. An all-zero status indicates that the MAC is in idle state (and FIFOs are empty) and no activity is going on in the data-paths.

Table 9.1-41 Register 9 (Debug Register)

Field		Description	Reset	Access
31:26	-	Reserved	00H	RO
25	TXSTSFSTS	MTL TxStatus FIFO Full Status When high, this bit indicates that the MTL TxStatus FIFO is full. Therefore, the MTL cannot accept any more frames for transmission. This bit is reserved in the ETH-AHB and ETH-DMA configurations.	0	RO
24	TXFSTS	MTL Tx FIFO Not Empty Status When high, this bit indicates that the MTL Tx FIFO is not empty and some data is left for transmission.	0	RO
23	-	Reserved	0	RO
22	TWCSTS	MTL Tx FIFO Write Controller Status When high, this bit indicates that the MTL Tx FIFO Write Controller is active and is transferring data to the Tx FIFO.	0	RO

21:20	TRCSTS	MTL Tx FIFO Read Controller Status This field indicates the state of the Tx FIFO Read Controller: ■ 00: IDLE state ■ 01: READ state (transferring data to the MAC transmitter) ■ 10: Waiting for TxStatus from the MAC transmitter ■ 11: Writing the received TxStatus or flushing the Tx FIFO	00	RO
19	TXPAUSED	MAC Transmitter in Pause When high, this bit indicates that the MAC transmitter is in the Pause condition (in the full-duplex-only mode) and hence does not schedule any frame for transmission.	0	RO
18:17	TFCSTS	MAC Transmit Frame Controller Status This field indicates the state of the MAC Transmit Frame Controller module: ■ 00: IDLE state ■ 01: Waiting for status of previous frame or IFG or backoff period to be over ■ 10: Generating and transmitting a Pause frame (in the full-duplex mode) ■ 11: Transferring input frame for transmission	00	RO
16	TPESTS	MAC GMII or MII Transmit Protocol Engine Status When high, this bit indicates that the MAC GMII or MII transmit protocol engine is actively transmitting data and is not in the IDLE state.	0	RO
15:10	-	Reserved	0H	RO
9:8	RXFSTS	MTL RxFIFO Fill-Level Status This field gives the status of the fill-level of the Rx FIFO: ■ 00: Rx FIFO Empty ■ 01: Rx FIFO fill-level below flow-control deactivate threshold ■ 10: Rx FIFO fill-level above flow-control activate threshold ■ 11: Rx FIFO Full	00	RO
7	-	Reserved	0	RO
6:5	RRCSTS	MTL RxFIFO Read Controller State This field gives the state of the Rx FIFO read Controller: ■ 00: IDLE state ■ 01: Reading frame data ■ 10: Reading frame status (or timestamp) ■ 11: Flushing the frame data and status	00	RO
4	RWCSTS	MTL Rx FIFO Write Controller Active Status When high, this bit indicates that the MTL Rx FIFO Write Controller is active and is transferring a received frame to the FIFO.	0	RO
3	-	Reserved	0	RO
2:1	RFCFCSTS	MAC Receive Frame FIFO Controller Status When high, this field indicates the active state of the small FIFO Read and Write controllers of the MAC Receive Frame Controller Module. ■ RFCFCSTS[1] represents the status of small FIFO Read controller. ■ RFCFCSTS[0] represents the status of small FIFO Write controller.	00	RO
0	RPESTS	MAC GMII or MII Receive Protocol Engine Status When high, this bit indicates that the MAC GMII or MII receive protocol engine is actively receiving data and not in IDLE state.	0	RO

109.1.5.2.11. Register 12 (LPI Control and Status Register)

The LPI Control and Status Register controls the LPI functions and provides the LPI interrupt status. The status bits are cleared when this register is read. This register is present only when you select the Energy Efficient Ethernet feature during core configuration.

Table 9.1-42 Register 12 (LPI Control and Status Register)

Field	Field Name	Description	Reset	Access
31:20	-	Reserved	00000H	RO
19	LPITXA	LPI TX Automate This bit controls the behavior of the MAC when it is entering or coming out of the LPI mode on the transmit side. This bit is not functional in the ETH- CORE configuration in which the Tx clock gating is done during the LPI mode. If the LPITXA and LPIEN bits are set to 1, the MAC enters the LPI mode only after all outstanding frames (in the core) and pending frames (in the application interface) have been transmitted. The MAC comes out of the LPI mode when the application sends any frame for transmission or the application issues a TX FIFO Flush command. In addition, the MAC automatically clears the LPIEN bit when it exits the LPI state. If TX FIFO Flush is set in Bit 20 of Register 6 (Operation Mode Register), when the MAC is in the LPI mode, the MAC exits the LPI mode. When this bit is 0, the LPIEN bit directly controls behavior of the MAC when it is entering or coming out of the LPI mode.	0	R_W
18	PLSEN	PHY Link Status Enable This bit enables the link status received on the RGMII, SGMII, or SMII receive paths to be used for activating the LPI LS TIMER. When set, the MAC uses the link-status bits of Register 54 (SGMII/RGMII/SMII Control and Status Register) and Bit 17 (PLS) for the LPI LS Timer trigger. When cleared, the MAC ignores the link-status bits of Register 54 and takes only the PLS bit. This bit is RO and reserved if you have not selected the RGMII, SGMII, or SMII PHY interface.	0	R_W
17	PLS	PHY Link Status This bit indicates the link status of the PHY. The MAC Transmitter asserts the LPI pattern only when the link status is up (okay) at least for the time indicated by the LPI LS TIMER. When set, the link is considered to be okay (up) and when reset, the link is considered to be down.	0	R_W
16	LPIEN	LPI Enable When set, this bit instructs the MAC Transmitter to enter the LPI state. When reset this bit instructs the MAC to exit the LPI state and resume normal transmission. This bit is cleared when the LPITXA bit is set and the MAC exits the LPI state because of the arrival of a new packet for transmission.	0	R_W_SC
15:10	-	Reserved	00H	RO

9	RLPIST	Receive LPI State When set, this bit indicates that the MAC is receiving the LPI pattern on the GMII or MII interface.	0	RO
8	TLPIST	Transmit LPI State When set, this bit indicates that the MAC is transmitting the LPI pattern on the GMII or MII interface.	0	RO
7:4	-	Reserved	0H	RO
3	RLPIEX	Receive LPI Exit When set, this bit indicates that the MAC Receiver has stopped receiving the LPI pattern on the GMII or MII interface, exited the LPI state, and resumed the normal reception. This bit is cleared by a read into this register. **Note:** This bit may not get set if the MAC stops receiving the LPI pattern for a very short duration, such as, less than 3 clock cycles of CSR clock.	0	R_SS_RC
2	RLPIEN	Receive LPI Entry When set, this bit indicates that the MAC Receiver has received an LPI pattern and entered the LPI state. This bit is cleared by a read into this register. **Note:** This bit may not get set if the MAC stops receiving the LPI pattern for a very short duration, such as, less than 3 clock cycles of CSR clock.	0	R_SS_RC
1	TLPIEX	Transmit LPI Exit When set, this bit indicates that the MAC transmitter has exited the LPI state after the user has cleared the LPIEN bit and the LPI TW Timer has expired. This bit is cleared by a read into this register.	0	R_SS_RC
0	TLPIEN	Transmit LPI Entry When set, this bit indicates that the MAC Transmitter has entered the LPI state because of the setting of the LPIEN bit. This bit is cleared by a read into this register.	0	R_SS_RC

109.1.5.2.12. Register 13 (LPI Timers Control Register)

The LPI Timers Control register controls the timeout values in the LPI states. It specifies the time for which the MAC transmits the LPI pattern and also the time for which the MAC waits before resuming the normal transmission. This register is present only when you select the Energy Efficient Ethernet feature during core configuration.

Table 9.1-43 Register 13 (LPI Timers Control Register)

Field	Field Name	Description	Reset	Access
31:26	-	Reserved	00H	RO
25:16	LST	LPI LS TIMER This field specifies the minimum time (in milliseconds) for which the link status from the PHY should be up (OKAY) before the LPI pattern can be transmitted to the PHY. The MAC does not transmit the LPI pattern even when the LPIEN bit is set unless the LPI LS Timer reaches the programmed terminal count. The default value of the LPI LS Timer is 1000 (1 sec) as defined in the IEEE standard.	0x3E8	R_W

15:0	TWT	LPI TW TIMER This field specifies the minimum time (in microseconds) for which the MAC waits after it stops transmitting the LPI pattern to the PHY and before it resumes the normal transmission. The TLPIEX status bit is set after the expiry of this timer.	0	R_W

109.1.5.2.13. Register 14 (Interrupt Status Register)

The Interrupt Status register identifies the events in the MAC that can generate interrupt. All interrupt events are generated only when the corresponding optional feature is selected during core configuration and enabled during operation. Therefore, these bits are reserved when the corresponding features are not present in the core.

Table 9.1-44 Register 14 (Interrupt Status Register)

Field	Field Name	Description	Reset	Access
31:12	-	Reserved	000000H	RO
11	GPIIS	GPI Interrupt Status When the GPIO feature is enabled, this bit is set when any active event (LL or LH) occurs on the GPIS field (Bits [3:0]) of Register 56 (General Purpose IO Register) and the corresponding GPIE bit is enabled. This bit is cleared on reading lane 0 (GPIS) of Register 56 (General Purpose IO Register). When the GPIO feature is not enabled, this bit is reserved.	0	RO
10	LPIIS	LPI Interrupt Status When the Energy Efficient Ethernet feature is enabled, this bit is set for any LPI state entry or exit in the MAC Transmitter or Receiver. This bit is cleared on reading Bit 0 of Register 12 (LPI Control and Status Register). In all other modes, this bit is reserved.	0	RO
9	TSIS	Timestamp Interrupt Status When the Advanced Timestamp feature is enabled, this bit is set when any of the following conditions is true: ■ The system time value equals or exceeds the value specified in the Target Time High and Low registers. ■ There is an overflow in the seconds register. ■ The Auxiliary snapshot trigger is asserted. This bit is cleared on reading Bit 0 of Register 458 (Timestamp Status Register). If default Timestamping is enabled, when set, this bit indicates that the system time value is equal to or exceeds the value specified in the Target Time registers. In this mode, this bit is cleared after the completion of the read of this bit. In all other modes, this bit is reserved.	0	RO/ R_SS_RC
8	-	Reserved	000H	RO
7	MMCRXIPIS	MMC Receive Checksum Offload Interrupt Status This bit is set high when an interrupt is generated in the MMC Receive Checksum Offload Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is valid only when you select the optional MMC module and Checksum Offload Engine (Type 2) during core configuration.	0	RO

6	MMCTXIS	MMC Transmit Interrupt Status This bit is set high when an interrupt is generated in the MMC Transmit Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is valid only when you select the optional MMC module during core configuration.	0	RO
5	MMCRXIS	MMC Receive Interrupt Status This bit is set high when an interrupt is generated in the MMC Receive Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is valid only when you select the optional MMC module during core configuration.	0	RO
4	MMCIS	MMC Interrupt Status This bit is set high when any of the Bits [7:5] is set high and cleared only when all of these bits are low. This bit is valid only when you select the optional MMC module during core configuration.	0	RO
3	PMTIS	PMT Interrupt Status This bit is set when a magic packet or remote wake-up frame is received in the power-down mode (see Bits 5 and 6 in the PMT Control and Status Register). This bit is cleared when both Bits[6:5] are cleared because of a read operation to the PMT Control and Status register. This bit is valid only when you select the optional PMT module during core configuration.	0	RO
2	PCSANCIS	PCS Auto-Negotiation Complete This bit is set when the Auto-negotiation is completed in the TBI, RTBI, or SGMII PHY interface (Bit 5 in Register 49 (AN Status Register)). This bit is cleared when you perform a read operation to the AN Status register. This bit is valid only when you select the optional TBI, RTBI, or SGMII PHY interface during core configuration and operation.	0	RO
1	PCSLCHGIS	PCS Link Status Changed This bit is set because of any change in Link Status in the TBI, RTBI, or SGMII PHY interface (Bit 2 in Register 49 (AN Status Register)). This bit is cleared when you perform a read operation on the AN Status register. This bit is valid only when you select the optional TBI, RTBI, or SGMII PHY interface during core configuration and operation.	0	RO
0	RGSMIIIS	RGMII or SMII Interrupt Status This bit is set because of any change in value of the Link Status of RGMII or SMII interface (Bit 3 in Register 54 (SGMII/RGMII/SMII Control and Status Register)). This bit is cleared when you perform a read operation on the SGMII/RGMII/SMII Control and Status Register. This bit is valid only when you select the optional RGMII or SMII PHY interface during core configuration and operation.	0	RO

109.1.5.2.14. Register 15 (Interrupt Mask Register)

The Interrupt Mask Register bits enable you to mask the interrupt signal because of the corresponding event in the Interrupt Status Register. The interrupt signal is sbd_intr_o in the

ETH-AHB, ETH-AXI, and ETH-DMA configuration and mci_intr_o in the ETH-MTL and ETH-CORE configuration.

Table 9.1-45 Register 15 (Interrupt Mask Register)

Field	Field Name	Description	Reset	Access
31:11	-	Reserved	00000H	RO
10	LPIIM	LPI Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of the LPI Interrupt Status bit in Register 14 (Interrupt Status Register). This bit is valid only when you select the Energy Efficient Ethernet feature during core configuration. In all other modes, this bit is reserved.	0	R_W
9	TSIM	Timestamp Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of Timestamp Interrupt Status bit in Register 14 (Interrupt Status Register). This bit is valid only when IEEE1588 timestamping is enabled. In all other modes, this bit is reserved.	0	R_W
8:4	-	Reserved	00H	RO
3	PMTIM	PMT Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of PMT Interrupt Status bit in Register 14 (Interrupt Status Register).	0	R_W
2	PCSANCIM	PCS AN Completion Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of PCS Auto-negotiation complete bit in Register 14 (Interrupt Status Register).	0	R_W
1	PCSLCHGIM	PCS Link Status Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of the PCS Link-status changed bit in Register 14 (Interrupt Status Register).	0	R_W
0	RGSMIIIM	RGMII or SMII Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of the RGMII or SMII Interrupt Status bit in Register 14 (Interrupt Status Register).	0	R_W

109.1.5.2.15. Register 16 (MAC Address0 High Register)

The MAC Address0 High register holds the upper 16 bits of the first 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress0 Register [47:0] is compared with 0x665544332211.

If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address0 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.

Table 9.1-46 Register 16 (MAC Address0 High Register)

Field	Field Name	Description	Reset	Access
31	AE	Address Enable This bit is always set to 1.	1	RO
30:16	-	Reserved	0000H	RO
15:0	ADDRHI	MAC Address0 [47:32] This field contains the upper 16 bits (47:32) of the first 6-byte MAC address. The MAC uses this field for filtering the received frames and inserting the MAC address in the Transmit Flow Control (Pause) Frames.	FFFFH	R_W

109.1.5.2.16. Register 17 (MAC Address0 Low Register)

The MAC Address0 Low register holds the lower 32 bits of the 6-byte first MAC address of the station.

Table 9.1-47 Register 17 (MAC Address0 Low Register)

Field	Field Name	Description	Reset	Access
31:0	ADDRLO	MAC Address0 [31:0] This field contains the lower 32 bits of the first 6-byte MAC address. This is used by the MAC for filtering the received frames and inserting the MAC address in the Transmit Flow Control (Pause) Frames.	FFFF_FFFFH	R_W

109.1.5.2.17. Register 18 (MAC Address1 High Register)

The MAC Address1 High register holds the upper 16 bits of the second 6-byte MAC address of the station.

If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address1 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.

Table 9.1-48 Register 18 (MAC Address1 High Register)

Field	Field Name	Description	Reset	Access
31	AE	Address Enable When this bit is set, the address filter module uses the second MAC address for perfect filtering. When this bit is reset, the address filter module ignores the address for filtering.	0	R_W

30	SA	Source Address When this bit is set, the MAC Address1[47:0] is used to compare with the SA fields of the received frame. When this bit is reset, the MAC Address1[47:0] is used to compare with the DA fields of the received frame.	0	R_W
29:24	MBC	Mask Byte Control These bits are mask control bits for comparison of each of the MAC Address bytes. When set high, the MAC does not compare the corresponding byte of received DA or SA with the contents of MAC Address1 registers. Each bit controls the masking of the bytes as follows: ■ Bit 29: Register 18[15:8] ■ Bit 28: Register 18[7:0] ■ Bit 27: Register 19[31:24] ■ ... ■ Bit 24: Register 19[7:0] You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address.	000000	R_W
23:16	-	Reserved	00H	RO
15:0	ADDRHI	MAC Address1 [47:32] This field contains the upper 16 bits (47:32) of the second 6-byte MAC address.	FFFFH	R_W

109.1.5.2.18. Register 19 (MAC Address1 Low Register)

The MAC Address1 Low register holds the lower 32 bits of the second 6-byte MAC address of the station.

Table 9.1-49 Register 19 (MAC Address1 Low Register)

Field	Field Name	Description	Reset	Access
31:0	ADDRLO	MAC Address1 [31:0] This field contains the lower 32 bits of the second 6-byte MAC address. The content of this field is undefined until loaded by the Application after the initialization process.	FFFF_FFFFH	R_W

109.1.5.2.19. Register 544 (MAC Address32 High Register)

The MAC Address32 High register holds the upper 16 bits of the 33rd 6-byte MAC address of the station.

You can configure the MAC address registers to be double-synchronized by selecting the Synchronize CSR MAC Address to Tx/Rx Clock Domain option in coreConsultant. If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address Low Register (Register 545) are written. For proper synchronization updates, you should perform the consecutive writes to the MAC Address Low Register (Register 545) after at least four clock cycles of the destination clock.

Table 9.1-50　Register 544 (MAC Address32 High Register)

Field	Field Name	Description	Reset	Access
31	AE	Address Enable When this bit is set, the Address filter module uses the 33rd MAC address for perfect filtering. When reset, the address filter module ignores the address for filtering.	0	R_W
30:16	-	Reserved	0000H	RO
15:0	ADDRHI	MAC Address32 [47:32] This field contains the upper 16 bits (47:32) of the 33rd 6-byte MAC address.	FFFFH	R_W

109.1.5.2.20.　Register 48 (AN Control Register)

The AN Control register enables and/or restarts auto-negotiation. It also enables PCS loopback. This register is optional and is present only when the MAC is configured for the TBI, SGMII, or RTBI PHY interface.

Table 9.1-51　Register 48 (AN Control Register)

Field	Field Name	Description	Reset	Access
31:19	-	Reserved	0000H	RO
18	SGMRAL	SGMII RAL Control When set, this bit forces the SGMII RAL block to operate in the speed configured in the Speed and Port Select bits of the MAC Configuration register. This is useful when the SGMII interface is used in a direct MAC to MAC connection (without a PHY) and any MAC must reconfigure the speed. When reset, the SGMII RAL block operates according to the link speed status received on SGMII (from the PHY). This bit is reserved (and RO) if the SGMII PHY interface is not selected during core configuration.	0	R_W
17	LR	Lock to Reference When set, this bit enables the PHY to lock its PLL to the 125 MHz reference clock. This bit controls the pcs_lck_ref_o signal on the TBI, RTBI, or SGMII interface.	0	R_W
16	ECD	Enable Comma Detect When set, this bit enables the PHY for comma detection and word resynchronization. This bit controls the pcs_en_cdet_o signal on the TBI, RTBI, or SGMII interface.	0	R_W
15	-	Reserved	0	RO
14	ELE	External Loopback Enable When set, this bit causes the PHY to loopback the transmit data into the receive path. The pcs_ewrap_o signal is asserted high when this bit is set.	0	R_W
13	-	Reserved	0	RO

12	ANE	Auto-Negotiation Enable When set, this bit enables the MAC to perform auto-negotiation with the link partner. Clearing this bit disables the auto-negotiation.	0	R_W
11:10	-	Reserved	00	RO
9	RAN	Restart Auto-Negotiation When set, this bit causes auto-negotiation to restart if Bit 12 (ANE) is set. This bi is self-clearing after auto-negotiation starts. This bit should be cleared for normal operation.	0	R_WS_SC
8:0	-	Reserved	00H	RO

109.1.5.2.21. Register 49 (AN Status Register)

The AN Status register indicates the link and the auto-negotiation status. This register is optional and is present only when the MAC is configured for the TBI, RTBI, or SGMII PHY interface.

Table 9.1-52 Register 49 (AN Status Register)

Field	Field Name	Description	Reset	Access
31:9	-	Reserved	00_0000H	RO
8	ES	Extended Status This bit is tied to high if the TBI or RTBI interface is selected during core configuration indicating that the MAC supports extended status information in Register 53 (TBI Extended Status Register). This bit is tied to low if the SGMII interface is selected and the TBI or RTBI interface is not selected during core configuration indicating that Register 53 is not present.	1: TBI or RTBI interface 0: SGMII interface withou TBI or RTBI interface	RO
7:6	-	Reserved	00	RO
5	ANC	Auto-Negotiation Complete When set, this bit indicates that the auto-negotiation process is complete. This bit is cleared when auto-negotiation is reinitiated.	0	RO
4	-	Reserved	0	RO
3	ANA	Auto-Negotiation Ability This bit is always high because the MAC supports auto-negotiation.	1	RO
2	LS	Link Status This bit indicates whether the data channel (link) is up or down. For the TBI, RTBI or SGMII interfaces, if ANEG is going on, data cannot be transferred across the link and hence the link is given as down.	0	R_SS_SC _LLO
1:0	-	Reserved	00	RO

109.1.5.2.22. Register 50 (Auto-Negotiation Advertisement Register)

The Auto-Negotiation Advertisement register indicates the link and the auto-negotiation status.

This register is optional and is present only when the MAC is configured for the TBI or RTBI PHY interface.

Table 9.1-53 Register 50 (Auto-Negotiation Advertisement Register)

Field	Field Name	Description	Reset	Access
31:16	-	Reserved	0000H	RO
15	NP	Next Page Support This bit is always low because the MAC does not support the next page.	0	RO
14	-	Reserved	0	RO
13:12	RFE	Remote Fault Encoding These bits provide a remote fault encoding, indicating to a link partner that a fault or error condition has occurred. The encoding of these bits is defined in IEEE 802.3z, *Section 37.2.1.5*.	00	R_W
11:9	-	Reserved	000	RO
8:7	PSE	Pause Encoding These bits provide an encoding for the Pause bits, indicating that the MAC is capable of configuring the Pause function as defined in IEEE 802.3x. The encoding of these bits is defined in IEEE 802.3z, *Section 37.2.1.4*.	11	R_W
6	HD	Half-Duplex When set high, this bit indicates that the MAC supports the half-duplex mode. This bit is always low (and RO) when the MAC is configured for the full-duplex-only mode.	1	R_W
5	FD	Full-Duplex When set high, this bit indicates that the MAC supports the full-duplex mode.	1	R_W
4:0	-	Reserved	00000	RO

109.1.5.2.23. Register 51 (Auto-Negotiation Link Partner Ability Register)

The Auto-Negotiation Link Partner Ability register contains the advertised ability of the link partner. This register is optional and present only when the MAC is configured for the TBI or RTBI PHY interface.

Table 9.1-54 Register 51 (Auto-Negotiation Link Partner Ability Register)

Field	Field Name	Description	Reset	Access
31:16	-	Reserved	0000H	RO
15	NO	Next Page Support When set, this bit indicates that more next page information is available. When cleared, this bit indicates that next page exchange is not desired.	0	RO
14	ACK	Acknowledge When set, the auto-negotiation function uses this bit to indicate that the link partner has successfully received the base page of the MAC. When cleared, it indicates that the link partner did not successfully receive the base page of the MAC.	0	RO

13:12	RFE	Remote Fault Encoding These bits provide a remote fault encoding, indicating a fault or error condition of the link partner. The encoding of these bits is defined in IEEE 802.3z, *Section 37.2.1.5*.	00	RO
11:9	-	Reserved	000	RO
8:7	PSE	Pause Encoding These bits provide an encoding for the Pause bits, indicating that the link partner's capability of configuring the Pause function as defined in the IEEE 802.3x specification The encoding of these bits is defined in IEEE 802.3z, *Section 37.2.1.4*.	00	RO
6	HD	Half-Duplex When set, this bit indicates that the link partner has the ability to operate in the half-duplex mode. When cleared, this bit indicates that the link partner does not have the ability to operate in the half-duplex mode.	0	RO
5	FD	Full-Duplex When set, this bit indicates that the link partner has the ability to operate in the full-duplex mode. When cleared, this bit indicates that the link partner does not have the ability to operate in the full-duplex mode.	0	RO
4:0	-	Reserved	00000	RO

109.1.5.2.24. Register 52 (Auto-Negotiation Expansion Register)

The Auto-Negotiation Expansion register indicates if the MAC received a new base page from the link partner. This register is optional and is present only when the MAC is configured for the TBI or RTBI PHY interface.

Table 9.1-55 Register 52 (Auto-Negotiation Expansion Register)

Field	Field Name	Description	Reset	Access
31:3	-	Reserved	0000_0000H	RO
2	NPA	Next Page Ability This bit is always low because the MAC does not support the next page function.	0	RO
1	NPR	New Page Received When set, this bit indicates that the MAC has received a new page. This bit is cleared when read.	0	RO
0	-	Reserved	0	RO

109.1.5.2.25. Register 53 (TBI Extended Status Register)

The TBI Extended Status register indicates all modes of operation of the MAC. This register is optional and is present only when the MAC is configured for the TBI or RTBI PHY interface.

Table 9.1-56 Register 53 (TBI Extended Status Register)

Field	Field Name	Description	Reset	Access

31:16	-	Reserved	0000H	RO
15	GFD	1000BASE-X Full-Duplex Capable This bit indicates that the MAC is able to perform the full-duplex and 1000BASE-X operations.	1	RO
14	GHD	1000BASE-X Half-Duplex Capable This bit indicates that the MAC is able to perform the half-duplex and 1000BASE-X operations. This bit is always low when the MAC is configured for the full-duplex-only operation during core configuration.	1	RO
13:0	-	Reserved	0000H	RO

109.1.5.2.26. Register 54 (SGMII/RGMII/SMII Control and Status Register)

The SGMII/RGMII/SMII Control and Status register indicates the status signals received by the SGMII, RGMII, or SMII interface (selected at reset) from the PHY. This register is optional and is present only when the MAC is configured for the SGMII, RGMII, or SMII PHY interface.

Table 9.1-57 Register 54 (SGMII/RGMII/SMII Control and Status Register)

Field	Field Name	Description	Reset	Access
31:17	-	Reserved	000H	RO
16	SMIDRXS	Delay SMII RX Data Sampling with respect to the SMII SYNC Signal When set, the first bit of the SMII RX data is sampled one cycle after the SMII SYNC signal. When reset, the first bit of the SMII RX data is sampled along with the SMII SYNC signal. If the SMII PHY Interface with source synchronous mode is selected during core configuration, this bit is reserved (RO with default value).	0	R_W
15:6	-	Reserved	000H	RO
5	FALSCARDET	False Carrier Detected This bit indicates whether the SMII PHY detected false carrier (1'b1). This bit is reserved when the MAC is configured for the SGMII or RGMII PHY interface.	0	RO
4	JABTO	Jabber Timeout This bit indicates whether there is jabber timeout error (1'b1) in the received frame. This bit is reserved when the MAC is configured for the SGMII or RGMII PHY interface.	0	RO
3	LNKSTS	Link Status This bit indicates whether the link between the local PHY and the remote PHY is up or down. It gives the status of the link between the SGMII of MAC and the SGMII of the local PHY. The status bits are received from the local PHY during ANEG betweent he MAC and PHY on the SGMII link.	0	RO

2:1	LNKSPEED	Link Speed This bit indicates the current speed of the link: ■ 00: 2.5 MHz ■ 01: 25 MHz ■ 10: 125 MHz Bit 2 is reserved when the MAC is configured for the SMII PHY interface.	10 for SGMII 00 for RGMII and SMII	RO
0	LNKMOD	Link Mode This bit indicates the current mode of operation of the link: ■ 1'b0: Half-duplex mode ■ 1'b1: Full-duplex mode	0	RO

109.1.5.2.27. Register 55 (Watchdog Timeout Register)

This register controls the watchdog timeout for received frames.

Table 9.1-58 Register 55 (Watchdog Timeout Register)

Field	Field Name	Description	Reset	Access
31:17		Reserved	0000H	RO
16	PWE	Programmable Watchdog Enable When this bit is set and Bit 23 (WD) of Register 0 (MAC Configuration Register) is reset, the WTO field (Bits[13:0]) is used as watchdog timeout for a received frame. When this bit is cleared, the watchdog timeout for a received frame is controlled by the setting of Bit 23 (WD) and Bit 20 (JE) in Register 0 (MAC Configuration Register).	0	R_W
15:14		Reserved	00	RO
13:0	WTO	Watchdog Timeout When Bit 16 (PWE) is set and Bit 23 (WD) of Register 0 (MAC Configuration Register) is reset, this field is used as watchdog timeout for a received frame. If the length of a received frame exceeds the value of this field, such frame is terminated and declared as an error frame. **Note**: When Bit 16 (PWE) is set, the value in this field should be more than 1,522 (0x05F2). Otherwise, the IEEE Std 802.3-specified valid tagged frames are declared as error frames and are dropped.	0000H	R_W

109.1.5.2.28. Register 56 (General Purpose IO Register)

This register provides the control to drive up to 4 bits of output ports (GPO) and the status of up to 4 input ports (GPIS). It also provides the control to generate interrupts on events occurring on the gpi_i pin.

Table 9.1-59 Register 56 (General Purpose IO Register)

Field	Field Name	Description	Reset	Access
31:28		Reserved	0H	RO

27:24	GPIT	GPI Type When set, this bit indicates that the corresponding GPIS is of latched-low (LL) type. When reset, this bit indicates that the corresponding GPIS is of latched-high (LH) type The number of bits available in this field depend on the GP Input Signal Width option. Other bits are not used (reserved and always reset).	0H	R_W
23:20		Reserved	0H	RO
19:16	GPIE	GPI Interrupt Enable When this bit is set and the programmed event (LL or LH) occurs on the corresponding GPIS bit, Bit 11 (GPIIS) of Register 14 (Interrupt Status Register) is set. Accordingly, the interrupt is generated on the mci_intr_o or sbd_intr_o. The GPIIS bit is cleared when the host reads the Bits[7:0] of this register. When reset, Bit 11 (GPIIS) of Register 14 (Interrupt Status Register) is not set when any event occurs on the corresponding GPIS bits. The number of bits available in this field depend on the GP Input Signal Width option. Other bits are not used (reserved and always reset).	0H	R_W
15:12		Reserved	0H	RO
11:8	GPO	General Purpose Output When this bit is set, it directly drives the gpo_o output ports. When this bit is reset, it does not directly drive the gpo_o output ports. The number of bits available in this field depend on the GP Output Signal Width option. Other bits are not used (reserved and always reset).	0H	R_W
7:4		Reserved	0H	RO
3:0	GPIS	General Purpose Input Status This field gives the status of the signals connected to the gpi_i input ports. This field is of the following types based on the setting of the corresponding GPIT field of this register: ■ Latched-low (LL): This field is cleared when the corresponding gpi_i input becomes low. This field remains low until the host reads this field. After this, this field reflects the current value of the gpi_i input. ■ Latched-high (LH): This field is set when the corresponding gpi_i input becomes high. This field remains high until the host reads this field. After this, this field reflects the current value of the gpi_i input. The number of bits available in this field depend on the GP Input Signal Width option. Other bits are not used (reserved and always reset).	0H	LL, LH

109.1.5.2.29. Register 256 (Layer 3 and Layer 4 Control Register 0)

This register controls the operations of the filter 0 of Layer 3 and Layer 4. This register is reserved if the
Layer 3 and Layer 4 Filtering feature is not selected during core configuration.

Table 9.1-60 Register 256 (Layer 3 and Layer 4 Control Register 0)

Field	Field Name	Description	Reset	Access
31:22	-	Reserved	000H	RO

21	L4DPIM0	Layer 4 Destination Port Inverse Match Enable When set, this bit indicates that the Layer 4 Destination Port number field is enabled for inverse matching. When reset, this bit indicates that the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when Bit 20 (L4DPM0) is set high.	0	R_W
20	L4DPM0	Layer 4 Destination Port Match Enable When set, this bit indicates that the Layer 4 Destination Port number field is enabled for matching. When reset, the MAC ignores the Layer 4 Destination Port number field for matching.	0	R_W
19	L4SPIM0	Layer 4 Source Port Inverse Match Enable When set, this bit indicates that the Layer 4 Source Port number field is enabled fo inverse matching. When reset, this bit indicates that the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when Bit 18 (L4SPM0) is set high.	0	R_W
18	L4SPM0	Layer 4 Source Port Match Enable When set, this bit indicates that the Layer 4 Source Port number field is enabled fo matching. When reset, the MAC ignores the Layer 4 Source Port number field for matching.	0	R_W
17	-	Reserved	0	RO
16	L4PEN0	Layer 4 Protocol Enable When set, this bit indicates that the Source and Destination Port number fields for UDP frames are used for matching. When reset, this bit indicates that the Source and Destination Port number fields for TCP frames are used for matching. The Layer 4 matching is done only when either L4SPM0 or L4DPM0 bit is set high.	0	R_W
15:11	L3HDBM0	Layer 3 IP DA Higher Bits Match IPv4 Frames: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 frames. The following list describes the values of this field: ■ 0: No bits are masked. ■ 1: LSb[0] is masked. ■ 2: Two LSbs [1:0] are masked. ■ ... ■ 31: All bits except MSb are masked. IPv6 Frames: Bits [12:11] of this field correspond to Bits [6:5] of L3HSBM0, which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 frames. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: ■ 0: No bits are masked. ■ 1: LSb[0] is masked. ■ 2: Two LSbs [1:0] are masked. ■ … ■ 127: All bits except MSb are masked. This field is valid and applicable only if L3DAM0 or L3SAM0 is set high.	00H	R_W

10:6	L3HSBM0	Layer 3 IP SA Higher Bits Match IPv4 Frames: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 frames. The following list describes the values of this field: ■ 0: No bits are masked. ■ 1: LSb[0] is masked. ■ 2: Two LSbs [1:0] are masked. ■ ... ■ 31: All bits except MSb are masked. IPv6 Frames: This field contains Bits [4:0] of the field that indicates the number of higher bits of IP Source or Destination Address matched in the IPv6 frames. This field is valid and applicable only if L3DAM0 or L3SAM0 is set high.	00H	R_W
5	L3DAIM0	Layer 3 IP DA Inverse Match Enable When set, this bit indicates that the Layer 3 IP Destination Address field is enabled for inverse matching. When reset, this bit indicates that the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when Bit 4 (L3DAM0) is set high.	0	R_W
4	L3DAM0	Layer 3 IP DA Match Enable When set, this bit indicates that Layer 3 IP Destination Address field is enabled for matching. When reset, the MAC ignores the Layer 3 IP Destination Address field for matching. **Note**: When Bit 0 (L3PEN0) is set, you should set either this bit or Bit 2 (L3SAM0) because either IPv6 DA or SA can be checked for filtering.	0	R_W
3	L3SAIM0	Layer 3 IP SA Inverse Match Enable When set, this bit indicates that the Layer 3 IP Source Address field is enabled for inverse matching. When reset, this bit indicates that the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when Bit 2 (L3SAM0) is set high.	0	R_W
2	L3SAM0	Layer 3 IP SA Match Enable When set, this bit indicates that the Layer 3 IP Source Address field is enabled for matching. When reset, the MAC ignores the Layer 3 IP Source Address field for matching. **Note**: When Bit 0 (L3PEN0) is set, you should set either this bit or Bit 4 (L3DAM0) because either IPv6 SA or DA can be checked for filtering.	0	R_W
1	-	Reserved	0	RO
0	L3PEN0	Layer 3 Protocol Enable When set, this bit indicates that the Layer 3 IP Source or Destination Address matching is enabled for the IPv6 frames. When reset, this bit indicates that the Layer 3 IP Source or Destination Address matching is enabled for the IPv4 frames. The Layer 3 matching is done only when either L3SAM0 or L3DAM0 bit is set high.	0	R_W

109.1.5.2.30. Register 257 (Layer 4 Address Register 0)

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option in coreConsultant. If the Layer 3 and Layer 4 Address Registers are configured to be double-synchronized to the Rx clock domains, then the synchronization is triggered only when

Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform the consecutive writes to the same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

If the Layer 3 and Layer 4 Filtering feature is not selected during core configuration, this register and registers 260 through 299 are reserved (RO with default value).

Table 9.1-61 Register 257 (Layer 4 Address Register 0)

Field	Field Name	Description	Reset	Access
31:16	L4DP0	Layer 4 Destination Port Number Field When Bit 16 (L4PEN0) is reset and Bit 20 (L4DPM0) is set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 frames. When Bit 16 (L4PEN0) and Bit 20 (L4DPM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 frames.	0000H	R_W
15:0	L4SP0	Layer 4 Source Port Number Field When Bit 16 (L4PEN0) is reset and Bit 20 (L4DPM0) is set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 frames. When Bit 16 (L4PEN0) and Bit 20 (L4DPM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 frames.	0000H	R_W

109.1.5.2.31. Register 260 (Layer 3 Address 0 Register 0)

For IPv4 frames, the Layer 3 Address 0 Register 0 contains the 32-bit IP Source Address field. For IPv6 frames, it contains Bits [31:0] of the 128-bit IP Source Address or Destination Address field.

Table 9.1-62 Register 260 (Layer 3 Address 0 Register 0)

Field	Field Name	Description	Reset	Access
31:0	L3A00	Layer 3 Address 0 Field When Bit 0 (L3PEN0) and Bit 2 (L3SAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [31:0] of the IP Source Address field in the IPv6 frames. When Bit 0 (L3PEN0) and Bit 4 (L3DAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [31:0] of the IP Destination Address field in the IPv6 frames. When Bit 0 (L3PEN0) is reset and Bit 2 (L3SAM0) is set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with the IP Source Address field in the IPv4 frames.	00000000H	R_W

109.1.5.2.32. Register 261 (Layer 3 Address 1 Register 0)

For IPv4 frames, the Layer 3 Address 1 Register 0 contains the 32-bit IP Destination Address field. For IPv6 frames, it contains Bits [63:32] of the 128-bit IP Source Address or Destination Address field.

Table 9.1-63 Register 261 (Layer 3 Address 1 Register 0)

Field	Field Name	Description	Reset	Access
31:0	L3A10	Layer 3 Address 1 Field When Bit 0 (L3PEN0) and Bit 2 (L3SAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [63:32] of the IP Source Address field in the IPv6 frames. When Bit 0 (L3PEN0) and Bit 4 (L3DAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [63:32] of the IP Destination Address field in the IPv6 frames. When Bit 0 (L3PEN0) is reset and Bit 4 (L3DAM0) is set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with the IP Destination Address field in the IPv4 frames.	00000000H	R_W

109.1.5.2.33. Register 262 (Layer 3 Address 2 Register 0)

For IPv4 frames, the Layer 3 Address 2 Register 0 is reserved. For IPv6 frames, it contains Bits [95:64] of the
128-bit IP Source Address or Destination Address field.

Table 9.1-64 Register 262 (Layer 3 Address 2 Register 0)

Field	Field Name	Description	Reset	Access
31:0	L3A20	Layer 3 Address 2 Field When Bit 0 (L3PEN0) and Bit 2 (L3SAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [95:64] of the IP Source Address field in the IPv6 frames. When Bit 0 (L3PEN0) and Bit 4 (L3DAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains value to be matched with Bits [95:64] of the IP Destination Address field in the IPv6 frames. When Bit 0 (L3PEN0) is reset in Register 256 (Layer 3 and Layer 4 Control Register 0), this register is not used.	00000000H	R_W

109.1.5.2.34. Register 263 (Layer 3 Address 3 Register 0)

For IPv4 frames, the Layer 3 Address 3 Register 0 is reserved. For IPv6 frames, it contains Bits [127:96] of the
128-bit IP Source Address or Destination Address field.

Table 9.1-65 Register 263 (Layer 3 Address 3 Register 0)

Field	Field Name	Description	Reset	Access
31:0	L3A30	Layer 3 Address 3 Field When Bit 0 (L3PEN0) and Bit 2 (L3SAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [127:96] of the IP Source Address field in the IPv6 frames. When Bit 0 (L3PEN0) and Bit 4 (L3DAM0) are set in Register 256 (Layer 3 and Layer 4 Control Register 0), this field contains the value to be matched with Bits [127:96] of the IP Destination Address field in the IPv6 frames. When Bit 0 (L3PEN0) is reset in Register 256 (Layer 3 and Layer 4 Control Register 0), this register is not used.	00000000H	R_W

109.1.5.2.35. Register 320 (Hash Table Register 0)

This register contains the first 32 bits of the hash table when the width of the Hash table is 128 bits or 256 bits. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The 128-bit or 256-bit Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming frame is passed through the CRC logic and the upper seven (eight in
256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register *X*), and the least significant five bits determine the bit within the register. For example, a hash value of 7b'1100000 (in 128-bit Hash) selects Bit 0 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

1. Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).

2. Perform bitwise reversal for the value obtained in Step 1.

3. Take the upper 7 (or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the frame is accepted. Otherwise, it is rejected. If the Bit
1 (Pass All Multicast) is set in Register 1 (MAC Frame Filter), then all multicast frames are accepted
regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register *X* registers are written.

Table 9.1-66 Register 320 (Hash Table Register 0)

Field	Field Name	Description	Reset	Access
31:0	HT31T0	First 32 bits of Hash Table This field contains the first 32 Bits (31:0) of the Hash table.	0000_0000H	R_W

109.1.5.2.36. Register 353 (VLAN Tag Inclusion or Replacement Register)

The VLAN Tag Inclusion or Replacement register contains the VLAN tag for insertion or replacement in the transmit frames. This register is present only when the Enable SA, VLAN, and CRC Insertion on TX option is selected during core configuration.

Table 9.1-67 Register 353 (VLAN Tag Inclusion or Replacement Register)

Field	Field Name	Description	Reset	Access
31:20	-	Reserved	000H	RO

19	CSVL	C-VLAN or S-VLAN When this bit is set, S-VLAN type (0x88A8) is inserted or replaced in the 13th and 14th bytes of transmitted frames. When this bit is reset, C-VLAN type (0x8100) is inserted or replaced in the transmitted frames.	0	R_W
18	VLP	VLAN Priority Control When this bit is set, the control Bits [17:16] are used for VLAN deletion, insertion, or replacement. When this bit is reset, the mti_vlan_ctrl_i control input is used, and Bits [17:16] are ignored.	0	R_W
17:16	VLC	VLAN Tag Control in Transmit Frames ■ 2'b00: No VLAN tag deletion, insertion, or replacement ■ 2'b01: VLAN tag deletion The MAC removes the VLAN type (bytes 13 and 14) and VLAN tag (bytes 15 and 16) of all transmitted frames with VLAN tags. ■ 2'b10: VLAN tag insertion The MAC inserts VLT in bytes 15 and 16 of the frame after inserting the Type value (0x8100/0x88a8) in bytes 13 and 14. This operation is performed on all transmitted frames, irrespective of whether they already have a VLAN tag. ■ 2'b11: VLAN tag replacement The MAC replaces VLT in bytes 15 and 16 of all VLAN-type transmitted frames (Bytes 13 and 14 are 0x8100/0x88a8). **Note**: Changes to this field take effect only on the start of a frame. If you write this register field when a frame is being transmitted, only the subsequent frame can use the updated value, that is, the current frame does not use the updated value.	00	R_W
15:0	VLT	VLAN Tag for Transmit Frames This field contains the value of the VLAN tag to be inserted or replaced. The value must only be changed when the transmit lines are inactive or during the initialization phase. Bits[15:13] are the User Priority, Bit 12 is the CFI/DEI, and Bits[11:0] are the VLAN tag's VID field.	0000H	R_W

109.1.5.2.37. Register 354 (VLAN Hash Table Register)

The 16-bit Hash table is used for group address filtering based on VLAN tag when Bit 19 (VTHM) of Register 7 (VLAN Tag Register) is set. For hash filtering, the content of the 16-bit VLAN tag or 12-bit VLAN ID (based on Bit 16 (ETV) of VLAN Tag Register) in the incoming frame is passed through the CRC logic and the upper four bits of the calculated CRC are used to index the contents of the VLAN Hash table. For example, a hash value of 4b'1000 selects Bit 8 of the VLAN Hash table.

The hash value of the destination address is calculated in the following way:

1. Calculate the 32-bit CRC for the VLAN tag or ID (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).

2. Perform bitwise reversal for the value obtained in Step 1.

3. Take the upper four bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the frame is accepted. Otherwise, it is rejected. If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the

synchronization is triggered only when Bits[15:8] (in little-endian mode) or Bits[7:0] (in big-endian mode) of this register are written
This register is valid and present only when VLAN Hash feature is enabled during core configuration.

Table 9.1-68 Register 354 (VLAN Hash Table Register)

Field	Field Name	Description	Reset	Access
31:16	-	Reserved	0000H	RO
15:0	VLHT	VLAN Hash Table This field contains the 16-bit VLAN Hash Table.	0000H	R_W

109.2. USB

109.2.1. Overview

One USB 2.0 High-Speed OTG Ports with Integrated PHY in the MCU subsystem.

109.2.2. Features

- Support for the following speed modes:
  - High-Speed (HS, 480-Mbps)
  - Full-Speed(FS,12-Mbps)
  - Low-Speed(LS, 1.5-Mbps)
- Support OTG1.3(SRP and HNP) (optional)
- Multiple DMA/non DMA mode access support on the application mode
- Supports up to8 bidirectional endpoints, including control endpoint 0
- Supports control transfer, bulk transfer, isochronous transfer, interrupt transfer
- Support 8 DMA Channels
- Support 1 interrupt, 1 USB DMA interrupt
- Support dynamic FIFO, max 16K Bytes FIFO depth
- Total 16K Bytes RAM inside for EP's FIFO

109.2.3. Function Description

109.2.3.1. Block Diagram

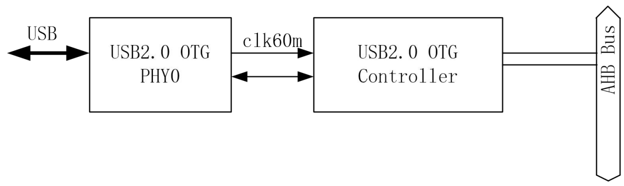


Figure 9.2- 3 USB2.0 High-Speed OTG Overview

USB2.0 High-Speed OTG Device is composed of two part: one USB2.0 High-Speed OTG controller and one USB2.0 OTG PHY. Clk60m is generated from USB PHY. The interface between USB PHY and USB controller is UTMI+ interface (use clk60m, synchronous interface).

109.2.3.2. Block Diagram of USB Controller.

Below picture describes the block diagram of the USB2.0 OTG Controller.

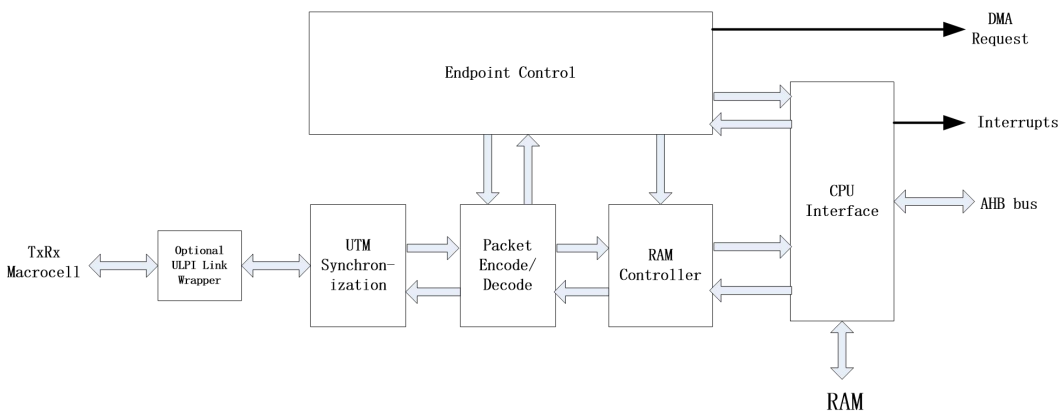


Figure 9.2- 4 Block Diagram of USB 2.0 Controller

109.2.3.3. Modes of Operation

The USB2.0 OTG CONTROLLER has two main modes of operation – Peripheral mode and Host mode.

In Peripheral mode, the USB2.0 OTG CONTROLLER encodes, decodes, checks and directs all USB packets sent and received. IN transactions are handled through the device's TX FIFOs, OUT transactions are handled through its Rx FIFOs. Control, Bulk, Isochronous and Interrupt transactions are supported.

In Host mode, the way in which the USB2.0 OTG CONTROLLER behaves depends on whether it is linked up for point-to-point communications with another USB function or whether it is attached to a hub. When attached to another USB function, the USB2.0 OTG CONTROLLER offers the range of capabilities needed in order to act as the host in point-to-point communications with this USB function. When attached to a hub, it provides the facilities required to act as the host to a number of devices, supported simultaneously.

When operating in Host mode and used for point-to-point communications with a single other USB device (which can be high-, full or low-speed), the USB2.0 OTG CONTROLLER can support Control, Bulk, Isochronous or Interrupt transactions. IN transactions are handled through the Rx FIFOs, OUT transactions are handled through the TX FIFOs. As well as encoding, decoding and checking the USB packets sent and received, the USB2.0 OTG CONTROLLER will also automatically schedule Isochronous endpoints and Interrupt endpoints to perform one transaction every n frames/micro frames (or up to three transactions the high-bandwidth option is selected), where n represents the polling interval that has been programmed for the endpoint. The remaining bus bandwidth is shared equally amongst the Control and Bulk endpoints.

When attached to a hub, the USB2.0 OTG CONTROLLER continues to offer the above facilities but it further needs to be programmed with details of:

- The function address of the target device.
- The operating speed of the target device (so that the appropriate speed conversion can be carried out).
- the target device is a full- or low-speed device that is accessed through a high-speed hub, the endpoint additionally needs to be programmed with the function address and port number of the hub.

The device may be required to power the VBus to 5V as the 'A' device of the connection (source of power and default host) or, as the 'B' device (default peripheral), to be able to wake the 'A' device by charging the VBus to 2V. Outputs from the USB2.0 OTG CONTROLLER indicate when these charging options are required.

Whether the USB2.0 OTG CONTROLLER initially operates in Host mode or in Peripheral mode depends on whether it is being used in an 'A' device or a 'B' device, which in turn depends on whether the IDDIG input is low or high. When the USB2.0 OTG CONTROLLER is operating as an 'A 'device, it is initially configured to operate in Host mode. When operating as a 'B' device, the USB2.0 OTG CONTROLLER is initially configured to operate in Peripheral mode. However, a "Host Req" bit is provided in the DevCtl register through which the CPU can request that the 'B' device becomes the Host the next time there is no activity on the USB bus.

The IDDIG input reflects the state of the ID pin of the device's mini-AB receptacle, with IDDIG being low indicating an 'A' plug i.e. operation as an 'A' device, and IDDIG being high indicating a 'B' plug and operation as a 'B' device. Information on whether the USB2.0 OTG CONTROLLER is acting as an 'A' device or as a 'B' device and on whether the device it is connected to is high-, full- or low-speed is also recorded in the DevCtl register, along with information about the level of the VBus relative to the high and low voltage thresholds used to signal Session Start and Session End.

109.2.4. Programming Guide

109.2.4.1. As Host

- Set the Session bit in the DevCtl register (D0) to start a session.
- Enable DIGID pin sensing automatically. If low level detected, the USB device is as A-Device. The device will enter host mode and turn on VBUS.
- Wait for a peripheral to be connected. When a peripheral is detected, a Connect interrupt (IntrUSB.D4) will be generated (if enabled) and either the FSDev or LSDev bit in the DevCtl register (D6/D5 respectively) will be set depending on whether a high-speed/full-speed peripheral or a low-speed peripheral was detected.
- Enter into HS/FS/LS speed mode, and start transaction by core.
- To end the session, the CPU should clear the Session bit (DevCtl.D0). The MUSBMHDRC will also automatically end the session if babble is detected.

109.2.4.2. As Peripheral

- Wait VBUS connected, then the VBUSVALID is high.
- Set soft_con bit(POWER.D6)/HS_Enab(POWER.D5), and enable USB interrupt.
- Wait reset interrupt generation, and configure EP0's registers.
- Wait Host transaction and communicate with it.

109.2.5. Register Summary

Table 9.2- 69 Register Summary

Groups	Addr offset	Name	Width	R/W	Default value	Description
Common USB Registers	0x0	Faddr	8	RW	0x00	Function address register.
	0x1	Power	8	RW	0x20	Power management register.
	0x2~0x3	IntrTx	16	R	0x0000	Interrupt register for Endpoint 0 plus TX Endpoints 1 to 15.
	0x4~0x5	IntrRx	16	R	0x0000	Interrupt register for Rx Endpoints 1 to 15.
	0x6~0x7	IntrTxE	16	RW	0xffff	Interrupt enable register for IntrTx.

	0x8~0x9	IntrRxE	16	RW	0xffff	Interrupt enable register for IntrRx.
	0xa	IntrUSB	8	R	0x00	Interrupt register for common USB interrupts.
	0xb	IntrUSBE	8	RW	0x06	Interrupt enable register for IntrUSB.
	0xc~0xd	Frame	16	R	0x0000	Frame number.
	0xe	Index	8	RW	0x00	Index register for selecting the endpoint status and control registers.
	0xf	Testmode	8	RW	0x00	Enables the USB 2.0 test modes.
Index Registers <Peripheral Mode>	0x10~0x11	TxMaxP	16	RW	0x0000	Maximum packet size for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x12	CSR0L	8	RW	0x00	Control Status register for 12,13 Endpoint 0. (Index register set to select Endpoint 0)
		TxCSRL	8	RW	0x00	Control Status register for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x13	CSR0H	8	RW	0x00	Control Status register for 12,13 Endpoint 0. (Index register set to select Endpoint 0)
		TxCSRH	8	RW	0x00	Control Status register for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x14~0x15	RxMaxP	16	RW	0x0000	Maximum packet size for peripheral Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x16	RxCSRL	8	RW	0x00	Control Status register for peripheral Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x17	RxCSRH	8	RW	0x00	Control Status register for peripheral Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x18~0x19	Count0	8	R	0x00	Number of received bytes in Endpoint 0 FIFO. (Index register set to select Endpoint 0)
		RxCount	16	R	0x0000	Number of bytes to be read from peripheral Rx endpoint FIFO. (Index register set to select Endpoints 1 – 15)
	0x1a~0x1b	Reserved				
	0x1c~0x1e	Reserved				
	0x1f	ConfigData	8	R	0x00	Returns details of core configuration. (Index register set to

						select Endpoint 0.)
		FIFOSize	8	RW	0x00	Returns the configured size of the selected Rx FIFO and TX FIFOs (Endpoints 1 – 15).
Index Registers <Host Mode>	0x10~0x11	TxMaxP	16	RW	0x0000	Maximum packet size for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x12	CSR0L	8	RW	0x00	Control Status register for 12,13 Endpoint 0. (Index register set to select Endpoint 0)
		TxCSRL	8	RW	0x00	Control Status register for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x13	CSR0H	8	RW	0x00	Control Status register for 12,13 Endpoint 0. (Index register set to select Endpoint 0)
		TxCSRH	8	RW	0x00	Control Status register for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x14~0x15	RxMaxP	16	RW	0x0000	Maximum packet size for host Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x16	RxCSRL	8	RW	0x00	Control Status register for host Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x17	RxCSRH	8	RW	0x00	Control Status register for host Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x18~0x19	Count0	8	RW	0x00	Number of received bytes in Endpoint 0 FIFO. (Index register set to select Endpoint 0)
		RxCount	14	RW	0x0000	Number of bytes to be read from host Rx endpoint FIFO. (Index register set to select Endpoints 1 – 15)
	0x1a	Type0	8	RW	0x00	Defines the speed of Endpoint 0. (Index register set to select Endpoint 0)
		TxType	8	RW	0x00	Sets the transaction protocol, speed and peripheral endpoint number for the host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x1b	NAKLimit	8	RW	0x00	Sets the NAK response timeout on

		0				Endpoint 0. (Index register set to select Endpoint 0)
		TxInterval	8	RW	0x00	Sets the polling interval for Interrupt/ISOC transactions or the NAK response timeout on Bulk transactions for host TX endpoint. (Index register set to select Endpoints 1 – 15)
	0x1c	RxType	8	RW	0x00	Sets the transaction protocol, speed and peripheral endpoint number for the host Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x1d	RxInterval	8	RW	0x00	Sets the polling interval for Interrupt/ISOC transactions or the NAK response timeout on Bulk transactions for host Rx endpoint. (Index register set to select Endpoints 1 – 15)
	0x1e	Unused				
	0x1f	ConfigData	8	RW	0x00	Returns details of core configuration. (Index register set to select Endpoint 0.)
		FIFOSize	8	RW	0x00	Returns the configured size of the selected Rx FIFO and TX FIFOs (Endpoints 1 – 15).
FIFOs	0x20~0x5f	FIFOx	8	RW	0x00	FIFOs for Endpoints 0 – 15.
Additional Control & Configuration Registers	0x60	DevCtl	8	RW	0x00	OTG device control register.
	0x61	MISC	8	RW	0x00	Miscellaneous Register
	0x62	TxFIFOsz	8	RW	0x00	TX Endpoint FIFO size (used Dynamic FIFO sizing option is selected)
	0x63	RxFIFOsz	8	RW	0x00	Rx Endpoint FIFO size (used Dynamic FIFO sizing option is selected)
	0x64~0x65	TxFIFOadd	16	RW	0x0000	TX Endpoint FIFO address (used Dynamic FIFO sizing option is selected)
	0x66~0x67	RxFIFOadd	16	RW	0x0000	Rx Endpoint FIFO address (used Dynamic FIFO sizing option is selected)
	0x68~0x6b	Vcontrol	32	W	0x00000000	UTMI+ PHY Vendor registers
		Vstatus	32	R	0x00000000	UTMI+ PHY Vendor registers
	0x6c~0x6d	HWVers	16	R	0x0000	Hardware Version Number Register

	0x6e~0x77	Reserved				
	0x78	EPInfo	8	R	Implement Dependent	Information about numbers of TX and Rx endpoints.
	0x79	RAMInfo	8	R	Implement Dependent	Information about the width of the RAM and the number of DMA channels.
	0x7a	LinkInfo	8	RW	0x5c	Information about delays to be applied.
	0x7b	VPLen	8	RW	0x3c	Duration of the VBus pulsing charge.
	0x7c	HS_EOF1	8	RW	0x80	Time buffer available on High-Speed transactions.
	0x7d	FS_EOF1	8	RW	0x77	Time buffer available on Full-Speed transactions.
	0x7e	LS_EOF1	8	RW	0x72	Time buffer available on Low-Speed transactions.
	0x7f	SOFT_RST	8	RW	0x00	Soft Reset.
Target Address Registers	0x80+8*n	TxFuncAddr	8	RW	0x00	Transmit Endpoint n Function Address (Host Mode)
	0x81+8*n	Reserved				
	0x82+8*n	TxHubAddr	8	RW	0x00	Transmit Endpoint n Hub Address (Host Mode)
	0x83+8*n	TxHubPort	8	RW	0x00	Transmit Endpoint n Hub Port (Host Mode)
	0x84+8*n	RxFuncAddr	8	RW	0x00	Receive Endpoint n Function Address (Host Mode)
	0x85+8*n	Reserved				
	0x86+8*n	RxHubAddr	8	RW	0x00	Receive Endpoint n Hub Address (Host Mode)
	0x87+8*n	RxHubPort	8	RW	0x00	Receive Endpoint n Hub Port (Host Mode)
	0x100~0x10f	CSR EP0	128	RW		Refer to the definition of Index Registers Group definition
	0x110~0x11f	CSR EP1	128	RW		Refer to the definition of Index Registers Group definition
	0x120~0x12f	CSR EP2	128	RW		Refer to the definition of Index Registers Group definition
	0x130~0x13f	CSR EP3	128	RW		Refer to the definition of Index Registers Group definition
	0x140~0x14f	CSR EP4	128	RW		Refer to the definition of Index Registers Group definition
	0x150~0x15f	CSR EP5	128	RW		Refer to the definition of Index Registers Group definition

	0x160~0x16f	CSR EP6	128	RW		Refer to the definition of Index Registers Group definition
	0x170~0x17f	CSR EP7	128	RW		Refer to the definition of Index Registers Group definition
	0x180~0x18f	CSR EP8	128	RW		Refer to the definition of Index Registers Group definition
	0x190~0x19f	CSR EP9	128	RW		Refer to the definition of Index Registers Group definition
	0x1a0~0x1af	CSR EP10	128	RW		Refer to the definition of Index Registers Group definition
	0x1b0~0x1bf	CSR EP11	128	RW		Refer to the definition of Index Registers Group definition
	0x1c0~0x1cf	CSR EP12	128	RW		Refer to the definition of Index Registers Group definition
	0x1d0~0x1df	CSR EP13	128	RW		Refer to the definition of Index Registers Group definition
	0x1e0~0x1ef	CSR EP14	128	RW		Refer to the definition of Index Registers Group definition
	0x1f0~0x1ff	CSR EP15	128	RW		Refer to the definition of Index Registers Group definition
Optional Extended Registers	0x200	DMA_INTR	8	RW	0x00	DMA Interupt register.
	0x204 +(n-1)*0x10	DMA_CNTL	16	RW	0x0000	DMA Control Register for DMA channel n. (channel 1 thru 8).
	0x208 +(n-1)*0x10	DMA_ADDR	32	RW	0x00000000	DMA Address Register for DMA channel n (channel 1 thru 8).
	0x20c +(n-1)*0x10	DMA_COUNT	32	RW	0x00000000	DMA Count Register for DMA channel n (channel 1 thru 8).
Extended Registers	0x300+4*n	RqPktCount	32	RW	0x00000000	Number of requested packets for Receive Endpoint n (Endpoints 1 – 15)
	0x340~0x341	Rx_DPktBufDis	16	RW	0x0000	Double Packet Buffer Disable register for Rx Endpoints 1 to 15
	0x342~0x343	TX_DPktBufDis	16	RW	0x0000	Double Packet Buffer Disable register for TX Endpoints 1 to 15
	0x344~0x345	C_T_UCH	16	RW		This register sets the Chirp Timeout Timer
	0x346~0x347	C_T_HSRTN	16	RW		This register sets the delay from the end of High Speed resume signaling to enable UTM normal operating mode.
LPM Registers	0x360	LPM_ATTR	16	R	0x0000	LPM Attribute Register

	0x362	LPM_CNTRL	8	RW	0x00	LPM Control Register
	0x363	LPM_INTREN	8	RW	0x00	LPM Interrupt Enable Register
	0x364	LPM_INTR	8	R	0x00	LPM Interrupt Register
	0x365	LPM_FADDR	8	RW	0x00	LPM Function Address Register

Table 9- 70 USB Register Summary

109.2.6. **Register Description**

109.2.6.1. **FADDR**

Table 9.2- 71 Faddr

Bit Field	Name	R/W	Default value	Description
6:0	Func Addr	RW	0x0	The function address.

109.2.6.2. **POWER**

Table 9.2- 72 Power

Bit Field	Name	R/W	Default value	Description
7	ISO Update	RW	0x0	When set by the CPU, the MUSBMHDRC will wait for an SOF token from the time TxPktRdy is set before sending the packet. an IN token is received before an SOF token, then a zero length data packet will be sent. Note: valid in Peripheral Mode. Also, this bit affects endpoints performing Isochronous transfers.
6	Soft Conn	RW	0x0	If Soft Connect/Disconnect feature is enabled, then the USB D+/D- lines are enabled when this bit is set by the CPU and tri-stated when this bit is cleared by the CPU. (See Section 8.2) Note: valid in Peripheral Mode
5	HS Enab	RW	0x1	When set by the CPU, the MUSBMHDRC will negotiate for High-speed mode when the device is reset by the hub. not set, the device will operate in Full-speed mode.
4	HS Mode	RW	0x0	When set, this read- bit indicates High-speed mode successfully negotiated during USB reset. In Peripheral Mode, becomes valid when USB reset completes (as

				indicated by USB reset interrupt). In Host Mode, becomes valid when Reset bit is cleared. Remains valid for the duration of the session. Note: Allowance is made for Tiny-J signaling in determining the transfer speed to select.
3	Reset	RW	0x0	This bit is set when Reset signaling is present on the bus. Note: This bit is Read/Write from the CPU in Host Mode but Read- in Peripheral Mode.
2	Resume	RW	0x0	Set by the CPU to generate Resume signaling when the device is in Suspend mode. In Peripheral mode, the CPU should clear this bit after 10 ms (a maximum of 15 ms), to end Resume signaling. In Host mode, the CPU should clear this bit after 20 ms.
1	Suspend Mode	RW	0x0	In Host mode, this bit is set by the CPU to enter Suspend mode. In Peripheral mode, this bit is set on entry into Suspend mode. It is cleared when the CPU reads the interrupt register, or sets the Resume bit above.
0	Enable SuspendM	RW	0x0	Set by the CPU to enable the SUSPENDM output.

109.2.6.3. **INTRTX**

Table 9.2- 73 IntrTx

Bit Field	Name	R/W	Default value	Description
15	EP15 TX	R	0x0	TX Endpoint 15 interrupt.
14	EP14 TX	R	0x0	TX Endpoint 14 interrupt.
13	EP13 TX	R	0x0	TX Endpoint 13 interrupt.
12	EP12 TX	R	0x0	TX Endpoint 12 interrupt.
11	EP11 TX	R	0x0	TX Endpoint 11 interrupt.
10	EP10 TX	R	0x0	TX Endpoint 10 interrupt.
9	EP9 TX	R	0x0	TX Endpoint 9 interrupt.
8	EP8 TX	R	0x0	TX Endpoint 8 interrupt.
7	EP7 TX	R	0x0	TX Endpoint 7 interrupt.
6	EP6 TX	R	0x0	TX Endpoint 6 interrupt.
5	EP5 TX	R	0x1	TX Endpoint 5 interrupt.
4	EP4 TX	R	0x0	TX Endpoint 4 interrupt.
3	EP3 TX	R	0x0	TX Endpoint 3 interrupt.
2	EP2 TX	R	0x0	TX Endpoint 2 interrupt.
1	EP1 TX	R	0x0	TX Endpoint 1 interrupt.
0	EP0 TX	R	0x0	TX Endpoint 0 interrupt.

109.2.6.4. INTRRX

Table 9.2- 74 IntrRx

Bit Field	Name	R/W	Default value	Description
15	EP15 RX	R	0x0	RX Endpoint 15 interrupt.
14	EP14 RX	R	0x0	RX Endpoint 14 interrupt.
13	EP13 RX	R	0x0	RX Endpoint 13 interrupt.
12	EP12 RX	R	0x0	RX Endpoint 12 interrupt.
11	EP11 RX	R	0x0	RX Endpoint 11 interrupt.
10	EP10 RX	R	0x0	RX Endpoint 10 interrupt.
9	EP9 RX	R	0x0	RX Endpoint 9 interrupt.
8	EP8 RX	R	0x0	RX Endpoint 8 interrupt.
7	EP7 RX	R	0x0	RX Endpoint 7 interrupt.
6	EP6 RX	R	0x0	RX Endpoint 6 interrupt.
5	EP5 RX	R	0x1	RX Endpoint 5 interrupt.
4	EP4 RX	R	0x0	RX Endpoint 4 interrupt.
3	EP3 RX	R	0x0	RX Endpoint 3 interrupt.
2	EP2 RX	R	0x0	RX Endpoint 2 interrupt.
1	EP1 RX	R	0x0	RX Endpoint 1 interrupt.

109.2.6.5. INTRTXE

Table 9.2- 75 IntrTxE

Bit Field	Name	R/W	Default value	Description
15	EP15 TXE	RW	0x1	TX Endpoint 15 interrupt Enable
14	EP14 TXE	RW	0x1	TX Endpoint 14 interrupt Enable
13	EP13 TXE	RW	0x1	TX Endpoint 13 interrupt Enable
12	EP12 TXE	RW	0x1	TX Endpoint 12 interrupt Enable
11	EP11 TXE	RW	0x1	TX Endpoint 11 interrupt Enable
10	EP10 TXE	RW	0x1	TX Endpoint 10 interrupt Enable
9	EP9 TXE	RW	0x1	TX Endpoint 9 interrupt Enable
8	EP8 TXE	RW	0x1	TX Endpoint 8 interrupt Enable
7	EP7 TXE	RW	0x1	TX Endpoint 7 interrupt Enable
6	EP6 TXE	RW	0x1	TX Endpoint 6 interrupt Enable
5	EP5 TXE	RW	0x1	TX Endpoint 5 interrupt Enable
4	EP4 TXE	RW	0x1	TX Endpoint 4 interrupt Enable
3	EP3 TXE	RW	0x1	TX Endpoint 3 interrupt Enable
2	EP2 TXE	RW	0x1	TX Endpoint 2 interrupt Enable
1	EP1 TXE	RW	0x1	TX Endpoint 1 interrupt Enable

0	EP0 TXE	RW	0x1	TX Endpoint 0 interrupt Enable

109.2.6.6. **INTRRXE**

Table 9.2- 76 IntrRxE

Bit Field	Name	R/W	Default value	Description
15	EP15 RXE	RW	0x1	RX Endpoint 15 interrupt Enable
14	EP14 RXE	RW	0x1	RX Endpoint 14 interrupt Enable
13	EP13 RXE	RW	0x1	RX Endpoint 13 interrupt Enable
12	EP12 RXE	RW	0x1	RX Endpoint 12 interrupt Enable
11	EP11 RXE	RW	0x1	RX Endpoint 11 interrupt Enable
10	EP10 RXE	RW	0x1	RX Endpoint 10 interrupt Enable
9	EP9 RXE	RW	0x1	RX Endpoint 9 interrupt Enable
8	EP8 RXE	RW	0x1	RX Endpoint 8 interrupt Enable
7	EP7 RXE	RW	0x1	RX Endpoint 7 interrupt Enable
6	EP6 RXE	RW	0x1	RX Endpoint 6 interrupt Enable
5	EP5 RXE	RW	0x1	RX Endpoint 5 interrupt Enable
4	EP4 RXE	RW	0x1	RX Endpoint 4 interrupt Enable
3	EP3 RXE	RW	0x1	RX Endpoint 3 interrupt Enable
2	EP2 RXE	RW	0x1	RX Endpoint 2 interrupt Enable
1	EP1 RXE	RW	0x1	RX Endpoint 1 interrupt Enable

109.2.6.7. **INTRUSB**

Table 9.2- 77 IntrUSB

Bit Field	Name	R/W	Default value	Description
7	VBus Error	RW	0x0	Set when VBus drops below the VBus Valid threshold during a session. valid when MUSBMHDRC is ‘A’ device
6	Sess Req	RW	0x0	Set when Session Request signaling has been detected. valid when MUSBMHDRC is ‘A’ device.
5	Discon	RW	0x0	Set in Host mode when a device disconnect is detected. Set in Peripheral mode when a session ends. Valid at all transaction speeds.
4	Conn	RW	0x0	Set when a device connection is detected. valid in Host mode. Valid at all transaction speeds.
3	SOF	RW	0x0	Set when a new frame starts
2	Reset	RW	0x0	Set in Peripheral mode when Reset signaling is detected on the bus.
	Babble	RW	0x0	Set in Host mode when babble is detected. Note: active after first SOF has been sent.

1	Resume	RW	0x0	Set when Resume signaling is detected on the bus while the MUSBMHDRC is in Suspend mode
0	Suspend	RW	0x0	Set when Suspend signaling is detected on the bus. valid in Peripheral mode.

109.2.6.8. INTRUSBE

Table 9.2- 78 IntrUSBE

Bit Field	Name	R/W	Default value	Description
7	VBus Error En	RW	0x0	VBus Error Enable
6	Sess Req En	RW	0x0	Sess Req Enable
5	Discon En	RW	0x0	Discon Enable
4	Conn En	RW	0x0	Conn Enable
3	SOF En	RW	0x0	SOF Enable
2	Reset En	RW	0x0	Reset Enable
	Babble En	RW	0x1	Babble Enable
1	Resume En	RW	0x1	Resume Enable
0	Suspend En	RW	0x0	Suspend Enable

109.2.6.9. FRAME

Table 9.2- 79 Frame

Bit Field	Name	R/W	Default value	Description
10:0	Frame Number	R	0x0000	Frame Number

109.2.6.10. INDEX

Table 9.2- 80 Index

Bit Field	Name	R/W	Default value	Description
3:0	Selected Endpoint	RW	0x0000	Selected Endpoint for TX and RX

109.2.6.11. TESTMODE

Table 9.2- 81 Testmode

Bit Field	Name	R/W	Default value	Description
7	Force_Host	RW	0x0	The CPU sets this bit to instruct the core to enter Host mode when the Session bit is set, regardless of whether it is connected to any peripheral. The state of the CID input, HostDisconnect and LineState signals are ignored. The core will then remain in Host mode until the Session bit is cleared, even a device is disconnected, and the Force_Host bit remains set, will re-enter Host mode the next time the Session bit is set. While in this mode, the status of the HOSTDISCON signal from the PHY may be read from bit 7 of the DevCtl register. The operating speed is determined from the Force_HS and Force_FS as follows: {Force_HS, Force_FS} = 2'b00 Low Speed {Force_HS, Force_FS} = 2'b01 Full Speed {Force_HS, Force_FS} = 2'b10 High Speed {Force_HS, Force_FS} = 2'b11 Undefined
6	FIFO_Access	RW	0x0	The CPU sets this bit to transfer the packet in the Endpoint 0 TX FIFO to the Endpoint 0 Rx FIFO. It is cleared automatically.
5	Force_FS	RW	0x0	The CPU sets this bit either in conjunction with bit 7 above or to force the MUSBMHDRC into Full-speed mode when it receives a USB reset.
4	Force_HS	RW	0x0	The CPU sets this bit either in conjunction with bit 7 above or to force the MUSBMHDRC into High-speed mode when it receives a USB reset.
3	Test_Packet	RW	0x0	(High-speed mode) The CPU sets this bit to enter the Test_Packet test mode. In this mode, the MUSBMHDRC repetitively transmits on the bus a 53-byte test packet, the form of which is defined in the Universal Serial Bus Specification Revision 2.0, Section 7.1.20 (and in section 28.4). Note: The test packet has a fixed format and must be loaded into the Endpoint 0 FIFO before the test mode is entered.
2	Test_K	RW	0x0	(High-speed mode) The CPU sets this bit to enter the Test_K test mode. In this mode, the MUSBMHDRC transmits a continuous K on the bus.
1	Test_J	RW	0x0	(High-speed mode) The CPU sets this bit to enter the Test_J test mode. In this mode, the MUSBMHDRC transmits a continuous J on the bus.
0	Test_SE0_NAK	RW	0x0	(High-speed mode) The CPU sets this bit to enter the Test_SE0_NAK test mode. In this mode, the MUSBMHDRC remains in High-speed mode but responds to any valid IN

				token with a NAK.

109.2.6.12. **TXMAXP**

Table 9.2- 82 TxMaxP

Bit Field	Name	R/W	Default value	Description
15/13:11	m-1	RW	0x0	multiplier m which can be 2 or 5
10:0	Max_Payload	RW	0x0	Maximum Payload/transaction

109.2.6.13. **CSR0L**

Table 9.2- 83 CSR0L in peripheral mode

Bit Field	Name	R/W	Default value	Description
7	ServicedSetupEnd	W	0x0	The CPU writes a 1 to this bit to clear the SetupEnd bit. It is cleared automatically.
6	ServicedRxPktRdy	W	0x0	The CPU writes a 1 to this bit to clear the RxPktRdy bit. It is cleared automatically
5	SendStall	W	0x0	The CPU writes a 1 to this bit to terminate the current transaction. The STALL handshake will be transmitted and then this bit will be cleared automatically.
4	SetupEnd	R	0x0	This bit will be set when a control transaction ends before the DataEnd bit has been set. An interrupt will be generated and the FIFO flushed at this time. The bit is cleared by the CPU writing a 1 to the ServicedSetupEnd bit.
3	DataEnd	W	0x0	The CPU sets this bit: 1. When setting TxPktRdy for the last data packet. 2. When clearing RxPktRdy after unloading the last data packet. 3. When setting TxPktRdy for a zero length data packet. It is cleared automatically.
2	SentStall	RC	0x0	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
1	TxPktRdy	RS	0x0	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (enabled).
0	RxPktRdy	R	0x0	This bit is set when a data packet has been received. An

				interrupt is generated when this bit is set. The CPU clears this bit by setting the ServicedRxPktRdy bit.

Table 9.2- 84 CSR0L in Host mode

Bit Field	Name	R/W	Default value	Description
7	NAK Timeout	RC	0x0	This bit will be set when Endpoint 0 is halted following the receipt of NAK responses for longer than the time set by the NAKLimit0 register. The CPU should clear this bit to allow the endpoint to continue
6	StatusPkt	RW	0x0	The CPU sets this bit at the same time as the TxPktRdy or ReqPkt bit is set, to perform a status stage transaction. Setting this bit ensures that the data toggle is set to 1 so that a DATA1 packet is used for the Status Stage transaction.
5	ReqPkt	RW	0x0	The CPU sets this bit to request an IN transaction. It is cleared when RxPktRdy is set.
4	Error	RC	0x0	This bit will be set when three attempts have been made to perform a transaction with no response from the peripheral. The CPU should clear this bit. An interrupt is generated when this bit is set.
3	SetupPkt	RC	0x0	The CPU sets this bit, at the same time as the TxPktRdy bit is set, to send a SETUP token instead of an OUT token for the transaction. Note: Setting this bit also clears the Data Toggle.
2	RxStall	RC	0x0	This bit is set when a STALL handshake is received. The CPU should clear this bit.
1	TxPktRdy	RS	0x0	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (enabled).
0	RxPktRdy	RC	0x0	This bit is set when a data packet has been received. An interrupt is generated (enabled) when this bit is set. The CPU should clear this bit when the packet has been read from the FIFO.

109.2.6.14. **CSR0H**

Table 9.2- 85 CSR0H in peripheral mode

Bit Field	Name	R/W	Default value	Description

0	FlushFIFO	W	0x0	The CPU writes a 1 to this bit to flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset and the TxPktRdy/RxPktRdy bit (below) is cleared. Note: FlushFIFO should be used when TxPktRdy/RxPktRdy is set. At other times, it may cause data to be corrupted.

Table 9.2- 86 CSR0H in Host mode

Bit Field	Name	R/W	Default value	Description
3	Dis Ping	RW	0x0	The CPU writes a 1 to this bit to instruct the core not to issue PING tokens in data and status phases of a high-speed Control transfer (for use with devices that do not respond to PING).
2	Data Toggle Write Enable	W	0x0	The CPU writes a 1 to this bit to enable the current state of the Endpoint 0 data toggle to be written (see Data Toggle bit, below). This bit is automatically cleared once the new value is written.
1	Data Toggle	RW	0x0	When read, this bit indicates the current state of the Endpoint 0 data toggle. D10 is high, this bit may be written with the required setting of the data toggle. D10 is low, any value written to this bit is ignored.
0	FlushFIFO	W	0x0	The CPU writes a 1 to this bit to flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset and the TxPktRdy/RxPktRdy bit (below) is cleared. Note: FlushFIFO should be used when TxPktRdy/RxPktRdy is set. At other times, it may cause data to be corrupted.

109.2.6.15. **TXCSRL**

Table 9.2- 87 TXCSRL in peripheral mode

Bit Field	Name	R/W	Default value	Description
7	IncompTx	RC	0x0	When the endpoint is being used for high-bandwidth Isochronous, this bit is set to indicate where a large packet has been split into 2 or 3 packets for transmission but insufficient IN tokens have been received to send all the parts. Note: In anything other than isochronous transfers, this bit will always return 0.
6	ClrDataTog	W	0x0	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.

5	SentStall	RC	0x0	This bit is set when a STALL handshake is transmitted. The FIFO is flushed and the TxPktRdy bit is cleared (see below). The CPU should clear this bit.
4	SendStall	RW	0x0	The CPU writes a 1 to this bit to issue a STALL handshake to an IN token. The CPU clears this bit to terminate the stall condition. Note: This bit has no effect where the endpoint is being used for Isochronous transfers.
3	FlushFIFO	W	0x0	The CPU writes a 1 to this bit to flush the latest packet from the endpoint TX FIFO. The FIFO pointer is reset, the TxPktRdy bit (below) is cleared and an interrupt is generated. May be set simultaneously with TxPktRdy to abort the packet that is currently being loaded into the FIFO. Note: FlushFIFO should be used when TxPktRdy is set. At other times, it may cause data to be corrupted. Also note that, the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO
2	UnderRun	RC	0x0	The USB sets this bit an IN token is received when TxPktRdy is not set. The CPU should clear this bit.
1	FIFONotEmpty	RC	0x0	The USB sets this bit when there is at least 1 packet in the TX FIFO.
0	TxPktRdy	RS	0x0	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (enabled). TxPktRdy is also automatically cleared prior to loading a second packet into a double-buffered FIFO.

Table 9.2- 88 TXCSRL in Host mode

Bit Field	Name	R/W	Default value	Description
7	NAK Timeout IncompTx	RC	0x0	Bulk endpoints : This bit will be set when the TX endpoint is halted following the receipt of NAK responses for longer than the time set as the NAK Limit by the TxInterval register. The CPU should clear this bit to allow the endpoint to continue. High-bandwidth Interrupt endpoints : This bit will be set no response is received from the device to which the packet is being sent.
6	ClrDataTog	W	0x0	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
5	RxStall	RC	0x0	This bit is set when a STALL handshake is received. When this bit is set, any DMA request that is in progress is stopped, the FIFO is completely flushed and the TxPktRdy bit is cleared (see below). The CPU should clear this bit.

4	SetupPkt	RW	0x0	The CPU sets this bit, at the same time as the TxPktRdy bit is set, to send a SETUP token instead of an OUT token for the transaction. Note: Setting this bit also clears the Data Toggle.
3	FlushFIFO	W	0x0	The CPU writes a 1 to this bit to flush the latest packet from the endpoint TX FIFO. The FIFO pointer is reset, the TxPktRdy bit (below) is cleared and an interrupt is generated. May be set simultaneously with TxPktRdy to abort the packet that is currently being loaded into the FIFO. Note: FlushFIFO should be used when TxPktRdy is set. At other times, it may cause data to be corrupted. Also note that, the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO
2	Error	RC	0x0	The USB sets this bit when 3 attempts have been made to send a packet and no handshake packet has been received. When the bit is set, an interrupt is generated, TxPktRdy is cleared and the FIFO is completely flushed. The CPU should clear this bit. Valid when the endpoint is operating in Bulk or Interrupt mode.
1	FIFONotEmpty	RC	0x0	The USB sets this bit when there is at least 1 packet in the TX FIFO.
0	TxPktRdy	RS	0x0	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (enabled). TxPktRdy is also automatically cleared prior to loading a second packet into a double-buffered FIFO.

109.2.6.16. TXCSRH

Table 9.2- 89 TXCSRH in peripheral mode

Bit Field	Name	R/W	Default value	Description
7	AutoSet	RW	0x0	If the CPU sets this bit, TxPktRdy will be automatically set when data of the maximum packet size (value in TxMaxP) is loaded into the TX FIFO. a packet of less than the maximum packet size is loaded, then TxPktRdy will have to be set manually. Note: Should not be set for either high-bandwidth Isochronous endpoints or high-bandwidth Interrupt endpoints.
6	ISO	RW	0x0	The CPU sets this bit to enable the TX endpoint for Isochronous transfers, and clears it to enable the TX endpoint for Bulk or Interrupt transfers. Note: This bit has any effect in Peripheral mode. In Host mode, it always returns zero.

5	Mode	RW	0x0	The CPU sets this bit to enable the endpoint direction as TX, and clears the bit to enable it as Rx. Note: This bit has any effect where the same endpoint FIFO is used for both TX and Rx transactions.
4	DMAReqEnab	RW	0x0	The CPU sets this bit to enable the DMA request for the TX endpoint.
3	FrcDataTog	R	0x0	The CPU sets this bit to force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received. This can be used by Interrupt TX endpoints that are used to communicate rate feedback for Isochronous endpoints
2	DMAReqMode	R	0x0	The CPU sets this bit to select DMA Request Mode 1 and clears it to select DMA Request Mode 0. Note: This bit must not be cleared either before or in the same cycle as the above DMAReqEnab bit is cleared.

Table 9.2- 90 TXCSRH in Host mode

Bit Field	Name	R/W	Default value	Description
7	AutoSet	RW	0x0	If the CPU sets this bit, TxPktRdy will be automatically set when data of the maximum packet size (value in TxMaxP) is loaded into the TX FIFO. a packet of less than the maximum packet size is loaded, then TxPktRdy will have to be set manually. Note: Should not be set for either high-bandwidth Isochronous endpoints or high-bandwidth Interrupt endpoints.
5	Mode	R	0x0	The CPU sets this bit to enable the endpoint direction as TX, and clears it to enable the endpoint direction as Rx. Note: This bit has any effect where the same endpoint FIFO is used for both TX and Rx transactions.
4	DMAReqEnab	RW	0x0	The CPU sets this bit to enable the DMA request for the TX endpoint.
3	FrcDataTog	RW	0x0	The CPU sets this bit to force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received. This can be used by Interrupt TX endpoints that are used to communicate rate feedback for Isochronous endpoints.
2	DMAReqMode	RW	0x0	The CPU sets this bit to select DMA Request Mode 1 and clears it to select DMA Request Mode 0. Note: This bit must not be cleared either before or in the same cycle as the above DMAReqEnab bit is cleared.
1	Data Toggle Write Enable	W	0x0	The CPU writes a 1 to this bit to enable the current state of the TX Endpoint data toggle to be written (see Data Toggle bit, below). This bit is automatically cleared once the new

				value is written
0	Data Toggle	RW	0x0	When read, this bit indicates the current state of the TX Endpoint data toggle. D1 is high, this bit may be written with the required setting of the data toggle. D1 is low, any value written to this bit is ignored

109.2.6.17. **RXMAXP**

Table 9.2- 91 RxMaxP

Bit Field	Name	R/W	Default value	Description
15/13:11	m-1	RW	0x0	multiplier m which can be 2 or 5
10:0	Max_Payload	RW	0x0	Maximum Payload/transaction

109.2.6.18. **RXCSRL**

Table 9.2- 92 RXCSRL in peripheral mode

Bit Field	Name	R/W	Default value	Description
7	ClrDataTog	W	0x0	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
6	SentStall	RC	0x0	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
5	SendStall	RW	0x0	The CPU writes a 1 to this bit to issue a STALL handshake. The CPU clears this bit to terminate the stall condition. Note: This bit has no effect where the endpoint is being used for Isochronous transfers.
4	FlushFIFO	W	0x0	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint Rx FIFO. The FIFO pointer is reset and the RxPktRdy bit (below) is cleared. Note: FlushFIFO should be used when RxPktRdy is set. At other times, it may cause data to be corrupted. Also note that, the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO.
3	DataError	R	0x0	This bit is set when RxPktRdy is set the data packet has a CRC or bit-stuff error. It is cleared when RxPktRdy is cleared. Note: This bit is valid when the endpoint is operating in ISO mode. In Bulk mode, it always returns zero.
2	OverRun	RC	0x0	This bit is set an OUT packet cannot be loaded into the Rx FIFO. The CPU should clear this bit.

				Note: This bit is valid when the endpoint is operating in ISO mode. In Bulk mode, it always returns zero.
1	FIFOFull	R	0x0	This bit is set when no more packets can be loaded into the Rx FIFO.
0	RxPktRdy	RC	0x0	This bit is set when a data packet has been received. The CPU should clear this bit when the packet has been unloaded from the Rx FIFO. An interrupt is generated when the bit is set.

Table 9.2- 93 RXCSRL in Host mode

Bit Field	Name	R/W	Default value	Description
7	ClrDataTog	W	0x0	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
6	RxStall	RC	0x0	When a STALL handshake is received, this bit is set and an interrupt is generated. The CPU should clear this bit.
5	ReqPkt	RW	0x0	The CPU writes a 1 to this bit to request an IN transaction. It is cleared when RxPktRdy is set.
4	FlushFIFO	W	0x0	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint Rx FIFO. The FIFO pointer is reset and the RxPktRdy bit (below) is cleared. Note: FlushFIFO should be used when RxPktRdy is set. At other times, it may cause data to be corrupted. Also note that, the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO.
3	DataError/NAK Timeout	R	0x0	When operating in ISO mode, this bit is set when RxPktRdy is set the data packet has a CRC or bit-stuff error and cleared when RxPktRdy is cleared. In Bulk mode, this bit will be set when the Rx endpoint is halted following the receipt of NAK responses for longer than the time set as the NAK Limit by the RxInterval register. The CPU should clear this bit to allow the endpoint to continue. However, double packet buffering is enabled this alone will not allow the transfer to continue. In this case, the reqpkt bit should also be set in the same cycle as this bit is cleared.
2	Error	RC	0x0	The USB sets this bit when 3 attempts have been made to receive a packet and no data packet has been received. The CPU should clear this bit. An interrupt is generated when the bit is set. Note: This bit is valid when the Rx endpoint is operating in Bulk or Interrupt mode. In ISO mode, it always returns zero.
1	FIFOFull	R	0x0	This bit is set when no more packets can be loaded into the Rx FIFO.

0	RxPktRdy	RC	0x0	This bit is set when a data packet has been received. The CPU should clear this bit when the packet has been unloaded from the Rx FIFO. An interrupt is generated when the bit is set.

109.2.6.19. **RXCSRH**

Table 9.2- 94 RXCSRH in peripheral mode

Bit Field	Name	R/W	Default value	Description
7	AutoClear	RW	0x0	the CPU sets this bit then the RxPktRdy bit will be automatically cleared when a packet of RxMaxP bytes has been unloaded from the Rx FIFO. When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually. When using a DMA to unload the RxFIFO, data is read from the RxFIFO in 4 byte chunks regardless of the RxMaxP.
6	ISO	RW	0x0	The CPU sets this bit to enable the Rx endpoint for Isochronous transfers, and clears it to enable the Rx endpoint for Bulk/Interrupt transfers
5	DMAReqEnab	RW	0x0	The CPU sets this bit to enable the DMA request for the Rx endpoint.
4	DisNyet/PID Error	RW	0x0	Bulk/Interrupt Transactions: The CPU sets this bit to disable the sending of NYET handshakes. When set, all successfully received Rx packets are ACK'd including at the point at which the FIFO becomes full. Note: This bit has any effect in High-speed mode, in which mode it should be set for all Interrupt endpoints. ISO Transactions: The core sets this bit to indicate a PID error in the received packet
3	DMAReqMode	RW	0x0	The CPU sets this bit to select DMA Request Mode 1 and clears it to select DMA Request Mode 0.
0	IncompRx	RC	0x0	This bit is set in a high-bandwidth Isochronous/Interrupt transfer the packet in the Rx FIFO is incomplete because parts of the data were not received. It is cleared when RxPktRdy is cleared. Note: In anything other than Isochronous transfer, this bit will always return 0

Table 9.2- 95 RXCSRH in Host mode

Bit Field	Name	R/W	Default	Description

			value	
7	AutoClear	RW	0x0	the CPU sets this bit then the RxPktRdy bit will be automatically cleared when a packet of RxMaxP bytes has been unloaded from the Rx FIFO. When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually. When using a DMA to unload the RxFIFO, data is read from the RxFIFO in 4 byte chunks regardless of the RxMaxP.
6	AutoReq	RW	0x0	the CPU sets this bit, the ReqPkt bit will be automatically set when the RxPktRdy bit is cleared. Note: This bit is automatically cleared when a short packet is received.
5	DMAReqEnab	RW	0x0	The CPU sets this bit to enable the DMA request for the Rx endpoint.
4	PID Error	R	0x0	ISO Transactions : The core sets this bit to indicate a PID error in the received packet. Bulk/Interrupt Transactions: The setting of this bit is ignored.
3	DMAReqMode	RW	0x0	The CPU sets this bit to select DMA Request Mode 1 and clears it to select DMA Request Mode 0.
2	Data Toggle/Write Enable	R	0x0	The CPU writes a 1 to this bit to enable the current state of the Endpoint 0 data toggle to be written (see Data Toggle bit, below). This bit is automatically cleared once the new value is written
1	Data Toggle	R	0x0	When read, this bit indicates the current state of the Endpoint 0 data toggle. D10 is high, this bit may be written with the required setting of the data toggle. D10 is low, any value written to this bit is ignored.
0	IncompRx	RC	0x0	This bit will be set in a high-bandwidth Isochronous or Interrupt transfer the packet received is incomplete. It will be cleared when RxPktRdy is cleared. Note: USB protocols are followed correctly, this bit should never be set. The bit becoming set indicates a failure of the associated Peripheral device to behave correctly. (In anything other than Isochronous transfer, this bit will always return 0.)

109.2.6.20. COUNT0

Table 9.2- 96 Count0

Bit Field	Name	R/W	Default value	Description
6:0	Endpoint 0	R	0x0	Endpoint 0 Rx Count

	Rx Count			

109.2.6.21. **RXCOUNT**

Table 9.2- 97 RxCount

Bit Field	Name	R/W	Default value	Description
13:0	Endpoint Rx Count	R	0x0	Endpoint Rx Count

109.2.6.22. **CONFIGDATA**

Table 9.2- 98 ConfigData

Bit Field	Name	R/W	Default value	Description
7	MPRxE	R	0x0	When set to ‘1’, automatic amalgamation of bulk packets is selected
6	MPTxE	R	0x0	When set to ‘1’, automatic splitting of bulk packets is selected
5	BigEndian	R	0x0	Always “0”. Indicates Little Endian ordering.
4	HBRxE	R	0x0	When set to ‘1’ indicates High-bandwidth Rx ISO Endpoint Support selected.
3	HBTxE	R	0x0	When set to ‘1’ indicates High-bandwidth TX ISO Endpoint Support selected
2	DynFIFO Sizing	R	0x0	When set to ‘1’ indicates Dynamic FIFO Sizing option selected.
1	SoftConE	R	0x0	Always ‘1’ . Indicates Soft Connect/Disconnect.
0	UTMI DataWidth	R	0x0	Indicates selected UTMI+ data width. Always 0 indicating 8 .

109.2.6.23. **FIFOSIZE**

Table 9.2- 99 FIFOSize

Bit Field	Name	R/W	Default value	Description
7:4	Rx FIFO Size	R	0x0	Rx FIFO Size
3:0	Tx FIFO Size	R	0x0	Tx FIFO Size

Table 9.2- 100 TYPE0

Bit Field	Name	R/W	Default value	Description
7:6	Speed	RW	0x0	Operating speed of the target device: 00: Unused (Note: selected, the target will be assumed to be using the same connection speed as the core.) 01: High 10: Full 11: Low

109.2.6.24. **TXTYPE**

Table 9.2- 101 TxType

Bit Field	Name	R/W	Default value	Description
7:6	Speed	RW	0x0	Operating speed of the target device when the core is configured with the multi-point option: 00: Unused (Note: selected, the target will be assumed to be using the same connection speed as the core.) 01: High 10: Full 11: Low When the core is not configured with the multipoint option these should not be accessed
5:4	Protocol	RW	0x0	The CPU should set this to select the required protocol for the TX endpoint: 00: Control 01: Isochronous 10: Bulk 11: Interrupt
3:0	Target Endpoint Number	RW	0x0	The CPU should set this value to the endpoint number contained in the TX endpoint descriptor returned to the MUSBMHDRC during device enumeration.

109.2.6.25. **NAKLIMIT0**

Table 9.2- 102 NAKLimit0

Bit Field	Name	R/W	Default value	Description
4:0	Endpoint 0 NAK Limit	RW	0x0	Endpoint 0 NAK Limit (m)

109.2.6.26. **TXINTERVAL**

Table 9.2- 103 TxInterval

Bit Field	Name	R/W	Default value	Description
7:0	TxInterval	RW	0x0	Tx Polling Interval/NAK Limit (m)

109.2.6.27. **RXTYPE**

Table 9.2- 104 RxType

Bit Field	Name	R/W	Default value	Description
7:6	Speed	RW	0x0	Operating speed of the target device when the core is configured with the multipoint option: 00: Unused (Note: selected, the target will be assumed to be using the same connection speed as the core.) 01: High 10: Full 11: Low When the core is not configured with the multipoint option these should not be accessed.
5:4	Protocol	RW	0x0	The CPU should set this to select the required protocol for the Rx endpoint: 00: Control 01: Isochronous 10: Bulk 11: Interrupt
3:0	Target Endpoint Number	RW	0x0	The CPU should set this value to the endpoint number contained in the Rx endpoint descriptor returned to the MUSBMHDRC during device enumeration.

109.2.6.28. **RXINTERVAL**

Table 9.2- 105 RxInterval

Bit Field	Name	R/W	Default value	Description
7:0	RxInterval	RW	0x0	Rx Polling Interval/NAK Limit (m)

109.2.6.29. **FIFOx**

Table 9.2- 106 FIFOx

Bit Field	Name	R/W	Default value	Description
31:0	FIFOx	RW	0x0	FIFOs for Endpoints 0 -15

109.2.6.30. **DEVCTL**

Table 9.2- 107 DevCtl

Bit Field	Name	R/W	Default value	Description
7	B-Device	R	0x1	This Read- bit indicates whether the MUSBMHDRC is operating as the 'A' device or the 'B' device. 0 ⇒ 'A' device; 1 ⇒ 'B' device. valid while a session is in progress. To determine the role when no session is in progress, set the session bit and read this bit. Note: the core is in Force_Host mode (i.e. a session has been started with Testmode.D7 = 1), this bit will indicate the state of the HOSTDISCON input signal from the PHY
6	FSDev	R	0x0	This Read- bit is set when a full-speed or high-speed device has been detected being connected to the port. (High-speed devices are distinguished from full-speed by checking for high-speed chirps when the device is reset.) valid in Host mode.
5	LSDev	R	0x0	This Read- bit is set when a low-speed device has been detected being connected to the port. valid in Host mode.
4:3	VBus[1:0]	R	0x0	These Read- encode the current VBus level as follows: {D4, D3} = 2'b00: Below SessionEnd {D4, D3} = 2'b01: Above SessionEnd, below AValid {D4, D3} = 2'b10: Above AValid, below VBusValid {D4, D3} = 2'b11: Above VBusValid
2	Host Mode	R	0x0	This Read- bit is set when the MUSBMHDRC is acting as a Host.
1	Host Req	RW	0x0	When set, the MUSBMHDRC will initiate the Host Negotiation when Suspend mode is entered. It is cleared when Host Negotiation is completed. See Section 15. ('B' device)
0	Session	RW	0x0	When operating as an 'A' device, this bit is set or cleared by the CPU to start or end a session. When operating as a 'B' device, this bit is set/cleared by the MUSBMHDRC when a session starts/ends.

				It is also set by the CPU to initiate the Session Request Protocol. When the MUSBMHDRC is in Suspend mode, the bit may be cleared by the CPU to perform a software disconnect. Note: Clearing this bit when the core is not suspended will result in undefined behavior.

109.2.6.31. MISC

Table 9.2- 108 MISC

Bit Field	Name	R/W	Default value	Description
1	tx_edma	RW	0x0	1'b0: DMA_REQ signal for all IN Endpoints will be de-asserted when MAXP bytes have been written to an endpoint. This is late mode. 1'b1: DMA_REQ signal for all IN Endpoints will be de-asserted when MAXP-8 bytes have been written to an endpoint. This is early mode
0	rx_edma	RW	0x0	1'b0: DMA_REQ signal for all OUT Endpoints will be de-asserted when MAXP bytes have been read to an endpoint. This is late mode. 1'b1: DMA_REQ signal for all OUT Endpoints will be de-asserted when MAXP-8 bytes have been read to an endpoint. This is early mode.

109.2.6.32. TXFIFOSZ

Table 9.2- 109 TxFIFOsz

Bit Field	Name	R/W	Default value	Description
4	DPB	RW	0x0	Defines whether double-packet buffering supported. When '1', double-packet buffering is supported. When '0', single-packet buffering is supported.
3:0	SZ[3:0]	RW	0x0	Maximum packet size to be allowed for (before any splitting within the FIFO of Bulk/High-Bandwidth packets prior to transmission – see Sections 8.4.1.3, 8.4.1.4 and 8.5.3) DPB = 0, the FIFO will also be this size; DPB = 1, the FIFO will be twice this size.

109.2.6.33. RXFIFOSZ

Table 9.2- 110 RxFIFOsz

Bit Field	Name	R/W	Default value	Description
4	DPB	RW	0x0	Defines whether double-packet buffering supported. When ‘1’, double-packet buffering is supported. When ‘0’, single-packet buffering is supported.
3:0	SZ[3:0]	RW	0x0	Maximum packet size to be allowed for (after any combination within the FIFO of Bulk/High- Bandwidth packets following their reception – see Sections 8.4.2.3, 8.4.2.4 and 8.5.2) DPB = 0, the FIFO will also be this size; DPB = 1, the FIFO will be twice this size.

109.2.6.34. TXFIFOADD

Table 9.2- 111 TxFIFOadd

Bit Field	Name	R/W	Default value	Description
12:0	AD[12:0]	RW	0x0	Start address of the endpoint FIFO in units of 8 bytes as follows:

109.2.6.35. RXFIFOADD

Table 9.2- 112 RxFIFOadd

Bit Field	Name	R/W	Default value	Description
12:0	AD[12:0]	RW	0x0	Start address of the endpoint FIFO in units of 8 bytes as follows:

109.2.6.36. VCONTROL

Table 9.2- 113 VControl

Bit Field	Name	R/W	Default value	Description
31:0	Vcontrol	W	0x0	VControl is a UTMI+ PHY Vendor register that may optionally be included in the core when the core is configured.

109.2.6.37. VSTATUS

Table 9.2- 114 VStatus

Bit Field	Name	R/W	Default	Description

			value	
31:0	Vstatus	R	0x0	VStatus is a UTMI+ PHY Vendor register that may optionally be included in the core when the core is configured.

109.2.6.38. **HWVERS**

Table 9.2- 115 HWVers

Bit Field	Name	R/W	Default value	Description
15	RC	R	Version Dependent	Set to '1' RTL used from a Release Candidate rather than from a full release of the core.
14:10	xx	R	Version Dependent	Major Version Number (Range 0 – 31).
9:0	yyy	R	Version Dependent	Minor Version Number (Range 0 – 999).

109.2.6.39. **EPINFO**

Table 9.2- 116 EPInfo

Bit Field	Name	R/W	Default value	Description
7:4	RxEndPoints	R	Implement Dependent	The number of Rx endpoints implemented in the design.
3:0	TxEndPoints	R	Implement Dependent	The number of TX endpoints implemented in the design.

109.2.6.40. **RAMINFO**

Table 9.2- 117 RAMInfo

Bit Field	Name	R/W	Default value	Description
7:4	DMAChans	R	Implement Dependent	The number of DMA channels implemented in the design.
3:0	Ram	R	Implement Dependent	The width of the RAM address bus.

109.2.6.41. **LINKINFO**

Table 9.2- 118 LinkInfo

Bit Field	Name	R/W	Default	Description

			value	
7:4	WTCON	RW	0x5	Sets the wait to be applied to allow for the user's connect/disconnect filter in units of 533.3ns. (The default setting corresponds to 2.667μs.)
3:0	WTID	RW	0xC	Sets the delay to be applied from IDPULLUP being asserted to IDDIG being considered valid in units of 4.369ms. (The default setting corresponds to 52.43ms.)

109.2.6.42. **VPLEN**

Table 9.2- 119 VPLen

Bit Field	Name	R/W	Default value	Description
7:0	VPLEN	RW	0x3c	Sets the duration of the VBus pulsing charge in units of 546.1 μs. (The default setting corresponds to 32.77ms.)

109.2.6.43. **HS_EOF1**

Table 9.2- 120 HS_EOF1

Bit Field	Name	R/W	Default value	Description
7:0	HS_EOF1	RW	0x80	Sets for High-speed transactions the time before EOF to stop beginning new transactions, in units of 133.3ns. (The default setting corresponds to 17.07μs.)

109.2.6.44. **FS_EOF1**

Table 9.2- 121 FS_EOF1

Bit Field	Name	R/W	Default value	Description
7:0	FS_EOF1	RW	0x77	Sets for Full-speed transactions the time before EOF to stop beginning new transactions, in units of 533.3ns. (The default setting corresponds to 63.46μs.)

109.2.6.45. **LS_EOF1**

Table 9.2- 122 LS_EOF1

Bit Field	Name	R/W	Default value	Description
7:0	LS_EOF1	RW	0x72	Sets for Low-speed transactions the time before EOF to stop

				beginning new transactions, in units of 1.067μs. (The default setting corresponds to 121.6μs.)

109.2.6.46. SOFT_RST

Table 9.2- 123 SOFT_RST

Bit Field	Name	R/W	Default value	Description
1	NRSTX	RW	0x0	The default value of this bit is 1'b0; When a 1 is written to this bit, the output NRSTXO will be asserted (LOW) within a minimum delay of 7 cycles of the CLK input. The output NRSTXO will be asynchronously asserted and synchronously de-asserted with respect to XCLK. This register is self clearing and will be reset by the input NRST.
0	NRST	RW	0x0	The default value of this bit is 1'b0; When a 1 is written to this bit, the output NRSTO will be asserted (LOW) within a minimum delay of 7 cycles of the CLK input. The output NRSTO will be asynchronously asserted and synchronously de-asserted with respect to CLK. This register is self clearing and will be reset by the input NRST.

109.2.6.47. TXFUNCADDR

Table 9.2- 124 TxFuncAddr

Bit Field	Name	R/W	Default value	Description
7:0	TXFUNCADDR	RW	0x0	Address of TX Target Function

109.2.6.48. TXHUBADDR

Table 9.2- 125 TxHubAddr

Bit Field	Name	R/W	Default value	Description
7	Multiple Translators	RW	0x0	The top bit should record whether the hub has multiple transaction translators (set to '0' single transaction translator; set to '1' multiple transaction translators).
6:0	Hub Address	RW	0x0	Record the address of this USB 2.0 hub

109.2.6.49. **TXFUNCPORT**

Table 9.2- 126 TxFuncPort

Bit Field	Name	R/W	Default value	Description
6:0	TXFUNCPort	RW	0x0	Record the port of that USB 2.0 hub through which the target associated with the endpoint is accessed.

109.2.6.50. **RXFUNCADDR**

Table 9.2- 127 RxFuncAddr

Bit Field	Name	R/W	Default value	Description
7:0	RXFUNCADDR	RW	0x0	Address of RX Target Function

109.2.6.51. **RXHUBADDR**

Table 9.2- 128 RxHubAddr

Bit Field	Name	R/W	Default value	Description
7	Multiple Translators	RW	0x0	The top bit should record whether the hub has multiple transaction translators (set to '0' single transaction translator; set to '1' multiple transaction translators).
6:0	Hub Address	RW	0x0	Record the address of this USB 2.0 hub

109.2.6.52. **RXFUNCPORT**

Table 9.2- 129 RxFuncPort

Bit Field	Name	R/W	Default value	Description
6:0	RXFUNCPort	RW	0x0	Record the port of that USB 2.0 hub through which the target associated with the endpoint is accessed.

109.2.6.53. **DMA_INTR**

Table 9.2- 130 DMA_INTR

Bit Field	Name	R/W	Default value	Description

7	CH8_DMA_INTR	RW	0x0	Channel 8 DMA Interrupt
6	CH7_DMA_INTR	RW	0x0	Channel 7 DMA Interrupt
5	CH6_DMA_INTR	RW	0x0	Channel 6 DMA Interrupt
4	CH5_DMA_INTR	RW	0x0	Channel 5 DMA Interrupt
3	CH4_DMA_INTR	RW	0x0	Channel 4 DMA Interrupt
2	CH3_DMA_INTR	RW	0x0	Channel 3 DMA Interrupt
1	CH2_DMA_INTR	RW	0x0	Channel 2 DMA Interrupt
0	CH1_DMA_INTR	RW	0x0	Channel 1 DMA Interrupt

109.2.6.54. **DMA_CNTL**

Table 9.2- 131 DMA_CNTL

Bit Field	Name	R/W	Default value	Description
10:9	DMA_BRSTM	RW	0x0	Burst Mode 00 = Burst Mode 0 : Bursts of unspecified length 01 = Burst Mode 1 : INCR4 or unspecified length 10 = Burst Mode 2 : INCR8, INCR4 or unspecified length 11 = Burst Mode 3 : INCR16, INCR8, INCR4 or unspecified length
8	DMA_ERR	RW	0x0	Bus Error Bit. Indicates that a bus error has been observed on the input AHB_HRESPM[1:0]. This bit is cleared by software.
7:4	DMAEP	RW	0x0	The endpoint number this channel is assigned to.
3	DMAIE	RW	0x0	DMA Interrupt Enable.
2	DMAMODE	RW	0x0	This bit selects the DMA Transfer Mode. 0 = DMA Mode0 Transfer 1 = DMA Mode1 Transfer
1	DMA_DIR	RW	0x0	This bit selects the DMA Transfer Direction. 0 = DMA Write (RX Endpoint) 1 = DMA Read (TX Endpoint)
0	DMA_ENAB	RW	0x0	This bit enables the DMA transfer and will cause the transfer to begin.

109.2.6.55. DMA_ADDR

Table 9.2- 132 DMA_ADDR

Bit Field	Name	R/W	Default value	Description
31:0	DMA_ADDR	RW	0x0	The DMA memory address. Note that the initial memory address written to this register must have a value such that it's modulo 4 value is equal to 0. That is, DMA_ADDR[1:0] must be equal to 2'b00. The lower two of this register are read and cannot be set by software.

109.2.6.56. DMA_COUNT

Table 9.2- 133 DMA_COUNT

Bit Field	Name	R/W	Default value	Description
31:0	DMA_COUNT	RW	0x0	The DMA memory address for the corresponding DMA channel. Note: DMA is enabled with a count of 0, the bus will not be requested and a DMA interrupt will be generated.

109.2.6.57. RQPKTCOUNT

Table 9.2- 134 RqPktCount

Bit Field	Name	R/W	Default value	Description
15:0	RqPktCount	RW	0x0	Sets the number of packets of size MaxP that are to be transferred in a block transfer. used in Host mode when AutoReq is set. Has no effect in Peripheral mode or when AutoReq is not set.

109.2.6.58. RX_DPKTBUFDIS

Table 9.2- 135 Rx DPktBufDis

Bit Field	Name	R/W	Default value	Description
15	EP15 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 15.
14	EP14 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 14.
13	EP13 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 13.
12	EP12 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 12.

11	EP11 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 11.
10	EP10 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 10.
9	EP9 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 9.
8	EP8 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 8.
7	EP7 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 7.
6	EP6 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 6.
5	EP5 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 5.
4	EP4 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 4.
3	EP3 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 3.
2	EP2 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 2.
1	EP1 RxDis	RW	0x1	Rx Double Packet Buffer Disable for Endpoint 1

109.2.6.59. **TX_DPKTBUFDIS**

Table 9.2- 136 Tx DPktBufDis

Bit Field	Name	R/W	Default value	Description
15	EP15 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 15.
14	EP14 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 14.
13	EP13 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 13.
12	EP12 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 12.
11	EP11 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 11.
10	EP10 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 10.
9	EP9 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 9.
8	EP8 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 8.
7	EP7 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 7.
6	EP6 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 6.
5	EP5 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 5.
4	EP4 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 4.
3	EP3 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 3.
2	EP2 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 2.
1	EP1 TxDis	RW	0x1	Tx Double Packet Buffer Disable for Endpoint 1

109.2.6.60. **C_T_UCH**

Table 9.2- 137 C_T_UCH

Bit Field	Name	R/W	Default value	Description
15:0	C_T_UCH	RW	0x0	Configurable Chirp Timeout timer; The default value is determined by compiler directive in musbhsfc_xcfg.v file. The default value is 203Ah the host PHY data width is 16

				(XCLK is 30MHz) and 4074h the PHY data width is 8 (XCLK is 60Mhz) corresponding to a delay of 1.1ms

109.2.6.61. **C_T_HSRTN**

Table 9.2- 138 C_T_HSRTN

Bit Field	Name	R/W	Default value	Description
15:0	C_T_HSRTN	RW	0x0	The delay from the end of High Speed resume signaling to enabling UTM normal operating mode. The default value is determined by compiler directive in musbhsfc_xcfg.v file. The default value is 2F3h the host PHY data width is 16 (XCLK is 30MHz) and 5E6h the PHY data width is 8 (XCLK is 60Mhz) corresponding to a delay of 100us.

109.2.6.62. **LPM_ATTR**

Table 9.2- 139 LPM_ATTR

Bit Field	Name	R/W	Default value	Description
15:12	EndPnt	R	0x0	This is the EndPnt that in the Token Packet of the LPM transaction.
8	RmtWak	R	0x0	This bit is the remote wakeup enable bit. RmtWak = 1'b0 : Remote wakeup is not enabled. RmtWak = 1'b1: Remote wakeup is enabled. This bit is applied on a temporary basis and is applied to the current suspend state. After the current suspend cycle, the remote wakeup capability that was negotiated upon enumeration will apply.
7:4	HIRD	R	0x0	This is the Host Initiated Resume Duration. This value is the minimum time the host will drive resume on the Bus. The value in this register corresponds to an actual resume time of : Resume Time = 50us + HIRD*75us. This results a range 50us to 1200us.
3:0	LinkState	R	0x0	This value is provided by the host to the peripheral to indicate what state the peripheral must transition to after the receipt and acceptance of a LPM transaction. LinkState = 4'h0 - Reserved LinkState = 4'h1 - Sleep State (L1) LinkState = 4'h2 - Reserved LinkState = 4'h3 - Reserved

109.2.6.63. **LPM_CNTRL**

Table 9.2- 140 LPM_CNTRL in Peripheral mode

Bit Field	Name	R/W	Default value	Description
4	LPMNAK	RW	0x0	This bit is used to place all end points in a state such that the response to all transactions other then an LPM transaction will be a NAK. This bit will take effect after the MUSBMHDRC has been LPM suspended. In this case, the MUSBMHDRC will continue to NAK until this bit has been cleared by software.
3:2	LPMEN	RW	0x0	This register is used to enable LPM in the MUSBMHDRC. There are three levels in which LPM can be enabled which will determine the response of the MUSBMHDRC to LPM transactions. The three levels are: • 2'b00, 2'b10 - LPM and Extended transactions are not supported. In this case, the MUSBMHDRC will not respond to LPM transactions and the transaction will timeout. • 2'b01 – LPM is not supported but extended transactions are supported. In this case, the MUSBMHDRC will respond to an LPM transaction with a STALL. • 2'b11 – The MUSBMHDRC supports LPM extended transactions. In this case, the MUSBMHDRC will respond with a NYET or an ACK as determined by the value of LPMXMT and other conditions.
1	LPMRES	RW	0x0	This bit is used by software to initiate resume (remote wake up). This bit differs from the classic RESUME bit in the POWER register (address 0x01.2) in that the RESUME signal timing is controlled by hardware. When software writes this bit, resume signaling will be asserted for 50us. This bit is self clearing.
0	LPMXMT	RW	0x0	This bit is set by software to instruct the MUSBMHDRC to transition to the L1 state upon the receipt of the next LPM transaction. This bit is effective LPMEN is set to 2'b11. This bit can be set in the same cycle as LPMEN. this bit is set to 1'b1 and LPMEN = 2'b11, the MUSBMHDRC can respond in the following ways: • no data is pending (All TX FIFOs are empty), the MUSBMHDRC will respond with an ACK. In this case this bit will self clear and a software interrupt will be generated. • data is pending (Data resides in at least one TX FIFO), the MUSBMHDRC will respond with a NYET. In this case, this bit will NOT self clear however a software interrupt will be

				generated.

Table 9.2- 141 LPM_CNTRL in Host mode

Bit Field	Name	R/W	Default value	Description
1	LPMRES	RW	0x0	This bit is used by software to initiate a RESUME from the L1 State. This bit dfers from the classic RESUME bit in the POWER register (address 0x01.2) in that the RESUME signal timing is controlled by hardware. When software writes this bit, resume signaling will be asserted for a time specified by the HIRD field in the LPM_ATTR register. This bit is self clearing.
0	LPMXMT	RW	0x0	Software should set this bit to transmit an LPM transaction. This bit is self clearing. This bit will be immediately cleared upon receipt of any Token or three timeouts have occurred

109.2.6.64. **LPM_INTREN**

Table 9.2- 142 LPM_INTREN

Bit Field	Name	R/W	Default value	Description
5	LPMERREN	R	0x0	1’b0 : Disable the LPMERR interrupt 1’b1 : Enable the LPMERR interrupt
4	LPMRESEN	R	0x0	1’b0 : Disable the LPMRES interrupt 1’b1 : Enable the LPMRES interrupt
3	LPMNCEN	RW	0x0	1’b0 : Disable the LPMNC interrupt 1’b1 : Enable the LPMNC interrupt
2	LPMACKEN	RW	0x0	1’b0 : Disable the LPMACK interrupt 1’b1 : Enable the LPMACK interrupt
1	LPMNYEN	RW	0x0	1’b0 : Disable the LPMNY interrupt 1’b1 : Enable the LPMNY interrupt
0	LPMSTEN	RW	0x0	1’b0 : Disable the LPMST interrupt 1’b1 : Enable the LPMST interrupt

109.2.6.65. **LPM_INTR**

Table 9.2- 143 LPM_INTR in Peripheral Mode

Bit Field	Name	R/W	Default value	Description
5	LkPMERR	R	0x0	This bit is set an LPM transaction is received that has a LinkState field that is not supported. In this case, the response to the transaction will be a STALL. However, the LPM_ATTR register will be updated so that software can observe the non compliant LPM packet payload.
4	LPMRES	R	0x0	This bit is set the MUSBMHDRC has been resumed for any reason. This bit is mutually exclusive from the RESUME bit in the Power register (address 0x01).
3	LPMNC	R	0x0	This bit is set when an LPM transaction is received and the MUSBMDHRC responds with a NYET due to data pending in the RX FIFOs. This can occur under following condition: • The LPMRESP field in the LMRESP register is set to 2'b11, the LPMXMT field is set to 1'b1 and there was data pending in the MUSBMHDRC TX FIFOs
2	LPMACK	R	0x0	This bit is set when an LPM transaction is received and the MUSBMDHRC responds with an ACK. This can occur under the following condition: • The LPMRESP field in the LMRESP register is set to 2'b11, the LPMXMT field is set to 1'b1 and there was no data pending in the MUSBMHDRC TX FIFOs.
1	LPMNY	R	0x0	This bit is set when an LPM transaction is received and the MUSBMDHRC responds with a NYET. This can occur under the following conditions: • The LPMRESP field in the LMRESP register is set to 2'b11 and the LPMXMT field is set to 1'b0.
0	LPMST	R	0x0	This bit is set when an LPM transaction is received and the MUSBMDHRC responds with a STALL. This can occur under the following condition: • The LPMRESP field in the LMRESP register is set to 2'b01.

Table 9.2- 144 LPM_INTR in Host Mode

Bit Field	Name	R/W	Default value	Description
5	LkPMERR	R	0x0	This bit is set a response to the LPM transaction is received with a Bit Stuff error or a PID error. In this case, no suspend occurs and the state of the device is now unknown.

4	LPMRES	R	0x0	This bit is set when the MUSBMHDRC has been resumed for any reason. This
3	LPMNC	R	0x0	This bit is set when an LPM transaction has been transmitted and has failed to complete. The transaction will have failed because a timeout occurred or there were bit errors in the response for three attempts.
2	LPMACK	R	0x0	This bit is set when an LPM transaction is transmitted and the device responds with an ACK.
1	LPMNY	R	0x0	This bit is set when an LPM transaction is transmitted and the device responds with a NYET.
0	LPMST	R	0x0	This bit is set when an LPM transaction is transmitted and the device responds with a STALL

109.2.6.66. **LPM_FADDR**

Table 9.2- 145 LPM_FADDR

Bit Field	Name	R/W	Default value	Description
6:0	LPM_FADDDR	RW	0x0	The LPM function address

109.3. SDIO Controller

109.3.1. **Overview**

2 MMC/SD/SDIO ports (SDIO0,SDIO1) available. One MMC/SD/SDIO port supports one card (SD card, SDIO card, or MMC Card)

109.3.2. **Features**

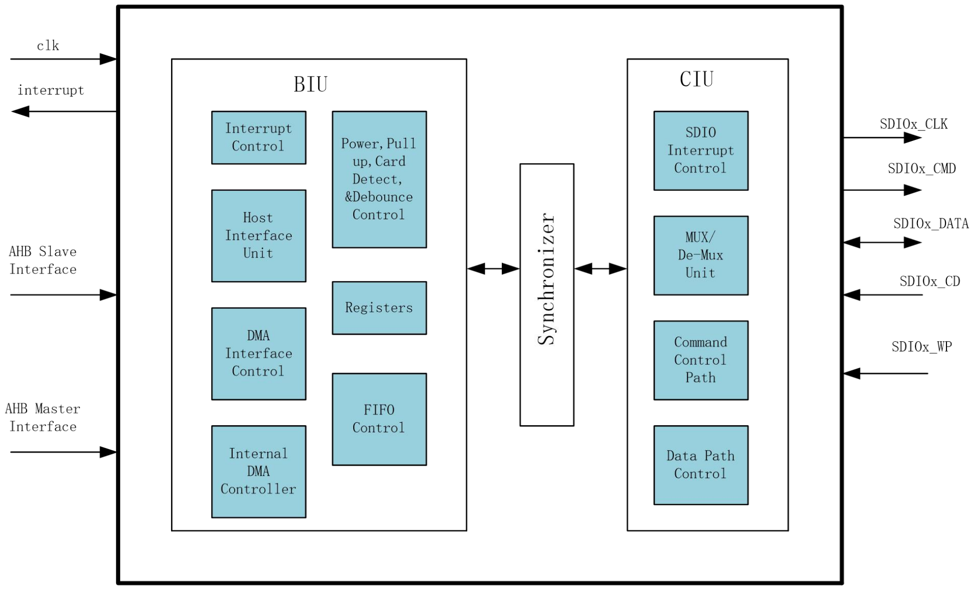
- 1, and 4 bit MMC, SD, SDIO modes.
- Each port (SDIO0 or SDIO1) has its own dedicated power rail for 1.8V or 3.3V operation.
- Up to 48 MHz data transfer rate.
- Supports card detect and write protect.
- Complies with MMC4.3, SD2.0, SDIO 2.0 specications.
- Card Interface Features
- o 1-/4-bit data operational mode
- o Sample Clock Tuning mechanism

- Signaling voltage level support
- Independent Interrupt and Wakeup outputs
- Integrated DMA Controllers
  - Single Operation DMA (SDMA) Controller
  - Advanced DMA Controller (ADMA)
  - Programmable burst length for DMA transfers
  - Programmable precise/imprecise burst support
- Data Buffering
  - 512*32-bit wide FO buffers

109.3.3. Function Description

109.3.3.1. Block diagram

Figure 9.3- 5 SD/SDIO/MMC Controller Block Diagram



109.3.3.2. **Bus Interface Unit (BIU)**

110. The BIU provides the following functions:

- Host interface
- DMA interface
- Interrupt control
- Register access
- FO access
- Power/pullup control and card detection

110.1.1.1. **Internal DMAC Controller (IDMAC)**

111. The Internal Direct Memory Access Controller (IDMAC) has a Control and Status Register (CSR) and a single Transmit/Receive engine, which transfers data from host memory to the device port and vice versa. The controller utilizes a descriptor to efficiently move data from source to destination with minimal Host CPU intervention. The controller can be programmed to interrupt the Host CPU in situations such as data Transmit and Receive transfer completion from the card, as well as other normal or error conditions.

112.

The IDMAC and the Host driver communicate through a single data structure. CSR addresses 0x80 to 0x98 are reserved for host programming.

113.

114. The IDMAC transfers the data received from the card to the Data Buffer in the Host memory, and it transfers Transmit data from the Data Buffer in the Host memory to the FO. Descriptors that reside in the Host memory act as pointers to these buffers.

115.

116. A data buffer resides in physical memory space of the Host and consists of complete data or partial data. Buffers contain data, while buffer status is maintained in the descriptor. Data chaining refers to data that spans multiple data buffers. However, a single descriptor cannot span multiple data.

117.

A single descriptor is used for both reception and transmission. The base address of the list is written into Descriptor List Base Address Register (DBADDR @0x88). A descriptor list is forward linked. The Last Descriptor can point back to the first entry in order to create a ring structure. The descriptor list resides in the physical memory address space of the Host. Each descriptor can point to a maximum of two data buffers.

118.

118.1.1.1. Card Interface Unit (CIU)

119. The Card Interface Unit (CIU) interfaces with the Bus Interface Unit (BIU) and the SD_MMC_CEATA cards or devices. The host writes command parameters to the BIU control registers, and these parameters are then passed to the CIU. Depending on control register values, the CIU generates SD_MMC_CEATA command and data traffic on a selected card bus according to SD_MMC_CEATA protocol. The control register values also decide whether the command and data traffic is destined for the CE-ATA device, and the accordingly controls the command and data path.

120.

121. The following software restrictions should be met for proper CIU operation:

- one card can be selected at a time for command or data transfer. For example, when data is transferred from a card, a new command should not be sent to another card; for the same card, a new command is allowed for read status or to stop the transfer.
- one data transfer command can be issued at a time.
- During an open-ended card write operation, the card clock is stopped because the FO is empty, the software must first fill the data into the FO and start the card clock. It can then issue a stop/abort command to the card.
- During an SDIO/COMBO card transfer, the card function is suspended and the software wants to resume the suspended transfer, it must first reset the FO and start the resume command as it were a new data transfer command.
- When issuing card reset commands (CMD0, CMD15 or CMD52_reset) while a card data transfer is in progress, the software must set the stop_abort_cmd bit in the Command register so that the SD/MMC controller can stop the data transfer after issuing the card reset command.
- When the data end bit error is set in the RINTSTS register, it does not guarantee SDIO interrupts. The software should ignore the SDIO interrupts and issue the stop/abort command to the card, so that the card stops sending the read data.
- the card clock is stopped because the FO is full during a card read, the software should read at least two FO locations to start the card clock.

- one CE-ATA device can be selected at a time for command or data transfer. For example, when data is transferred from a CE-ATA device, a new command should not be sent to another CE_ATA device.

- CE-ATA device interrupts are enabled (nIEN=0), a new RW_BLK command should not be sent to the same device there is a pending RW_BLK command in progress (the RW_BLK command used in this databook is the RW_MULTIPLE_BLOCK MMC command defined by the CE-ATA specication). the CCSD can be sent while waiting for the CCS.

- For the same device, a new command is allowed for reading status information, interrupts are disabled in the CE-ATA device (nIEN=1).

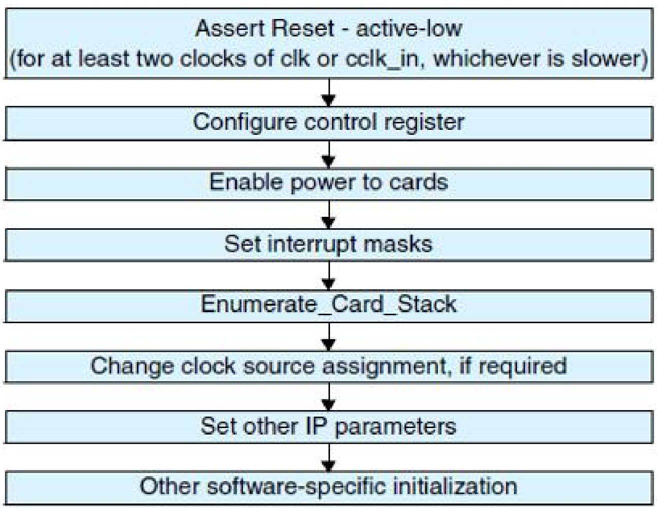
- Open-ended transfers are not supported for the CE-ATA devices.

- The send_auto_stop signal is not supported (software should not set the send_auto_stop bit) for CE-ATA transfers.

121.1.1. Programming Guide

121.1.1.1. Initialization

Figure 9.3-6 illustrates the initialization flow.



Once the power and clocks are stable, reset_n should be asserted (active-low) for at least two clocks of clk or cclk_in, whichever is slower. The reset initializes the registers, ports, FIFO-pointers, DMA interface controls, and state-machines in the design. After power-on reset, the software should do the following:

1. Configure control register – For MMC-Ver3.3-only mode, enable the open-drain pullup by setting
enable_OD_pullup (bit 24) in the control register.

2. Enable power to cards – Before enabling the power, confirm that the voltage setting to the voltage
regulator(s) is correct. Enable power to the connected cards by setting the corresponding bit to 1 in the Power Enable register at @0x04. To enable maximum cards in MMC-Ver3.3-only mode – that is, 30 cards – write 0x03fffffff in the Power Enable register. To enable maximum cards in SD_MMC_CEATA mode – that is, 16 cards – write 0x0000ffff in the Power Enable register. Wait for the power ramp-up time.

3. Set masks for interrupts by clearing appropriate bits in the Interrupt Mask register @0x024. Set the global int_enable bit of the Control register @0x00. It is recommended that you write 0xffff_ffff to the Raw Interrupt register @0x044 in order to clear any pending interrupts before setting the int_enable bit.

4. Enumerate card stack – Each card is enumerated according to card type. For enumeration, you should restrict the clock frequency to 400 KHz in accordance with SD_MMC_CEATA standards.

5. Changing clock source assignment – Required in only SD_MMC_CEATA mode. Set the card frequency using the clock-divider and lock-source registers. For MMC-Ver3.3-only mode, only clock divider0 is used. MMC cards operate at a maximum of 20 MHz (at maximum of 52 MHz in high-speed mode). SD_MMC_CEATA mode operates at a maximum of 25 MHz (at maximum of 50 MHz in high-speed mode).

6. Set other IP parameters, which normally do not need to be changed with every command, with a typical value such as timeout values in cclk_out according to SD_MMC_CEATA specifications.

- ResponseTimeOut = 0x64
- DataTimeOut = highest of one of the following:
  - (10 * ((TAAC*Fop) + (100*NSAC))
  - Host FIFO read/write latency from FIFO empty/full

a. Set the debounce value to 25 ms (default: 0x0fffff) in host clock (clk) cycle units in the DEBNCE
register @0x064.

- FIFO threshold value in bytes in the FIFOTH register @0x04C. Typically, the threshold value can be set to half the FIFO depth; that is:
  - RX_WMark = (FIFO_DEPTH/2) - 1;
  - TX_WMark = FIFO_DEPTH/2

7. According to MMC standards, the open-drain pullup resistor is required during only the enumeration phase. Therefore for MMC-Ver3.3-only mode, disable the open-drain pullup by clearing the enable_OD_pullup (bit 24) in the Control register.

8. If the software decides to handle the interrupts provided by the IP core, you should create another
thread to handle interrupts.

121.1.1.2. Power Control

You can implement power control using the following registers, along with external circuitry:

- Control register bits card_voltage_a (16-19) and card_voltage_b (20-23) – Status of these bits is reflected at the IO pins. The bits can be used to generate or control the supply voltage that

the
memory cards require.

■ Power enable register @0x04 – Status of these bits is reflected at the IO pins. The pins can control power to individual cards.

Programming these two registers depends on the implemented external circuitry. While turning on or off the power enable, you should confirm that power supply settings are correct. Power to all cards usually should be disabled while switching off the power.

121.1.1.3. Clock Programming

The Mobile storage supports four clock sources, each of which can be programmed with a different frequency; software can select the clock source for each card. The clock to an individual card can be enabled or disabled. Registers that support this are:

■ CLKDIV @0x08 – Programs individual clock source frequency.

■ CLKSRC @0x0C – Assigns clock source for each card.

■ CLKENA @0x10 – Enables or disables clock for individual card and enables low-power mode, which automatically stops the clock to a card when the card is idle for more than 8 clocks.

The Mobile storage loads each of these registers only when the start_cmd bit and the Update_clk_regs_only bit in the CMD register are set. When a command is successfully loaded, the Mobile storage clears this bit, unless the Mobile storage already has another command in the queue, at which point it gives an HLE (Hardware Locked Error).

Software should look for the start_cmd and the Update_clk_regs_only bits, and should also set the wait_prvdata_complete bit to ensure that clock parameters do not change during data transfer. Note that even though start_cmd is set for updating clock registers, the Mobile storage does not raise a command_done signal upon command completion.

For MMC-Ver3.3-only mode, only clock divider0 is used. Therefore, the CLKSRC register cannot be used in MMC-Ver3.3-only mode. The following shows how to program these registers:

1. Confirm that no card is engaged in any transaction; if there is a transaction, wait until it finishes.
2. Stop all clocks by writing xxxx0000 to the CLKENA register. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
3. Program the CLKDIV and CLKSRC registers, as required. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
4. Re-enable all clocks by programming the CLKENA register. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.

121.1.1.4. Switching the Card Number

The Mobile storage can be configured to have multiple card slots (up to 16) when the CARD_TYPE parameter is configured to 1. In such a configuration, the Mobile storage Host

Controller provides an option to select any of the card slots for operation. This section discusses procedures to be followed while switching the operation from one card slot to another card slot when ASYNC_SAMPLE_DATA_PATH = 1. The user must confirm that no card is engaged in any transaction, and if there is a transaction, the switching operation must wait until that transaction completes.

Follow this procedure if the change in clock frequency is required:

1. Stop all clocks by writing xxxx0000 to the CLKENA register.
2. Set the start_cmd, update_clk_regs_only, card_number, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
3. Program the CLKDIV and CLKSRC registers, as required.
4. Set the start_cmd, update_clk_regs_only, card_number, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
5. Re-enable the required clocks by programming the CLKENA register.
6. Set the start_cmd, update_clk_regs_only, card_number, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.

Follow this procedure if the change in clock frequency is not required

1. Stop the corresponding clock of the card number (to be switched) by writing the corresponding bit to 0 in the CLKENA register.
2. Set the start_cmd, update_clk_regs_only, card_number, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
3. Re-enable the corresponding card clocks by programming the CLKENA register.
4. Set the start_cmd, update_clk_regs_only, card_number, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.

121.1.1.5. **No-Data Command With or Without Response Sequence**

To send any non-data command, the software needs to program the CMD register @0x2C and the CMDARG register @0x28 with appropriate parameters. Using these two registers, the Mobile storage forms the command and sends it to the command bus. The Mobile storage reflects the errors in the command response through the error bits of the RINTSTS register.

When a response is received – either erroneous or valid – the Mobile storage sets the command_done bit in the RINTSTS register. A short response is copied in Response Register0, while a long response is copied to all four response registers @0x30, 0x34, 0x38, and 0x3C. The Response3 register bit 31 represents the MSB, and the Response0 register bit 0 represents the LSB of a long response.

For basic commands or non-data commands, follow these steps:

1. Program the Command register @0x28 with the appropriate command argument parameter.
2. Program the Command register @0x2C

Table 9.3-146 Command Register Settings for No-Data Command

Parameter	Value	Comment
Default		
Start`_cmd	1	–
use_hold_reg	1/0	Choose value based on speed mode being used
Update_clk_regs_only	0	No clock parameters update command
data_expected	0	No data command
card_number	*n*CardNo	Actual card number
cmd_index	command index	–
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
response_length	0	Can be 1 for R2 (long) response
response_expect	1	Can be 0 for commands with no response; for example CMD0, CMD4, CMD15, and so on
er-selectable		
it_prvdata_complete		fore sending command on command line, host should wai completion of any data command in process, if any commended to always set this bit, unless the curren mmand is to query status or stop data transfer when transfe
eck_response_crc		iost should crosscheck CRC of response received

3. Wait for command acceptance by host. The following happens when the command is loaded into the Mobile storage:

- ❑ Mobile storage accepts the command for execution and clears the start_cmd bit in the CMD register, unless one command is in process, at which point the Mobile storage can load and keep the second command in the buffer.
- ❑ If the Mobile storage is unable to load the command – that is, a command is already in progress, a second command is in the buffer, and a third command is attempted – then it generates an HLE (hardware-locked error).

4. Check if there is an HLE.

5. Wait for command execution to complete. After receiving either a response from a card or response timeout, the Mobile storage sets the command_done bit in the RINTSTS register. Software can either poll for this bit or respond to a generated interrupt.

6. Check if response_timeout error, response_CRC error, or response error is set. This can be done either by responding to an interrupt raised by these errors or by polling bits 1, 6, and 8 from the RINTSTS register @0x44. If no response error is received, then the response is valid. If required, the software can copy the response from the response registers @0x30-0x3C.

Software should not modify clock parameters while a command is being executed.

121.1.1.6. Single-Block or Multiple-Block Read

Steps involved in a single-block or multiple-block read are:

1. Write the data size in bytes in the BYTCNT register @0x20.

2. Write the block size in bytes in the BLKSIZ register @0x1C. The Mobile storage expects data from the card in blocks of size BLKSIZ each.

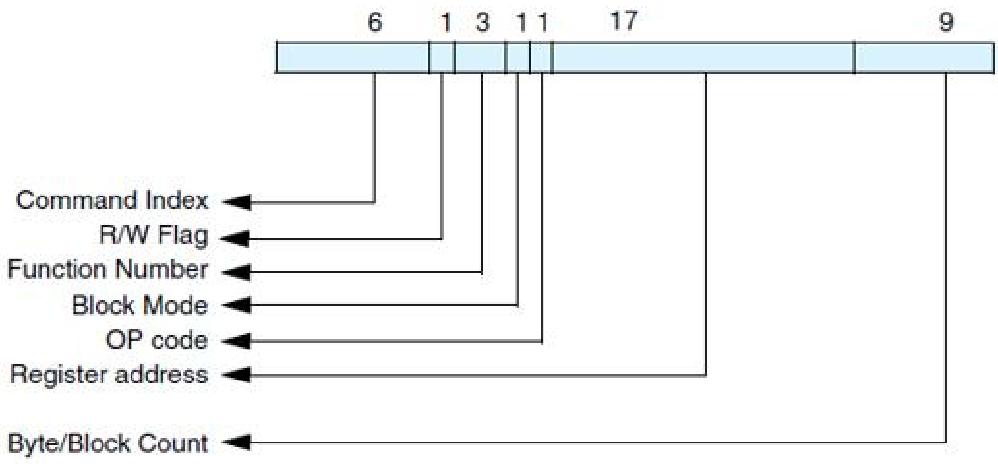
3. For cards that have a round trip delay of more than 0.5 cclk_in period, program the Card Read Threshold to ensure that the card clock does not stop in the middle of a block of data being transferred from the card to the Host.

If the Card Read Threshold feature is not enabled for such cards, then the Host system should ensure

that the Rx FIFO does not become full during a Read Data transfer by ensuring that the Rx FIFO is drained out at a rate faster than that at which data is pushed into the FIFO; alternatively, a sufficiently large FIFO can be provided.

4. Program the CMDARG register @0x28 with the data address of the beginning of a data read.

Figure 9.3-7 Command Argument for CMD53



Program the Command register. For SD and MMC cards, use CMD17 for a single-block read and CMD18 for a multiple-block read. For SDIO cards, use CMD53 for both single-block and multiple-block transfers.

Table 9.3-147 Command Register Settings for Single-Block or Multiple-Block Read

Parameter	**Value**	**Comment**
Default		
start_cmd	1	–
use_hold_reg	1/0	Choose value based on speed mode being used
Update_clk_regs_only	0	No clock parameters update command
card_number	*n*CardNo	Actual card number

send_initialization	*0*	Can be 1, but only for card reset commands, such a
stop_abort_cmd	*0*	Can be 1 for commands to stop data transfer, such a
send_auto_stop	*0 or 1*	
transfer_mode	*0*	Block transfer
read_write	*0*	Read from card
data_expected	*1*	Data command
response_length	*0*	Can be 1 for R2 (long) response
response_expect	*1*	Can be 0 for commands with no response; for example CMD0, CMD4, CMD15, and so on
er-selectable		
d_index	mmand ex	
it_prvdata_complete		sends command immediately sends command after previous data transfer ends
eck_response_crc		– DWC_mobile_storage should not check response C DWC_mobile_storage should check response CRC

After writing to the CMD register, the Mobile storage starts executing the command; when the command is sent to the bus, the command_done interrupt is generated.

1. Software should look for data error interrupts; that is, bits 7, 9, 13, and 15 of the RINTSTS register. If required, software can terminate the data transfer by sending a STOP command.
2. Software should look for Receive_FIFO_Data_request and/or data starvation by host timeout conditions. In both cases, the software should read data from the FIFO and make space in the FIFO for receiving more data.
3. When a Data_Transfer_Over interrupt is received, the software should read the remaining data from the FIFO.

121.1.1.7. Single-Block or Multiple-Block Write

Steps involved in a single-block or multiple-block write are:

1. Write the data size in bytes in the BYTCNT register @0x20.
2. Write the block size in bytes in the BLKSIZ register @0x1C; the Mobile storage sends data in blocks of size BLKSIZ each.
3. Program CMDARG register @0x28 with the data address to which data should be written.

The command argument for SDIO card has to be programmed.

1. Write data in the FIFO; it is usually best to start filling data the full depth of the FIFO.
2. Program the Command register. For SD and MMC cards, use CMD24 for a single-block write

and CMD25 for a multiple-block write. For SDIO cards, use CMD53 for both single-block and multiple-block transfers.

Table 9.3-148 Command Register Settings for Single-Block or Multiple-Block Write

Parameter	**Value**	**Comment**
Default		
start_cmd	1	–
use_hold_reg	1 or 0	Choose value based on speed mode.
Update_clk_regs_only	0	No clock parameters update command
card_number	*n*CardNo	Actual card number
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
send_auto_stop	0 or 1	
transfer_mode	0	Block transfer
read_write	1	Write to card
data_expected	1	Data command
response_length	0	Can be 1 for R2 (long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4 CMD15, and so on
User-selectable		
cmd_index	command index	–
wait_prvdata_complete	1	0 – sends command immediately 1 – sends command after previous data transfer ends
check_response_crc	1	0 – DWC_mobile_storage should not check response CRC 1 – DWC_mobile_storage should check response CRC

After writing to the CMD register, Mobile storage starts executing a command; when the command is sent to the bus, a command_done interrupt is generated.

3. Software should look for data error interrupts; that is, for bits 7, 9, and 15 of the RINTSTS register. If required, software can terminate the data transfer by sending the STOP command.

4. Software should look for Transmit_FIFO_Data_request and/or timeout conditions from data starvation by the host. In both cases, the software should write data into the FIFO.

5. When a Data_Transfer_Over interrupt is received, the data command is over. For an open-ended block transfer, if the byte count is 0, the software must send the STOP command. If the byte count is not 0, then upon completion of a transfer of a given number of bytes, the Mobile storage should send the STOP command, if necessary. Completion of the AUTO-STOP command is reflected by the Auto_command_done interrupt – bit 14 of the RINTSTS register. A response to

AUTO_STOP is stored in RESP1 @0x34.

6. Wait for Busy Clear Interrupt. Card may drive busy on the DAT line; the host controller generates the interrupt after the busy is completed.

121.1.1.8. Stream Read

A stream read is like the block read mentioned in "Single-Block or Multiple-Block Read", except for the following bits in the Command register:
transfer_mode= 1; //Stream transfer
cmd_index = CMD20;
A stream transfer is allowed for only a single-bit bus width.

121.1.1.9. Stream Write

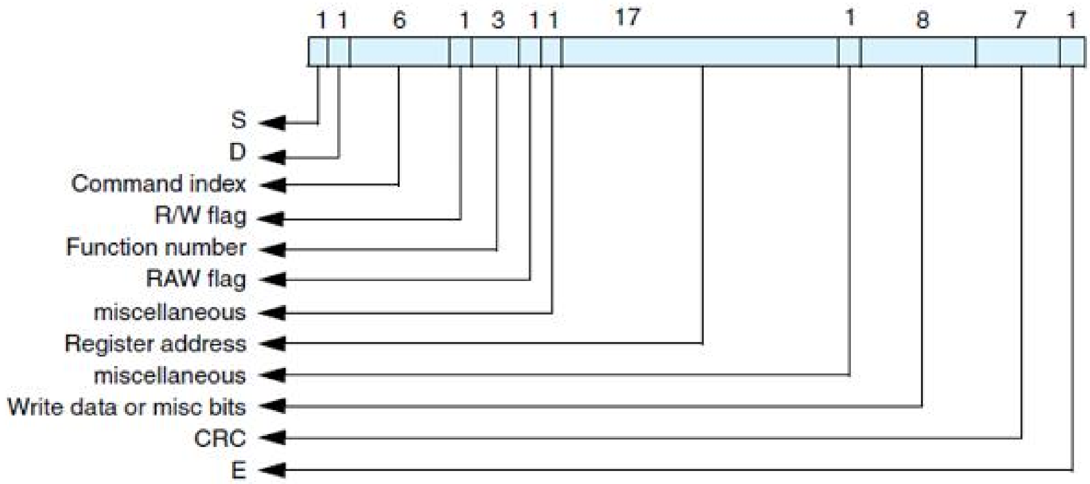
A stream write is exactly like the block write mentioned in "Single-Block or Multiple-Block Write", except for the following bits in the Command register:
transfer_mode= 1;//Stream transfer
cmd_index = CMD11;
In a stream transfer, if the byte count is 0, then the software must send the STOP command. If the byte count is not 0, then when a given number of bytes completes a transfer, the Mobile storage sends the STOP command. Completion of this AUTO_STOP command is reflected by the Auto_command_done interrupt. A response to an AUTO_STOP is stored in the RESP1 register @0x34.
A stream transfer is allowed for only a single-bit bus width.

121.1.1.10. Sending Stop or Abort in Middle of Transfer

The STOP command can terminate a data transfer between a memory card and the Mobile storage, while the ABORT command can terminate an I/O data transfer for only the SDIO_IOONLY and SDIO_COMBO cards.

- Send STOP command – Can be sent on the command line while a data transfer is in progress; this command can be sent at any time during a data transfer. For information on sending this command, refer to "No-Data Command With or Without Response Sequence". You can also use an additional setting for this command in order to set the Command register bits (5-0) to CMD12 and set bit 14 (stop_abort_cmd) to 1. If stop_abort_cmd is not set to 1, the Mobile storage does not know that the user stopped a data transfer. Reset bit 13 of the Command register (wait_prvdata_complete) to 0 in order to make the Mobile storage send the command at once, even though there is a data transfer in progress.
- Send ABORT command – Can be used with only an SDIO_IOONLY or SDIO_COMBO card. To abort the function that is transferring data, program the function number in ASx bits (CCCR register of card, address 0x06, bits (0-2) using CMD52. This is a non-data command. For information on sending this command.

Figure 9.3-8 Command Argument for CMD52



a. Program the CMDARG register @0x28 with the appropriate command argument parameters

Table 9.3-149 Parameters for CMDARG Register

CMDARG Bits	Contents	Value
31	R/W flag	1
30-28	Function Number	0, for CCCR access
27	RAW flag	1, if needed to read after write
26	Don't care	–
25-9	Register address	0x06
8	Don't care	–
7-0	Write Data	Function number to be aborted

b. Program the Command register using the command index as CMD52. Similar to the STOP command described on 228, set bit 14 of the Command register (stop_abort_cmd) to 1, which must be done in order to inform the Mobile storage that the user aborted the data transfer. Reset bit 13 (wait_prvdata_complete) of the Command register to 0 in order to make the Mobile storage send the command at once, even though a data transfer is in progress.

c. Wait for command_transfer_over.

d. Check response (R5) for errors.

121.1.1.11. **Suspend or Resume Sequence**

In an SDIO card, the data transfer between an I/O function and the Mobile storage can be temporarily halted using the SUSPEND command; this may be required in order to perform a high-priority data transfer with another function. When desired, the data transfer can be resumed using the RESUME command.

The following functions can be implemented by programming the appropriate bits in the CCCR register (Function 0) of the SDIO card. To read from or write to the CCCR register, use the CMD52 command.

1. SUSPEND data transfer – Non-data command.

a. Check if the SDIO card supports the SUSPEND/RESUME protocol; this can be done through the SBS bit in the CCCR register @0x08 of the card.

b. Check if the data transfer for the required function number is in process; the function number that is currently active is reflected in bits 0-3 of the CCCR register @0x0D. Note that if the BS bit (address 0xc::bit 0) is 1, then only the function number given by the FSx bits is valid.

c. To suspend the transfer, set BR (bit 2) of the CCCR register @0x0C.

d. Poll for clear status of bits BR (bit 1) and BS (bit 0) of the CCCR @0x0C. The BS (Bus Status) bit is 1 when the currently-selected function is using the data bus; the BR (Bus Release) bit remains 1 until the bus release is complete. When the BR and BS bits are 0, the data transfer from the selected function has been suspended.

e. During a read-data transfer, the Mobile storage can be waiting for the data from the card. If the data transfer is a read from a card, then the Mobile storage must be informed after the successful completion of the SUSPEND command. The Mobile storage then resets the data state machine and comes out of the wait state. To accomplish this, set abort_read_data (bit 8) in the Control register.

f. Wait for data completion. Get pending bytes to transfer by reading the TCBCNT register @0x5C.

g. For a write data transfer, wait for the Busy Clear Interrupt after the Data Transfer Over (DTO) interrupt.

2. RESUME data transfer – This is a data command.

a. Check that the card is not in a transfer state, which confirms that the bus is free for data transfer.

b. If the card is in a disconnect state, select it using CMD7. The card status can be retrieved in response to CMD52/CMD53 commands.

c. Check that a function to be resumed is ready for data transfer; this can be confirmed by reading the RFx flag in CCCR @0x0F. If RF = 1, then the function is ready for data transfer.

d. To resume transfer, use CMD52 to write the function number at FSx bits (0-3) in the CCCR register @0x0D. Form the command argument for CMD52 and write it in CMDARG @0x28.

Table 9.3-150 CMDARG Bit Values

CMDARG Bits	**Content**	**Value**
31	R/W flag	1
30-28	Function Number	0, for CCCR access
27	RAW flag	1, read after write
26	Don’t care	–
25-9	Register address	0x0D
8	Don’t care	–
7-0	Write Data	Function number that is to be resumed

e. Write the block size in the BLKSIZ register @0x1C; data will be transferred in units of this block size.

f. Write the byte count in the BYTCNT register @0x20. This is the total size of the data; that is, the remaining bytes to be transferred. It is the responsibility of the software to handle the data.
g. Program Command register; similar to a block transfer. For details, refer to "Single-Block or Multiple-Block Read" and "Single-Block or Multiple-Block Write".
h. When the Command register is programmed, the command is sent and the function resumes data transfer. Read the DF flag (Resume Data Flag). If it is 1, then the function has data for the transfer and will begin a data transfer as soon as the function or memory is resumed. If it is 0, then the function has no data for the transfer.
i. If the DF flag is 0, then in case of a read, the Mobile storage waits for data. After the data timeout period, it gives a data timeout error.

121.1.1.12. Read_Wait Sequence

Read_wait is used with only the SDIO card and can temporarily stall the data transfer—either from function or memory—and allow the host to send commands to any function within the SDIO device. The host can stall this transfer for as long as required. The Mobile storage provides the facility to signal this stall transfer to the card. The steps for doing this are:
1. Check if the card supports the read_wait facility; read SRW (bit 2) of the CCCR register @0x08. If this bit is 1, then all functions in the card support the read_wait facility. Use CMD52 to read this bit.
2. If the card supports the read_wait signal, then assert it by setting the read_wait (bit 6) in the CTRL register @0x00.
3. Clear the read_wait bit in the CTRL register.

121.1.1.13. Transmission and Reception with Internal DMAC (IDMAC)

The general sequence of events for transmit and receive is as follows:
1. Program the required programming in the Bus Mode Register. If IDMAC enable is disabled during the middle of an IDMAC transfer, it has no effect. It will only take effect for a new data transfer command. Issuing a Software reset will immediately terminate the transfer. It is recommended that the host issue a reset to DMA interface by setting dma_reset bit of the CTRL register and then issue a IDMAC software reset. The PBL contents are read-only and a direct reflection of the DW_DMA_Mutiple_Transaction_Size contents in FIFOTH register. The Fixed burst bit has to be programmed appropriately for system performance.
2. In case a Descriptor Unavailable Interrupt is asserted, the host needs to form the descriptor, appropriately set its own bit and then write to poll demand register for the IDMAC to re-fetch the descriptor.
3. It is always appropriate for the host to enable abnormal interrupts since any errors related to the transfer are reported to the host.
4. For handling scenarios like Fatal Bus Error, Abort and FIFO overrun/under-run refer to the Interrupts section of IDMAC chapter.

For more information, refer to “Transmission” and “Reception”.

121.1.1.14. **Descriptors**

The IDMAC uses these types of descriptor structures:

■ Dual-Buffer Structure – The distance between two descriptors is determined by the Skip Length value programmed in the Descriptor Skip Length (DSL) field of the Bus Mode Register (BMOD@0x80).

■ Chain Structure – Each descriptor points to a unique buffer and the next descriptor.

Figure 9.3- 9 Dual-Buffer Descriptor Structure

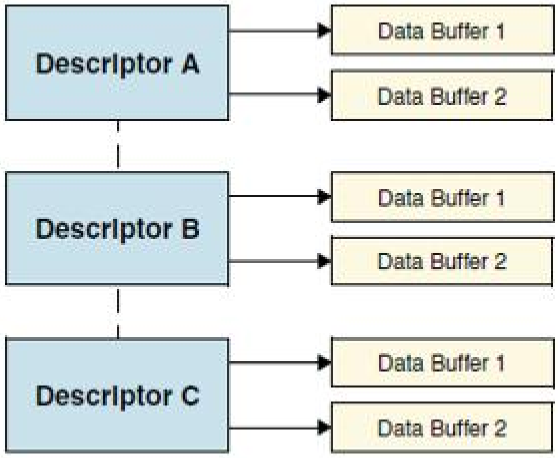
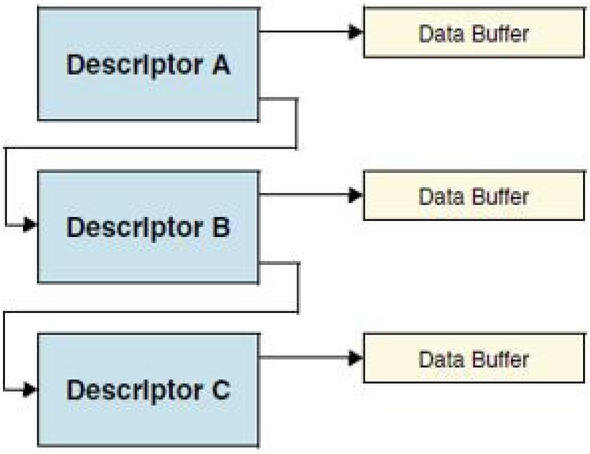
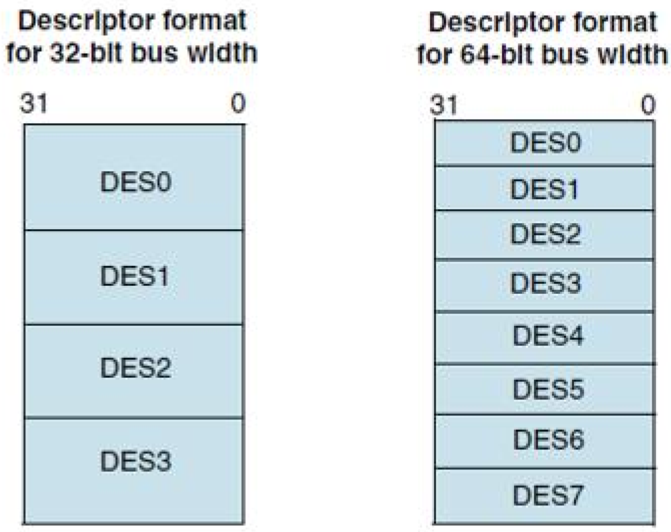


Figure 9.3- 10 Chain Descriptor Structure



The descriptor addresses must be aligned to the bus width used for 16/32/64-bit AHB data buses. Each descriptor contains 16 bytes of control and status information. DES0 is a notation used to denote the [31:0] bits, DES1 to denote [63:32] bits, DES2 to denote [95:64] bits, DES3 to denote [127:96] bits, DES4 to denote [159:128] bits, DES5 to denote [191:160] bits, DES6 to denote [223:192] bits, DES7 to denote [255:224] bits in a descriptor.

Figure 9.3- 11 Descriptor Formats for 32-bit and 64-bit AHB Address Bus Widths



The DES0 element in the IDMAC contains control and status information (for 32-bit address configuration).

Table 9.3-151 Bits in IDMAC DES0 Element (32-bit Address Configuration)

Bits	Name	Description
31	OWN	When set, this bit indicates that the descriptor is owned by the IDMAC. When this bit is reset, it indicates that the descriptor is owned by the Host. The IDMAC clears this bit when it completes the data transfer.
30	Card Error Summary (CES)	These error bits indicate the status of the transaction to or from the card. These bits are also present in RINTSTS Indicates the logical OR of the following bits: ■ EBE: End Bit Error ■ RTO: Response Time out ■ RCRC: Response CRC ■ SBE: Start Bit Error ■ DRTO: Data Read Timeout ■ DCRC: Data CRC for Receive ■ RE: Response Error
29:6	Reserved	
5	End of Ring (ER)	When set, this bit indicates that the descriptor list reached its final descriptor. The IDMAC returns to the base address of the list, creating a Descriptor Ring. This is meaningful for only a dual-buffer descriptor structure.

Table 9.3-152 Bits in IDMAC DES0 Element (32-bit Address Configuration) (Cont.)

Bits	Name	Description
4	Second Address Chained (CH)	When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When this bit is set, BS2 (DES1[25:13]) should be all zeros.
3	First Descriptor (FS)	When set, this bit indicates that this descriptor contains the first buffer of the data. If the size of the first buffer is 0, next Descriptor contains the beginning of the data.
2	Last Descriptor (LD)	This bit is associated with the last block of a DMA transfer. When set, the bit indicates that the buffers pointed to by this descriptor are the last buffers of the data. After this descriptor is completed, the remaining byte count is 0. In other words, after the descriptor with the LD bit set is completed, the remaining byte count should be 0.
1	Disable Interrupt on Completion (DIC)	When set, this bit will prevent the setting of the TI/RI bit of the IDMAC Status Register (IDSTS) for the data that ends in the buffer pointed to by this descriptor.
0	Reserved	

The DES1 element contains the buffer size (for 32-bit address configuration).

Table 9.3-153 Bits in IDMAC DES1 Element (32-bit Address Configuration)

Bits	Name	Description
31:13	Reserved	
25:13	Buffer 2 Size (BS2)	These bits indicate the second data buffer byte size. The buffer size must be a multiple of 2, 4, or 8, depending upon the bus widths—16, 32, and 64, respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and proceeds to the next buffer in case of a dual-buffer structure. This field is not valid for chain structure; that is, if DES0[4] is set. If Buffer 2 Size is set to 0 in any of the descriptor, the remaining descriptor cannot have non zero value for Buffer 2 size until the end of the descriptor.
12:0	Buffer 1 Size (BS1)	Indicates the data buffer byte size, which must be a multiple of 2, 4, or 8 bytes, depending upon the bus widths—16, 32, and 64, respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. This field should not be zero. **Note:** If there is only one buffer to be programmed, you need to use only the Buffer 1, and *not* Buffer 2.

The DES2 element contains the address pointer to the data buffer (for 32-bit address configuration).

Table 9.3-154 Bits in IDMAC DES2 Element (32-bit Address Configuration)

Bits	Name	Description
31:0	Buffer Address Pointer 1	These bits indicate the physical address of the first data buffer when the dual-buffer structure is used. In case of chain descriptor, these bits indicate the physical address of the data buffer.

The DES3 element contains the address pointer to the next descriptor if the present descriptor is not the last descriptor in a chained descriptor structure or the second buffer address for a dual-buffer structure (for 32-bit address configuration).

Table 9.3-155 Bits in IDMAC DES3 Element (32-bit Address Configuration)

Bits	Name	Description
31:0	Buffer Address Pointer 2/ Next Descriptor Address (BAP2)	These bits indicate the physical address of the second buffer when the dual-buffer structure is used. If the Second Address Chained (DES0[4]) bit is set, then this address contains the pointer to the physical memory where the Next Descriptor is present. If this is not the last descriptor, then the Next Descriptor address pointer must be bus-width aligned (DES3[2/1/0:0] = 0 corresponding to bus-width of 64/32/16, Internally the LSBs are ignored).

121.1.1.15. **Transmission**

IDMAC transmission occurs as follows:

1. The Host sets up the elements (DES0-DES3) for transmission and sets the OWN bit (DES0[31]). The Host also prepares the data buffer.
2. The Host programs the write data command in the CMD register in BIU.
3. The Host will also program the required transmit threshold level (TX_WMark field in FIFOTH register).
4. The IDMAC determines that a write data transfer needs to be done as a consequence of step 2.
5. The IDMAC engine fetches the descriptor and checks the OWN bit. If the OWN bit is not set, it means that the host owns the descriptor. In this case the IDMAC enters suspend state and asserts the Descriptor Unable interrupt in the IDSTS register (IDSTS64 register, in case of 64-bit address configuration). In such a case, the host needs to release the IDMAC by writing any value to the poll demand register.
6. It will then wait for Command Done (CD) bit and no errors from BIU which indicates that a transfer can be done.
7. The IDMAC engine will now wait for a DMA interface request (dw_dma_req) from BIU. This request will be generated based on the programmed transmit threshold value. For the last bytes of data which can’t be accessed using a burst, SINGLE transfers are performed on AHB Master Interface.
8. The IDMAC fetches the Transmit data from the data buffer in the Host memory and transfers to the FIFO for transmission to card.
9. When data spans across multiple descriptors, the IDMAC will fetch the next descriptor and continue with its operation with the next descriptor. The Last Descriptor bit in the descriptor indicates whether the data spans multiple descriptors or not.
10. When data transmission is complete, status information is updated in IDSTS register (IDSTS64 register, in case of 64-bit address configuration) by setting Transmit Interrupt, if enabled. Also, the OWN bit is cleared by the IDMAC by performing a write transaction to DES0.

121.1.1.16. **Reception**

IDMAC reception occurs as follows:

1. The Host sets up the element (DES0-DES3) for reception, sets the OWN (DES0[31]).
2. The Host programs the read data command in the CMD register in BIU.
3. The Host will program the required receive threshold level (RX_WMark field in FIFOTH register).
4. The IDMAC determines that a read data transfer needs to be done as a consequence of step 2.
5. The IDMAC engine fetches the descriptor and checks the OWN bit. If the OWN bit is not set, it means that the host owns the descriptor. In this case the DMA enters suspend state and asserts the Descriptor Unable interrupt in the IDSTS register (IDSTS64 register, in case of 64-bit address configuration). In such a case, the host needs to release the IDMAC by writing any value to the poll demand register.
6. It will then wait for Command Done (CD) bit and no errors from BIU which indicates that a transfer can be done.
7. The IDMAC engine will now wait for a DMA interface request (dw_dma_req) from BIU. This request will be generated based on the programmed receive threshold value. For the last bytes of data which can't be accessed using a burst, SINGLE transfers are performed on AHB.
8. The IDMAC fetches the data from the FIFO and transfer to Host memory.
9. When data spans across multiple descriptors, the IDMAC will fetch the next descriptor and continue with its operation with the next descriptor. The Last Descriptor bit in the descriptor indicates whether the data spans multiple descriptors or not.
10. When data reception is complete, status information is updated in IDSTS register (IDSTS64 register, in case of 64-bit address configuration) by setting Receive Interrupt, if enabled. Also, the OWN bit is cleared by the IDMAC by performing a write transaction to DES0.

121.1.2. Register Address Map

Table 9.3-156 Memory Map

Addr offset	Name	Width	R/W	Reset value	Description
0x00	CTRL	32	RW	0x0	Controller Register
0x04	PWREN	32	RW	0x0	Power-enable Register
0x08	CLKDIV	32	RW	0x0	Clk-diver Register
0x0c	CLKSRC	32	RW	0x0	Clk-source Register
0x10	CLKENA	32	RW	0x0	Clk-enable Register
0x14	TMOUT	32	RW	0xFFFFFF40	Time-out Register (number of card clock output clocks-cclk_out)
0x18	CTYPE	32	RW	0x0	Clk-type Register
0x1c	BLKSIZ	32	RW	0x200	Block-size Register
0x20	BYTCNT	32	RW	0x200	Byte-count Register
0x24	INTMASK	32	RW	0x0	Interrupt-mask Register
0x28	CMDARG	32	RW	0x0	Command-argument Register

0x2c	CMD	32	RW	0x0	Command Register
0x30	RESP0	32	R	0x0	Response-0 Register
0x34	RESP1	32	R	0x0	Response-1 Register
0x38	RESP2	32	R	0x0	Response-2 Register
0x3c	RESP3	32	R	0x0	Response-3 Register
0x40	MINTSTS	32	R	0x0	Masked interrupt-status Register
0x44	RINTSTS	32	RW	0x0	Raw interrupt-status Register
0x48	STATUS	32	R	{20'h00000,4'b 00xx,8'h06}	Status Register, mainly for debug purposes
0x4c	FIFOTH	32	RW	{4'h0,12'h1ff,16 'h0}	FIFO threshold Register
0x50	CDETECT	32	R	0x0	Card-detect Register
0x54	WRTPRT	32	R	0x0	Write-protect Register
0x58	GPIO	32	RW	0x0	GPIO Register
0x5c	TCBCNT	32	R	0x0	Transferred CIU card byte count
0x60	TBBCNT	32	R	0x0	Transferred host/DMA to/from BIU-FIFO byte count
0x64	DEBNCE	32	RW	0x00ffffff	Card detect debounce Register(number of host closks-clk)
0x68	USRID	32	RW	0x7967797	User ID Register
0x6c	VERID	32	R	0x5342270a	Synopsys version ID Register
0x70	HCON	32	R	0xc44cc1	Hardware configuration Register
0x74	UHS_REG	32	RW	0x0	UHS-1 Register
0x78	RST-n	32	RW	0x0000FFFF	Hardware reset Register
0x80	BMOD	32	RW	0x0	Bus mode Register;controls the HOST interface Mode
0x84	PLDMND	32	W	0x0	Poll Demand Register. Used by the host to instruct the IDMAC to poll the Descripor List while in suspend
0x88	DBADDR(For 32-bit Address Configuration Only)	32	RW	0x0	Descriptor List Base Address Lower Register

0x8C	IDSTS	32	RW	0x0	Internal DMAC Status Register. Software driver application reads this register during interrrupt service routine or polling to determine the status of the IDMAC
0x90	IDINTEN	32	RW	0x0	Internal DMAC Interrupt Enable Register. Enables the interrupts reported by the IDMAC Status Register-IDSTS
0x94	DSCARDDR(For 32-bit Address Configuration Only)	32	R	0x0	Current Host Descriptor Address Register. Points to the start of current Descriptor read by the IDMAC
0x98	BUFADDR(For 32-bit Address Configuration Only)	32	R	0x0	Current Host Buffer Address Register. Points to the current Buffer address by the IDMAC
0x9C0~0xFF	Reserved				
0x100	CardThrCtl	32	RW	0x0	Card Read Threshold Enable (CardRdThrEn) 1'b0 - Card Read Threshold disabled 1'b1 - Card Read Threshold enabled Card Write Threshold Enable (CardWrThrEn) 1'b0 - Card Write Threshold disabled 1'b1 - Card Write Threshold enabled
0x104	Back_end_power	32	RW	0x0	Back-end Power 0 - Off; Reset 1 - Back-end Power is supplied
0x108	UHS_REG_EXT	32	RW	0x0	eMMC 4.5 1.2V register
0x10c	EMMC_DDR_REG	32	RW	0x0	eMMC DDR START bit

					detection or HS400 mode enable control register
0x110	ENABLE_SHIFT	32	RW	0x0	Phase shit control register
0x200	DATA	32	RW	32'hx	Data FIFO read/write; if address is equal or greater than 0x200, then FIFO is selected as long as device is selected (hsel/psel active)

121.1.3. **Register and Field Descriptions**

Table 9.3-157 CTRL

Bit Field	Name	Width	R/W	Default value	Description
31:26	Reserved	6			
25	use_internal_dmac	1	RW	0	Present only for the Internal DMAC configuration; else, it is reserved. 0 – The host performs data transfers through the slave interface 1 – Internal DMAC used for data transfer
24	enable_OD_pullup	1	RW	1	External open-drain pullup 0 – Disable 1 – Enable
23:20	Card_voltage_b	4	RW	0	Card regulator-B voltage setting; output to card_volt_b port. Optional feature; ports can be used as general-purpose outputs
19:16	Card_voltage_a	4	RW	0	Card regulator-A voltage setting; output to card_volt_a port Optional feature; ports can be used as general-purpose outputs
15:12	Reserved	4			

11	ceata_device_interrupt_status	1	RW	0	0 – Interrupts not enabled in CE-ATA device(nIEN = 1 in ATA control register) 1 – Interrupts are enabled in CE-ATA device(nIEN = 0 in ATA control register)
10	send_auto_stop_ccsd	1	RW	0	0 – Clear bit if Mobile storage does not reset the bit 1 – Send internally generated STOP after sending CCSD to CE-ATA device
9	send_ccsd	1	RW	0	0 – Clear bit if Mobile storage does not reset the bit. 1 – Send Command Completion Signal Disable (CCSD) to CE-ATA device
8	abort_read_data	1	RW	0	0 – No change 1 – After suspend command is issued during read-transfer, software polls card to find when suspend happened. Once suspend occurs, software sets bit to reset data state-machine, which is waiting for next block of data. Bit automatically clears once data state- machine resets to idle
7	send_irq_response	1	RW	0	0 – No change 1 – Send auto IRQ response
6	read_wait	1	RW	0	0 – Clear read wait 1 – Assert read wait
5	dma_enable	1	RW	0	0 – Disable DMA transfer mode

					1 – Enable DMA transfer mode
4	int_enable	1	RW	0	Global interrupt enable/disable bit: 0 – Disable interrupts 1 – Enable interrupts
3	Reserved	1			
2	dma_reset	1	RW	0	0 – No change 1 – Reset internal DMA interface control logic
1	fifo_reset	1	RW	0	0 – No change 1 – Reset to data FIFO To reset FIFO pointers
0	controller_reset	1	RW	0	0 – No change 1 – Reset Mobile storage controller

Table 9.3-158 PWREN

Bit Field	Name	Width	R/W	Default value	Description
31:30	Reserved	2			
29:0	power_enable	30	RW	0x0	Power on/off switch for up to 16 cards; for example, bit[0] controls card 0. Once power is turned on, firmware should wait for regulator/switch ramp-up time before trying to initialize card. 0 – power off 1 – power on

Table 9.3-159 CLKDIV

Bit Field	Name	Width	R/W	Reset value	Description
31:24	clk_divider3	8	RW	0x00	Clock divider-3 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), a value of 1 means divide by 2*1 = 2, a

					value of “ff” means divide by 2*255 = 510, and so on. In MMC-Ver3.3-only mode, bits not implemented because only one clock divider is supported.
23:16	clk_divider2	8	RW	0x00	Clock divider-2 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of “ff” means divide by 2*255 = 510, and so on. In MMC-Ver3.3-only mode, bits not implemented because only one clock divider is supported
15:8	clk_divider1	8	RW	0x00	Clock divider-1 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of “ff” means divide by 2*255 = 510, and so on. In MMC-Ver3.3-only mode, bits not implemented because only one clock divider is supported
7:0	clk_divider0	8	RW	0x00	Clock divider-0 value. Clock division is 2*n. For example, value of 0 means

					divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of "ff" means divide by 2*255 = 510, and so on

Table 9.3-160 CLKSRC

Bit Field	Name	Width	R/W	Default value	Description
31:0	clk_source	32	RW	0x00	Clock divider source for up to 16 SD cards supported. Each card has two bits assigned to it. For example, bits[1:0] assigned for card-0, which maps and internally routes clock divider[3:0] outputs to cclk_out[15:0] pins, depending on bit value. 00 – Clock divider 0 01 – Clock divider 1 10 – Clock divider 2 11 – Clock divider 3

Table 9.3-161 CLKENA

Bit Field	Name	Width	R/W	Default value	Description
31:16	cclk_low_power	16	RW	0x0	Low-power control for up to 16 SD card clocks and one MMC card clock supported. 0 – Non-low-power mode 1 – Low-power mode; stop clock when card in IDLE (should be normally set to only MMC and SD memory

					cards; for SDIO cards, if interrupts must be detected, clock should not be stopped)
15:0	cclk_enable	16	RW	0x0	Clock-enable control for up to 16 SD card clocks and one MMC card clock supported. 0 – Clock disabled 1 – Clock enabled

Table 9.3-162 TMOUT

Bit Field	Name	Width	R/W	Default value	Description
31:8	data_timeout	24	RW	0xffffff	Value for card Data Read Timeout; same value also used for Data Starvation by Host timeout. The timeout counter is started only after the card clock is stopped. Value is in number of card output clocks – cclk_out of selected card
23:0	response_timeout	8	RW	0x40	Response timeout value. Value is in number of card output clocks – cclk_out

Table 9.3-163 CTYPE

Bit Field	Name	Width	R/W	Default value	Description
31:16	card_width	16	RW	0x0	One bit per card indicates if card is 8-bit: 0 – Non 8-bit mode 1 – 8-bit mode Bit[31] corresponds to card[15]; bit[16] corresponds to card[0].

15:0	card_width	16	RW	0x0	One bit per card indicates if card is 1-bit or 4-bit: 0 – 1-bit mode 1 – 4-bit mode Bit[15] corresponds to card[15], bit[0] corresponds to card[0]. Only NUM_CARDS*2 number of bits are implemented.

Table 9.3-164 BLKSIZ

Bit Field	Name	Width	R/W	Default value	Description
31:16	Reserved	16	RW		
15:0	block_size	16	RW	0x200	Block size

Table 9.3-165 BYTCNT

Bit Field	Name	Width	R/W	Default value	Description
31:0	byte_count	32	RW	0x200	Number of bytes to be transferred; should be integer multiple of Block Size for block transfers. For undefined number of byte transfers, byte count should be set to 0. When byte count is set to 0, it is responsibility of host to explicitly send stop/abort command to terminate data transfer.

Table 9.3-166 INTMASK

Bit Field	Name	Width	R/W	Default value	Description
31:16	sdio_int_mask	16	RW	0x0	Mask SDIO interrupts One bit for each card. Bit[31] corresponds to card[15], and bit[16]

					corresponds to card[0]. When masked, SDIO interrupt detection for that card is disabled. A 0 masks an interrupt, and 1 enables an interrupt.
15:0	int_mask	16	RW	0x0	Bits used to mask unwanted interrupts. Value of 0 masks interrupt; value of 1 enables interrupt. bit 15 – End-bit error (read)/Write no CRC (EBE) bit 14 – Auto command done (ACD) bit 13 – Start Bit Error(SBE)/Busy Clear Interrupt (BCI) bit 12 – Hardware locked write error (HLE) bit 11 – FIFO underrun/overrun error (FRUN) bit 10 – Data starvation-by-host timeout (HTO) /Volt_switch_int bit 9 – Data read timeout (DRTO) bit 8 – Response timeout (RTO) bit 7 – Data CRC error (DCRC) bit 6 – Response CRC error (RCRC) bit 5 – Receive FIFO data request (RXDR) bit 4 – Transmit FIFO data request (TXDR) bit 3 – Data transfer

					over (DTO) bit 2 – Command done (CD) bit 1 – Response error (RE) bit 0 – Card detect (CD)

Table 9.3-167 CMDARG

Bit Field	Name	Width	R/W	Default value	Description
31:0	cmd_arg	32	RW	0x0	Value indicates command argument to be passed to card

Table 9.3-168 CMD

Bit Field	Name	Width	R/W	Default value	Description
31	start_cmd	1	RW	0	Start command. Once command is taken by CIU, bit is cleared. When bit is set, host should not attempt to write to any command registers. If write is attempted, hardware lock error is set in raw interrupt register. Once command is sent and response is received from SD_MMC_CEATA cards, Command Done bit is set in raw interrupt register.
30	Reserved	1			

29	use_hold_reg	1	RW	1	Use Hold Register 0 - CMD and DATA sent to card bypassing HOLD Register 1 - CMD and DATA sent to card through the HOLD Register
28	volt_switch	1	RW	0	Voltage switch bit 0 - No voltage switching 1 - Voltage switching enabled; must be set for CMD11 only
27	boot_mode	1	RW	0	Boot Mode 0 - Mandatory Boot operation 1 - Alternate Boot operation
26	disable_boot	1	RW	0	Disable Boot. When software sets this bit along with start_cmd, CIU terminates the boot operation. Do NOT set disable_boot and enable_boot together
25	expect_boot_ack	1	RW	0	Expect Boot Acknowledge. When Software sets this bit along with enable_boot, CIU expects a boot acknowledge start pattern of 0-1-0 from the selected card
24	enable_boot	1	RW	0	Enable Boot—this bit should be set only for mandatory boot mode. When Software sets this bit along with start_cmd, CIU starts the boot sequence for the corresponding card by asserting the CMD line low. Do NOT set disable_boot and

					enable_boot together
23	ccs_expected	1	RW	0	0 – Interrupts are not enabled in CE-ATA device (nIEN = 1 in ATA control register), or command does not expect CCS from device 1 – Interrupts are enabled in CE-ATA device (nIEN = 0), and RW_BLK command expects command completion signal from CE-ATA device
22	read_ceata_device	1	RW	0	0 – Host is not performing read access (RW_REG or RW_BLK) towards CE-ATA device 1 – Host is performing read access (RW_REG or RW_BLK) towards CE-ATA device
21	update_clock_registers_only	1	RW	0	0 – Normal command sequence 1 – Do not send commands, just update clock register value into card clock domain Following register values transferred into card clock domain: CLKDIV, CLRSRC, CLKENA. Changes card clocks (change frequency, truncate off or on, and set low-frequency mode); provided in order to change clock frequency or stop clock without having to send command

					to cards. During normal command sequence, when update_clock_registers_only = 0, following control registers are transferred from BIU to CIU: CMD, CMDARG, TMOUT, CTYPE, BLKSIZ, BYTCNT. CIU uses new register values for new command sequence to card(s). When bit is set, there are no Command Done interrupts because no command is sent to SD_MMC_CEATA cards
20:16	card_number	5	RW	0	Card number in use. Represents physical slot number of card being accessed. In MMC-Ver3.3-only mode, up to 30 cards are supported; in SD-only mode, up to 16 cards are supported. Registered version of this is reflected on dw_dma_card_num and ge_dma_card_num ports, which can be used to create separate DMA requests, if needed. In addition, in SD mode this is used to mux or demux signals from selected card because each card is interfaced to Mobile storage by separate bus.

15	send_initialization	1	RW	0	0 – Do not send initialization sequence (80 clocks of 1) before sending this command 1 – Send initialization sequence before sending this command After power on, 80 clocks must be sent to card for initialization before sending any commands to card. Bit should be set while sending first command to card so that controller will initialize clocks before sending command to card. This bit should not be set for either of the boot modes (alternate or mandatory).
14	stop_abort_cmd	1	RW	0	0 – Neither stop nor abort command to stop current data transfer in progress. If abort is sent to function-number currently selected or not in data-transfer mode, then bit should be set to 0. 1 – Stop or abort command intended to stop current data transfer in progress. When open-ended or predefined data transfer is in progress, and host issues stop or abort command to stop data transfer, bit should be set so that command/data state-machines of CIU

					can return correctly to idle state. This is also applicable for Boot mode transfers. To Abort boot mode, this bit should be set along with CMD[26] = disable_boo
13	wait_prvdata_complete	1	RW	0	0 – Send command at once, even if previous data transfer has not completed 1 – Wait for previous data transfer completion before sending command The wait_prvdata_complete = 0 option typically used to query status of card during data transfer or to stop current data transfer; card_number should be same as in previous command
12	send_auto_stop	1	RW	0	0 – No stop command sent at end of data transfer 1 – Send stop command at end of data transfer Don't care if no data expected
11	transfer_mode	1	RW	0	0 – Block data transfer command 1 – Stream data transfer command Don’t care if no data expected.
10	read/write	1	RW	0	0 – Read from card 1 – Write to card Don’t care if no data

					expected from card.
9	data_expected	1	RW	0	0 – No data transfer expected (read/write) 1 – Data transfer expected (read/write)
8	check_response_crc	1	RW	0	0 – Do not check response CRC 1 – Check response CRC Some of command responses do not return valid CRC bits. Software should disable CRC checks for those commands in order to disable CRC checking by controller
7	response_ length	1	RW	0	0 – Short response expected from card 1 – Long response expected from card
6	response_expect	1	RW	0	0 – No response expected from card 1 – Response expected from card
5:0	cmd-index	6	RW	0	Command index

Table 9.3-169 RESP0

Bit Field	Name	Width	R/W	Default value	Description
31:0	response 0	32	R	0	Bit[31:0] of response

Table 9.3-170 RESP1

Bit Field	Name	Width	R/W	Default value	Description
31:0	response 1	32	R	0	Register represents bit[63:32] of long response. When CIU sends auto-stop command, then response is saved in register. Response for previous

					command sent by host is still preserved in Response 0 register. Additional auto-stop issued only for data transfer commands, and response type is always "short" for them

Table 9.3-171 RESP2

Bit Field	Name	Width	R/W	Default value	Description
31:0	response 2	32	R	0	Bit[95:64] of long response

Table 9.3-172 RESP3

Bit Field	Name	Width	R/W	Default value	Description
31:0	response 3	32	R	0	Bit[127:96] of long response

Table 9.3-173 MINTSTS

Bit Field	Name	Width	R/W	Default value	Description
31:16	sdio_interrupt	16	R	0	Interrupt from SDIO card; one bit for each card. Bit[31] corresponds to Card[15], and bit[16] is for Card[0]. SDIO interrupt for card enabled only if corresponding sdio_int_mask bit is set in Interrupt mask register (mask bit 1 enables interrupt; 0 masks interrupt). 0 – No SDIO interrupt from card 1 – SDIO interrupt from card

15:0	int_status	16	R	0	Interrupt enabled only if corresponding bit in interrupt mask register is set. bit 15 – End-bit error (read)/write no CRC (EBE) bit 14 – Auto command done (ACD) bit 13 – Start Bit Error(SBE)/Busy Clear Interrupt (BCI) bit 12 – Hardware locked write error (HLE) bit 11 – FIFO underrun/overrun error (FRUN) bit 10 – Data starvation by host timeout (HTO)/Volt_switch_int bit 9 – Data read timeout (DRTO) bit 8 – Response timeout (RTO) bit 7 – Data CRC error (DCRC) bit 6 – Response CRC error (RCRC) bit 5 – Receive FIFO data request (RXDR) bit 4 – Transmit FIFO data request (TXDR) bit 3 – Data transfer over (DTO) bit 2 – Command done (CD) bit 1 – Response error (RE) bit 0 – Card detect (CD)

Table 9.3-174 RINTSTS

Bit Field	Name	Width	R/W	Default value	Description

31:16	sdio_interrupt	16	RW	0	Interrupt from SDIO card; one bit for each card. Bit[31] corresponds to Card[15], and bit[16] is for Card[0]. Writes to these bits clear them. Value of 1 clears bit and 0 leaves bit intact. 0 – No SDIO interrupt from card 1 – SDIO interrupt from card In MMC-Ver3.3-only mode, bits always 0. Bits are logged regardless of interrupt-mask status.
15:0	int_status	16	RW	0	Writes to bits clear status bit. Value of 1 clears status bit, and value of 0 leaves bit intact. Bits are logged regardless of interrupt mask status. bit 15 – End-bit error (read)/write no CRC (EBE) bit 14 – Auto command done (ACD) bit 13 – Start-bit error (SBE) /Busy Clear Interrupt (BCI) bit 12 – Hardware locked write error (HLE) bit 11 – FIFO underrun/overrun error (FRUN) bit 10 – Data starvation-by-host timeout (HTO) /Volt_switch_int bit 9 – Data read timeout (DRTO)/Boot Data Start (BDS) bit 8 – Response timeout (RTO)/Boot Ack Received (BAR) bit 7 – Data CRC error

					(DCRC) bit 6 – Response CRC error (RCRC) bit 5 – Receive FIFO data request (RXDR) bit 4 – Transmit FIFO data request (TXDR) bit 3 – Data transfer over (DTO) bit 2 – Command done (CD) bit 1 – Response error (RE) bit 0 – Card detect (CD)

Table 9.3-175 STATUS

Bit Field	Name	Width	R/W	Default value	Description
31	dma_req	1	R	0	DMA request signal state; either dw_dma_req or ge_dma_req, depending on DW-DMA or Generic-DMA selection.
30	dma_ack	1	R	0	DMA acknowledge signal state; either dw_dma_ack or ge_dma_ack, depending on DW-DMA or Generic-DMA selection.
29:17	fifo_count	13	R	0	FIFO count – Number of filled locations in FIFO.
16:11	response_index	6	R	0	Index of previous response, including any auto-stop sent by core
10	data_state_mc_busy	1	R	1	Data transmit or receive state-machine is busy
9	data_busy	1	R	1 or 0; depend on cdata_in	Inverted version of raw selected card_data[0] 0 – card data not busy 1 – card data bus
8	ata_3_status	1	R	1 or 0; depend on cdata_in	Raw selected card_data[3]; checks whether card is present

					0 – card not present 1 – card present
7:4	command fsm states	4	R	0	Command FSM states: 0 – Idle 1 – Send init sequence 2 – Tx cmd start bit 3 – Tx cmd tx bit 4 – Tx cmd index + arg 5 – Tx cmd crc7 6 – Tx cmd end bit 7 – Rx resp start bit 8 – Rx resp IRQ response 9 – Rx resp tx bit 10 – Rx resp cmd idx 11 – Rx resp data 12 – Rx resp crc7 13 – Rx resp end bit 14 – Cmd path wait NCC 15 – Wait; CMD-to-response turnaround NOTE: The command FSM state is represented using 19 bits. The STATUS Register(7:4) has 4 bits to represent the command FSM states. Using these 4 bits, only 16 states can be represented. Thus three states cannot be represented in the STATUS(7:4) register. The three states that are not represented in the STATUS Register(7:4) are: * Bit 16 – Wait for CCS * Bit 17 – Send CCSD * Bit 18 – Boot Mode Due to this, while command FSM is in "Wait for CCS state" or "Send CCSD" or "Boot Mode", the Status register indicates status

					as 0 for the bit field 7:4.
3	fifo_full	1	R	0	FIFO is full status
2	fifo_empty	1	R	1	FIFO is empty status
1	fifo_tx_watermark	1	R	1	FIFO reached Transmit watermark level; not qualified with data transfer.
0	fifo_rx_watermark	1	R	0	FIFO reached Receive watermark level; not qualified with data transfer

Table 9.3-176 FIFOTH

Bit Field	Name	Width	R/W	Default value	latency for the controller.
31	Reserved	1			

30:28	DW_DMA_Mutiple_Transaction_Size	3	RW	3'b000	Burst size of multiple transaction; should be programmed same as DW-DMA controller multiple-transaction-size SRC/DEST_MSIZE. 000 – 1 transfers 001 – 4 010 – 8 011 – 16 100 – 32 101 – 64 110 – 128 111 – 256 The units for transfers is the H_DATA_WIDTH parameter. A single transfer (dw_dma_single assertion in case of Non DW DMA interface) would be signalled based on this value. Value should be sub-multiple of (RX_WMark + 1)*(F_DATA_WIDTH/H_DATA_WIDTH) and (FIFO_DEPTH - TX_WMark)*(F_DATA_WIDTH/H_DATA_WIDTH) For example, if FIFO_DEPTH = 16, FDATA_WIDTH == H_DATA_WIDTH Allowed combinations for MSize and TX_WMark are: MSize = 1, TX_WMARK = 1-15 MSize = 4, TX_WMark = 8 MSize = 4, TX_WMark = 4 MSize = 4, TX_WMark = 12 MSize = 8, TX_WMark = 8

					MSize = 8, TX_WMark = 4 Allowed combinations for MSize and RX_WMark are: MSize = 1, RX_WMARK = 0-14 MSize = 4, RX_WMark = 3 MSize = 4, RX_WMark = 7 MSize = 4, RX_WMark = 11 MSize = 8, RX_WMark = 7 Recommended: MSize = 8, TX_WMark = 8, RX_WMark = 7

27:16	RX_WMark	12	RW	FIFO_DEPTH - 1	FIFO threshold watermark level when receiving data to card. When FIFO data count reaches greater than this number, DMA/FIFO request is raised. During end of packet, request is generated regardless of threshold programming in order to complete any remaining data. In non-DMA mode, when receiver FIFO threshold (RXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, interrupt is not generated if threshold programming is larger than any remaining data. It is responsibility of host to read remaining bytes on seeing Data Transfer Done interrupt. In DMA mode, at end of packet, even if remaining bytes are less than threshold, DMA request does single transfers to flush out any remaining bytes before Data Transfer Done interrupt is set. 12 bits – 1 bit less than FIFO-count of status register, which is 13 bits.

					Limitation: RX_WMark <= FIFO_DEPTH-2 Recommended: (FIFO_DEPTH/2) - 1; (means greater than (FIFO_DEPTH/2) - 1) NOTE: In DMA mode during CCS time-out, the DMA does not generate the request at the end of packet, even if remaining bytes are less than threshold. In this case, there will be some data left in the FIFO. It is the responsibility of the application to reset the FIFO after the CCS timeout.
15:12	Reserved	4	RW		
11:0	TX_WMak	12	RW	0	FIFO threshold watermark level when transmitting data to card. When FIFO data count is less than or equal to this number, DMA/FIFO request is raised. If Interrupt is enabled, then interrupt occurs. During end of packet, request or interrupt is generated, regardless of threshold programming. In non-DMA mode, when transmit FIFO threshold (TXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, on last interrupt, host is

					responsible for filling FIFO with only required remaining bytes (not before FIFO is full or after CIU completes data transfers, because FIFO may not be empty). In DMA mode, at end of packet, if last transfer is less than burst size, DMA controller does single cycles until required bytes are transferred. 12 bits – 1 bit less than FIFO-count of status register, which is 13 bits. Limitation: TX_WMark >= 1; Recommended: FIFO_DEPTH/2; (means less than or equal to FIFO_DEPTH/2)

Table 9.3-177 CDETECT

Bit Field	Name	Width	R/W	Default value	Description
31:30	Reserved	2			
29:0	TRAS_MIN_F0_0	30	R	card_detect_n inputs	Value on card_detect_n input ports (1 bit per card); read-only bits. 0 represents presence of card. Only NUM_CARDS number of bits are implemented.

Table 9-178 WRTPRT

Bit Field	Name	Width	R/W	Default value	Description
31:30	Reserved	2	R		
29:0	write_protect	30	R	card_write_prt inputs	Value on card_write_prt input ports (1 bit per card). 1 represents write protection.

					Only NUM_CARDS number of bits are implemented.

Table 9.3-179 GPIO

Bit Field	Name	Width	R/W	Default value	Description
31:24	Reserved	8			
23:8	gpo	16	Partly write/ read and partly read-only	0	Value needed to be driven to gpo pins; this portion of register is read/write. Valid only when AREA_OPTIMIZED parameter is 0.
7:0	gpi	8	Partly write/ read and partly read-only	gpi input	Value on gpi input ports; this portion of register is read-only. Valid only when AREA_OPTIMIZED parameter is 0.

Table 9.3-180 TCBCNT

Bit Field	Name	Width	R/W	Default value	Description
31:0	trans_card_byte_count	32	R	0	Number of bytes transferred by CIU unit to card. In 32-bit or 64-bit AMBA data-bus-width modes, register should be accessed in full to avoid read-coherency problems. In 16-bit AMBA data-bus-width mode, internal 16-bit coherency register is implemented. User should first read lower 16 bits and then higher 16 bits. When reading lower 16 bits, higher 16 bits of counter are stored in temporary register. When higher 16 bits are read, data from temporary register is supplied. Both TCBCNT and TBBCNT share

					same coherency register. When AREA_OPTIMIZED parameter is 1, register should be read only after data transfer completes; during data transfer, register returns 0.

Table 9.3-181 TBBCNT

Bit Field	Name	Width	R/W	Default value	Description
31:0	trans_fifo_byte_count	32	R	0	Number of bytes transferred between Host/DMA memory and BIU FIFO. In 32-bit or 64-bit AMBA data-bus-width modes, register should be accessed in full to avoid read-coherency problems. In 16-bit AMBA data-bus-width mode, internal 16-bit coherency register is implemented. User should first read lower 16 bits and then higher 16 bits. When reading lower 16 bits, higher 16 bits of counter are stored in temporary register. When higher 16 bits are read, data from temporary register is supplied. Both TCBCNT and TBBCNT share same coherency register.

Table 9.3-182 DEBNCE

Bit Field	Name	Width	R/W	Default value	Description
31:24	Reserved	8			
23:0	debounce_count	24	RW	24’hff_ffff	Number of host clocks (clk) used by debounce filter logic; typical debounce time is 5-25 ms.

Table 9.3-183 USRID

Bit Field	Name	Width	R/W	Default value	Description
31:0	USRID	32	RW	Configuration Value	User identification register; value set by user. Default reset value can be pickedby user while configuring core before synthesis. Can also be used as scratch pad register by user.

Table 9.3-184 VERID

Bit Field	Name	Width	R/W	Default value	Description
31:0	VERID	32	R	32'h5342_270a	Synopsys version identification register; register value is hard-wired. Can be read by firmware to support different versions of core.

Table 9.3-185 HCON

Bit Field	Name	Width	R/W	Default value	Description

31:0	HCON	32	R	Configuration Dependent	Hardware configurations selected by user before synthesizing core. Register values can be used to develop configuration-independent software drivers bit 0 – CARD_TYPE 0 – MMC_ONLY 1 – SD_MMC bit [5:1] – NUM_CARDS - 1 bit 6 – H_BUS_TYPE 0 – APB 1 – AHB bit [9:7] H_DATA_WIDTH 000 – 16 bits 001 – 32 bits 010 – 64 bits others – reserved bit [15:10] – H_ADDR_WIDTH 0 to 7 – reserved 8 – 9 bits 9 – 10 bits … 31 – 32 bits 32 to 63 – reserved bit[17:16] – DMA_INTERFACE 00 – none 01 – DW_DMA 10 – GENERIC_DMA 11 – NON-DW-DMA bit [20:18] GE_DMA_DATA_WIDTH 000 – 16 bits 001 – 32 bits 010 – 64 bits others – reserved bit 21 FIFO_RAM_INSIDE

					0 – outside 1 – inside bit 22 – IMPLEMENT_HOLD_REG 0 – no hold register 1 – hold register bit 23 – SET_CLK_FALSE_PATH 0 – no false path 1 – false path set bit [25:24] – NUM_CLK_DIVIDER-1 bit 26 - AREA_OPTIMIZED 0 – no area optimization 1 – Area optimization For 64-bit Address Configuration only bit 27 - ADDR_CONFIG 0 – 32-bit addressing supported 1 – 64-bit addressing supported For 32-bit Address Configuration only: bit [31:27] – reserved (0) For 64-bit Address Configuration only: bit [31:28] – reserved (0) For FIFO_DEPTH parameter, power-on value of RX_WMark value of FIFO Threshold Watermark Register represents FIFO_DEPTH - 1 .

Table 9.3-186 UHS_REG

Bit Field	Name	Width	R/W	Default value	Description

31:16	DDR_REG	16	RW	0	DDR mode. These bits indicate DDR mode of operation to the core for the data transfer. 0 – Non-DDR mode 1 – DDR mode UHS_REG [16] should be set for card number 0, UHS_REG [17] for card number 1 and so on. Note: Clear these bits in HS400 mode.
15:0	VOLT_REG	16	RW	0	High Voltage mode. Determines the voltage fed to the buffers by an external voltage regulator. 0 – Buffers supplied with 3.3V Vdd 1 – Buffers supplied with 1.8V Vdd These bits function as the output of the host controller and are fed to an external voltage regulator. The voltage regulator must switch the voltage of the buffers of a particular card to either 3.3V or 1.8V, depending on the value programmed in the register. VOLT_REG[0] should be set to 1'b1 for card number 0 in order to make it operate for 1.8V.

Table 9.3-187 RST_n

Bit Field	Name	Width	R/W	Default value	Description
31:16	Reserved	16	RW		
15:0	CARD_RESET	16	RW	1	Hardware reset. 1 – Active mode 0 – Reset These bits cause the cards to enter pre-idle state, which requires them to be re-initialized. CARD_RESET[0] should be set to 1’b0 to reset card number 0 CARD_RESET[15] should be set to 1'b0 to reset card number 15. The number of bits implemented is restricted to NUM_CARDS.

Table 9.3-188 BMOD

Bit Field	Name	Width	R/W	Default value	Description
31:11	Reserved	21			
10:8	PBL	3	RW	3'h0	Programmable Burst Length. These bits indicate the maximum number of beats to be performed in one IDMAC transaction. The IDMAC will always attempt to burst as specified in PBL each time it starts a Burst transfer on the host bus. The permissible values are 1, 4, 8, 16, 32, 64, 128 and 256. This value is the mirror of MSIZE of FIFOTH register. In order to change this value, write the required value to FIFOTH register. This is

					an encode value as follows. 000 – 1 transfers 001 – 4 transfers 010 – 8 transfers 011 – 16 transfers 100 – 32 transfers 101 – 64 transfers 110 – 128 transfers 111 – 256 transfers Transfer unit is either 16, 32, or 64 bits, based on HDATA_WIDTH. PBL is a read-only value and is applicable only for Data Access; it does not applyto descriptor accesses
7	DE	1	RW	0	IDMAC Enable. When set, the IDMAC is enabled. DE is read/write
6:2	DSL	5	RW	5'h0	Descriptor Skip Length. Specifies the number of HWord/Word/Dword (depending on 16/32/64-bit bus) to skip between two unchained descriptors. This is applicable only for dual buffer structure. DSL is read/write.
1	FB	1	RW	0	Fixed Burst. Controls whether the AHB Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 or INCR16 during start of normal burst transfers. When reset, the AHB will use

					SINGLE and INCR burst transfer operations
0	SWR	1	RW	0	Software Reset. When set, the DMA Controller resets all its internal registers. SWR is read/write. It is automatically cleared after 1 clock cycle.

Table 9.3-189 PLDMND

Bit Field	Name	Width	R/W	Default value	Description
31:0	PD	32	W	32’h0000_0000	Poll Demand. If the OWN bit of a descriptor is not set, the FSM goes to the Suspend state. The host needs to write any value into this register for the IDMAC FSM to resume normal descriptor fetch operation. This is a write only register.

Table 9.3-190 DBADDR

Bit Field	Name	Width	R/W	Default value	Description
31:0	SDL	32	RW	32’h0000_0000	Start of Descriptor List. Contains the base address of the First Descriptor. The LSB bits [0/1/2:0] for 16/32/64-bit bus-width) are ignored and taken as all-zero by the IDMAC internally. Hence these LSB bits are read-only.

Table 9.3-191 IDSTS

Bit Field	Name	Width	R/W	Default value	Description

31:17	Reserved	15			
16:13	FSM	4	R	32'h0	DMAC FSM present state. 0 – DMA_IDLE 1 – DMA_SUSPEND 2 – DESC_RD 3 – DESC_CHK 4 – DMA_RD_REQ_WAIT 5 – DMA_WR_REQ_WAIT 6 – DMA_RD 7 – DMA_WR 8 – DESC_CLOSE This bit is read-only
12:10	FBE_CODE	3	R	3'b000	Fatal Bus Error Code. Indicates the type of error that caused a Bus Error. Valid only with Fatal Bus Error bit—IDSTS[2] (IDSTS64[2], in case of 64-bit address configuration) set. This field does not generate an interrupt. 3'b001 – Host Abort received during transmission 3'b010 – Host Abort received during reception Others: Reserved EB is read-only.
9	AIS	1	RW	0	Abnormal Interrupt Summary. Logical OR of the following: IDSTS[2] (IDSTS64[2], in case of 64-bit address configuration) – Fatal Bus Interrupt IDSTS[4] (IDSTS64[4], in case of 64-bit address configuration) – DU bit Interrupt Only unmasked bits affect

					this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared. Writing a 1 clears this bit.
8	NIS	1	RW	0	Normal Interrupt Summary. Logical OR of the following: IDSTS[0] (IDSTS64[0], in case of 64-bit address configuration) – Transmit Interrupt IDSTS[1] (IDSTS64[1], in case of 64-bit address configuration) – Receive Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes NIS to be set is cleared. Writing a 1 clears this bit.
7:6	Reserved	2			
5	CES	1	RW	0	Card Error Summary. Indicates the status of the transaction to/from the card; also present in RINTSTS. Indicates the logical OR of the following bits: EBE – End Bit Error RTO – Response Timeout/Boot Ack Timeout RCRC – Response CRC SBE – Start Bit Error DRTO – Data Read Timeout/BDS timeout DCRC – Data CRC for Receive RE – Response Error

					Writing a 1 clears this bit. The abort condition of the IDMAC depends on the setting of this CES bit. If the CES bit is enabled, then the IDMAC aborts on a “response error”; however, it will not abort if theCES bit is cleared.
4	DU	1	RW	0	Descriptor Unavailable Interrupt. This bit is set when the descriptor is unavailable due to OWN bit = 0 (DES0[31] =0). Writing a 1 clears this bi
3	Reserved	1			
2	FBE	1	RW	0	Fatal Bus Error Interrupt. Indicates that a Bus Error occurred (IDSTS[12:10]) (IDSTS64[12:10], in case of 64-bit address configuration). When this bit is set, the DMA disables all its bus accesses. Writing a 1 clears this bit.
1	RI	1	RW	0	Receive Interrupt. Indicates the completion of data reception for a descriptor. Writing a1 clears this bit.
0	TI	1	RW	0	Transmit Interrupt. Indicates that data transmission is finished for a descriptor. Writing a ‘1’ clears this bit.

Table 9.3-192 IDINTEN

Bit Field	Name	Width	R/W	Default value	Description
31:10	Reserved	22			

9	AI	1	RW	0	Abnormal Interrupt Summary Enable. When set, an abnormal interrupt is enabled. This bit enables the following bits: IDINTEN[2] (IDINTEN64[2], in case of 64-bit address configuration) – Fatal Bus Error Interrupt IDINTEN[4] (IDINTEN64[4], in case of 64-bit address configuration) – DU Interrupt
8	NI	1	RW	0	Normal Interrupt Summary Enable. When set, a normal interrupt is enabled. When reset, a normal interrupt is disabled. This bit enables the following bits: IDINTEN[0] (IDINTEN64[0], in case of 64-bit address configuration) – Transmit Interrupt IDINTEN[1] (IDINTEN64[1], in case of 64-bit address configuration) – Receive Interrupt
7:6	Reserved	2			
5	CES	1	RW	0	Card Error summary Interrupt Enable. When set, it enables the Card Interrupt summary
4	DU	1	RW	0	Descriptor Unavailable Interrupt. When set along with Abnormal Interrupt Summary Enable, the DU interrupt is enabled
3	Reserved	1			
2	FBE	1	RW	0	Fatal Bus Error Enable. When set, the Fatal Bus Error Interrupt is enabled. When reset, Fatal Bus Error Enable Interrupt is disabled.

1	RI	1	RW	0	Receive Interrupt Enable. When set, Receive Interrupt is enabled. When reset,Receive Interrupt is disabled.
0	TI	1	RW	0	Transmit Interrupt Enable. When set, Transmit Interrupt is enabled. When reset,Transmit Interrupt is disabled

Table 9.3-193 DSCADDR

Bit Field	Name	Width	R/W	Default value	Description
31:0	HAD	32	R	32’h0000_0000	Host Descriptor Address Pointer. Cleared on reset. Pointer updated by IDMAC during operation. This register points to the start address of the current descriptor read by the IDMAC.

Table 9.3-194 BUFADDR

Bit Field	Name	Width	R/W	Default value	Description
31:0	HBA	32	R	32’h0000_0000	Host Buffer Address Pointer. Cleared on Reset. Pointer updated by IDMAC during operation. This register points to the current Data Buffer Address being accessed by the IDMAC.

Table 9.3-195 CardThrCtl

Bit Field	Name	Width	R/W	Default value	Description
31:28	Reserved	4			
N:16	CardThreshold	12	RW	12h'000	Card Threshold size; N depends on the FIFO size: N = 27, FIFO Size = 512 N = 26, FIFO Size = 256 N = 25, FIFO Size = 128 N = 24, FIFO Size = 64 N = 23, FIFO Size = 32 N = 22, FIFO Size = 16

					N = 21, FIFO Size = 8 Note: This register is applicable when CardWrThrEn is set to '1' or CardRdThrEn is set to '1'
15:3	Reserved	13			
2	CardWrThrEn	1	RW	0	Card Write Threshold Enable. Applicable when HS400 mode is enabled 0 - Card write Threshold disabled 1 - Card write Threshold enabled
1	BsyClrIntEn	1	RW	0	Busy Clear Interrupt generation: 1'b0 - Busy Clear Interrupt disabled 1'b1 - Busy Clear Interrupt enabled Note: The application can disable this feature if it does not want to wait for a Busy Clear Interrupt. For example, in a multi-card scenario, the application can switch to the other card without waiting for a busy to be completed. In such cases, the application can use the polling method to determine the status of busy. By default this feature is disabled and backward-compatible to the legacy drivers where polling is used.
0	CardRdThrEn	1	RW	0	Card Read Threshold Enable 1'b0 - Card Read Threshold disabled 1'b1 - Card Read Threshold

					enabled. Host Controller initiates Read

Table 9.3-196 Back_end_power

Bit Field	Name	Width	R/W	Default value	Description
31:16	Reserved	16			
15:0	Back End power	16	RW	0	Back end power 1'b0 – Off; Reset 1'b1 – Back-end Power supplied to card application; one pin per card

Table 9.3-197 UHS_REG_EXT

Bit Field	Name	Width	R/W	Default value	Description
31:30	EXT_CLK_MUX_CTRL	2	RW	0	Input clock control for cclk_in. The MUX controlled by these bits exists outside Mobile storage IP
29:23	CLK_DRV_PHASE_CTRL	7	RW	0	Control for amount of phase shift on cclk_in_drv clock. Can choose three MSBs to control delay lines and four LSBs to control phase shift; alternatively, use only LSBs
22:16	CLK_SMPL_PHASE_CTRL	7	RW	0	Control for amount of phase shift on cclk_in_sample clock. Can choose three MSBs to control delay lines and four LSBs to control phase shift; alternatively, use only LSB
15:0	MMC_VOLT_REG	16	RW	0	Support for 1.2V. MMC_VOLT_REG bits; must be read in combination with UHS_VOLT_REG to decode output selected voltage. The biu_volt_reg_1_2[NUM_CA

					RD_BUS-1:0] signal decodes the voltage combination selected for the I/O voltage logic. Host controllers that support only SD standard or standard versions before eMMC4.41 do not program MMC_VOLT_REG. Only host controllers that support all three versions—3.3,1.8,1.2 V—can program MMC_VOLT_REG and connect biu_volt_reg_1_2

Table 9.3-198 EMMC_DDR_REG

Bit Field	Name	Width	R/W	Default value	Description
31	hs400_mode	1	RW	0	HS400 Mode enable 1 - Enable 0 - Disable Note: The application is required to set this bit to '1' before initiating any data transfer CMD in HS400 mode. This bit shall be cleared by the host on exiting HS400 mode. In non HS400 mode, this bit shall be set to '0'.
30:16	Reserved	15			
15:0	HALF_START_BIT	16	RW	0	Control for start bit detection mechanism inside Mobile storage based on duration of start bit; each bit refers to one slot. For eMMC 4.5, start bit can be: Full cycle (HALF_START_BIT = 0) Less than one full cycle (HALF_START_BIT = 1) Set HALF_START_BIT=1

					for eMMC 4.5 and above; set to 0 for SD applications. Note: This bit is not applicable for HS400 mode. This bit is also ignored for a DDR operation when either of the parameter START_BIT_SAMPLING_DDR or ASYNC_SAMPLE_DATA_PATH is set to 1

Table 9.3-199 ENABLE_SHIFT

Bit Field	Name	Width	R/W	Default value	Description
31:0	Enable_ 00_shift	32	RW	0	ontrol for the amount of phase shift provided on the default enables in the design. Two bits are assigned for each card/slot. For example, bits[1:0] control slot0 and indicate the following. 00 – Default phase shift 01 – Enables shifted to next immediate positive edge 10 – Enables shifted to next immediate negative edge 11 – Reserved

121.2. UART

121.2.1. Overview

The uart is modeled after the industry-standard 16550. However, the register address space is relocated to 32-bit data boundaries for APB bus implementation. Data is written from a master (CPU) over the APB bus to the UART, and it is converted to serial form and transmitted to the destination device. Serial data is also received by the UART and stored for the master (CPU) to read back.

121.2.2. Features

AMBA APB interface allows integration into AMBA SoC implementations

- 9-bit serial data support
- False start bit detection
- Programmable fractional baud rate support
- Configurable parameters for the following:
    - APB data bus widths of 32
    - Additional DMA interface signals for compatibility with DesignWare DMA interface
    - DMA interface signal polarity
- Configuration identification registers present
- Programmable FIFO enable/disable
- Programmable serial data baud rate as calculated by the following:
  baud rate = (serial clock frequency)/(16×divisor)
- Modem and status lines are independently controlled

121.2.3. Function Description

121.2.3.1. Uart Block Diagram

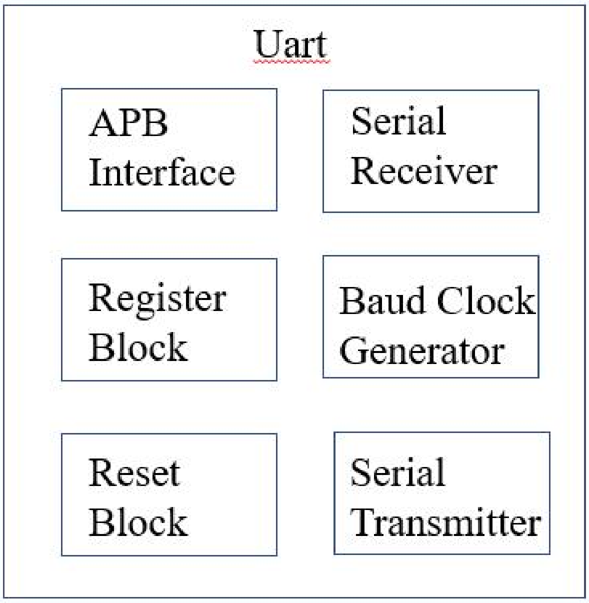


Figure 9- 12 uart Functional Block Diagram

121.2.3.2. Timing Description

Because the serial communication between the uart and a selected device is asynchronous, additional bits (start and stop) are added to the serial data to indicate the beginning and end. Utilizing these bits allows two devices to be synchronized. This structure of serial data—accompanied by start and stop bits—is referred to as a character, as shown in Figure1.1.2:

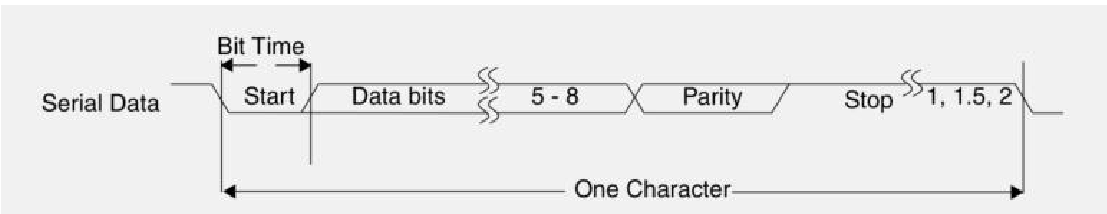


Figure 9- 13 Serial Data Format

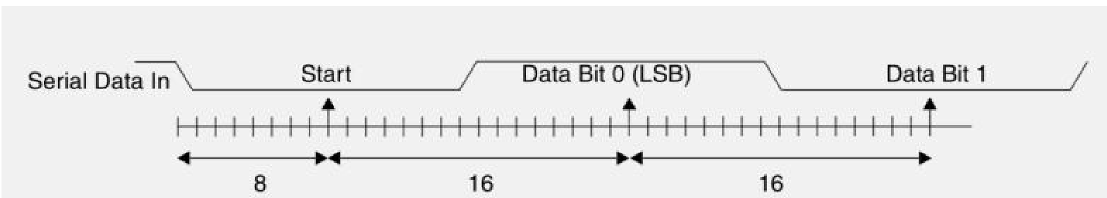


Figure 9- 14 Receiver Serial Data Sample Points

The uart can be configured to have 9-bit data transfer in both transmit and receive mode. The 9th bit in the character appears after the 8th bit and before the parity bit in the character.Figure1.1.4 shows the serial transmission for a character in which D8 represents the 9th bit and also shows general serial transmission in the 9-bit mode.

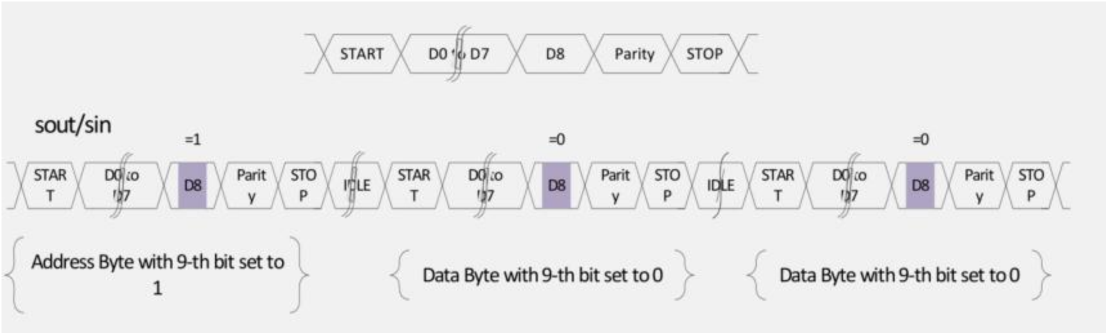


Figure 9- 15 9-Bit Character

121.2.4. **Programming Guide**

121.2.4.1. **Configuration of the uart for 9-bit data transfer following:**

1. LCR_EXT[0] bit is used to enable or disable the 9-bit data transfer.
2. LCR_EXT[1] bit is used to choose between hardware and software based address match in the case of receive.
3. LCR_EXT[2] bit is used to enable to send the address in the case of transmit.
4. LCR_EXT[3] bit is used to choose between hardware and software based address transmission.
5. TAR and RAR registers are used to transmit address and to match the received address, respectively.
6. THR, RBR, STHR and SRBR registers are of 9-bit which is used to do the data transfers in 9-bit mode.
7. LSR[8] bit is used to indicate the address received interrupt.

121.2.4.2. **About Interrupt**

Assertion of the uart interrupt output signal (intr)—a positive-level interrupt—occurs whenever one of the several prioritized interrupt types are enabled and active. When an interrupt occurs, the master accesses the IIR register. The following interrupt types can be enabled with the IER register:
1. Receiver Error
2. Receiver Data Available
3. Character Timeout (in FIFO mode only)
4. Transmitter Holding Register Empty at/below threshold (in Programmable THRE interrupt mode)
5. Modem Status
6. Busy Detect Indication

121.2.4.3. Example DMA Flow

The extra handshaking signals are explained in the following DMA flow for a uart that is configured with FIFOs and Programmable THRE interrupt mode. As a block flow control device, the DMA Controller is programmed by the processor with the number of data items (block size) that are to be transmitted or received by the uart; this is programmed into the BLOCK_TS field of the CTLx register. The block is broken into a number of transactions, each initiated by a request from the uart. The DMA Controller must also be programmed with the number of data items (in this case, uart FIFO entries) to be transferred for each DMA request. This is also known as the burst transaction length, and is programmed into the SRC_MSIZE/DEST_MSIZE fields of the dmac CTLx register for source and destination, respectively.

121.2.5. Register Summary

Table 9- 200 UART Register Summary

Addr offset	Name	Width	R/W	Default value	Description
0x0	RBR	32	R	0x00000000	Receive Buffer Register (Dependencies: LCR[7] bit = 0)
	THR	32	W	0x00000000	Transmit Holding Register (Dependencies: LCR[7] bit = 0)
	DLL	32	RW	0x00000000	Divisor Latch (Low) (Dependencies: LCR[7] bit = 1)
0x4	DLH	32	RW	0x00000000	Divisor Latch (High) (Dependencies: LCR[7] bit = 1)
	IER	32	RW	0x00000000	Interrupt Enable Register (Dependencies: LCR[7] bit = 0)
0x8	IIR	32	R	0x00000001	Interrupt Identification Register
	FCR	32	W	0x00000000	FIFO Control Register
0xC	LCR	32	RW	0x00000000	Line Control Register
0x10	MCR	32	RW	0x00000000	Modem Control Register
0x14	LSR	32	R	0x00000060	Line Status Register
0x18	MSR	32	R	0x00000000	Modem Status Register
0x1C	SCR	32	RW	0x00000000	Scratchpad Register
0x20	LPDLL	32	RW	0x00000000	Low Power Divisor Latch (Low) Register (Dependencies: LCR[7] bit = 1)
0x24	LPDLH	32	RW	0x00000000	Low Power Divisor Latch (High) Register (Dependencies: LCR[7] bit = 1)
0x30-0x6C	SRBR	32	R	0x00000000	Shadow Receive Buffer Register (Dependencies: LCR[7] bit = 0)

	STHR	32	W	0x00000000	Shadow Transmit Holding Register (Dependencies: LCR[7] bit = 0)
0x70	FAR	32	RW	0x00000000	FIFO Access Register
0x74	TFR	32	R	0x00000000	Transmit FIFO Read
0x78	RFW	32	W	0x00000000	Receive FIFO Write
0x7C	USR	32	R	0x00000006	UART Status Register
0x80	TFL	32	R	0x00000000	Transmit FIFO Level
0x84	RFL	32	R	0x00000000	Receive FIFO Level
0x88	SRR	32	W	0x00000000	Software Reset Register
0x8C	SRTS	32	RW	0x00000000	Shadow Request to Send
0x90	SBCR	32	RW	0x00000000	Shadow Break Control Register
0x94	SDMAM	32	RW	0x00000000	Shadow DMA Mode
0x98	SFE	32	RW	0x00000000	Shadow FIFO Enable
0x9C	SRT	32	RW	0x00000000	Shadow RCVR Trigger
0xA0	STET	32	RW	0x00000000	Shadow TX Empty Trigger
0xA4	HTX	32	RW	0x00000000	Halt TX
0xA8	DMASA	32	W	0x00000000	DMA Software Acknowledge
0xC0	DLF	32	W	0x00000000	Divisor Latch Fraction Register
0xC4	RAR	32	W	0x00000000	Receive Address Register
0xC8	TAR	32	W	0x00000000	Transmit Address Register
0xCC	LCR_EXT	32	RW	0x00000000	Line Extended Control Register
0xF4	CPR	32	R	0x00043f32	Component Parameter Register
0xF8	UCV	32	R	0x3430302a	UART Component Version
0xFC	CTR	32	R	0x44570110	Component Type Register

121.2.6. **Register Description**

121.2.6.1. **RBR**

Table 9- 201 Fields for Register: RBR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				

7:0	Receive Buffer Register	8	R	0x00	Data byte received on the serial input port (sin) in UART mode, or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line Status Register (LSR) is set. If in non-FIFO mode (FIFO_MODE = NONE) or FIFOs are disabled (FCR[0] set to 0), the data in the RBR must be read before the next data arrives, otherwise it is overwritten, resulting in an over-run error. If in FIFO mode (FIFO_MODE != NONE) and FIFOs are enabled (FCR[0] set to 1), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO is preserved, but any incoming data are lost and an over-run error occurs.

121.2.6.2. **THR**

Table 9- 202 Fields for Register: THR

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
7:0	Transmit Holding Register	8	W	0x00	Data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set. If in non-FIFO mode or FIFOs are disabled (FCR[0] = 0) and THRE is set, writing a single character to the THR clears the THRE. Any additional writes to the THR before the THRE is set again causes the THR data to be overwritten. If in FIFO mode and FIFOs are enabled (FCR[0] = 1) and THRE is set, x number of characters of data may be written to the THR before the FIFO is full. The number x (default=16) is determined by the value of FIFO Depth that you set during configuration. Any attempt to write data when the FIFO is full results in the

					write data being lost.

121.2.6.3. DLL

Table 9- 203 Fields for Register: DLL

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
7:0	Divisor Latch (Low)	8	RW	0x00	Lower 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. The output baud rate is equal to the serial clock (pclk if one clock design, sclk if two clock design (CLOCK_MODE = Enabled) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to 0, the baud clock is disabled and no serial communications occur. Also, once the DLL is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

121.2.6.4. DLH

Table 9- 204 Fields for Register: DLH

Bit Field	Name	Width	R/W	Default value	Description

31:1	Reserved				
7:0	Divisor Latch (High)	8	RW	0x00	Upper 8-bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. The output baud rate is equal to the serial clock (pclk if one clock design, sclk if two clock design (CLOCK_MODE = Enabled) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to 0, the baud clock is disabled and no serial communications occur. Also, once the DLH is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

121.2.6.5. **IER**

Table 9- 205 Fields for Register: IER

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7	PTIME	1	RW	0x0	Programmable THRE Interrupt Mode Enable that can be written to only when THRE_MODE_USER = Enabled, always readable. This is used to enable/disable the generation of THRE Interrupt. 0 – disabled 1 – enabled
6:4	Reserved				
3	EDSSI	1	RW	0x0	Enable Modem Status Interrupt. This is used to enable/disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt. 0 – disabled 1 – enabled
2	ELSI	1	RW	0x0	Enable Receiver Line Status Interrupt. This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt. 0 – disabled 1 – enabled

1	ETBEI	1	RW	0x0	Enable Transmit Holding Register Empty Interrupt. This is used to enable/disable the generation of Transmitter Holding Register Empty Interrupt. This is the third highest priority interrupt. 0 – disabled 1 – enabled
0	ERBFI	1	RW	0x0	Enable Received Data Available Interrupt. This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode and FIFOs enabled). These are the second highest priority interrupts. 0 – disabled 1 – enabled

121.2.6.6. **IIR**

Table 9-206 Fields for Register: IIR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7:6	FIFOs Enabled	2	R	0x0	FIFOs Enabled. This is used to indicate whether the FIFOs are enabled or disabled. 00 – disabled 11 – enabled
5:4	Reserved				
3:0	Interrupt ID	4	R	0x0	Interrupt ID. This indicates the highest priority pending interrupt which can be one of the following types: 0000 – modem status 0001 – no interrupt pending 0010 – THR empty 0100 – received data available 0110 – receiver line status 0111 – busy detect 1100 – character timeout

121.2.6.7. **FCR**

Table 9- 207 Fields for Register: FCR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7:6	RCVR Trigger	2	W	0x0	RCVR Trigger. This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode, this trigger is used to determine when the rts_n signal is de-asserted only when RTC_FCT is disabled. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation. The following trigger levels are supported: 00 – 1 character in the FIFO 01 – FIFO ¼ full 10 – FIFO ½ full 11 – FIFO 2 less than full
5:4	TX Empty	2	W	0x0	TX Empty Trigger. Writes have no effect when THRE_MODE_USER = Disabled. This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation.The following trigger levels are supported: 00 – FIFO empty 01 – 2 characters in the FIFO 10 – FIFO ¼ full 11 – FIFO ½ full
3	DMA Mode	1	W	0x0	DMA Mode. This determines the DMA signalling mode used for the dma_tx_req_n and dma_rx_req_n output signals when additional DMA handshaking signals are not selected (DMA_EXTRA = No). 0 = mode 0 1 = mode 1

2	XMIT FIFO Reset	1	W	0x0	XMIT FIFO Reset. This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request and single signals when additional DMA handshaking signals are selected (DMA_EXTRA = YES). Note that this bit is 'self-clearing'. It is not necessary to clear this bit.
1	RCVR FIFO Reset	1	W	0x0	RCVR FIFO Reset. This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request and single signals when additional DMA handshaking signals are selected (DMA_EXTRA = YES). Note that this bit is 'self-clearing'. It is not necessary to clear this bit.
0	FIFO Enable	1	W	0x0	FIFO Enable. This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. Whenever the value of this bit is changed both the XMIT and RCVR controller portion of FIFOs is reset.

121.2.6.8. **LCR**

Table 9- 208 Fields for Register: LCR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7	DLAB	1	RW	0x0	Divisor Latch Access Bit. This bit is used to enable reading and writing of the Divisor Latch register (DLL and DLH/LPDLL and LPDLH) to set the baud rate of the UART. This bit must be cleared after initial baud rate setup in order to access other registers.

6	Break	1	RW	0x0	Break Control Bit.This is used to cause a break condition to be transmitted to the receiving device. If set to 1, the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared. If SIR_MODE = Enabled and active (MCR[6] set to 1) the sir_out_n line is continuously pulsed. When in Loopback Mode, the break condition is internally looped back to the receiver and the sir_out_n line is forced low.
5	Stick Parity	1	RW	0x0	Stick Parity.This bit is used to force parity value. When PEN, EPS, and Stick Parity are set to 1, the parity bit is transmitted and checked as logic 0. If PEN and Stick Parity are set to 1 and EPS is a logic 0, then parity bit is transmitted and checked as a logic 1. If this bit is set to 0, Stick Parity is disabled.
4	EPS	1	RW	0x0	Even Parity Select. otherwise always writable, always readable. This is used to select between even and odd parity, when parity is enabled (PEN set to 1). If set to 1, an even number of logic 1s is transmitted or checked. If set to 0, an odd number of logic 1s is transmitted or checked.
3	PEN	1	RW	0x0	Parity Enable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively. 0 – parity disabled 1 – parity enabled
2	STOP	1	RW	0x0	Number of stop bits. This is used to select the number of stop bits per character that the peripheral transmits and receives. If set to 0, one stop bit is transmitted in the serial data. If set to 1 and the data bits are set to 5 (LCR[1:0] set to 0) one and a half stop bits is transmitted. Otherwise, two stop bits are transmitted. Note that regardless of the number of stop bits selected, the receiver checks only the first stop bit. 0 – 1 stop bit 1 – 1.5 stop bits when DLS (LCR[1:0]) is 0,

					else 2 stop bit [Note] The STOP bit duration implemented by UART may appear longer due to idle time inserted between characters for some configurations and baud clock divisor values in the transmit direction; for details on idle time between transmitted transfers.
1:0	DLS	2	RW	0x0	Data Length Select. When DLS_E in LCR_EXT is set to 0, this register is used to select the number of data bits per character that the peripheral transmits and receives. The number of bits that may be selected are as follows: 00 – 5 bits 01 – 6 bits 10 – 7 bits 11 – 8 bits

121.2.6.9. MCR

Table 9- 209 Fields for Register: MCR

Bit Field	Name	Width	R/W	Default value	Description
31:7	Reserved				
6	SIRE	1	R	0x0	SIR Mode Enable. Read as zero. Because we do not support IrDA SIR Mode.
5	AFCE	1	RW	0x0	Auto Flow Control Enable. When FIFOs are enabled and the Auto Flow Control Enable (AFCE) bit is set. 0 – Auto Flow Control Mode disabled 1 – Auto Flow Control Mode enabled
4	LoopBack	1	RW	0x0	LoopBack Bit. This is used to put the UART into a diagnostic mode for test purposes. If MCR[6] set to 0, data on the sout line is held high, while serial data output is looped back to the sin line, internally. In this mode all the interrupts are fully functional. Also, in loopback mode, the modem control inputs (dsr_n, cts_n, ri_n, dcd_n) are disconnected and the modem control outputs (dtr_n, rts_n, out1_n, out2_n) are looped back to the inputs, internally.

3	OUT2	1	RW	0x0	OUT2. This is used to directly control the user-designated Output2 (out2_n) output. The value written to this location is inverted and driven out on out2_n, that is: 0 – out2_n de-asserted (logic 1) 1 – out2_n asserted (logic 0) Note that in Loopback mode (MCR[4] set to 1), the out2_n output is held inactive high while the value of this location is internally looped back to an input.
2	OUT1	1	RW	0x0	OUT1. This is used to directly control the user-designated Output1 (out1_n) output. The value written to this location is inverted and driven out on out1_n, that is: 0 – out1_n de-asserted (logic 1) 1 – out1_n asserted (logic 0) Note that in Loopback mode (MCR[4] set to 1), the out1_n output is held inactive high while the value of this location is internally looped back to an input.
1	RTS	1	RW	0x0	Request to Send. This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data. When Auto RTS Flow Control is not enabled (MCR[5] set to 0), the rts_n signal is set low by programming MCR[1] (RTS) to a high.In Auto Flow Control, AFCE_MODE = Enabled and active (MCR[5] set to 1) and FIFOs enable (FCR[0] set to 1), the rts_n output is controlled in the same way, but is also gated with the receiver FIFO threshold trigger (rts_n is inactive high when above the threshold) only when the RTC Flow Trigger is disabled; otherwise it is gated by the receiver FIFO almost-full trigger, where “almost full” refers to two available slots in the FIFO (rts_n is inactive high when above the threshold). The rts_n signal is de-asserted when MCR[1] is set low. Note that in Loopback mode (MCR[4] set to 1), the rts_n output is held inactive high while the value of this location is internally looped back to an

					input.
0	DTR	1	RW	0x0	Data Terminal Ready. This is used to directly control the Data Terminal Ready (dtr_n) output. The value written to this location is inverted and driven out on dtr_n, that is: 0 – dtr_n de-asserted (logic 1) 1 – dtr_n asserted (logic 0) The Data Terminal Ready output is used to inform the modem or data set that the UART is ready to establish communications. Note that in Loopback mode (MCR[4] set to 1), the dtr_n output is held inactive high while the value of this location is internally looped back to an input.

121.2.6.10. **LSR**

Table 9- 210 Fields for Register: LSR

Bit Field	Name	Width	R/W	Default value	Description
31:9	Reserved				
8	ADDR_RCVD	1	RW	0x0	Address Received bit If 9-bit data mode (LCR_EXT[0]=1) is enabled, this bit is used to indicate that the 9th bit of the receive data is set to 1. This bit can also be used to indicate whether the incoming character is an address or data. 1 - Indicates that the character is an address. 0 - Indicates that the character is data. In the FIFO mode, since the 9th bit is associated with the received character, it is revealed when the character with the 9th bit set to 1 is at the top of the FIFO list. Reading the LSR clears the 9th bit
7	RFE	1	R	0x0	Receiver FIFO Error bit. This bit is only relevant when FIFO_MODE != NONE AND FIFOs are enabled (FCR[0] set to 1). This is used to indicate if there is at

					least one parity error, framing error, or break indication in the FIFO. 0 – no error in RX FIFO 1 – error in RX FIFO This bit is cleared when the LSR is read and the character with the error is at the top of the receiver FIFO and there are no subsequent errors in the FIFO.
6	TEMT	1	R	0x1	Transmitter Empty bit. If FIFOs enabled (FCR[0] set to 1), this bit is set whenever the Transmitter Shift Register and the FIFO are both empty. If in non-FIFO mode or FIFOs are disabled, this bit is set whenever the Transmitter Holding Register and the Transmitter Shift Register are both empty.
5	THRE	1	R	0x1	Transmit Holding Register Empty bit. If THRE_MODE_USER = Disabled or THRE mode is disabled (IER[7] set to 0) and regardless of FIFO's being implemented/enabled or not, this bit indicates that the THR or TX FIFO is empty. This bit is set whenever data is transferred from the THR or TX FIFO to the transmitter shift register and no new data has been written to the THR or TX FIFO. This also causes a THRE Interrupt to occur, if the THRE Interrupt is enabled. If THRE_MODE_USER = Enabled AND FIFO_MODE != NONE and both modes are active (IER[7] set to 1 and FCR[0] set to 1 respectively), the functionality is switched to indicate the transmitter FIFO is full, and no longer controls THRE interrupts, which are then controlled by the FCR[5:4] threshold setting.

4	BI	1	R	0x0	Break Interrupt bit. This is used to indicate the detection of a break sequence on the serial input data. If in UART mode (SIR_MODE = Disabled), it is set whenever the serial input, sin, is held in a logic '0' state for longer than the sum of start time + data bits + parity + stop bits. If in infrared mode (SIR_MODE = Enabled), it is set whenever the serial input, sir_in, is continuously pulsed to logic '0' for longer than the sum of start time + data bits + parity + stop bits. A break condition on serial input causes one and only one character, consisting of all 0s, to be received by the UART. In FIFO mode, the character associated with the break condition is carried through the FIFO and is revealed when the character is at the top of the FIFO. Reading the LSR clears the BI bit. In non-FIFO mode, the BI indication occurs immediately and persists until the LSR is read.
3	FE	1	R	0x0	Framing Error bit. This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data. In the FIFO mode, since the framing error is associated with a character received, it is revealed when the character with the framing error is at the top of the FIFO. When a framing error occurs, the UART tries to resynchronize. It does this by assuming that the error was due to the start bit of the next character and then continues receiving the other bit; that is, data, and/or parity and stop. It should be noted that the Framing Error (FE) bit (LSR[3]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]). This happens because the break character implicitly generates a

					framing error by holding the sin input to logic 0 for longer than the duration of a character. 0 – no framing error 1 – framing error Reading the LSR clears the FE bit.
2	PE	1	R	0x0	Parity Error bit. This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set. In the FIFO mode, since the parity error is associated with a character received, it is revealed when the character with the parity error arrives at the top of the FIFO. It should be noted that the Parity Error (PE) bit (LSR[2]) can be set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]). In this situation, the Parity Error bit is set if parity generation and detection is enabled (LCR[3]=1) and the parity is set to odd (LCR[4]=0). 0 – no parity error 1 – parity error Reading the LSR clears the PE bit.
1	OE	1	R	0x0	Overrun error bit. This is used to indicate the occurrence of an overrun error. This occurs if a new data character was received before the previous data was read. In the non-FIFO mode, the OE bit is set when a new character arrives in the receiver before the previous character was read from the RBR. When this happens, the data in the RBR is overwritten. In the FIFO mode, an overrun error occurs when the FIFO is full and a new character arrives at the

					receiver. The data in the FIFO is retained and the data in the receive shift register is lost. 0 – no overrun error 1 – overrun error Reading the LSR clears the OE bit.
0	DR	1	R	0x0	Data Ready bit. This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO. 0 – no data ready 1 – data ready This bit is cleared when the RBR is read in non-FIFO mode, or when the receiver FIFO is empty, in FIFO mode.

121.2.6.11. MSR

Table 9- 211 Fields for Register: MSR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7	DCD	1	R	0x0	Data Carrier Detect. This is used to indicate the current state of the modem control line dcd_n. This bit is the complement of dcd_n. When the Data Carrier Detect input (dcd_n) is asserted it is an indication that the carrier has been detected by the modem or data set. 0 – dcd_n input is de-asserted (logic 1) 1 – dcd_n input is asserted (logic 0) In Loopback Mode (MCR[4] set to 1), DCD is the same as MCR[3] (Out2).
6	RI	1	R	0x0	Ring Indicator. This is used to indicate the current state of the modem control line ri_n. This bit is the complement of ri_n. When the Ring Indicator input (ri_n) is asserted it is an indication that a telephone ringing signal has been received by the modem or data set. 0 – ri_n input is de-asserted (logic 1) 1 – ri_n input is asserted (logic 0) In Loopback Mode (MCR[4] set to 1), RI is the same as MCR[2] (Out1).

5	DSR	1	R	0x0	Data Set Ready. This is used to indicate the current state of the modem control line dsr_n. This bit is the complement of dsr_n. When the Data Set Ready input (dsr_n) is asserted it is an indication that the modem or data set is ready to establish communications with the UART. 0 – dsr_n input is de-asserted (logic 1) 1 – dsr_n input is asserted (logic 0) In Loopback Mode (MCR[4] set to 1), DSR is the same as MCR[0] (DTR).
4	CTS	1	R	0x0	Clear to Send. This is used to indicate the current state of the modem control line cts_n. This bit is the complement of cts_n. When the Clear to Send input (cts_n) is asserted it is an indication that the modem or data set is ready to exchange data with the UART. 0 – cts_n input is de-asserted (logic 1) 1 – cts_n input is asserted (logic 0) In Loopback Mode (MCR[4] = 1), CTS is the same as MCR[1] (RTS).
3	DDCD	1	R	0x0	Delta Data Carrier Detect. This is used to indicate that the modem control line dcd_n has changed since the last time the MSR was read. 0 – no change on dcd_n since last read of MSR 1 – change on dcd_n since last read of MSR Reading the MSR clears the DDCD bit. In Loopback Mode (MCR[4] = 1), DDCD reflects changes on MCR[3] (Out2). Note, if the DDCD bit is not set and the dcd_n signal is asserted (low) and a reset occurs (software or otherwise), then the DDCD bit is set when the reset is removed if the dcd_n signal remains asserted.
2	TERI	1	R	0x0	Trailing Edge of Ring Indicator. This is used to indicate that a change on the input ri_n (from an active-low to an inactive-high state) has occurred since the last time the MSR was read. 0 – no change on ri_n since last read of MSR 1 – change on ri_n since last read of MSR Reading the MSR clears the TERI bit. In

					Loopback Mode (MCR[4] = 1), TERI reflects when MCR[2] (Out1) has changed state from a high to a low.
1	DDSR	1	R	0x0	Delta Data Set Ready. This is used to indicate that the modem control line dsr_n has changed since the last time the MSR was read. 0 – no change on dsr_n since last read of MSR 1 – change on dsr_n since last read of MSR Reading the MSR clears the DDSR bit. In Loopback Mode (MCR[4] = 1), DDSR reflects changes on MCR[0] (DTR). Note, if the DDSR bit is not set and the dsr_n signal is asserted (low) and a reset occurs (software or otherwise), then the DDSR bit is set when the reset is removed if the dsr_n signal remains asserted.
0	DCTS	1	R	0x0	Delta Clear to Send. This is used to indicate that the modem control line cts_n has changed since the last time the MSR was read. 0 – no change on cts_n since last read of MSR 1 – change on cts_n since last read of MSR Reading the MSR clears the DCTS bit. In Loopback Mode (MCR[4] = 1), DCTS reflects changes on MCR[1] (RTS). Note, if the DCTS bit is not set and the cts_n signal is asserted (low) and a reset occurs (software or otherwise), then the DCTS bit is set when the reset is removed if the cts_n signal remains asserted.

121.2.6.12. **SCR**

Table 9- 212 Fields for Register: SCR

Bit	Name	Width	R/W	Default	Description

Field				value	
31:8	Reserved				
7:0	Scratchpad Register	8	RW	0x00	This register is for programmers to use as a temporary storage space. It has no defined purpose in the UART.

121.2.6.13. **LPDLL**

Table 9-213 Fields for Register: LPDLL

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7:0	LPDLL	8	RW	0x00	This register makes up the lower 8-bits of a 16-bit, read/write, Low Power Divisor Latch register that contains the baud rate divisor for the UART, which must give a baud rate of 115.2K. This is required for SIR Low Power (minimum pulse width) detection at the receiver. The output low-power baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: Low power baud rate = (serial clock frequency)/(16* divisor) Therefore, a divisor must be selected to give a baud rate of 115.2K. [Note] When the Low Power Divisor Latch registers (LPDLL and LPDLH) are set to 0, the low-power baud clock is disabled and no low-power pulse detection (or any pulse detection) occurs at the receiver. Also, once the LPDLL is set, at least eight clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

121.2.6.14. **LPDLH**

Table 9-214 Fields for Register: LPDLH

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				

7:0	LPDLH	8	RW	0x00	This register makes up the higher 8-bits of a 16-bit, read/write, Low Power Divisor Latch register that contains the baud rate divisor for the UART, which must give a baud rate of 115.2K. This is required for SIR Low Power (minimum pulse width) detection at the receiver. The output low-power baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: Low power baud rate = (serial clock frequency)/(16* divisor) Therefore, a divisor must be selected to give a baud rate of 115.2K. [Note] When the Low Power Divisor Latch registers (LPDLL and LPDLH) are set to 0, the low-power baud clock is disabled and no low-power pulse detection (or any pulse detection) occurs at the receiver. Also, once the LPDLL is set, at least eight clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

121.2.6.15. **SRBR**

Table 9- 215 Fields for Register: SRBR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7:0	Shadow Receive Buffer Register (LSB 8 bits)	8	R		This is a shadow register for the RBR and has been allocated sixteen 32-bit locations so as to accommodate burst accesses from the master. This register contains the data byte received on the serial input port (sin) in UART mode or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line status Register (LSR) is set. If in non-FIFO mode (FIFO_MODE = NONE) or FIFOs are disabled (FCR[0] set to 0), the data in the RBR must be read before the next data arrives, otherwise it is overwritten, resulting in an overrun error. If in FIFO mode

					(FIFO_MODE != NONE) and FIFOs are enabled (FCR[0] set to 1), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO are preserved, but any incoming data is lost. An overrun error also occurs.

121.2.6.16. **STHR**

Table 9- 216 Fields for Register: STHR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
7:0	Shadow Transmit Holding Register	8	W	0x00	This is a shadow register for the THR and has been allocated sixteen 32-bit locations so as to accommodate burst accesses from the master. This register contains data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set. If in non-FIFO mode or FIFOs are disabled (FCR[0] set to 0) and THRE is set, writing a single character to the THR clears the THRE. Any additional writes to the THR before the THRE is set again causes the THR data to be overwritten. If in FIFO mode and FIFOs are enabled (FCR[0] set to 1) and THRE is set, x number of characters of data may be written to the THR before the FIFO is full. The number x (default=16) is determined by the value of FIFO Depth that you set during configuration. Any attempt to write data when the FIFO is full results in the write data being

					lost.

121.2.6.17. **FAR**

Table 9- 217 Fields for Register: FAR

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
0	FIFO Access Register	1	RW	0x0	This register is use to enable a FIFO access mode for testing, so that the receive FIFO can be written by the master and the transmit FIFO can be read by the master when FIFOs are implemented and enabled. When FIFOs are not implemented or not enabled it allows the RBR to be written by the master and the THR to be read by the master. 0 – FIFO access mode disabled 1 – FIFO access mode enabled Note that when the FIFO access mode is enabled/disabled, the control portion of the receive FIFO and transmit FIFO is reset and the FIFOs are treated as empty.

121.2.6.18. **TFR**

Table 9- 218 Fields for Register: TFR

Bit Field	Name	Width	R/W	Default value	Description

31:8	Reserved				
7:0	Transmit FIFO Read	8	R	0x00	Transmit FIFO Read. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to 1). When FIFOs are implemented and enabled, reading this register gives the data at the top of the transmit FIFO. Each consecutive read pops the transmit FIFO and gives the next data value that is currently at the top of the FIFO. When FIFOs are not implemented or not enabled, reading this register gives the data in the THR.

121.2.6.19. **RFW**

Table 9- 219 Fields for Register: RFW

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
9	RFFE	1	W	0x0	Receive FIFO Framing Error. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to 1). When FIFOs are implemented and enabled, this bit is used to write framing error detection information to the receive FIFO. When FIFOs are not implemented or not enabled, this bit is used to write framing error detection information to the RBR.
8	RFPE	1	W	0x0	Receive FIFO Parity Error. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to 1). When FIFOs are implemented and enabled, this bit is used to write parity error detection information to the receive FIFO. When FIFOs are not implemented or not enabled, this bit is used to write parity error detection information to the RBR.
7:0	RFWD	8	W	0x00	Receive FIFO Write Data. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to 1). When FIFOs are implemented and enabled, the data that is written to the RFWD is pushed into the receive FIFO. Each consecutive write pushes the new

					data to the next write location in the receive FIFO. When FIFOs are not implemented or not enabled, the data that is written to the RFWD is pushed into the RBR.

121.2.6.20. **USR**

Table 9- 220 Fields for Register: USR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
4	RFF	1	R	0x0	Receive FIFO Full. This is used to indicate that the receive FIFO is completely full. 0 – Receive FIFO not full 1 – Receive FIFO Full This bit is cleared when the RX FIFO is no longer full.
3	RFNE	1	R	0x0	Receive FIFO Not Empty.This is used to indicate that the receive FIFO contains one or more entries. 0 – Receive FIFO is empty 1 – Receive FIFO is not empty This bit is cleared when the RX FIFO is empty.
2	TFE	1	R	0x1	Transmit FIFO Empty. This is used to indicate that the transmit FIFO is completely empty. 0 – Transmit FIFO is not empty 1 – Transmit FIFO is empty This bit is cleared when the TX FIFO is no longer empty.
1	TFNF	1	R	0x1	Transmit FIFO Not Full.This is used to indicate that the transmit FIFO in not full. 0 – Transmit FIFO is full 1 – Transmit FIFO is not full This bit is cleared when the TX FIFO is full.
0	Reserved				

121.2.6.21. **TFL**

Table 9- 221 Fields for Register: TFL

Bit Field	Name	Width	R/W	Default value	Description
31:6	Reserved				
5:0	Transmit FIFO level	6	R	0x0	Transmit FIFO Level. This indicates the number of data entries in the transmit FIFO.

121.2.6.22. **RFL**

Table 9- 222 Fields for Register: RFL

Bit Field	Name	Width	R/W	Default value	Description
31:6	Reserved				
5:0	Receive FIFO level	6	R	0x0	Receive FIFO Level. This indicates the number of data entries in the receive FIFO.

121.2.6.23. **SRR**

Table 9- 223 Fields for Register: SRR

Bit Field	Name	Width	R/W	Default value	Description
31:8	Reserved				
2	XFR	1	W	0x0	XMIT FIFO Reset. This is a shadow register for the XMIT FIFO Reset bit (FCR[2]). This can be used to remove the burden on software having to store previously written FCR values (which are pretty static) just to reset the transmit FIFO. This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request and single signals.Note that this bit is 'self-clearing'. It is not necessary to clear this bit.

1	RFR	1	W	0x0	RCVR FIFO Reset. This is a shadow register for the RCVR FIFO Reset bit (FCR[1]). This can be used to remove the burden on software having to store previously written FCR values (which are pretty static) just to reset the receive FIFO This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request and single signals. Note that this bit is 'self-clearing'. It is not necessary to clear this bit.
0	UR	1	W	0x0	UART Reset. This asynchronously resets the UART and synchronously removes the reset assertion. For a two clock implementation both pclk and sclk domains are reset.

121.2.6.24. **SRTS**

Table 9- 224 Fields for Register: SRTS

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
0	Shadow Request to Send	1	RW	0x0	Shadow Request to Send. This is a shadow register for the RTS bit (MCR[1]), this can be used to remove the burden of having to performing a read-modify-write on the MCR. This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data. When Auto RTS Flow Control is not enabled (MCR[5] = 0), the rts_n signal is set low by programming MCR[1] (RTS) to a high. In Auto Flow Control, MCR[5] = 1 and FIFOs enable (FCR[0] = 1), the rts_n output is controlled in the same way, but is also gated with the receiver FIFO threshold trigger (rts_n is inactive high when above the threshold) only when RTC Flow Trigger is disabled; otherwise it is gated by the receiver FIFO almost-full trigger, where “almost full” refers to two available slots in the FIFO (rts_n

					is inactive high when above the threshold). Note that in Loopback mode (MCR[4] = 1), the rts_n output is held inactive-high while the value of this location is internally looped back to an input.

121.2.6.25. **SBCR**

Table 9- 225 Fields for Register: SBCR

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
0	Shadow Break Control Register	1	RW	0x0	Shadow Break Control Bit. This is a shadow register for the Break bit (LCR[6]), this can be used to remove the burden of having to performing a read modify write on the LCR. This is used to cause a break condition to be transmitted to the receiving device. If set to 1, the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared.

121.2.6.26. **SDMAM**

Table 9- 226 Fields for Register: SDMAM

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				

0	Shadow DMA Mode	1	RW	0x0	Shadow DMA Mode. This is a shadow register for the DMA mode bit (FCR[3]). This can be used to remove the burden of having to store the previously written value to the FCR in memory and having to mask this value so that only the DMA Mode bit gets updated. 0 – mode 0 1 – mode 1

121.2.6.27. **SFE**

Table 9- 227 Fields for Register: SFE

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
0	Shadow FIFO Enable	1	RW	0x0	Shadow FIFO Enable. This is a shadow register for the FIFO enable bit (FCR[0]). This can be used to remove the burden of having to store the previously written value to the FCR in memory and having to mask this value so that only the FIFO enable bit gets updated.This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. If this bit is set to 0 (disabled) after being enabled then both the XMIT and RCVR controller portion of FIFOs are reset.

121.2.6.28. **SRT**

Table 9- 228 Fields for Register: SRT

Bit Field	Name	Width	R/W	Default value	Description
31:2	Reserved				

1:0	Shadow RCVR Trigger	2	RW	0x0	Shadow RCVR Trigger. This is a shadow register for the RCVR trigger bits (FCR[7:6]). This can be used to remove the burden of having to store the previously written value to the FCR in memory and having to mask this value so that only the RCVR trigger bit gets updated. This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. It also determines when the dma_rx_req_n signal is asserted when DMA Mode (FCR[3]) = 1. The following trigger levels are supported: 00 – 1 character in the FIFO 01 – FIFO ¼ full 10 – FIFO ½ full 11 – FIFO 2 less than full

121.2.6.29. **STET**

Table 9- 229 Fields for Register: STET

Bit Field	Name	Width	R/W	Default value	Description
31:2	Reserved				
1:0	Shadow TX Empty Trigger	2	RW	0x0	Shadow TX Empty Trigger. This is a shadow register for the TX empty trigger bits (FCR[5:4]). This can be used to remove the burden of having to store the previously written value to the FCR in memory and having to mask this value so that only the TX empty trigger bit gets updated. This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. The following trigger levels are supported: 00 – FIFO empty 01 – 2 characters in the FIFO 10 – FIFO ¼ full 11 – FIFO ½ full

121.2.6.30. **HTX**

Table 9- 230 Fields for Register: HTX

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
0	Halt TX	1	RW	0x0	This register is use to halt transmissions for testing, so that the transmit FIFO can be filled by the master when FIFOs are implemented and enabled. 0 – Halt TX disabled 1 – Halt TX enabled

121.2.6.31. **DMASA**

Table 9- 231 Fields for Register: DMASA

Bit Field	Name	Width	R/W	Default value	Description
31:1	Reserved				
0	DMA Software Acknowledge	1	W	0x0	This register is use to perform a DMA software acknowledge if a transfer needs to be terminated due to an error condition. For example, if the DMA disables the channel, then the UART should clear its request. This causes the TX request, TX single, RX request and RX single signals to de-assert. Note that this bit is 'self-clearing'. It is not necessary to clear this bit.

121.2.6.32. **DLF**

Table 9- 232 Fields for Register: DLF

Bit Field	Name	Width	R/W	Default value	Description
31:6	RSVD_DLF	25	R	0x0	DLF 31 to DLF_SIZE Reserved bits read as 0.

5:0	DLF	6	W	0x0	The fractional value is added to integer value set by DLH,DLL. Fractional value is determined by (Divisor Fraction value)/(2^DLF_SIZE). For information on DLF values to be programmed for DLF_SIZE=4, see the 'Fractional Baud Rate Support' section in the uart Databook.

121.2.6.33. **RAR**

Table 9- 233 Fields for Register: RAR

Bit Field	Name	Width	R/W	Default value	Description
31:8	RSVD_RAR_31to8	23	R	0x0	RAR 31to8 Reserved bits read as 0.
7:0	RAR	8	W	0x0	This is an address matching register during receive mode. If the 9-th bit is set in the incoming character then the remaining 8-bits will be checked against this register value. If the match happens then sub-sequent characters with 9-th bit set to 0 will be treated as data byte until the next address byte is received.

121.2.6.34. **TAR**

Table 9- 234 Fields for Register: TAR

Bit Field	Name	Width	R/W	Default value	Description
31:8	RSVD_TAR_ 31to8	23	R	0x0	TAR 31to8 Reserved bits read as 0.
7:0	TAR	8	W	0x0	This is an address matching register during transmit mode. If DLS_E (LCR_EXT[0]) bit is enabled, then uart will send the 9-bit character with 9-th bit set to 1 and remaining 8-bit address will be sent from this register provided 'SEND_ADDR' (LCR_EXT[2]) bit is set to 1.

121.2.6.35. **LCR_EXT**

Bit Field	Name	Width	R/W	Default value	Description
31:4	RSVD_LCR_ EXT	27	R	0x0	LCR_EXT 31to4 Reserved bits read as 0.
3	TRANSMIT_ MODE	1	R/W	0x0	Transmit mode control bit. This bit is used to control the type of transmit mode during 9-bit data transfers. 1: In this mode of operation, Transmit Holding Register (THR) and Shadow Transmit Holding Register (STHR) are 9-bit wide. The user needs to ensure that the THR/STHR register is written correctly for address/data. Address: 9th bit is set to 1, Data : 9th bit is set to 0. Note: Transmit address register (TAR) is not applicable in this mode of operation. 0: In this mode of operation, Transmit Holding Register (THR) and Shadow Transmit Holding register (STHR) are 8-bit wide. The user needs to program the address into Transmit Address Register (TAR) and data into the THR/STHR register. SEND_ADDR bit is used as a control knob to indicate the uart on when to send the address.

Table 9- 235 Fields for Register: LCR_EXT

Bit Field	Name	Width	R/W	Default value	Description
31:24	Reserved				
2	SEND_ADDR	1	R/W	0x0	Send address control bit. This bit is used as a control knob for the user to determine when to send the address during transmit 1 = 9-bit character will be transmitted with 9-th bit set to 1 and the remaining 8-bits will match to what is being programmed in 'Transmit Address Register'. 0 = 9-bit character will be transmitted with 9-th bit set to 0 and the remaining 8-bits will be taken from the TXFIFO which is programmed through 8-bit wide THR/STHR register.
1	ADDR_MATCH	1	R/W	0x0	Address Match Mode.This bit is used to enable the address match feature during receive. 1 = Address match mode; DW_apb_uart will wait until the incoming character with 9-th bit set to 1. And further checks to see if the address matches with what is programmed in 'Receive Address Match Register'. If match is found, then sub-sequent characters will be treated as valid data and DW_apb_uart starts receiving data. 0 = Normal mode; DW_apb_uart will start to receive the data and 9-bit character will be formed and written into the receive RXFIFO. User is responsible to read the data and differentiate b/n address and data.
0	DLS_E	1	R/W	0x0	Extension for DLS. This bit is used to enable 9-bit data for transmit and receive transfers.

121.2.6.36. **CPR**

Table 9- 236 Fields for Register: CPR

23:16	FIFO mode	8	R	0x04	0x00 = 0 0x01 = 16 0x02 = 32 to 0x80 = 2048 0x81- 0xff = reserved
15:14	Reserved				
13	DMA_EXTRA	1	R	0x1	0 – FALSE 1 – TRUE
12	UART_ADD_ENCODED_PARAMS	1	R	0x1	0 – FALSE 1 – TRUE
11	SHADOW	1	R	0x1	0 – FALSE 1 – TRUE
10	FIFO_STAT	1	R	0x1	0 – FALSE 1 – TRUE
9	FIFO_ACCESS	1	R	0x1	0 – FALSE 1 – TRUE
8	ADDITIONAL_FEAT	1	R	0x1	0 – FALSE 1 – TRUE
7	SIR_LP_MODE	1	R	0x0	0 – FALSE 1 – TRUE
6	SIR_MODE	1	R	0x0	0 – FALSE 1 – TRUE
5	THRE_MODE	1	R	0x1	0 – FALSE 1 – TRUE
4	AFCE_MODE	1	R	0x1	0 – FALSE 1 – TRUE
3:2	Reserved				
1:0	APB_DATA_WIDTH	2	R	0x2	00 – 8 bits 01 – 16 bits 10 – 32 bits 11 – reserved

121.2.6.37. **UCV**

Table 9-237 Fields for Register: UCV

Bit Field	Name	Width	R/W	Default value	Description
31:0	UART Component Version	32	R		ASCII value for each number in the version, followed by *. For example 32_30_31_2A

					represents the version 2.01*

121.2.6.38. CTR

Table 9- 238 Fields for Register: CTR

Bit Field	Name	Width	R/W	Default value	Description
31:0	Peripheral ID	32	R	0x44570110	This register contains the peripherals identification code.

121.3. I2C

121.3.1. Overview

I2C Master/Slave interfaces capable of supporting standard mode (up to 100 KHz), and Fast Mode (up to 400 KHz).The I2C interfaces can be multiplexed with GPIO pins.

121.3.2. Feature

- Compliant with Philips I2C specification version 2.1
- Compliance to AMBA APB interface
- Support for standard mode (up to 100K bits/s) and fast mode (up to 400K bits/s).
- Multi-master transmitter/slave receiver mode
- Multi-master receiver/slave transmitter mode
- Combined master transmit/receive and slave receive/transmit modes
- 7-bit and 10-bit device addressing modes
- Programmable clock generation
- Built-in 32-byte FIFO for buffered read or writes in each module
- Two DMA channels per PHY, one interrupt line
- I2C clock and data pins shall have configurable drive strengths of 4mA and 8mA

121.3.3. Function Description

121.3.3.1. Block Description

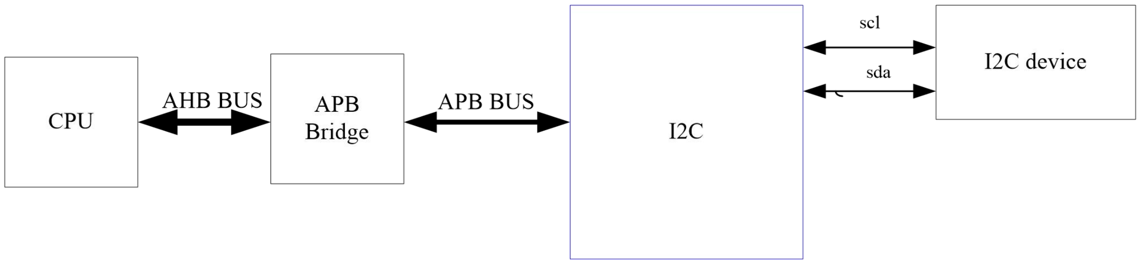


Figure 9- 16 Block Diagram of I2C

The I2C bus is a two-wire serial interface, consisting of a serial data line (SDA) and a serial clock (SCL). These wires carry information between the devices connected to the bus. Each device is recognized by a unique address and can operate as either a "transmitter" or "receiver," depending on the function of the device. Devices can also be considered as masters or slaves when performing data transfers. A master is a device that initiates a data transfer on the bus and generates the clock signals to permit that transfer. At that time, any device addressed is considered a slave.

The I2C controller module can operate in standard mode (with data rates 0 to 100 Kb/s), fast mode (with data rates less than or equal to 400 Kb/s), fast mode plus (with data rates less than or equal to 1000 Kb/s), high-speed mode (with data rates less than or equal to 3.4 Mb/s).

The I2C controller is made up of an AMBA APB slave interface, an I2C interface, and FIFO logic to maintain coherency between the two interfaces. A simplified block diagram of the component is illustrated in below picture.

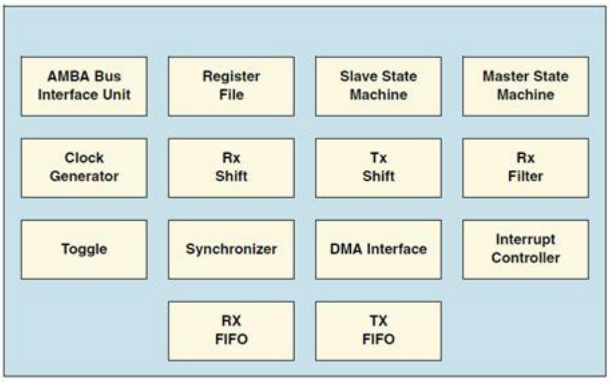


Figure 9- 17 I2C Simplified Block Diagram of Components

121.3.4. Programming Guide

121.3.4.1. I2C Terminology

The following terms relate to how the role of the I2C device and how it interacts with other I2C devices on the bus.

Transmitter – the device that sends data to the bus. A transmitter can either be a device that initiates the data transmission to the bus (a master-transmitter) or responds to a request from the master to send data to the bus (a slave-transmitter).

Receiver – the device that receives data from the bus. A receiver can either be a device that receives data on its own request (a master-receiver) or in response to a request from the master (a slave-receiver).

Master – the component that initializes a transfer (START command), generates the clock (SCL) signal and terminates the transfer (STOP command). A master can be either a transmitter or a receiver.

Slave – the device addressed by the master. A slave can be either receiver or transmitter. These concepts are illustrated in below figure.

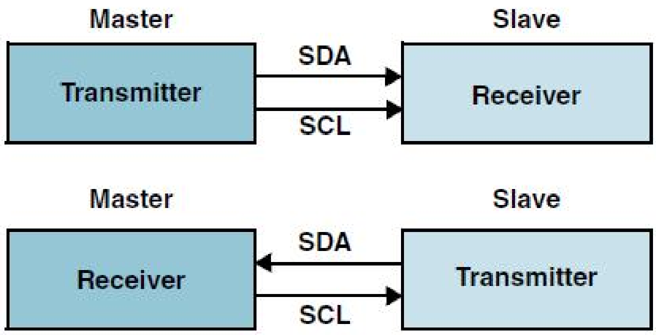


Figure 9- 18 I2C Master-Slave Connectivity

The following terms are specific to data transfers that occur to/from the I2C bus.

START (RESTART) – data transfer begins with a START or RESTART condition. The level of the SDA data line changes from high to low, while the SCL clock line remains high. When this occurs, the bus becomes busy.

STOP – data transfer is terminated by a STOP condition. This occurs when the level on the SDA data line passes from the low state to the high state, while the SCL clock line remains high. When the data transfer has been terminated, the bus is free or idle once again. The bus stays busy if a RESTART is generated instead of a STOP condition.

121.3.4.2. I2C Behavior

The master is responsible for generating the clock and controlling the transfer of data. The slave is responsible for either transmitting or receiving data to/from the master. The acknowledgement of data is sent by the device that is receiving data, which can be either a master or a slave. As mentioned previously, the I2C protocol also allows multiple masters to reside on the I2C bus and uses an arbitration procedure to determine bus ownership.

Each slave has a unique address that is determined by the system designer. When a master wants to communicate with a slave, the master transmits a START/RESTART condition that is then followed by the slave's address and a control bit (R/W) to determine if the master wants to transmit data or receive data from the slave. The slave then sends an acknowledge (ACK) pulse after the address.

If the master (master-transmitter) is writing to the slave (slave-receiver), the receiver gets one byte of data. This transaction continues until the master terminates the transmission with a STOP condition. If the master is reading from a slave (master-receiver), the slave transmits (slave-transmitter) a byte of data to the master, and the master then acknowledges the transaction with the ACK pulse. This transaction continues until the master terminates the transmission by not acknowledging (NACK) the transaction after the last byte is received, and then the master issues a STOP condition or addresses another slave after issuing a RESTART condition. This behavior is illustrated in below figure.

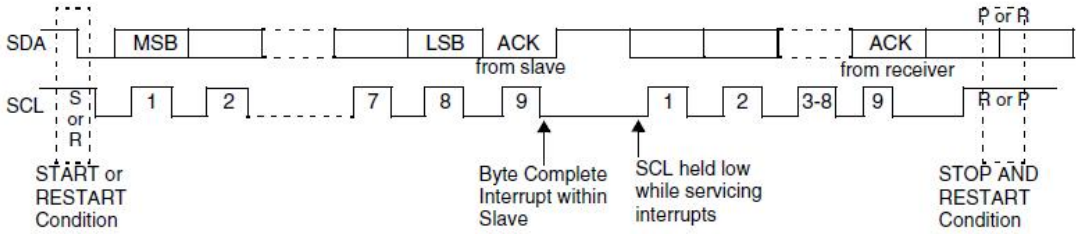


Figure 9- 19 I2C Behavior

The I2C controller is a synchronous serial interface. The SDA line is a bidirectional signal and changes only while the SCL line is low, except for STOP, START, and RESTART conditions. The output drivers are open-drain or open-collector to perform wire-AND functions on the bus. The maximum number of devices on the bus is limited by only the maximum capacitance specification of 400 pF. Data is transmitted in byte packages.

As an I2C Master, putting data into the transmit FIFO causes the I2C controller to generate a START condition on the I2C bus. Writing a 1 to IC_DATA_CMD[9] causes the I2C controller to generate a STOP condition on the I2C bus; a STOP condition is not issued if this bit is not set, even if the transmit FIFO is empty.

The I2C controller supports mixed read and write combined format transactions in both 7-bit and 10-bit addressing modes.The I2C controller does not support mixed address and mixed address format—that is, a 7-bit address transaction followed by a 10-bit address transaction or vice versa—combined format transactions. To initiate combined format transfers, IC_CON.IC_RESTART_EN should be set to 1. With this value set and operating as a master, when the I2C controller completes an I2C transfer, it checks the transmit FIFO and executes the next transfer. If the direction of this transfer differs from the previous transfer, the combined format is used to issue the transfer.

121.3.4.3. TX FIFO Management and START, STOP and RESTART Generation

The component does not generate a STOP if the Tx FIFO becomes empty; in this situation the component holds the SCL line low, stalling the bus until a new entry is available in the Tx FIFO. A STOP condition is generated only when the user specifically requests it by setting bit 9 (Stop bit) of the command written to IC_DATA_CMD register. Below figure shows the bits in the IC_DATA_CMD register.

121.3.4.4. **Master Operation Mode**

To use the I2C controller as a master, user needs to perform the following steps:
1. Disable the I2C controller by writing 0 to bit 0 of the IC_ENABLE register.
2. Write to the IC_CON register to set the maximum speed mode supported (bits 2:1) and the desired speed of the I2C controller master-initiated transfers, either 7-bit or 10-bit addressing (bit 4). Ensure that bit 6 (IC_SLAVE_DISABLE) is written with a '1' and bit 0 (MASTER_MODE) is written with a '1'.
3. Write to the IC_TAR registers the address of the I2C device to be addressed (bits 9:0). This register also indicates whether a General Call or a START BYTE command is going to be performed by I2C.
4. Write to the IC_HS_MADDR register the desired master code for the I2C controller (Only applicable for high-speed mode). The master code is programmer-defined.
5. Enable the I2C controller by writing a 1 to bit 0 of the IC_ENABLE register. 6. Now write transfer direction and data to be sent to the IC_DATA_CMD register. If the IC_DATA_CMD register is written before the I2C controller is enabled, the data and commands are lost as the buffers are kept cleared when I2C controller is disabled.
6. Now write transfer direction and data to be sent to the IC_DATA_CMD register. If the IC_DATA_CMD register is written before the I2C controller is enabled, the data and commands are lost as the buffers are kept cleared when I2C controller is disabled. This step generates the START condition and the address byte on the I2C controller. Once I2C controller is enabled and there is data in the TX FIFO, I2C controller starts reading the data.

In fast mode plus, the I2C controller allows the fast mode operation to be extended to support speeds up to 1000 Kb/s. To enable the I2C controller for fast mode plus operation, perform the following steps before initiating any data transfer:
1. Configure the Maximum Speed mode of I2C controller Master or Slave to Fast Mode or High Speed mode (IC_MAX_SPEED_MODE> = 2).
2. Set ic_clk frequency greater than or equal to 32 MHz.
3. Program the IC_CON register [2:1] = 2'b10 for fast mode or fast mode plus.
4. Program IC_FS_SCL_LCNT and IC_FS_SCL_HCNT registers to meet the fast mode plus SCL.
5. Program the IC_FS_SPKLEN register to suppress the maximum spike of 50ns.
6. Program the IC_SDA_SETUP register to meet the minimum data setup time (tSU; DAT).

121.3.4.5. **Spike Suppression**

The I2C controller contains programmable spike suppression logic that match requirements imposed by the I2C Bus Specification for SS/FS (tSP, Table 9), HS (tSP, Table 11), and UFm (tSP, Table 13) modes. This logic is based on counters that monitor the input signals (SCL and SDA), checking if they remain stable for a predetermined amount of ic_clk cycles before they are sampled internally. There is one separate counter for each signal (SCL and SDA). The number of ic_clk cycles can be programmed by the user and should be calculated taking into account the frequency of ic_clk and the relevant spike length specification.

Each counter is started whenever its input signal changes its value. Depending on the behavior of the input signal, one of the following scenarios occurs:

- The input signal remains unchanged until the counter reaches its count limit value. When this happens, the internal version of the signal is updated with the input value, and the counter is reset and stopped. The counter is not restarted until a new change on the input signal is detected.
- The input signal changes again before the counter reaches its count limit value. When this happens, the counter is reset and stopped, but the internal version of the signal is not updated. The counter remains stopped until a new change on the input signal is detected.

The below timing diagram in illustrates the behavior described above.

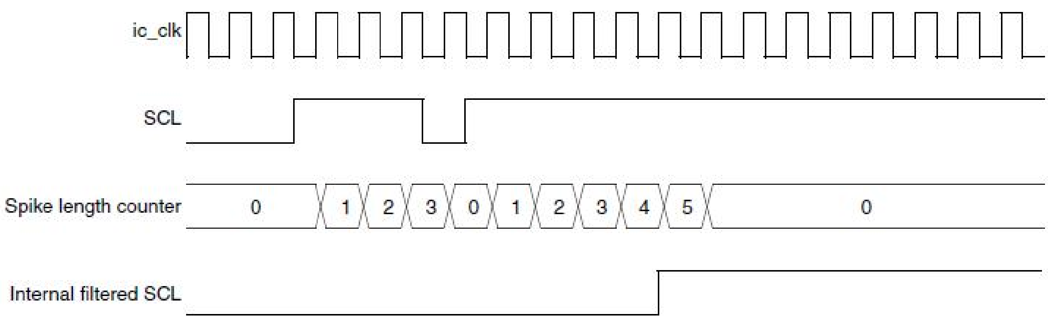


Figure 9- 20 I2C Spike Suppression Example

The count limit value used in this example is 5 and was calculated for a 10 ns ic_clk period and for SS/FS operation (50 ns spike suppression).The I2C Bus Specification calls for different maximum spike lengths according to the operating mode—50 ns for SS and FS; 10 ns for HS, 10 ns for UFm, so three registers are required to store the values needed for each case:

- Register IC_FS_SPKLEN holds the maximum spike length for SS and FS modes
- Register IC_HS_SPKLEN holds the maximum spike value for HS mode.
- Register IC_UFM_SPKLEN holds the maximum spike value for UFm.

These registers are 8 bits wide and accessible through the APB interface for read and write purposes; however, they can be written to only when the I2C controller is disabled. The minimum value that can be programmed into these registers is 1; attempting to program a value smaller than 1 result in the value 1 being written

121.3.5. Register Summary

Table 9- 239 Register List for I2C

Addr offset	Name	Width	R/W	Description
0x0	IC_CON	32	RW	I2C Control
0x4	IC_TAR	32	RW	I2C Target Address
0x8	IC_SAR	32	RW	I2C Slave Address
0xC	IC_HS_MADDR	32	RW	I2C HS Master Mode Code Address
0x10	IC_DATA_CMD	32	RW	I2C Rx/Tx Data Buffer

				and Command
0x14	IC_SS_SCL_HCNT	32	RW	Standard speed I2C Clock SCL High Count
0x18	IC_SS_SCL_LCNT	32	RW	Standard speed I2C Clock SCL Low Count
0x1C	IC_FS_SCL_HCNT	32	RW	Fast Mode and Fast Mode Plus I2C Clock SCL High Count
0x20	IC_FS_SCL_LCNT	32	RW	Fast Mode and Fast Mode Plus I2C Clock SCL Low Count
0x24	IC_HS_SCL_HCNT	32	RW	High speed I2C Clock SCL High Count
0x28	IC_HS_SCL_LCNT	32	RW	High speed I2C Clock SCL Low Count
0x2c	IC_INTR_STAT	32	R	I2C Interrupt Status
0x30	IC_INTR_MASK	32	R	I2C Interrupt Mask
0x34	IC_RAW_INTR_STAT	32	RW	I2C Raw Interrupt Status
0x38	IC_RX_TL	32	RW	I2C Receive FIFO Threshold
0x3C	IC_TX_TL	32	RW	I2C Transmit FIFO Threshold
0x40	IC_CLR_INTR	32	R	Clear Combined and Individual Interrupts
0x44	IC_CLR_RX_UNDER	32	R	Clear RX_UNDER Interrupt
0x48	IC_CLR_RX_OVER	32	R	Clear RX_OVER Interrupt
0x4C	IC_CLR_TX_OVER	32	R	Clear TX_OVER Interrupt
0x50	IC_CLR_RD_REQ	32	R	Clear RD_REQ Interrupt
0x54	IC_CLR_TX_ABRT	32	R	Clear TX_ABRT Interrupt
0x58	IC_CLR_RX_DONE	32	R	Clear RX_DONE Interrupt
0x5C	IC_CLR_ACTIVITY	32	R	Clear ACTIVITY Interrupt
0x60	IC_CLR_STOP_DET	32	R	Clear STOP_DET Interrupt
0x64	IC_CLR_START_DET	32	R	Clear START_DET Interrupt

0x68	IC_CLR_GEN_CALL	32	R	Clear GEN_CALL Interrupt
0x6C	IC_ENABLE	32	RW	I2C Enable
0x70	IC_STATUS	32	R	I2C Status register
0x74	IC_TXFLR	32	R	Transmit FIFO Level Register
0x78	IC_RXFLR	32	R	Receive FIFO Level Register
0x7C	IC_SDA_HOLD	32	RW	SDA hold time length register
0x80	IC_TX_ABRT_SOURCE	32	R	I2C Transmit Abort Status Register
0x84	IC_SLV_DATA_NACK_ONLY	32	RW	Generate SLV_DATA_NACK Register
0x88	IC_DMA_CR	32	RW	DMA Control Register for transmit and receive handshaking interface
0x8C	IC_DMA_TDLR	32	RW	DMA Transmit Data Level
0x90	IC_DMA_RDLR	32	RW	DMA Receive Data Level
0x94	IC_SDA_SETUP	32	RW	I2C SDA Setup Register
0x98	IC_ACK_GENERAL_CALL	32	RW	I2C ACK General Call Register
0x9C	IC_ENABLE_STATUS	32	R	I2C Enable Status Register
0xA0	IC_FS_SPKLEN	32	RW	ISS and FS spike suppression limit
0xA4	IC_HS_SPKLEN	32	RW	HS spike suppression limit
0xA8	IC_CLR_RESTART_DET	32	R	Clear RESTART_DET Interrupt
0xAC	IC_SCL_STUCK_AT_LOW_TIMEOUT	32	RW	I2C SCL stuck at low timeout register
0xB0	IC_SDA_STUCK_AT_LOW_TIMEOUT	32	RW	I2C SDA Stuck at Low Timeout
0xB4	IC_CLR_SCL_STUCK_DET	32	R	Clear SCL Stuck at Low Detect Interrupt Register
0xB8	IC_DEVICE_ID	32	R	I2C Device ID
0xBC-0xF0	Reserved			
0xF4	IC_COMP_PARAM_1	32	R	Component Parameter Register

0xF8	IC_COMP_VERSION	32	R	Component Version ID

121.3.6. Register Description

121.3.6.1. IC_CON

Table 9- 240 Fields for Register: IC_CON

Bit Field	Name	R/W	Default value	Description
31:11	Reserved			
10	STOP_DET_IF_MASTER_ACTIVE	RW	0x0	In Master mode 1'b1: Issues the STOP_DET interrupt only when the master is active 1'b0: Issues the STOP_DET irrespective of whether the master is active
9	RX_FIFO_FULL_HLD_CTRL	RW	0x0	This bit controls whether i2c should hold the bus when the Rx FIFO is physically full to its RX_BUFFER_DEPTHWhen the Rx FIFO is physically full to its RX_BUFFER_DEPTH, it provides a hardware method to hold the bus till RX FIFO data is read out and there is a space available in the FIFO
8	TX_EMPTY_CTRL	RW	0x0	This bit controls the generation of the TX_EMPTY interrupt, as described in the IC_RAW_INTR_STAT register.
7	STOP_DET_IFADDRESSED	RW	0x0	In slave mode: 1'b1 – issues the STOP_DET interrupt only when it is addressed. 1'b0 – issues the STOP_DET irrespective of whether it's addressed or not.
6	IC_SLAVE_DISABLE	RW	0x1	This bit controls whether I2C has its slave disabled, which means once the presetn signal is applied, then this bit takes on the value of the configuration parameter IC_SLAVE_DISABLE. You have the choice of having the slave enabled or disabled after reset is applied, which means software does not have to configure the slave. By default, the slave is always enabled (in reset state as well). If you need to disable it after reset, set this bit to 1. If this bit is set (slave is disabled), I2c

				functions only as a master and does not perform any action that requires a slave. 0: slave is enabled 1: slave is disabled
5	IC_RESTART_EN	RW	0x1	Determines whether RESTART conditions may be sent when acting as a master. Some older slaves do not support handling RESTART conditions; however, RESTART conditions are used in several I2c operations. 0: disable 1: enable When the RESTART is disabled, the I2c master is incapable of performing the following functions: Sending a START BYTE Performing any high-speed mode operation Performing direction changes in combined format mode Performing a read operation with a 10-bit address By replacing RESTART condition followed by a STOP and a subsequent START condition, split operations are broken down into multiple I2c transfers. If the above operations are performed, it will result in setting bit 6 (TX_ABRT) of the IC_RAW_INTR_STAT register.
4	IC_10BITADDR_MASTER	RW	0x1	Controls whether the I2c starts its transfers in 7-or 10-bit addressing mode when acting as a master. 0: 7-bit addressing 1: 10-bit addressing
3	IC_10BITADDR_SLAVE	RW	0x1	When acting as a slave, this bit controls whether the I2c responds to 7- or 10-bit addresses. 0: 7-bit addressing. The I2c ignores transactions that involve 10-bit addressing; for 7-bit addressing, only the lower 7 bits of the IC_SAR register are compared. 1: 10-bit addressing. The I2c responds to only 10-bit addressing transfers that match the full

				10 bits of the IC_SAR register.
2:1	SPEED	RW	0x3	These bits control at which speed the I2c operates. Hardware protects against illegal values being programmed by software. register These bits must be programmed appropriately for slave mode also, as it is used to capture correct value of spike filter as per the speed mode. This register should be programmed only with a value in the range of 1 to 3; otherwise, hardware updates this register with the value of 3. 1: standard mode (0 to 100 Kb/s) 2: fast mode (≤ 400 Kb/s) or fast mode plus (≤ 1000 Kb/s) 3: high speed mode (≤ 3.4 Mb/s)
0	MASTER_MODE	RW	0x1	This bit controls whether the I2c master is enabled. 0: master disabled 1: master enabled

121.3.6.2. **IC_TAR**

Table 9- 241 Fields for Register: IC_TAR

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
11	SPECIAL	RW	0x0	This bit indicates whether software performs a Device-ID, General Call or START BYTE command. 0: ignore bit 10 GC_OR_START and use IC_TAR normally 1: perform special I2C command as specified in Device-ID or GC_OR_START bit

10	GC_OR_START	RW	0x0	If bit 11 (SPECIAL) is set to 1 and bit 13 (Device-ID) is set to 0, then this bit indicates whether a General Call or START byte command is to be performed by the I2c. 0: General Call Address – after issuing a General Call, only writes may be performed. Attempting to issue a read command results in setting bit 6 (TX_ABRT) of the IC_RAW_INTR_STAT register. The I2c remains in General Call mode until the SPECIAL bit value (bit 11) is cleared. 1: START BYTE
9:0	IC_TAR	RW	0x055	This is the target address for any master transaction. When transmitting a General Call, these bits are ignored. To generate a START BYTE, the CPU needs to write only once into these bits.

121.3.6.3. IC_SAR

Table 9- 242 Fields for Register: IC_SAR

Bit Field	Name	R/W	Default value	Description
31:10	Reserved			
9:0	IC_SAR	RW	0x055	The IC_SAR holds the slave address when the I2C is operating as a slave. For 7-bit addressing, only IC_SAR[6:0] is used. This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. [NOTE] The default values cannot be any of the reserved address locations: that is, 0x00 to 0x07, or 0x78 to 0x7f. The correct operation of the device is not guaranteed if you program the IC_SAR or IC_TAR to a reserved value.

121.3.6.4. IC_HS_MADDR

Table 9- 243 Fields for Register: IC_HS_MADDR

Bit Field	Name	R/W	Default	Description

			value	
31:3	Reserved			
2:0	IC_HS_MAR	RW	0x1	This bit field holds the value of the I2C HS mode master code. HS-mode master codes are reserved 8-bit codes (00001xxx) that are not used for slave addressing or other purposes. Each master has its unique master code; up to eight highspeed mode masters can be present on the same I2C bus system. Valid values are from 0 to 7. This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect.

121.3.6.5. **IC_DATA_CMD**

Table 9- 244 Fields for Register: IC_DATA_CMD

Bit Field	Name	R/W	Default value	Description
31:11	Reserved			
10	RESTART	W	0x0	This bit controls whether a RESTART is issued before the byte is sent or received. This bit is available only if IC_EMPTYFIFO_HOLD_MASTER_EN is configured to 1. 1 – If IC_RESTART_EN is 1, a RESTART is issued before the data is sent/received (according to the value of CMD), regardless of whether or not the transfer direction is changing from the previous command; if IC_RESTART_EN is 0, a STOP followed by a START is issued instead. 0 – If IC_RESTART_EN is 1, a RESTART is issued only if the transfer direction is changing from the previous command; if IC_RESTART_EN is 0, a STOP followed by a START is issued instead.

9	STOP	W	0x0	This bit controls whether a STOP is issued after the byte is sent or received. This bit is available only if IC_EMPTYFIFO_HOLD_MASTER_EN is configured to 1. 1 – STOP is issued after this byte, regardless of whether or not the Tx FIFO is empty. If the Tx FIFO is not empty, the master immediately tries to start a new transfer by issuing a START and arbitrating for the bus. 0 – STOP is not issued after this byte, regardless of whether or not the Tx FIFO is empty. If the Tx FIFO is not empty, the master continues the current transfer by sending/receiving data bytes according to the value of the CMD bit. If the Tx FIFO is empty, the master holds the SCL line low and stalls the bus until a new command is available in the Tx FIFO.
8	CMD	W	0x0	This bit controls whether a read or a write is performed. This bit does not control the direction when the I2c acts as a slave. It controls only the direction when it acts as a master. 1 = Read 0 = Write When a command is entered in the TX FIFO, this bit distinguishes the write and read commands. In slave-receiver mode, this bit is a “don’t care” because writes to this register are not required. In slave-transmitter mode, a “0” indicates that the data in IC_DATA_CMD is to be transmitted. When programming this bit, you should remember the following: attempting to perform a read operation after a General Call command has been sent results in a TX_ABRT interrupt (bit 6 of the IC_RAW_INTR_STAT register), unless bit 11 (SPECIAL) in the IC_TAR register has been cleared. If a “1” is written to this bit after receiving a RD_REQ interrupt, then a TX_ABRT interrupt occurs.
7:0	DAT	RW	0x00	This register contains the data to be transmitted or received on the I2C bus. If you are writing to this register and want to perform a read, bits 7:0 (DAT) are ignored by the I2c. However, when you read this register, these bits return the value of data received on the I2c interface.

121.3.6.6. IC_SS_SCL_HCNT

Table 9- 245 Fields for Register: IC_SS_SCL_HCNT

Bit Field	Name	R/W	Default value	Description
31:16	Reserved			
15:0	IC_SS_SCL_HCNT	RW	0x190	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for standard speed. This register can be written only when the I2C interface is disabled which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. For designs with APB_DATA_WIDTH = 8, the order of programming is important to ensure the correct operation of the I2c. The lower byte must be programmed first. Then the upper byte is programmed.[NOTE] This register must not be programmed to a value higher than 65525

121.3.6.7. IC_SS_SCL_LCNT

Table 9- 246 Fields for Register: IC_SS_SCL_LCNT

Bit Field	Name	R/W	Default value	Description
31:8	Reserved			

15:0	IC_SS_SCL_LCNT	RW	0x1d6	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for standard speed. This register can be written only when the I2C interface is disabled which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. For designs with APB_DATA_WIDTH = 8, the order of programming is important to ensure the correct operation of the I2c. The lower byte must be programmed first. Then the upper byte is programmed.

121.3.6.8. **IC_FS_SCL_HCNT**

Table 9- 247 Fields for Register: IC_FS_SCL_HCNT

Bit Field	Name	R/W	Default value	Description
31:16	Reserved			
15:0	IC_FS_SCL_HCNT	RW	0x3c	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for fast mode or fast mode plus. It is used in high-speed mode to send the Master Code and START BYTE or General CALL.This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. For designs with APB_DATA_WIDTH == 8 the order

				of programming is important to ensure the correct operation of the I2c. The lower byte must be programmed first. Then the upper byte is programmed.

121.3.6.9. **IC_FS_SCL_LCNT**

Table 9- 248 Fields for Register: IC_FS_SCL_LCNT

Bit Field	Name	R/W	Default value	Description
31:16	Reserved			
15:0	IC_FS_SCL_LCNT	RW	0x82	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for fast mode or fast mode plus. It is used in high-speed mode to send the Master Code and START BYTE or General CALL.This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. For designs with APB_DATA_WIDTH == 8 the order of programming is important to ensure the correct operation of the I2c. The lower byte must be programmed first. Then the upper byte is programmed.

121.3.6.10. IC_HS_SCL_HCNT

Table 9- 249 Fields for Register: IC_HS_SCL_HCNT

Bit Field	Name	R/W	Default value	Description
31:16	Reserved			
15:0	IC_HS_SCL_HCNT	RW	0x6	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high period count for high speed.The SCL High time depends on the loading of the bus. For 100pF loading, the SCL High time is 60ns; for 400pF loading, the SCL High time is 120ns. This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect.

121.3.6.11. IC_HS_SCL_LCNT

Table 9- 250 Fields for Register: IC_HS_SCL_LCNT

Bit Field	Name	R/W	Default value	Description
31:16	Reserved			
15:0	IC_HS_SCL_LCNT	RW	0x10	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high period count for high speed.The SCL High time depends on the loading of the bus. For 100pF loading, the SCL High time is 60ns; for 400pF loading, the SCL High time is 120ns. This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect.

121.3.6.12. IC_INTR_STAT

Table 9- 251 Fields for Register: IC_INTR_STAT

Bit Field	Name	R/W	Default value	Description
31:15	Reserved			
14	R_SCL_STUCK_AT_LOW	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
13	R_MST_ON_HOLD	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
12	R_RESTART_DET	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
11	R_GEN_CALL	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
10	R_START_DET	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
9	R_STOP_DET	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
8	R_ACTIVITY	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
7	R_RX_DONE	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
6	R_TX_ABRT	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
5	R_RD_REQ	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
4	R_TX_EMPTY	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
3	R_TX_OVER	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
2	R_RX_FULL	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
1	R_RX_OVER	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.
0	R_RX_UNDER	R	0x0	See IC_RAW_INTR_STAT for a detailed description of this bit.

121.3.6.13. **IC_INTR_MASK**

Table 9- 252 Fields for Register: IC_INTR_MASK

Bit Field	Name	R/W	Default value	Description
31:15	Reserved			
14	M_SCL_STUCK_AT_LOW	R	0x0	This bit masks the R_SCL_STUCK_AT_LOW interrupt bit in the IC_INTR_STAT register
13	M_MST_ON_HOLD	R	0x0	This bit masks the R_MST_ON_HOLD interrupt bit in the IC_INTR_STAT register.
12	M_RESTART_DET	RW	0x0	This bit masks the R_RESTART_DET interrupt status bit in the IC_INTR_STAT register.
11	M_GEN_CALL	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
10	M_START_DET	RW	0x0	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
9	M_STOP_DET	RW	0x0	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
8	M_ACTIVITY	RW	0x0	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
7	M_RX_DONE	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
6	M_TX_ABRT	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
5	M_RD_REQ	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
4	M_TX_EMPTY	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
3	M_TX_OVER	RW	0x1	These bits mask their corresponding interrupt status bits

				in the IC_INTR_STAT register.
2	M_RX_FULL	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
1	M_RX_OVER	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.
0	M_RX_UNDER	RW	0x1	These bits mask their corresponding interrupt status bits in the IC_INTR_STAT register.

121.3.6.14. IC_RAW_INTR_STAT

Table 9- 253 Fields for Register: IC_RAW_INTR_STAT

Bit Field	Name	R/W	Default value	Description
31:15	Reserved			
14	SCL_STUCK_AT_LOW	R	0x0	Indicates whether the SCL Line is stuck at low for the IC_SCL_STUCK_LOW_TIMOUT number of ic_clk periods.
13	MST_ON_HOLD	R	0x0	Indicates whether a master is holding the bus and the Tx FIFO is empty.
12	RESTART_DET	R	0x0	Indicates whether a RESTART condition has occurred on the I2C interface when I2c is operating in slave mode and the slave is the addressed slave. [NOTE] However, in high-speed mode or during a START BYTE transfer, the RESTART comes before the address field as per the I2C protocol. In this case, the slave is not the addressed slave when the RESTART is issued, therefore I2c does not generate the RESTART_DET interrupt.
11	GEN_CALL	R	0x0	Set only when a General Call address is received and it is acknowledged. It stays set until it is cleared either by disabling I2c or when the CPU reads bit 0 of the IC_CLR_GEN_CALL register. I2c stores the received data in the Rx

				buffer.
10	START_DET	R	0x0	Indicates whether a START or RESTART condition has occurred on the I2C interface regardless of whether I2c is operating in slave or master mode.
9	STOP_DET	R	0x0	Indicates whether a STOP condition has occurred on the I2C interface regardless of whether I2c is operating in slave or master mode. In Slave Mode: If IC_CON[7]=1'b1 (STOP_DET_IFADDRESSED), the STOP_DET interrupt is generated only if the slave is addressed. Note: During a general call address, this slave does not issue a STOP_DET interrupt if STOP_DET_IF_ADDRESSED=1'b1, even if the slave responds to the general call address by generating ACK. The STOP_DET interrupt is generated only when the transmitted address matches the slave address (SAR). If IC_CON[7]=1'b0 (STOP_DET_IFADDRESSED), the STOP_DET interrupt is issued irrespective of whether it is being addressed. In Master Mode: If IC_CON[10]=1’b1 (STOP_DET_IF_MASTER_ACTIVE), the STOP_DET interrupt is issued only if the master is active. If IC_CON[10]=1’b0 (STOP_DET_IFADDRESSED), the STOP_DET interrupt is issued irrespective of whether the master is active.

8	ACTIVITY	R	0x0	This bit captures I2c activity and stays set until it is cleared. There are four ways to clear it: Disabling the I2c Reading the IC_CLR_ACTIVITY register Reading the IC_CLR_INTR register System reset Once this bit is set, it stays set unless one of the four methods is used to clear it. Even if the I2c module is idle, this bit remains set until cleared, indicating that there was activity on the bus.
7	RX_DONE	R	0x0	When the I2c is acting as a slave-transmitter, this bit is set to 1 if the master does not acknowledge a transmitted byte. This occurs on the last byte of the transmission, indicating that the transmission is done.
6	TX_ABRT	R	0x0	This bit indicates if I2c, as an I2C transmitter, is unable to complete the intended actions on the contents of the transmit FIFO. This situation can occur both as an I2C master or an I2C slave, and is referred to as a “transmit abort”. When this bit is set to 1, the IC_TX_ABRT_SOURCE register indicates the reason why the transmit abort takes places. [NOTE] The I2c flushes/resets/empties only the TX_FIFO whenever there is a transmit abort caused by any of the events tracked by the IC_TX_ABRT_SOURCE register. The Tx FIFO remains in this flushed state until the register IC_CLR_TX_ABRT is read. Once this read is performed, the Tx FIFO is then ready to accept more data bytes from the APB interface

5	RD_REQ	R	0x0	This bit is set to 1 when I2c is acting as a slave and another I2C master is attempting to read data from I2c. The I2c holds the I2C bus in a wait state (SCL=0) until this interrupt is serviced, which means that the slave has been addressed by a remote master that is asking for data to be transferred. The processor must respond to this interrupt and then write the requested data to the IC_DATA_CMD register. This bit is set to 0 just after the processor reads the IC_CLR_RD_REQ register.
4	TX_EMPTY	R	0x0	The behavior of the TX_EMPTY interrupt status differs based on the TX_EMPTY_CTRL selection in the IC_CON register. When TX_EMPTY_CTRL = 0: This bit is set to 1 when the transmit buffer is at or below the threshold value set in the IC_TX_TL register. When TX_EMPTY_CTRL = 1: This bit is set to 1 when the transmit buffer is at or below the threshold value set in the IC_TX_TL register and the transmission of the address/data from the internal shift register for the most recently popped command is completed. It is automatically cleared by hardware when the buffer level goes above the threshold. When IC_ENABLE[0] is set to 0, the TX FIFO is flushed and held in reset. There the TX FIFO looks like it has no data within it, so this bit is set to 1, provided there is activity in the master or slave state machines. When there is no longer any activity, then with ic_en=0, this bit is set to 0.
3	TX_OVER	R	0x0	Set during transmit if the transmit buffer is filled to IC_TX_BUFFER_DEPTH and the processor attempts to issue another I2C command by writing to the

				IC_DATA_CMD register. When the module is disabled, this bit keeps its level until the master or slave state machines go into idle, and when ic_en goes to 0, this interrupt is cleared.
2	RX_FULL	R	0x0	Set when the receive buffer reaches or goes above the RX_TL threshold in the IC_RX_TL register. It is automatically cleared by hardware when buffer level goes below the threshold. If the module is disabled (IC_ENABLE[0]=0), the RX FIFO is flushed and held in reset; therefore the RX FIFO is not full. So this bit is cleared once IC_ENABLE[0] is set to 0, regardless of the activity that continues.
1	RX_OVER	R	0x0	Set if the receive buffer is completely filled to IC_RX_BUFFER_DEPTH and an additional byte is received from an external I2C device. The I2c acknowledges this, but any data bytes received after the FIFO is full are lost. If the module is disabled (IC_ENABLE[0]=0), this bit keeps its level until the master or slave state machines go into idle, and when ic_en goes to 0, this interrupt is cleared. bit 9 of the IC_CON register (RX_FIFO_FULL_HLD_CTRL) is programmed to HIGH, then the RX_OVER interrupt never occurs, because the Rx FIFO never overflows.
0	RX_UNDER	R	0x0	Set if the processor attempts to read the receive buffer when it is empty by reading from the IC_DATA_CMD register. If the module is disabled (IC_ENABLE[0]=0), this bit keeps its level until the master or slave state machines go into idle, and when ic_en goes to 0, this interrupt is cleared.

121.3.6.15. IC_RX_TL

Table 9- 254 Fields for Register: IC_RX_TL

Bit Field	Name	R/W	Default value	Description
31:8	Reserved			
7:0	RX_TL	RW	0x00	Receive FIFO Threshold Level Controls the level of entries (or above) that triggers the RX_FULL interrupt (bit 2 in IC_RAW_INTR_STAT register). The valid range is 0-255, with the additional restriction that hardware does not allow this value to be set to a value larger than the depth of the buffer. If an attempt is made to do that, the actual value set will be the maximum depth of the buffer. A value of 0 sets the threshold for 1 entry, and a value of 255 sets the threshold for 256 entries.

121.3.6.16. IC_TX_TL

Table 9- 255 Fields for Register: IC_TX_TL

Bit Field	Name	R/W	Default value	Description
31:8	Reserved			
7:0	TX_TL	RW	0x00	Transmit FIFO Threshold Level Controls the level of entries (or below) that trigger the TX_EMPTY interrupt (bit 4 in IC_RAW_INTR_STAT register). The valid range is 0-255, with the additional restriction that it may not be set to value larger than the depth of the buffer. If an attempt is made to do that, the actual value set will be the maximum depth of the buffer. A value of 0 sets the threshold for 0 entries, and a value of 255 sets the threshold for 255 entries.

121.3.6.17. IC_CLR_INTR

Table 9- 256 Fields for Register: IC_CLR_INTR

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_INTR	R	0x0	Read this register to clear the combined interrupt, all individual interrupts, and the IC_TX_ABRT_SOURCE register. This bit does not clear hardware clearable interrupts but software clearable interrupts. Refer to Bit 9 of the IC_TX_ABRT_SOURCE register for an exception to clearing IC_TX_ABRT_SOURCE.

121.3.6.18. IC_CLR_RX_UNDER

Table 9- 257 Fields for Register: IC_CLR_RX_UNDER

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_RX_UNDER	R	0x0	Read this register to clear the RX_UNDER interrupt (bit 0) of the IC_RAW_INTR_STAT register.

121.3.6.19. IC_CLR_RX_OVER

Table 9- 258 Fields for Register: IC_CLR_RX_OVER

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_RX_OVER	R	0x0	Read this register to clear the RX_OVER interrupt (bit 1) of the IC_RAW_INTR_STAT register.

121.3.6.20. **IC_CLR_TX_OVER**

Table 9- 259 Fields for Register: IC_CLR_TX_OVER

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_TX_OVER	R	0x0	Read this register to clear the TX_OVER interrupt (bit 3) of the IC_RAW_INTR_STAT register.

121.3.6.21. **IC_CLR_RD_REQ**

Table 9- 260 Fields for Register: IC_CLR_RD_REQ

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_RD_REQ	R	0x0	Read this register to clear the RD_REQ interrupt (bit 5) of the IC_RAW_INTR_STAT register.

121.3.6.22. **IC_CLR_TX_ABRT**

Table 9- 261 Fields for Register: IC_CLR_TX_ABRT

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_TX_ABRT	R	0x0	Read this register to clear the TX_ABRT interrupt (bit 6) of the IC_RAW_INTR_STAT register, and the IC_TX_ABRT_SOURCE register. This also releases the Tx FIFO from the flushed/reset state, allowing more writes to the Tx FIFO. Refer to Bit 9 of the IC_TX_ABRT_SOURCE register for an exception to clearing IC_TX_ABRT_SOURCE.

121.3.6.23. **IC_CLR_RX_DONE**

Table 9- 262 Fields for Register: IC_CLR_RX_DONE

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_RX_DONE	R	0x0	Read this register to clear the RX_DONE interrupt (bit 7) of the IC_RAW_INTR_STAT register.

121.3.6.24. **IC_CLR_ACTIVITY**

Table 9- 263 Fields for Register: IC_CLR_ACTIVITY

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_ACTIVITY	R	0x0	Reading this register clears the ACTIVITY interrupt if the I2C is not active anymore. If the I2C module is still active on the bus, the ACTIVITY interrupt bit continues to be set. It is automatically cleared by hardware if the module is disabled and if there is no further activity on the bus. The value read from this register to get status of the ACTIVITY interrupt (bit 8) of the IC_RAW_INTR_STAT register.

121.3.6.25. **IC_CLR_STOP_DET**

Table 9- 264 Fields for Register: IC_CLR_STOP_DET

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_STOP_DET	R	0x0	Read this register to clear the STOP_DET interrupt (bit 9) of the IC_RAW_INTR_STAT register.

121.3.6.26. IC_CLR_START_DET

Table 9- 265 Fields for Register: IC_CLR_START_DET

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_START_DET	R	0x0	Read this register to clear the START_DET interrupt (bit 10) of the IC_RAW_INTR_STAT register.

121.3.6.27. IC_CLR_GEN_CALL

Table 9- 266 Fields for Register: IC_CLR_GEN_CALL

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_GEN_CALL	R	0x0	Read this register to clear the GEN_CALL interrupt (bit 11) of IC_RAW_INTR_STAT register.

121.3.6.28. IC_ENABLE

Table 9- 267 Fields for Register: IC_ENABLE

Bit Field	Name	R/W	Default value	Description
31:3	Reserved			
2	TX_CMD_BLOCK	RW	0x1	In Master mode 1’b1: Blocks the transmission of data on I2C bus even if Tx FIFO has data to transmit. 1’b0: The transmission of data starts on I2C bus automatically, as soon as the first data is available in the Tx FIFO. [Note] To block the execution of Master commands, set the TX_CMD_BLOCK bit only when Tx FIFO is empty (IC_STATUS[2]=1) and the master is in the Idle state

				(IC_STATUS[5] == 0). Any further commands put in the Tx FIFO are not executed until TX_CMD_BLOCK bit is unset.
1	ABORT	RW	0x0	When set, the controller initiates the transfer abort. 0: ABORT not initiated or ABORT done 1: ABORT operation in progress The software can abort the I2C transfer in master mode by setting this bit. The software can set this bit only when ENABLE is already set; otherwise, the controller ignores any write to ABORT bit. The software cannot clear the ABORT bit once set. In response to an ABORT, the controller issues a STOP and flushes the Tx FIFO after completing the current transfer, then sets the TX_ABORT interrupt after the abort operation. The ABORT bit is cleared automatically after the abort operation.
0	ENABLE	RW	0x0	Controls whether the I2c is enabled. 0: Disables I2c (TX and RX FIFOs are held in an erased state) 1: Enables I2c Software can disable I2c while it is active. However, it is important that care be taken to ensure that I2c is disabled properly. When I2c is disabled, the following occurs: The TX FIFO and RX FIFO get flushed. Status bits in the IC_INTR_STAT register are still active until I2c goes into IDLE state. If the module is transmitting, it stops as well as deletes the contents of the transmit buffer after the current

				transfer is complete. If the module is receiving, the I2c stops the current transfer at the end of the current byte and does not acknowledge the transfer.

121.3.6.29. **IC_STATUS**

Table 9- 268 Fields for Register: IC_STATUS

Bit Field	Name	R/W	Default value	Description
31:11	Reserved			
10	SLV_HOLD_RX_FIFO_FULL	R	0x0	This bit indicates the BUS Hold in Slave mode due to the Rx FIFO being Full and an additional byte being received
9	SLV_HOLD_TX_FIFO_EMPTY	R	0x0	This bit indicates the BUS Hold in Slave mode for the Read request when the Tx FIFO is empty. The Bus is in hold until the Tx FIFO has data to Transmit for the read request.
8	MST_HOLD_RX_FIFO_FULL	R	0x0	This bit indicates the BUS Hold in Master mode due to Rx FIFO is Full and additional byte has been received
7	MST_HOLD_TX_FIFO_EMPTY	R	0x0	the I2c master stalls the write transfer when Tx FIFO is empty, and the the last byte does not have the Stop bit set. This bit indicates the BUS hold when the master holds the bus because of the Tx FIFO being empty, and the the previous

				transferred command does not have the Stop bit set.
6	SLV_ACTIVITY	R	0x0	Slave FSM Activity Status. When the Slave Finite State Machine (FSM) is not in the IDLE state, this bit is set. 0: Slave FSM is in IDLE state so the Slave part of I2c is not Active 1: Slave FSM is not in IDLE state so the Slave part of I2c is Active
5	MST_ACTIVITY	R	0x0	Master FSM Activity Status. When the Master Finite State Machine (FSM) is not in the IDLE state, this bit is set. 0: Master FSM is in IDLE state so the Master part of I2c is not Active 1: Master FSM is not in IDLE state so the Master part of I2c is Active [NOTE] IC_STATUS[0]—that is, ACTIVITY bit—is the OR of SLV_ACTIVITY and MST_ACTIVITY bits.
4	RFF	R	0x0	Receive FIFO Completely Full. When the receive FIFO is completely full, this bit is set. When the receive FIFO contains one or more empty location, this bit is cleared. 0: Receive FIFO is not full 1: Receive FIFO is full
3	RFNE	R	0x0	Receive FIFO Not Empty. This bit is set when the receive FIFO contains one or more entries; it is cleared when the receive FIFO is empty. 0: Receive FIFO is empty

				1: Receive FIFO is not empty
2	TFE	R	0x1	Transmit FIFO Completely Empty. When the transmit FIFO is completely empty, this bit is set. When it contains one or more valid entries, this bit is cleared. This bit field does not request an interrupt. 0: Transmit FIFO is not empty 1: Transmit FIFO is empty
1	TFNF	R	0x1	Transmit FIFO Not Full. Set when the transmit FIFO contains one or more empty locations, and is cleared when the FIFO is full. 0: Transmit FIFO is full 1: Transmit FIFO is not full
0	ACTIVITY	R	0x0	I2C Activity Status.

121.3.6.30. **IC_TXFLR**

Table 9- 269 Fields for Register: IC_TXFLR

Bit Field	Name	R/W	Default value	Description
31:5	Reserved			
4:0	TXFLR	R	0x00	Transmit FIFO Level. Contains the number of valid data entries in the transmit FIFO.

121.3.6.31. **IC_RXFLR**

Table 9- 270 Fields for Register: IC_RXFLR

Bit Field	Name	R/W	Default value	Description
31:5	Reserved			
4:0	RXFLR	R	0x00	Receive FIFO Level. Contains the number of valid data entries in the receive FIFO.

121.3.6.32. **IC_SDA_HOLD**

Table 9- 271 Fields for Register: IC_SDA_HOLD

Bit Field	Name	R/W	Default value	Description
31:24	Reserved			
23:16	IC_SDA_RX_HOLD	RW	0x1	Sets the required SDA hold time in units of ic_clk period, when I2c acts as a reciever.
15:0	IC_SDA_TX_HOLD	RW	0x1	Sets the required SDA hold time in units of ic_clk period, when I2c acts as a transmitter.

121.3.6.33. **IC_TX_ABRT_SOURCE**

Table 9- 272 Fields for Register: IC_TX_ABRT_SOURCE

Bit Field	Name	R/W	Default value	Description
31:23	TX_FLUSH_CNT	R	0x000	This field indicates the number of Tx FIFO data commands that are flushed due to TX_ABRT interrupt. It is cleared whenever I2C is disabled.
22:21	Reserved			
20	ABRT_DEVICE_WRITE	R	0x0	This is a master-mode-only bit. Master is initiating the DEVICE_ID transfer and the Tx- FIFO consists of write commands.
19	ABRT_DEVICE_SLVADDR_NOACK	R	0x0	This is a master-mode-only bit. Master is initiating the DEVICE_ID transfer and the slave address sent was not acknowledged by any slave.
18	ABRT_DEVICE_NOACK	R	0x0	This is a master-mode-only bit. Master initiates the DEVICE_ID transfer and the device ID sent is not acknowledged by any slave.
17	ABRT_SDA_STUCK_AT_LOW	R	0x0	This is a master-mode-only bit. Master detects the SDA is Stuck at low for the

				IC_SDA_STUCK_AT_LOW_TIMEOUT value of ic_clks.
16	ABRT_USER_ABRT	R	0x0	This is a master-mode-only bit. Master has detected the transfer abort (IC_ENABLE[1]).
15	ABRT_SLVRD_INTX	R	0x0	1: When the processor side responds to a slave mode request for data to be transmitted to a remote master and user writes a 1 in CMD (bit 8) of IC_DATA_CMD register.
14	ABRT_SLV_ARBLOST	R	0x0	1: Slave lost the bus while transmitting data to a remote master. IC_TX_ABRT_SOURCE[12] is set at the same time.[NOTE] Even though the slave never “owns” the bus, something could go wrong on the bus. This is a fail safe check. For instance, during a data transmission at the low-to-high transition of SCL, if what is on the data bus is not what is supposed to be transmitted, then I2c no longer own the bus.
13	ABRT_SLVFLUSH_TXFIFO	R	0x0	1: Slave has received a read command and some data exists in the TX FIFO so the slave issues a TX_ABRT interrupt to flush old data in TX FIFO.
12	ARB_LOST	R	0x0	1: Master has lost arbitration, or if IC_TX_ABRT_SOURCE[14] is also set, then the slave transmitter has lost arbitration.
11	ABRT_MASTER_DIS	R	0x0	1: User tries to initiate a Master operation with the Master mode disabled.
10	ABRT_10B_RD_NORSTRT	R	0x0	1: The restart is disabled (IC_RESTART_EN bit (IC_CON[5]) = 0) and the master sends a read command in 10-bit addressing mode.

9	ABRT_SBYTE_NORSTRT	R	0x0	To clear Bit 9, the source of the ABRT_SBYTE_NORSTRT must be fixed first; restart must be enabled (IC_CON[5]=1), the SPECIAL bit must be cleared (IC_TAR[11]), or the GC_OR_START bit must be cleared (IC_TAR[10]). Once the source of the ABRT_SBYTE_NORSTRT is fixed, then this bit can be cleared in the same manner as other bits in this register. If the source of the ABRT_SBYTE_NORSTRT is not fixed before attempting to clear this bit, bit 9 clears for one cycle and then gets re-asserted. 1: The restart is disabled (IC_RESTART_EN bit (IC_CON[5]) = 0) and the user is trying to send a START Byte.
8	ABRT_HS_NORSTRT	R	0x0	1: The restart is disabled (IC_RESTART_EN bit (IC_CON[5]) = 0) and the user is trying to use the master to transfer data in High Speed mode.
7	ABRT_SBYTE_ACKDET	R	0x0	1: Master has sent a START Byte and the START Byte was acknowledged (wrong behavior).
6	ABRT_HS_ACKDET	R	0x0	1: Master is in High Speed mode and the High Speed Master code was acknowledged (wrong behavior).
5	ABRT_GCALL_READ	R	0x0	1: I2c in master mode sent a General Call but the user programmed the byte following the General Call to be a read from the bus (IC_DATA_CMD[9] is set to 1).
4	ABRT_GCALL_NOACK	R	0x0	1: I2c in master mode sent a General Call and no slave on the bus acknowledged the General Call.
3	ABRT_TXDATA_NOACK	R	0x0	1: This is a master-mode only bit. Master has received an acknowledgement for the address, but when it sent data byte(s) following the address, it did not

				receive an acknowledge from the remote slave(s).
2	ABRT_10ADDR2_NOACK	R	0x0	1: Master is in 10-bit address mode and the second address byte of the 10-bit address was not acknowledged by any slave.
1	ABRT_10ADDR1_NOACK	R	0x0	1: Master is in 7-bit addressing mode and the address sent was not acknowledged by any slave.
0	ABRT_7B_ADDR_NOACK	R	0x0	1: Master is in 7-bit addressing mode and the address sent was not acknowledged by any slave.

121.3.6.34. **IC_SLV_DATA_NACK_ONLY**

Table 9- 273 Fields for Register: IC_SLV_DATA_NACK_ONLY

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	NACK	RW	0x0	Generate NACK. This NACK generation only occurs when I2c is a slavereceiver. If this register is set to a value of 1, it can only generate a NACK after a data byte is received; hence, the data transfer is aborted and the data received is not pushed to the receive buffer. When the register is set to a value of 0, it generates NACK/ACK, depending on normal criteria. 1 = generate NACK after data byte received 0 = generate NACK/ACK normally

121.3.6.35. **IC_DMA_CR**

Table 9- 274 Fields for Register: IC_DMA_CR

Bit Field	Name	R/W	Default value	Description
31:2	Reserved			

1	TDMAE	RW	0x0	Transmit DMA Enable. This bit enables/disables the transmit FIFO DMA channel. 0 = Transmit DMA disabled 1 = Transmit DMA enabled
0	RDMAE	RW	0x0	Receive DMA Enable. This bit enables/disables the receive FIFO DMA channel. 0 = Receive DMA disabled 1 = Receive DMA enabled

121.3.6.36. **IC_DMA_TDLR**

Table 9- 275 Fields for Register: IC_DMA_TDLR

Bit Field	Name	R/W	Default value	Description
31:6	Reserved			
5:0	DMATDL	RW	0x00	Transmit Data Level. This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the transmit FIFO is equal to or below this field value, and TDMAE = 1.

121.3.6.37. **IC_DMA_RDLR**

Table 9- 276 Fields for Register: IC_DMA_RDLR

Bit Field	Name	R/W	Default value	Description
31:6	Reserved			
5:0	DMARDL	RW	0x00	Receive Data Level. This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or more than this field value + 1, and RDMAE = 1.

				For instance, when DMARDL is 0, then dma_rx_req is asserted when 1 or more data entries are present in the receive FIFO.

121.3.6.38. IC_SDA_SETUP

Table 9- 277 Fields for Register: IC_SDA_SETUP

Bit Field	Name	R/W	Default value	Description
31:8	Reserved			
7:0	SDA_SETUP	RW	0x64	SDA Setup. It is recommended that if the required delay is 1000ns, then for an ic_clk frequency of 10 MHz, IC_SDA_SETUP should be programmed to a value of 11. IC_SDA_SETUP must be programmed with a minimum value of 2.

121.3.6.39. IC_ACK_GENERAL_CALL

Table 9- 278 Fields for Register: IC_ACK_GENERAL_CALL

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	ACK_GEN_CALL	RW	0x1	ACK General Call. When set to 1, I2c responds with a ACK (by asserting ic_data_oe) when it receives a General Call. When set to 0, the I2c does not generate General Call interrupts.

121.3.6.40. IC_ENABLE_STATUS

Table 9- 279 Fields for Register: IC_ENABLE_STATUS

Bit Field	Name	R/W	Default value	Description

31:3	Reserved			
2	SLV_RX_DATA_LOST	R	0x0	Slave Received Data Lost. This bit indicates if a Slave-Receiver operation has been aborted with at least one data byte received from an I2C transfer due to setting IC_ENABLE[0] from 1 to 0. When read as 1, I2c is deemed to have been actively engaged in an aborted I2C transfer (with matching address) and the data phase of the I2C transfer has been entered, even though a data byte has been responded with a NACK. [NOTE] If the remote I2C master terminates the transfer with a STOP condition before the I2c has a chance to NACK a transfer, and IC_ENABLE[0] has been set to 0, then this bit is also set to 1. When read as 0, I2c is deemed to have been disabled without being actively involved in the data phase of a Slave- Receiver transfer.
1	SLV_DISABLED_WHILE_BUSY	R	0x0	Slave Disabled While Busy (Transmit, Receive). This bit indicates if a potential or active Slave operation has been aborted due to setting bit 0 of the IC_ENABLE register from 1 to 0. This bit is set when the CPU writes a 0 to bit 0 of IC_ENABLE while: (a) I2c is receiving the address byte of the Slave-Transmitter operation from a remote master; OR, (b) address and data bytes of the Slave-Receiver operation from a remote master. When read as 1, I2c is deemed to have forced a NACK during any part of an I2C transfer, irrespective of whether

				the I2C address matches the slave address set in I2c (IC_SAR register) OR if the transfer is completed before bit 0 of IC_ENABLE is set to 0, but has not taken effect. NOTE: If the remote I2C master terminates the transfer with a STOP condition before the I2c has a chance to NACK a transfer, and bit 0 of IC_ENABLE has been set to 0, then this bit will also be set to 1. When read as 0, I2c is deemed to have been disabled when there is master activity, or when the I2C bus is idle. [NOTE] The CPU can safely read this bit when IC_EN (bit 0) is read as 0.
0	IC_EN	R	0x0	ic_en Status. This bit always reflects the value driven on the output port ic_en. When read as 1, I2c is deemed to be in an enabled state. When read as 0, I2c is deemed completely inactive. [NOTE] The CPU can safely read this bit anytime. When this bit is read as 0, the CPU can safely read SLV_RX_DATA_LOST (bit 2) and SLV_DISABLED_WHILE_BUSY (bit 1).

121.3.6.41. **IC_FS_SPKLEN**

Table 9- 280 Fields for Register: IC_FS_SPKLEN

Bit Field	Name	R/W	Default value	Description
31:8	Reserved			

7:0	IC_HS_SPKLEN	RW	0x5	This register must be set before any I2C bus transaction can take place to ensure stable operation. This register sets the duration, measured in ic_clk cycles, of the longest spike in the SCL or SDA lines that are filtered out by the spike suppression logic;This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. The minimum valid value is 1; hardware prevents values less than this being written, and if attempted, results in 1 being set.

121.3.6.42. IC_HS_SPKLEN

Table 9- 281 Fields for Register: IC_HS_SPKLEN

Bit Field	Name	R/W	Default value	Description
31:8	Reserved			
7:0	IC_HS_SPKLEN	RW	0x5	This register must be set before any I2C bus transaction can take place to ensure stable operation. This register sets the duration, measured in ic_clk cycles, of the longest spike in the SCL or SDA lines that are filtered out by the spike suppression logic;This register can be written only when the I2C interface is disabled, which corresponds to IC_ENABLE[0] being set to 0. Writes at other times have no effect. The minimum valid value is 1; hardware prevents values less than this being written, and if attempted, results in 1 being set.

121.3.6.43. IC_CLR_RESTART_DET

Table 9- 282 Fields for Register: IC_CLR_RESTART_DET

Bit Field	Name	R/W	Default	Description

			value	
31:1	Reserved			
0	CLR_RESTART_DET	R	0x0	Read this register to clear the RESTART_DET interrupt (bit 12) of the IC_RAW_INTR_STAT register.

121.3.6.44. IC_SCL_STUCK_AT_LOW_TIMEOUT

Table 9-283 Fields for Register: IC_SCL_STUCK_AT_LOW_TIMEOUT

Bit Field	Name	R/W	Default value	Description
31:0	SCL_STUCK_AT_LOW_TIMEOUT	RW	0xffffffff	i2c generates the interrupt to indicate SCL stuck at low if it detects the SCL stuck at low for the IC_SCL_STUCK_LOW_TIMEOUT in units of ic_clk period.

121.3.6.45. IC_SDA_STUCK_AT_LOW_TIMEOUT

Table 9-284 Fields for Register: IC_SDA_STUCK_AT_LOW_TIMEOUT

Bit Field	Name	R/W	Default value	Description
31:0	SDA_STUCK_AT_LOW_TIMEOUT	RW	0xffffffff	I2c initiates the recovery of SDA line through enabling the SDA_STUCK_RECOVERY_EN (IC_ENABLE[3]) register bit, if it detects the SDA stuck at low for the IC_SDA_STUCK_LOW_TIMEOUT in units of ic_clk period.

121.3.6.46. IC_CLR_SCL_STUCK_DET

Table 9-285 Fields for Register: IC_CLR_SCL_STUCK_DET

Bit Field	Name	R/W	Default value	Description
31:1	Reserved			
0	CLR_SCL_STUCK	R	0x0	Read this register to clear the

				SCL_STUCK_DET interrupt (bit 14) of the IC_RAW_INTR_STAT register.

121.3.6.47. IC_DEVICE_ID

Table 9-286 Fields for Register: IC_DEVICE_ID

Bit Field	Name	R/W	Default value	Description
31:24	Reserved			
23:0	DEVICE-ID	R	0x000000	Contains the Device-ID of the component

121.3.6.48. IC_COMP_PARAM_1

Table 9-287 Fields for Register: IC_COMP_PARAM_1

Bit Field	Name	R/W	Default value	Description
31:24	Reserved			
23:16	TX_BUFFER_DEPTH	R	0x1f	Depth of transmit buffer
15:8	RX_BUFFER_DEPTH	R	0x1f	Depth of receive buffer
7	ADD_ENCODED_PARAMS	R	0x1	Reading 1 in this bit means that the capability of reading these encoded parameters via software has been included. Otherwise, the entire register is 0 regardless of the setting of any other parameters that are encoded in the bits.
6	HAS_DMA	R	0x1	Include DMA handshaking interface signals? 0: False 1: True
5	INTR_IO	R	0x1	Single Interrupt output port present? 0: Individual 1: Combined
4	HC_COUNT_VALUES	R	0x0	Hard code the count values for each mode? 0: False 1: True

3:2	MAX_SPEED_MODE	R	0x3	Highest speed I2C mode supported 0x0 = Reserved 0x1 = Standard 0x2 = Fast 0x3 = High
1:0	APB_DATA_WIDTH	R	0x2	APB data bus width 0x0 = 8 bits 0x1 = 16 bits 0x2 = 32 bits 0x3 = Reserved

121.3.6.49. **IC_COMP_VERSION**

Table 9-288 Fields for Register: IC_COMP_VERSION

Bit Field	Name	R/W	Default value	Description
31:0	IC_COMP_VERSION	R		Component Version ID

121.4. GPIO

121.4.1. Overview

GPIO controller is integrated on the APB bus. Totally it can provide 32-bit channels. Each channel can be enabled and configured as input or output. Interrupt can be programmed to be generated when input channel change the value.

121.4.2. Features

GPIO supports the following features:

- Compliance with the AMBA2.0 APB interface
- Total 32 input/output pins, setup by software
- Alternative control modes set by software
- After reset, all the GPIO lines are configured as default inputs in software mode
- Programmable interrupt generation capability on any number of pins
- Two interrupts (GPIO_INT0 and GPIO_INT1) per GPIO bank, and by default the lower 16- of one bank will be routed to GPIO_INT0 of that bank while the higher 16- of one bank will be routed

to GPIO_INT1 of that bank

- Edge sensitive conditional interrupts configurable on each GPIO_in pins
- De-bounce circuits can be enabled/disabled for each data line
- The definition of filtered glitch (de-bounce pulse width) is programmable
- At least 4 channels GPIO can work when SOC is in low power mode

121.4.3. Function Description

121.4.3.1. Block Description

The APB interface accesses the registers to read/write data and status, the Data_ctrl block decides the data input or output, and Intr_detect block compare data to defined values. To avoid mistake, the DBCLK block is used to eliminate the bounce generated in the setup phase.

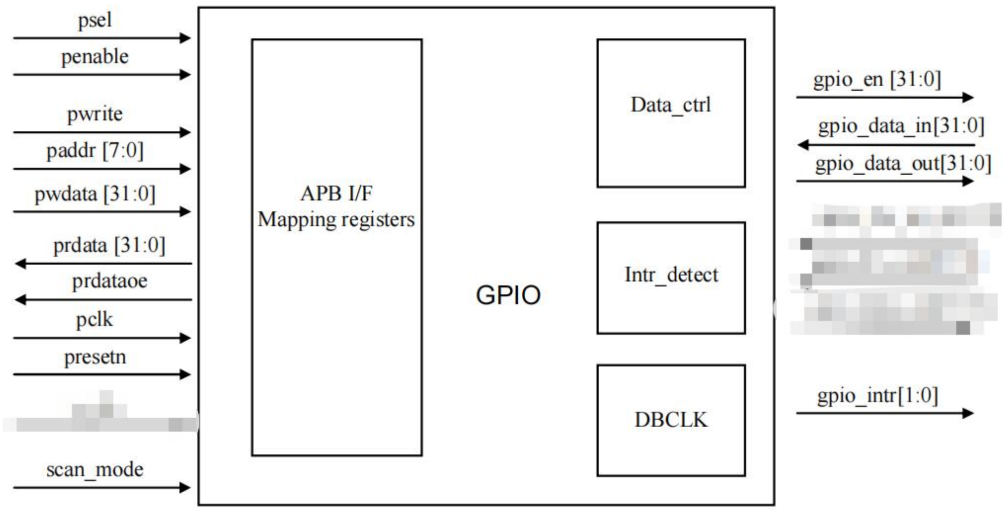


Figure 9-1 Block Structure Diagram

121.4.3.2. Timing Description

GPIO will filter the bounce of data_in by default. While the signal latched twice or more, keeping valid more than one clock pluses, will be passed and delayed for sync output.

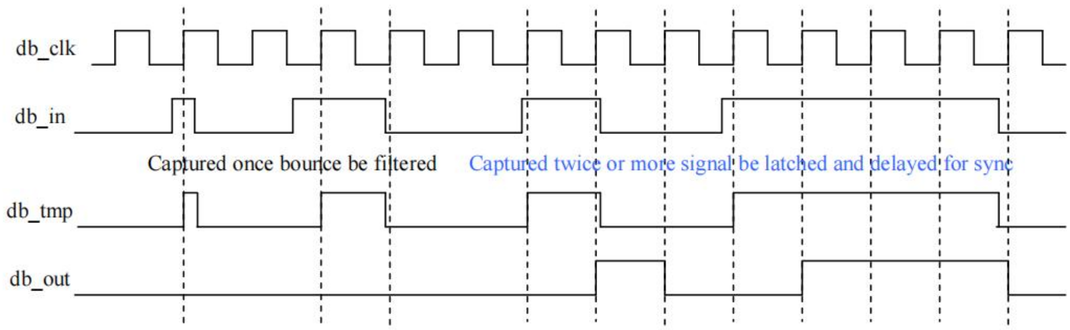


Figure 9-2 De-bounce timing diagram

The input signal will be output with more stability after sync as s_clk.

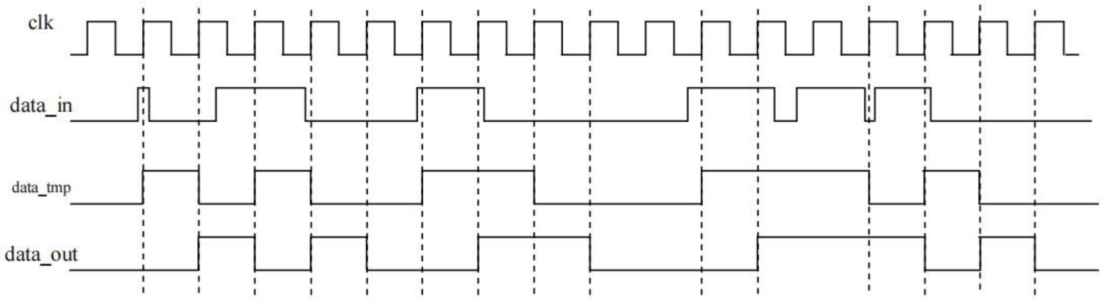


Figure 9-3 Meta-stability sync timing diagram

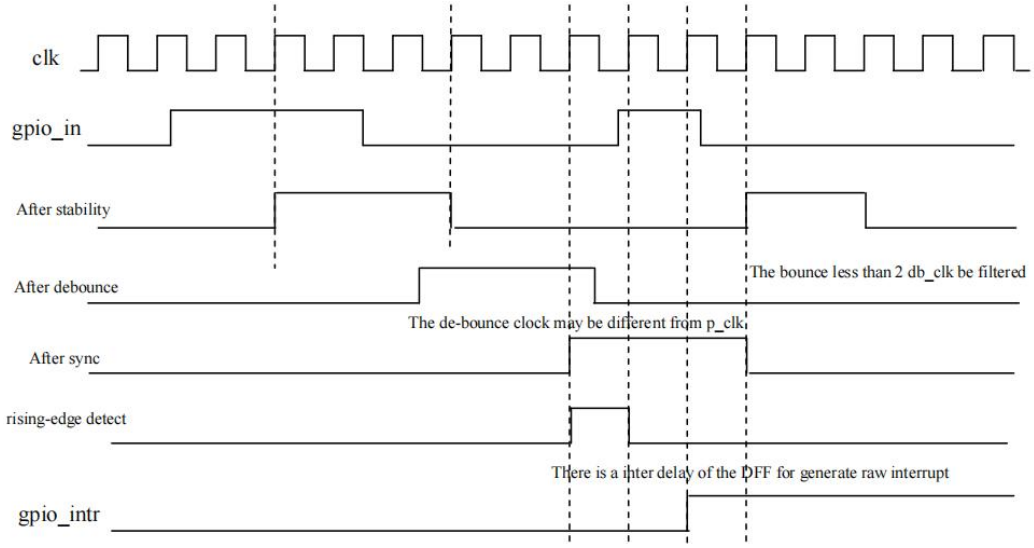


Figure 9-4 Interrupt timing diagram

121.4.4. **Programming Guide**

121.4.4.1. **Setup the data flow**

1. Program the register "gpio_dir", to setup the direction of data stream for each pin. Write "0" to “gpio_dir” means data flow is from gpio_in to gpio_ext; write "1" means data flow is from gpio_data to gpio_out and gpio_ext.
2. Keep gpio_in unchangeable during four setup clock cycles (2 de-bounce clock cycles and 2 meta-stability clock cycles.)
3. In all modes the read only register, ie "gpio_ext", shares the same value as the gpio_in register when gpio_dir is LOW; or the same value as the gpio_data register when gpio_dir is HIGH.

121.4.4.2. Setup the interrupt

To work in edge-triggered interrupts mode, the software must perform the following initialization sequence to prevent spurious interrupt from being interpreted by the system:

1. To trigger interrupt with gpio_ext in setup condition, it is necessary to clear gpio_dir in all modes.
2. It is strongly recommended to clear gpio_ien to avoid the abnormity;
3. Program "gpio_db" to enable or disable de-bounce the glitch;
4. Program the register: "gpio_ibe", appropriately as individual or both-edge detection;
5. Program the register: "gpio_iev", if have selected individual edge transactions previously;
6. Program the register: "gpio_is", to select edge-triggered path;
7. Program the "gpio_dfg" and define Filtered Glitch to select the source for db_clk: from pclk or divided from pclk;
8. Program the register "gpio_ien" to enable the interrupt;
9. Ensure the pin of "gpio_in" register remains stable throughout this operation;
10. Read the “gpio_ris” to wait for the interrupt;11. Program the register "gpio_ic" to clear the edge-detected interrupt.

121.4.4.3. Clear the interrupt

ISR write the gpio_ic to clear the interrupt status only for edge detecting interrupt. If new edge sensitivity interrupt detected at the same clock as clear interrupt status register, the new interrupt will keep active until next clear.

121.4.4.4. About the de-bounce

At first gpio_db should be asserted to delete bounce;

The glitches should be filtered to avoid the disturb bounce and secure correct interrupts. Choose right db_clk can minimize the bounce to less than 2 clock cycles of db_clk;

To setup accurate db_clk, please program the value of the register “gpio_dfg” (Define Filter-ed Glitch) as per the following table:

Table 9-1 Definition of db_clk

The value of gpio_dfg (Hex)	The definition of db_clk
32’h0000_0000	pclk, as systemsystem clk
32’hFFFF_FFFF	De-bounce clk tie to 0
Any other value else	gpio_dfg+1

There is only one db_clk for each GPIO implementation.

121.4.5. Register Summary

Table 9-2 Register List for GPIO

Name	Offset	Description
GPIO_DATA	0x00	Data of Input/Output register
GPIO_DIR	0x04	Direction of data flow register
GPIO_CTRL	0x08	Control mode select register (reserved and not in use)
GPIO_EXT	0x0C	Port data extend register
GPIO_IEN	0x10	Interrupt enable register
GPIO_IS	0x14	Interrupt sense register
GPIO_IBE	0x18	Interrupt both edge register
GPIO_IEV	0x1C	Interrupt event register
GPIO_RIS	0x20	Raw interrupt status register
GPIO_IM	0x24	Interrupt mask register
GPIO_MIS	0x28	Masked interrupt status register
GPIO_IC	0x2C	Interrupt clear register
GPIO_DB	0x30	Delete Bounce flag
GPIO_DFG	0x34	Define Filter-ed Glitch
GPIO_IG	0x38	Define Interrupt Group

121.4.6. **Register Description**

121.4.6.1. **GPIO_DATA**

Table 9- 3 GPIO_DATA Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	Each bit driving the output data of corresponding gpio_out in software mode.

121.4.6.2. **GPIO_DIR**

Table 9- 4 GPIO_DIR Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	Each bit mapping to the gpio_en_n and control the direction of corresponding gpio_xp in software mode: "0" for input from gpio_in, "1" for output to gpio_out.

121.4.6.3. GPIO_CTRL

Table 9- 5 GPIO_CTRL Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	"0" for software mode, each bit of gpio_dir is used to setup the direction for corresponding pin; "1" for hardware mode, auxiliary hardware signal, gpio_hw_en_n, is used to configure the direction for the corresponding pin. Hardware is disable now.

121.4.6.4. GPIO_EXT

Table 9- 6 GPIO_EXT Register

Bits	Access	Default	Description
[31:0]	RO	32’h0000_0000	For each bit, the value is the corresponding pin of gpio_in in input direction; or the same as the corresponding bit of gpio_data in output direction.

121.4.6.5. GPIO_IEN

Table 9- 7 GPIO_IEN Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	For each bit, "0" is disable the interrupt event on the corresponding pin, while "1" to enable the corresponding pins trigger interrupt.

121.4.6.6. GPIO_IS

Table 9- 8 GPIO_IS Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	For each bit, "0" is edge-detecting and "1" is level-sensitive on corresponding pin.

121.4.6.7. GPIO_IBE

Table 9- 9 GPIO_IBE Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	For each bit, "0" is single edge-sensitive (either falling edge or rising edge), decided by the bit of gpio_iev. "1" is both edges on corresponding pin trigger an interrupt.

121.4.6.8. GPIO_IEV

Table 9- 10 GPIO_IEV Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	For each bit, "0" means that interrupt is triggered in falling edge of corresponding pins, "1" means interrupt is triggered in rising edge of corresponding pins.

121.4.6.9. GPIO_RIS

Table 9- 11 GPIO_RIS Register

Bits	Access	Default	Description
[31:0]	RO	32’h0000_0000	Raw Interrupt status register before masked. “1”: Raw Interrupt “0”: No raw Interrupt

121.4.6.10. GPIO_IM

Table 9- 12 GPIO_IM Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	For each bit, "0" means that the interrupt on the corresponding gpio_ext bit is unmasked while "1" is masked.

121.4.6.11. GPIO_MIS

Table 9- 13 GPIO_MIS Register

Bits	Access	Default	Description
[31:0]	RO	32’h0000_0000	For each bit, this is the masked interrupt status on the corresponding gpio_ext bit.

121.4.6.12. GPIO_IC

Table 9- 14 GPIO_IC Register

Bits	Access	Default	Description
[31:0]	WO	32’h0000_0000	For each bit, write "1" to clear the edge-sensitive interrupt status on the corresponding pin, write "0" to a bit has no effects on the corresponding interrupt.

121.4.6.13. GPIO_DB

Table 9- 15 GPIO_DB Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	For each bit, write "1" to enable the function of debounce for each line, write "0" to disable the debounce for each line.

121.4.6.14. GPIO_DFG

Table 9- 16 GPIO_DFG Register

Bits	Access	Default	Description
[31:0]	RW	32’h0000_0000	The value is 32'h0000_0000, the db_clk, which is used to define the glitch, will be pclk. The value is 32'hFFFF_FFFF, the db_clk will tie to 0. Otherwise the db_clk will be gpio_dfg+1. Anyway, it has no effect on de-bounce when gpio_db is 0.

121.4.6.15. GPIO_IG

Table 9- 17 GPIO_IG Register

Bits	Access	Default	Description

[31:0]	RW	32'hffff_0000	Determines the interrupt group for 32 GPIO pins, each bit represents one GPIO pin. It will be associated with intr[1] when the bit is programmed to 1, and associated with intr[0] when the bit is programmed to 0.

121.5. SPI

Two Master/Slave SPI Interfaces, with at least two chip selects (channels) per interface. There are two general SPI, SPI0 and SPI1s.

121.5.1. Features

SPI0 and SPI1 support the following features:

- Buffered receive/transmit data register per channel (16 words deep)
- Multiple SPI word access with one channel using a FIFO
- Two DMA requests per channel, one interrupt line
- Support for double buffering, ping-pong ability is needed for each channel
- Single interrupt line, for multiple interrupt source events
- Serial link interface should support:
- Full duplex / half duplex
- Multi-channel master or single channel slave operation
- Programmable 1-32 bit transmit/receive shift operations
- Wide selection of SPI word lengths continuous from 4 to 32
- SPI word Transmit / Receive slot assignment based on round robin arbitration
- SPI configuration per channel (clock definition, enable polarity and word width)
- Clock generation should support:
- Programmable master clock generation of up to 48MHz or greater
- Select-able clock phase and clock polarity per chip select
- Two chip selects(2 channels) per SPI interface.
- Legacy (X1) SPI for SPI0 and SPI1

121.5.2. Function Description

121.5.2.1. Function Overview

SPI module including the following basic sub-modules, as shown in below figure:

- Bus Interface Logic
- Register Logic

- Transmitter FIFO
- Receiver FIFO
- Control Logic

122.

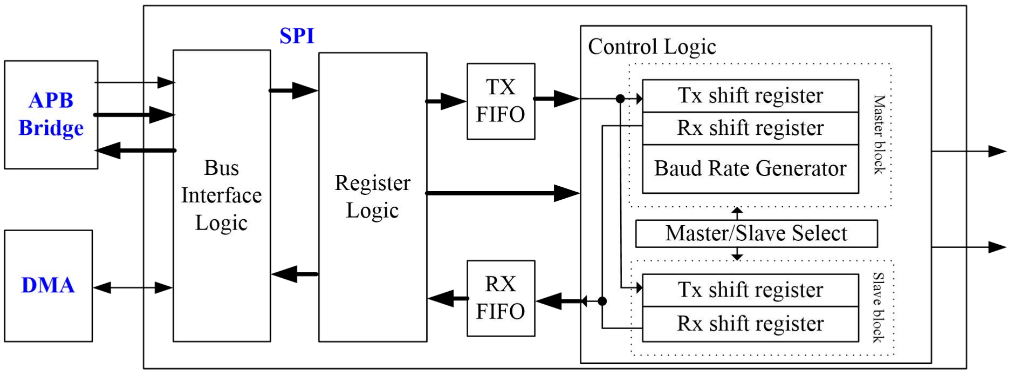


Figure 9- 5 SPI Basic Sub-modules

122.1.1.1. **Baud Rate Generator**

123. The Baud Rate Generator (BRG) is a dedicated16-bit baud rate generator. It is applied in the SPI master mode. It controls the transmit or receive frequency.

The baud rate is the frequency of the generated SCLK, which divided by the PCLK.

124. Given the desired baud rate and fpclk (APB clock frequency), the nearest integer value for the SPBRG register can be calculated using the formula by Baud Rate = fpclk / X, where X equals the value in the SPBRG register (2 to65535).

124.1.1.1. Bus Interface Logic

125. CPU can access control register, transmit data register and receive data register through APB or AHB bus interface. The DMA can access this block read receive data register and write transmit data register, and cannot access all other memory mapped register.

126. CPU access is prior to DMA when they are writing the same register simultaneously. CPU and DMA can get register data at the same time when CPU and DMA read the same register simultaneously.

126.1.1.1. Register logic

127. The register logic block contains all memory mapped register Bus Interface Logic

127.1.1.1. Transmitter FIFO

128. The transmitter FIFO is 32 words level FIFO. CPU or DMA write data to TX FIFO. If transmit, TX FIFO shift data to transmit shift register.

129. A transmit operation begins when data is written to the transmit data register (TXREG). When the transmit shift register is empty, TX FIFO output data to the transmitter shift register. Then the shift register shifts data to the serial port (TX).

130. The "txbreq" signal is set when TX FIFO is not full and need data to transmit, the signal sends to DMA and request DMA transmits data to TX FIFO.

131. When transmit, the interrupt "tx_intf" will generate when TX FIFO is empty (reach FIFO interrupt trigger-level).

131.1.1.1. Receiver FIFO

The receiver FIFO is 32 words level FIFO. CPU or DMA receive data from RX FIFO.

The receive shift register is used to receive data from the serial port (**RX**). Incoming data is placed into the receive shift register. When all bits are shifted in, data is transferred to receiver FIFO. The receive register ready bit (**rxavl**) in the **CSTAT** register is set when data is transferred from the shift register to the data register. The bit corresponding to the serial port interrupt (**rx_intf**) is also set in **INTSTAT** register. CPU then receives data from RXREG register. The same time, the "**rxbreq**" is set and DMA read data from RXREG register.

131.1.1.2. Control logic

SPI allows **4-32 bits** of data to be synchronously transmitted and received simultaneously. The SPI can be configured as a slave or a master in single-master environment. All four SPI modes are available using the clock polarity "**ckpl**" and phase "**ckph**" settings. The data bit order can also be set to either **LSB first** or **MSB first**.

The transmitter and receiver sections use the same clock, data is only output during the rising or falling edge of clock, and data is latched during the opposite edge of **SCLK**. Since SPI is an exchange of data, data must be read at the end of the transfer. This is true even if the data is not used in the program. In SPI mode, clock phase and polarity should be identical for the master SPI device and the communicating slave device.

The SPI system consists of two data lines and one clock line:

- Master Out Slave In (MOSI/TX/SDA): This data line supplies the output data from the master shifted into the input(s) of the slave(s).
- Master In Slave Out (MISO/RX): This data line supplies the output data from a slave to the input of the master. There may be no more than one slave transmitting data during any particular transfer.
- Serial Clock (SCLK/SCL): This clock line is driven by the master and regulates the flow of the data bits. The master may transmit data at a variety of baud rates; the SCLK line cycles once for each bit that is transmitted.

132.

133. Master Mode:

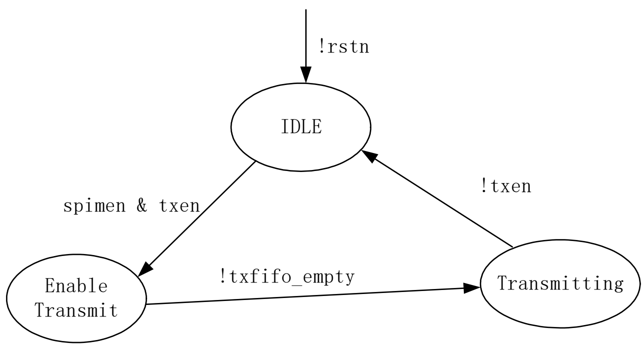
In master mode, the module only supports single-master environment. Only a master SPI module can initiate transmissions and receptions.

The SPI transmitter block diagram is shown in **Figure 11-35**. The heart of the transmitter is the transmit (serial) shift register (**TSR[7:0]**). The shift register obtains its data from **TX FIFO,** and the write transmit buffer register **TXREG** sends data to **TX FIFO**. The **TXREG** register is loaded with data in software from APB peripheral bus or DMA bus. If the **TSR** register is empty, the byte data of **TX FIFO** immediately transfers to the shift register **TSR**. Once the **TX FIFO** is empty and interrupt bit, **"tx_intf"** is set. The interrupt can be enabled/disabled by setting/clearing enable bit "**txien**". Flag bit **"tx_intf"** will be set regardless of the state of enable bit "**txien**" and only be cleared in software. The TSR is not mapped in data memory so it is not available to the user.
The master can initiate the data transfer at any time because it controls the **SCLK**. The master determines when the slave is to broadcast data by the software protocol.

In master mode the data is transmitted/received as soon as the **TXREG** register is written to. If the SPI is only going to receive, the **MOSI** output could be disabled. The **RSR** register will continue to shift in the signal present on the **MISO** pin at the programmed clock rate. As each byte is received, it will be loaded into the **RXREG** register as if a normal received byte (interrupts and status bits appropriately set). This could be useful in receiver applications as a "line activity monitor" mode.

The clock polarity is selected by appropriately programming the **"ckph" "ckpl"** bits. **SCLK** is a gated clock, active only during data transfers.

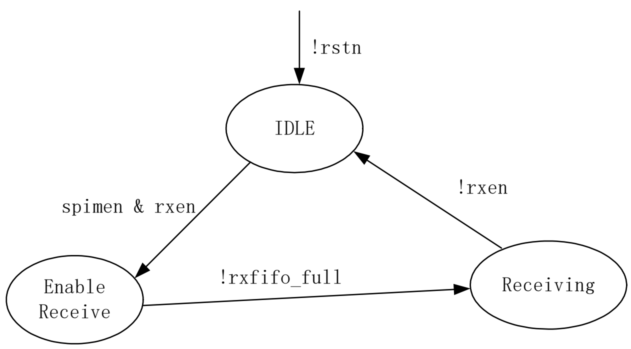
In SPI master mode, the transmit and receive FSM as following:



134.

Figure 9- 6 SPI Master Mode Transmit FSM

135.



136.

Figure 9- 7 SPI Master Mode Receive FSM

Slave Mode:

137. In slave mode, the SPI receives messages from an SPI master. In slave mode, the data is transmitted and received as the external clock pulses appear on SCLK.

138.

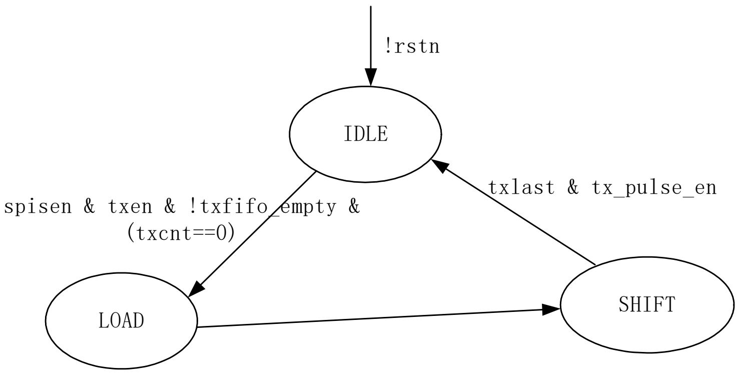
139. While in slave mode the external clock is supplied by the external clock source on the SCLK pin. This external clock must meet the minimum high and low times as specified in the electrical specifications.

140.

141. To prepare for data transfers, the slave SPI writes data to be sent into TXREG register, write data to TX FIFO, and then transfer to TSR register. Once SCLK is enabled, the slave shifts data out from MISO and in through MOSI.

142.

143. In SPI slave mode, the transmit and receive FSM are as following:

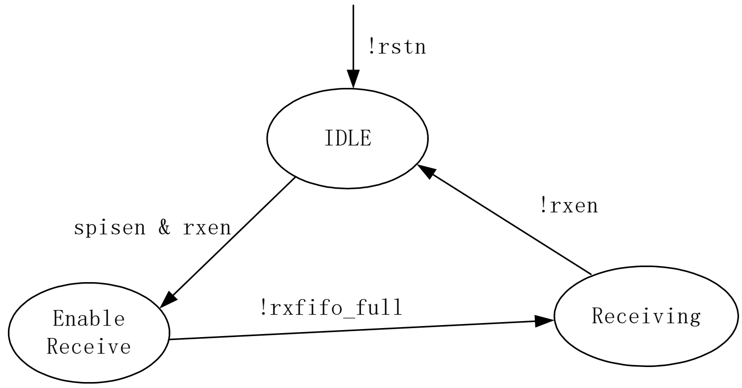


144.

145.

Figure 9- 8 SPI Slave Mode Transmit FSM

146.



147.

148.

Figure 9- 9 SPI Slave Mode Receive FSM

Data Stitch:

When enable data stitch by set rx_stitch and tx_stitch bits, you can stitch short datas into word in TX fifo and RX fifo. Stitch format is shown below:
Spilen < =5'h7, the spi datas will be stitched into a word with 4 bytes.
5'h8<=Spilen <=5'hf, the spi datas will be stitched into a word with 2 half words.
5'h10<=spilen<=5'h1f, the spi datas will be stitched into a word.
Some examples are shown below:

149.

Example1: spi_len = 5'h5

150.

31	30	29	28	27	26	25	24	……	7	6	5	4	3	2	1	0
		3									0					
		7									4					

151.

Figure 9- 10 SPI Data Stitch Example 1

152.

Example2: spi_len = 5'hd

153.

31	30	29	28	……	17	16	15	14	13	12	……	2	1	0
		1							0					
		3							2					

154.

Figure 9- 11 SPI Data Stitch Example 2

155.

Example3: spi_len = 5'h1c

31	30	29	28	……	4	3	2	1	0
					0				
					1				

Figure 9- 12 SPI Data Stitch Example 3

156.

156.1.1. Programming Guide

Master Mode：

1. Configure spilen, lsbfe,ckph,ckpl in CCTL register
2. Configure SPBRG register
3. Configure SCSR register
4. Configure rxtlf, txtlf,txen, rxen, mm,int_en,spien in GCTL register
5. Write TXREG to transmit data
6. Enable INTENA register to enable needed interrupt
7. Read RXREG when get interrupt

Slave Mode：

1. Configure spilen, lsbfe,ckph,ckpl in CCTL register
2. Configure rxtlf, txtlf,txen, rxen, mm,int_en,spien in GCTL register
3. Write TXREG to transmit data
4. Enable INTENA register to enable needed interrupt
5. Read RXREG when get interrupt

156.1.2. Register Summary

Table 9- 18 SPI Register Summary

Name	Address Offset	R/W	Description
TXREG	0x0	RW	Transmit Data Register
RXREG	0x4	R	Receive Data Register
CSTAT	0x8	R	Current Status Register
INTSTAT	0xC	R	Interrupt Status Register
INTEN	0x10	RW	Interrupt Enable Register
INTCLR	0x14	W	Interrupt Clear Register
GCTL	0x18	RW	Global Control Register
CCTL	0x1C	RW	Common Control Register
SPBRG	0x20	RW	Baud Rate Generator
RXDNR	0x24	RW	Receive Data Number Register
TXDNR	0x28	RW	Transmit Data Number Register
SCSR	0x2C	RW	Slave Chip Select Register
Reserved	0x30 ~ 0x3FF		Transmit Data Register

156.1.3. Register Description

156.1.3.1. TXREG

Table 9- 19 TXREG

Bit Field	Name	Width	R/W	Default value	
31:0	TxData	32	RW	0x00000000	The transmit data register contains the next complete data byte to transmit FIFO, and then be shifted out the output shift register. The valid bits count in this reg depend on the value of spilen and data stitch.

156.1.3.2. RXREG

Table 9- 20 RXREG

Bit Field	Name	Width	R/W	Default value	

31:0	RxData	32	R	0x00000000	The receive data register low byte contains the byte value of receiver FIFO that shifted in using the input shift register. The register is read-only. The valid bits count in this reg depend on the value of spilen and data stitch.

156.1.3.3. **CSTAT**

Table 9- 21 CSTAT

Bit Field	Name	Width	R/W	Default value	
31:4	Reserved				
3	rxavl_4	1	R	0x0	Receive available 4 word data message in RX fifo. The bit is set when Receiver FIFO has received more than 4 available data. 1 = receiver FIFO has more than 4 available data 0 = receiver FIFO empty
2	txfull	1	R	0x0	Transmitter FIFO Full Status bit 1 = Transmitter FIFO is full 0 = Transmitter FIFO is not full
1	rxavl	1	R	0x0	The bit is set when Receiver FIFO has received a complete word data. 1 = receiver FIFO has a complete available word data 0 = receiver FIFO empty
0	txept	1	R	0x1	Transmitter empty bit. 1 = transmitter FIFO and TX shift register are empty 0 = transmitter not empty

156.1.3.4. **INTSTAT**

Table 9- 22 INTSTAT

Bit Field	Name	Width	R/W	Default value	
31:8	Reserved				
7	txmatch_intf	1	R	0x0	The bit is set when SPI master completes a transmit task in transmit mode or duplex mode and SPI slave completes a transmit task in transmit

					mode. When transmit data match the TXDNR number, the transmit process will be completed and generate the interrupt.
6	txept_intf	1	R	0x0	Transmitter empty interrupt flag bit. 1 = transmitter FIFO and TX shift register are empty 0 = transmitter not empty Note: the bit is interrupt status signal. “txept” is a status signal.
5	rxfo_full_intf	1	R	0x0	RX FIFO Full interrupt flag bit 1 = Receive FIFO full 0 = Receive FIFO not full
4	rxmatch_intf	1	R	0x0	The bit is set when SPI master completes a receive task in receive mode and SPI slave completes a receive task in receive mode or duplex mode. When receive data match the RXDNR number, the receive process will be completed and generate the interrupt.
3	rxoerr_intf	1	R	0x0	Receiver Overrun Error interrupt Flag bit 1 = Overrun error 0 = No overrun error
2	underrun_intf	1	R	0x0	SPI slave transmitter underrun flag bit 1 = Underrun error 0 = No underrun error
1	rx_intf	1	R	0x0	Receiver data available interrupt flag bit. This bit is set when Receiver FIFO has received enough data (depend on rxtlf) 1 = receiver FIFO has available data 0 = receiver FIFO not available
0	tx_intf	1	R	0x0	Transmit FIFO available interrupt flag bit (depend on txtlf) 1 = transmitter FIFO is available 0 = transmitter FIFO is not available

156.1.3.5. INTENA

Bit Field	Name	Width	R/W	Default value	
31:8	Reserved				
7	txmatchen	1	RW	0x0	Transmit data complete interrupt signal. 1 = Interrupt enabled

					0 = Interrupt not enabled
6	txept_ien	1	RW	0x0	Transmit empty interrupt enable bit. 1 = interrupt enabled 0 = interrupt not enabled
5	rxfifo_full_ien	1	RW	0x0	Receiver FIFO Full interrupt enable bit. 1 = Interrupt enabled 0 = Interrupt not enabled
4	rxmatchen	1	RW	0x0	Receive data complete interrupt signal. 1 = Interrupt enabled 0 = Interrupt not enabled
3	rxoerren	1	RW	0x0	Overrun Error interrupt enable bit 1 = Interrupt enabled 0 = Interrupt not enabled
2	underrunen	1	RW	0x0	Transmitter underrun interrupt enable bit (SPI slave mode) 1 = Interrupt enabled 0 = Interrupt not enabled
1	rxien	1	RW	0x0	Receiver FIFO interrupt enable bit. 1 = Interrupt enabled 0 = Interrupt not enabled
0	txien	1	RW	0x0	Transmitter FIFO empty interrupt enable bit. 1 = Interrupt enabled 0 = Interrupt not enabled

Table 9- 23 INTENA

156.1.3.6. **INTCLR**

Table 9- 24 INTCLR

Bit Field	Name	Width	R/W	Default value	
31:8	Reserved				
7	txmatchclr	1	RW	0x0	Transmit completed interrupt clear bit 1 = Interrupt cleared 0 = Interrupt not cleared
6	txept_iclr	1	W	0x0	Transmitter empty interrupt clear bit 1 = interrupt cleared 0 = interrupt not cleared
5	rxfifo_full_iclr	1	RW	0x0	Receiver FIFO Full interrupt clear bit. 1 = Interrupt cleared 0 = Interrupt not cleared

4	rxmatchclr	1	RW	0x0	Receive completed interrupt clear bit 1 = Interrupt cleared 0 = Interrupt not cleared
3	rxoerrclr	1	RW	0x0	Overrun Error interrupt clear bit 1 = Interrupt cleared 0 = Interrupt not cleared
2	underrunclr	1	RW	0x0	Transmitter under-run interrupt clear bit (SPI slave mode) 1 = Interrupt cleared 0 = Interrupt not cleared
1	rxiclr	1	RW	0x0	Receiver interrupt clear bit. 1 = Interrupt cleared 0 = Interrupt not cleared
0	txiclr	1	RW	0x0	Transmitter FIFO empty interrupt clear bit. 1 = Interrupt cleared 0 = Interrupt not cleared

156.1.3.7. **GCTL**

Table 9- 25 GCTL

Bit Field	Name	Width	R/W	Default value	
31:16	Reserved				
15	oe_ssn_ctrl	1	W/R	0x0	1： ssn gate oe When ssn=1, miso_oe & mosi_oe=0, output High Z. When ssn=0, miso_oe & mosi_oe bypass to port. 0： Ignol ssn.
14	TI_mod	1	RW	0x0	1:TI mode 0:normal mode
13:10	Reserved				
9	dmamode	1	RW	0x0	DMA access mode bit 1 = DMA access mode (DMA access TX FIFO and RX FIFO) 0 = normal access mode (CPU access TX FIFO and RX FIFO)
8:7	txtlf	2	RW	0x0	TX FIFO trigger level bit 00 = one or more valid space in TX FIFO 01 = 4 or more valid space in TX FIFO 10 = 8 or more valid space in TX FIFO 11 = 16 or more valid space in TX FIFO

6:5	rxtlf	2	RW	0x0	RX FIFO trigger level bit 00 = One or more valid data in RX FIFO 01 = more than 4 data in RX FIFO 10 = more than 8 data in RX FIFO 11 = more than 16 data in RX FIFO
4	rxen	1	RW	0x0	Receive Enable bit 1 = Enable receive 0 = Disable receive. It can clear Receiver FIFO. Note: when receive and transmit exist simultaneously in master mode, receive must be completed before transmit, that is, it relies on the transmit.
3	txen	1	RW	0x0	Transmit Enable bit 1 = Transmit enabled 0 = Transmit disabled. It can clear Transmitter FIFO Note: when SPI is applied in master receive mode, txen must be set in zero.
2	mm	1	RW	0x0	Master Mode bit 1 = Master mode (Clock generated internally from BRG) 0 = Slave mode (Clock from external source)
1	int_en	1	RW	0x0	SPI interrupt enable bit 1 = enable SPI interrupt 0 = disable SPI interrupt
0	spien	1	RW	0x0	SPI Select 0 = SPI disable (held in Reset) 1 = SPI enable

156.1.3.8. **CCTL**

Table 9-26 CCTL

Bit Field	Name	Width	R/W	Default value	
31:15	Reserved				
14	rx_stitch	1	RW	0x0	Receive data stitch: 0: No stitch 1: Stitch received data
13	tx_stitch	1	RW	0x0	Transmit data stitch: 0: No stitch 1: Stitch transmitted data

12:8	spilen	5	RW	0x7	spi length (ti_mode support 4 to 31 bit) 0~2: Reserved 3: 4 bits mode 4: 5 bits mode … ... 31: 32 bits mode
7:5	Reserved				
4	rxedge	1	RW	0x0	Master mode rx data sample edge select 1:sample the edge of the rxed data, high speed use 0: sample the middle of the rxed data, low speed use
3	Reserved				
2	lsbfe	1	RW	0x0	LSB First Enable 1= Data is transferred or received least significant bit (LSB) first 0= Data is transferred or received most significant bit (MSB) first
1	ckpl	1	RW	0x0	SPI Clock Polarity Select bit. It selects an inverted or non-inverted SPI clock. 1=Active-low clocks selected. In idle state SCLK is high. 0=Active-high clocks selected. In idle state SCLK is low.
0	ckph	1	RW	0x0	Clock Phase Select bit

156.1.3.9. **SPBRG**

Table 9- 27 SPBRG

Bit Field	Name	Width	R/W	Default value	
31:16	Reserved				
15:0	BaudRate	16	RW	0x2	The SPI Baud Rate Control register for baud rate. Baud Rate Formula: Baud Rate = fpclk / SPBRG, where fpclk is APB clock frequency Note: don’t write '0' to the register

156.1.3.10. RXDNR

Table 9- 28 RXDNR

Bit Field	Name	Width	R/W	Default value	
31:16	Reserved				
15:0	RxDataNum	16	RW	0x1	The register is used to hold a count of to be received beats in next receive process. SPI Master Mode: The register is valid in SPI master single receive mode (simplex mode). SPI Slave Mode: The register is valid in both simplex mode and duplex mode. In duplex mode (receive and transmit), it's also a count of to be transmitted beats. The register must be set in stitch mode. The default value is 1. The value will not change until writing a new value to it. Note don't write "0" to the register.

156.1.3.11. TXDNR

Table 9- 29 TXDNR

Bit Field	Name	Width	R/W	Default value	
31:16	Reserved				
15:0	TxDataNum	16	RW	0x1	The register is used to hold a count of to be transmitting beats in next transmitter process. The register must be set in stitch mode. SPI Master Mode: The register is valid in both simplex mode and duplex mode. In duplex mode (receive and transmit), it's also a count of to be received beats. SPI Slave Mode: The register is valid in simplex mode. The default value is 1. The value will not

					change until MCU writes a new value to it. Note don't write "0" to the register.

156.1.3.12. SCSR

Table 9- 30 SCSR

Bit Field	Name	Width	R/W	Default value	
31:8	Reserved				
7:0	csn	8	RW	0xFF	Chip select output signal in Master mode. Low active. In SPI slave mode, the bit is not used. 0: slave selected 1: slave not selected Note: support 8 slave SPI device from 0 ~7

156.2. SAR ADC Controller

156.2.1. Overview

The SAR(Successive Approximation Register)ADC Controller module implements communication between CPU and on-chip analog SAR-ADC block. .

156.2.2. Features

- APB compatible interface
- Support DMA operation
- 16bit Programmable Divider to generate A/D clock
- Support 12 bit resolution A/D input data with max sample rate up to 2MHz)
- Support 8-channel ADC input, Storm in QFN package has CH0 and CH1 available only.
- Support power down function

- Support poll and interrupt transfer mode
- Support single scan or continuous scan transfer mode
- Interrupt source: channel data valid interrupt

156.2.3. SAR ADC

The SAR ADC block is designed with 8-channel 12-bits pipelined SAR A/D converter. It is optimized for low-power, low-leakage, high dynamic performance, medium speed applications. Multi single ended / Differential input: 8/4 channels， The reference voltage VREF has a large dynamic range up to VDDA. The 16-channel single-ended analog input signals CH [7:0] range from VSSA to VREF*. In practice, the analog input CH [7:0] should be limited within the range of 0.01*VREF~0.99*VREF to avoid overflow.

156.2.4. Function Description

156.2.4.1. Block Description

ADC Controller module can include the following basic sub-modules:

- APB Bus Interface Logic
- Register Logic
- DMA Handshaking Signal
- Control Logic (receiver logic, A/D clock generator)

The Block Diagram as the following:

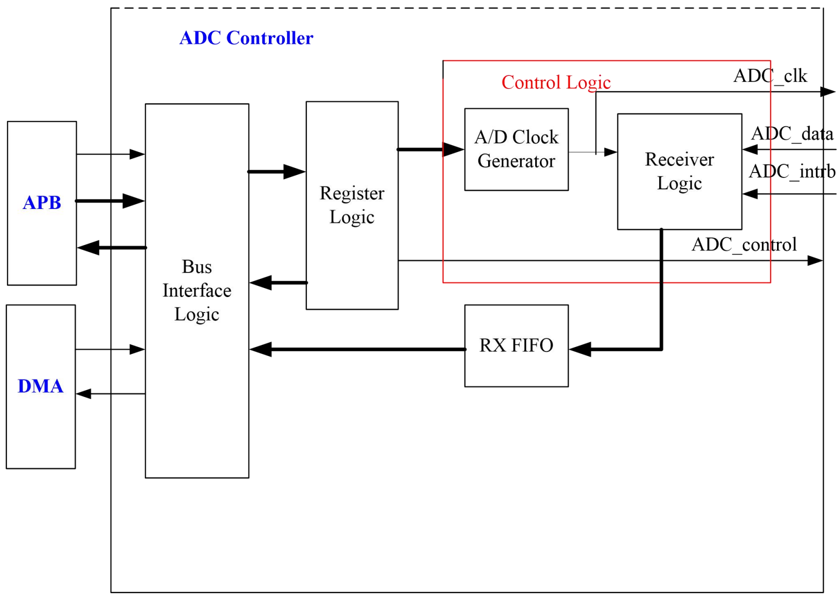


Figure 9- 13 Block Diagram of SAR-ADC controller

156.2.4.2. Single Scan Mode

In Single Mode (GCR. Continuous = 0), it will select channel N to do AD Conversion once.
The ADC Conversion program as following:
After system get the valid data, the conversion finished. It uses the following state machine:

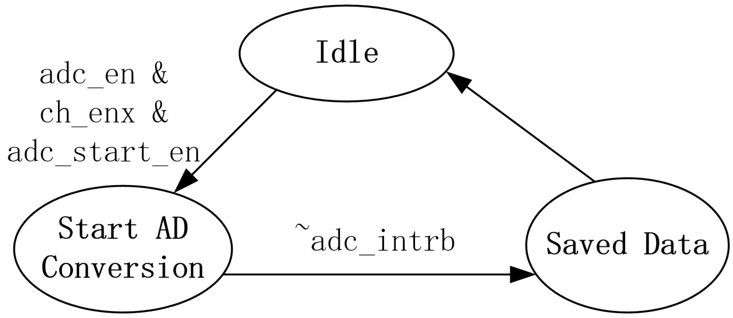


Figure 9- 14 state machine

(1) Single mode with one channel selected.

- set "adc_en"
- select channel0, adc_ch_en [7:0]=8'h0001
- set "adc_start_en" from low to high to start conversion,
- After few clock cycles, CPU read the data registers (ADCx_DR-.Data_Validx, if high, the data is valid). If interrupt enabled, get data when interrupt is active. Write"chx_int_clr" to clear the interrupt
- Disable AD Conversion, set "ADC_GCR.adc_start_en" and adc_en to low. .

NOTE:

1. When ADC_GCR.adc_rclr_enis enabled, the DATA_VALIDx and CHx_DATA will be cleared after the read operation to ADCx_DR register.
2. The interval between channel switch can be configured via ADC_COUNT register. This register can be used in both single & continuous mode.

(2) Single mode with multiple channels selected.

If "adc_ch_en [7:0]"=8'h25, there are 3 channels(channel0,channel3,channel6) selected. In single mode, the channel0 will first do AD converstion once, then channel3, then channel6, then the operation end and back to idle state.

156.2.4.3. Continuous Scan Mode

In continuous mode(GCR.Continuous = 1), it can select N channels to do AD Conversion in repeat mode. The conversion starts from converting the channel with lowest channel number.
It will use the following state machine:
Such as :
If "adc_ch_en [7:0]"=16'h0025, there are 3 channels(channel0,channel3,channel6) selected. In continuous mode, the channel0 will first do AD Conversion, then channel3, then channel6, then loop back to channel0.
Note:

- When channel 6 gets the valid data, the channel6 interrupt "ADC_ISR.ch6_intf" will be asserted.

And the interrupt could be used as a loop end interrupt.

- When "ADC_GCR.adc_start_en" or "ADC_GCR.adc_en" goes to 0, the state machine will be idle and conversion end.
- The interval between channels can be configured via ADC_COUNT register. And it applies to both of single and continuous mode.

NOTE:

It is assumed that there is no read operation to ADCx_DR register, so the data_valid will only be cleared when "adc_start_en=0".

156.2.4.4. Receiver fifo

The receiver FIFO is 32 words depth FIFO. CPU or DMA retrieves converted data from RX FIFO. The receive register ready bit (ADC_CSTAT.rxavl) in the CSTAT register is asserted when data is transferred from the shift register to the data register. The bit corresponding to the serial port interrupt (ADC_ISR.rx_intf) is also asserted. CPU then receives data from ADC_RXREG register. AT the same time, DMA request "rxbreq" is asserted to triger DMA read data from RXREG register.

156.2.5. Register Summary

Table 9- 31 SAR_ADC Register Summary

Name	Offset Address	Access	Width	Reset Value	Description
ADC_GCR	12'h000	RW	32	0x0000	ADC Global Control Register
ADC_MCR	12'h004	RW	32	0x8030	ADC Module Control Register
ADC0_DR	12'h008	R	32	0x0000	A/D Channel 0 data register
ADC1_DR	12'h00c	R	32	0x0000	A/D Channel 1 data register
ADC2_DR	12'h010	R	32	0x0000	A/D Channel 2 data register
ADC3_DR	12'h014	R	32	0x0000	A/D Channel 3 data register
ADC_CDR	12'h028	R	32	0x00FF	A/D clock divider register
ADC_ISR	12'h02c	R	32	0x0000	A/D interrupt status register
ADC_IER	12'h30	RW	32	0x0000	A/D interrupt enable register
ADC_ICR	12'h34	W	32	0x0000	A/D interrupt clear register
ADC_COUNT	12'h38	RW	32	0x001E	A/D switch interval count

					register
ADC_RXREG	12'h3c	R	32	0x0	A/D receiver data register
ADC_CSTAT	12'h40	R	32	0x0	ADC Status Register

156.2.6. **Register Description**

156.2.6.1. **ADC_GCR**

offset address : 12’h000

Table 9- 32 SAR_ADC Register ADC_GCR

Name	Bits	Access	Reset Value	Description
--	31:7	--	--	Reserved
adc_clk_sel	6	RW	0	A/D clock source select signal. 0: clock generated from internal clock divider 1: clock generated from system clock generator, NOT supported
adc_rclr_en	5	RW	0	ADC data register (ADCx_DR) read clear enable. 0: disable read clear feature of ADC data register 1: enable read clear feature of ADC data register
rxtlf	4	RW	0	RX FIFO trigger level bit 0 = One or more valid data in RX FIFO 1 = more than 16 data in RX FIFO
rxfifo_en	3	RW	0	Enable RX FIFO bit 1: Enable RX FIFO 0: RX FIFO disable, flush the data in RX FIFO
dmamode	2	RW	0	DMA access mode bit 1: DMA access mode (Only DMA access RX FIFO) 0:CPU access mode (Only CPU access RX FIFO)
continuous	1	RW	0	ADC work mode select signal. Default:0 0: Single Mode 1: Continuous Mode
adc_en	0	RW	0	ADC Controller enable signal. Default: 0 0: ADC controller module disable 1:ADC controller module enable

156.2.6.2. ADC_MCR

offset address : 12’h004

Table 9- 33 SAR_ADC Register ADC_MCR

Name	Bits	Access	Reset Value	Description
--	31:24	R	0	Reserved
adc_ch_en	23:16	RW	0x0	Enable the related ADC channel to do AD conversion. Default:0 0: channel disable 1: channel enable SAR Input Selection, If Single-Ended Mode: 8’b00: ADCINN[0] 8’b01:ADCINN[1] 8’b10: ADCINP[0] 8’b11: ADCINP[1] Else If Differential Mode: 8’h00: ADCINN[0], ADCINP[0] 8’h01: ADCINN[1], ADCINP[1]
adc_rst	15	RW	1	ADC Logic Reset Input: 0: Digital Part Reset 1: Not Reset
adc_samctrl	14:12	RW	0x00	ADC Sample Time Control (Clock Cycles): 000: 2.5 001: 4.5 010: 6.5 011: 10.5 100: 18.5 101: 34.5 110: 66.5 111: 130.5
--	11:10	R	0	Reserved
adc_srctl	9:8	R	0	Common Reference Control to save Power: 00: For <250KSPS 01: For <500KSPS 10: For <1MSPS 11: For <2MSPS
adc_ref_sel	7	RW	0	Reference Selection: 1’b0: Select VREFP1 1’b1: Select VREFP2

adc_sdsel	6	RW	0	Single or Differential Input Selection： 1’b0: Single-Ended 1’b1: Differential Inputs
adc_pdbias _en	5	RW	1	Bias Power Down Control: 0: Power On 1: Power Down
adc_pd_en	4	RW	1	SAR Analog Part Power Down Control: 0: Power On 1: Power Down
--	3:1	R	0	Reserved
adc_start_en	0	RW	0x0	ADC Conversion start enable signal. When the signal has a low to high transition, the ADC conversion start. When the signal has a high to low transition, ADC conversion operation complete. And it will be cleared when “ADC_GCR.adc_en”=0.

Note:

1) In single mode, only 1 loop AD Conversion of N channel “adc_ch_en[7:0]”can be valid. Such as, if select channel 6 to do ADConversion, set “adc_ch_en[7:0]=8’h20”.

If “adc_ch_en[7:0]”=8’h25, there are 3 channels(channel0,channel3,channel6) selected. The operation finished after channel6 conversion end.

2) In continuous mode, a few channels will be enabled and the AD Conversion will occur by the channel order (From low channel to high channel, then loop back).Such as :If “adc_ch_en[7:0]”=8’b01001001, there are channels(channel0,channel3,channel6) selected.

3) In continuous mode, the channel0 will first do AD Converstion, then channel3, then channel6, then loop back to channel0.

156.2.6.3. ADC0_DR

offset address: 12’h008

Table 9-34 SAR_ADC Register ADC0_DR

Name	Bits	Access	Reset value	Description
	31:16	R	0	Reserved.
Data_Valid0	15	R	0	Data valid signal (Only used for single mode when poll operation). It will be cleared when “ADC_GCR.adc_en”=0, or after read operation when “ADC_GCR.adc_rclr_en=1”. 0: data not valid 1: active after get the valid data
---	14:12	R	0x0	Reserved

CH0_DATA	11:0	R	0x0000	The A/D Channel 0 received data register. It will be cleared after read operation by software when “ADC_GCR.adc_rclr_en=1”.

156.2.6.4. **ADC1_DR**

offset address: 12’h00c

Table 9-35 SAR_ADC Register ADC1_DR

Name	Bits	Access	Reset value	Description
	31:16	R	0	Reserved.
Data_Valid1	15	R	0	Data valid signal(Only used for single mode when poll operation). It will be cleared when “ADC_GCR.adc_en”=0, or after read operation when “ADC_GCR.adc_rclr_en=1”. 0: data not valid 1: active after get the valid data
---	14:12	R	0x0	Reserved
CH1_DATA	11:0	R	0x0000	The A/D Channel 1 received data register. It will be cleared after read operation by software when “ADC_GCR.adc_rclr_en=1”.

156.2.6.5. **ADC2_DR**

offset address: 12’h010

Table 9-36 SAR_ADC Register ADC2_DR

Name	Bits	Access	Reset value	Description
	31:16	R	0	Reserved.
Data_Valid2	15	R	0	Data valid signal(Only used for single mode when poll operation). It will be cleared when “ADC_GCR.adc_en”=0, or after read operation by software when “ADC_GCR.adc_rclr_en=1”. 0: data not valid 1: active after get the valid data
---	14:12	R	0x0	Reserved
CH2_DATA	11:0	R	0x0000	The A/D Channel 2 received data register. It will be cleared after read operation by software when “ADC_GCR.adc_rclr_en=1”.

156.2.6.6. ADC3_DR

offset address: 12’h014

Table 9- 37 SAR_ADC Register ADC3_DR

Name	Bits	Access	Reset value	Description
	31:16	R	0	Reserved.
Data_Valid3	15	R	0	Data valid signal(Only used for single mode when poll operation). It will be cleared when “ADC_GCR.adc_en”=0, or after read operation by software when “ADC_GCR.adc_rclr_en=1”. 0: data not valid 1: active after get the valid data
---	14:12	R	0x0	Reserved
CH3_DATA	11:0	R	0x0000	The A/D Channel 3 received data register. It will be cleared after read operation by software when “ADC_GCR.adc_rclr_en=1”.

156.2.6.7. ADC_CDR

offset address: 12’h028

Table 9- 38 SAR_ADC Register ADC_CDR

Name	Bits	Access	Reset value	Description
---	31:16	R	0x0	Reserved
CLKDIV	15:0	RW	0x00FF	The DAC Channel clock divider register. adc_clk = f_{pclk} / CLKDIV (f_{pclk} is APB clock frequency) Note: It suggests the CLKDIV >= 1. The adc clock frequency must satisfy f_{clk} >= 11x Sample rate

156.2.6.8. ADC_ISR

offset address: 12’h02c

Table 9- 39 SAR_ADC Register ADC_ISR

Name	Bits	Access	Reset value	Description
--	31:10	R	0x0	Reserved
rxfifo_full_intf	9	R	0x0	RX FIFO Full interrupt flag bit(only used in fifomode enable) 1 = Receive FIFO full 0 = Receive FIFO not full
rx_intf	8	R	0x0	Receiver data available interrupt flag bit(only used in fifomode enable). This bit is set when Receiver FIFO has received enough data(depend on rxtlf) 1 = receiver FIFO has available data 0 = receiver FIFO not available
reseved	7:4			
ch3_intf	3	R	0x0	channel 3 data interrupt, active high 1 = interrupt active 0 = no interrupt
ch2_intf	2	R	0x0	channel 2 data interrupt, active high 1 = interrupt active 0 = no interrupt
ch1_intf	1	R	0x0	channel 1 data interrupt, active high 1 = interrupt active 0 = no interrupt
ch0_intf	0	R	0x0	channel 0 data interrupt, active high 1 = interrupt active 0 = no interrupt

156.2.6.9. **ADC_IER**

offset address: 12’h030

Table 9- 40 SAR_ADC Register ADC_IER

Name	Bits	Access	Reset Value	Description
--	31:10	R	0x0	Reserved
rxfifo_full_ien	9	W/R	0x0	Receiver FIFO Full interrupt enable bit(only used in fifomode enable). 1 = Interrupt enabled 0 = Interrupt not enabled
rxien	8	W/R	0x0	Receiver FIFO interrupt enable bit(only used in fifomode enable). 1 = Interrupt enabled

				0 = Interrupt not enabled
reseved	7:4			
ch3_int_en	3	W/R	0x0	Channel 3 data interrupt enable, active high. 1 = Interrupt enabled 0 = Interrupt not enabled
ch2_int_en	2	W/R	0x0	Channel 2 data interrupt enable, active high. 1 = Interrupt enabled 0 = Interrupt not enabled
ch1_int_en	1	W/R	0x0	Channel 1 data interrupt enable, active high. 1 = Interrupt enabled 0 = Interrupt not enabled
ch0_int_en	0	W/R	0x0	Channel 0 data interrupt enable, active high. 1 = Interrupt enabled 0 = Interrupt not enabled

156.2.6.10. **ADC_ICR**

offset address : 12’h034

Table 9-41 SAR_ADC Register ADC_ICR

Name	Bits	Access	Reset Value	Description
--	31:10	R	0x0	Reserved
rxfifo_full_iclr	9	W/R	0x0	Receiver FIFO Full interrupt clear bit(only used in fifomode enable). 1 = Interrupt cleared 0 = Interrupt not cleared
rxiclr	8	W/R	0x0	Receiver interrupt clear bit(only used in fifomode enable). 1 = Interrupt cleared 0 = Interrupt not cleared
reseved	7:4			
ch3_int_clr	3	W	0x0	Channel 3 data interrupt clear, active high. 1 = Interrupt cleared 0 = Interrupt not cleared
ch2_int_clr	2	W	0x0	Channel 2 data interrupt clear, active high. 1 = Interrupt cleared 0 = Interrupt not cleared
ch1_int_clr	1	W	0x0	Channel 1 data interrupt clear, active high. 1 = Interrupt cleared 0 = Interrupt not cleared

ch0_int_clr	0	W	0x0	Channel 0 data interrupt clear, active high. 1 = Interrupt cleared 0 = Interrupt not cleared

156.2.6.11. ADC_COUNT

offset address : 12'h038

Table 9- 42 SAR_ADC Register ADC_COUNT

Name	Bits	Access	Reset value	Description
--	31:8	R	0x0	Reserved
adc_count	7:0	RW	0x1E	Channel switch interval time, the unit of this value is ADC clock cycle. The actual channel switch time = (adc_count+16) * ADC clock cycle time. Note: this register can only be configured before ADC controller is enabled.

156.2.6.12. ADC_RXREG

offset address: 12'h03c

Table 9- 43 SAR_ADC Register ADC_XREG

Name	Bits	Access	Reset value	Description
RXREG	31:0	R	0x0000	The receive data register. Bit[11:0] 10bits contains the converted data from SAR ADC Bit[14:12] Index of Analog input for received data The register is read-only.

156.2.6.13. ADC_CSTAT

offset address: 12'h040

Table 9- 44 SAR_ADC Register ADC_CSTAT

Name	Bits	Access	Reset value	Description
Reserved	15:1	R	0x0	Reserved

rxavl	0	R	0x0	Receive available data status The bit is set when Receiver has received an valid data. 1 = receiver FIFO has an available data 0 = receiver FIFO empty